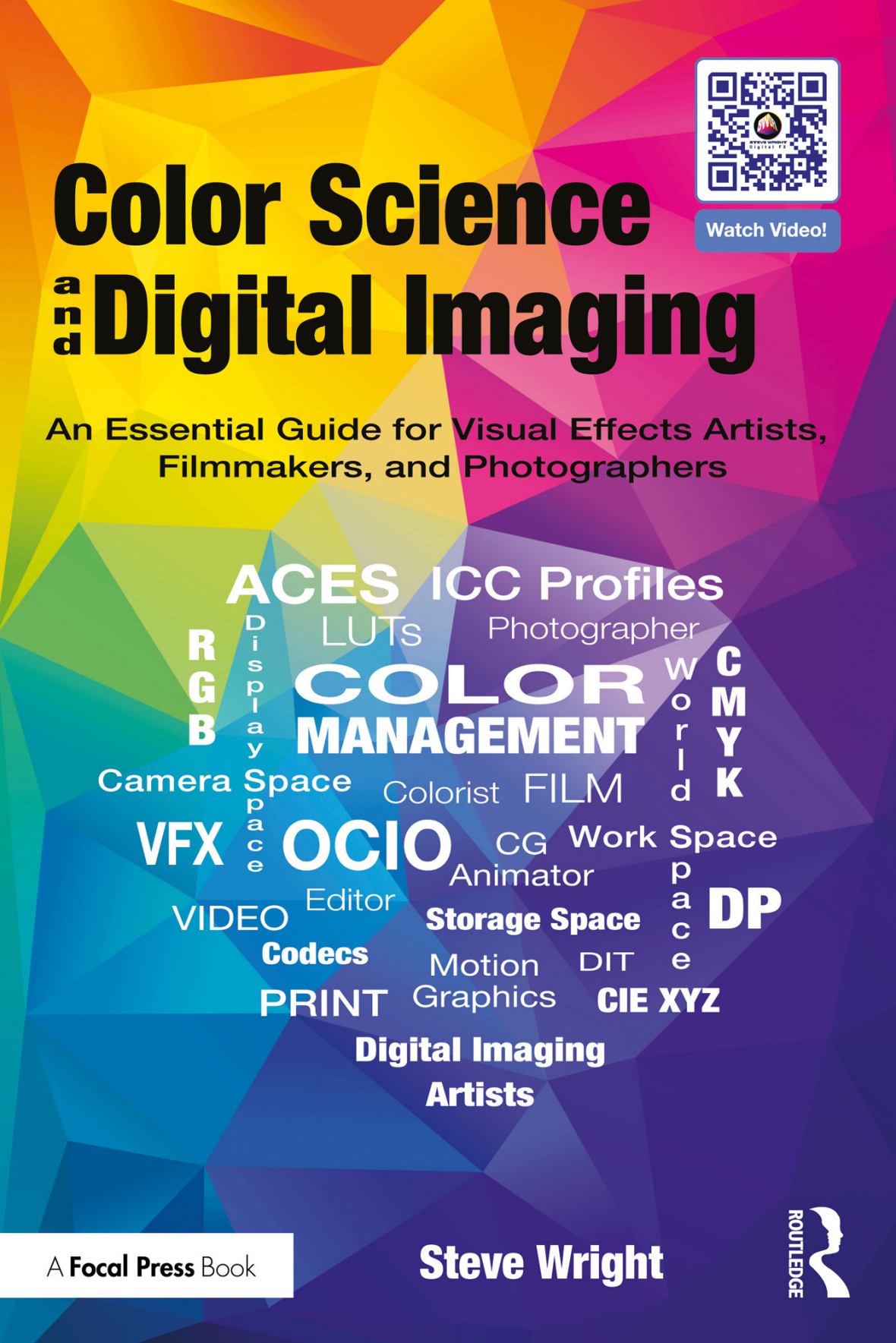
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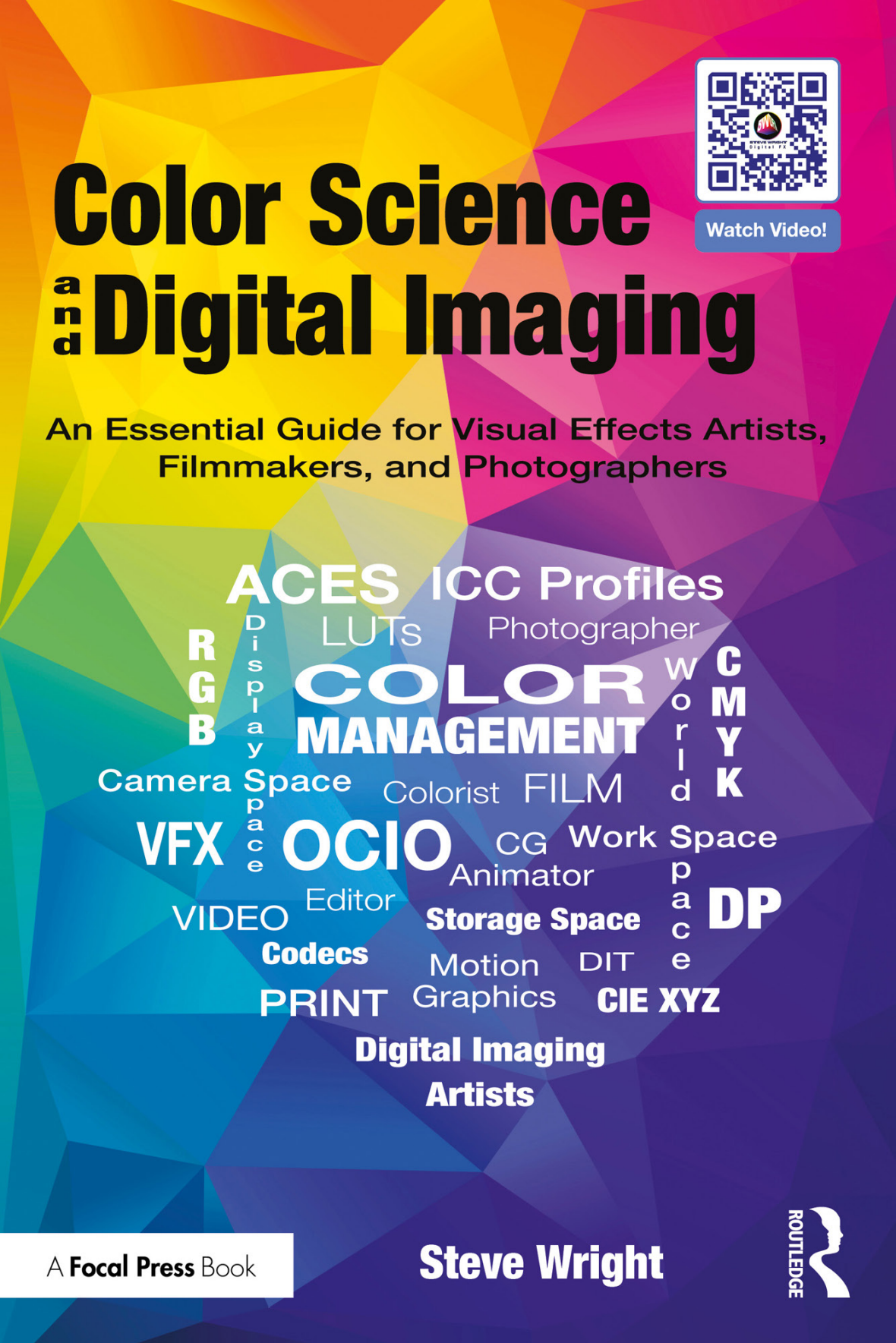
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Color Science and Digital Imaging

This book, written by a leading expert in the field of visual effects (VFX), demystifies the complex subject of color science and how it should be managed from project ideation to completion.

Readers will learn not only how things work, their capabilities and limitations, but also the color science that goes along with them, in an accessible and informative manner. Starting in the real world, with an understanding of light and our human perception of it, the book then explores how digital cameras "see" and digitize the scene. From the capture of these images, the author then explores the fundamental concerns of storage, editing, and managing color images, including with OpenColorIO and ACES, the Academy Color Encoding System.

Written for professional photographers, cinematographers, VFX, motion graphics and computer graphics artists, editors, and colorists, this book will provide you with knowledge of the upstream and downstream technology of your job that will not only give you a competitive advantage and help you to make better images, but also give you a real-world working knowledge of color science.

Steve Wright, an award-winning VFX pioneer and renowned educator, is the published author of two popular books: *Digital Compositing for Film and Video* and *Compositing Visual Effects*. In 2005, his attention turned full-time to educating the next generation of VFX practitioners, imparting his vast technical knowledge gleaned from 30-plus years of production experience. His love of digital imaging technology and color science is the mainstay of his training website, FX Ecademy.

色彩科学与数字成像

本书由视觉特效（VFX）领域的顶尖专家撰写，深入浅出地阐释了色彩科学这一复杂主题，并系统介绍了从项目构思到完成阶段应如何对其进行管理。

读者不仅将了解事物的工作原理、其能力与局限，还将以通俗易懂且信息丰富的方式学习相关的色彩科学。本书从现实世界出发，首先阐释光以及人类对光的感知，进而探讨数码相机如何“看见”并数字化场景。从图像采集出发，作者深入探讨了色彩图像的存储、编辑与管理等核心问题，其中涵盖了OpenColorIO与学院色彩编码系统（ACES）的应用。

本书专为专业摄影师、电影摄影师、视觉特效师、动态图形与计算机图形艺术家、剪辑师及调色师撰写，旨在为您提供行业上下游技术的深度认知。这些知识不仅能赋予您竞争优势、助您创作更出色的影像，更能让您掌握色彩科学在实际工作中的真实应用。

史蒂夫·赖特，一位屡获殊荣的视觉特效先驱和知名教育家，是两本畅销书《电影与视频数字合成》和《视觉特效合成》的作者。2005年起，他全身心投入培养新一代视觉特效从业者，传授其三十余年制作经验中积累的深厚技术知识。他对数字影像技术与色彩科学的热爱，构成了其培训网站FX Ecademy的核心支柱。

"This book is a treasure trove of knowledge about the technology and color science used for creating digital images. I wish this had been available when I was starting out."

Grant McKean, *Senior Flame VFX Artist, London*

"The author conducts the reader through a journey of the entire digital imaging pipeline providing essential information all the way. By the end of the book, the reader will have painlessly accumulated an astonishing depth of knowledge of color science."

Eberhard Hasche, *Senior Color Researcher at the Department of Computer Science and Media Technology at Technical University of Applied Sciences, Brandenburg, Germany*

"The topic of color science is difficult to teach but Steve Wright has a succinct way of communicating this complex subject matter making it easy even for the layperson to comprehend. The visual impact of the animated videos will prove to be an invaluable learning tool for my courses. As an educator, I highly recommend this book!"

Shaun Peyman, *Senior Lecturer of VFX at the Media Design School of Auckland, New Zealand*

"As a supervisor of Visual Effects and CG artists, I see my team struggle with the issues covered by this book all the time. It will be a boon to their careers and the industry as a whole when everybody knows this stuff by heart."

Isa Alsup, *Senior VFX and CG Supervisor*

这本书是关于创建数字图像所用技术与色彩科学的知识宝库。真希望在我起步时就有这样的资源。

格兰特·麦基恩，高级火焰特效艺术家，伦敦

作者引导读者踏上一段贯穿整个数字成像流程的旅程，全程提供关键信息。读完本书后，读者将能轻松积累令人惊叹的色彩科学知识深度。

埃伯哈德·哈舍，德国勃兰登堡应用技术大学计算机科学与媒体技术系高级色彩研究员

色彩科学这一主题难以教授，但史蒂夫·怀特有一种简洁的方式来传达这一复杂主题，使得即便是外行也能轻松理解。动画视频的视觉冲击力将被证明是我课程中宝贵的学习工具。作为一名教育工作者，我强烈推荐这本书！

肖恩·佩曼，新西兰奥克兰媒体设计学院视觉特效高级讲师

作为视觉特效和CG艺术家的主管，我经常看到我的团队为本书所涵盖的问题而挣扎。当每个人都能熟记这些知识时，这将成为他们职业生涯乃至整个行业的福音。

伊萨·阿尔苏普，高级视觉特效与CG总监

Color Science and Digital Imaging

An Essential Guide for Visual Effects Artists, Filmmakers, and Photographers

Steve Wright

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色彩科学与数字成像

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视觉特效艺术家、电影制作人和摄影师必备指南

史蒂夫·赖特

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Foreword

Technical color themes are often neglected or underestimated in media production. When I watch a live webinar now and then and ask a question about the color workflow, I immediately get a decent number of upvotes, which underlines my interest in color topics. Recently, I confronted ChatGPT several times with the same question about calculating the transfer function (gamma) in the sRGB color space. All ten answers were wrong, but at least one contained a correct formula. Since artificial intelligence is not intelligent, it adopts the sponginess of different statements anywhere and does not necessarily make them better. This is also often seen in tutorials that only touch on the color problem. The statements are not wholly wrong, but mostly they are not entirely correct either. What is missing is a trustworthy source. This is why this book was written. It combines all the scattered information— entailing physics, optics, vision science, mathematics, and computer science—and assembles it into a single comprehensive knowledge base.

When you take a photo with a cell phone, the stages from capture to display are completed in the blink of an eye. The photo is finished and ready to use. These steps can take weeks in the film industry, from live-action footage to final rendering. This is mainly because countless sources have to be linked together, in addition to all kinds of testing and experimenting with the content. There are primarily a dozen film cameras, all with their different color spaces. Still, action and consumer cameras, negative film, photos of all kinds, and even 3D models with their textures have to be integrated. Renderings from various software packages are then added during further processing. These elements also have different color spaces that need to be adapted.

Visual effects (VFX) are at the heart of this way of working in the film industry. These are often perceived as the creators of dinosaurs, spaceships,

技术色彩主题在媒体制作中常常被忽视或低估。如今，当我偶尔观看在线研讨会并询问有关色彩工作流程的问题时，总能立即获得不少点赞，这凸显了人们对色彩话题的兴趣。最近，我多次向ChatGPT提出关于计算sRGB色彩空间中传递函数（伽马值）的同一问题。十次回答全部错误，但至少有一次包含了一个正确的公式。由于人工智能并非真正的智能，它只是吸收各处不同的模糊陈述，未必能使其更准确。这种情况在仅浅谈色彩问题的教程中也屡见不鲜——这些说法并非全错，但大多也不完全正确。缺失的正是可靠的信息来源。本书的编写正是为了填补这一空白。它汇聚了物理学、光学、视觉科学、数学和计算机科学等领域零散的知识，并将其整合成一个全面统一的知识体系。

当你用手机拍照时，从拍摄到显示的各个阶段都在眨眼间完成。照片即刻生成，随时可用。而在电影行业中，从实拍镜头到最终渲染，这些步骤可能需要数周时间。这主要是因为除了对各种内容进行测试和实验外，还需要将无数素材源连接在一起。光是电影摄像机就有十几种，每种都有不同的色彩空间。此外，动作相机和消费级相机拍摄的素材、负片、各类照片，甚至带有纹理的3D模型都需要整合进来。在后续处理中，还会加入来自不同软件包的渲染效果。这些元素同样拥有各异的色彩空间，需要进行适配调整。

视觉效果（VFX）是电影行业这种工作方式的核心。它们常被视为恐龙、宇宙飞船的创造者，

explosions, monsters, and heroes. What is overlooked is that VFX are frequently invisible. Since they have to integrate seamlessly into the film shots, the requirements for the color workflow are high. Considering that a dozen or more companies share the VFX work on blockbuster films, ensuring that the results are congruent is essential. In addition, departments are becoming increasingly specialized, which means that the artist or technical director involved loses sight of the big picture—and the color workflow is part of this.

The constant evolution of working methods leads to the incorporation of new technologies. A modern example is the usage of massive light-emitting diode walls for virtual production that play real-time images from game engines like Epic Unreal. Here, too, color management has a decisive role. A completely independent approach to dealing with color is necessary to solve the tasks efficiently and with high quality.

Many artists know the psychological component of color but don't understand the technical part. This is mainly caused by the fact that nearly all levels of color education rely on the sRGB color space and the display-referred workflow alone. But this is a castle built on quicksand. For years, generating content in the sRGB color space (or the corresponding Rec709 video color space) was sufficient because the displays could not display colors outside it. But this is changing rapidly. The DCI-P3 color space for reproducing color values larger than sRGB is already being used for film projection and Apple notebooks. However, a technology dramatically impacting media production is just around the corner: high dynamic content for moving images. Dolby Vision Cinemas' high contrast during playback is a harbinger of what will come. These tasks cannot be accomplished with conventional color pipelines.

The structure of this book takes all these challenges into account and follows the practical workflow of film production. It covers digital camera capture, compression schemes, linear and sRGB workspace, plus attributes and inner workings of display devices like ultra-high-definition TV and digital cinema projectors. So, the reader always knows the framework conditions for the color issues discussed. This approach preserves practical relevance and makes it easier to follow the explanation because you always know where you are and don't get lost in theory. In most color books, the discussion of color issues is mostly too specific and too much into color science. This book focuses explicitly on the digital imaging artist and the tools

爆炸、怪物和英雄。常被忽视的是，视觉特效往往是无形的。由于它们必须无缝融入电影镜头，对色彩工作流程的要求极高。考虑到一部大片通常由十几家甚至更多公司分担视觉特效工作，确保成果的一致性至关重要。此外，各部门日益专业化，这意味着参与的艺术家或技术指导容易忽视整体画面——而色彩工作流程正是其中的一环。

工作方法的不断演进促使新技术的融入。一个现代实例是使用大型发光二极管墙进行虚拟制作，这些墙面实时播放来自Epic Unreal等游戏引擎的图像。在此过程中，色彩管理同样具有决定性作用。要高效且高质量地解决相关任务，必须采用一套完全独立的色彩处理方案。

许多艺术家了解色彩的心理成分，却不理解其技术部分。这主要是由于几乎所有层级的色彩教育都仅依赖于sRGB色彩空间和显示参考工作流程。但这无异于在流沙上筑城堡。多年来，在sRGB色彩空间（或对应的Rec709视频色彩空间）中制作内容已足够，因为显示器无法呈现该范围外的色彩。然而这一现状正迅速改变。用于再现比sRGB更广色域的DCI-P3色彩空间已应用于电影放映和苹果笔记本电脑。但真正将深刻影响媒体制作的技术已近在眼前：动态影像的高动态范围内容。杜比视界影院播放时的高对比度正是未来变革的先声。这些任务无法通过传统的色彩处理流程实现。

本书的结构充分考虑了所有这些挑战，并遵循电影制作的实际工作流程。它涵盖了数字相机捕捉、压缩方案、线性和sRGB工作空间，以及超高清电视和数字电影投影仪等显示设备的特性和内部工作原理。因此，读者始终清楚所讨论色彩问题的框架条件。这种方法保持了实际相关性，并使解释更容易理解，因为你始终知道自己所处的位置，不会迷失在理论中。在大多数色彩书籍中，对色彩问题的讨论往往过于具体，过多深入色彩科学。本书则明确聚焦于数字影像艺术家及其工具。

and technologies used for creating digital images. The author, with many years of experience in teaching, has written two books on this topic and offers high-quality training online. The books are well written—often with his unique sense of humor—and easy to understand. The author has excellent background knowledge. His practical experiences and overview of the entire VFX workflow are significant advantages. With this knowledge, he can better relate the color science issues to the practical realities than an author specializing in pure color topics.

One innovation by the author that I particularly appreciated was the embedding of some two dozen QR codes in the text to access videos in line with the narrative. Just hover a cell phone over the QR code and it plays a video. No logins, no passwords. The quality of these videos is high because more than 20 years of production experience on blockbusters and commercials and 20 years of teaching are incorporated into these short instructive videos and all the figures in the book.

All this accompanies students in their journey through the digital imaging pipeline. Along the way, they will be gently taught the relevant color science. As a result, they will be able to produce better pictures faster by understanding the underlying technology of the digital imaging pipeline. Other benefits include avoiding costly color mistakes and gaining a competitive advantage relative to other artists in the field. By the end of the book, the student will have painlessly accumulated an astonishing depth of knowledge of color science.

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以及用于创建数字图像的技术。作者拥有多年的教学经验，就此主题撰写了两本书，并提供高质量的在线培训。这些书籍文笔流畅——常带有他独特的幽默感——且易于理解。作者具备出色的背景知识。他的实践经验和对整个VFX工作流程的概述是显著优势。凭借这些知识，他能够比专攻纯色彩主题的作者更好地将色彩科学问题与实际应用联系起来。

作者的一项创新让我特别欣赏，就是在文本中嵌入了大约二十几个二维码，用于访问与叙述内容相符的视频。只需将手机悬停在二维码上，视频就会播放。无需登录，无需密码。这些视频质量很高，因为制作团队拥有超过20年的大片和广告制作经验，以及20年的教学经验，这些都被融入了这些简短的教学视频和书中的所有图表中。

所有这一切都将伴随学生走过数字成像流程的旅程。在此过程中，他们将逐步学习相关的色彩科学知识。因此，通过理解数字成像流程的底层技术，他们将能够更快地制作出更好的图片。其他好处包括避免代价高昂的色彩错误，并获得相对于该领域其他艺术家的竞争优势。到本书结束时，学生将在不知不觉中积累起惊人的色彩科学知识深度。

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Preface

In the beginning, there was only black-and-white print, photography, movies, and television. But today's world presents the digital imaging artist with an ever-evolving panoply of color media workspaces, from image capture to image display and everything in between, each with its own characteristics and rules for working in those spaces. When we inevitably cross these Work Spaces, this creates the sometimes confusing and almost certainly daunting task that digital artists face to keep the correct color space without introducing issues as they traverse the color and digital imaging pipelines no matter their line of work, whether film, television, photography, video editing, print, graphic arts, or my area of expertise—visual effects (VFX).

Well, you may be asking yourself: who is Steve Wright and why did he decide to write this book? I have been entertaining people with pixels since the early 1980s, programming videogames at Atari using the Stella chip and later managing the Home Video Game division. In those days, you had to do a lot with little! I took several things away from that experience that have proved to be my "superpower" in life. One of the most important things is to read the manual, and when there isn't one—write it! Documentation is "great to have," but when it is presented as disparate pieces of information with no structure it is difficult to understand the complexities with which you are presented. That's why I wrote the Stella Programmer's Guide, on the graphic chip that drove the Atari 2600—my first foray into writing. And I have been a prolific writer ever since!

I owned my own VFX studio, Sidley Wright & Associates, in the 1990s, worked at several Fortune 500 companies, and have had the immense privilege of working with and learning from the movers and shakers in my

起初，只有黑白印刷、摄影、电影和电视。但当今世界为数字影像艺术家呈现出一套不断演变的彩色媒体工作空间全景，从图像采集到图像显示以及其间的一切环节，每个空间都有其独特的特性和操作规则。当我们不可避免地跨越这些工作空间时，便产生了数字艺术家面临的挑战——有时令人困惑，且几乎必然艰巨——即无论从事电影、电视、摄影、视频剪辑、印刷、平面艺术，还是我的专业领域视觉特效（VFX），都必须在穿梭于色彩与数字影像流程时保持正确的色彩空间，避免引入问题。

那么，你可能会问自己：史蒂夫·赖特是谁？他为何决定撰写这本书？自20世纪80年代初以来，我一直在用像素为人们带来欢乐——最初在雅达利公司使用Stella芯片编写电子游戏程序，后来负责家用视频游戏部门。在那个年代，你不得不用有限的资源完成大量工作！那段经历让我收获了几样东西，后来证明它们成了我人生中的“超能力”。最重要的一点是：阅读手册，如果没有手册——就自己写！文档“有总比没有好”，但如果信息零散、缺乏结构，就很难理解所面对的复杂问题。正因如此，我撰写了《Stella程序员指南》，这本关于驱动雅达利2600游戏机的图形芯片的指南，是我写作生涯的首次尝试。而自那以后，我便成了一位多产的作家！

我在1990年代拥有自己的视觉特效工作室Sidley Wright & Associates，曾任职于多家《财富》500强企业，并有幸与行业内的领军人物共事并从中学习。

industry like Glenn Kennel, the Cinesite Hollywood Kodak color scientist who developed the Cineon log file format and designed the Kodak film scanners and film recorders. My second foray into color science was with Dr. Mitch Bogdanowicz, the Kodak color scientist who developed the Digital Intermediate process. I was also blessed with the outstanding friendships of Ray Feeney, ACES Project Committee Co-chair, who reviewed my writings on ACES and answered any color science question put to him, and Doug Walker, Technology Lead for Color Science at Autodesk, who also serves as the Chief Architect for OpenColorIO (OCIO) and is on the Academy's advisory board for the ACES project.

Over my extensive VFX compositing career, I became the "go-to guy" that fellow artists would consult with to solve their knotty digital imaging and color space problems. The reason for this is simple: I learned about the jobs upstream and downstream of my job to better understand what part I played and how I could help my colleagues solve their issues. Knowledge is powerful! So, in 2002, I began my transition from practitioner to educator with the initial release of my book *Digital Compositing for Film and Video*, sharing the knowledge and insights gleaned from several decades of experience compositing visual effects. Through my interactions with a vast number of aspiring compositors and VFX artists across the globe, a common challenge emerged in their knowledge: a profound gap in understanding and applying color theory.

This book is my effort to bridge that educational gap, making the information accessible and immediately applicable. To this end, I have pulled together all of that disparate color science and digital imaging information you need to understand in this essential guide, explaining how World Space or the real world creates color, how we humans perceive it, and how that is different from Camera Space or how the camera sees and digitizes the real world. Then, we will explore Storage Space and what file formats are best suited to protect your images. Let's not forget about our Work Space, where you will learn about preserving your image quality, as well as the Display Space technologies of today and tomorrow. To round out our incredible journey, we will look at the all-important aspects of color management like ACES and OCIO that can make or break a well-run pipeline!

像格伦·肯内尔这样的行业先驱，他是Cinesite好莱坞柯达色彩科学家，开发了Cineon对数文件格式，并设计了柯达胶片扫描仪和胶片记录仪。我第二次涉足色彩科学领域是与米奇·博格达诺维奇博士合作，他是柯达色彩科学家，开发了数字中间片工艺。我还很幸运地与雷·菲尼建立了深厚的友谊，他是ACES项目委员会联合主席，审阅了我关于ACES的著作，并回答了我提出的任何色彩科学问题；以及道格·沃克，他是欧特克的色彩科学技术负责人，同时担任OpenColorIO（OCIO）的首席架构师，并任职于美国电影艺术与科学学院的ACES项目顾问委员会。

在我漫长的视觉特效合成职业生涯中，我逐渐成为同行艺术家们解决棘手数字成像与色彩空间问题时首选的咨询对象。原因很简单：我深入了解了工作流程中上下游环节的职责，从而更清晰地认识到自身在整体中的定位，以及如何协助同事解决问题。知识就是力量！因此，在2002年，随着我的著作《影视数字合成技术》首次出版，我开始了从实践者到教育者的转型，分享数十年视觉特效合成经验中积累的知识与洞见。通过与全球大量 aspiring 合成师和视觉特效艺术家的交流，我发现他们普遍面临一个知识层面的挑战：对色彩理论的理解与应用存在显著断层。

本书旨在弥合这一教育鸿沟，让相关知识变得触手可及且能即时应用。为此，我汇集了所有你必须掌握的色彩科学与数字成像知识，在这本核心指南中阐释：世界空间（即现实世界）如何生成色彩，人类如何感知色彩，以及这与相机空间（即相机如何观察并数字化现实世界）有何不同。接着，我们将探索存储空间，分析哪些文件格式最能保护你的图像。当然不会忽略工作空间——你将在此学会如何保持图像质量，并了解当下与未来的显示空间技术。最后，我们将审视色彩管理中至关重要的环节，例如ACES与OCIO——这些要素足以成就或摧毁一条高效运行的制作管线！

Whether you are a visual effects artist, computer graphics animator, colorist, photographer, cinematographer, motion graphic artist, editor, or anyone who works with digital images, join me on this user-friendly deep dive into color science to help you, the digital artist, understand and cope with the intricacies, the pitfalls, and the brilliance of color management and digital imaging pipelines!

无论你是视觉特效师、计算机图形动画师、调色师、摄影师、电影摄影师、动态图形艺术家、剪辑师，还是任何与数字图像打交道的人，请与我一同踏上这场用户友好的色彩科学深度探索之旅，帮助你——这位数字艺术家——理解并应对色彩管理与数字成像流程中的复杂细节、潜在陷阱与卓越魅力！

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Acknowledgments

Writing a technical book, especially one this technical, is a daunting task and requires the assistance of able colleagues and supportive professionals. At the top of the list of supportive professionals is Ray Feeney, my go-to guy for ACES questions. And why not? Ray is a four-time Academy Award winner for Scientific and Engineering Achievement; *plus,* he is the ACES Project Committee Co-chair. He is the ace of ACES!

I understand scopes but wanted the wisdom of someone who uses them daily, so I turned to Jon Alvord, a previous Professor of Film and Television at Savannah College of Art & Design and a VFX Editor for Troublemaker Studios, Match Frame, CafeFX, and Digital Domain. He knows editing and lives and loves scopes.

Then there was the small army of book reviewers to whom I sent a draft copy of the book, that they might read and comment. They represent a cross-section of senior artists and educators that kept me focused on exactly what artists need to know, and I have them to thank for that:

- Eberhard Hasche, Senior Color Researcher at the Department of Computer Science and Media Technology, Technical University of Applied Sciences, Brandenburg, Germany;
- Ricardo Ferreira, VFX Professor at the School of Arts, *Universidade Católica Portuguesa*, Porto, Portugal;
- Shaun Peyman, Senior Lecturer of VFX at the Media Design School of Auckland, New Zealand;

撰写一本技术书籍，尤其是如此专业的技术书籍，是一项艰巨的任务，需要得力同事和支持性专业人士的协助。在支持性专业人士名单中，首屈一指的是雷·菲尼，他是我请教ACES问题的首选专家。这不足为奇！雷曾四次荣获奥斯卡科学与工程成就奖；此外，他还是ACES项目委员会的联合主席。他堪称ACES领域的王牌！

我理解示波器，但想请教日常使用者的真知灼见，于是找到了乔恩·阿尔沃德——他曾是萨凡纳艺术与设计学院的影视教授，也是Troublemaker Studios、Match Frame、CafeFX和Digital Domain等公司的视觉特效剪辑师。他精通剪辑，对示波器不仅了如指掌，更怀有深切的热爱。

然后还有一小群书评人，我向他们寄送了书籍的草稿副本，以便他们阅读并提出意见。他们代表了资深艺术家和教育家的一个横截面，让我始终专注于艺术家真正需要了解的内容，为此我要感谢他们：

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- 里卡多·费雷拉，葡萄牙波尔图葡萄牙天主教大学艺术学院视觉特效教授；
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- Grant McKean, Senior Flame VFX Artist in London;
- Isa Alsup, Senior VFX and CG Supervisor; and
- Jarmila Seflova, Senior Nuke VFX Compositor and Colorist, Studio 11211.

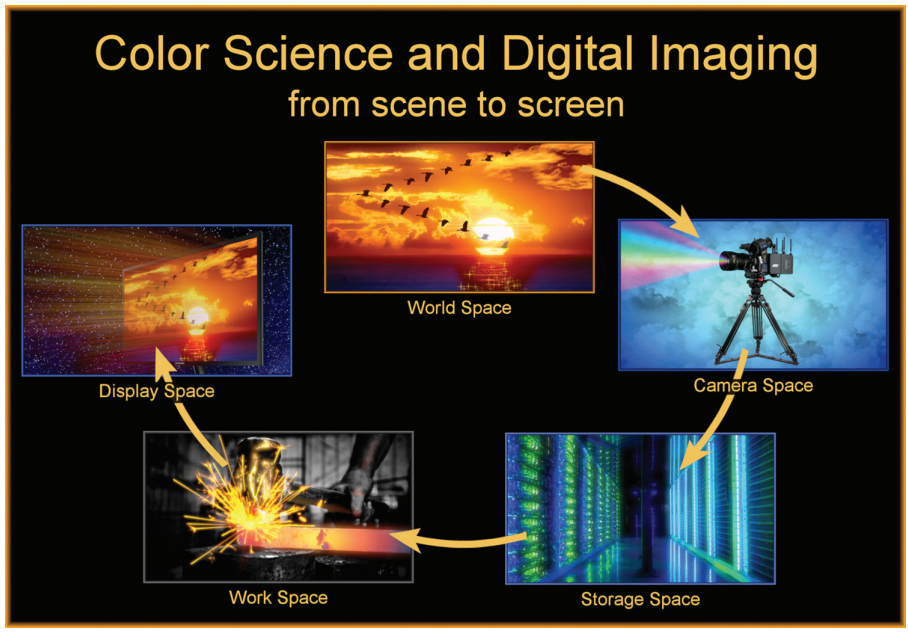
The biggest acknowledgment of all goes to my amazing wife, Diane, whose support was manifested in so many ways. She proofread every word of the book to fix my bad word usements, collected artwork for illustrations, prepared the glossary, indexed the book, wrangled the videos, and created their QR codes—all while keeping the world at bay so I could focus on writing. How did I get so lucky to have her as my wife?

•格兰特·麦基恩，伦敦高级火焰视觉特效艺术家；

•Isa Alsup，高级视觉特效与CG总监；以及

•Jarmila Seflova，高级核弹视觉特效合成师与调色师，11211工作室。

最深的感谢献给我了不起的妻子黛安，她的支持体现在方方面面。她校对了书中的每一个字，修正我的用词不当，为插图收集艺术作品，准备了词汇表，编制了索引，整理了视频，并生成了它们的二维码——与此同时，她为我抵挡了外界的纷扰，让我能专注于写作。我是何其幸运，能娶她为妻?



Introduction

The Five Spaces of the Digital Imaging Pipeline

An image passes through five distinct “spaces” as it travels through the digital imaging pipeline. Along the way, its original artistic intent must be preserved as it passes through sometimes several color spaces. This book traces that path, and at each stage reveals the technology and the color science that underpin it. This is the book for digital artists who want to produce better images by understanding the tech and color science used in the modern digital imaging pipeline. Designed for easy reading, it includes several short videos accessible with a cell phone through QR codes embedded in the text to enhance the learning experience even more.

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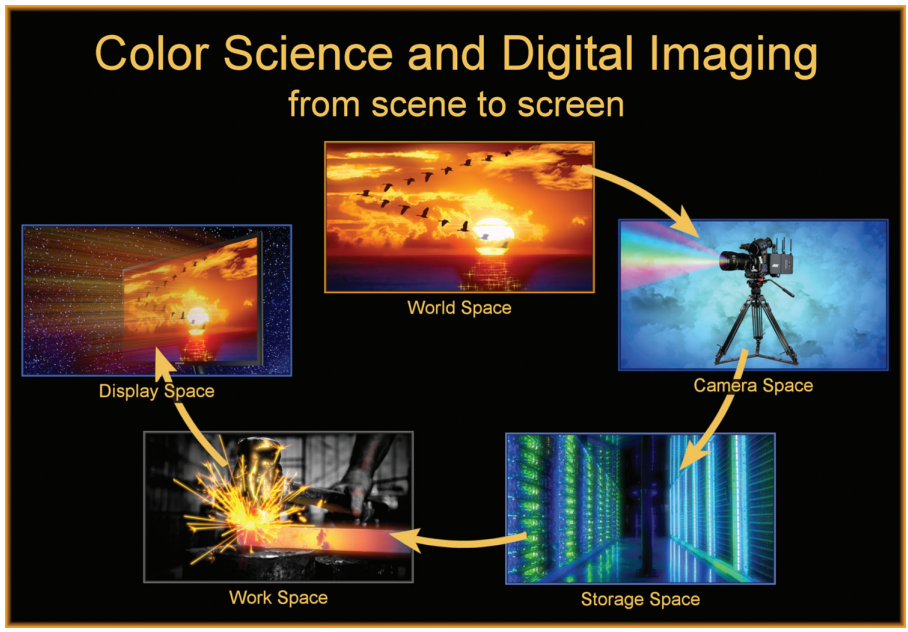

引言

数字成像流水线的五个空间

一幅图像在数字成像流程中会穿越五个不同的“空间”。在此过程中，其原始艺术意图必须得以保留，尽管它有时会经过多个色彩空间。本书追溯了这一路径，并在每个阶段揭示了支撑其运作的技术与色彩科学。这本专为数字艺术家打造的书籍，旨在帮助他们通过理解现代数字成像流程中使用的技术与色彩科学，创作出更优质的作品。本书设计便于阅读，内嵌多个可通过手机扫描二维码观看的短视频，进一步提升了学习体验。

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World Space is where our journey begins, to learn about the light in the real world that makes our pictures—where it comes from, how we measure it, how it interacts with surfaces, and how we see it. Along the way, we will learn key color science concepts such as emission spectrums, color temperature, and standard illuminants. Since color science is based on the human visual system, we will explore how we perceive color—the standard measure of the human visual system—and how we can see a world of color when the universe is in fact black and white. And we see all of this color by sampling just three tiny segments of the entire electromagnetic spectrum.

Camera Space is where we learn about the cameras that capture images by turning light into numbers. We will see how the Bayer array turns the black-and-white light of the real world into RGB colors using math and magic. But each model sees the world a bit differently, and they bring their own baggage to the imaging pipeline. They clip, distort both image and color, add noise, exhibit rolling shutter and introduce vignetting—two different ways, no less. Along the way, the color science for digital cameras is revealed, along with their performance specs.

Storage Space covers where digital images go when they are exported from the camera and saved to a file. Understanding file formats, with their codecs and compressions that can mutilate them, becomes the key to avoiding damaging your beautiful images. Log compression of images has become a standard industry practice that few understand the how and why of, so log images will be demystified so you can take full advantage of their virtues and avoid their pitfalls. Then there is the long list of file formats for cinematography, visual effects (VFX), and photography; the right one to use is situation dependent and the wrong one can damage your project.

Work Space examines the environment in which images are read in from the disk and manipulated. We start with a little color theory and then move to the difference between color models and color spaces, with their gamuts, white points, and transfer functions. Should we work in linear space or display space? Should the math be integer or float? As a photographer, should you work in a super-sized gamut; and if you do, how do you remap it back to normal? If you want to preserve the work for the future, you will need the essential information in the archiving section of this chapter.

世界空间是我们旅程的起点，在这里我们将了解真实世界中构成我们图像的光线——它从何而来，我们如何测量它，它如何与表面相互作用，以及我们如何看见它。沿途，我们将学习关键的颜色科学概念，例如发射光谱、色温和标准光源。由于颜色科学基于人类视觉系统，我们将探索我们如何感知颜色——人类视觉系统的标准度量——以及当宇宙实际上是黑白时，我们如何能看见一个多彩的世界。而我们能看见所有这些颜色，仅仅是通过对整个电磁频谱中三个微小片段进行采样。

相机空间是我们了解相机如何将光线转化为数字以捕捉图像的地方。我们将看到拜耳阵列如何运用数学与魔法，将现实世界的黑白光线转化为RGB色彩。但每款相机模型观察世界的方式略有不同，它们各自为成像流程带来独特的“包袱”——它们会裁剪画面、扭曲图像与色彩、添加噪点、产生滚动快门效应，并带来两种不同的暗角效果。在此过程中，数码相机的色彩科学及其性能参数也将逐一展现。

存储空间涉及数字图像从相机导出并保存为文件后的去向。理解文件格式及其可能损坏图像的编解码器和压缩方式，成为避免破坏精美图像的关键。图像对数压缩已成为行业标准实践，但鲜有人了解其原理与缘由，因此我们将揭开对数图像的神秘面纱，助您充分利用其优势并规避缺陷。此外，电影摄影、视觉特效（VFX）和摄影领域存在众多文件格式；正确选择需视具体情况而定，而错误选择可能会损害您的项目。

工作空间探讨了从磁盘读取并处理图像的环境。我们首先简要介绍色彩理论，然后转向色彩模型与色彩空间之间的区别，包括它们的色域、白点和传递函数。我们应该在线性空间还是显示空间中工作？数学计算应使用整数还是浮点数？作为摄影师，你是否应在超大色域中工作；如果这样做，如何将其重新映射回正常范围？若想为未来保存工作成果，你需要本章归档部分中的关键信息。

Display Space looks at the last link in the color science chain, where we project our images in a theater or view them on a monitor or high-definition television (HDTV). This is where we learn about the attributes of display devices such as their contrast ratios, gamuts, white points, brightness, frame rates, and the correct viewing surround. We then take a chainsaw to flat-panel displays to see their inner workings and review the specs for HDTV, ultra-high-definition television (UHDTV), and digital cinema. We also peer into the future to get a heads-up on the new specs and standards for the awesome next-gen displays.

The last chapter, **Color Management,** is where we learn about its mission both to preserve the original artistic intent and to provide a consistent appearance across all display platforms. We will learn the strengths and weaknesses of one-dimensional (1D) and three-dimensional (3D) look-up tables (LUTs) and take a rare look inside the four key color grading operations of lift, gamma, gain, and offset to see how each op affects the image data and the picture look. There is also a section on scopes and a grand tour of the Digital Intermediate (DI) color grading suite. The big finish is all about color management, featuring International Color Consortium (ICC) profiles, OpenColorIO (OCIO), and, of course, the Academy Color Encoding System (ACES).

显示空间关注色彩科学链的最后一环，即我们在影院投影图像或在显示器、高清电视（HDTV）上观看图像的环节。在此，我们将了解显示设备的各项特性，例如对比度、色域、白点、亮度、帧率以及正确的观看环境。接着，我们将拆解平板显示器以探究其内部结构，并审视高清电视（HDTV）、超高清电视（UHDTV）和数字影院的技术规格。同时，我们也将展望未来，提前了解下一代卓越显示设备的新规格与标准。

最后一章《色彩管理》将探讨其双重使命：既要保留原始艺术意图，也要确保所有显示平台呈现一致的外观。我们将了解一维（1D）和三维（3D）查找表（LUT）的优缺点，并深入解析四大核心调色操作——提升（lift）、伽马（gamma）、增益（gain）和偏移（offset），观察每种运算如何影响图像数据与画面效果。本章还包含示波器解析环节，并全景展示数字中间片（DI）调色系统。压轴部分全面聚焦色彩管理体系，涵盖国际色彩联盟（ICC）配置文件、OpenColorIO（OCIO），以及美国电影艺术与科学学院的色彩编码系统（ACES）。

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1 World Space

Our story begins in the real world, of course, where the original scene is captured. The real world is deceptively complex and does not work at all like our eyes, cameras, or display devices, so we need to start with an understanding of how it creates light and how we perceive it. This will be key for learning about why cameras see the real world differently than we do and why, when photographing a scene, what you see is not necessarily what you get. Understanding this difference is essential if we want to capture the real-world scene with a camera in a manner that matches our perceptions.

We begin with the nature of light itself, what it's made of, and how it behaves. After reviewing light measuring units, we tackle the dynamic range of the universe and how that affects our image capture. Along the way, there will be a few sober color science moments. Then we dash off to see the puzzling difference between the smooth spectrums of incandescent light sources like the sun vs. the spiky spectrums of luminescent light sources like LEDs. We then follow the photons down to surfaces and see both the mundane and exotic interactions light can have with materials, and even itself in the case of constructive and destructive interference.

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1 世界空间

我们的故事当然始于现实世界，那里是原始场景被捕捉的地方。现实世界看似简单实则复杂，其运作方式与我们的眼睛、相机或显示设备截然不同，因此我们首先需要理解现实世界如何产生光线，以及我们如何感知光线。这将是理解相机为何以不同于人类的方式观察现实世界的关键，也是理解为何拍摄场景时“所见非所得”的核心。若想用相机以符合人类感知的方式捕捉现实场景，理解这种差异至关重要。

我们从光的本质说起，探讨它的构成与行为规律。在回顾光的测量单位后，我们将解析宇宙的动态范围及其对图像捕捉的影响。过程中会穿插几个严谨的色彩科学知识点。接着，我们将快速探究一个令人困惑的现象：太阳等白炽光源的平滑光谱与LED等发光光源的尖峰光谱之间的差异。随后，我们将追踪光子抵达物体表面的过程，观察光与材料之间平凡而奇特的相互作用——甚至包括光自身在相长干涉与相消干涉中产生的奇妙现象。

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While some may disagree, I consider human beings and our visual system to be part of the real world, so the human perceptual system is part of this chapter. We take a close look at how we see, the nature of color perception, the very non-linear response of the eye, and how we measure colors in "eye space." Our vision is highly adaptable to viewing conditions, so it presents a moving target for understanding color science, but we will develop coping mechanisms to deal with that. So, join me to begin an adventure with light that will require some physics, perceptual science, a pinch of optics, and just a dash of quantum mechanics—all presented with a user-friendly narrative and some delightful surprises along the way with brief embedded videos. Bring your cell phone.

1.1 The Nature of Light

Our story begins with light. Capturing and digitizing it is the foundational principle of all our digital imaging powers, so understanding its nature is core to understanding digital images. Light is a perturbation of the electromagnetic (EM) spectrum, a wave of electrical and magnetic disturbance, and is made of the same stuff as radio waves. The only difference between radio waves and light is their wavelength, one of the attributes of light illustrated in Figure 1.1. Keep in mind that we can use wavelength and frequency interchangeably because the wavelength determines the frequency. As a case in point, when talking about radio waves, we most often refer to their frequencies (i.e., 99.7 on your FM dial is 99.7 MHz); but when we talk about its microscopic cousin, visible light, we refer to its wavelength.

Attributes of light include direction and speed (known as the speed of light) but light has many more interesting attributes, and every one of them affects our pictures in some way. Here we will closely examine our friend the photon, as it is the central actor in our pictures.

1.1.1 Wavelength

The wavelength of light is the distance from peak to peak in a light wave and is measured in nanometers (nm). When talking about visible light (yes, there is invisible light!), the wavelength is roughly between 400 nm and 700 nm, depending on your human test subject. Visible light is, of course, those

尽管有人可能持不同意见，但我认为人类及其视觉系统是现实世界的一部分，因此人类的感知系统也属于本章的讨论范畴。我们将深入探讨视觉的工作原理、色彩感知的本质、眼睛高度非线性的响应特性，以及如何在“视觉空间”中测量颜色。人类的视觉对观看环境具有极强的适应能力，这为色彩科学的研究带来了动态挑战，但我们将逐步建立应对机制。现在，请与我一同开启这场光的探索之旅——其中将涉及物理学、感知科学、少许光学知识，以及一丝量子力学，所有内容都将通过友好的叙述方式和途中穿插的简短趣味视频娓娓道来。请带上您的手机。

1.1 光的本质

我们的故事始于光。捕捉并数字化光是所有数字成像能力的基本原则，因此理解光的本质是理解数字图像的核心。光是电磁波谱的一种扰动，是一种电和磁的波动，与无线电波由相同的物质构成。无线电波与光之间的唯一区别在于它们的波长，这是图1.1中所示光的属性之一。请记住，我们可以互换使用波长和频率，因为波长决定了频率。例如，在谈论无线电波时，我们通常指的是它们的频率（即调频收音机上的99.7指的是99.7兆赫）；但当我们讨论其微观近亲——可见光时，我们则指的是它的波长。

光的属性包括方向和速度（即光速），但光还有许多更有趣的属性，其中每一个都会以某种方式影响我们的照片。在此我们将仔细审视我们的朋友——光子，因为它是我们照片中的核心角色。

1.1.1 波长

光的波长是光波中波峰到波峰的距离，以纳米（nm）为单位测量。当谈论可见光时（是的，还有不可见光！），波长大约在400纳米到700纳米之间，具体取决于您的人类测试对象。可见光当然是那些

wavelengths of light that we humans have evolved photoreceptors for, so this is a human-centric definition. Just how small is the wavelength of light? For a middle wavelength—say, 550 nm—about 130 would fit across the width of a human hair. Radio waves can be meters to miles in length; shorter they become microwaves, then infrared light, then visible light, ultraviolet light, x-rays, and finally gamma rays.

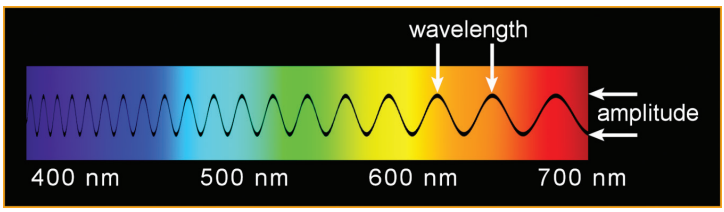


Figure 1.1 Light wave attributes

Some creatures can see light wavelengths that we cannot—so-called invisible light. And so can cameras. But that story must wait for Chapter 2: Camera Space. When the wavelength gets too short (below 400 nm), we call it ultraviolet (beyond violet), and our eyes just cannot perceive it.

> FUN FACT: While we can't see ultraviolet light, some birds and bees can, along with some reptiles, fish, amphibians, bats, and rodents.

Above 700 nm, the wavelength is too long, so we call it infrared (below red), and our eyes cannot perceive that either. Oddly, we can feel it on our skin, creating the sensation we call heat. While no earthly creature can see infrared, some can sense it. Pit vipers (rattlesnakes, copperheads, and cottonmouths) and blood-sucking insects can sense it and can use it to find you, even in the dark. Sweet dreams!

1.1.2 Amplitude

Another attribute of light from Figure 1.1 is amplitude, the height of the EM wave. This represents the wave's brightness, which is independent of the wavelength—also, the greater the amplitude, the greater the energy of the photon. A low-energy photon may not trip off the imaging sensor in a camera. Nobody likes feeble photons.

我们人类进化出感光器所能感知的光的波长，因此这是一个以人类为中心的定义。光的波长究竟有多小？以中等波长——比如550纳米——为例，大约需要130个这样的波长才能横跨一根人类头发的宽度。无线电波的波长可达数米至数英里；波长更短时，它们会变成微波，然后是红外光、可见光、紫外光、X射线，最终成为伽马射线。

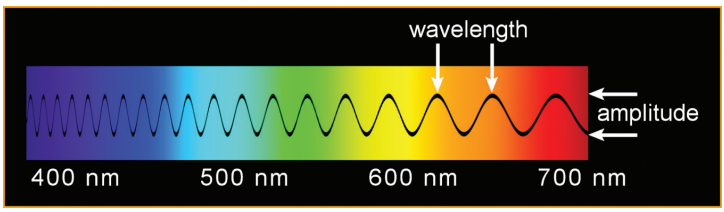


图1.1 光波属性

有些生物能看到我们看不见的光波长——即所谓的不可见光。相机也能做到这一点。但这个故事必须留到第二章：相机空间再讲。当波长变得太短（低于400纳米）时，我们称之为紫外线（在紫色之外），我们的眼睛根本无法感知到它。

> 有趣的事实：虽然我们看不见紫外线，但一些鸟类和蜜蜂可以，还有一些爬行动物、鱼类、两栖动物、蝙蝠和啮齿动物也能看见。

超过700纳米，波长就太长了，我们称之为红外线（红色以下），我们的眼睛也无法感知。奇怪的是，我们的皮肤却能感觉到它，产生我们称之为热的感觉。虽然地球上的生物都无法看到红外线，但有些生物能感知到它。蝮蛇（响尾蛇、铜头蛇和水蝮蛇）以及吸血昆虫能感知红外线，并利用它在黑暗中找到你。祝你好梦！

1.1.2 振幅

图1.1中光的另一个属性是振幅，即电磁波的高度。这代表了波的亮度，它与波长无关——此外，振幅越大，光子的能量就越大。低能量的光子可能无法触发相机中的成像传感器。没有人喜欢微弱的光子。

1.1.3 Phase

Like all waves, light waves can shift in phase, as shown in Figure 1.2, where the light wave on the left is illustrated in three phases. You can think of the phase as timing because it can be advanced or retarded. The phase is measured in degrees, with 360 degrees shifting the phase to exactly one full wavelength.

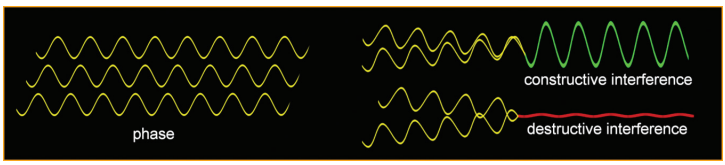


Figure 1.2 The phase attribute of light with interference

Also, phase plays a critical role in interference—the interaction of two light waves. In Figure 1.2, constructive interference is illustrated by two light waves that are in phase so their energy "stacks up" when they merge to create a greater amplitude photon (green). Conversely, destructive interference is illustrated by two light waves that are 180 degrees out of phase, thereby canceling each other out (red). We will encounter constructive and destructive interference again when we look at light reflecting off materials.

Scan this QR code to watch a brief animation about the wavelength, amplitude, and phase of visible light.

1.1.4 Polarization

Light waves oscillate, and this oscillation has an orientation called polarization (Figure 1.3). Of course, the oscillation can be in any direction, not just the three shown. The polarization of light from a normal light source is oriented randomly so it has no pattern. However, some reflection processes tend to orient the reflected photons with the same orientation, resulting in a polarized reflection that can throw a flare over your photograph. There is even circular polarizing. I don't even know what that is—but somebody does because circular polarized glasses are used for viewing RealD 3D stereo films.

1.1.3 相位

与所有波一样，光波也会发生相位移动，如图1.2所示，左侧光波展示了三种相位状态。你可以将相位理解为时间上的调整，因为它可以提前或延后。相位以度为单位进行测量，360度的相位移动恰好对应一个完整的波长。

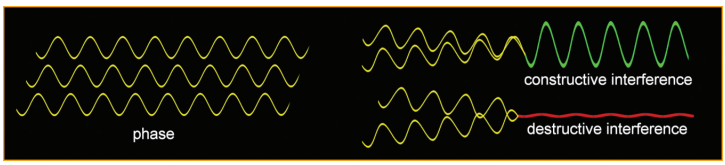


图1.2 具有干涉的光的相位属性

此外，相位在干涉——即两束光波的相互作用中——起着关键作用。在图1.2中，相长干涉由两束同相的光波演示，当它们合并时能量“叠加”，产生振幅更大的光子（绿色）。相反，相消干涉由两束相位相差180度的光波演示，从而相互抵消（红色）。当我们研究光从材料表面反射时，将再次遇到相长干涉和相消干涉。

扫描此二维码，观看关于可见光波长、振幅和相位的简短动画。

1.1.4 极化

光波会振荡，这种振荡具有一种称为偏振的取向（图1.3）。当然，振荡可以发生在任何方向，而不仅仅是图中所示的三种。来自普通光源的光的偏振方向是随机的，因此没有固定模式。然而，某些反射过程倾向于使反射的光子具有相同的取向，从而产生偏振反射，这可能会在你的照片上造成眩光。甚至还有圆偏振。我甚至不知道那是什么——但有人知道，因为观看RealD 3D立体电影时会使用圆偏振眼镜。

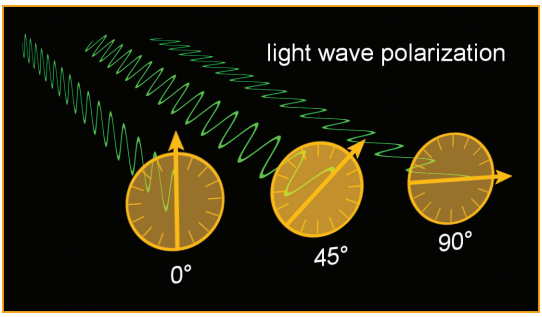


Figure 1.3 Light wave polarization

1.1.5 Monochromatic Light

Monochromatic means "one color," where all the photons are the same wavelength, producing a very pure colored light; whereas white light is a mix of wavelengths, which is what makes it white. Monochromatic light is what you get with LEDs as well as low-pressure sodium lamps, which produce a yellow light used for street lighting. While the photons are all the same wavelength, their phases and polarization are random so their energies cannot build up.

1.1.6 Coherent Light

Coherent light is where all the photons are marching in synchrony, with all their phases lined up, and is not really found in nature. There are very few natural phenomena that can produce some degree of coherency, like moonlight, some surface reflections, and sunset or sunrise scattering. But all these examples can produce only weak examples of coherency.

By far the dominant source of coherent light is *LASERS!* Lasers create light that is both monochromatic and coherent. In fact, this is the very definition of a laser—a device that emits monochromatic coherent light. And lasers are so powerful precisely because their light is monochromatic, of one frequency, and coherent—all-in-lockstep synchrony. Remember constructive interference in Figure 1.2? Just imagine trillions of photons, all the same wavelength, marching in perfect synchrony and piling up one upon the other until the amplitude is so great that it can burn through steel.

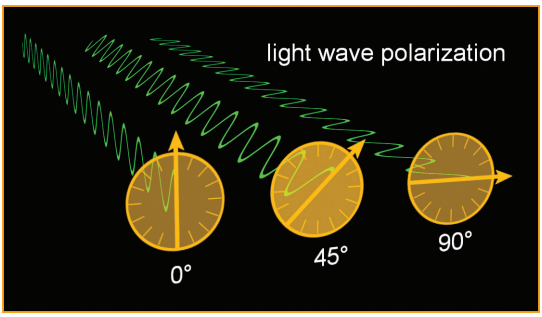


图1.3 光波偏振

1.1.5 单色光

单色意味着“一种颜色”，其中所有光子都具有相同的波长，产生非常纯净的彩色光；而白光则是多种波长的混合，这正是它呈现白色的原因。单色光可以从LED灯以及低压钠灯中获得，后者产生用于街道照明的黄光。尽管所有光子都具有相同的波长，但它们的相位和偏振是随机的，因此它们的能量无法累积。

1.1.6 相干光

相干光是指所有光子同步行进，相位完全对齐，这在自然界中并不真正存在。能够产生一定程度相干性的自然现象非常少，比如月光、某些表面反射以及日落或日出时的散射。但所有这些例子都只能产生微弱的相干性。

迄今为止，相干光最主要的来源是激光！激光产生的光既是单色的又是相干的。事实上，这正是激光的定义——一种发射单色相干光的设备。激光之所以如此强大，正是因为它们的光是单色的、单一频率且相干的——完全同步锁定。还记得图1.2中的相长干涉吗？想象一下数万亿个光子，所有光子波长相同，以完美的同步性行进并层层叠加，直到振幅大到足以烧穿钢铁。

1.1.7 Particles

So far, we have been exploring light as a wave, but light has also been proven to exhibit particle behavior by multiple experiments. This dual wave/particle behavior has vexed researchers so much that some have resorted to referring to light as a "wavicle" in frustration. But the particle nature of light is responsible for the photoelectric effect which is the foundational physics of our most important imaging device—digital cameras.

The photoelectric effect occurs when photons of light strike a metal surface and eject electrons. It was discovered by Heinrich Hertz in the 1880s (we will hear more from Hertz later) and this discovery vexed the physics community for decades. Einstein finally explained it in 1905, won the Nobel Prize in physics for it, set the foundation for quantum theory, and made possible the imaging sensor used in every digital camera today. Just think: all those selfies are made possible by quantum mechanics.

1.1.8 The Inverse Square Law

One of the facts of physics for light is the inverse square law. It regulates the falloff in light intensity with the distance from the light source. Simply put, if the distance from the light source is doubled, the falloff in light will be quartered. So, double the distance (2x), invert it (1/2), then square it ($1/2^2$)—meaning square the inverse, or the inverse square law.

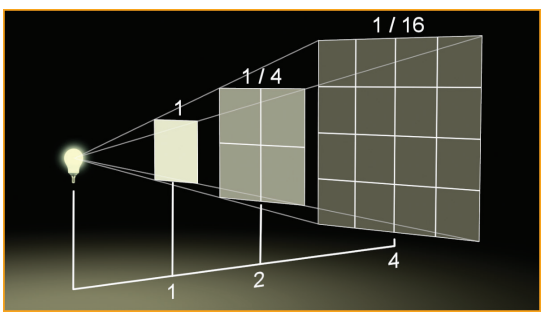


Figure 1.4 The inverse square law of light

For you mathletes:

$$I = 1 / d^2$$

Where I is the intensity of the light and d is the distance. Figure 1.4 neatly illustrates this amazing fact—the further from the light, the dimmer it gets—but

1.1.7 粒子

到目前为止，我们一直将光视为一种波，但多项实验也已证明光具有粒子行为。这种波粒二象性令研究人员十分困扰，以至于有些人无奈地将光称为“波粒子”。然而，光的粒子特性正是光电效应的成因，而光电效应构成了我们最重要的成像设备——数码相机的基础物理原理。

光电效应发生在光子撞击金属表面并击出电子时。它由海因里希·赫兹在19世纪80年代发现（我们稍后还会听到更多关于赫兹的内容），这一发现困扰了物理学界数十年。爱因斯坦最终在1905年解释了这一现象，并因此获得了诺贝尔物理学奖，为量子理论奠定了基础，并使如今每台数码相机中使用的图像传感器成为可能。试想：所有那些自拍照都得益于量子力学。

1.1.8 平方反比定律

关于光的一个物理事实是平方反比定律。它规定了光强随光源距离增加而衰减的规律。简而言之，若与光源的距离加倍，光的衰减将变为原来的四分之一。因此，距离加倍（2倍），取其倒数（1/2），再平方（$1/2^2$）——意即

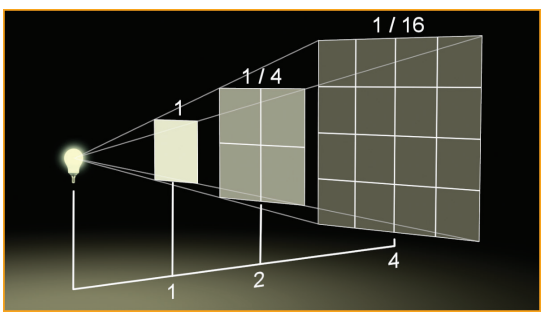


图 1.4 光的平方反比定律

平方反比，或反平方定律。

致各位数学爱好者：

$$I = 1 / d^2$$

其中 I 表示光强，d 表示距离。图 1.4 清晰地展示了这一奇妙现象——距离光源越远，光线越暗——但

with numbers. Along the bottom are the distance numbers and along the top are the brightness numbers. One unit distance from the light gets one unit of infalling light. Double that to a distance of 2 and you get 1/4 the light; then four times the distance, 1/16 the light. Why is this important? To set you up for the following discussion on interior vs. exterior lighting.

1.1.9 Interior vs. Exterior Lighting

It turns out that the inverse square law has a profound effect on lighting simulations such as CG (computer graphics), matte paintings, interactive lighting for composited shots, Photoshop work, and so on. If the scene in question has interior lighting, the work should reflect the falloff in lighting from the distance from the light source. This can be quite significant under many circumstances. A person lit with a nearby lamp, candle, or fireplace will have a noticeable falloff in lighting across the body. In-room lighting, placing an object near the room's light source vs. far from it, will also exhibit a noticeable difference.

However, if it's outdoor lighting (the sun and sky), there is no falloff in lighting with distance. So, what's this—is the sun able to break the inverse square law? Not at all. The key is the distance to the light source. If you want to double the distance from the sun, you are going to have to move 93 million miles. Moving a character 10 feet further from the sun is only one fifty-trillionth further away producing a negligible change in illumination. In fact, CG artists use a parallel light model which places the light source at infinity when lighting outdoor scenes. Keep these light distance rules in mind when creating lighting in your work if you want photorealistic results.

1.2 Measuring Light

Light is so important that it has its own unique units of measure developed by its own specialty within optics, called photometry. Here is a boiled-down, simplified, homogenized, and pasteurized version of *some* of the key units of measure of light brightness, immortalized in Figure 1.5. The left image measures the brightness of the light source; the middle image measures the amount of incident light falling onto a surface; and the right image measures

以数字为例。底部是距离数字，顶部是亮度数字。距离光源一个单位时，接收一个单位的入射光。距离翻倍至2时，接收的光量变为1/4；距离增至四倍时，光量则变为1/16。这为何重要？这是为了让你理解接下来关于内部照明与外部照明的讨论。

1.1.9 室内照明与室外照明

事实证明，平方反比定律对CG（计算机图形学）、哑光绘画、合成镜头的交互式照明、Photoshop工作等照明模拟有着深远影响。如果所讨论的场景涉及室内照明，作品应反映出光线随距离光源远近而产生的衰减。这在许多情况下可能相当显著。被附近台灯、蜡烛或壁炉照亮的人，其身体上的光照会有明显的衰减。室内照明中，将物体放置在房间光源附近与远离光源处，也会表现出明显的差异。

然而，如果是室外照明（太阳和天空），光照不会随距离衰减。那么这是怎么回事——太阳难道能打破平方反比定律吗？完全不是。关键在于光源的距离。如果你想将距离太阳的距离加倍，你需要移动9300万英里。让一个角色离太阳再远10英尺，仅仅增加了五十万亿分之一的距离，对光照产生的影响微乎其微。实际上，CG艺术家在制作室外场景照明时，会使用平行光模型，将光源置于无限远处。如果你想获得逼真的效果，在作品中创建照明时，请牢记这些光源距离的规则。

1.2 测量光线

光的重要性如此之高，以至于在光学领域内发展出了专门用于测量光的独特单位体系，称为光度学。以下是关于光亮度关键测量单位的简化、统一且标准化的概述，如图1.5所示。左图测量光源的亮度；中图测量照射到表面的入射光量；右图则测量

the actual brightness of the lit surface. They are all important because you will encounter every one of these units at different times with the equipment and software we use in movies, television, and photography. The following definitions are in SI units, which is the International System of Units used for scientific measurements worldwide.

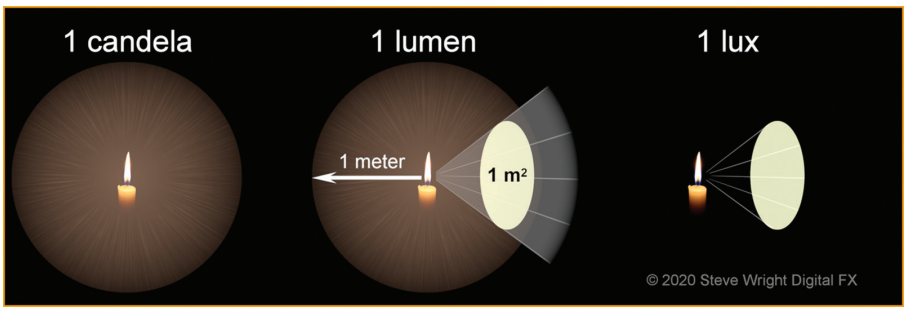


Figure 1.5 Light units of measure

1.2.1 Candela (cd): The Brightness of a Light Source

So how do we measure how bright a light source is? With candles, of course. Seriously. Take one standard-issued medieval wax candle and call that one **candela** (giving it a Latin name makes it more scientific)—also called **candlepower**. It is the unit of measure for the brightness of a light source, such as a stage light, the sun, or a villager's torch.

1.2.2 Lumen (lm): The Total Amount of Incident Light

This quantifies how much light is available to illuminate a scene and is measured in candela per square meter (cd/m^2). Also known as incident light, the lumen is perhaps the most important measurement for photography. This is what the incident light meters of photographers and cinematographers are measuring.

1.2.3 Lux (lx): How Brightly a Scene is Lit

This is concerned with how intense the light is in a unit area and is measured in lumens per square meter (lm/m^2). When photographers use incident light meters to measure a scene, they are measuring the lux - in terms of lumens per square meter.

被照亮表面的实际亮度。这些单位都很重要，因为在电影、电视和摄影中使用的设备和软件中，你会在不同时间遇到每一种单位。以下定义采用国际单位制（SI），这是全球科学测量通用的国际单位体系。

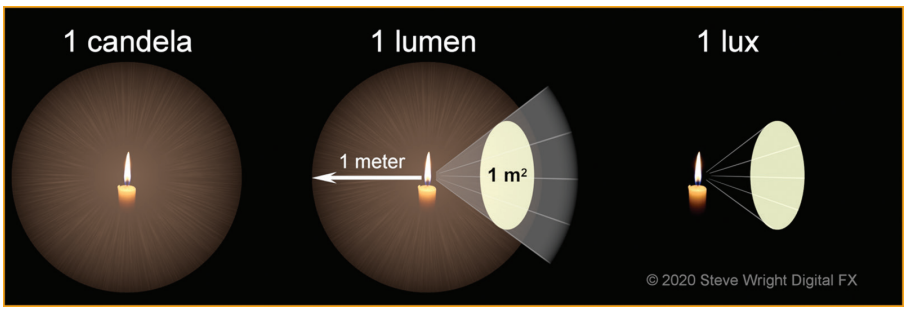


图1.5 光测量单位

1.2.1 坎德拉（cd）：光源的亮度

那么，我们如何衡量光源的亮度呢？当然是用蜡烛。不开玩笑。取一支标准的中世纪蜡制蜡烛，将其亮度定义为一坎德拉（给它起个拉丁名显得更科学）——也称为烛光。这是衡量光源亮度的单位，比如舞台灯光、太阳，或是村民手中的火把。

1.2.2 流明（lm）：入射光的总量

这量化了可用于照亮场景的光量，以坎德拉每平方米（cd/m²）为单位进行测量。也称为入射光，流明或许是摄影中最重要的测量指标。这正是摄影师和电影摄影师使用的入射式测光表所测量的内容。

1.2.3 勒克斯（lx）：场景的照明亮度

这关系到单位面积内光的强度，以每平方米流明（lm/m²）为单位进行测量。当摄影师使用入射式测光表测量场景时，他们测量的就是照度——即每平方米的流明数。

1.2.4 Other Units

While the previous SI units established a firm scientific basis for quantifying light, they are in fact the foundational units of measure for our familiar CG lighting parameters. This is the bridge between the arcane rigor of color science and our daily lives as digital imaging artists:

- **Luminance** - A measure of the amount of light reflecting from a surface projected in a particular direction such as a camera or the eye. It is measured in lumens, which are candela per square meter (cd/m^2).
- **Illuminance** - The intensity of light infalling onto a surface, measured in lux (lm/m^2), which indicates how brightly a scene is lit.
- **Radiance** - The amount of light passing through or emitted from a surface, as seen at a particular viewing angle such as a camera or the eye. It measures the total energy from *all* wavelengths, not just the "human readable" wavelengths used for luminance measures. Its units are watts per square meter per steradian (W/m^2*sr)—don't ask!
- **Footcandles** - The "American lux," meaning lumens per square foot rather than the square meter used for lux. Light meters may report their readings in either lux or footcandles. It's rather like measuring the distance between two cities in meters vs. miles. The numbers will be very different, but the distance is the same.
- **Nits** - The most common unit for measuring the brightness of a monitor, this indicates how much light the screen emits. One nit is equivalent to one cd/m^2. Most desktop LCDs are around 200-300 nits; the average HDTV will be around 200-600 nits; while UHDTVs range from 400-10000 nits. So why do we have nits? Because people got tired of saying "candela per square meter," of course.

1.3 Dynamic Range

Now that we have mastered the units of measuring light, we are ready to measure something. Let us start with dynamic range because it presents the single-greatest challenge to our digital images, for the simple reason that the dynamic range of the real world is wildly greater than that of our little

1.2.4 其他单位

虽然之前的国际单位制为量化光建立了坚实的科学基础，但它们实际上是我们熟悉的CG照明参数的基础度量单位。这正是色彩科学的神秘严谨性与我们作为数字影像艺术家的日常生活之间的桥梁：

- 亮度 - 测量从表面反射的光量在特定方向（如相机或眼睛）上的投影。它以流明为单位进行测量，即每平方米坎德拉（cd/m^2）。

- 照度 - 光线照射到表面的强度，以勒克斯（lm/m^2）为单位，表示场景被照亮的明亮程度。

- 辐射度 - 指从特定观察角度（如相机或人眼）看到通过或从表面发出的光量。它测量所有波长的总能量，而不仅限于用于亮度测量的“人眼可感知”波长。其单位为瓦特每平方米每球面度（$W/m^2 \cdot sr$）——别问为什么！

- 英尺烛光——相当于“美国版的勒克斯”，指每平方英尺的流明数，而非勒克斯所用的平方米。照度计可能以勒克斯或英尺烛光为单位显示读数。这就像用米和英里测量两座城市之间的距离——数值差异很大，但实际距离相同。

- 尼特 - 测量显示器亮度的最常见单位，它表示屏幕发出的光量。一尼特相当于一 cd/m^2。大多数桌面液晶显示器的亮度约为200-300尼特；普通高清电视约为200-600尼特；而超高清电视的范围则在400-10000尼特之间。那么为什么我们要用尼特呢？当然是因为人们厌倦了说“每平方米坎德拉”。

1.3 动态范围

既然我们已经掌握了测量光线的单位，现在就可以开始测量一些东西了。让我们从动态范围开始，因为它对我们的数字图像构成了最大的挑战，原因很简单：现实世界的动态范围远远大于我们小小的

cameras or display devices. We must somehow squeeze the huge dynamic range of the real world down to the limited dynamic range of our cameras and displays and yet have it still look natural. This will take some real color science. But first, we need to clear up some confusion.

1.3.1 Brightness vs. Luminance

Here is our first color science moment: brightness vs. luminance—a key issue that is chronically confused. Luminance is a measurement of the amount of light, so it is physical and repeatable. Brightness is our subjective perception of luminance, so it is biological and unreliable. Give ten color scientists the task of measuring the luminance of a surface and they will all return the same number. Give ten human observers the same task for the brightness and you will get 11 different responses. The reasons for this include variations in each person's visual system, other light sources in the scene, the observer's degree of alertness, and what they had for lunch. All you really have to remember is that luminance is a physical measurement of light, while brightness is the subjective human perception of it. Succinctly put, luminance is science, and brightness is psychology.

1.3.2 What is Dynamic Range?

Answer: Dynamic range is the luminance range from the darkest to the brightest part of a scene.

For a sunny outdoor scene, the dynamic range can be 10,000:1 measured by the darkest shadow compared to the brightest objects in the scene. This taxes our finest cameras. Overcast days lower the contrasts to around 1000:1—one-tenth of a sunny day. Ever wonder why a television studio has lights hanging from the ceiling everywhere? So they can throw light into every nook and cranny and on every shadow to lower the dynamic range of the scene for the studio cameras.

Once the scene is captured as an image, we switch terms and speak of the contrast ratio of the image or of a display device, which is the ratio of the darkest pixel to the brightest pixel. The problem we have with the real world is that it easily contains scenes with such high dynamic range (HDR) that it far exceeds our feeble little display devices. You can easily have a scene

相机或显示设备。我们必须设法将现实世界巨大的动态范围压缩到我们相机和显示器有限的动态范围内，同时还要让它看起来自然。这需要一些真正的色彩科学。但首先，我们需要澄清一些混淆。

1.3.1 亮度与发光度

这是我们色彩科学的第一个重要概念：亮度与光亮度——一个长期被混淆的关键问题。光亮度是对光线数量的测量，因此它是物理的、可重复的。亮度则是我们对光亮度的主观感知，因此它是生物的、不可靠的。让十位色彩科学家测量一个表面的光亮度，他们会得出相同的数值；让十位人类观察者评估同一表面的亮度，你却会得到十一种不同的回答。造成这种差异的原因包括：每个人视觉系统的差异、场景中的其他光源、观察者的警觉程度，甚至他们午餐吃了什么。你只需记住：光亮度是光的物理测量，而亮度是人类对其的主观感知。简而言之，光亮度是科学，亮度是心理学。

1.3.2 什么是动态范围?

答案：动态范围是指场景中最暗部分到最亮部分的亮度范围。

对于阳光明媚的户外场景，动态范围可达10,000:1，这是通过比较场景中最暗的阴影与最亮的物体测量得出的。这对我们最精良的相机也是一种考验。阴天会将对比度降低到大约1000:1——仅为晴天的十分之一。有没有想过为什么电视演播室的天花板上到处都挂着灯？这样他们就能将光线投射到每个角落和每处阴影中，以降低演播室摄像机拍摄场景的动态范围。

一旦场景被捕捉为图像，我们便转换术语，谈论图像或显示设备的对比度，即最暗像素与最亮像素的比值。我们在现实世界中遇到的难题是，它很容易包含动态范围（HDR）极高的场景，这远远超出了我们脆弱的小型显示设备的能力。你很容易就能遇到一个场景

with a 10,000:1 dynamic range that has to be displayed on an HDTV monitor with a 500:1 contrast ratio. This can be done and still look natural if you know how.

Figure 1.6 Examples of real-world contrast ratios. Courtesy of ILM

A high dynamic range scene is captured in Figure 1.6 to illustrate the contrast ratio at several points in the scene. In order to retain the high dynamic range of the original scene, it was captured as this High Dynamic Range (HDR) image. This is a sunlit outdoor scene, so we have a high dynamic range from the darkest shady tree root pixel, marked as 2:1 on the right, to the bright blue sky at 9000:1 on the left, where the 1 represents the blackest pixel in the image. This is by no means an extreme example. What if the sun were in the frame? That might be 100,000:1 or more.

This disparity between the HDR of the real world and our imaging technology can be confusing because it contradicts our natural daily experience. You are sitting indoors with normal interior lighting and glance at the daylight scene outside a window. Your visual system immediately compensates for the brightness of the daylight scene without changing the apparent room brightness. We see both of them in our field of view and they both look fine. Our cameras cannot do that—that is, adjust to the brightness of just one part of the scene without disturbing the rest. All we can do with a camera is increase or decrease the exposure for the entire scene. This disparity between how we see a scene and how the camera sees it explains why photographers and cinematographers need to use light meters—more on this in Chapter 2: Camera Space.

具有10,000:1动态范围的画面需要显示在对比度仅为500:1的HDTV监视器上。只要掌握正确方法，仍可实现自然逼真的显示效果。

图1.6 真实世界对比度示例。由工业光魔（ILM）提供。

图1.6拍摄了一个高动态范围场景，用以说明场景中多个位置的对比度。为了保留原始场景的高动态范围，该场景被拍摄为这张高动态范围（HDR）图像。这是一个阳光照射的户外场景，因此动态范围极高——从右侧标记为2:1的最暗树荫根部像素，到左侧亮度达9000:1的明亮蓝天（其中1代表图像中最黑的像素）。这绝非极端案例。若太阳进入画面会怎样？动态比可能达到100,000:1甚至更高。

现实世界的高动态范围与我们成像技术之间的这种差异可能会令人困惑，因为它与我们日常的自然体验相矛盾。当你坐在室内，处于正常的室内光照下，瞥一眼窗外的日光场景时，你的视觉系统会立即对日光场景的亮度进行补偿，而不会改变房间的明显亮度。我们在视野中同时看到两者，它们看起来都很好。但我们的相机做不到这一点——也就是说，它无法在不影响其他部分的情况下，仅针对场景中某一部分的亮度进行调整。我们使用相机所能做的，只是对整个场景增加或减少曝光。这种我们观看场景的方式与相机捕捉场景的方式之间的差异，解释了为什么摄影师和电影摄影师需要使用测光表——更多相关内容将在第二章：相机空间中进行讨论。

1.3.3 Stops

The HDR photo in Figure 1.6 neatly illustrates the dynamic range of several regions of the original scene based on their pixel values. But there is another way to quantify dynamic range—a better way that works more harmoniously with cameras and the human eye—and that is in **stops**. One stop is a doubling (or halving) of the exposure; so in Figure 1.6, the 2:1 pixels on the lower right could also be represented as 1 stop over black, black being code value 1, our blackest pixel in the photo. Therefore the 2:1 pixel is twice as bright or double the brightness of the black value, so 1 stop over black. The nearby 43:1 region exposure could be referenced as 5.43 stops, and on the left 4250:1 = 12.053 stops, 1000:1 = 9.966 stops, and 9000:1 = 13.136 stops. If you ever feel compelled to calculate the dynamic range from stops, grab a calculator and simply enter 2 to the power of the stops like this: $2^{\wedge 15}$ = 32,768. So, 15 stops have a dynamic range of 32,768:1. However, most can resist this compulsion.

Stops are a very convenient shorthand for quantifying relative exposure. Surely you would rather say the sky was about 13 stops brighter than the blackest pixel, rather than calling it a 9000:1 contrast ratio. Another advantage to stops is that they are non-linear and get "larger" as the numbers go up. The 2:1 pixels means that 1 stop is worth 1 unit of light but in the sky, 13.1 stops = 9000 while 12.1 stops = 4390, so one stop here is (9000 - 4390), which is 4610! Note that stops are relative—this pixel is 700 times brighter than that pixel—but this does not speak to the absolute brightness of the pixel.

1.4 A Color Science Moment

It is time to pause our simplified storytelling to pay homage to the Color Science Police, who are anxious to detain me for oversimplifying it. However, I believe getting lost in the weeds of technical correctness would detract from the practical gains for the reader; so while the next section is a bit technical, I have toned it down somewhat to be endurable.

This is foundational stuff, so you should sit up straight and pay close attention for the next few minutes. Here we are going to trace the light from a light source to a colored surface to the human eye—or camera, as the case may be. It tells the story of how the light entering a camera is a function of the

1.3.3 停止

图1.6中的HDR照片根据像素值清晰地展示了原始场景中多个区域的动态范围。但还有另一种量化动态范围的方式——一种与相机和人眼更协调的更好方法——那就是用档位表示。一档意味着曝光量加倍（或减半）；因此在图1.6中，右下角2:1的像素也可以表示为比黑色高1档，黑色即代码值1，也就是照片中最暗的像素。所以2:1像素的亮度是黑色的两倍，即比黑色高1档。附近43:1区域的曝光可记为5.43档，左侧4250:1 =为12.053档，1000:1 =为9.966档，9000:1 =为13.136档。如果你曾迫切想从档位计算动态范围，只需拿出计算器输入2的档位次方，例如：$2^{\wedge 15} = 32,768$。也就是说，15档对应的动态范围是32,768:1。不过，大多数人能克制这种冲动。

挡位是量化相对曝光的一种非常方便的简写方式。你肯定更愿意说天空比最暗的像素亮了大约13挡，而不是称其为9000:1的对比度。挡位的另一个优势在于它们是非线性的，并且随着数值增大而“扩大”。2:1的像素意味着1挡相当于1个单位的光量，但在天空中，13.1挡对应9000，而12.1挡对应4390，因此这里的一挡差值就是（9000 - 4390），即4610！需要注意的是，挡位是相对的——这个像素比那个像素亮700倍——但这并不代表像素的绝对亮度。

1.4 色彩科学时刻

是时候暂停我们简化的叙述，向色彩科学警察致敬了，他们正急于以过度简化之名拘留我。然而，我认为陷入技术正确性的细枝末节会削弱读者的实际收获；因此，尽管下一节内容略显技术性，我已稍作缓和，使其易于接受。

这是基础内容，因此接下来几分钟请坐直身子、集中注意力。我们将追溯从光源到彩色表面，再到人眼（或相机）的光线路径。它揭示了进入相机的光线如何作为{v*}的函数这一过程。

spectrum of the light source and the spectrum of reflected colors from the surface. As a color scientist, you need to know this to determine the precise colors that will be perceived by your human subjects or the response of the digital camera you are designing. But for us artists, it's all about understanding how to make prettier pictures.

1.4.1 Emission Spectrums

Technically, this section should be titled **Spectral Power Distribution,** but I was afraid you wouldn't read it if it were. Spectral power distribution can be understood with the aid of Figure 1.7, which graphs the energy output of a hypothetical light source for each wavelength. The X-axis indicates the wavelength of light, with some attractive colors to indicate the visible spectrum. What is glossed over is the Y-axis. It indicates how much power or energy is invested in each wavelength. This particular light source radiates a lot of energy at around 620 nm (yellow), somewhat less at 450 nm (blue), and precious little at 520 nm (green). The total power output of this light source is roughly the area under the entire curve.

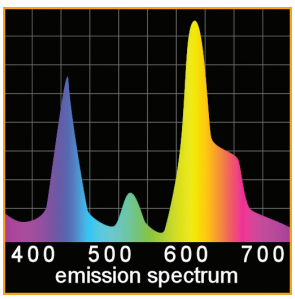


Figure 1.7 The emission spectrum of a hypothetical light source

For a truly accurate power assessment, we would have to take into account Planck's constant at each frequency because shorter wavelengths of light (i.e., 400 nm) have more energy than longer wavelengths (i.e., 700 nm). But that sounds like a lot more work than we need right now. Instead, this illustration is simply graphing which wavelengths are radiating and how much, so I am just calling it an emission spectrum.

Going forward, when you see graphs like this in this book (and there are a lot of them), keep in mind that the height of the curve indicates how much power is invested in each wavelength in that spectrum. Or you could just think of it as how bright the light is at that wavelength.

1.4.2 Spectral Reflectance

Next, we now turn our steely gaze to an object's surface receiving the rainbow of white light infalling from a light source. The surface is going to reflect

光源的光谱与表面反射颜色的光谱。作为色彩科学家，你需要了解这一点，以确定人类受试者将感知到的精确颜色，或你正在设计的数码相机的响应。但对我们艺术家来说，这一切都是为了理解如何创作出更美的画面。

1.4.1 发射光谱

严格来说，这一节的标题应该是光谱功率分布，但我担心那样写你们就不会读了。借助图1.7可以理解光谱功率分布，该图绘制了一个假设光源在每个波长上的能量输出。X轴表示光的波长，并附有吸引人的颜色来指示可见光谱。而Y轴则被轻描淡写地带过——它表示每个波长投入了多少功率或能量。这个特定光源在620纳米（黄色）附近辐射大量能量，在450纳米处则稍少一些。

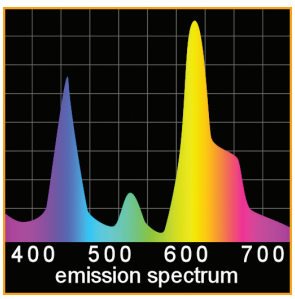


图1.7 假设光源的发射光谱

nm（蓝色），而在520 nm（绿色）处则微乎其微。该光源的总输出功率大致等于整个曲线下方的面积。

要进行真正精确的功率评估，我们必须考虑每个频率下的普朗克常数，因为较短波长的光（即400纳米）比较长波长的光（即700纳米）具有更高的能量。但这听起来比我们目前所需的工作量要大得多。相反，这个图示只是简单地绘制了哪些波长正在辐射以及辐射的强度，因此我仅将其称为发射光谱。

在本书后续内容中，当你看到此类图表（书中会出现大量类似图示）时，请记住曲线的高度代表该光谱中每个波长所分配的能量强度。或者你也可以简单理解为：该波长处光线明暗程度的直观呈现。

1.4.2 光谱反射率

接下来，我们将锐利的目光转向物体表面，它正接收来自光源的白色光彩虹。表面将反射

selected wavelengths based on its color, and most importantly, it will reflect those wavelengths regardless of the color of the light source. Figure 1.8 illustrates just such a reflected spectrum, officially known as spectral reflectance. Because the surface absorbs some wavelengths and reflects others, spectral reflectance is nature's subtractive color model of the real world and produces the vast majority of the color that we see. Not to worry—we will revisit subtractive color models at the top of Chapter 4: Work Space. As it turns out, subtractive color is not just the stuff of print and fingerpainting. It pervades our world to make up most of the color we see.

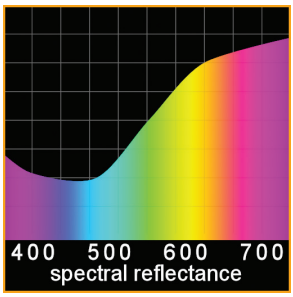


Figure 1.8 Spectral reflectance of a hypothetical color surface

The spectral reflectance illustration in Figure 1.8 indicates that the surface will weakly reflect blue light but strongly reflect yellow and red wavelengths. Again, this is regardless of what the light source is radiating. We now have two pieces of the puzzle that will allow us to calculate the color stimulus.

1.4.3 Color Stimulus

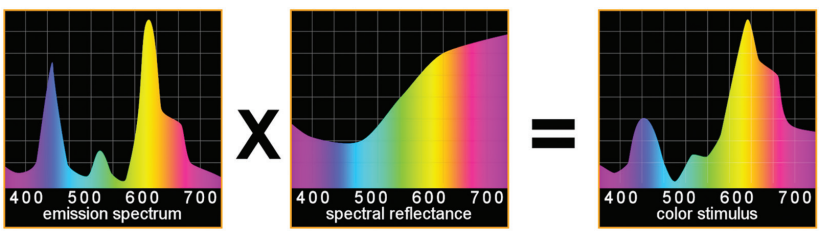


Figure 1.9 Emission spectrum multiplied by the spectral reflectance produces the color stimulus

Color stimulus is the color science term for the observed light reflected from a colored surface lit by a light source (Figure 1.9). It is the simple product of the spectral power distribution (here, the emission spectrum) of the light source and the spectral reflectance of that surface. The takeaway is that changing the color of the light source will change the color stimulus—that is,

根据其颜色选择波长，最重要的是，无论光源的颜色如何，它都会反射这些波长。图1.8展示了这样一种反射光谱，正式名称为光谱反射率。由于表面吸收某些波长并反射其他波长，光谱反射率是自然界对现实世界的减色模型，并产生了我们所看到的大部分颜色。不必担心——我们将在第四章《工作空间》的开头重新讨论减色模型。事实证明，减色不仅是印刷和手指绘画的范畴，它遍布我们的世界，构成了我们所看到的大部分颜色。

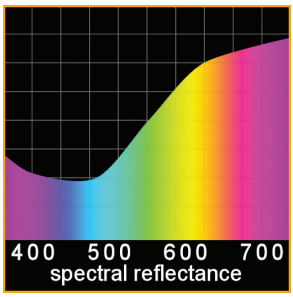


图1.8 假设彩色表面的光谱反射率 {v*}

图1.8中的光谱反射率示意图表明，该表面会微弱地反射蓝光，但强烈反射黄色和红色波长的光。同样，这与光源辐射何种光线无关。我们现在掌握了两个关键线索，可以用于计算色刺激。

1.4.3 颜色刺激

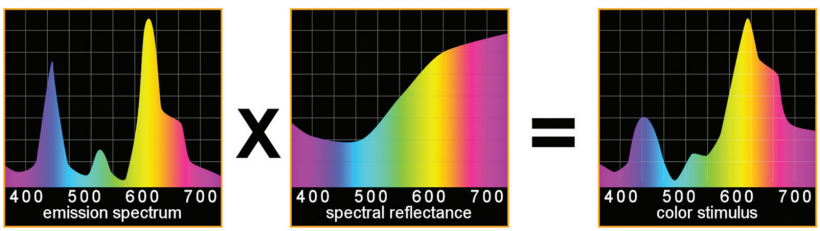


图1.9 发射光谱乘以光谱反射率产生颜色刺激

颜色刺激是颜色科学术语，指在光源照射下从有色表面反射的观察光（图1.9）。它是光源的光谱功率分布（此处为发射光谱）与该表面的光谱反射率的简单乘积。其核心在于：改变光源颜色将改变颜色刺激——也就是说，

how it appears to an observer—or a camera. This completes the color science moment. We can now return to our regularly scheduled program.

1.5 Light Sources

So far, we have been looking at the attributes of light, units of measure, and the dynamic range of the real world. The next step in our understanding of the real world is the *spectrum* of that light. The spectrum of a light source is simply what wavelengths of light it emits, which in turn defines the emission spectrum of that light source.

Here we will see that there are two fundamental types of emission spectrums. The first is the continuous spectrum of incandescent light sources like the sun, incandescent lightbulbs, and village torches—all of which derive their light from heat. The second is the "spiky" emission spectrums of luminescent light sources such as fluorescent lights, monitors, and LEDs, which derive their light by fooling Mother Nature with technology. And boy, are they different—even though they don't appear to be different to the eye.

We work in the limited digital imaging world of discrete RGB colors, but we live in the real world with continuous spectrums of light. The importance of this is that the spectrum of light emitted by a light source dramatically affects the appearance of the lit surfaces and hence the pictures we take of them. When a scene is lit for photography, the emission spectrum and color temperature of the lighting have a profound effect on the photographed results, so here we will see what emission spectrums and color temperature are all about.

1.5.1 Emission Spectrums

Emission spectrums are the wavelengths and intensities of light emitted by light sources. Let's consider the biggest incandescent light in the solar system, the sun, under which our vision evolved and was shaped. The sun emits light in a continuous range of frequencies (called a "continuous spectrum"), with some wavelengths too long and some too short for us to see.

它如何呈现在观察者或相机面前。至此，色彩科学的时刻告一段落。现在我们可以回归常规节目了。

1.5 光源

到目前为止，我们一直在探讨光的属性、度量单位以及现实世界的动态范围。我们理解现实世界的下一步是分析该光的光谱。光源的光谱简单来说就是它发射的光的波长，这进而定义了该光源的发射光谱。

这里我们将看到，发射光谱有两种基本类型。第一种是白炽光源（如太阳、白炽灯泡和乡村火炬）的连续光谱——所有这些光源的光都来自热量。第二种是发光光源（如荧光灯、显示器和LED）的“尖峰状”发射光谱，它们通过技术手段“欺骗”自然法则来产生光。天哪，它们截然不同——尽管在肉眼看来似乎并无差异。

我们在有限的数字成像世界中处理离散的RGB色彩，却生活在拥有连续光谱的真实世界里。关键在于光源发出的光谱会显著影响被照射表面的外观，进而影响我们拍摄的照片效果。当为摄影布光时，照明光源的发射光谱与色温会对拍摄成果产生深远影响，因此接下来我们将探讨发射光谱与色温的本质。

1.5.1 发射光谱

发射光谱是光源发出的光的波长和强度。让我们考虑太阳系中最大的白炽灯——太阳，我们的视觉正是在太阳下进化并成形的。太阳发出的光频率范围连续（称为“连续光谱”），其中部分波长过长或过短，超出了我们的可见范围。

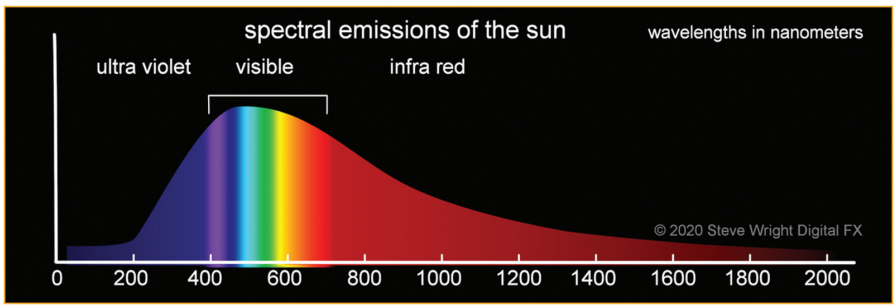


Figure 1.10 Visible light in the spectral emissions of the sun

Referring to Figure 1.10, the frequencies are on the horizontal axis labeled with their wavelengths and their intensity is on the vertical axis. Our visual system only perceives a narrow range of those frequencies between 400 and 700 nm that we call "visible light," where the sun concentrates most of its output energy. The point here is that the visible light from the sun emits a smooth, continuous spectrum of frequencies. But what about luminescent light sources?

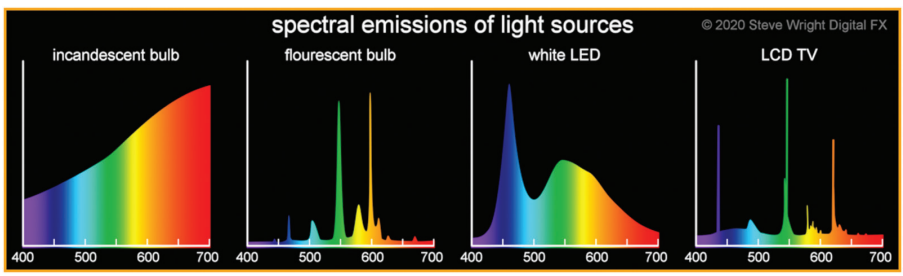


Figure 1.11 Incandescent continuous vs. luminescent spiky emission spectrums

As mentioned previously, the lofty-sounding term "spectral emission" is simply geek-speak for what frequencies of light are emitted in what strength by a light source. Figure 1.11 illustrates the spectral emission of several common types of light sources. Starting with the incandescent bulb on the left, its spectral emission is continuous, like the sun (Figure 1.10), because it generates light with the same type of physics: incandescence, or heating something until it glows. This is also known as a "full-spectrum light source."

However, the fluorescent bulb next to it uses luminescence, so it is very "spiky," with lots of light in only a few narrow frequencies. Next is a white LED—a prevalent type of lighting on the set now. It too has a very weird

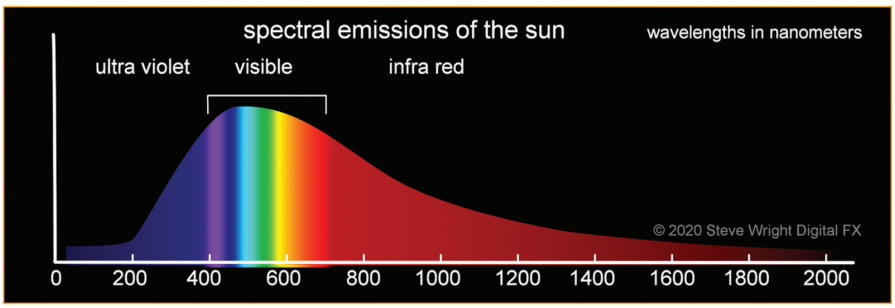


图1.10 太阳光谱发射中的可见光

参考图1.10，频率位于横轴上，标有其波长，而强度位于纵轴上。我们的视觉系统只能感知400到700纳米之间的狭窄频率范围，我们称之为“可见光”，太阳的大部分输出能量都集中于此。这里的重点是，太阳发出的可见光具有平滑、连续的频率谱。但发光光源的情况又如何呢?

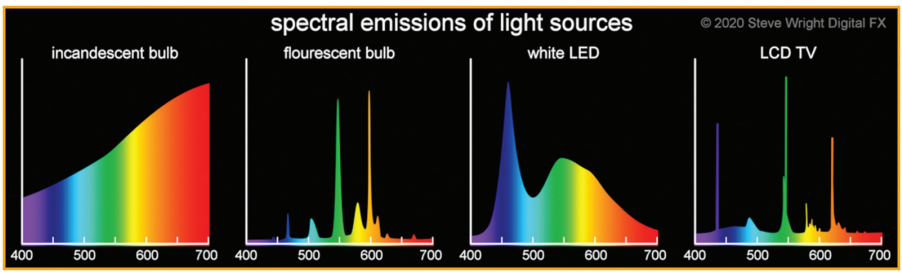


图1.11 白炽连续光谱与发光尖峰发射光谱对比

如前所述，听起来高大上的术语“光谱发射”其实就是极客们用来描述光源以何种强度发射哪些频率的光。图1.11展示了几种常见光源的光谱发射情况。从左边的白炽灯开始，它的光谱发射是连续的，就像太阳一样（图1.10），因为它产生光的物理原理相同：白炽现象，即通过加热物体使其发光。这也被称为“全光谱光源”。

然而，旁边的荧光灯管使用的是发光原理，因此它的光谱非常“尖锐”，只在少数几个狭窄的频率上有大量光线。接下来是白色LED灯——如今片场普遍使用的一种照明类型。它的光谱也同样非常奇特。

shape, not at all like the incandescent bulb. When we get to the LCD TV, we can see that it is even spikier than the fluorescent bulb. By design, its spikes are in the RGB light frequencies that tickle our cones.

FUN FACT: There are no "white" LEDs. They are really blue LEDs coated with phosphors that absorb some of the blue light and re-radiate it as yellow, red, and green. So blue + red + yellow + green = white.

1.5.2 Color Temperature

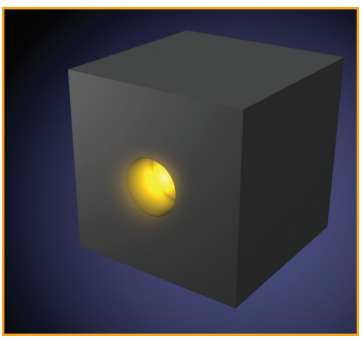


Figure 1.12 Black-body radiation

I know what you are thinking. No discussion of light emitters is complete without touching on black-body radiation. You are no doubt aware of the color temperature of lights and how that affects scene exposure by casting a hue, or color, over the picture. To deal with this, we have white balance settings in our cameras. But what is this color temperature thing anyway? And why does it ruin our photographs if we get it wrong? It's all based on a quantum mechanics phenomenon called black-body radiation.

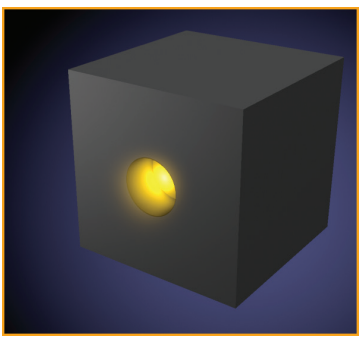
Join me for a moment in a thought experiment. Referring to Figure 1.12, assume that you have built a magical box out of carbon nanotubes that is totally black and does not reflect any light. Next, imagine drilling a hole in the side and then heating the interior of the box to a high temperature. As expected, the light coming out of the hole will soon start to glow red. Continue heating it and the red glow becomes orange, then yellow, then white, then blue (it never glows green, for some reason). This also explains why stars come in those same colors, ranging from red to blue. Now back to our story.

FUN FACT: Actually, we do know the reason. The spectrum of light from stars forms a wide bell curve that shifts left and right with the temperature of the star. If the star's temperature puts the peak of the bell curve over green, the curve is so wide that it includes all the other colors, washing it out to white.

形状，完全不像白炽灯泡。当我们看到液晶电视时，可以发现它甚至比荧光灯还要多尖峰。其设计使这些尖峰恰好位于刺激我们视锥细胞的RGB光频率上。

有趣的事实：实际上并没有“白色”LED。它们本质上是蓝色LED，表面涂有荧光粉，这些荧光粉会吸收部分蓝光并重新辐射出黄光、红光和绿光。因此蓝色 + 红色 + 黄色 + 绿色 = 白色。

1.5.2 色温



我知道你在想什么。不谈黑体辐射，任何关于光源的讨论都不完整。你肯定知道灯光的色温，以及它如何通过给画面投射色调或颜色来影响场景曝光。为此，我们的相机中设有白平衡设置。但这个色温到底是什么？为什么一旦弄错就会毁掉我们的照片？这一切都基于一种名为黑体辐射的量子力学现象。

图1.12 黑体辐射

和我一起做个思想实验。参考图1.12，假设你用碳纳米管造了一个完全漆黑、不反射任何光线的魔法盒子。接着，想象在盒子侧面钻一个小孔，然后将盒子内部加热到高温。不出所料，从小孔中透出的光很快会开始泛红。继续加热，红光会变成橙色，然后是黄色、白色，最后是蓝色（出于某种原因，它永远不会变成绿色）。这也解释了为什么恒星会呈现从红到蓝的相同颜色序列。现在回到我们的故事。

有趣的事实：实际上，我们确实知道原因。来自恒星的光谱形成一个宽钟形曲线，随着恒星温度的变化而左右移动。如果恒星的温度使钟形曲线的峰值落在绿色区域，那么曲线会宽到包含所有其他颜色，从而将其冲淡成白色。

We can now associate a specific color to each temperature—hence "color temperature." When we say a light source has a color temperature of 3200 degrees Kelvin (K) (again, it *has* to be Kelvin to be scientific), we do not mean that the light source has a thermal temperature of 3200 degrees Kelvin. We mean that it radiates the same color of light as a black body heated to 3200K—namely yellow. Fortunately, the color temperature is not the physical temperature of things because the color temperature of daytime blue sky-light is around 9000K. Yikes!

FUN FACT: Kelvin has the same-sized units as Celsius, but Celsius sets 0 at water freezing while Kelvin sets it to absolute zero (-273 C).

The white balance of a camera is all about color temperature. You are dialing in the color balance to remove the color temperature cast and make it look white. Here again, we crash into our visual system. Given any non-white light source, after a few moments, our eyes will undergo chromatic adaptation to see it as white and "color balance" all objects in the scene accordingly. So, if you don't do the color balance on a camera, you can stand there staring at a perfectly normal-looking scene (thanks to chromatic adaptation) but when you view the captured image, it will have a hideous color cast to it. If you shoot a scene with light sources of different color temperatures, such as a mix of incandescent (3200K) and fluorescent lights (5000K), you will find different areas of the picture have different color casts. This is not only bad photographic form, but also very difficult to fix in post.

If you shoot a TV screen which normally has a 6500K white point with a camera color balanced for 3200K (standard indoor incandescent lights) the resulting image will have a blue cast on the screen because 6500K is a bluer light than 3200K. If you adjust the white balance of the camera to 6500K, the television will look great, but the rest of the scene will turn reddish because it is lit with 3200K incandescent lights. We have just crashed into the mixed color temperature light source problem cited in the previous paragraph. Interestingly, some upscale TVs will let you set the color temperature of the screen in their setup menu.

Watch out for artists when talking about color temperature, though. When we talk tech about a cooler color temperature like 3200K, which is a yellow light, the artist will insist that it's "warm." When we talk about a warmer color temperature like 9000K, which is a bluish light, the artist will insist it is "cool."

现在我们可以将特定颜色与每个温度关联起来——因此有了“色温”这个概念。当我们说一个光源的色温为3200开尔文（K）（重申，必须使用开尔文才科学），并不是指该光源的实际热力学温度达到3200开尔文，而是指它辐射出的光色与加热到3200K的黑体所发出的光色相同——即黄色。幸运的是，色温并非物体的实际温度，因为白天蓝色天空光的色温约为9000K。天哪！

> 有趣的事实：开尔文温标与摄氏温标单位大小相同，但摄氏温标将0度设为水的冰点，而开尔文温标将其设为绝对零度（-273°C）。

相机的白平衡完全关乎色温。你调整色彩平衡是为了消除色温偏色，使其呈现白色。这里我们再次与自身的视觉系统产生冲突。在任何非白光源下，经过短暂时间后，我们的眼睛会进行色彩适应，将其视为白色，并据此“色彩平衡”场景中的所有物体。因此，如果不对相机进行色彩平衡，你可能会站在那里凝视着看起来完全正常的场景（得益于色彩适应），但当你查看拍摄的图像时，它会带有难看的偏色。如果你拍摄的场景包含不同色温的光源，例如白炽灯（3200K）和荧光灯（5000K）混合照明，你会发现图片的不同区域存在不同的偏色。这不仅是不良的摄影习惯，而且在后期处理中也极难修正。

如果你用一台白平衡设为3200K（标准室内白炽灯色温）的摄像机去拍摄一台通常拥有6500K白点的电视屏幕，那么生成的图像中屏幕会偏蓝，因为6500K的光比3200K更偏蓝。如果你将摄像机的白平衡调整为6500K，电视屏幕看起来会非常棒，但场景的其他部分会偏红，因为它们是由3200K的白炽灯照亮的。我们刚刚就撞上了前文提到的混合色温光源问题。有趣的是，一些高端电视会在设置菜单中允许你调整屏幕的色温。

不过，在谈论色温时要注意艺术家的说法。当我们从技术角度谈论像3200K这样偏黄的冷色温时，艺术家会坚称那是“暖色”。而当我们谈论像9000K这样偏蓝的暖色温时，艺术家又会坚持说那是“冷色”。

This is just all part of the merry mix-up of color science. And remember—this is the super-simplified version of the story!

1.5.3 Standard Illuminants

Now that we have the whole color temperature thing clear, we can move on to the next exciting topic—standard illuminants. This is a way of describing the color of a light source in terms of its color temperature and is a method used *everywhere* in our world of color equipment. Higher temperatures are bluer, cooler ones are yellower. Several color temperatures have been designated "standard illuminants" as shorthand to be used by industry and science.

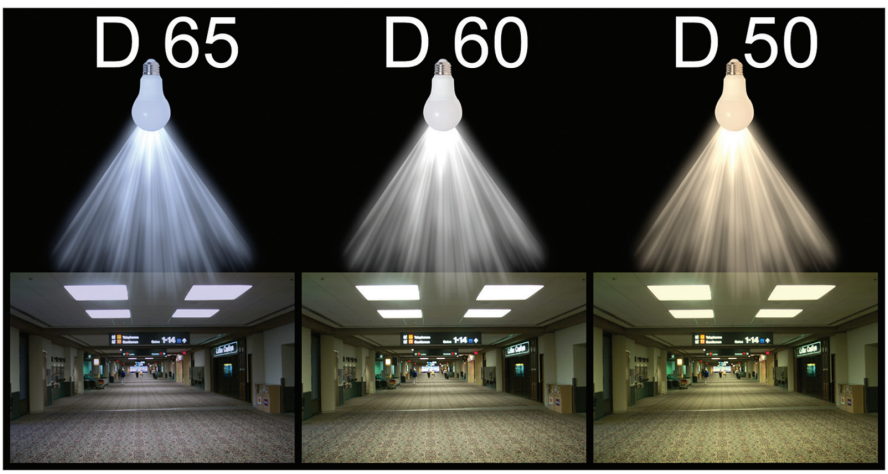


Figure 1.13 Standard Illuminants (Colors exaggerated for illustration purposes)

Figure 1.13 illustrates the three most important standard illuminants you are likely to encounter in the digital imagery world. The first is D65 (6500K), the white point for television, which is considered to correspond to daylight near noon and is heavily influenced by the blue sky. The second is D60 (6000K), the white point for ACES Digital Cinema P3 color. When not using ACES, the P3 white point is D65. The third is D50 (5000K), the white point used in photography and print, which is considered to correspond to the warm daylight of late afternoon.

A mnemonic to help you to remember all this is: "Hotter is bluer." When comparing two color temperatures, the hotter one will be bluer, and conversely, the cooler one will be redder. These white points are critical because they influence our perception of color, so if we are to be consistent with our color, we need to properly manage our white points.

这只是色彩科学中欢乐的混乱的一部分。记住——这是故事的超级简化版！

1.5.3 标准光源

既然我们已经弄清了色温这回事，就可以继续探讨下一个激动人心的话题——标准光源。这是一种通过色温来描述光源颜色的方法，在我们色彩设备的世界里无处不在。温度越高越偏蓝，温度越低越偏黄。业界和科学领域将几种色温指定为“标准光源”作为简称使用。

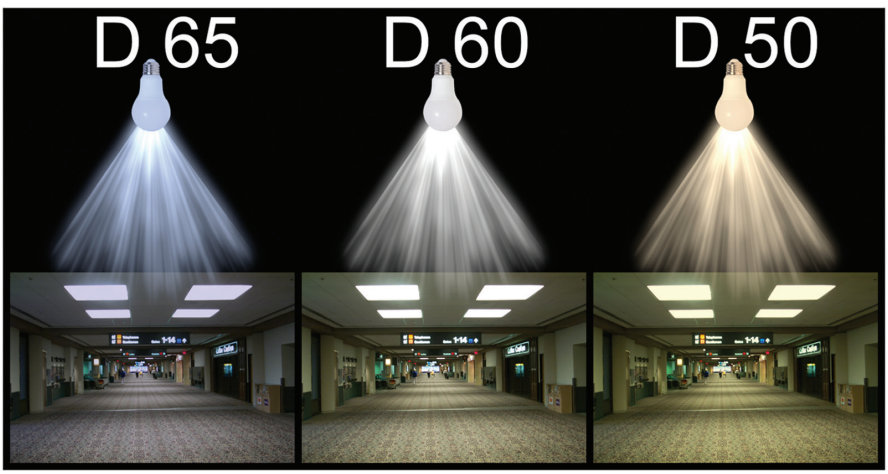


图1.13 标准光源（为说明目的，颜色已做夸张处理） s)

图1.13展示了数字图像领域中最可能遇到的三种重要标准光源。第一种是D65（6500K），即电视白点，被认为对应正午日光，并深受蓝天影响。第二种是D60（6000K），是ACES数字影院P3色彩的白点；若不使用ACES，P3白点则为D65。第三种是D50（5000K），是摄影与印刷领域采用的白点，被认为对应午后温暖的日光。

一个助记口诀帮你记住这一切是：“越热越蓝。”比较两种色温时，温度更高的会偏蓝，反之，温度更低的会偏红。这些白点至关重要，因为它们影响我们对色彩的感知，因此如果我们要保持色彩的一致性，就需要妥善管理白点。

This video will help you to visualize color temperatures, black-body radiation, and the standard illuminants.

1.6 Surface Interactions with Light

Now that we have a bunch of photons bouncing around the room, it is time to think about the surfaces in the real world that the light illuminates. When light strikes a surface, it can generally interact in three ways: transmit, absorb, or reflect, as illustrated in Figure 1.14. Of course, just to keep things interesting, surfaces can do all three at the same time to different wavelengths of light. For CG artists trying to simulate the real world, this is a truly annoying complication. In this section, we will consider what happens to the incoming light when it reflects off surfaces to be photographed.

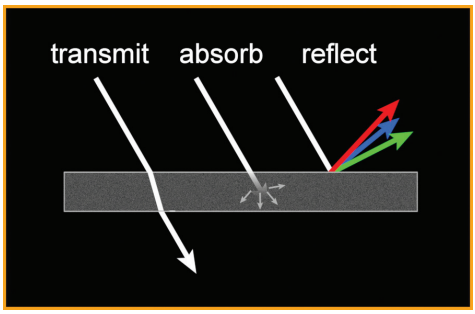


Figure 1.14 Surface interactions with light

1.6.1 Spectral Reflectance

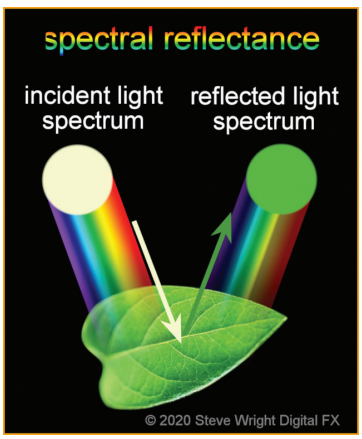


Figure 1.15 Spectral Reflectance

Figure 1.15 illustrates a subtractive event with your basic green leaf. Assuming a natural light source like the sun, the incoming white light (scientifically labeled "incident light spectrum") contains a full spectrum of visible light. The white light reflects off the surface of the green leaf, but most of the spectrum is absorbed (subtracted) and it is mostly the green wavelengths that are strongly reflected. Leaves appear green because they absorb red and blue wavelengths to power their photosynthesis and

本视频将帮助您直观理解色温、黑体辐射以及标准光源。

1.6 光与表面的相互作用

既然我们已经有了大量光子在房间里四处弹射，现在就该思考现实世界中光线所照亮的表面了。当光线照射到表面时，通常会产生三种相互作用方式：透射、吸收或反射，如图1.14所示。当然，为了保持问题的复杂性，表面可以同时以不同比例进行这三种作用。

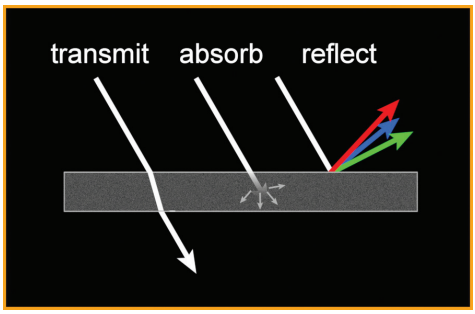


图 1.14 表面与光的相互作用

光的波长。对于试图模拟真实世界的CG艺术家来说，这确实是一个令人烦恼的复杂因素。在本节中，我们将探讨入射光在从待拍摄表面反射时会发生什么变化。

1.6.1 光谱反射率

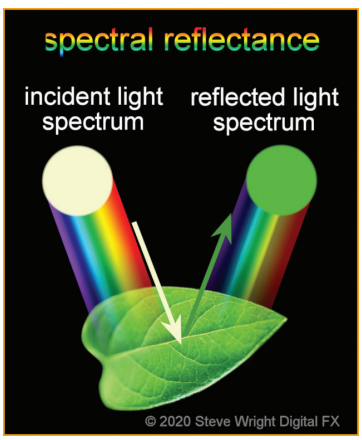


图 1.15 光谱反射率

图 1.15 以基础绿叶为例说明了减色现象。假设存在太阳这样的自然光源，入射白光（科学上标记为“入射光谱”）包含完整的可见光谱。白光从绿叶表面反射，但大部分光谱被吸收（减去），强烈反射的主要是绿色波长。叶子呈现绿色是因为它们吸收红色和蓝色波长以驱动光合作用，

reflect the green for us to see. It demonstrates two out of three of the surface interactions in Figure 1.14—namely absorption and reflection, and if you hold a leaf up to the sun and see light coming through (transmission), you will have all three interactions.

This is just the first approximation explanation of reflecting surfaces. In the real world, there are additional physical effects that affect the apparent color of a surface. For example, human skin is partially transparent and has sub-surface scattering, giving it a complex and difficult look to replicate—which is why it took CG decades to produce convincing human skin.

Life was simpler when the light sources were just the sun, candles, and incandescent bulbs. Referring to Figure 1.11, they produce continuous spectrums of light, so all wavelengths are well represented. However, our modern "techno-lights" (e.g., LEDs, CFLs, fluorescent lights) emit very non-continuous spectrums of light that introduce a new problem to manage, namely the spectral emission of light sources that can discolor objects in the scene.

The effect of the emission spectrum on the spectral reflectance of a surface is illustrated in the simplified example in Figure 1.16. The left panel shows that the spectrums are aligned so the spectral emission (yellow spike) matches the spectral reflectance (white curve) of the surface. As a result, a great deal of that color is reproduced in the resulting color chip and the resulting banana is a yummy yellow.

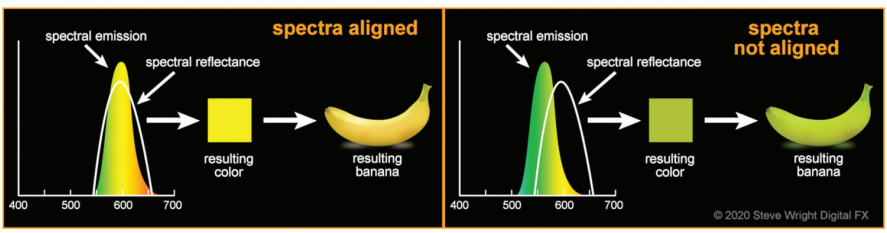


Figure 1.16 Effect of emission spectrum on the captured image

The panel on the right, however, shows the spectral emission shifted to the left, which no longer aligns very well with the spectral reflectance of the surface. As a result, much less of the light color is reflected and it is also greener, so the resulting color chip is not only darker but also hue-shifted. The resulting banana is a yucky green. Broad spectrum emitters like the sun

为我们反射出绿色以供我们看见。它展示了图1.14中三种表面相互作用中的两种——即吸收和反射，如果你将一片叶子举向阳光并看到光线透过（透射），你就看到了所有三种相互作用。

这只是反射表面的初步近似解释。在现实世界中，还有额外的物理效应会影响表面的表观颜色。例如，人类皮肤具有部分透明性并存在次表面散射，这使其外观复杂且难以复现——这也是计算机图形学花费数十年才实现逼真人肤效果的原因。

当光源仅仅是太阳、蜡烛和白炽灯泡时，生活要简单得多。参考图1.11，它们产生连续的光谱，因此所有波长都得到了很好的呈现。然而，我们现代的“技术光源”（如LED、节能灯、荧光灯）发出的光谱非常不连续，这带来了一个新的管理问题，即光源的光谱发射可能导致场景中的物体变色。

图1.16的简化示例说明了发射光谱对表面光谱反射率的影响。左图显示光谱经过校准，使光谱发射（黄色尖峰）与表面的光谱反射率（白色曲线）相匹配。因此，该颜色在生成的色卡中得到大量再现，最终香蕉呈现出诱人的黄色。

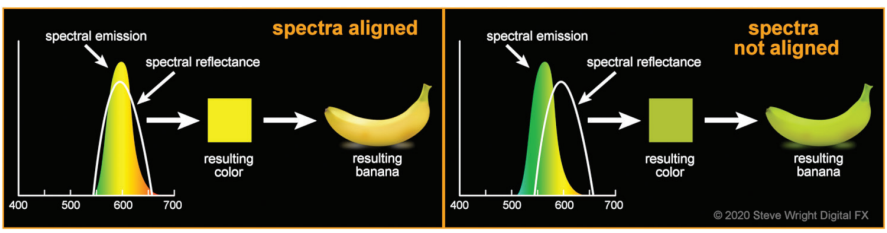


图 1.16 发射光谱对捕获图像的影响

然而，右侧面板显示光谱发射向左偏移，不再与表面的光谱反射率良好匹配。因此，反射的浅色光大大减少，且颜色更偏绿，所以最终的颜色芯片不仅更暗，色调也发生了偏移。生成的香蕉呈现出一种难看的绿色。像太阳这样的宽光谱发射体

and incandescent bulbs fully cover the spectrum that surfaces reflect, while spiky light sources can miss the target resulting in hue shifts.

1.6.2 Diffuse vs. Specular Reflections

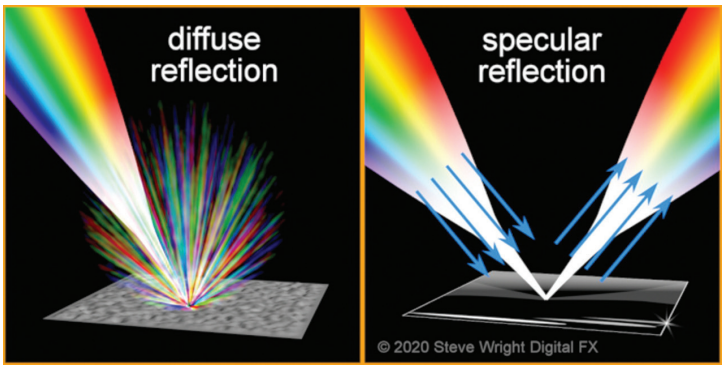


Figure 1.17 Two types of surface reflections

Our next consideration of real-world lighting is the way it reflects off the surface—is it diffuse or specular? Referring to Figure 1.17, diffuse reflections are what we get from dull, non-shiny surfaces like cloth. When the incoming light strikes a diffuse surface the light rays "splash" and scatter in all directions. While some frequencies are absorbed, others are reflected, giving the surface its color.

The specular reflection, on the other hand, is what you get with a shiny, mirrored surface. The incoming light is not scattered but instead reflects off the surface at the same angle that it came in on. If your eye is in the flight path of the reflected light, you will see a reflected image of the original light source. As we saw in Section 1.3 Dynamic Range, the original light source can have a brightness of 10,000 or more while the diffuse surface might have a brightness of 0.2 or 0.5. These specular highlights create problems for cinematography and VFX because they have an incredibly high dynamic range but must somehow be preserved in the shot.

Assuming a white light source, metals like aluminum, silver, and chrome reflect the full spectrum of the incoming light to get white specular highlights. Some metals, like gold and copper, also absorb certain wavelengths of light and reflect others, which gives them their characteristic colors.

而白炽灯泡能完全覆盖表面反射的光谱，而尖峰光源可能会错过目标，导致色调偏移。

1.6.2 漫反射与镜面反射

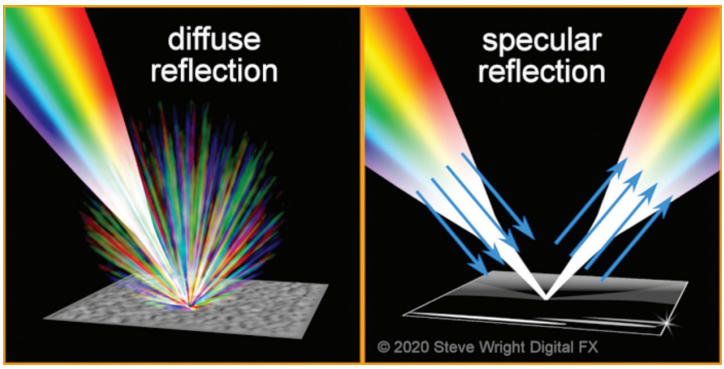


图 1.17 两种类型的表面反射

我们对现实世界光照的下一个考量是它从表面反射的方式——是漫反射还是镜面反射？参考图1.17，漫反射来自像布料这样暗淡、无光泽的表面。当入射光照射到漫反射表面时，光线会“溅射”并向各个方向散射。虽然某些频率的光被吸收，但其他频率的光被反射，从而赋予表面颜色。

另一方面，镜面反射是你在闪亮、如镜的表面所看到的现象。入射光不会被散射，而是以与入射角相同的角度从表面反射出去。如果你的眼睛位于反射光的光路上，你会看到原始光源的反射影像。正如我们在第1.3节“动态范围”中所见，原始光源的亮度可能高达10,000或更高，而漫反射表面的亮度可能只有0.2或0.5。这些镜面高光给电影摄影和视觉特效带来了难题，因为它们具有极高的动态范围，但又必须在镜头中以某种方式保留下来。

假设使用白光光源，铝、银、铬等金属会反射入射光的全光谱，从而产生白色的镜面高光。而金、铜等金属则会吸收特定波长的光并反射其他波长，这赋予了它们特有的颜色。

FUN FACT: Mirrors are made by depositing a thin aluminum film on the back of a piece of glass.

Of course, the real world is never so simple, so we have many materials that exhibit both diffuse *and* specular reflections. Glossy paint is one example. Some of the light rays reflect off the surface of the glossy paint to give it specular shininess, while other light rays penetrate the surface and reflect diffusely with some wavelengths being absorbed, giving the paint its color.

1.7 Exotic Surface Interactions with Light

There can be more complex interactions between light and surfaces than just reflecting or absorbing specific wavelengths. Light can be scattered by particles, suspended in either the atmosphere or water. Reflected light can become polarized. The reflectivity of a surface can change with the viewing angle. And in the case of thin membranes, the light can interact with itself to create iridescent colors. All these effects show up in the pictures that we make and take.

1.7.1 Scattering

Scattering is a ubiquitous light phenomenon that affects the appearance of most pictures. Illustrated in Figure 1.18, scattering occurs in any transparent medium—like air, water, or transparent aluminum—that has particles suspended in it. The incoming light travels through the medium until a light wave hits a particle,

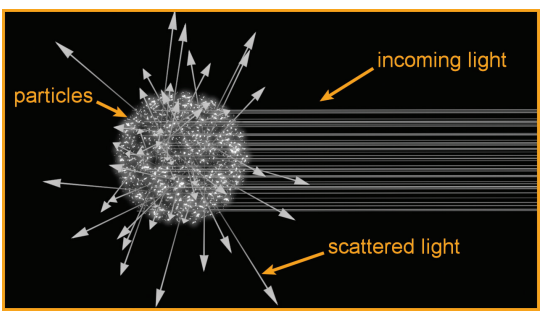


Figure 1.18 Light rays scattered by particles

有趣的事实：镜子是通过在玻璃片背面沉积一层薄铝膜制成的。

当然，现实世界从不会如此简单，因此我们有许多材料同时表现出漫反射和镜面反射。光泽涂料便是一例。部分光线从光泽涂料表面反射，赋予其镜面光泽；而其他光线则穿透表面，进行漫反射，其中某些波长被吸收，从而赋予涂料色彩。

1.7 光与表面的奇异相互作用

光与表面的相互作用可能比仅仅反射或吸收特定波长更为复杂。光线可被悬浮于大气或水中的粒子散射；反射光可能发生偏振；表面的反射率会随观察角度而变化；而在薄膜情况下，光线甚至能与自身相互作用产生虹彩。所有这些效应都会在我们拍摄和制作的图像中显现。

1.7.1 散射

散射是一种普遍存在的光学现象，它影响着大多数图像的视觉效果。如图1.18所示，散射发生在任何含有悬浮粒子的透明介质中——例如空气、水或透明铝材。入射光穿过介质，直到光波撞击到粒子，

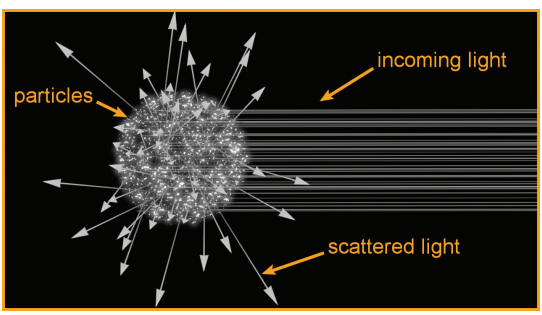


图 1.18 被粒子散射的光线

then ricochets off in a random direction. The scattering is omnidirectional, which is why the light can be seen from the side, from above, and even from behind the projecting light source. While it is much more visible in the dark, this phenomenon does not go away in daylight. In the dark, the scattered light might be 1,000 times brighter than the surrounding blackness; yet in the daylight, it will be the same intensity as the surrounding area.

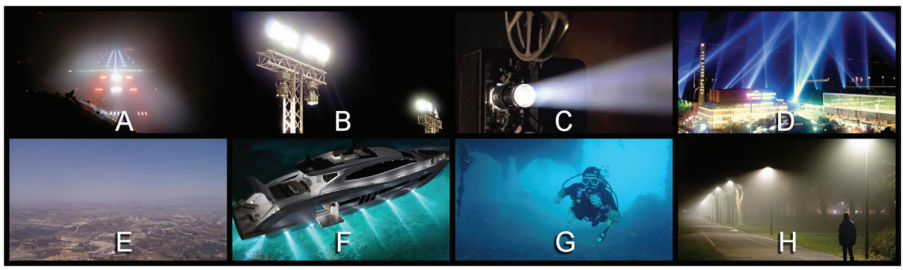


Figure 1.19 Examples of scattering

Figure 1.19 shows several examples of the omnidirectional nature of scattered light. Pictures A and B show how the omnidirectional scattering adds a halo around a light source. The size of the halo is caused by light waves traveling some distance from the light source before hitting a particle and then bouncing off in the camera's direction. The most common scattering is when the light waves are seen from the side, as in pictures C, D, and H. Underwater scattering can be seen with the yacht in picture F. My yacht does not have underwater lights, but perhaps yours does. Without yacht lights, underwater scattering would appear more like picture G.

Perhaps the most common scattering example is atmospheric haze, exemplified by picture E. In VFX, we often need to add atmospheric haze (some call it depth haze) to a shot, so it is important to know how it affects the picture. As we go deeper into the scene, the light scattering causes the loss of saturation (colorfulness), the lifting of the blacks, the loss of detail, and a convergence toward white or light gray.

Another attribute of scattering is that not all wavelengths are scattered equally and depend on the particle size. Now we can explain why the sky is blue and sunsets are red. The sky is blue because the sun's multi-spectral light waves pass through the atmosphere, and mostly the blue wavelengths

然后以随机方向反弹。散射是全方位的，这就是为什么可以从侧面、上方甚至投影光源后方看到光线。虽然在黑暗中更为明显，但这种现象在白天也不会消失。在黑暗中，散射光可能比周围的黑暗亮1000倍；而在白天，它的强度将与周围区域相同。

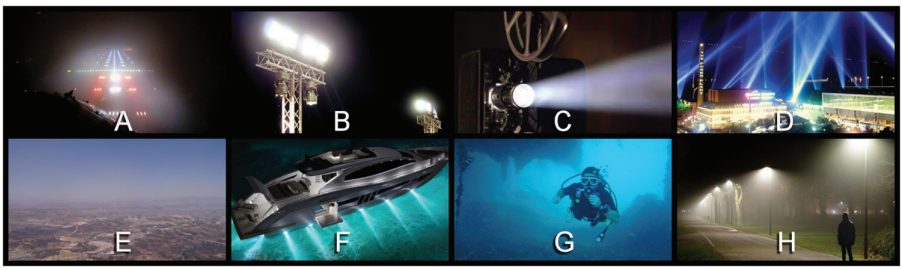


图 1.19 散射示例

图1.19展示了散射光全向特性的几个示例。图片A和B显示了全向散射如何在光源周围形成光晕。光晕的大小是由光波从光源传播一段距离后撞击粒子，然后反弹至相机方向造成的。最常见的散射现象是从侧面观察光波，如图C、D和H所示。水下散射可以在图片F中的游艇上观察到。我的游艇没有水下灯光，但您的或许有。若无游艇灯光，水下散射会更接近图片G的效果。

也许最常见的散射例子就是大气雾霾，如图E所示。在视觉特效中，我们经常需要为镜头添加大气雾霾（有些人称之为深度雾霾），因此了解它如何影响画面至关重要。随着我们深入场景，光线散射会导致饱和度（色彩鲜艳度）降低、暗部提亮、细节丢失，并逐渐趋近于白色或浅灰色。

散射的另一个特性是并非所有波长都受到同等程度的散射，这取决于粒子的大小。现在我们可以解释为什么天空是蓝色的，而日落是红色的。天空之所以呈现蓝色，是因为太阳的多光谱光波穿过大气层时，其中蓝色波长的光

are scattered and seen by an observer. As a result, a shower of blue light descends on us from all directions in the sky, making it appear blue. Without this scattering, the sunlight would pass right through the atmosphere, the sky would appear black, and the stars would be out during the day.

The sun is actually white (even though it is categorized as a yellow dwarf) but scattering much of the blue light away from it makes it appear yellow in the sky. At sunset, the sun's light travels through much more of the atmosphere than when overhead, so more blue is removed, resulting in lovely red sunsets as the remaining red light is scattered by the atmosphere. Add some extra dust or smoke to the atmosphere and even more blue light is scattered, making even more vibrant red sunsets. Viewed from space, the sky is black and the sun is white. Who knew that we get a blue sky, yellow sun, and red sunsets all due to scattering?

1.7.2 Polarized Light

We met polarized light earlier in Section 1.1.4 Polarization, where we learned that light waves have an orientation called polarization and that some reflecting surfaces can create polarized light. When light reflects off water, plastics, glass, and many other surfaces, much of it gets polarized. These polarized reflections introduce glare into our photography and cinematography, so polarized filters are used to knock down the glare, as illustrated in Figure 1.20. Hot spots are often polarized, so the polarizing filter can knock these down too, which helps to lower the dynamic range of the picture. Since it is extremely difficult to remove from the captured image, prudence dictates that we try to block it from getting into the picture in the first place. And that is exactly what polarized filters do for cameras.

Figure 1.20 Example of polarized glare

Since the polarized light waves all share the same orientation, they can be blocked by a filter that selects against that orientation. Figure 1.21 illustrates a few light waves of different polarization. The polarized filters are all oriented

被散射并被观察者看到。因此，一阵蓝光从天空的各个方向洒向我们，使天空呈现蓝色。如果没有这种散射，阳光将直接穿过大气层，天空会显得漆黑，而星星在白天也会显现出来。

太阳实际上是白色的（尽管它被归类为黄矮星），但大量蓝光被散射离开太阳，使得它在天空中看起来是黄色的。日落时，太阳光穿过的大气层比在头顶时厚得多，因此更多的蓝光被移除，剩余的红光被大气散射，形成了美丽的红色日落。如果大气中有额外的尘埃或烟雾，更多的蓝光会被散射，从而产生更加鲜艳的红色日落。从太空看，天空是黑色的，太阳是白色的。谁能想到，我们之所以拥有蓝天、黄色的太阳和红色的日落，全都是因为散射呢?

1.7.2 偏振光

我们在1.1.4节偏振中已经接触过偏振光，了解到光波具有一种称为偏振的取向，并且某些反射表面可以产生偏振光。当光从水面、塑料、玻璃和许多其他表面反射时，大部分光会变为偏振光。这些偏-

图 1.20 偏振眩光示例

镜面反射会给我们的摄影和摄像带来眩光，因此如图1.20所示，我们使用偏振滤镜来抑制眩光。高光区域通常具有偏振性，所以偏振滤镜也能有效减弱这些光斑，从而帮助降低画面的动态范围。由于眩光一旦进入拍摄画面就极难消除，谨慎的做法是从源头阻止其进入画面——这正是偏振滤镜为相机实现的功能。

由于偏振光波都具有相同的方向，它们可以被一个筛选掉该方向的滤光片阻挡。图1.21展示了几种不同偏振方向的光波。偏振滤光片均朝向

horizontally and are covered with micro-polymer "slots," all oriented in one direction like sub-micron Venetian blinds. If the polarization of the light wave matches the polarization of the filter, the light wave passes through, as in the example on the right. If the polarization of the light wave is perpendicular, as in the left example, the filter will block almost all of the light. The middle example shows a midway point where the polarization is off angle just a bit, so some of the light gets through.

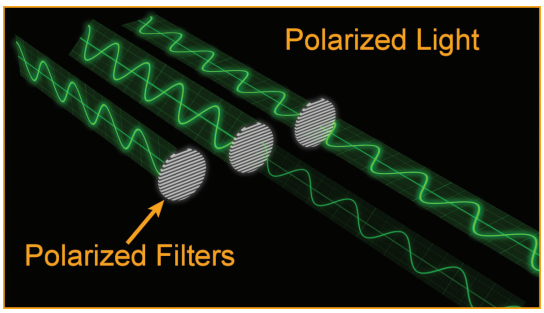


Figure 1.21 Horizontally polarized filters

This is how a linear polarized filter on a camera lens works. It is also how LCD displays modulate the light brightness of each pixel, as we will learn in Chapter 5: Display Space. You can have fun with polarized light by taking two pairs of polarized sunglasses and holding one lens in front of the other at a 90-degree angle, which will block almost all light. As you rotate one of the lenses, you will see the brightness change. If you don't, then one or both are simply tinted, not polarized.

Check out this video about polarized light and how polarized filters work. There might even be a sunglasses demonstration.

1.7.3 The Fresnel Effect

A material attribute that hovers between the diffuse and specular reflection models is the Fresnel effect, which is pervasive in the real world. The short story of the Fresnel (pronounced "fray-NEL") effect is that at grazing angles, an otherwise dull, diffuse surface suddenly becomes shiny and reflective

水平排列，并覆盖有微聚合物“狭缝”，全部朝一个方向排列，如同亚微米级的威尼斯百叶窗。如果光波的偏振方向与滤光片的偏振方向匹配，光波就能通过，如右侧示例所示。如果光波的偏振方向与之垂直，如左侧示例所示，滤光片将阻挡几乎所有的光。中间示例展示了一个偏振方向稍有偏差的中间状态，因此部分光得以通过。

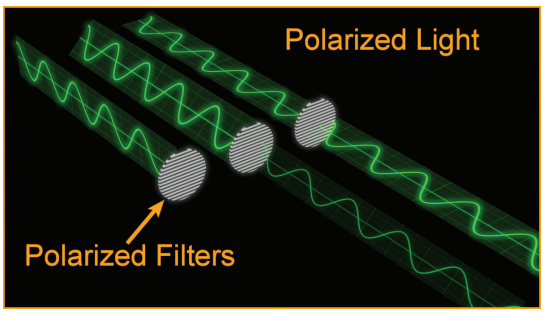


图 1.21　水平偏振滤光片

这就是相机镜头上的线性偏振滤镜的工作原理。正如我们将在第五章：显示空间中学习到的，这也是液晶显示器调节每个像素亮度的方式。你可以用两副偏光太阳镜来玩转偏振光，将其中一副镜片以90度角放在另一副前面，这样几乎会阻挡所有光线。当你旋转其中一副镜片时，你会看到亮度发生变化。如果没有变化，那么其中一副或两副只是染色镜片，而非偏振镜片。

看看这个关于偏振光以及偏振滤镜如何工作的视频。说不定还有太阳镜的演示呢。

1.7.3 菲涅尔效应

一种在漫反射和镜面反射模型之间徘徊的材料属性是菲涅耳效应，这在现实世界中无处不在。菲涅耳（发音为“fray-NEL”）效应的简要描述是：在掠射角度下，原本暗淡、漫反射的表面会突然变得闪亮且具有反射性。

(specular), as illustrated in Figure 1.22. As the POV (Point Of View) moves down from the high angle in Frame 1, the table surface remains stubbornly diffuse. However, starting with Frame 4, we start to see a bit of a reflection near the base of the bust; then on Frame 5, the tabletop sports a very shiny reflection, as if suddenly polished by the Fresnel fairy.

Figure 1.22 At grazing angles, surfaces suddenly become reflective

The Fresnel effect is caused by a variable surface refractive index coefficient (blah blah technobabble), thus making it highly shiny at grazing angles. Referring to Figure 1.23, the first panel illustrates how the observer's high viewing angle sees a dull, flat, diffuse reflected light from the tabletop. The middle panel shows how the low viewing angle sees a highly reflective surface with a reflection of the bust. The last panel shows a typical Fresnel effect curve for a typical dull surface material and how its reflectance (specularity) stays dull for most viewing angles and then suddenly spikes upwards at shallow, grazing angles.

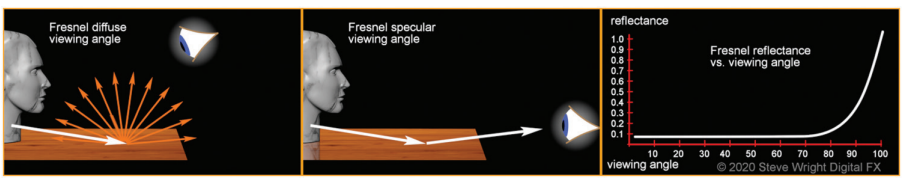


Figure 1.23 How surface reflectivity changes at the Fresnel angle

The Fresnel effect is pervasive and shows up everywhere. A marvelous example of this is seen in Figure 1.24, where the camera has made the appropriate grazing angle to the water surface. Keep in mind that the same water would appear nearly transparent if the viewing angle was up higher, looking down. In the world of CG, a Fresnel lighting model is now a standard option and is an absolute requirement for photorealistic renders. In compositing, we can use a Fresnel AOV (Arbitrary Output Variable) to add or enhance our own Fresnel effect.

(镜面反射)，如图1.22所示。当视角从帧1的高角度向下移动时，桌面始终保持顽固的漫反射状态。然而，从帧4开始，我们开始在半身像底座附近看到一点反射；接着在帧5中，桌面呈现出非常闪亮的反射，仿佛突然被菲涅耳精灵抛光过一般。

图1.22 在掠射角下，表面会突然变得具有反射性

菲涅尔效应是由可变的表面折射率系数引起的（等等技术术语），因此在掠射角度下会显得非常闪亮。参考图1.23，第一幅图展示了观察者从高视角看到桌面反射的暗淡、平坦、漫射的光线。中间图展示了低视角如何看到高度反光的表面以及半身像的倒影。最后一幅图展示了典型暗淡表面材质的菲涅尔效应曲线，其反射率（镜面反射强度）在大多数视角下保持暗淡，随后在浅掠射角度突然急剧上升。

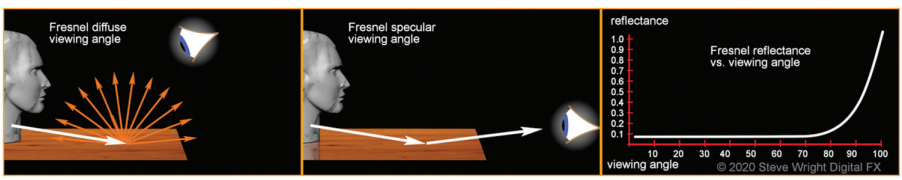


图 1.23 菲涅尔角处表面反射率的变化规律

菲涅尔效应无处不在，随处可见。图1.24展示了一个绝妙的例子，其中相机与水面形成了恰当的掠射角。请记住，如果视角更高、向下俯视，同样的水看起来几乎是透明的。在CG领域，菲涅尔光照模型如今已成为标准选项，也是实现照片级真实感渲染的绝对要求。在合成中，我们可以使用菲涅尔AOV（任意输出变量）来添加或增强我们自己的菲涅尔效果。

Figure 1.24 Example of the Fresnel effect on water

1.7.4 Color from Interference

Another way to get colored reflections from white light hitting surfaces and then interfering with itself is what is called structural color. This means the color comes from the structure of the material on a microscopic scale rather than from pigments or dyes. A prime example of this is iridescence, with several examples shown in Figure 1.25. It is caused by light waves reflecting off two very close surfaces, like the membrane of a soap bubble, and a few other real-world examples, both natural and man-made.

Figure 1.25 Iridescence

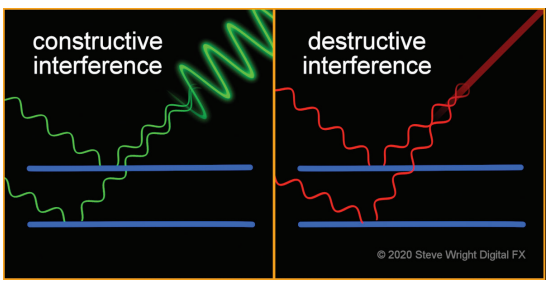


Figure 1.26 Color from interference

In a thin film membrane, like a soap bubble, some light waves reflect off the first surface while others penetrate to the second surface and then reflect. Figure 1.26 shows how the two sets of reflected light waves then interact with each other with both constructive and destructive interference to

图1.24 水面上菲涅尔效应的示例

1.7.4 干涉色

另一种从白光照射表面并随后与自身干涉产生彩色反射的方式被称为结构色。这意味着颜色来源于材料在微观尺度上的结构，而非来自颜料或染料。这种现象的一个典型例子是虹彩，图1.25展示了几个实例。它是由光波在两个非常接近的表面上反射引起的，比如肥皂泡的薄膜，以及其他一些自然界和人造的实际例子。

图 1.25 虹彩

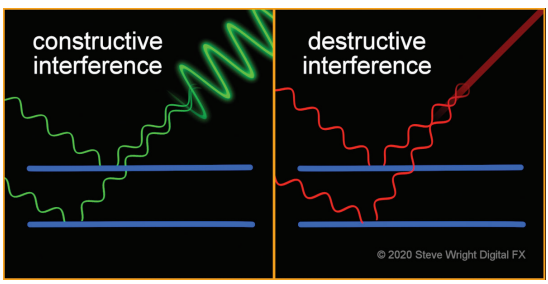


图 1.26 干涉产生的色彩

在薄膜中，比如肥皂泡，一些光波从第一表面反射，而另一些则穿透到第二表面然后反射。图1.26展示了两组反射光波如何相互作用，产生相长干涉和相消干涉，从而

produce the iridescence. If the wavelengths are in phase together, like the green waves, they sum together with constructive interference, making that wavelength brighter. Other wavelengths will be out of phase with each other, like the red waves, so they cancel each other out, with destructive interference damping out that color. This can produce very narrow wavelengths of light, making the color very pure and vibrant. Further, the viewing angle affects which frequencies interfere with each other, and this interference can give iridescent surfaces a lovely rainbow coloring.

1.7.5 Dichroic Filters

Dichroic filters are very pure and precise color filters used in lighting systems, projectors, photography and cinematography, spectroscopy, optical instruments, and laser systems. They make use of this constructive/destructive interference phenomenon to "sort" light waves into two paths: one that is transmitted through the filter and one that is reflected by it. They can be made very precise, allowing only a very narrow range of wavelengths of light to pass through. Another virtue is that since they reflect the unwanted light instead of absorbing it, they don't heat up, as regular dye filters do. Their name, "dichroic," might look like the Greek roots for "two" (*di*) and "color" (*chroma*) but alas, it actually refers to the two paths of light—transmitted and reflected. Looks like I need to brush up on my Greek!

1.8 How We See the World

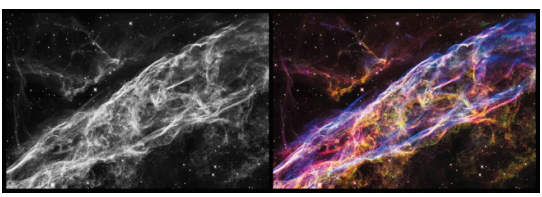


Figure 1.27 Our colorless universe

We now switch from the real world to us humans—since we are part of the World Space—to explore the process of image perception. Whether man or machine, perception involves three steps: initial perception, signal processing, and image formation. For humans, initial perception involves the photons impacting the retina. Signal processing is performed in the retina, optic nerve, and visual cortex. The final "display" occurs in the occipital lobe. We think. We will learn about machine perception in Chapter 2: Camera Space.

产生虹彩。如果波长同相，比如绿色波，它们会通过相长干涉叠加在一起，使该波长更亮。其他波长则会彼此异相，比如红色波，因此它们会通过相消干涉相互抵消，抑制该颜色。这可以产生非常窄的波长，使颜色非常纯净和鲜艳。此外，视角会影响哪些频率相互干涉，这种干涉可以赋予虹彩表面美丽的彩虹色。

1.7.5 二向色滤光片

二向色滤光片是一种非常纯净且精确的滤色片，广泛应用于照明系统、投影仪、摄影与电影摄影、光谱学、光学仪器以及激光系统中。它们利用这种相长/相消干涉现象将光波“分离”成两路：一路透过滤光片，另一路被其反射。这类滤光片可制作得极为精密，仅允许极窄波长范围的光通过。另一个优点是，由于它们反射不需要的光而非吸收，因此不会像普通染料滤光片那样发热。其名称“dichroic”看似源自希腊语词根“二”（di）和“颜色”（chroma），但遗憾的是，它实际上指的是光的两种路径——透射与反射。看来我得好好温习一下希腊语了！

1.8 我们如何看待世界

现在我们从现实世界转向我们人类——因为我们是世界空间的一部分——来探索图像感知的过程。无论是人还是机器，感知都涉及三个步骤：初始感知、信-

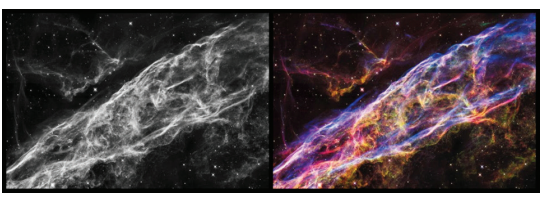


图1.27 我们无色的宇宙

nal处理，以及图像形成。对人类而言，初始感知涉及光子冲击视网膜。信号处理在视网膜、视神经和视觉皮层中进行。最终的“显示”发生在枕叶。我们思考。我们将在第2章：相机空间中学习机器感知。

You may be disappointed to learn that the universe is an utterly colorless grayscale. Color is an invention of our brains to distinguish between different wavelengths of light because it was a terrific survival advantage to be able to tell the difference between a green strawberry and a ripe one, or to spot an orange tiger hiding in the green undergrowth. Figure 1.27 illustrates the monochrome universe on the left as it actually appears, while the colorful image on the right is our brain-colorized version of it. As it turns out, the way that our brains create color for us is what makes it possible for us to have cameras, television, and movies in color. So, all of our color science is based on and pegged to the specific responses of the human visual system. No wonder color science is such a squishy subject.

FUN FACT: While your vision may be steady, the eye is constantly twitching with what are called "saccades." While the tempo varies with different activities, it is normally around three to five times per second.

1.8.1 Color Perception

I'm sure you've seen this many times before, but a quick recap will be helpful to refresh our terminology for the upcoming topics. As you no doubt know, our eyes respond to specific frequency ranges of light that our brains turn into RGB color sensations. We call these three colors the "primaries" and mix (add) them together to create secondary colors like yellow, cyan, and magenta with the additive color system of our vision. The familiar RGB color response curves are illustrated in Figure 1.28 (somewhat simplified). What you may not know is the incredible amount of image processing that goes on behind our perceptions.

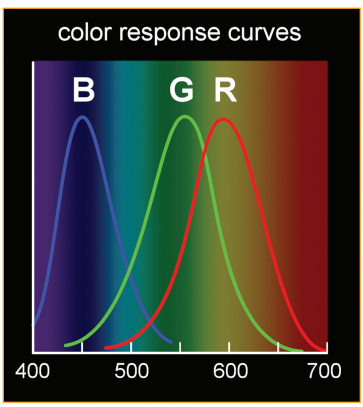


Figure 1.28 RGB color response curves

To take one small example of our image processing, consider how close together the green and red curves are in Figure 1.28. It makes no sense. Any reasonably designed color system would have the red curve much further

你可能会失望地发现，宇宙其实是毫无色彩的灰度世界。颜色只是大脑为区分不同波长光线而创造的感知——因为辨别青涩草莓与成熟果实、或发现藏身绿丛的橙色老虎，能带来巨大的生存优势。图1.27左侧展示了宇宙真实的单色面貌，而右侧绚丽的图像则是我们大脑为其渲染的彩色版本。事实上，正是大脑创造色彩的方式，才使得彩色相机、电视和电影成为可能。因此，我们所有的色彩科学都基于并受限于人类视觉系统的特定反应机制。难怪色彩科学是如此难以捉摸的领域。

有趣的事实：虽然你的视线可能很稳定，但眼睛会不断进行所谓的“扫视”微颤。尽管节奏因活动不同而变化，但通常每秒大约三到五次。

1.8.1 色彩感知

我确信你之前已经多次见过这个内容，但快速回顾一下将有助于为接下来的主题刷新我们的术语。毫无疑问，你知道我们的眼睛对特定频率范围的光线做出反应，大脑将这些光线转化为RGB颜色感觉。我们将这三种颜色称为“原色”，并通过我们视觉的加色系统将它们混合（相加）在一起，以创建如黄色、青色和品红色等二次色。熟悉的RGB颜色响应曲线如图1.28所示（略有简化）。你可能

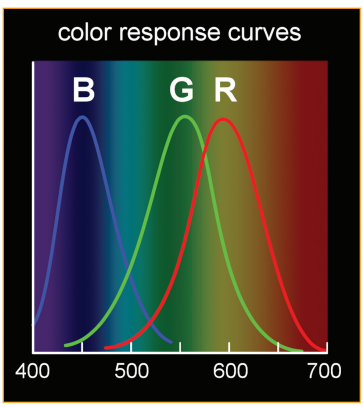


图 1.28 RGB 颜色响应曲线

不知道的是，在我们感知背后进行着数量惊人的图像处理。

以我们的图像处理为例，看看图1.28中绿色和红色曲线有多接近。这毫无道理。任何合理设计的色彩系统都会让红色曲线远得多

away from the green curve so we can easily tell the difference between red and green. Our image processing secret here is that the curves are not used directly to detect the color, but the *difference* between the two curves is used. The full extent of the image processing that goes on in our visual cortex is so complicated that we are very far from understanding it even to this day.

As we saw above, the sun and other "natural" light sources give off a wide spectrum of light, not just a few narrow frequencies. Since it was not practical to evolve a similarly wide-spectrum visual system to match sunlight, Mother Nature settled on the three-color "sub-sample" scheme we all know and love from color theory. The fact that we can see the full spectrum of color with our limited RGB eyes makes it possible for us to build cameras, television sets, movies, and color photographs based on those systems matching our RGB sensibilities. While it is true that print and film use CMYK inks and dyes, they use them to create RGB colors. This will be covered in detail in the subtractive color section of Chapter 4: Work Space. So, film and color photographs are just as RGB as HDTV.

FUN FACT: Humans have photoreceptors for three colors of light, while the mantis shrimp can see 12 colors plus polarized light!

1.8.2 CIE xy-chromaticity Diagram

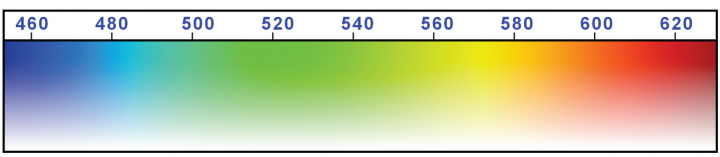


Figure 1.29 Chromaticity wavelengths

One of the foundational references for color is the CIE xy-chromaticity diagram, so understanding it is key. We start with a diagram of visible light in Figure 1.29, with the wavelength in nanometers marked with blue text. The colors start at the top totally pure at 100% saturation, then fade off to white at the bottom. If we put a grid over this diagram, we could identify any color as an xy location on that grid.

远离绿色曲线，这样我们就能轻易区分红色和绿色的差异。我们图像处理的秘诀在于，并非直接利用曲线检测颜色，而是利用两条曲线之间的差异。我们视觉皮层中进行的图像处理过程极其复杂，其完整机制至今仍远未被人类理解。

如上所述，太阳和其他“自然”光源发出的光谱范围很广，而不仅仅是少数几个狭窄的频率。由于进化出类似宽光谱的视觉系统来匹配阳光并不现实，大自然选择了我们熟知的色彩理论中的三色“子采样”方案。尽管我们的眼睛仅能感知有限的RGB色彩，却能看到完整的色谱，这使得我们能够基于匹配人类RGB感知的系统来制造相机、电视机、电影和彩色照片。虽然印刷和胶片确实使用CMYK油墨和染料，但它们是用这些材料来创造RGB色彩的。这将在第4章《工作空间》的减色法色彩部分详细阐述。因此，胶片和彩色照片与高清电视一样，本质上都是基于RGB原理的。

有趣的事实：人类拥有感知三种颜色光的光感受器，而螳螂虾能看到12种颜色外加偏振光！

1.8.2 CIE xy色度图

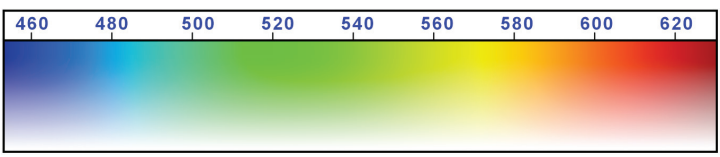


图1.29 色度波长

色彩的一个基础参考是CIE xy色度图，因此理解它至关重要。我们从图1.29中的可见光图谱开始，波长以纳米为单位用蓝色文字标注。颜色从顶部完全纯净的100%饱和度开始，然后逐渐向下褪为白色。若在此图上叠加网格，我们便可将任何颜色定位为该网格上的一个xy坐标位置。

In its current shape, Figure 1.29 is inconvenient for certain math operations that only color scientists care about. To make it more mathematically accessible, the diagram is carefully bent into a horseshoe shape, and a new xy grid is placed over it, producing the CIE xy-chromaticity diagram in Figure 1.30. This gives us a coordinate system to identify any color by its xy location plus much more elegant mathematical manipulations. The chart represents just chromaticity, pure hue, and saturation without luminance. While it is a 2D slice out of a 3D color space, the luminance has been factored out using a clever color space transformation.

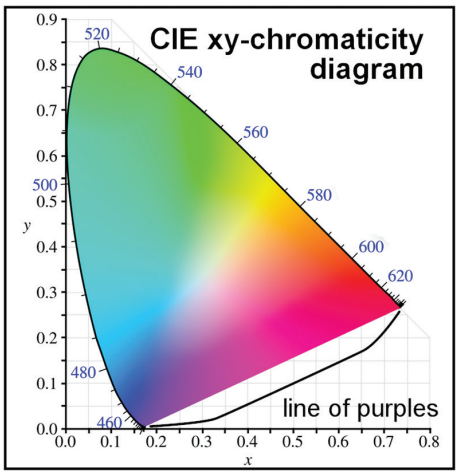


Figure 1.30 CIE xy-chromaticity diagram

Even though our visual system is complex beyond imagining and we may never fully understand it, we can do the next best thing, which is to measure the heck out of it so at least we understand what its capabilities are. So, in 1931, the International Commission on Illumination published the CIE xy-chromaticity diagram shown in Figure 1.30. Based on tests with a group of volunteers, it maps out the entire range of colors visible to the human eye. The xy location in the diagram provides a precise mathematical specification for any visible color. It is an essential tool for color management and is used to specify the color of printing inks, illuminated displays (monitors), and recording devices like scanners and digital cameras. It is used everywhere in our work and, even more importantly, throughout this book.

Referring to Figure 1.30, the blue numbers around the perimeter mark the wavelength in nanometers of the color. They transition from 100% saturated at the perimeter to completely desaturated in the center. The bottom edge of the graph without numbers is known as the "line of purples," which are not "natural" colors. There is no single frequency of light that creates those colors like there is for the other colors, and they are only made possible by mixing red and blue primaries. Some call purple an "unnatural color" or say it isn't real. Try explaining that to a purple honeycreeper.

在目前的形式下，图1.29对于某些只有色彩科学家关心的数学运算来说并不方便。为了使它在数学上更易于处理，该图被精心弯曲成马蹄形，并添加了新的xy网格。

置于其上，便得到了图1.30中的CIE x y色度图。这为我们提供了一个坐标系，可以通过颜色的xy位置来识别任何颜色，并且能进行更为优雅的数学运算。该图表仅表示色度、纯色调和饱和度，不包含亮度。虽然它是从三维色彩空间中截取的一个二维切片，但通过巧妙的色彩空间变换，亮度已被分离出去。

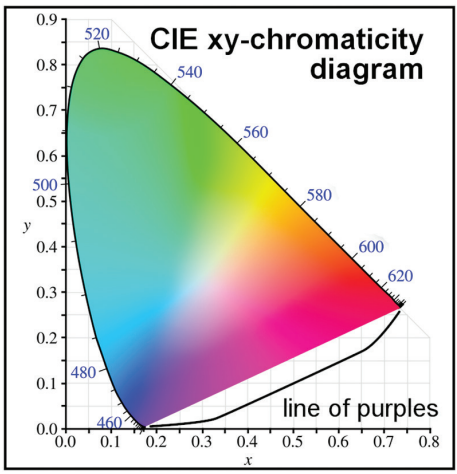


图 1.30 CIE xy 色度图

尽管我们的视觉系统复杂到难以想象，我们可能永远无法完全理解它，但我们能做次优选择，即对其进行详尽测量，至少让我们了解其能力范围。于是，1931年国际照明委员会发布了如图1.30所示的CIE xy色度图。基于一组志愿者的测试数据，该图完整绘制了人眼可见的全部色彩范围。图中的xy坐标位置为任何可见色彩提供了精确的数学定义。它是色彩管理的重要工具，被用于规定印刷油墨、发光显示器（监视器）以及扫描仪和数码相机等记录设备的色彩标准。它不仅在我们的工作中无处不在，更重要的是，本书通篇都将使用这一体系。

参考图1.30，边缘的蓝色数字表示颜色的波长（单位：纳米）。颜色从边缘的100%饱和度逐渐过渡到中心的完全非饱和状态。图表底部未标注数字的边缘被称为“紫色线”，这些并非“自然”颜色。不存在

单一频率的光无法像其他颜色那样生成紫色，紫色只能通过混合红蓝两种原色来实现。有些人称紫色为“非自然色”或说它不真实。试着对一只紫蜜雀解释这个道理吧。

Many find the odd shape of the CIE xy-chromaticity diagram hard to understand. This video reveals why it has that shape and much more.

1.8.3 Our Visual Response

Okay, so that's color. What about sensitivity? How do our eyes respond to different levels of luminance? Our visual system has a logarithmic response, as illustrated in Figure 1.31. This means that in the darks, very small changes in luminance result in large changes in our brightness perception; while in the brights, large changes in luminance result in small changes in perception.

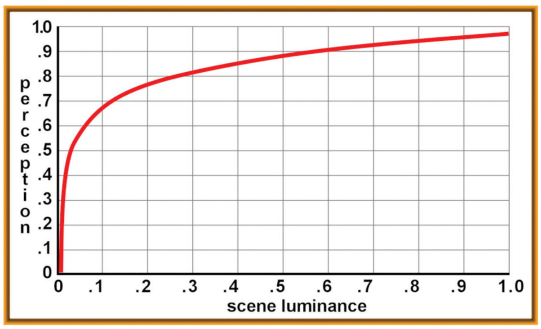


Figure 1.31 Log response of the visual system

Our visual system dramatically increases sensitivity in the darks to the huge degree indicated by the steep slope of the log curve at the left end between the scene luminance values of 0 and .1. When the luminance gets out to around .3, the curve starts to level off making our eyes progressively less sensitive to bright light. One interesting side effect of this log response of our visual sensitivity is that if the luminance is doubled, to our visual system the perceived brightness has only increased by 18%. This should also serve as a stark reminder of the difference between luminance and brightness.

We evolved this log response to light to satisfy two survival conditions: first, at the dark end, to see the tiger hidden in the shady bushes, and second, at the bright end, to protect our eyes so full sunlight would not blind us. Having a log response like this allows the brightness-adapted eye to span a huge dynamic range of 24 stops, which is a contrast ratio of almost 17 million to one!

许多人觉得CIE xy色度图的奇特形状难以理解。本视频揭示了它为何具有这种形状，以及更多奥秘。

1.8.3 我们的视觉响应

好的，颜色部分就是这样。那么敏感度呢？我们的眼睛如何响应不同的亮度水平？我们的视觉系统具有对数响应特性，如图1.31所示。这意味着在暗处，亮度的微小变化会导致我们感知到的亮度发生巨大变化。

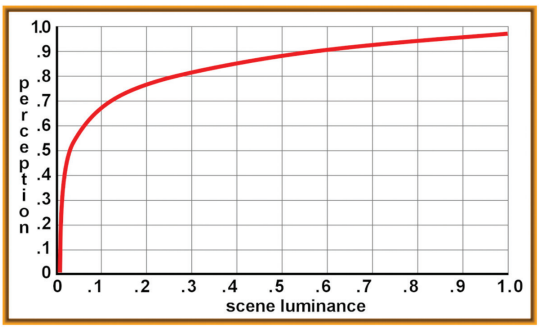


图 1.31 视觉系统的对数响应

感知；而在明亮处，亮度的巨大变化只会引起感知上的微小变化。

我们的视觉系统在暗光下的敏感度显著增强，其提升幅度之大，正如对数曲线左端在场景亮度值0到0.1之间的陡峭斜率所示。当亮度达到约0.3时，曲线开始趋于平缓，使得我们的眼睛对强光的敏感度逐渐降低。这种视觉敏感度的对数响应带来一个有趣的副作用：如果亮度增加一倍，我们的视觉系统感知到的亮度仅提升18%。这也应作为一个鲜明的提醒，让我们注意亮度与感知明度之间的差异。

我们进化出这种对数光响应是为了满足两个生存条件：首先，在黑暗端，要能看见隐藏在阴暗灌木丛中的老虎；其次，在明亮端，要保护我们的眼睛，以免强光致盲。拥有这样的对数响应，使得亮度适应后的眼睛能够跨越高达24档的动态范围，这相当于近1700万比一的对比度！

So here is the basic conundrum of digital images: the universe is linear, but our vision is log! As we will learn in Chapter 2, the camera captures the scene linearly. The real world is linear, CG is linear, and compositing must be done in linear space. All of this is true to get results that match the real world. But in the end, we are going to view the linear results with our log eyes. Managing this difference is the root of all evil in color science.

Another well-established characteristic of our visual response is that it can detect a difference in luminance between two adjacent patches of about 1%. Interestingly, this is consistent from the darks up through the brights. What this means is that in the darks, the 1% difference represents a very small change in luminance but in the brights, that same 1% represents a large change in luminance. This makes sense based on our understanding of Figure 1.31, the log response of our vision, where the eye is much more sensitive in the darks so a very small luminance change there will be easily seen but in the brights, the eye is much less sensitive so a larger luminance change is required to trigger the 1% threshold. Keep this 1% difference rule in mind for Chapter 3: Storage Space. It's important.

1.8.4 Adaptation

Adaptation is the astonishing ability of the eye to adjust to different viewing conditions to maintain a constancy of color regardless of the lighting conditions. Our eyes can adapt to not only huge ranges of brightness but also the color of light. This is particularly vexing to those designing display systems because they are designed to appeal to the human visual system, but adaptation makes that a moving target. The answer is to control the viewing conditions so the eye is in a known state allowing the display designers to at least have a stable target.

The purpose of all this adaptation is to maintain a consistent appearance of an object to the eye even under varying lighting conditions. This is a helpful survival skill that allows us to identify the poisonous blueberries of the deadly nightshade (Altropa belladonna) regardless of the lighting conditions. In this section, we will take a brief look at three of the most influential adaptations: general brightness adaptation, lateral brightness adaptation, and chromatic adaptation. The exciting thing here is that this is not even a complete list, so we will be skipping photobleaching recovery, focus adaptation,

这就是数字图像的基本难题：宇宙是线性的，但我们的视觉是对数的！正如我们将在第二章学到的，相机以线性方式捕捉场景。真实世界是线性的，计算机图形是线性的，合成也必须在线性空间中进行。所有这些都符合现实世界的规律。但最终，我们将用对数视觉来观看这些线性结果。处理这种差异，正是色彩科学中一切问题的根源。

我们视觉反应的另一个公认特征是，它能够检测出两个相邻区域之间约1%的亮度差异。有趣的是，这一特性从暗部到亮部都保持一致。这意味着在暗部，1%的差异代表亮度的极小变化，而在亮部，同样的1%却代表亮度的巨大变化。这基于我们对图1.31的理解是合理的——视觉的对数响应表明，眼睛在暗部更为敏感，因此微小的亮度变化也容易被察觉；而在亮部，眼睛的敏感度大幅降低，需要更大的亮度变化才能触发1%的差异阈值。请牢记这1%差异规则，这在第三章“存储空间”中至关重要。

1.8.4 自适应

适应是眼睛令人惊奇的能力，它能调整到不同的观看条件，以保持颜色的恒定性，无论光照条件如何。我们的眼睛不仅能适应巨大的亮度范围，还能适应光的颜色。这对设计显示系统的人来说尤其令人烦恼，因为这些系统旨在吸引人类视觉系统，但适应能力使其成为一个移动的目标。解决办法是控制观看条件，使眼睛处于已知状态，这样显示设计师至少能有一个稳定的目标。

所有这些适应的目的是为了在不同光照条件下，保持物体在视觉上的一致性外观。这是一种有益的生存技能，使我们能够在任何光照条件下识别出致命颠茄（Altropa belladonna）的有毒蓝莓。在本节中，我们将简要介绍三种最具影响力的适应：整体亮度适应、侧向亮度适应和色彩适应。令人兴奋的是，这甚至不是一个完整的列表，因此我们将跳过光漂白恢复、焦点适应等。

visual learning adaptation, and peripheral adaptation for the sake of brevity. Perhaps in the second edition.

1.8.4.1 General Brightness Adaptation

This refers to the eye's adaptation to the overall level of brightness of the scene: broad daylight, moonlight, etc. This entails the familiar pupil size change, of course, but it also entails changes in the retina's light sensitivity. The astonishing thing about our visual system is its consistency under these varying brightness extremes. A white flower will be perceived as white whether seen in broad daylight or dim moonlight.

However, there are known degradations of the visual system under low light conditions. Things will appear to have a lower contrast and objects become less colorful. Take a colorful photo outdoors in sunlight and the colors will appear vibrant and the blacks quite dark. Walk indoors to a lower light level and (after adaptation) the photo will lose saturation and the blacks will appear less black. These perception changes come from the adaptation of the eye to the lower light conditions.

1.8.4.2 Lateral Brightness Adaptation

Photoreceptors in the retina change their sensitivity based on what their neighbors are seeing. The sensitivity of a photoreceptor may increase or decrease depending on the amount of light seen by its neighbors in a process known as lateral brightness adaptation. The manifestation of that is that the apparent contrast of an image will be lowered if the area immediately surrounding it is dark. As you compare the gray chips in Figure 1.32 left and right, each square on the black background appears lighter than its partner on the white background.

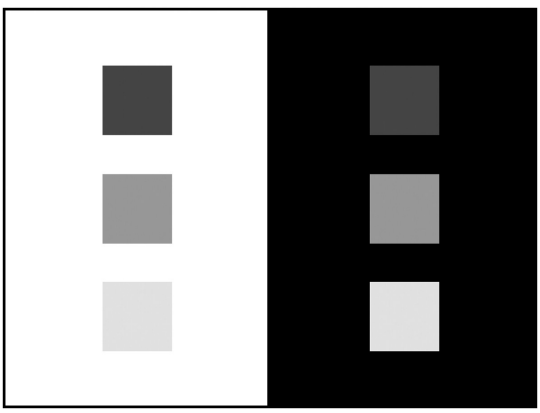


Figure 1.32 Lateral Brightness Adaptation

视觉学习适应，以及为了简洁起见的外围适应。也许在第二版中。

1.8.4.1 总体亮度适应

这指的是眼睛对场景整体亮度水平的适应：如白昼、月光等。这当然包括我们熟悉的瞳孔大小变化，但也涉及视网膜光敏感度的调整。我们视觉系统令人惊叹之处在于，它在这些极端亮度变化下仍能保持一致性。无论是在明亮的日光下还是在昏暗的月光下，一朵白花都会被感知为白色。

然而，在低光照条件下，视觉系统会出现已知的退化现象。物体看起来对比度会降低，色彩也会变得不那么鲜艳。在阳光下拍摄一张色彩丰富的户外照片，颜色会显得鲜艳，黑色也会非常深邃。走进室内光线较暗的环境（经过适应后），照片会失去饱和度，黑色看起来也不再那么深黑。这些感知变化源于眼睛对低光照条件的适应。

1.8.4.2 横向亮度适应

视网膜中的光感受器会根据其邻近细胞的视觉信息调整自身的灵敏度。这一过程被称为侧向亮度适应，即光感受器的灵敏度可能增加或减少，取决于其邻近细胞接收到的光量。其表现是，如果图像周围区域——

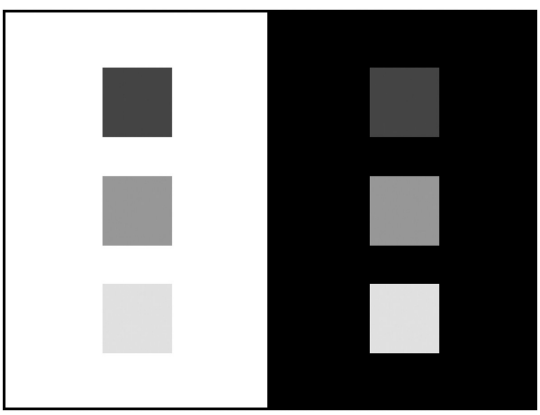


图 1.32 横向亮度适应

天色已暗。当你比较图1.32左右两侧的灰色色块时，每个黑色背景上的方块看起来都比白色背景上的对应方块更亮。

This adaptation is considered in the design of display systems and the environment in which they are viewed. It dictates the gamma of HDTV as well as the dark surround of theaters. You will (or should) find the room where colorists work with walls that are painted 18% gray, which is mid-gray for the human eye.

1.8.4.3 Chromatic Adaptation

The eye adapts to the color of a light source to turn it to white to maintain color constancy. In a room with 3200-degree tungsten lighting, there will be a preponderance of red and yellow wavelengths. The longer wavelength photoreceptors become somewhat desensitized, while the shorter wavelength receptors become more sensitive, resulting in the apparent color of the light shifting from yellow to white—our own personal built-in white balance. Once again, the white flower will appear white under different colored lighting after the eye has adapted to the color.

In addition to photoreceptors adjusting to the light color, we also have a neural processing response. One of the components of maintaining color consistency is that the brain compares an object's color to color memories of the object to offer a "color correction" to the perception, to bring it in line with past experience. Designers of display systems account for chromatic adaptation by including room light sensors that analyze the chromaticity of the ambient light and adjust the RGB values accordingly. This also points out the importance of using standardized light conditions when engaging in color-critical work. Artists like graphic designers, colorists, photographers, and editors need to use a standard D65 illuminant to avoid influencing their color perception and degrading their work product.

1.8.5 Calculating Luminance

Our visual response to luminance outlined above is a general statement of the response of the eye to light overall. But when color is thrown into the mix, things get more complicated. It turns out that our sensitivity to luminance depends on the color of the light. Calculating the luminance of an image is a fundamental operation pervasive throughout digital imaging software. It is

这种适应性在显示系统的设计及其观看环境中被考虑。它决定了高清电视的伽马值以及影院的暗环境。你会发现（或应该发现）调色师工作的房间墙壁被涂成18%灰色，这对人眼来说是中灰色。

1.8.4.3 色适应

人眼会适应光源的颜色，将其视为白色以保持色彩恒常性。在3200K色温的白炽灯照明房间内，红光和黄光波长会占据主导。长波感光细胞会变得相对不敏感，而短波感光细胞则更加敏感，导致光的表观颜色从黄色转变为白色——这是我们自身内置的白平衡功能。当眼睛适应颜色后，那朵白花在不同色光下将再次呈现白色。

除了感光细胞会适应光线颜色外，我们还有神经处理反应。维持色彩一致性的要素之一，是大脑会将物体的颜色与对该物体的色彩记忆进行比较，从而对感知进行“色彩校正”，使其与过去的经验保持一致。显示系统的设计者通过加入环境光传感器来考虑色适应，这些传感器分析环境光的色度并相应调整RGB值。这也指出了在进行色彩关键工作时使用标准化光照条件的重要性。像平面设计师、调色师、摄影师和编辑这样的艺术家需要使用标准的D65光源，以避免影响他们的色彩感知并降低其工作成果的质量。

1.8.5 计算亮度

我们上述对亮度的视觉反应是对眼睛对光线整体反应的一般性陈述。但当色彩加入其中时，情况就变得更加复杂。事实证明，我们对亮度的敏感度取决于光的颜色。计算图像的亮度是贯穿数字成像软件的基本操作。

needed for data compression, color correction, color space transformations, and much more. The goal is to create a grayscale image that has the same perceived brightness as the original color image.

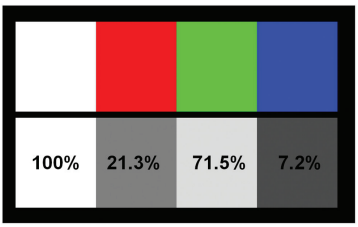


Figure 1.33 Color luminance

As we know, our eyes are only sensitive to three colors: red, green, and blue. Oddly, our sensitivity to each color is quite different. Our eyes respond very strongly to green, less so to red, and hardly at all to blue light. What this means is that if each color is presented at the same level of brightness, we do not see them as equally bright. Our perception of brightness for each primary color is illustrated in Figure 1.33, where the white chip gets 100% luminance, of course, and each of the colors gets its appropriate fraction. The official formula for creating a luminance image for humans is:

$$\text{Luminance} = 0.213*R + 0.715*G + 0.072*$$

Notice how it mixes them in the same proportions as the color luminance in Figure 1.33. If we tried to create a luminance image simply by adding 1/3 R + 1/3 G + 1/3 B, the green grass would be underrepresented (0.333 instead of 0.715) and appear too dark, and the blue sky would be overrepresented (0.333 instead of 0.072) and appear too bright. The colors must be mixed in the right proportions for the resulting luminance image to appear correctly to the eye. Figure 1.34 shows the difference. Starting with the original RGB image on the left, the proper luminance version in the center looks reasonable. However, the straight 1/3 + 1/3 + 1/3 version on the right looks most unnatural with its over-bright sky and too-dark lawn.

Figure 1.34 Creating a luminance image the right way and the wrong way

用于数据压缩、色彩校正、色彩空间转换等诸多方面。其目标是生成一幅与原始彩色图像具有相同感知亮度的灰度图像。

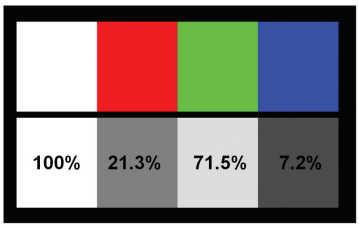


图 1.33 色彩亮度

众所周知，我们的眼睛只对三种颜色敏感：红、绿、蓝。奇怪的是，我们对每种颜色的敏感度差异很大。眼睛对绿色的反应非常强烈，对红色的反应较弱，而对蓝光的反应几乎微乎其微。这意味着，如果每种颜色以相同的亮度呈现，我们并不会觉得它们一样亮。我们对每种原色亮度的感知颜色如图1.33所示，其中白色芯片获得100%的亮度，当然，每种颜色都获得其适当的比例。为人类创建亮度图像的官方公式为：

$$Luminance = 0.213*R + 0.715*G + 0.072*$$

注意它是如何按照图1.33中颜色亮度的相同比例混合它们的。如果我们试图简单地通过1/3 R + 1/3 G + 1/3 B相加来创建亮度图像，绿色的草地会因比例不足（0.333而非0.715）而显得过暗，蓝色的天空则会因比例过高（0.333而非0.072）而显得过亮。必须按正确比例混合颜色，生成的亮度图像在人眼中才能呈现正确效果。图1.34展示了这种差异。从左边的原始RGB图像开始，中间正确的亮度版本看起来是合理的。然而，右侧直接使用1/3 + 1/3 + 1/3混合的版本则显得极不自然——天空过亮，草坪过暗。

图 1.34 正确与错误的亮度图像创建方法

To summarize, if you want to make a grayscale image from an RGB image that will look natural to the eye, you need to sum together (in round numbers) 21% of the red plus 72% of the green plus 7% of the blue. This will produce results very close to the previous luminance equation.

1.8.6 Metamers

Figure 1.35 illustrates another human visual idiosyncrasy called a metamer. A metamer is where two different light source emission spectrums produce the same perceived color. In this example, the yellow ball on the left is illuminated by a narrow frequency of pure yellow light while the right ball is lit with a mix of RGB colors, and yet the two balls are perceived to be the same yellow color—a metamer.

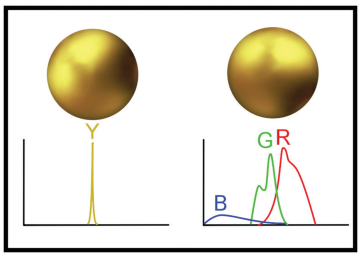


Figure 1.35 Metamer

Metamers are important to us for two very different reasons. There are "evil" metamers, where a photographed item will appear as one color to the photographer but a different color in the exposed image. This is because the photographer had a metamer but the camera did not. They are particularly notorious for virtual productions using LED walls. Then there are "good" metamers, which make it possible for us to have color photographs, RGB monitors, and digital cinema projectors. Allow me to illustrate.

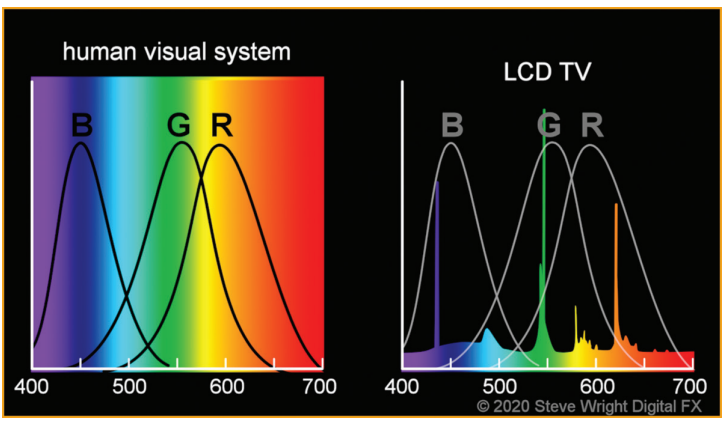


Figure 1.36 The essence of metamers

总而言之，若要从RGB图像生成视觉上自然的灰度图像，你需要将大约21%的红色、72%的绿色与7%的蓝色相加。这样得到的结果会非常接近先前的亮度公式。

1.8.6 同色异谱色

图1.35展示了另一种人类视觉特性，称为同色异谱。同色异谱是指两种不同光源的发射光谱产生相同感知颜色的现象。在此示例中，左侧黄球由窄频纯黄光照射，而右侧球体则由混合RGB光照明，但两个球体却被感知为相同的黄色——这正是同色异谱现象。

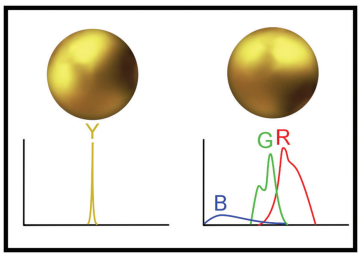


图 1.35 同色异谱体

同色异谱对我们来说非常重要，原因有二且截然不同。一种是“有害”同色异谱，即被摄物体在拍摄者眼中呈现一种颜色，但在曝光图像中却显示为另一种颜色。这是因为拍摄者感知到的光谱匹配与相机传感器不匹配。这种现象在使用LED墙的虚拟制作中尤为棘手。另一种则是“有益”同色异谱，它使得彩色照片、RGB显示器和数字电影投影成为可能。请容我举例说明。

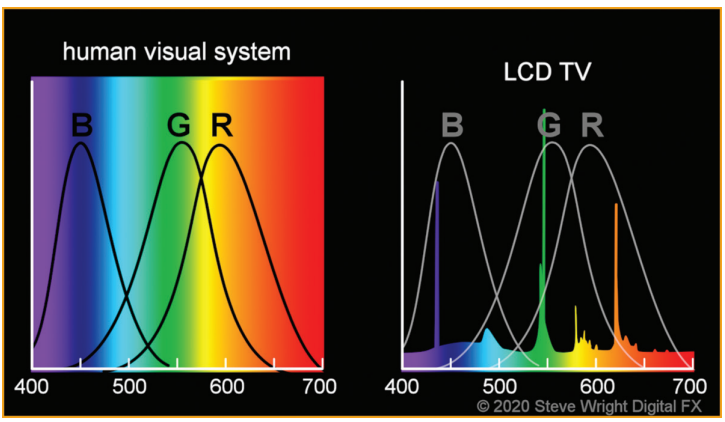


图 1.36 同色异谱的本质

Figure 1.36 illustrates how an LCD TV puts out a three-color "spiky" emission spectrum carefully aligned with our color sensitivities. Now consider the color yellow. We get that color from the combination of the green and red emissions of the TV screen. However, we can also see yellow with a single frequency of light of around 580 nanometers. If we had to make monitors that put out the full spectrum of color, such as the yellow frequency and all the rest, it becomes technically impossible to make a color monitor. The fact that we can "stimulate" yellow with red and green (a good metamer) makes it possible for us to use RGB monitors to replicate the full spectrum. This additive nature of our vision allows us to make monitors that only require three primary colors to reproduce all the colors of our perceptual rainbow. Without this three-color "subsampling" of our visual system, it would not be possible to have color movies, television, or photographs.

FUN FACT: Metamerism demonstrates that our perception of color is determined not solely by the physical properties of light, but also by how our visual system interprets these properties. This has major implications for color theory and the importance of considering the viewer and viewing conditions.

1.9 Conclusion

As we have seen, the real world is immensely complex and generates light far in excess of our limited image capture and display devices, in both scope and intensity. Real-world scenes reflect continuous spectrums of light, while our imaging devices can only capture limited RGB subsamples—yet somehow the images still look natural. Surfaces can reflect light with subtractive color in diffuse or specular ways, or a combination of both, in addition to mixing things up with physical effects to reflect light in even more complex ways, such as interference. Our color perceptions have been documented by the CIE xy-chromaticity chart, which provides a unifying method of quantifying human perception and is used throughout the entire digital imaging pipeline—and this book.

The next chapter, Camera Space, explores what happens when we try to capture a high dynamic range world with a limited dynamic range camera,

图1.36展示了液晶电视如何输出与人类色彩感知精确匹配的三色"尖峰状"发射光谱。现在考虑黄色：我们通过电视屏幕的绿色和红色光组合获得这种颜色，但也能通过约580纳米的单一频率光线感知黄色。若显示器需直接发出完整光谱（包括黄色频率及其他所有颜色），技术上将无法实现彩色显示器的制造。而利用红绿光"激发"黄色感知（这是一种优良的同色异谱现象），使得RGB显示器能够复现全光谱成为可能。人类视觉的这种加色特性，让我们仅需三种原色就能重现感知中的所有色彩。若视觉系统不具备这种三色"子采样"机制，彩色电影、电视和摄影都将无法实现。

有趣的事实：同色异谱现象表明，我们对颜色的感知不仅取决于光的物理特性，还取决于我们的视觉系统如何解读这些特性。这对色彩理论以及考虑观察者和观察条件的重要性具有重大意义。

1.9 结论

正如我们所知，现实世界极其复杂，无论在范围还是强度上，其产生的光线都远超我们有限的图像采集与显示设备的能力。真实场景反射的是连续光谱，而我们的成像设备只能捕捉有限的RGB子样本——然而图像看起来依然自然。表面可以通过漫反射或镜面反射的方式以减色模式反射光线，或两者结合，此外还会与物理效应混合，以更复杂的方式（如干涉）反射光线。我们的色彩感知已通过CIE xy色度图得以记录，该图提供了一种量化人类感知的统一方法，并贯穿整个数字成像流程——以及本书。

下一章《相机空间》将探讨当我们尝试用动态范围有限的相机捕捉高动态范围世界时会发生什么，

and the continuous spectrum of real-world light becomes subsampled to just three RGB colors. There is a detailed exploration of how digital cameras work and their most inner secrets are laid bare. There is also a revelation of the destructive things that the cameras themselves can do to our images with their rolling shutters, anamorphic compression, cropped image sensors, and the addition of digital noise. And yet they still take lovely pictures. It is in the very next chapter that you will learn how.

现实世界中的连续光谱被下采样为仅有的三种RGB颜色。书中详细探讨了数码相机的工作原理，揭示了其最核心的秘密。同时，也揭露了相机自身可能对图像造成的破坏性影响——如滚动快门、变形压缩、裁剪式图像传感器以及数字噪点的添加。然而，它们依然能拍出美丽的照片。就在接下来的章节中，你将学到其中的奥秘。

2 Camera Space

The digital imaging pipeline starts with the camera, which sets the upper limit on the quality of the captured images. Even if you are not a shooter (cinematographer or photographer), you should understand how cameras capture images because you must live with the consequences and may even have to fix them. Here, in Camera Space, there are many surprises.

We start this chapter with a discussion of how the camera "sees" the world and embosses its fingerprint on the captured image, which also explains why the same scene is seen differently by different cameras. We will trace the light path through the Bayer array and imaging sensor, and then through the camera's imaging calculations to produce the output image—only to discover that the camera's resolution is less than advertised and that much of the color is made up during "color reconstruction." Along the way, we will pick up some information on camera gamuts, the effect of different light sources, and how they affect the camera differently than us.

On the seamier side of cinematography, we do a deep dive into camera and lens artifacts with an eye toward preventing or curing them. We cannot discuss Camera Space without exploring the role of the digital imaging technician

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2 相机空间

数字成像流程始于相机，它设定了所捕获图像质量的上限。即使你不是拍摄者（电影摄影师或摄影师），也应当理解相机如何捕捉图像，因为你必须面对其带来的结果，甚至可能需要修复问题。在相机空间这里，存在着许多意想不到的情况。

本章开篇，我们将探讨相机如何“看见”世界，并在捕获的图像上留下其独特的印记，这也解释了为何不同相机对同一场景的呈现会有所差异。我们将追踪光线穿过拜耳阵列和图像传感器的路径，随后通过相机的成像计算生成输出图像——结果却发现相机的实际分辨率低于宣传数值，且大量色彩是在“色彩重建”过程中合成的。在此过程中，我们还将了解相机色域、不同光源的影响，以及它们对相机与对人眼产生差异的作用机制。

在电影摄影较为阴暗的一面，我们深入探讨相机和镜头伪影，着眼于预防或修复它们。我们无法在不探讨数字影像技师作用的情况下讨论相机空间。

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(DIT)—the human bridge between the tech and the screen. We will also get a succinct introduction to the new *wunderkind* of VFX, the virtual production stage, then wrap things up with a look back on 35-millimeter (mm) film.

2.1 Camera Perceptions of Color

In this first section, we will examine how cameras see the world and convert photons to RGB values. We will start by delving into the spectral sensitivity of cameras and how it is determined by the Bayer array, which we can then use to determine the camera's gamut. We will then reveal the "secret sauce" that camera manufacturers use to "sweeten" the captured image to produce the beauty that is photography. From there, we will look at how light sources affect the camera's view of the world, then close off the chapter with the great CCD vs. CMOS debate.

2.1.1 Spectral Sensitivity

Just as humans have differences in color perception, so do cameras—not just differences in their sensitivity to light, but in their perception of color itself. This means that different cameras will "see" the same scene slightly differently, as depicted in Figure 2.1. What this means to colorists is that when multiple cameras are used for a production (which they always are), they must color grade the clips to make them all appear to be shot with the same camera. Those of us in visual effects hope that the colorist has applied a "base grade" to all the clips so that they all appear uniform and well exposed regardless of which camera was used before we lay in our VFX. What we don't want to do is comp in a CG character to one shot that is overexposed, but the next shot is too green and the following shot has a low blue channel. It becomes nearly impossible to match our elements to wildly differing plates.

Figure 2.1 The same scene as "seen" by three different cameras

(DIT)——技术与银幕之间的人类桥梁。我们还将简要介绍视觉特效领域的新秀——虚拟制作舞台，最后回顾35毫米胶片，以此作为收尾。

2.1 相机对颜色的感知

在第一部分中，我们将探讨相机如何观察世界并将光子转换为RGB值。我们将从深入分析相机的光谱灵敏度开始，探讨其如何由拜耳阵列决定，并借此确定相机的色域范围。随后，我们将揭示相机制造商用来“润色”捕获图像、成就摄影之美的“秘制配方”。接着，我们会研究光源如何影响相机对世界的观察，最后以CCD与CMOS传感器的经典之争作为本章的收尾。

2.1.1 光谱灵敏度

正如人类对色彩的感知存在差异，相机亦是如此——不仅对光的敏感度不同，对色彩本身的感知也各不相同。这意味着不同相机"看到"的同一场景会略有差异，如图2.1所示。对于调色师而言，当制作中使用多台摄像机时（实际拍摄总是如此），他们必须对素材进行色彩分级，使所有镜头看起来像是用同一台摄像机拍摄的。我们视觉特效工作者希望调色师已对所有素材应用了"基础调色"，这样无论前期使用何种摄像机，在我们添加视觉特效前，所有镜头都能呈现统一且曝光良好的状态。我们最不希望看到的是：将CG角色合成到过度曝光的镜头后，下一镜却偏绿，紧接着的镜头又出现蓝色通道不足的情况。若要与差异巨大的实拍画面匹配元素，这几乎是不可能完成的任务。

图2.1 同一场景被三台不同相机“看到”的效果

There are two causes for these perceptual differences: different spectral sensitivities and different camera imaging math. Spectral sensitivity refers to how sensitive the imaging sensor is to specific wavelengths of light. For example, camera A in Figure 2.1 has a very sensitive blue sensor giving it a blue cast. Camera B lacks low light sensitivity, so is heavy in the darks. Camera C is more sensitive in the reds and greens giving it a yellow cast. We will see how this issue is addressed with the ACES workflow in Chapter 6: Color Management.

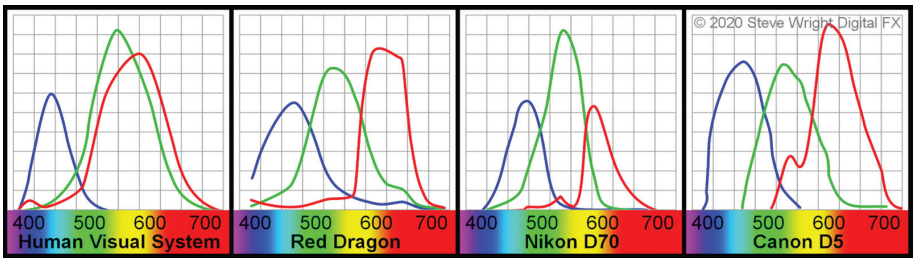


Figure 2.2 Comparison of the human visual system to three camera spectral sensitivities

Figure 2.2 compares the spectral sensitivity of the human visual system to the imaging sensors of one high-end digital cinema camera (Red Dragon) and two popular professional DSLR cameras (Nikon and Canon). The first odd thing here is that the cameras don't look very much like the human visual system, so how can they make pictures that look right to us? The second thing to note is that the cameras don't look very much like each other, but they still produce images that look very similar.

Part of the answer to these questions is metamers—that phenomenon of the human visual system described in Chapter 1 where the eye can perceive different color values as the same color. The other part of the answer is the camera imaging math where the raw image sensor data is processed to produce the color image. Note that all imaging sensors are linear in their response to light. However, when the image is exported, there are options to convert it to other color spaces on the way out that might be linear, log, or gamma-corrected. More on this later.

这些感知差异有两个原因：不同的光谱敏感度和不同的相机成像算法。光谱敏感度指的是成像传感器对特定波长光的敏感程度。例如，图2.1中的相机A拥有非常敏感的蓝色传感器，使其呈现蓝色调。相机B缺乏低光敏感度，因此在暗部表现沉重。相机C对红色和绿色更为敏感，赋予其黄色调。我们将在第六章：色彩管理中看到ACES工作流程如何解决这一问题。

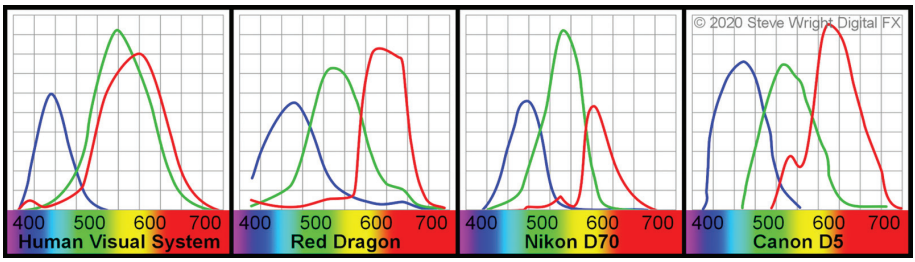


图2.2 人类视觉系统与三种相机光谱灵敏度的比较

图2.2比较了人类视觉系统的光谱敏感度与一台高端数字电影摄像机（Red Dragon）和两台流行的专业单反相机（尼康和佳能）的成像传感器。这里第一个奇怪之处是，这些相机看起来与人类视觉系统不太相似，那么它们如何能制作出在我们看来正确的图像呢？第二个需要注意的是，这些相机彼此之间也不太相似，但它们仍然能产生看起来非常相似的图像。

这些问题的部分答案在于同色异谱现象——即第一章中描述的人类视觉系统现象，即眼睛能将不同的颜色值感知为同一种颜色。另一部分答案在于相机成像的数学原理，即原始图像传感器数据经过处理生成彩色图像的过程。需要注意的是，所有成像传感器对光的响应都是线性的。然而，当图像导出时，可以选择在输出过程中将其转换为其他色彩空间，这些空间可能是线性、对数或经过伽马校正的。关于这一点，后续会有更详细的讨论。

2.1.2 Camera Gamuts

An irritated color scientist snarled at me once for asking about the color space of a camera. He insisted that they do not have color spaces, while others argue that they do. What cannot be argued is that even if a camera does not have a true color space, it most definitely has a gamut. The gamut of a color system is the full range of colors that it can produce and this can be visualized by graphing it on a CIE diagram, which typically forms a triangle.

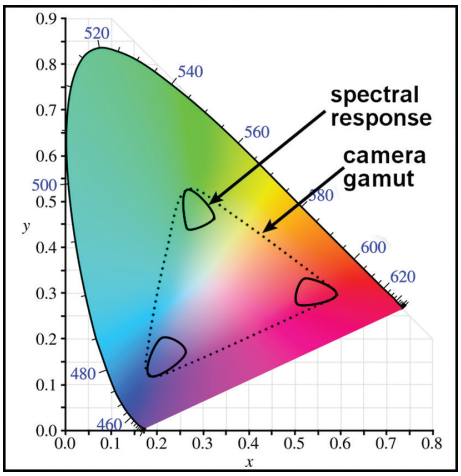


Figure 2.3 Camera color gamut

Referring to the camera spectral sensitivities in Figure 2.2, the area under each curve is outlined in a spectral response region of the CIE diagram illustrated in Figure 2.3. They are regions instead of points because the camera's spectral sensitivities cover a short range of wavelengths, as shown in Figure 2.2. If we then draw a line around the three regions, it forms a triangle that defines the color gamut of that camera, which demarcates the full range of colors that it will respond to. Keep in mind that these are not necessarily the colors the camera will output since the actual output colors are computed by the camera's internal software during image reconstruction. Both ARRI and RED cameras have wide gamut color spaces that are larger than Rec. 709 and sRGB. I wonder what would happen if their wide gamut images were displayed on standard gamut monitors, like our workstations? I guess we will have to wait until Chapter 5: Display Space to find out.

2.1.3 Converting Light to Digits

So how does the camera sensor distinguish the different light wavelengths that make up its spectral sensitivity? After all, the photosites on the image sensors are colorblind and will fire off when hit by all wavelengths of light. To solve the colorblind photosite problem, colored glasses are used in the form of a Bayer array,

2.1.2 相机色域

一位被激怒的色彩科学家曾因我询问相机的色彩空间而对我咆哮。他坚称相机没有色彩空间，而其他人则认为它们有。无可争辩的是，即使相机没有真正的色彩空间，它也绝对有色域。色彩系统的色域是其能够产生的全部颜色范围，这可以通过在CIE图表上绘制来可视化，通常形成一个三角形。

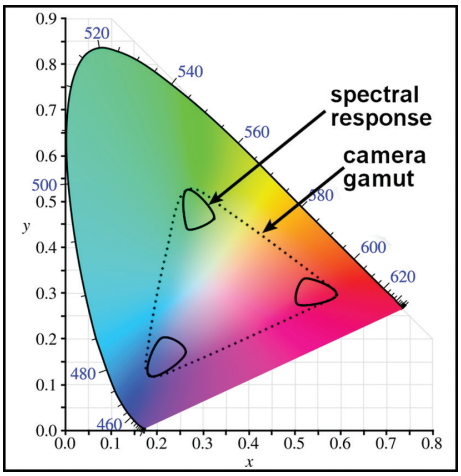


图2.3 相机色域

参考图2.2中的相机光谱灵敏度，该区域

每条曲线下方都对应着CIE色度图中的一个光谱响应区域，如图2.3所示。这些区域之所以呈现为面状而非点状，是因为相机的光谱敏感度覆盖了一段波长范围（如图2.2所示）。若将这三个区域用边界线连接，便会形成一个三角形，该三角形定义了相机的色域范围，标明了相机能够响应的全部色彩。需要注意的是，这些并非相机最终输出的色彩——实际输出色彩由相机内部软件在图像重建过程中计算生成。ARRI和RED相机均拥有比Rec.709和sRGB更宽广的色域空间。我不禁思考：若将这些广色域图像放在标准色域显示器（比如我们的工作站屏幕）上显示，会发生什么？或许我们要等到第五章《显示空间》才能找到答案。

2.1.3 将光转换为数字

那么相机传感器是如何区分构成其光谱灵敏度的不同光波长呢？毕竟，图像传感器上的感光点是色盲的，会被所有波长的光触发。为了解决感光点色盲的问题，人们采用了拜耳阵列形式的彩色滤光片，

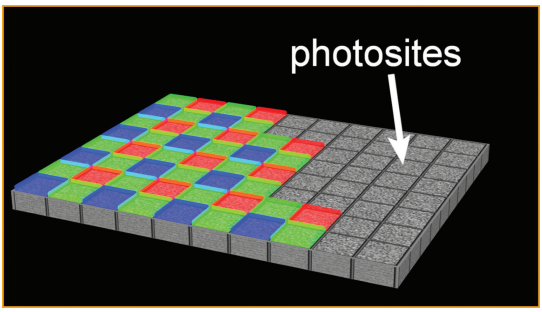


Figure 2.4 The Bayer array

shown in Figure 2.4. A Bayer array is essentially a checkerboard of color filters that sits over the image sensor to let only selected wavelengths of light through to each photosite. The range of wavelengths passed by each color filter determines the spectral response of the camera.

A digital camera would ideally "see" a scene exactly like the human eye. You may recall from Chapter 1 that the eye perceives luminance by combining (roughly) 72% of the green with 21% of the red and 7% of the blue light waves. For the camera to see the scene the same as we do, it needs to match this sensitivity. The Bayer array comes fairly close by using two green, one red, and one blue for each set of four photosites. This arrangement collects 50% green + 25% red + 25% blue, which is then juggled with clever color processing in the camera imaging math to produce a balanced color image.

The green record is produced by scanning the photosites that are under the green filter in Figure 2.4, the red record by scanning the photosites under the red filter, and the same for the blue. You now have a raw RGB image where the color of a pixel is determined by the spectral response of the Bayer array as stimulated by the spectral reflectance of an object in world space as illuminated by the spectral emission of a light source. Got all that? Excellent. You may proceed.

2.1.4 The Debayer Process

Converting the raw Bayer RGB values into a meaningful RGB image entails backing out the effect of the Bayer filter in a process called **debayering** (or **demosaicing**), which is also known as color reconstruction.

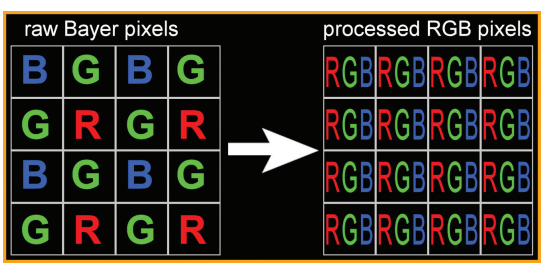
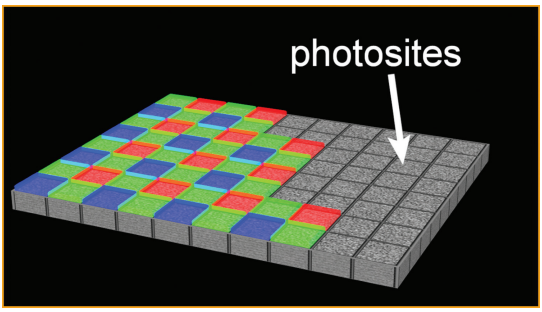


Figure 2.5 Debayering the raw image



如图2.4所示。拜耳阵列本质上是一层覆盖在图像传感器上的彩色滤光片棋盘格，它仅允许特定波长的光通过每个感光点。每种彩色滤光片所透过的波长范围决定了相机的光谱响应。

图2.4 拜耳阵列

理想的数码相机应该能像人眼一样“看见”场景。你可能还记得第一章提到，人眼感知亮度时大致会混合72%的绿光、21%的红光和7%的蓝光。为了让相机与我们所见一致，它需要匹配这种敏感度。拜耳阵列通过每组四个感光元件使用两个绿色、一个红色和一个蓝色单元，相当接近这一目标。这种排列能采集50%绿色+、25%红色+和25%蓝色，随后通过相机成像算法中巧妙的色彩处理技术进行调配，最终生成色彩平衡的图像。

绿色记录是通过扫描图2.4中绿色滤光片下的感光点生成的，红色记录通过扫描红色滤光片下的感光点生成，蓝色记录同理。现在你得到了一个原始RGB图像，其中像素的颜色由拜耳阵列的光谱响应决定——这种响应由世界空间中物体在光源光谱照射下的光谱反射所激发。都明白了吗？很好。你可以继续了。

2.1.4 去马赛克处理

将原始拜耳RGB值转换为有意义的RGB图像，需要在称为去拜耳化（或去马赛克）的过程中消除拜耳滤镜的影响，这一过程也被称为色彩重建。

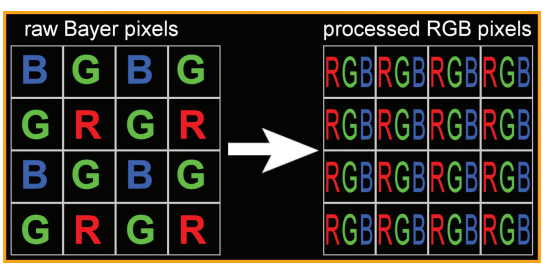


图2.5 对原始图像进行去拜耳处理

Why it's called "reconstruction" will become clear shortly. The camera can be instructed to perform the debayering operation internally or the camera raw may be exported and a colorist can supervise the debayer process offline. Because the image sensors are linear, this produces a linear image that can be exported as an OpenEXR file. Cameras offer further export options such as log, ProRes, JPEG, and others.

As we saw in the previous section, the Bayer filter captures too little green and too much red and blue. The debayer algorithm performs color reconstruction by remixing the colors and guessing the missing pixel colors. Yes, you read that correctly—guessing. Assumptions are made about the missing colors and estimates are made to compute the final RGB code values. Some debayering software even has an edge reconstruction feature. Edge reconstruction, assuming, guessing, estimating—with all this color chicanery, it is amazing that the cameras take as good a picture as they do. It also makes clear why different cameras see the scene differently.

But there is something amiss here. Check out Figure 2.5, which shows each pixel in the raw image has only one color: R, G, or B. But suddenly, after debayering, there are now *three* colors in each pixel (RGB). What is going on here? If this were an 8K imaging sensor, for example, we would have 4K green pixels, 2K red pixels, and 2K blue pixels because each pixel captures just one color. But after debayering, we have 8K RGB pixels with 8K green, 8K red, and 8K blue. How is this possible?

It's possible because the debayering process interpolates the missing colors to perform the color reconstruction. The software considers a pixel's neighbors and then filters them together in the right proportions to calculate the RGB value of the output pixel. An illustration of this conversion process is depicted in Figure 2.6 with a tight close-up of a small 980-pixel wide region. The raw Bayer version is shown on the left, with each of the 980 pixels getting only one color. The image on the right shows the same image after debayering to get the final RGB image, also 980 pixels wide. The bottom line is that the output image is a computational estimate produced from the raw Bayer image, which is why it is called color reconstruction.

为什么称之为“重建”很快就会明了。相机可以被指示在内部执行去马赛克操作，或者相机原始数据可以被导出，并由调色师离线监督去马赛克过程。由于图像传感器是线性的，这会生成一个线性图像，可以导出为OpenEXR文件。相机还提供进一步的导出选项，如log、ProRes、JPEG等。

正如我们在上一节所见，拜耳滤镜捕捉的绿色过少，红色和蓝色过多。去马赛克算法通过重新混合颜色并猜测缺失的像素颜色来进行色彩重建。没错，您没看错——是猜测。算法会对缺失的颜色做出假设，并通过估算计算出最终的RGB编码值。有些去马赛克软件甚至具备边缘重建功能。边缘重建、假设、猜测、估算——经历了这么多色彩“戏法”，相机最终能拍出如此优质的照片着实令人惊叹。这也清楚地解释了为何不同相机对同一场景的呈现会存在差异。

但这里有些不对劲。看看图2.5，它显示原始图像中的每个像素只有一种颜色：R、G或B。但突然之间，经过去马赛克处理后，每个像素现在有了三种颜色（RGB）。这到底是怎么回事？举例来说，如果这是一个8K成像传感器，由于每个像素仅捕捉一种颜色，我们会有4K个绿色像素、2K个红色像素和2K个蓝色像素。但去马赛克后，我们却得到了8K个RGB像素，包含8K个绿色、8K个红色和8K个蓝色。这怎么可能呢?

之所以能够实现，是因为去马赛克过程通过插值缺失的颜色来进行色彩重建。软件会考虑像素的相邻像素，然后以适当的比例将它们一起滤波，以计算输出像素的RGB值。图2.6以980像素宽的小区域特写展示了这一转换过程。左侧显示的是原始拜耳版本，其中每个980像素仅获得一种颜色。右侧图像展示了去马赛克后得到的最终RGB图像，宽度同样为980像素。关键在于，输出图像是从原始拜耳图像计算得出的估计结果，因此这一过程被称为色彩重建。

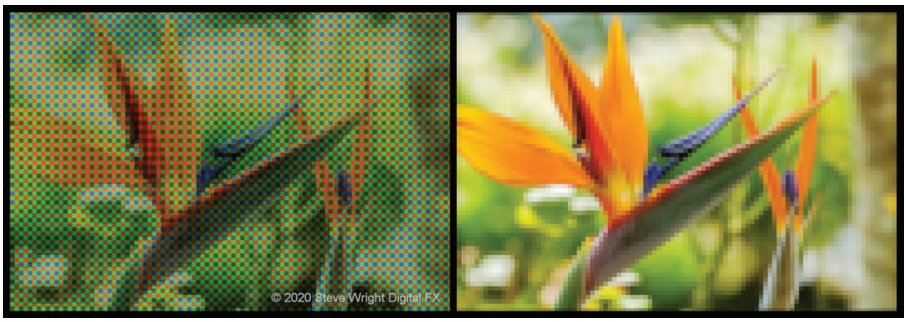


Figure 2.6 The raw Bayer image The debayered image

Of course, there is a price to pay for this computational image and that is a loss of resolution. You may think you have an 8K camera, but the manual proudly declares 8K, and the output images are indeed 8K. However, each of those 8K pixels started with just one color: R, G, or B, as we saw in Figure 2.5. When the entire image is debayered and the filtering and processing are all done, you do end up with 8K images—but they are not a sharp 8K. The image is softened by all the filtering, averaging, and interpolating, so its *effective* resolution is only about 70% of the resolution of the image sensor. It is as if you took a 5.6K image and stretched it up to 8K to fulfill the promise of the sales brochure. Sharpening is then added to "crisp up" the image—yet more pixel chicanery.

Watch the Bayer array in action in this video and see how the image sensor turns boring monochrome photons into exciting color images.

2.1.5 Camera Imaging Math

The second cause of the perceptual differences between cameras is the color science used by the camera manufacturer. All of them take the camera's raw captured image after debayering and put it through a LUT (Look Up Table) to "adjust" it to make it "look right." They are not trying to make scientific instruments that faithfully measure the chromatic illumination of a real-world scene. That device is called a spectrophotometer. They are instead

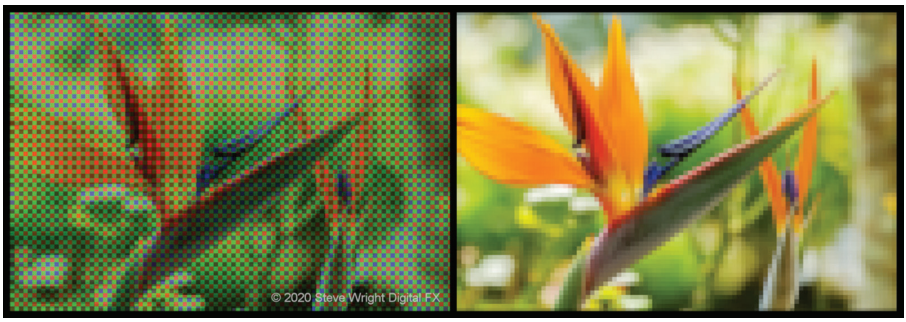


图2.6 原始拜耳图像　　去马赛克后的图像

当然，这种计算成像技术是有代价的，那就是分辨率的损失。你可能以为自己拥有一台8K相机，说明书也自豪地宣称支持8K，输出的图像确实也是8K规格。但正如我们在图2.5中所见，这些8K像素中的每一个最初都只包含一种颜色信息：红、绿或蓝。当完成整个图像的去马赛克处理以及所有滤波和计算后，最终确实会得到8K图像——但它们并非锐利的8K。图像经过层层滤波、平均化和插值处理后会变得柔和，其实际有效分辨率仅达到图像传感器分辨率的70%左右。这就像将一张5.6K的图像拉伸至8K，以实现宣传手册上的承诺。随后通过锐化处理让图像显得“清晰”——这不过是又一种像素层面的视觉戏法。

在这段视频中观看拜耳阵列的实际运作，看看图像传感器如何将单调的单色光子转换成激动人心的彩色图像。

2.1.5 相机成像数学

相机之间感知差异的第二个原因是相机制造商使用的色彩科学。它们都会在去马赛克后获取相机的原始捕获图像，并通过查找表（LUT）进行“调整”，使其“看起来正确”。它们并非试图制造能忠实测量真实世界场景色度照度的科学仪器，那种设备被称为分光光度计。相反，它们

making an artistic device that takes beautiful pictures, and beauty is in the eye of the beholder.

A great example of beauty in the eye of the beholder comes from my days at Kodak's Cinesite Hollywood working with the Kodak film engineers. I learned of confidential (at the time) color research that Kodak did on what they called "color memory." They found that people's recollections of a blue sky, green grass, and fleshy skin tones were much more vibrant (saturated) than real life. So, to make film images beautiful in the eye of the beholder in the theater, they adjusted the film's chemistry to create bluer blues for the sky, greener greens for grass, and fleshier flesh tones. People loved it and it is one of the key ingredients in the "film look." In fact, digital cinema cameras are measured by how well they match the film look—which is, as we will see later when we delve into ACES, the visual reference for what pictures should look like. The consensus is they should look like film. Like digital cinema cameras, film is not a scientific imaging tool. It is an artistic imaging tool.

2.1.6 Light Sources

One more messy issue that wildly affects the camera's perception of color is the light source. In Chapter 1: World Space, we covered the spectral emission of light sources, as well as color temperature. Here we will see how these two issues affect the captured image.

2.1.6.1 Color Temperature

Film a yellow banana under different light sources without adjusting the camera to compensate and the banana will hue shift due to the different color temperatures of the light sources. Speaking artistically, if the light is "warmer"—say, 3200K—the banana will look yellower/redder. If the light source is "cooler"—say, 6500K—the banana will look bluer. Never mind that 3200K is physically cooler than 6500K. Artists still call it "warmer" because it is perceptually warmer to the eye when the picture looks yellower/redder compared to a perceptually cooler picture that looks bluer. This is art, not physics. Well actually, it's both—which is one reason color science is such a nettlesome subject.

制作一种能够拍摄美丽照片的艺术设备，而美在于观者的眼中。

一个关于“情人眼里出西施”的绝佳例子，来自我在柯达好莱坞Cinesite工作室与柯达胶片工程师共事的经历。当时我了解到柯达进行了一项（在当时属机密的）色彩研究，他们称之为“色彩记忆”。研究发现，人们对蓝天、绿草和肌肤色调的记忆，远比现实中的色彩更加鲜活（饱和度更高）。因此，为了让影院观众眼中的胶片影像更富美感，他们通过调整胶片化学配方，创造出更湛蓝的天空、更翠绿的草地以及更饱满的肤色调。人们对此喜爱有加，这也成为“胶片质感”的关键要素之一。实际上，数字电影摄影机的评判标准，正是其还原胶片质感的能力——正如我们后续深入探讨ACES时将看到的，这种质感已成为影像视觉表现的参照基准。业界共识是：影像应当呈现胶片般的质感。与数字电影摄影机一样，胶片并非科学成像工具，而是一种艺术创作工具。

2.1.6 光源

另一个严重影响相机对颜色感知的混乱问题是光源。在第一章：世界空间中，我们讨论了光源的光谱发射以及色温。这里我们将看到这两个问题如何影响捕获的图像。

2.1.6.1 色温

在不同光源下拍摄黄色香蕉而不调整相机进行补偿，香蕉会因光源色温不同而产生色调偏移。从艺术角度来说，如果光线更“暖”——比如3200K——香蕉会显得更黄/更红。如果光源更“冷”——比如6500K——香蕉则会显得更蓝。不必在意3200K在物理上其实比6500K更冷。艺术家仍称之为“暖”，因为当画面看起来更黄/更红时，相较于视觉上更冷（偏蓝）的画面，人眼感知上会觉得更温暖。这是艺术，不是物理学。不过实际上两者皆是——这也正是色彩科学如此棘手的缘由之一。

The confounding fact for us humans is that after our eyes have become light-adapted, the banana will look pretty much the same under all those different lighting conditions, but the camera-captured image will not. Because the camera sees the scene differently than we do, we require instrumentation (light meters, waveform monitors, etc.) to help with our photography and cinematography. White balancing the camera compensates for different color temperatures of the light source, so the captured image will appear the same as the original scene did to the light-adapted observer.

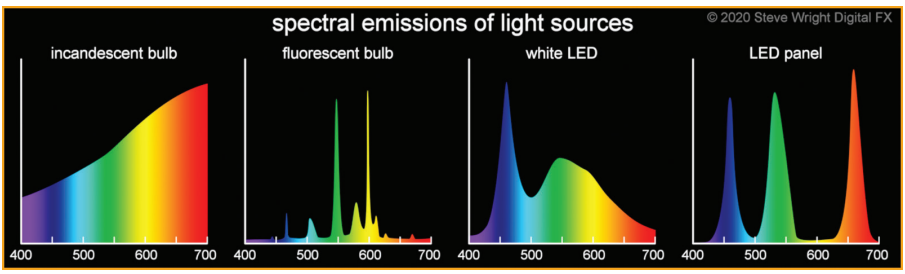


Figure 2.7 The spectral emissions of four different light sources

2.1.6.2 Spectral Emissions

The second issue that affects the captured image is the spectral emissions of the light source, as illustrated in Figure 2.7. The sun and incandescent lightbulbs put out a continuous spectrum of light waves between 400 and 700 nanometers (nm). As we saw in Chapter 1, they radiate light waves both far above and below these limits; we just can't see them—but a camera can. Oops! Yet another issue. So, the camera manufacturers put in filters to block UV (below 400 nm) and IR (above 700 nm) to prevent exposing the image to these unwanted wavelengths.

The other disturbing thing about the spectral emissions illustrated here is the spiky nature of the fluorescent bulb, the irregular output of the white LED, and the precariously narrow spectrum of the LED panel. They don't match the spectral sensitivities of the camera sensor. Both the incandescent bulb and the sun put out a continuous spectrum, so they will obviously have full coverage of the spectral sensitivity of any camera with no gaps. Fluorescent bulbs and LEDs, however, pile a lot of energy in some wavelengths and little in others, which may or may not line up with the camera's sensitivities. This

令我们人类困惑的是，当眼睛适应光线后，香蕉在各种不同光照条件下看起来几乎一样，但相机捕捉到的图像却并非如此。由于相机观察场景的方式与我们不同，我们需要借助仪器（如测光表、波形监视器等）来辅助摄影和电影拍摄。相机的白平衡功能可以补偿光源的不同色温，从而使捕捉到的图像与光线适应观察者所见的原始场景保持一致。

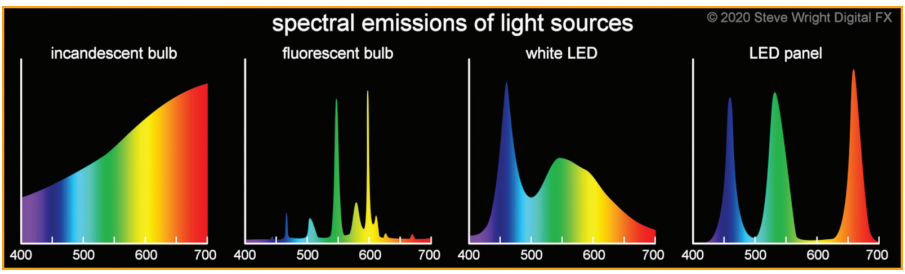


图2.7 四种不同光源的光谱辐射

2.1.6.2 光谱发射

影响捕获图像的第二个问题是光源的光谱发射，如图2.7所示。太阳和白炽灯泡会发出400至700纳米（nm）之间的连续光谱。正如我们在第1章中所见，它们辐射的光波远高于和低于这些限制；我们只是看不见——但相机可以。哎呀！这又是一个问题。因此，相机制造商加入了滤镜来阻挡紫外线（低于400 nm）和红外线（高于700 nm），以防止图像暴露在这些不需要的波长下。

此处展示的光谱发射的另一令人不安之处在于荧光灯泡的尖峰特性、白色LED的不规则输出以及LED面板岌岌可危的狭窄光谱。它们与相机传感器的光谱灵敏度并不匹配。白炽灯泡和太阳都发出连续光谱，因此显然能完整覆盖任何相机的光谱灵敏度范围，不存在缺口。然而荧光灯泡和LED在某些波长处聚集大量能量，在其他波长处却能量微弱，这些波长可能与相机的灵敏度范围重合，也可能不重合。

has the effect of shifting the hue of the captured image. Hopefully, the camera's white balance can compensate.

LED panels are gaining popularity for both stage lighting and virtual production stages. They are very cool for stage lighting because you can dial in the color temperature you would like to shoot with. Virtual production has revolutionized visual effects where entire sound stages are surrounded on up to three sides by a huge circular wall of LED panels displaying the surrounding scenery. Drop in the talent, props, and a tracked camera, and you are good to go. Of course, this glib summary overlooks the incredibly complex technology, hardware, software, and color science that goes into setting up the virtual stage in the first place. More on virtual production shortly.

The problem with the LED panels is that they have narrow spiky spectral emissions that can introduce metameric discoloration on some objects in the scene. These are difficult to fix in post because a yellow banana (for example) may have turned a disturbing orange under an LED panel, but other objects in the scene look fine.

2.1.7 CCD vs. CMOS

Historically, there have been two camera sensor technologies competing for market share, CCD and CMOS. The early victory went to CCD (Charged Couple Device) when digital photography was introduced in the 1990s. CCD's advantages were much less noise and greater sensitivity to light compared to the CMOS (Complementary Metal Oxide Semiconductor). However, they were much more expensive and consumed a great deal more power.

Fast forward 20 years and the addition of digital cameras to cell phones spawned a flurry of research and development (R&D) to improve CMOS sensors, which started with a price and power advantage to begin with. And, of course, price and power consumption were core issues for cell phones. All they had to do was reduce noise and improve the light sensitivity—a mission accomplished by the early 2000s when CMOS started to become the dominant imaging technology.

CCD's price penalty comes from the fact that its chip fabrication requires a costly special manufacturing process. Its exquisite light sensitivity is due to

具有改变捕获图像色调的效果。希望相机的白平衡能够进行补偿。

LED面板在舞台灯光和虚拟制作舞台上都越来越受欢迎。对于舞台灯光来说，它们非常酷，因为你可以直接设定拍摄所需的色温。虚拟制作彻底改变了视觉效果制作方式，整个摄影棚被高达三面的巨大环形LED面板墙包围，屏幕上显示着周围的场景。只需加入演员、道具和追踪摄像机，一切就准备就绪了。当然，这种轻描淡写的总结忽略了搭建虚拟舞台背后极其复杂的技术、硬件、软件和色彩科学。关于虚拟制作的更多内容稍后介绍。

LED面板的问题在于其光谱发射窄而尖锐，可能导致场景中某些物体出现同色异谱的变色现象。这类问题在后期处理中难以修正，因为（举例来说）一根黄色的香蕉在LED面板下可能变成令人不适的橙色，而场景中的其他物体看起来却很正常。

2.1.7 CCD 与 CMOS 对比

历史上，CCD和CMOS两种相机传感器技术一直在争夺市场份额。20世纪90年代数码摄影兴起时，CCD（电荷耦合器件）凭借更低的噪点和更高的感光灵敏度，较CMOS（互补金属氧化物半导体）取得早期优势。然而，CCD的成本更高且功耗显著更大。

20年转瞬即逝，手机加入数码相机功能催生了一波研究与开发（R&D）热潮，旨在改进CMOS传感器——这种技术本就具备价格和功耗的先天优势。而价格与功耗恰恰是手机的核心关切。研发只需攻克降低噪声、提升感光度两大难题，这一使命在21世纪初已基本实现，CMOS也由此开始成为主导的成像技术。

CCD的价格劣势源于其芯片制造需要昂贵特殊工艺。其卓越的感光性能则得益于

the fact that most of the imaging surface is covered with light-sensitive cells; whereas the CMOS surface is cluttered with additional support transistors that block some of the light to the light-sensitive cells. The CCD chip clocks out the image data as an analog signal that goes to an off-chip A/D (Analog-to-Digital) converter to become digital data. One happy consequence of this arrangement is that most CCD chips have a global shutter, meaning that the entire sensor is read at the same time.

CMOS's price advantage comes from the fact that it is manufactured with common and mature chip fabrication technology. However, because of the requirement for support transistors adjacent to the light-sensitive cells, it not only blocks some incoming light but also reduces the size of the sensor cell. This reduces its light sensitivity, so the weak captured signal has to be amplified so much that it introduces noise into the image.

A couple of decades of intense R&D have largely solved these problems but have not addressed one remaining issue, namely the sensor scanning strategy. Most CMOS sensors are scanned from top to bottom, producing the rolling shutter artifact for certain moving image content. We will learn more about rolling shutter shortly, but in the meantime, CMOS's improved image quality, low cost, and intrinsic low power consumption have made it today's dominant image sensor technology.

The heated debate over whether CCD or CMOS takes a prettier picture is misdirected. Today, all cameras take spectacularly beautiful pictures, whether they are CMOS or CCD. The difference in picture rendering is mostly due to the camera manufacturers' post-capture image processing rather than the capabilities of the imaging sensor. CCD sensors still retain an advantage in low-light photography as well as the Near Infra-Red (NIR) photography used in agriculture, astronomy, and environmental studies. But you will probably want a CMOS camera unless you are a farming astronomical environmentalist.

FUN FACT: The first digital camera was developed in 1975 at Eastman Kodak. It weighed 8 pounds and captured a 0.01-megapixel black-and-white image that it downloaded to a cassette tape in "only" 23 seconds.

大多数成像表面覆盖着光敏细胞；而CMOS表面则杂乱地布满了额外的支持晶体管，这些晶体管会阻挡部分光线到达光敏细胞。CCD芯片以模拟信号形式输出图像数据，该信号传输至片外A/D（模数）转换器转换为数字数据。这种设计带来的一个可喜结果是：大多数CCD芯片具备全局快门功能，这意味着整个传感器可同时读取数据。

CMOS的价格优势源于其采用通用且成熟的芯片制造技术。然而，由于需要在感光单元旁设置辅助晶体管，这不仅会阻挡部分入射光，还会减小传感器单元的尺寸。这降低了其感光灵敏度，因此微弱的捕获信号必须被大幅放大，以至于在图像中引入了噪点。

几十年的密集研发已基本解决了这些问题，但尚未解决一个遗留问题，即传感器扫描策略。大多数CMOS传感器采用从上至下的扫描方式，这导致某些动态图像内容出现滚动快门效应。我们稍后将详细探讨滚动快门，但与此同时，CMOS传感器在图像质量提升、低成本及固有的低功耗方面的优势，已使其成为当今主流的图像传感技术。

关于CCD和CMOS哪个能拍出更美照片的激烈争论其实偏离了重点。如今，无论是CMOS还是CCD，所有相机都能拍出极其精美的照片。画面呈现的差异主要源于相机厂商的后期图像处理，而非影像传感器的性能。CCD传感器在弱光摄影以及农业、天文学和环境研究中使用的近红外（NIR）摄影方面仍具优势。但除非你是位从事农业研究的天文环保主义者，否则你很可能还是会选择CMOS相机。

有趣的事实：第一台数码相机于1975年在柯达公司诞生。它重达8磅，能拍摄0.01兆像素的黑白图像，并将数据下载到盒式磁带中——整个过程“仅需”23秒。

2.2 Space and Time

The introduction of digital cameras cut cinematography and photography loose from the constraints and cost of the chemical photography process of 35-millimeter (mm) film and has also introduced many more options and features that can be overwhelming. Here we will try to organize the chaos about camera resolution, frame rates, sensor cropping, and anamorphic formats into some useful information. Since all cameras are now digital, to say "digital cinema camera" is a bit silly so going forward we will just use "cinema camera."

2.2.1 Resolution

Cinema camera resolution has reached astonishing levels. The Blackmagic Ursa Mini Pro will shoot 12K at 60 frames per second (fps)! I suspect that the reason for this push for high resolution is that it is relatively easy to increase the resolution of the image sensors, as it does not require fundamental re-engineering or new color science. Just shrink the size of the photosites even more. It looks really good on the spec sheets. Seriously, what do you need 8K for? Any camera worth its pixels will now do 4K or better. Remember, all that fabulous resolution requires ever more disk space, file transfer time, and image processing time. A 4K image has four times the pixels than a 2K. For extreme resolution, you cannot beat the Blackmagic URSA Mini Pro 12K with a sensor resolution of 12,288 x 6480 for an astonishing 80 megapixels per frame - until another manufacturer comes out with a newer model next week.

4K is catching on for theatrical and broadcast to "future-proof" projects. The latest TV sets support UHD (Ultra High Definition), which is called 4K but is really double HD at 3840 x 2160. For movies, there is a "cine 4K" or, more properly, DCP 4K (Digital Cinema Package), which is a true 4K at 4096 x 2160—a simple doubling of DCP 2K at 2048 x 1080. Any of these high resolutions will end up on your workstation soon if they haven't already. Thank goodness for proxies, the low-resolution version of these huge frames.

While consumer TVs are all dashing toward 4K, the movie theaters have been slow to adopt it because of the capital equipment costs. While a lot of movies are produced at 4K, the vast majority are released at 2K, as few theaters have 4K digital cinema projectors. The movie industry is falling behind the consumer television industry in display resolution. One reason for this is the added burden of the movie file size at 4K. The other, I suspect, is that the theater owners know that unless you are in the first five rows, you cannot

2.2 空间与时间

数码相机的出现使电影摄影和摄影摆脱了35毫米胶片化学摄影工艺的约束与成本，同时也带来了更多令人眼花缭乱的选择和功能。在此，我们将尝试梳理关于相机分辨率、帧率、传感器裁切和变形宽银幕格式的混乱信息，将其转化为实用参考。鉴于所有相机现已数字化，“数字电影摄影机”的提法略显多余，因此下文将统一使用“电影摄影机”。

2.2.1 分辨率

电影摄影机的分辨率已经达到了惊人的水平。Blackmagic Ursa Mini Pro 能以每秒60帧的速度拍摄12K视频！我怀疑这种追求高分辨率的原因在于，提高图像传感器的分辨率相对容易，因为它不需要根本性的重新设计或新的色彩科学。只需进一步缩小感光元件的尺寸。这在规格表上看起来确实很吸引人。说真的，你需要8K来做什么呢？任何称职的相机现在都能拍摄4K或更高分辨率。请记住，所有这些惊人的分辨率都需要更多的磁盘空间、文件传输时间和图像处理时间。4K图像的像素是2K的四倍。在极致分辨率方面，Blackmagic URSA Mini Pro 12K凭借12,288 x 6480的传感器分辨率，每帧高达8000万像素的表现无人能及——直到下周另一家制造商推出更新的型号。

4K技术正在影院和广播领域普及，以“面向未来”的项目。最新的电视机支持UHD（超高清），这被称为4K，但实际上是3840 x 2160的双倍高清。对于电影，有“影院4K”或更准确地说，DCP 4K（数字电影包），这是真正的4K，分辨率为4096 x 2160——简单地将DCP 2K的2048 x 1080翻倍。如果这些高分辨率尚未出现在您的工作站上，它们很快就会出现。幸好有代理文件，这些巨大帧的低分辨率版本。

当消费级电视纷纷奔向4K时，电影院却因高昂的设备成本而迟迟未能普及。尽管许多电影以4K规格制作，但绝大多数仍以2K格式上映，因为配备4K数字电影放映机的影院寥寥无几。在显示分辨率方面，电影行业正落后于消费级电视产业。原因之一是4K电影文件体积带来的额外负担。另一个原因，我猜想是影院经营者深知——除非坐在前五排，否则观众根本无法

even see the difference between 2K and 4K. To a degree, this is also true of home theater, but the consumer is more willing to pay extra for 4K than the theater owner because it's a cool new thing. But be advised, Netflix and Amazon are both aggressively pursuing 4K production, delivery, and display, so if you work in broadcast, you'd better get some more disk space.

2.2.2 ISO

In case you are not a shutterbug, ISO refers to the sensitivity or "speed" of the imaging sensor. The term is a holdover from the days of 35 mm film when you had to change film stocks to change ISO. It is also not true that changing the ISO changes the sensitivity of the sensor, even though everybody says it does. The sensitivity of the sensor is fixed when the chip is fabricated, and that's it. What is really going on is that when the ISO is increased to shoot in low light, the sensor's signal is simply amplified. Of course, that amplifies the noise, too. Hence, low-light or high-speed photography is noisy.

Low-light photography can occur in two different ways: low-light conditions (of course) or operating the camera at a high frame rate, as when "over-cranking" the camera to shoot slow motion. The fast shutter is open for less time, therefore, less light gets to the sensor, so you need to crank up the ISO to compensate—but that also cranks up the noise. You can't fool Mother Nature—or Father Tech, in this case.

2.2.3 Speed

Camera frame rates are also climbing. Virtually all cinema cameras will do 120 fps, with one Black Magic model clipping along at a breath-taking 600 fps. Of course, when you use the higher frame rates, you usually have to drop the camera's resolution. There is also the issue of shutter speed. Referring to the shutter speed in the exposure triangle in Figure 2.8, you can see that as the shutter speed increases, the duration of light decreases. This makes sense since a faster shutter

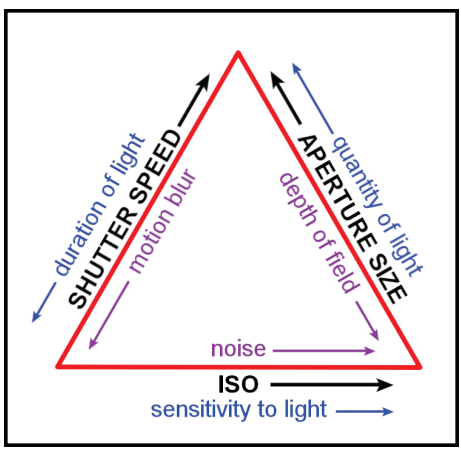


Figure 2.8 Classic exposure triangle

甚至看不出2K和4K的区别。在某种程度上，家庭影院也是如此，但消费者比影院老板更愿意为4K支付额外费用，因为它是个很酷的新事物。但请注意，Netflix和Amazon都在积极追求4K制作、传输和显示，所以如果你从事广播行业，最好多准备一些磁盘空间。

2.2.2 ISO

如果你不是摄影发烧友，ISO指的是成像传感器的感光度或“速度”。这个术语是35毫米胶片时代的遗留，那时你必须更换胶卷才能改变ISO。尽管大家都这么说，但改变ISO并不会改变传感器的感光度，这种说法并不正确。传感器的感光度在芯片制造时就已固定，无法改变。实际上，当提高ISO在弱光下拍摄时，只是放大了传感器的信号。当然，这也会放大噪点。因此，弱光或高速摄影会产生噪点。

低光摄影可能以两种不同方式发生：低光条件（当然）或以高帧率操作相机，例如在“过卷”相机拍摄慢动作时。快速快门开启时间较短，因此到达传感器的光线较少，所以你需要提高ISO来补偿——但这也会增加噪点。你无法欺骗大自然母亲——或者在这种情况下，是科技之父。

2.2.3 速度

相机帧率也在不断攀升。几乎所有电影摄影机都能达到120帧/秒，其中一款Black Magic型号更是以惊人的600帧/秒疾驰。当然，使用更高帧率时通常需要降低相机分辨率。快门速度也是需要考虑的问题。参考图2.8中曝光三角里的快门速度，你会发现随着快门速度提升，进光时间会缩短。这很合理，因为更快的快门

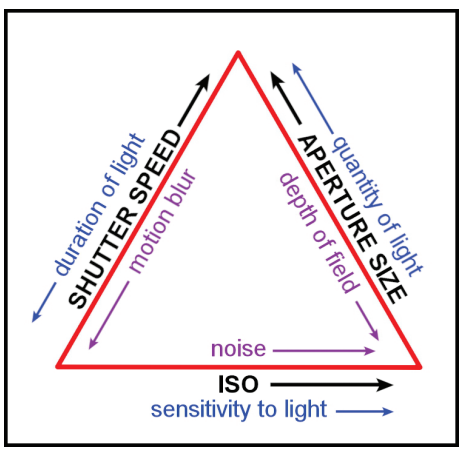


图 2.8 经典曝光三角

is open for shorter intervals. To compensate for the reduced light, the ISO might be increased, but then the noise is also increased. You might prefer to increase the aperture size to let in more light, but you will lose depth of field. Even modern digital cameras are bound by this exposure "iron triangle."

Fortunately, these higher frame rates don't necessarily mean more frames for us to work on, because it is typically done to produce a slow-motion shot without using a speed-changer in post. Of course, 120 fps is only a modest slowdown of the action; so for a serious slo-mo shot, we get to use our motion vector software in post, then repair the artifacts. It is surprising how fast the camera has to be "over-cranked" in order to produce a seriously slow-motion shot. I remember working on a beer commercial where they wanted to show the bubbles rising as a graceful ballet, so they had to run the camera at 1500 fps. This required so much light that the beer started to smolder after a few minutes! This, of course, was in the days before cool LED lighting was available.

Do keep in mind that a camera running at 24 fps does not mean that the shutter is open 1/24th of a second. Cinema cameras and video cameras use what is called a "180-degree" shutter, just like film cameras in order to introduce a pleasing amount of motion blur. The term "180 degrees" comes from the mechanical shutter of film cameras, which consists of a rotating flat disk that covers the film aperture with half the disk (180 degrees) gone. As it rotates, the open section lets light into the film aperture, then the solid section blocks the light. Most film cameras allow adjustment of the opening for greater or less than 180 degrees. A shutter greater than 180 degrees is open longer so creates more motion blur, and less than 180 degrees reduces it. Digital cameras offer the same feature, but the shutter is electronic rather than mechanical. It is, however, still expressed in terms of "180 degrees," in deference to the old film cameras.

Watch the exposure triangle in action as you witness the effects on an image while the parameters are adjusted.

在较短的间隔内保持开启。为了补偿减少的光线，ISO可能会被提高，但随之噪点也会增加。你可能更倾向于增大光圈以引入更多光线，但这会损失景深。即使是现代数码相机，也受限于这个曝光的“铁三角”法则。

幸运的是，这些更高的帧率并不一定意味着我们需要处理更多帧，因为这通常是为了在后期制作中不使用变速器就能生成慢动作镜头。当然，120帧每秒只能实现适度的动作放缓；因此，要制作真正的慢动作镜头，我们必须在后期使用运动矢量软件，然后修复伪影。令人惊讶的是，为了制作出真正的慢动作镜头，摄影机必须“超频”到多快的速度。我记得曾参与一个啤酒广告的制作，他们希望展示气泡如优雅芭蕾般上升，因此不得不将摄影机设置为每秒1500帧。这需要如此强的光线，以至于几分钟后啤酒就开始冒烟了！当然，那还是在凉爽的LED照明设备尚未普及的年代。

请务必记住，以24帧每秒运行的相机并不意味着快门开启时间为1/24th秒。电影摄影机和摄像机采用所谓的“180度”快门，就像胶片相机一样，以引入适量的运动模糊。“180度”这一术语源于胶片相机的机械快门，它由一个旋转的扁平圆盘构成，圆盘的一半（180度）缺失，用于覆盖胶片孔径。当圆盘旋转时，开口部分让光线进入胶片孔径，然后实心部分阻挡光线。大多数胶片相机允许调整开口角度大于或小于180度。大于180度的快门开启时间更长，因此会产生更多运动模糊；小于180度则会减少模糊。数码相机提供相同的功能，但快门是电子式的而非机械式。不过，出于对传统胶片相机的尊重，这一功能仍以“180度”来表示。

观看曝光三角的实际应用，当参数调整时，您将亲眼见证其对图像产生的影响。

2.2.4 Sensor Crop

Life was a lot simpler in the days of 35 mm film. The standard image size for the film back was 36 x 24 mm and that was that. Enter digital cameras and everything changed. Ideally, the imaging sensor would be the same size as 35 mm film, so all those 35 mm lenses could be used. However, the cinematographer now has the option of applying a sensor crop to use only a portion of the imaging sensor. Further, most cameras will automatically enforce a sensor crop when shooting high frame rates, simply because the electronics cannot shovel the image data out fast enough at high resolution simultaneously with a high frame rate.

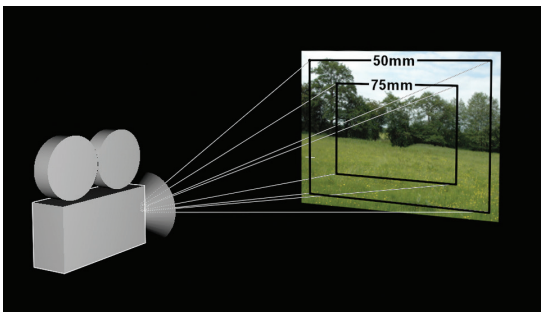


Figure 2.9 A 50 mm lens becomes 75 mm with a 1.5x sensor crop

The problem is that introducing a sensor crop alters the effective focal length of the lens. Per Figure 2.9, suddenly a 50 mm lens is now operating as a 75 mm lens. This may also be done deliberately to create a "poor man's" zoom lens. However, the downside is that the cropped sensor uses fewer pixels, so the finished shot will have a lower resolution. Of course, if you are using 6K pixels out of an 8K sensor for a 4K job this may not be a problem. Fortunately, the cameras come with tables that show how the crop factor affects the focal length.

However, large image sensors are expensive, so many digital cameras have smaller sensor sizes designed to lower cost and sow confusion. The confusion comes from the fact that we now have to factor in the sensor crop in our camera work and visual effects camera tracking because the listed focal length of a lens applies only if it is mounted over a standard sensor size. There are, however, lenses available for the smaller sensor sizes that avoid this problem.

Using the 35 mm film back size of 36 x 24 mm as the standard reference for a full-frame sensor, the sensor crop factor of a camera is how much smaller the camera's sensor is compared to it. The virtue of smaller sensors is a cheaper camera and smaller, lighter, cheaper lenses. The smaller sensor

2.2.4 传感器裁切

在35毫米胶片时代，生活要简单得多。胶片后背的标准图像尺寸是36 x 24毫米，仅此而已。数码相机的出现改变了一切。理想情况下，成像传感器应与35毫米胶片尺寸相同，这样所有35毫米镜头都能继续使用。然而，如今摄影师可以选择应用传感器裁切，只使用成像传感器的一部分。此外，大多数相机在拍摄高帧率时会自动启用传感器裁切，这仅仅是因为电子元件无法同时以高分辨率和高帧率快速处理图像数据。

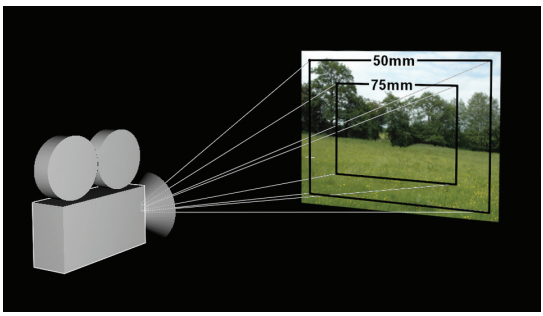


图2.9 一个50毫米镜头在1.5x倍率下变为75毫米传感器裁切

问题在于，引入传感器裁切会改变镜头的有效焦距。如图2.9所示，一个50毫米镜头突然变成了75毫米镜头。这种做法也可能被刻意用来打造“廉价版”变焦镜头。然而，其缺点在于裁切后的传感器使用的像素更少，因此最终拍摄的画面分辨率会较低。当然，如果你是从8K传感器中提取6K像素用于4K工作，这可能不是问题。幸运的是，相机附带的表格显示了裁切系数如何影响焦距。

然而，大型图像传感器价格昂贵，因此许多数码相机采用较小的传感器尺寸以降低成本并制造混淆。这种混淆源于一个事实：由于镜头标注的焦距仅适用于安装在标准尺寸传感器上的情况，我们在摄像工作和视觉特效相机追踪中必须考虑传感器裁切系数。不过，市面上也有专为较小传感器尺寸设计的镜头，能够避免这一问题。

以35毫米胶片后部尺寸36 x 24毫米作为全画幅传感器的标准参考，相机的传感器裁剪系数是指相机传感器比之小多少。较小传感器的优点在于相机更便宜，镜头更小、更轻、更便宜。较小的传感器

shrinks the field of view of the lens effectively increasing the focal length. Figure 2.9 illustrates a 50 mm lens with a 1.5x crop sensor multiplier resulting in an effective focal length of 75 mm (50 mm x 1.5x = 75 mm).

For accurate camera tracking results, the VFX supervisor will have collected camera information on location, including the focal length of the lens and the aperture size. The camera tracking software uses this information in solving for the camera. The problem arises when we are informed of the focal length but there is a sensor crop for the shot that we don't know about and that changes the effective focal length of the lens. We have now misinformed our camera tracker, so the results will be off. Of course, we can always tell the camera tracker the focal length is unknown and let it figure this out for itself, but that gives less accurate results. The best tracking results happen when you provide accurate lens information, especially the aperture size.

2.2.5 Sensor Size

The imaging sensor in your Hasselblad has a profound effect on camera performance. Here's a quick survey of the popular sensor sizes today:

- **Full-frame 36 x 24 mm** - The same as a frame of 35 mm film, this is the go-to professional standard with the highest all-around performance complete with professional features.
- **APS-C 23.5 x 15.6 mm** - Large enough to offer most of the advantages of full-frame sensors but small enough for smaller, lighter, and cheaper lenses.
- **Four-thirds 18 x 13.5 mm** - Designed for uncommitted beginners, these are noisier at higher ISO settings with noticeably poorer image quality, but their zoom lenses are small and inexpensive.
- **One-inch sensor 12.8 x 9.6 mm** - Just the thing for your compact cameras and drones.
- **1/2.55 6.17 x 4.55 mm** - Your standard-issue cell phone camera. Of necessity, low power and long battery life.

The depth of field is influenced by the size of the sensor. Large (or full-frame 36 x 24 mm) sensors allow for a shallow depth of field, while smaller sensors

缩小了镜头的视野，有效增加了焦距。图2.9展示了一个50毫米镜头配合1.5倍裁切传感器倍率，实现了75毫米的有效焦距（50毫米 x 1.5倍 = 75毫米）。

为了获得精确的摄像机跟踪结果，视觉特效总监会在拍摄现场收集摄像机信息，包括镜头的焦距和光圈大小。摄像机跟踪软件利用这些信息来解算摄像机运动。问题出现在我们被告知焦距，但镜头存在我们未知的传感器裁切，这会改变镜头的有效焦距。此时我们向摄像机跟踪器提供了错误信息，导致结果出现偏差。当然，我们总可以告诉摄像机跟踪器焦距未知，让它自行计算，但这会降低结果的准确性。当提供准确的镜头信息——尤其是光圈大小时，才能获得最佳的跟踪效果。

2.2.5 传感器尺寸

您的哈苏相机中的成像传感器对相机性能有着深远影响。以下是当今流行传感器尺寸的快速概览：

- 全画幅 36 x 24 毫米——与 35 毫米胶片画幅相同，这是专业标准的首选，具备最高的综合性能，并配备齐全的专业功能。

- APS-C 23.5 x 15.6 毫米传感器——尺寸足够大，能提供大部分全画幅传感器的优势，同时又足够小巧，可适配更小、更轻、更经济的镜头。

- 四分之三 18 x 13.5 毫米 - 专为尚未确定方向的初学者设计，这些相机在高ISO设置下噪点更明显，图像质量显著较差，但其变焦镜头体积小巧且价格低廉。

- 一英寸传感器 12.8 x 9.6 毫米 - 正是您紧凑型相机和无人机的理想选择。

- 1/2.55 英寸 6.17 x 4.55 毫米 - 您手机的标准配置摄像头。出于必要，它功耗低且电池续航长。

景深受传感器尺寸的影响。大（或全画幅36 x 24 mm）传感器可实现较浅的景深，而较小的传感器

lead to greater depth of field. The shallow depth of field has advantages when the director wants a closeup of the talent with the background out of focus as a method of focusing the audience's attention on the talent. For long panoramic shots, a deep depth of field is appropriate so that everything in the frame is in focus. You can still get everything in focus with the large sensor using the right lens, but the smaller sensor size introduces limitations. And directors hate limitations.

To maintain the same field of view between a large and small sensor, the large sensor will require a longer focal length lens and longer focal length lenses have a shallower depth of field. To maintain the same viewing angle and level of light, the large sensor will want a larger aperture, and a large aperture further reduces the depth of field. The larger the sensor, the closer the camera can be placed to the subject while maintaining the same field of view. The closer the camera gets, the shallower the depth of field.

Large sensors have several other advantages beyond a shallow depth of field. Because of their physically large size, they can capture a lot more light, which has several positive consequences. The first is image quality, with greater detail and color accuracy. Capturing more light obviously also leads to better low-light performance for those popular dimly lit shots. The large sensor size also provides for a greater dynamic range; and because they capture so much light, they require less amplification, resulting in much less noise. With a large sensor being the same size as 35 mm film, cinematographers can use their old 35 mm film lens sets, which can cost tens of thousands of dollars. Directors of photography (DPs) know and love their lens sets.

However, the large sensor cameras are larger, heavier, and therefore more expensive. Cameras with small sensors are smaller, cheaper, and lighter than their big brothers. They also offer a greater depth of field, which is sometimes important. Longer zoom ranges are also possible because the zoom lens assemblies can be made smaller and more compact. With fewer pixels than their big brothers, they can have faster readout speeds, which is just the thing for high-speed photography.

导致更大的景深。当导演想要特写演员而背景虚化，以此引导观众注意力集中于演员时，浅景深具有优势。对于长镜头全景拍摄，深景深更为合适，以确保画面中的所有元素都清晰对焦。使用大传感器配合合适的镜头仍可实现全画面清晰，但较小的传感器尺寸会带来限制。而导演们最讨厌的就是限制。

为了在大传感器和小传感器之间保持相同的视野，大传感器需要更长的焦距镜头，而更长的焦距镜头景深更浅。为了保持相同的视角和光线水平，大传感器需要更大的光圈，而大光圈会进一步减小景深。传感器越大，相机在保持相同视野的同时可以离拍摄对象越近。相机离得越近，景深就越浅。

大型传感器除了浅景深之外，还有其他几个优势。由于其物理尺寸较大，它们能够捕捉更多的光线，这带来了多方面的积极影响。首先是图像质量，细节更丰富，色彩更准确。捕捉更多光线显然也意味着在那些流行的昏暗场景中，低光表现更出色。大尺寸传感器还提供了更大的动态范围；并且因为它们捕捉的光线如此之多，所需的放大倍数更小，从而产生的噪点也少得多。由于大型传感器的尺寸与35毫米胶片相同，电影摄影师可以使用他们昂贵的旧35毫米胶片镜头组，这些镜头组可能价值数万美元。摄影指导（DPs）熟悉并热爱他们的镜头组。

然而，大尺寸传感器的相机体积更大、重量更重，因此价格也更昂贵。相比之下，采用小尺寸传感器的相机则更为小巧、轻便且价格亲民。它们还能提供更大的景深，这在某些情况下至关重要。此外，由于变焦镜头组件可以设计得更小更紧凑，这类相机还能实现更长的变焦范围。虽然像素数量不及大尺寸传感器相机，但它们拥有更快的读取速度，这正是高速摄影所需要的。

2.2.6 Anamorphic Formats

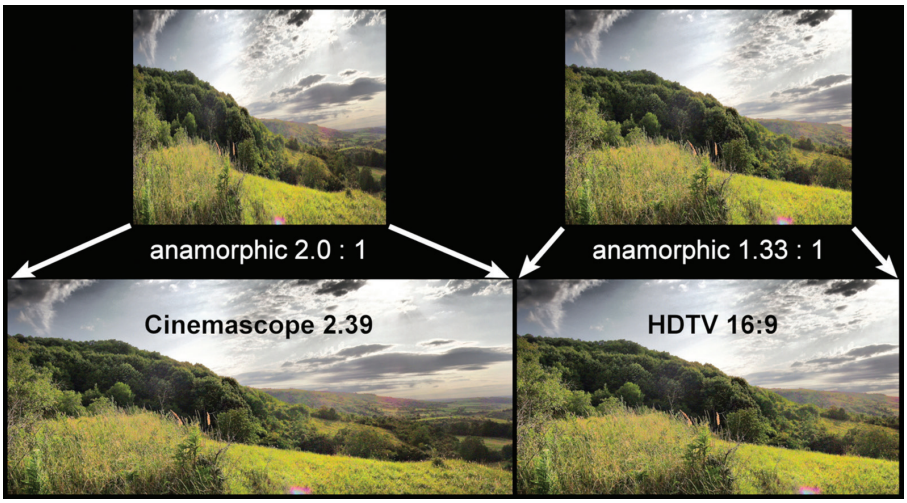


Figure 2.10 Anamorphic 2.0 and 1.33

Anamorphic images are squeezed horizontally when captured, then stretched horizontally for projection or display. In the production pipeline between capture and display, the image is squeezed so the pixels are not square. There are two main anamorphic values—2.0:1 and 1.33:1—illustrated in Figure 2.10. These numbers do not represent how much they are squeezed, but rather the "stretch factor" to display them properly. For example, anamorphic 2.0:1 images are squeezed by 50% horizontally in the camera, then stretched by 2 upon display. This means the pixel aspect ratio of the captured image is 0.5.

2.2.6.1 Cinemascope 2.39

Anamorphic 2:1 is the Cinemascope (Cscope or "scope") feature film standard with an on-screen projected aspect ratio of 2.39 (changed from 2.35 in 1970), but some refer to it as 2.4. The anamorphic lens squeezes the image optically by 50% horizontally before it hits the imaging sensor. It was developed for widescreen theatrical projection and some directors still love it, especially the unique lens flares that the "scope" lenses produce. It was originally done to record a larger image onto the 35 mm film frame, which improved sharpness and reduced grain compared to projecting just a small

2.2.6 Anamorphic Formats

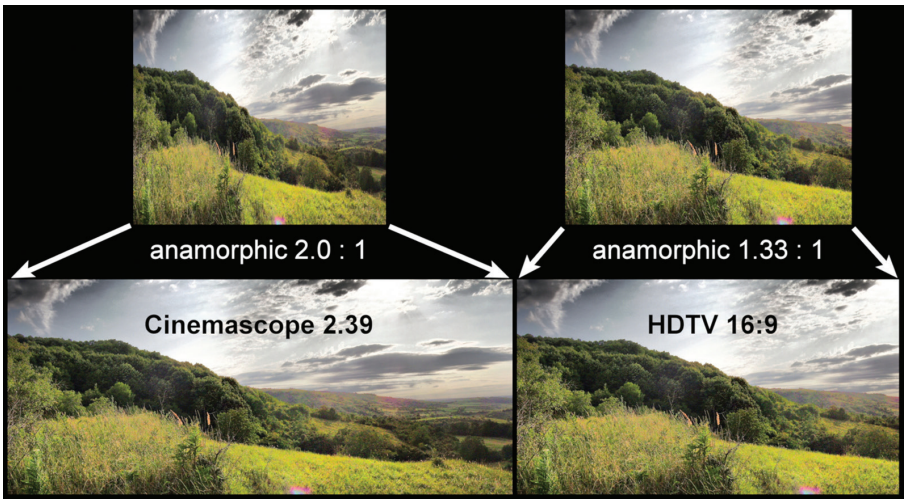


图2.10 变形2.0与1.33

变形宽银幕图像在拍摄时被水平压缩，随后在投影或显示时水平拉伸。在拍摄与显示之间的制作流程中，图像经过压缩处理，因此像素并非正方形。主要有两种变形宽银幕比例值——2.0:1和1.33:1，如图2.10所示。这些数字并不代表压缩程度，而是指为正确显示所需的“拉伸系数”。例如，2.0:1变形宽银幕图像在摄像机中水平压缩50%，显示时则拉伸2倍。这意味着拍摄图像的像素宽高比为0.5。

2.2.6.1 宽银幕电影 2.39

变形宽银幕2:1是电影宽银幕（Cscope或“scope”）故事片的标准格式，其银幕投影宽高比为2.39（1970年由2.35调整而来），但也有人称之为2.4。变形镜头在图像到达成像传感器前，会将其在水平方向上光学压缩50%。这项技术最初是为宽银幕影院投影而开发，至今仍受一些导演青睐，尤其是“scope”镜头产生的独特眩光效果。其初衷是在35毫米胶片画幅上记录更大尺寸的图像，与仅投影小尺寸画面相比，这提升了清晰度并减少了颗粒感。

2.39 window out of a regular frame of film. However, modern film stocks have such improved grain structure that we often actually shoot full aperture 35 mm without an anamorphic lens, crop the 2.39 image out of it, and then stretch it vertically by two to conform to the Cinemascope format.

2.2.6.2 HDTV 1.33

Anamorphic 1.33:1, however, has a much more mundane purpose—to make cheap cameras. HDTV was introduced in 1998 with a resolution of 1920 x 1080 and a 16:9 aspect ratio. In those days, video cameras recorded a full 1920 x 1080 frame to tape with over 2 million pixels, which required faster high-quality (more expensive) components. So, the camera manufacturers came up with HDV a 1440 x 1080 capture spec with about 1.5 million pixels using slower, cheaper electronics. When the 1440 x 1080 HDV image is displayed, it is stretched by 1.33 horizontally to 1920 x 1080 to conform to the HDTV spec.

The important thing about HDTV anamorphic footage is that it is marked in-camera with metadata indicating it is anamorphic for the software downstream in the pipeline: editing, color timing, VFX, etc. The software then "automagically" unsqueezes the anamorphic image stretching it to the proper width for display without human intervention. You could be working on an anamorphic shot and not even realize it because the software has made it transparent to the user. Even visual effects software supports anamorphic formats and automatically compensates for the non-square pixels.

2.3 Camera Dynamic Range

The dynamic range of cameras is measured in "stops," where each stop is a doubling of the amount of light for exposure. So, a camera with 8 stops of dynamic range would have the exposure doubled eight times (2^8), which is 256. This means the camera could capture a scene where the brightest part of the scene was 256 times brighter than the darkest part. As we saw in Chapter 1: World Space, we need to deal with scenes that have a dynamic range of thousands to one, even tens of thousands to one. So, today's modern digital cinema cameras—like RED Epic and ARRI Alexa—boast dynamic ranges of 15 or 16 stops, depending on whether you ask the camera manufacturer or

2.39 窗口是从常规电影画幅中截取出来的。然而，现代电影胶片的颗粒结构已大幅改进，因此我们实际上经常直接拍摄全画幅35毫米胶片（不使用变形镜头），从中裁剪出2.39的画面，再将其垂直拉伸两倍以符合西尼玛斯科普格式。

2.2.6.2 HDTV 1.33

变形1.33:1却有着更为实际的目的——降低摄像机制造成本。高清电视于1998年问世时，其分辨率为1920×1080，宽高比为16:9。当时摄像机需将完整1920×1080画面（超过200万像素）录制到磁带，这要求使用更高速的高品质（也更昂贵）元件。因此制造商推出了HDV标准，采用1440×1080采集规格（约150万像素），通过更慢速、廉价的电子元件实现。当1440×1080的HDV图像播放时，会通过1.33倍水平拉伸至1920×1080以符合高清电视标准。

关于HDTV变形宽银幕素材的重要之处在于，它会在摄像机内标记元数据，表明其为变形宽银幕格式，以供后续流程中的软件使用：剪辑、调色、视觉特效等。随后，软件会“自动”解除变形宽银幕图像的挤压，将其拉伸至适合显示的宽度，无需人工干预。你可能正在处理变形宽银幕镜头却浑然不觉，因为软件已为用户实现了无缝操作。即便是视觉特效软件也支持变形宽银幕格式，并能自动补偿非方形像素。

2.3 相机动态范围

相机的动态范围以“档”为单位衡量，每档意味着曝光所需的光量增加一倍。因此，一台具有8档动态范围的相机会将曝光量翻倍八次（2^8），即256倍。这意味着相机能够捕捉到场景中最亮部分比最暗部分亮256倍的画面。正如我们在第1章：世界空间中所见，我们需要处理的场景动态范围可达数千比一，甚至数万比一。因此，当今现代数字电影摄像机——如RED Epic和ARRI Alexa——宣称拥有15或16档的动态范围，具体数值取决于询问对象是相机制造商还是

experienced cinematographers. Top-of-the-line DSLR still cameras go up to 13 to 14 stops—a bit less than digital cinema cameras.

> FUN FACT: The term "stop" comes from the *very* early days of photography when metal plates with different-sized aperture holes, called "stops," were physically inserted into the camera lens to reduce the amount of light.

2.3.1 Clipping

So how does this affect us in production? A modern digital cinema camera, with its (realistically) 15 stops of dynamic range, can capture a contrast ratio of 32,000:1. But the big question here is what happens when the scene luminance exceeds the image sensor's ability to digitize it? The answer is, it gets clipped. When the scene is clipped in the camera, the clipped data is unrecoverable. Clipping is forever, but there is something that you can do to make it a little less awful, and that is to introduce a soft clip in post. How to create a quality soft clip is revealed later in Chapter 4: Work Space.

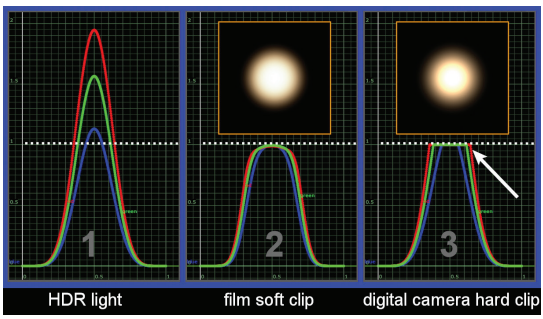


Figure 2.11 Comparison of clipping strategies

Panel 1 of Figure 2.11 shows the cross-section color curves of a reddish high dynamic range light captured in the camera. All three curves range above the clipping point of the camera sensor, represented by the white dotted line. Everything above that will be clipped at the sensor. Panel 2 shows how a 35 mm film stock will respond by introducing a soft clip due to the reciprocity failure of the film when overexposed. Note that while the clipped light source has lost its color in the center, it is a smooth blend out to the colored edges.

Panel 3 shows how the imaging sensor of a digital camera will simply hard clip the image, introducing unattractive color rings. The colors of the rings can be understood by examining the color curves at the clipping line.

经验丰富的电影摄影师。顶级的单反静态相机可达13到14档动态范围——略低于数字电影摄影机。

> 趣闻：术语“光圈”源于摄影的早期时代，当时人们会将带有不同大小孔径孔的金属板（称为“光圈片”）实际插入相机镜头，以减少进光量。

2.3.1 裁剪

那么，这在制作中对我们有何影响？一台现代数字电影摄像机，凭借其（实际可达的）15档动态范围，能够捕捉32,000:1的对比度。但这里的关键问题是，当场景亮度超过图像传感器数字化能力时会发生什么？答案是，它会被裁剪。当场景在摄像机中被裁剪时，被裁剪的数据将无法恢复。裁剪是永久性的，但你可以采取一些措施让它不那么糟糕，那就是在后期引入软裁剪。如何创建高质量的软裁剪将在第四章：工作空间中揭晓。

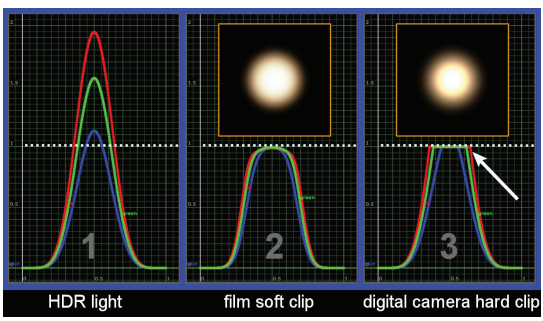


图2.11的面板1展示了相机捕捉到的偏红色高动态范围光源的横截面色彩曲线。三条曲线均位于相机传感器的裁剪点（以白色虚线表示）之上。高于此线的所有信息都将在传感器处被裁剪。面板2显示了

图2.11 裁剪策略比较

35毫米胶片在过度曝光时，由于互易律失效会如何通过引入软裁剪来响应。请注意，虽然裁剪后的光源在中心失去了颜色，但它会平滑地过渡到彩色的边缘。

面板3展示了数码相机的成像传感器如何简单地硬性裁剪图像，从而引入不美观的色环。通过检查裁剪线上的颜色曲线，可以理解这些色环的颜色。

At the very center, all three colors are clipped to 1.0, so we see white. Moving away from the center, the blue drops away leaving red and green, which makes the orange ring. Moving further, the green drops away, leaving just red for the outer ring.

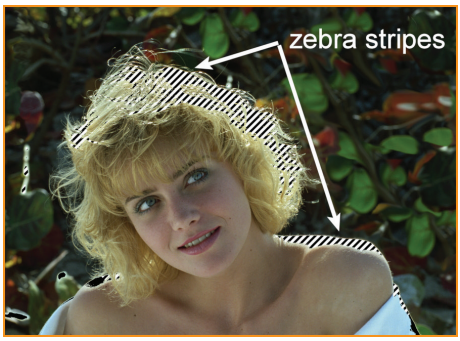


Figure 2.12 Overexposed areas

Some prosumer and most professional video cameras have a display feature called **zebra stripes** (Figure 2.12), where overexposed or out-of-gamut colors are marked in the viewfinder so you can correct the exposure to fix it before recording. Remember—clipping is forever.

2.3.2 Bit Depth

The imaging sensors used in cinema cameras come in a variety of bit depths, from a low of 10 bits per channel for the RED Scarlet up to 16 bits per channel for the Panavision DXL2. A "channel" is the technical term for one color record: the red channel, the green channel, and the blue channel. The DSLR cameras used by photographers also vary in bit depth, like the 12 bits for the Nikon D7500 or the 14 bits for the Canon 5D. What bit depth gets you is finer and finer steps in the color gradient from black to white as the bit depth is increased. 10 bits get you 1,024 "shades of gray" per color channel, while 16 bits provide a whopping 65,535 steps. For this reason, bit depth is also referred to as "color depth." The interesting part of this story is how those bits might be allocated.

Figure 2.13 illustrates the two strategies for allocating the camera's bit depth (a very low bit depth was used to make the banding more visible). The maximum dynamic range case at the top shows the bit depth spread out to capture the

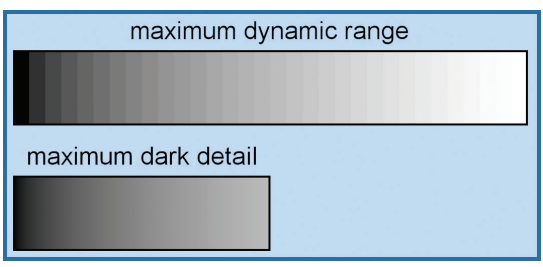


Figure 2.13 Bit depth allocation

在最中心处，三种颜色均被裁剪至1.0，因此我们看到白色。从中心向外移动时，蓝色逐渐消退，留下红色和绿色，从而形成橙色的光环。继续向外移动，绿色也逐渐消退，最终仅剩红色构成最外层的圆环。

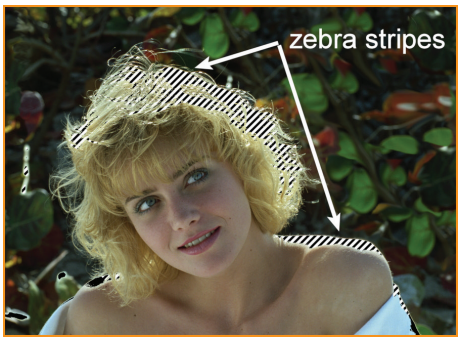


图2.12 曝光过度区域

一些准专业和大多数专业摄像机都配备了一种称为斑马纹的显示功能。（图 2.12），取景器中会标记出曝光过度或超出色域的颜色，以便您在录制前调整曝光来修正此问题。请记住——一旦发生裁切就不可逆转。

2.3.2 位深度

电影摄影机中使用的成像传感器具有多种位深度，从RED Scarlet的每通道10位低值，到Panavision DXL2的每通道16位高值。“通道”是单色记录的技术术语：红色通道、绿色通道和蓝色通道。摄影师使用的数码单反相机位深也各不相同，例如尼康D7500的12位或佳能5D的14位。位深度越高，从黑色到白色的色彩渐变阶梯就越精细。10位深度为每个颜色通道提供1,024种“灰度阶”，而16位深度则能提供高达65,535级阶梯。因此，位深度也被称为“色彩深度”。这个故事的有趣之处在于这些比特位可能如何分配。

图2.13展示了分配相机位深的两种策略（此处使用极低位深以使条带效应更明显）。顶部的最大动态范围案例显示了位深被展开以捕捉

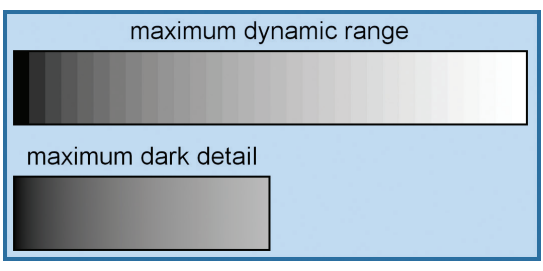


图 2.13 位深度分配

most stops of scene brightness. The problem here is that the large steps in luminance can introduce banding, most noticeably in the darks.

The maximum dark detail case below shows the same bit depth optimized for dark detail by squeezing the bits together, which means the luminance "steps" are much closer together so there is no banding in the darks. The problem here is that dynamic range has to be sacrificed for the dark detail, meaning this camera will not cover as many stops as the top one, so it might start clipping the image in the brights.

Each camera manufacturer allocates the color depth for the best fit for its intended customers, so a high bit depth may not necessarily mean a high dynamic range camera. However the bits are allocated, the more bit depth in the image, the more image processing may be done on it before showing banding.

2.4 Video Standards

So far, we have been looking at cinema cameras, so it's now time to look at video cameras. While cinema cameras each have their own sense of image resolution and aspect ratio, the world of video is much more buttoned-down, with carefully specified standards. There is a list of pre-defined video formats that you should know about so you don't get that "deer caught in the headlights" look when someone asks you if your clip is 720p or 1080i. Oddly, all of these video standards are named after their vertical resolution and scan modes.

2.4.1 Resolutions

The HDTV television standards support two resolutions and the new UHDTV standards add two more for ever-bigger viewing experiences. Here are the specs:

2.4.1.1 HDTV Standards

- **1080p** - 1920 x 1080 resolution with progressive scan. A higher-quality picture than 1080i and widely used in Blu-Ray, streaming, and gaming.

场景亮度的大部分档位。这里的问题在于，亮度的大幅阶跃可能导致条带效应，在暗部区域尤为明显。

下面的最大暗部细节案例展示了通过将比特位紧密压缩来优化暗部细节的相同比特深度，这意味着亮度“阶梯”更加紧密，因此在暗部没有条带。这里的问题是，动态范围必须为暗部细节做出牺牲，意味着这款相机无法覆盖像顶部相机那样多的档位，因此可能在亮部开始出现图像裁剪。

每家相机制造商都会根据其目标客户群分配色彩深度，因此高比特深度不一定意味着高动态范围相机。无论比特如何分配，图像中的比特深度越高，在显示条带之前可进行的图像处理就越多。

2.4 视频标准

到目前为止，我们一直在探讨电影摄影机，现在该来看看摄像机了。电影摄影机各有其图像分辨率和宽高比的独特设定，而视频领域则更为规范，遵循严格制定的标准。有一系列预定义的视频格式需要了解，这样当有人问起你的视频片段是720p还是1080i时，你就不会露出那种“车灯前不知所措的鹿”般的表情。奇怪的是，所有这些视频标准都以垂直分辨率和扫描模式命名。

2.4.1 分辨率

高清电视标准支持两种分辨率，而新的超高清电视标准又增加了两种，以提供更宏大的观看体验。以下是具体规格：

2.4.1.1 HDTV标准

- 1080p - 1920 x 1080 分辨率，采用逐行扫描。相比 1080i 画质更佳，广泛应用于蓝光、流媒体和游戏领域。

- **1080i** - 1920 x 1080 resolution with interlaced scan. More common in early HDTV broadcasts, it is being replaced by the more popular 1080p progressive scan.
- **720p** - 1280 x 720 resolution with progressive scan. Popular for lower-bandwidth applications.

2.4.1.2 UHDTV Standards

- **2160p (4K)** - 3840 x 2160 resolution with progressive scan. This is becoming the standard for advanced streaming services, UHD Blu-ray, and more.
- **4320p (8K)** - 7680 x 4320 resolution with progressive scan. It offers 16 times the resolution of HDTV and is just the thing for large screens and close viewing distances.

By now, you have no doubt figured out that the little "p" stands for "progressive scan" and the little "i" stands for "interlaced scan." So perhaps it's time to discuss these scan modes.

2.4.2 Scan Modes

The video scan mode describes how the video image is read out of the camera, as well as how it is drawn in the display memory. There are two standards, the progressive scan, which makes perfect sense, and the interlaced scan, which does not.

2.4.2.1 Progressive Scan

The image is read out of the camera's image sensor one scan line at a time from top to bottom. It is also read into the TV frame buffer in the same manner. A sensible scheme, it does suffer from one obnoxious artifact - rolling shutter, which we shall dive deeper into in Section 2.6.2 Rolling Shutter. No peeking!

2.4.2.2 Interlaced Scan

It's time to have "the talk"—about interlaced video. This is a dirty little subject that some of you may not have encountered before, especially if you entered the industry after the introduction of HDTV in 1998. Both HDTV and UHDTV support interlaced scanning which is used for fast-action content like sports

- 1080i - 1920 x 1080分辨率，采用隔行扫描。早期高清电视广播中较为常见，现正逐渐被更流行的1080p逐行扫描所取代。

- 720p - 1280 x 720 分辨率，采用逐行扫描。在低带宽应用中较为流行。

2.4.1.2 超高清电视标准

- 2160p（4K） - 3840 x 2160分辨率，采用逐行扫描。这正逐渐成为高级流媒体服务、超高清蓝光等领域的标准。

- 4320p（8K）——7680 x 4320分辨率，采用逐行扫描。它提供了比高清电视高16倍的解析度，非常适合大屏幕和近距离观看。

到现在，你无疑已经明白小写的“p”代表“逐行扫描”，而小写的“i”代表“隔行扫描”。那么，或许是时候讨论这些扫描模式了。

2.4.2 扫描模式

视频扫描模式描述了视频图像如何从相机中读出，以及如何在显示内存中绘制。有两种标准：逐行扫描，这完全合理；以及隔行扫描，这并不合理。

2.4.2.1 逐行扫描

图像从摄像机的图像传感器中逐行自上而下读出，并以相同方式读入电视帧缓冲区。这一合理方案确实存在一个令人不快的缺陷——滚动快门效应，我们将在第2.6.2节《滚动快门》中深入探讨。请勿提前翻阅！

2.4.2.2 隔行扫描

是时候来谈谈“隔行扫描视频”这个话题了。对于部分从业者而言，这可能是个陌生而棘手的小众领域——尤其当您是在1998年高清电视问世后才进入行业时。无论是高清电视还是超高清电视，至今仍保留着隔行扫描技术，这种技术常用于体育赛事等高速运动内容的呈现。

and NASCAR races. The reason it's used for fast action content is because it takes two scans per frame, so producing 60 "pictures" per second, which blends fast action much better than a straight 30 fps. This does, however, create ugly problems for visual effects.

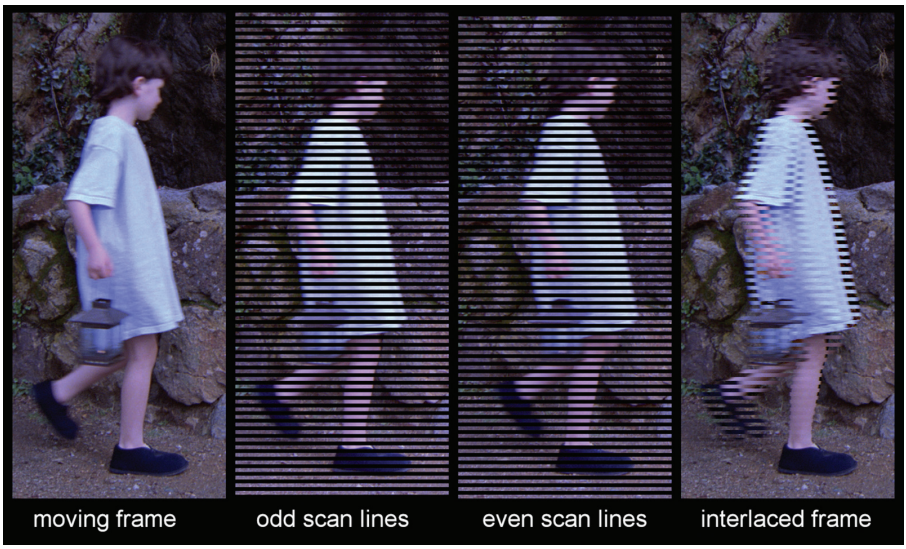


Figure 2.14 Interlaced video frame

Following the action in Figure 2.14, the moving frame shows the character walking from left to right. The camera starts scanning the image sensor at the top scan line and scans every other scan line - lines 1, 3, 5, etc. (odd scan lines). Reaching the bottom of the image sensor, it starts back at the top with lines 2, 4, 6, etc. (even scan lines). The key point here is that these two scans were made at two slightly different moments in time, so the character has moved in those moments. The odd scan lines (called field 1) capture the character in one position, while the even scan lines (field 2) capture him slightly to the right. When the two fields are merged, they produce the interlaced frame on the right. And that's how an interlaced frame will appear on a workstation monitor.

While the results look hideous here, at speed on an interlaced monitor things look fine because only one field is displayed at a time at 29.97 frames per second, while the eye interpolates the action using persistence of vision. The ugly problems interlaced images create are because they are actually two different pictures from two different moments in time interlaced together into

以及NASCAR赛车比赛。它被用于快速动作内容的原因在于，每帧需要两次扫描，因此每秒能生成60张“图片”，这比直接的30帧每秒能更好地融合快速动作。然而，这确实给视觉效果带来了棘手的问题。

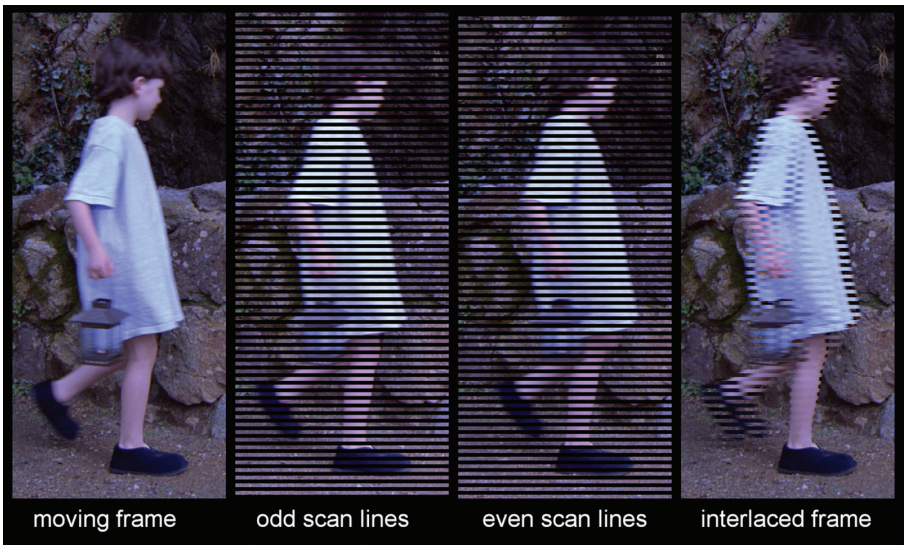


图 2.14 隔行扫描视频帧

如图2.14所示，移动画面显示角色从左向右行走。相机从顶部扫描线开始扫描图像传感器，并每隔一条扫描线进行扫描——即第1、3、5等行（奇数扫描线）。到达图像传感器底部后，它从顶部重新开始扫描第2、4、6等行（偶数扫描线）。关键在于这两次扫描发生在两个略微不同的时间点，因此角色在这段时间内发生了移动。奇数扫描线（称为场1）捕捉到角色在一个位置，而偶数扫描线（场2）则捕捉到他略微偏右的位置。当两个场合并时，它们生成了右侧的交错帧。这就是交错帧在工作站监视器上呈现的方式。

虽然这里的结果看起来糟糕，但在隔行扫描显示器上以一定速度播放时，画面效果尚可，因为每秒29.97帧的速率下每次只显示一个场，而人眼会通过视觉暂留现象对动作进行插值补全。隔行图像产生的难看问题，本质上是因为它们实际是将两个不同时间点的画面交织合并而成的

one frame. The field scan lines are only one pixel high, so when transformations are applied to interlaced frames the two fields' scan lines are filtered together, creating quite a mess.

To work with interlaced images, the frames must first be de-interlaced into their respective fields, the transformations or filtering must be performed on each field, and then the fields must be re-interlaced back into frames. Not fun. The old NTSC video standard only supported interlaced video. When HDTV was developed, they still needed to support interlaced video for sports channels but added a progressive scan mode to make a more sensible TV standard. They can still shoot and broadcast interlaced video to this day. Just hope they don't give it to you to add visual effects.

2.4.2.3 History

Interlaced video was the bane of my existence in my early compositing career, simply because all video at the time (analog NTSC) was interlaced. The reason it was interlaced was to provide a smoother picture without judder (motion strobing) with fast action. The video frame rate of 30 fps (okay—29.97 fps) just didn't hack it if the action was too swift. "But wait a minute," I can hear you thinking, "35 mm film runs at 24 fps without flicker and NTSC is faster than that—so what gives?"

What gives, my thinking-out-loud friend, is that the 35 mm projector is running at 48 fps by pulling the shutter down twice while advancing the film frame once. Before they figured this out in the 1920s, the projector ran at an honest 24 fps (it was only 18 fps in the *very* early days!), so all movies flickered—hence the nickname "the flicks." So the 35 mm projector is advancing the film at 24 fps but pulling the shutter down 48 frames per second to beat the "flicker fusion threshold" (FFT) of 60 to 90 fps. The fact is that if a light flickers faster than the human FFT the brain perceives it as a steady light.

> FUN FACT: Animals have flicker rates too: dogs, 70-80 hertz (Hz); cats, 100 Hz; birds, 100-120 Hz; rats, 60-80 Hz; and insects, a blazing 200-300 Hz.

This was, of course, initially a serious problem for the CRTs (Cathode Ray Tubes) of early TV because after the picture was drawn on the screen from

一帧。场扫描线只有一像素高，因此当对隔行扫描帧应用变换时，两个场的扫描线会被一起过滤，造成相当混乱的效果。

要处理隔行扫描图像，首先必须将帧去隔行扫描为各自的场，在每个场上执行变换或过滤，然后再将场重新隔行扫描回帧。这并不有趣。旧的NTSC视频标准仅支持隔行扫描视频。当高清电视开发时，他们仍需为体育频道支持隔行扫描视频，但增加了逐行扫描模式以制定更合理的电视标准。直到今天，他们仍然可以拍摄和广播隔行扫描视频。只希望他们不会把这种视频交给你来添加视觉效果。

2.4.2.3 历史

交错视频曾是我早期合成生涯中的噩梦，原因很简单：当时所有的视频（模拟NTSC制式）都是交错的。采用交错扫描是为了在快速动作场景中提供更流畅的画面，避免抖动（动态频闪）。30帧/秒（好吧——实际上是29.97帧/秒）的视频帧率在动作过快时根本不够用。“等等，”我仿佛能听到你的疑问，“35毫米胶片以24帧/秒播放都不会闪烁，NTSC帧率明明更高——这怎么解释呢？”

怎么回事，我那位爱自言自语的朋友，35毫米放映机是通过在胶片每前进一帧时让快门下拉两次，以48帧/秒的速度运行的。在20世纪20年代人们搞明白这一点之前，放映机是以实打实的24帧/秒运行的（在非常早期甚至只有18帧/秒！），所以所有电影都会闪烁——因此得了“the flicks”（闪烁片）这个绰号。所以35毫米放映机是以24帧/秒的速度推进胶片，但以每秒48次的速度下拉快门，以超越60到90帧/秒的“闪烁融合阈值”（FFT）。事实上，如果光线闪烁的速度超过了人类的FFT，大脑就会将其感知为稳定的光线。

> 趣闻：动物也有闪烁融合频率：狗，70-80赫兹（Hz）；猫，100赫兹；鸟类，100-120赫兹；老鼠，60-80赫兹；而昆虫则高达惊人的200-300赫兹。

当然，这最初对早期电视的阴极射线管（CRT）来说是一个严重问题，因为当画面从屏幕上绘制完成后，

top to bottom, the screen went blank as the raster snapped back to the top for the next frame. This would introduce a 30 fps flicker—well inside the FFT of the average viewer. Keep in mind that the problem is transmission limited, not TV display speed limited. The solution was to transmit the odd scan lines in 1/60th of a second followed by the even scan lines in the second 1/60th of a second, resulting in 60 fields per second interlaced (60i), making 30 frames per second. The frame rate was still 30 fps, but the refresh rate was 60 fps—safely outside the FFT. For most people. There are a few that can still see a little flicker even at 60i, usually only with their peripheral vision.

When the specs for HDTV were being developed, a high priority was to phase out interlaced TV and the non-square pixels of PAL and NTSC television. So the new digital HDTV was spec'd out with square pixels (praise the Lord!) and a progressive scan (yahoo!). The interlaced standard (1080i) was important in the early days of digital TV as the network connections were not as fast as today's. Also, the modern flat-panel HDTV does not go black between frames, as the old CRTs did. A frame is loaded into the TV set's buffer and sits there until replaced.

Even though there is precious little interlaced video produced today, that will not save you from the billion hours of old interlaced video on tape and in storage waiting since the 1950s for restoration someday.

This video will give you a slow-motion demonstration of how interlaced video works for both horizontal and vertical motion.

2.4.3 Frame Rates

Due to the long, rich history of television, it has inherited several frame rates from its analog days. The transition to modern digital television has added some more, and being digital, we can now indulge all of them. Enjoy:

- **23.98 fps** - This is what you get when you take a 24 fps movie on film and transfer it to 29.97 fps video. We won't go into exactly why this is so, because the explanation is arduous and ultimately unrewarding.

从上到下，屏幕在光栅弹回顶部准备下一帧时变为空白。这会导致30帧每秒的闪烁——恰好落在普通观看者的FFT感知范围内。需注意，问题在于传输速率受限，而非电视显示速度受限。解决方案是在1/60秒内传输奇数扫描线，随后在下一个1/60秒内传输偶数扫描线，从而形成每秒60场交错扫描（60i），构成30帧每秒的画面。帧率仍为30 fps，但刷新率提升至60 fps——安全超出了FFT感知阈值。对大多数人而言如此。仍有少数人能在60i下察觉到轻微闪烁，通常仅通过周边视觉感知。

当高清电视的规格正在制定时，一个高度优先事项是逐步淘汰隔行扫描电视以及PAL和NTSC电视的非方形像素。因此，新的数字高清电视被设计为采用方形像素（感谢上帝！）和逐行扫描（太棒了！）。隔行扫描标准（1080i）在数字电视早期非常重要，因为当时的网络连接速度不如现在快。此外，现代平板高清电视不会像老式CRT电视那样在帧之间变黑。一帧图像被加载到电视的缓冲区中，并在那里停留直到被替换。

尽管如今生产的隔行扫描视频已经少之又少，但这并不能让你免于面对自20世纪50年代以来存储在磁带和仓库中、等待有朝一日修复的数十亿小时老旧隔行扫描视频。

本视频将以慢动作演示隔行扫描视频在水平和垂直运动中的工作原理。

2.4.3 帧率

由于电视悠久而丰富的历史，它从模拟时代继承了几种帧率。向现代数字电视的过渡又增加了一些，并且由于数字化，我们现在可以尽情享受所有这些帧率。请欣赏：

- 23.98 帧/秒——这是将 24 帧/秒的电影胶片转换为 29.97 帧/秒视频时得到的结果。我们不会深入探讨其具体成因，因为解释过程繁琐且最终并无太大意义。

- **24 fps** - The traditional cinema frame rate. Since the modern HDTV can operate at any frame rate this is often used when showing theatrical movies to preserve the original feel of the film.
- **25 fps** - The standard television frame rate for PAL countries in Europe and Asia.
- **29.97 fps** - The standard television frame rate for the NTSC countries of the US, Canada, and Japan. The odd frame rate is inherited from the days of adding color to black and white TV, which introduced a broadcast signal interference at exactly 30 fps, so the entire system was slightly "de-tuned" to 29.97 to eliminate the interference. Again, you really don't want to know why.
- **30 fps** - An honest 30 fps used for some digital content production and online streaming.
- **50 fps** - Used for high-frame-rate action content like sports in PAL 25 fps countries.
- **59.97 fps** - Used for high frame-rate action content like sports in NTSC 29.97 countries.
- **60 fps** - Used for high frame-rate content like sports, gaming, and super high-quality streaming in NTSC countries.

These standard video frame rates are, of course, utilized by HDTV sets, which we will visit in loving detail in Chapter 5: Display Space, where we will "refresh" these numbers.

2.5 Output Options

Modern cameras, being digital, offer a wide range of file formats and color space options—which, of course, complicates the whole color management story. Fortunately, video cameras just output Rec. 709 video, which immediately answers their color space question. DSLR cameras are just a bit more complicated, in that they will typically offer output options of RAW, JPEG, or TIFF. The photographer may choose to export a TIFF or JPEG in sRGB, Adobe RGB, or ProPhotoRGB (if supported) color spaces; so already we

- 24 fps - 传统电影帧率。由于现代高清电视可在任意帧率下运行，因此在播放影院电影时常用此帧率以保留影片的原始观感。

- 25 fps - 欧洲和亚洲PAL制式国家的标准电视帧率。

- 29.97 fps - 美国、加拿大和日本等NTSC制式国家的标准电视帧率。这一特殊帧率源于黑白电视向彩色电视过渡的时期：当时在精确30帧/秒的广播信号中出现了干扰，因此整个系统被轻微“调偏”至29.97帧/秒以消除干扰。重申一次，您确实不必深究其技术成因。

- 30 fps - 用于某些数字内容制作和在线流媒体的真实30帧每秒。

- 50 fps - 用于PAL制式25 fps国家中体育等需要高帧率的动作内容。

- 59.97 fps - 用于NTSC 29.97国家/地区的高帧率动作内容，如体育节目。

- 60 fps - 用于NTSC国家的高帧率内容，如体育赛事、游戏和超高质量流媒体。

这些标准视频帧率当然也被高清电视机所采用，我们将在第五章“显示空间”中详细探讨，届时会“刷新”这些数据。

2.5 输出选项

现代相机由于是数字化的，提供了多种文件格式和色彩空间选项——这当然使整个色彩管理变得更加复杂。幸运的是，摄像机直接输出Rec. 709视频，这立即解决了它们的色彩空间问题。数码单反相机则稍微复杂一些，因为它们通常提供RAW、JPEG或TIFF的输出选项。摄影师可以选择以sRGB、Adobe RGB或ProPhotoRGB（如果支持）色彩空间导出TIFF或JPEG文件；因此，我们已经

have some color space questions—all of which are answered in Chapter 4: Work Space.

And then there are the cinema cameras, such as ARRI and RED, that offer a long list of output options. These might include Camera RAW, 16-bit half-float linear OpenEXR files, 10-bit log DPX files, various ProRes video formats, Avid DNxHR, and even JPEG. Again, these are all explained in Chapter 4. The point here is to realize that there is a dizzying array of options for the cinematographer, and therefore for you when you get the clips.

You may hear a cinematographer say, "I shot it in log" or ProRes or whatever—but they didn't really. Cameras only shoot in linear light space because that is how their imaging sensors work. The log, ProRes, or whatever was generated from the debayered Camera RAW upon output with color transformations built into the camera.

At the high end, the colorist may choose to export the Camera RAW to supervise the debayering process themselves to make creative tweaks. The output of that will likely be an OpenEXR file. To save some money and time, the camera export may be OpenEXR or DPX, letting the camera make all the debayering decisions. Down from that, we have compressed video file formats such as ProRes and DNxHR, and I refuse to even discuss JPEG, for reasons that are made clear in Chapter 4: Work Space.

2.6 Camera Artifacts

And now for the bad news. Digital cameras introduce their own set of issues and artifacts into their captured images, some of which are bizarre and darned difficult to fix in post. Here we take a look at how the camera itself damages and degrades the image.

2.6.1 Noise

First up on the hit parade of artifacts is noise. Film has grain; digital cameras have noise. To be honest, digital cameras on the whole produce a much cleaner lower noise image than film. I have degrained hundreds of film plates

有一些色彩空间的问题——所有这些问题都在第4章：工作空间中得到了解答。

然后是电影摄影机，比如ARRI和RED，它们提供一长串输出选项。这些可能包括Camera RAW、16位半浮点线性OpenEXR文件、10位对数DPX文件、各种ProRes视频格式、Avid DNxHR，甚至JPEG。同样，这些都在第4章中有详细解释。这里的重点是认识到，对于电影摄影师来说，因此当你拿到片段时，可供选择的选项多得令人眼花缭乱。

你可能会听到摄影师说，“我用log模式拍的”或者ProRes格式之类的——但其实并非如此。摄像机只能在线性光空间拍摄，因为这是其图像传感器的工作原理。所谓的log、ProRes或其他格式，都是在输出时通过相机内置的色彩转换，从去马赛克后的相机RAW文件生成的。

在高端领域，调色师可能会选择导出Camera RAW文件，亲自监督去马赛克过程以进行创意调整。其输出结果很可能是OpenEXR文件。为了节省成本和时间，相机导出的文件可能是OpenEXR或DPX格式，由相机完成所有去马赛克决策。再往下则是压缩视频文件格式，如ProRes和DNxHR，至于JPEG格式——基于第四章"工作空间"中阐明的原因——我甚至拒绝讨论。

2.6 相机伪影

现在来说说坏消息。数码相机在拍摄的图像中引入了自身的一系列问题和伪影，其中一些非常怪异，后期修复起来也相当棘手。这里我们来看看相机本身是如何损害和降低图像质量的。

2.6.1 噪声

首先登上伪影排行榜的是噪点。胶片有颗粒感；数码相机则有噪点。老实说，数码相机整体产生的图像比胶片更干净、噪点更低。我曾为数百张胶片底片进行过降颗粒处理

and denoised hundreds of digital cinema plates and the digital images win the clean contest hands down. However, they do have noise—to such a degree that you will usually want to invest shot development time in noise management. Interestingly, the blue channel is the noisiest for both film and digital. The reason blue is the noisiest is because the imaging sensor is less sensitive to blue light, so its signal has to be amplified more than the other two channels. When the blue channel is amplified, so is its noise.

There are two types of noise: shot noise and digital noise. Shot noise is due to random photon hits (or statistical quantum fluctuations, if you prefer) from the scene itself and is, therefore, both pervasive and unavoidable. The fact that the shutter is open only briefly means that fewer photons enter the imaging sensor, so the quantum fluctuations become noticeable. The shorter the exposure and the higher the f-stop both mean less light and therefore more shot noise. Take a one-minute exposure and I can guarantee you a noise-free image.

Digital noise, however, is generated by thermal noise in the image sensor and the surrounding electronics and is manageable. Several high-end digital cinema cameras have digital "chiller plates" to cool the imaging electronics, which dramatically reduces this noise. Solid-state thermoelectric cooling plates use the Peltier effect to lower the temperature. Very cool.

I saw this for myself in 2011, when I was asked by ARRI to evaluate their new ARRI Alexa camera for greenscreen shots. ARRI provided me with 38 ARRI RAW bluescreen and greenscreen clips over a range of exposures showing two people dancing, plus a few steam shots, for keying tests. While they were more interested in my evaluation of how suitable the footage was for keying (it was fantastic!), I was struck by the low noise levels. So, I asked the ARRI engineers about it—which is when I found out that the ARRI Alexa had one of those thermoelectric cooling plates for the image sensor.

The other thing about digital noise is how the DP can screw it all up by increasing the ISO to get those low-light level shots or to lower the exposure for some creative reason. Increasing the ISO in a digital camera simply means amplifying the imaging sensor signal. When you amplify the signal, you amplify the noise. The noise will be worse in the darks where there is much less light relative to the noise. If the ISO is cranked up high enough, the footage can become unusable. The wise DP will shoot the scene with

并且，经过降噪处理的数百张数字电影底片和数字图像在清洁度竞赛中轻松胜出。然而，它们确实存在噪点——其程度通常让你愿意在镜头开发阶段投入时间进行噪点管理。有趣的是，无论是胶片还是数字影像，蓝色通道都是噪点最多的。蓝色通道噪点最多的原因在于成像传感器对蓝光的敏感度较低，因此其信号需要比其他两个通道进行更大程度的放大。当蓝色通道被放大时，其噪点也随之放大。

噪声分为两种：散粒噪声和数字噪声。散粒噪声源于场景本身随机光子撞击（或称为统计量子涨落），因此既普遍存在又无法避免。快门开启时间短暂意味着进入图像传感器的光子数量减少，从而使量子涨落变得明显。曝光时间越短、光圈f值越高，都意味着进光量减少，进而导致更明显的散粒噪声。若进行一分钟长曝光，我敢保证你能获得毫无噪声的图像。

然而，数字噪声是由图像传感器及周边电子设备的热噪声产生的，并且是可控制的。多款高端数字电影摄影机配备了数字“冷却板”来冷却成像电子元件，这显著降低了此类噪声。固态热电冷却板利用帕尔贴效应来降低温度。非常酷。

2011年，我亲眼见证了这一点。当时ARRI邀请我评估他们新款ALEXA摄影机在绿幕拍摄中的表现。ARRI向我提供了38段ARRI RAW格式的蓝幕与绿幕素材片段，涵盖不同曝光条件下两名舞者的动作以及少量蒸汽场景，供抠像测试使用。虽然他们更关注我对素材抠像适用性的评估（效果极其出色！），但令我震撼的是其极低的噪点水平。于是我向ARRI工程师咨询了此事——这才得知ARRI ALEXA的图像传感器配备了热电冷却板。

数字噪点的另一个问题在于，摄影师如何通过提高ISO来获得低光照条件下的拍摄效果，或出于某些创意考虑降低曝光，从而可能搞砸一切。在数码相机中提高ISO仅仅意味着放大成像传感器的信号。当你放大信号时，噪点也会被放大。在暗部区域，由于光线相对于噪点要少得多，噪点会更加明显。如果ISO调得过高，拍摄的素材可能会变得无法使用。明智的摄影师会以

adequate light and then rely on the powerful tools in post-production to give the shot the final look they want.

2.6.2 Rolling Shutter

This is a bizarre problem unique to digital cameras. To transfer the data out of the image sensor, the camera scans the image sensor array one line at a time—top to bottom, for example. As a result, there is a brief time delay between the top and bottom scan lines of the sensor, and if an object has moved significantly during that time delay, it will appear in a slightly different place at the bottom than the top. For horizontal motions, this will introduce a shear in the image.

Figure 2.15 Types of rolling shutter artifacts

Figure 2.15 illustrates several types of rolling shutter artifacts. The car outlined in red in the first panel has an action shear created by rapid object motion with a locked-off camera. The second panel shows what happens with a static scene and rapid camera motion. This could be either a camera car drive-by or a rapid horizontal pan. The third panel shows how bizarre rotating objects can get from rolling shutter with the curved golf club. The last panel with the bent prop blades is there because the rotation distortion is totally insane.

The problem with rolling shutter artifacts is that they are complicated (i.e., expensive) to fix. Take the sheared car in the first panel. First, the car would have to be isolated and straightened up, which would then open up holes behind the car where the background would be missing. Then the missing pieces of the background would have to be found in nearby frames and tracked in. The distorted propeller in the right panel is hopeless. It would have to be completely removed, the background repaired, and a CG propeller rendered and comped in. Yikes!

A few—and only a few—cameras have a global shutter that exports the entire frame simultaneously without scanning from top to bottom, so they have no

充足的光线，然后依靠后期制作中的强大工具，赋予镜头他们想要的最终外观。

2.6.2 卷帘快门

这是数码相机特有的一个奇怪问题。为了将数据从图像传感器传输出来，相机会逐行扫描图像传感器阵列——例如从上到下。因此，传感器顶部和底部扫描线之间存在短暂的时间延迟，如果物体在这段时间内发生了显著移动，它在底部出现的位置会与顶部略有不同。对于水平移动，这会在图像中引入剪切变形。

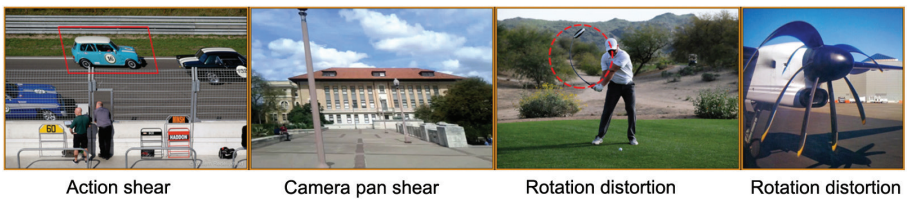


图2.15 滚动快门伪影的类型

图2.15展示了多种类型的滚动快门伪影。第一幅图中用红色勾勒的汽车，在相机固定不动的情况下，因物体快速运动产生了动作剪切。第二幅图展示了静态场景下相机快速移动时的效果——这可能是相机车驶过或快速水平摇摄所致。第三幅图显示了旋转物体在滚动快门下的怪异变形，例如弯曲的高尔夫球杆。最后一幅图中螺旋桨叶片弯曲，是因为旋转畸变达到了极其夸张的程度。

卷帘快门伪影的问题在于修复起来复杂（即昂贵）。以第一幅图中扭曲的汽车为例。首先需要将汽车分离并扶正，这会在汽车后方留下背景缺失的空洞。随后必须从邻近帧中寻找缺失的背景片段并进行追踪处理。而右侧画面中变形的螺旋桨则更为棘手——它需要被完全移除，修补背景，再渲染一个CG螺旋桨并进行合成。真是棘手！

少数——并且只有少数——相机拥有全局快门，能够同时导出整个帧而无需从上到下扫描，因此它们没有

rolling shutter. However, this is a more expensive image sensor to build and lowers the frame rate that a camera can support, so not many cameras use this approach.

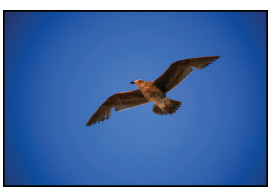
2.6.3 DSLR Rolling Shutter

Most DSLR cameras have a rolling shutter because most use CMOS sensors. Most CCD sensors use a global shutter, but CCD sensors are expensive. While even a 4K Digital Cinema camera will only have around 9 megapixels, the average DSLR resolution is around 25 megapixels, with some reaching 40 or even 50 megapixels! A CCD sensor of that size would be prohibitively expensive, so most DSLRs use CMOS to save cost and simply endure the rolling shutter.

Stills photography has a better chance of avoiding rolling shutter artifacts than cinematography because stills photographers have more control over their camera setups and composition. For example, taking a longer exposure of a moving subject will convert a rolling shutter into motion blur. Cinematographers have limited options for longer exposures. Still photographers normally don't have to take whip pans. However, if tracking a fast-moving target like a race car, the background may get rolling shutter. Zooming in closer to the subject and lowering the depth of field to put the background out of focus can make it less noticeable.

FUN FACT: The **Jell-O Effect** is the name given to the rolling shutter artifact produced by a camera mounted to a shaky base, which produces footage that wobbles like a bowl of Jell-O.

2.6.4 Vignetting



We all know about vignetting caused by the lens body, which results in darkening at the corners of an image like the seagull picture in Figure 2.16. What you may not know is that there is a second cause of vignetting specific to digital cameras—namely

Figure 2.16 Vignetting

卷帘快门。然而，这是一种制造成本更高的图像传感器，并且会降低相机支持的帧率，因此很少有相机采用这种方法。

2.6.3 数码单反相机卷帘快门

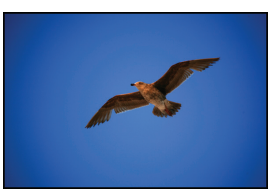
大多数单反相机采用滚动快门，因为多数使用CMOS传感器。大多数CCD传感器采用全局快门，但CCD传感器价格昂贵。尽管即便是4K数字电影机通常只有约900万像素，普通单反相机的分辨率却能达到约2500万像素，部分机型甚至可达4000万或5000万像素！而CCD传感器——

那种尺寸的传感器成本会高得令人望而却步，因此大多数单反相机使用CMOS来节省成本，并只能忍受滚动快门的问题。

静态摄影比电影摄影更有机会避免滚动快门伪影，因为静态摄影师对相机设置和构图有更多控制。例如，对移动主体进行更长的曝光会将滚动快门转化为运动模糊。电影摄影师在延长曝光时间方面选择有限。静态摄影师通常不需要进行快速摇摄。然而，如果追踪像赛车这样的快速移动目标，背景可能会出现滚动快门。通过拉近拍摄主体并降低景深使背景失焦，可以减少其明显程度。

> 有趣的事实：果冻效应是指安装在晃动基座上的相机产生的滚动快门伪影，这种伪影会使拍摄的画面像一碗果冻一样晃动。

2.6.4 渐晕



我们都知道由镜头本体引起的暗角，这会导致图像角落变暗，就像图2.16中的海鸥照片那样。你可能不知道的是，数码相机还存在第二种暗角成因——即

图2.16 渐晕

"pixel vignetting," illustrated in Figure 2.17. Here's the story; the light rays traveling from the lens to the center of the sensor array are entering the photosite cells straight on. However, the light rays out near the edges of the array are slightly off-angle. Being off-angle, some of the photons are blocked by the surrounding silicon, which weakens the signal and darkens the image. It is a circular vignette like the lens vignette and is added to it. Your image now has a double vignette, each with its separate characteristics of location and fall-off. Joy.

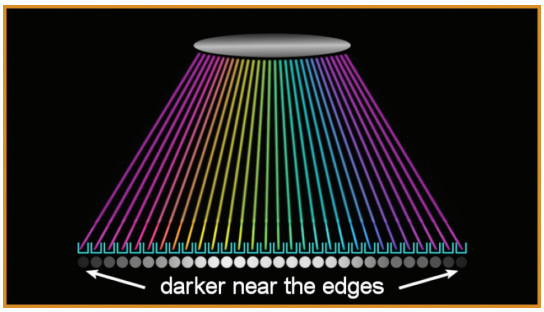


Figure 2.17 Sensor pixel vignetting

2.7 Lens Artifacts

We all know the story about how lenses bend light rays to converge at the focal point, so I won't bore you with it here. But I'll bet you never heard the story about *why* the light rays bend. It's all about the speed of light. As you know, the speed of light in a vacuum is 186,000 miles per second (300,000 kilometers per second), but when light travels through a medium other than a vacuum, it slows down. How much it slows down is determined by the medium's refractive index. In fact, the refractive index of a material is simply the measure of how much it slows light.

I'm not going to draw the 10-millionth illustration of light rays refracting through a lens, but we can do an interesting thought experiment to understand lensing. As mentioned, a light ray slows down when entering a transparent medium like air, glass, water, or plastic. If the light ray strikes that surface perpendicularly, it just slows down. But if it strikes that surface at an angle (called the "angle of incidence"), it both slows down *and* changes direction to a sharper angle because of the slowing. The greater the angle of incidence, the greater the change in angle. Light rays at the outer edge of a lens have a greater angle of incidence than those near the center, so they get refracted more. With a carefully shaped lens, you can get the light rays to

“像素渐晕”，如图2.17所示。情况是这样的：从镜头传至传感器阵列中心的光线是垂直进入感光单元内的。然而，靠近阵列边缘的光线则存在轻微的角度偏移。由于存在角度偏移，部分光子被

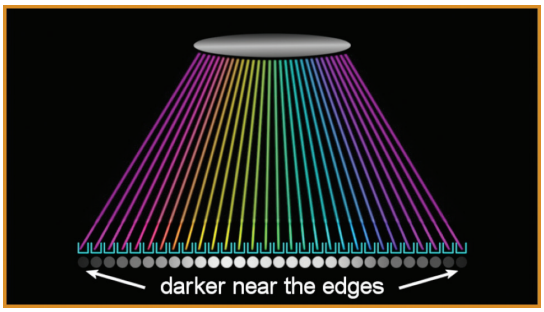


图2.17 传感器像素渐晕

环绕硅层会削弱信号并使图像变暗。这是一种类似镜头暗角的圆形渐晕，并叠加其上。现在你的图像具有双重渐晕，各自拥有独特的位置和衰减特性。妙哉。

2.7 镜头伪影

我们都知道透镜如何使光线弯曲并汇聚于焦点的故事，这里就不再赘述了。但我敢打赌，你从未听说过光线为何会弯曲的原因。这其实与光速有关。众所周知，光在真空中的速度为每秒186,000英里（每秒300,000公里），但当光穿过真空以外的介质时，速度会减慢。减慢的程度由介质的折射率决定。实际上，材料的折射率就是衡量其使光减速程度的指标。

我并不打算绘制第1000万张光线通过透镜折射的示意图，但我们可以做一个有趣的思想实验来理解透镜成像原理。如前所述，光线进入透明介质（如空气、玻璃、水或塑料）时会减速。如果光线垂直射入介质表面，它只会减速；但如果以一定角度（称为“入射角”）射入，则不仅会减速，还会因速度变化而转向更锐利的角度。入射角越大，角度变化越显著。透镜边缘的光线比中心区域的光线具有更大的入射角，因此折射程度更强。通过精密设计的透镜形状，可以使光线

bend more from center to edge such that they all converge at one point—the focal point. You now have a focusing lens.

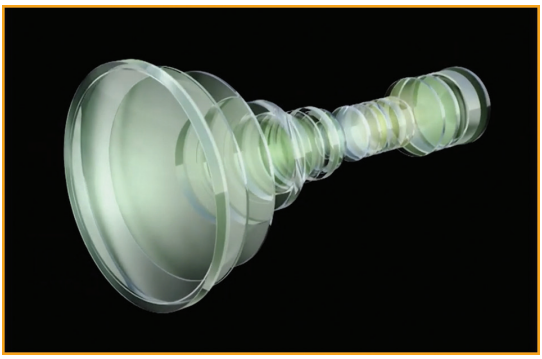


Figure 2.18 Sophisticated zoom lens assembly. Courtesy of Canon

The problem we get with focusing lenses is that they have issues, seven of which we will meet shortly. To compensate for these issues, correcting lenses must be added, which fix one thing but introduce another. So, a compensating lens must also be added to the correcting lens and then the zoom elements must be added until you end up with the glass marvel shown in Figure 2.18. Now you know why zoom lenses are so heavy and expensive.

Lenses are an integral part of the camera, so lens artifacts must be added to the camera artifacts listed above. Some artifacts, like lens distortion, can alter the shape of the captured image. Other lens artifacts disturb the focused light rays and can either defocus the image or shift the color channels relative to each other. Lenses can also bounce stray light rays to add flare and glare on top of the captured image.

See the wonders of light refraction and how it is affected by wavelength and angle of incidence to really understand lensing.

There are lens artifacts—such as depth of field and bokeh—that some will argue are not artifacts at all, but in fact artistic assets. I included them as artifacts because my definition of an artifact is any unintended distortion of the perfect picture introduced by a lens. The fact that some may like those artifacts and even use them creatively does not change the fact that they are still unintended distortions of the image. Lenses are complex devices, which

从中心到边缘弯曲得更多，使得它们都汇聚于一点——焦点。现在你有了一个聚焦透镜。

聚焦透镜面临的问题在于它们存在缺陷，我们很快会探讨其中七点。为弥补这些缺陷，必须添加校正透镜，但这往往解决了旧问题又引发新问题。因此，校正透镜本身也需要配备补偿透镜，接着还需加入变焦组件，直至最终形成{v*}

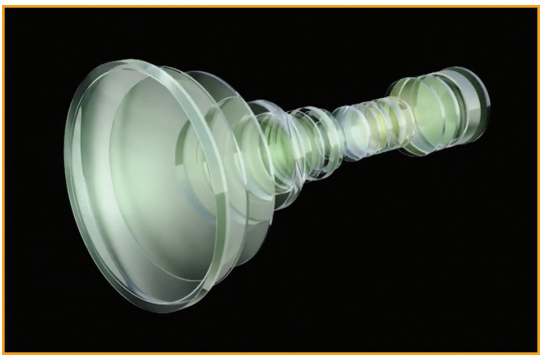


图2.18 精密的变焦镜头组件。由佳能提供

图2.18中展示的玻璃奇迹。现在你明白为什么变焦镜头如此沉重且昂贵了。

镜头是相机不可或缺的一部分，因此必须将镜头伪影添加到上述相机伪影列表中。一些伪影，如镜头畸变，会改变捕获图像的形状。其他镜头伪影会干扰聚焦光线，可能导致图像失焦或使颜色通道相互偏移。镜头还可能反射杂散光线，在捕获的图像上增添光晕和眩光。

观察光的折射奇迹，了解波长和入射角如何影响这一现象，从而真正理解透镜效应。

镜头存在一些光学伪影——例如景深和散景——有些人会争辩说这些根本不是伪影，而是艺术资产。我将它们归为伪影，是因为我对伪影的定义是镜头引入的对完美画面的任何非预期失真。有些人可能喜欢这些伪影，甚至创造性地使用它们，但这并不能改变它们仍然是非预期图像失真的事实。镜头是复杂的设备，

means their artifacts are complex so here we will take a look at what they are, how they work, and if there is anything we can do to fix them.

2.7.1 Lens Distortion

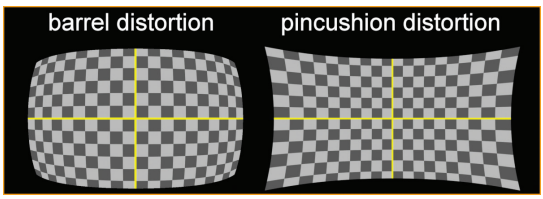


Figure 2.19 Lens distortion

Lens distortion is an artifact that we wrestle with all the time in visual effects. It appears in the form of a barrel or pincushion distortion (Figure 2.19). While usually not a problem for editors or colorists, in visual effects, we have to do camera tracking for some shots and they must be undistorted to get accurate results. Another VFX case is when combining two plates such as a greenscreen shot or a monitor insert. If the lens distortion of the two layers does not match, then when the camera moves, the two elements can drift and squirm against each other. In extreme cases, lens distortion can even be aesthetically objectionable. Fortunately, we now have commercially available software that can reduce or eliminate lens distortion entirely.

2.7.2 Spherical Aberration

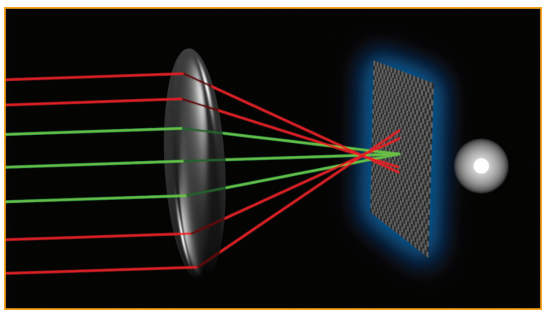


Figure 2.20 Spherical aberration

Spherical aberration imparts a defocused glow around bright elements in the scene and Figure 2.20 shows why. Light rays entering the center of a convex lens (green) converge perfectly at the focal plane, which is where the image sensor is located to form a sharp image. However, light rays entering the outer edges of the lens (red) converge closer than the image sensor and then spread out to form a fuzzy image, illustrated to the right of the sensor. To correct for this, a concave corrective lens is placed in the light path. This may go unnoticed for the main elements of the shot, but

意味着他们的制品很复杂，因此这里我们将探讨它们是什么、如何运作，以及我们是否能做些什么来修复它们。

2.7.1 镜头畸变

镜头畸变是我们在视觉效果中一直需要处理的一种人为现象。它表现为桶形或枕形畸变（图2.19）。虽然这对剪辑师或调色师通常不构成问题，但在视觉效果中，

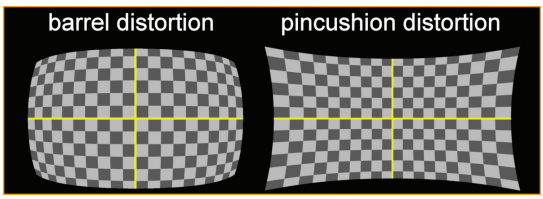


图 2.19 镜头畸变

我们必须对某些镜头进行摄像机跟踪，并且必须消除镜头畸变以获得准确结果。另一个视觉特效案例是合成两个画面，比如绿幕镜头或屏幕插入镜头。如果两个图层的镜头畸变不匹配，那么当摄像机移动时，两个元素可能会相互漂移和扭曲。在极端情况下，镜头畸变甚至可能在审美上令人不适。幸运的是，我们现在有商业软件可以完全减少或消除镜头畸变。

2.7.2 球面像差

球面像差会在场景中的明亮元素周围产生一种失焦的光晕，图2.20展示了其原因。光线从凸透镜中心（绿色）进入时，会在焦平面上完美聚焦——图像传感器正是位于此处以形成清晰的图像。然而，

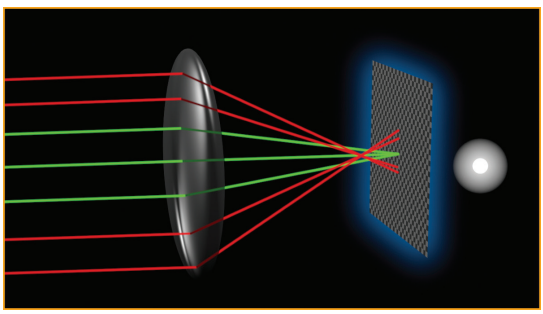


图 2.20 球面像差

进入镜头外缘的光线（红色）会聚于比图像传感器更近的位置，然后扩散形成模糊图像，如传感器右侧所示。为纠正此现象，需在光路中放置凹面矫正镜片。这对于拍摄主体元素可能不易察觉，但

bright light sources will show a fuzzy halo. Normal VFX tools cannot fix this artifact, however, if you have a model of the lens plus the right Fast Fourier Transforms, this artifact can be removed from images.

2.7.3 Chromatic Aberration

Chromatic aberration causes the color channels to fan out near the edges of the frame. Figure 2.21 illustrates how the light paths for each color diverge, producing the colored spot shown to the right of the image sensor. This is caused by the fact that the degree to which the light is refracted in the lens depends on its wavelength, so the different wavelengths in the white light refract differently causing them to converge at slightly different distances. To correct for this, another element is added to the lens assembly.

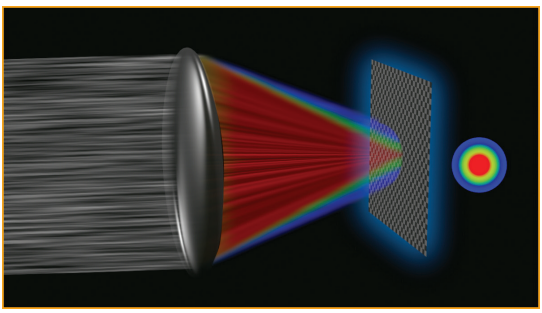


Figure 2.21 Chromatic aberration

The good news is that this lens aberration can be corrected in post (post-production) or Photoshop for a single frame rather easily, simply by slightly scaling down each color channel by a different amount. For the pattern in Figure 2.21, the blue channel might be scaled down by, say, 0.993, while the green channel gets 0.997—a slightly less scale factor—while the red channel is used as is.

2.7.4 Depth of Field

Some will be surprised to see the depth of field listed as a lens artifact, but it is. Lenses were developed to gather light and focus it on a small image sensor. Unfortunately, it was quickly discovered that only objects a certain distance from the

Figure 2.22 Depth of field

明亮的光源会显示模糊的光晕。通常的视觉特效工具无法修复这种伪影，然而，如果你拥有镜头模型以及正确的快速傅里叶变换{v*}，这种伪影可以从图像中移除。

2.7.3 色差

色差导致颜色通道在画面边缘附近呈扇形散开。图2.21展示了每种颜色的光路如何发散，从而在图像传感器右侧产生彩色光斑。这是由于光线在镜头中的折射程度不同所致。

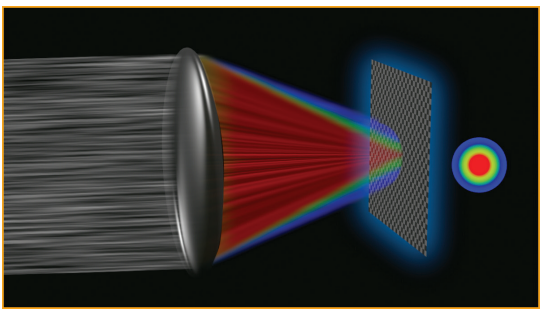


图2.21 色差

取决于其波长，因此白光中不同波长的光折射程度不同，导致它们会聚在略微不同的距离上。为了纠正这一点，在镜头组件中加入了另一个元件。

好消息是，这种镜头像差可以在后期（后期制作）或Photoshop中对单帧图像进行相对容易的校正，只需将每个颜色通道按不同比例略微缩小即可。对于图2.21中的图案，蓝色通道可能需要缩小，比如缩小到0.993倍，而绿色通道则缩小到0.997倍——缩放比例稍小一些——红色通道则保持不变。

2.7.4 景深

有些人会惊讶地发现景深被列为镜头的一种缺陷，但事实确实如此。镜头的设计初衷是收集光线并将其聚焦在小型图像传感器上。然而，人们很快发现只有与传感器保持特定距离的物体

图 2.22 景深

lens could be in sharp focus, while everything in front or behind that zone was out of focus. Figure 2.22 illustrates this nicely, with the cluster of leaves in sharp focus with the fence in front and the bushes behind seriously out of focus. Fortunately, creative cinematographers and photographers quickly realized that they could use depth of field to focus the viewer's attention on any part of the picture they wanted. The fact that depth of field is used as a powerful creative tool does not change the fact that it is an unavoidable artifact of lenses. The fix for this artifact requires very sophisticated Fast Fourier Transforms plus a model of the lens. Not available in commercial software yet, but I would look for an AI solution soon.

2.7.5 Veiling Glare

Veiling glare adds an overall haze to a picture, as in Figure 2.23. A bright light source just out of frame throws light rays into the lens assembly that bounce around and spread out to cover most of the plate. This has the effect of washing out the image by imparting an overall lift operation to the entire plate, which lowers contrast and saturation. A talented colorist or VFX artist can remove most of the veiling glare manually with commercially available software.

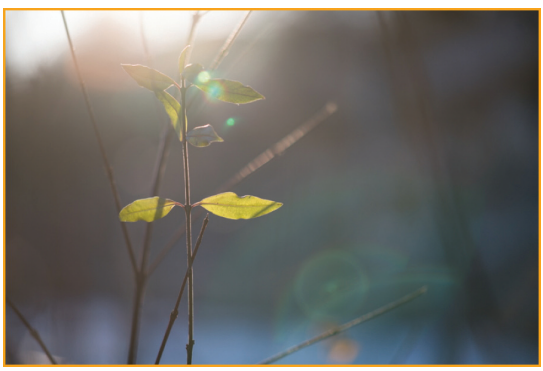


Figure 2.23 Veiling glare

2.7.6 Lens Flare

Lens flare adds geometric light elements to the image, such as rings, rays, and circles, as in Figure 2.24. It is caused by a strong light source in frame that bounces and scatters stray light rays off internal lens elements.

Figure 2.24 Lens flare

镜头可以对焦清晰，而该区域前后的一切则处于失焦状态。图2.22生动展示了这一现象：一簇树叶清晰对焦，而前方的栅栏和后方灌木丛则严重失焦。幸运的是，富有创造力的电影摄影师和摄影师很快意识到，他们可以利用景深将观众的注意力引导至画面中任意想要突出的部分。尽管景深已被用作一种强大的创作工具，但这并未改变它本质上是镜头不可避免的成像特性。要修正这一特性，需要运用非常精密的快速傅里叶变换并结合镜头模型进行演算。目前商业软件尚未实现此功能，但我预计不久后便会出现基于人工智能的解决方案。

2.7.5 杂散光

眩光会给图片增添一层整体的朦胧感，如图2.23所示。恰好在画框外的明亮光源将光线射入镜头组件，光线在其中反复反射并扩散开来，覆盖了底片的大部分区域。这相当于对整个底片施加了整体提亮操作，导致图像细节被冲淡，从而降低了对比度和饱和度。一位才华横溢的调色师——

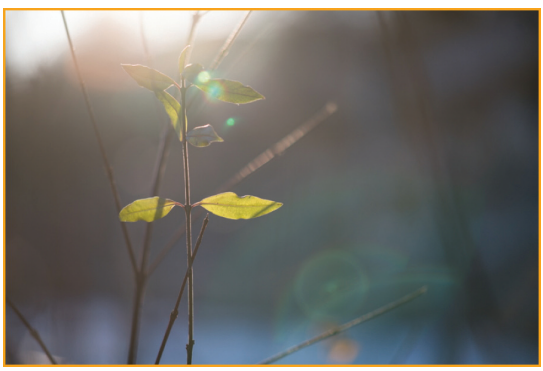


图2.23 杂散光

ist或VFX艺术家可以使用市售软件手动去除大部分眩光。

2.7.6 镜头光晕

镜头光晕为图像添加了几何光元素，例如光环、光线和圆环，如图2.24所示。它是由画面中的强光源在镜头内部元件间反射和散射杂散光线造成的。

图2.24 镜头光晕

Often considered a nuisance artifact, some cinematographers embrace it as a creative element. When creating a CG scene in visual effects, a lens flare will often be added to give it a more photorealistic look or to match the look of surrounding shots. A talented paint artist can paint out the lens flare frame by frame with commercially available software.

2.7.7 Bokeh

Figure 2.25 Bokeh

Bokeh is what you get with small bright lights out of focus in the background. The blurry spot takes on the shape of the aperture, so bokehs can be multi-sided or round. Different lenses produce different appearing bokeh; some are considered "good" bokeh, because they are esthetically pleasing, while others are cursed with "bad" bokeh. Shooting a shallow depth of field so that the far background is very out of focus will increase the amount of bokeh. While there is no practical technology for removing bokeh from a shot, there are many tools available for adding it.

2.7.8 The Swirly Bokeh

Figure 2.26 The beautiful Petzval lens swirly bokeh

Invented in 1840 by Joseph Petzval, an Austrian mathematician, this was one of the earliest lenses designed specifically for photography with reduced exposure times and the characteristic "swirly bokeh" intended for portrait photography (see Figure 2.26). The design was sharp in the center of the image, with a rapid falloff in sharpness toward the edges, accompanied by its signature swirl effect from

常被视为令人讨厌的瑕疵，一些电影摄影师却将其作为创意元素加以运用。在视觉效果中创建CG场景时，通常会添加镜头光晕以增强照片真实感，或与周围镜头的视觉效果保持一致。技艺高超的绘景师可以使用市售软件逐帧擦除镜头光晕。

2.7.7 散景

散景是背景中失焦的小亮点所形成的效果。模糊的光斑会呈现出光圈形状，因此散景可能是多边形的或圆形的。不同镜头会产生不同外观的散景；有些被认为是“优质”散景，因为它们

图 2.25 散景

有些散景在美学上令人愉悦，而另一些则被“糟糕”的散景所困扰。拍摄浅景深使远处背景高度失焦，会增加散景的量。虽然目前尚无实用技术能从拍摄中移除散景，但有许多工具可用于添加散景效果。

2.7.8 旋焦散景

图 2.26 美丽的匹兹瓦镜头旋焦散景

由奥地利数学家约瑟夫·佩茨瓦尔于1840年发明，这是最早专门为摄影设计的镜头之一，其特点是缩短曝光时间，并具备为人像摄影而设的标志性“旋焦散景”（见图2.26）。该设计在图像中心区域成像锐利，但边缘画质迅速衰减。

边缘锐利，伴随着其标志性的漩涡效果。

the circular blurring of the out-of-focus areas generated by significant field curvature and spherical aberration.

Despite its 19th-century origins, there has been a 21st-century resurgence in the popularity of this unique lens so that today, one can be purchased that blends modern lens-making technology with its classical optical design. Besides, it was such a unique lens effect that I just had to share it with you.

See the inner workings of lens aberrations as their effect on the captured image is revealed in these unique animations.

2.8 The DIT

The DIT (Digital Imaging Technician) is a key player in the modern digital imaging pipeline. They bridge the gap between the director's vision and the bits, bytes, and color spaces that make up the digital database that the torrent of data from the camera is creating. From camera capture to final color grading, the DIT plays a key role in managing the digital database that is the movie—a non-trivial undertaking, as we will soon see.

2.8.1 On-Set Support

The DIT's job starts with the camera itself. They set up the cinema camera to the specs of the cinematographer (Director of Photography or DP) for exposure, color temperature, and ISO settings. They advise the DP of over or under-exposures, as these differ significantly from their experiences with film.

They will set up the "video village" in a tent on location for their computers and monitors, where they will monitor the camera data being captured to ensure it contains usable picture information and will even perform a preliminary color grade to rough in the film look. They will convert the day's takes into "digital dailies" that can be played on the DP's iPad.

由显著的场曲和球差产生的离焦区域的圆形模糊。

尽管其起源可追溯至19th世纪，但这款独特镜头在21st世纪重新流行起来，如今已能购得融合现代镜头制造技术与经典光学设计的版本。此外，它的镜头效果如此独特，我实在忍不住要与您分享。

在这些独特的动画中，镜头像差对拍摄图像的影响被揭示出来，让您一窥其内部工作原理。

2.8 DIT

数字影像技师（DIT）是现代数字影像流程中的关键角色。他们连接着导演的视觉构想与构成数字数据库的比特、字节和色彩空间——这些数据正由摄像机源源不断地生成。从摄像机捕捉到最终调色，DIT在管理作为电影核心的数字数据库方面发挥着关键作用，而这项工作绝非易事，我们将在后文详述。

2.8.1 现场支持

DIT的工作始于摄影机本身。他们根据摄影指导（DP）的规格设置电影摄影机的曝光、色温和ISO参数。他们会就曝光过度或不足向DP提出建议，因为这些与胶片拍摄经验存在显著差异。

他们将在拍摄现场的帐篷里搭建“视频村”，放置电脑和显示器，用于监控拍摄的摄像机数据，确保画面信息可用，甚至进行初步调色以粗略呈现影片视觉效果。他们还会将当天的拍摄素材转换为“数字样片”，以便摄影指导在iPad上播放。

2.8.2 Data Management

With terabytes of camera raw data to manage, log, and protect, the DIT is a serious data wrangler, doing the backups and preparing viewable clips for a long list of downstream clients. The advertising department wants JPEGs of key moments in the film. The DP wants a QuickTime movie of today's takes for their iPad. The colorist wants a debayered Camera RAW for preliminary grading. And the list goes on.

On a more global scale, the DIT works with the downstream post-production team to ensure that the data is prepared and presented to them in file formats compatible with their workflows—all while checking to maintain quality and ensure that somebody doesn't do an unrecoverable compression of the data in a misbegotten desire to save disk space. Image quality and pipeline efficiency are the priorities of the modern DIT.

2.8.3 Color Grading

The DIT is not the colorist, but one of their on-set duties is to apply a preliminary color grade. Working with the director and colorist, they will develop "look LUTs" that are applied to the raw camera footage before review. This also entails a look management role designed to maintain a consistent look to the footage regardless of the time of day, weather conditions, or camera settings.

2.8.4 Post-Production Liaison

The DIT is the liaison between the on-set production team and the post-production team. They are the go-to guy for post-production when they want to know something about the shoot—which lens was used for Scene 29, what was the sensor crop in Scene 101, and what in God's name is all the noise in Scene 411? The director wanted to shoot Scene 411 using available light (for creative emphasis, of course), but the light levels were so low that the camera ISO had to be jacked up to 3200, which baked a bunch of noise into the shot. The DIT warned the DP about this, but the DP had their vision, you know. Now we have to fix it in post.

2.8.2 数据管理

面对需要管理、记录和保护的海量相机原始数据，DIT（数字影像技师）是至关重要的数据管理者，负责备份工作并为众多下游客户准备可查看的片段。广告部门需要影片关键帧的JPEG图像；摄影指导想要今日拍摄镜头的QuickTime影片以便在iPad上查看；调色师则需要去拜耳处理的Camera RAW文件进行初步调色——需求清单还在不断延伸。

在全球范围内，数字影像技师与后期制作团队紧密合作，确保数据以符合其工作流程的文件格式进行准备和交付——同时全程检查以维持质量，防止有人为节省磁盘空间而错误地对数据进行不可恢复的压缩。图像质量与流程效率是现代数字影像技师的首要任务。

2.8.3 色彩分级

DIT并非调色师，但其在片场的职责之一便是进行初步的色彩分级。他们与导演和调色师协作，共同开发“风格LUT”，并将其应用于原始摄影素材以供审阅。这还涉及一项视觉管理职责，旨在确保无论拍摄时间、天气条件或相机设置如何变化，画面风格都能保持统一。

2.8.4 后期制作联络

DIT是现场制作团队与后期制作团队之间的联络人。当后期团队想了解拍摄细节时——比如第29场用了什么镜头、第101场的传感器裁切比例是多少，以及第411场那些莫名其妙的噪点究竟怎么回事——他们就是对接人。导演原本想用自然光拍摄第411场（当然是为了艺术效果），但光照实在太暗，相机ISO不得不调到3200，导致画面噪点泛滥。DIT曾就此提醒过摄影指导，但摄影指导坚持自己的艺术构想，你懂的。现在这些全得靠后期来修补了。

2.9 Virtual Production

The biggest paradigm shift in the production of visual effects has been virtual production using LED walls to create synthetic sets. A quantum leap over rear-projection, it is sweeping through the production of VFX as the bulk of them can now be done in-camera. Buoyed by the success of *The Mandalorian*, virtual sets are now all the rage and on the rise, with over 100 of them in development around the world. They offer unprecedented versatility and realism in creating strange new worlds that the entire industry now sees as a major new component in the future of cinemaphotography. Virtual production takes a lot of tech and talent, so here we will see how virtual production works, its promises, and its problems. The problems are of special interest to those of us in post-production because that is where we get to fix them.

Figure 2.27 A virtual production set with LED walls

2.9.1 How it Works

The short form of the story is that a CG world is displayed on a huge LED wall that wraps around the on-set talent by up to 270 degrees, along with an overhead light panel (see Figure 2.27). The images are generated in real-time with powerful gaming engines that are linked to the camera's position. Wherever the camera points, the CG world is rendered from a matching point

2.9 虚拟制作

视觉特效制作领域最大的范式转变是使用LED墙创建合成场景的虚拟制作。作为对背投技术的巨大飞跃，它正席卷整个视觉特效制作领域，因为大部分特效现在都可以在摄像机内完成。得益于《曼达洛人》的成功，虚拟场景如今风靡一时且日益兴起，全球有超过100个虚拟场景正在开发中。它们在创造奇异新世界时提供了前所未有的灵活性和真实感，整个行业现在将其视为电影摄影未来的重要新组成部分。虚拟制作需要大量技术和人才，因此我们将探讨虚拟制作的运作方式、前景及其存在的问题。这些问题对我们后期制作人员尤其值得关注，因为那正是我们需要解决它们的地方。

图 2.27 采用 LED 墙的虚拟制作场景

2.9.1 工作原理

故事的简略版本是，一个CG世界显示在一个巨大的LED墙上，这个墙环绕着现场演员，可达270度，同时还有一个顶部的灯光面板（见图2.27）。图像是通过强大的游戏引擎实时生成的，这些引擎与摄像机的位置相连。无论摄像机指向哪里，CG世界都会从一个匹配的点进行渲染。

of view. The talent is filmed in front of the LED wall, which does a real-time match-move as the camera follows the talent.

As you can imagine, there is a tremendous amount of tech and color science behind this rig. The LED walls are driven by multiple synchronized Unreal Engines running on massively parallel gaming engines at 4K or 8K resolution, all designed to put out real-time CG. The on-set camera is motion tracked in real time and its position is updated to the gaming engines so their CG cameras can match the move. As the camera is moved around the set, it maintains perfect perspective and parallax at all times.

To keep all this running is the "brain bar," a collection of artists and technicians behind the virtual set to keep the disk drives, GPUs, and scene databases humming in synchrony and harmony. To change the scene from desert to jungle, they no longer pack up the entire production company and fly to Brazil. The data wrangler just loads the rainforest database. If there is a tree in the way of the director's favorite camera angle, the digital greensman just moves it over a few feet. When the director wants sunset lighting, the lighting supervisor just dials in presets for that lighting condition. It's a director's creative paradise.

2.9.2 The Promise

Virtual production makes three main promises: the ability for the director to see the finished shot immediately after shooting; the addition of interactive lighting; and improved performances from the talent. The director can view the "composited" shot immediately and not wait for it to be turned around by a VFX studio three weeks later. The huge wraparound LED lighting walls include an overhead panel, so the on-screen talent is awash in interactive lighting, which furthers the realism of the effects shots. And finally, the talent can deliver a far more compelling performance to a "real" T-Rex trying to eat their face in contrast to attempting to perform against a pair of ping-pong balls glued to a green stick.

Directors love the God-like creative control (who wouldn't!) where they can, on command, part the seas or smother the sun in broad daylight. Sets can be modified and entire mountain ranges added or removed at the whim of the director. Need a dark stormy sky? No problem. Want a blazing red sunrise? We've got that. How about a cold, frosty morning? Got you covered. Producers love the virtual production set too, because 50 or 60 stagehands,

视角。人才在LED墙前拍摄，当摄像机跟随人才时，LED墙会实时进行匹配移动。

正如你所想象，这套设备背后蕴含着大量的技术和色彩科学。LED墙由多个同步运行的虚幻引擎驱动，这些引擎运行在支持4K或8K分辨率的大规模并行游戏引擎上，全部设计用于实时输出CG图像。现场摄像机通过实时运动追踪，其位置数据会同步更新至游戏引擎，使虚拟CG摄像机能够匹配真实摄像机的运动轨迹。当摄像机在片场移动时，系统始终保持精准的透视关系和视差效果。

维持这一切运转的是“大脑吧”——虚拟场景背后的一群艺术家和技术人员，他们确保磁盘驱动器、GPU和场景数据库同步和谐地运行。要将场景从沙漠切换为丛林，他们不再需要打包整个制作公司飞往巴西。数据管理员只需加载雨林数据库。如果导演偏爱的拍摄角度被树木遮挡，数字园艺师只需将树移动几英尺。当导演想要日落光线效果时，灯光总监只需调出该光照条件的预设参数。这里是导演的创意天堂。

2.9.2 承诺

虚拟制作主要承诺实现三点：导演能够在拍摄后立即看到成品镜头；增加互动式灯光；以及提升演员的表演效果。导演可以即时查看“合成”镜头，无需等待三周后由视觉特效工作室返工。巨大的环绕式LED灯光墙包含顶置面板，使屏幕中的演员沐浴在互动灯光下，从而增强了特效镜头的真实感。最后，与对着粘在绿棒上的两颗乒乓球表演相比，演员能够面对一只试图吞噬他们脸庞的“真实”霸王龙，呈现出更具感染力的表演。

导演们热爱这种如神祇般的创作掌控力（谁不爱呢！），他们可以一声令下，让大海分开，或在光天化日下遮蔽太阳。布景可以随意修改，整条山脉也能依导演的意愿增添或移除。需要一片昏暗的暴风雨天空？没问题。想要一轮炽热的红色日出？我们也能实现。那寒冷霜冻的清晨呢？同样不在话下。制片人也钟情于虚拟制作场景，因为五六十名舞台工作人员，

greensmen, gaffers, and electricians are replaced by a dozen tech wizards—not to mention the whole "not flying the film crew to Brazil" thing.

2.9.3 The Problems

Like any powerful new technology, virtual production has its special list of issues, limitations, and problems. Fortunately, my fellow compositors have a vast array of tools, tricks, and techniques that can solve virtually any imaging problem—even virtual ones. Here are just some of the exciting challenges that virtual production can produce:

- **Moiré** - Because the LED wall being photographed is made up of discrete point light sources instead of the smooth continuous light source of a real mountain scene, at the right camera distance, the size of the LEDs becomes a match to the size of the camera's imaging sensor photosites. When this happens, moiré patterns erupt all over the frame, which are difficult to control and impossible to fix in post.
- **Metamers** - Because of the narrow-band spiky spectral emissions of the LED wall, various props and articles of clothing are prone to harrowing color shifts.
- **Practical floors** - The floor of the virtual set runs right up to the LED wall and when it is off-color, there can be a discontinuity between the floor and the wall that will have to be fixed in post. Imagine a desert scene where the practical sand on the floor of the set does not match the sand in the CG wall.
- **Additional lighting** - An overhead LED panel, if present, solves the top lighting, but lights still have to be set up to illuminate the front of the talent. This can be problematic because you don't want those front lights hitting the LED wall and washing them out.
- **Latency** - The game engines that render the CG environment in real-time can fall behind if the scene content is too heavy, resulting in a delay when the camera moves (latency). It is not just embarrassing when the talent stays frozen in place for several frames at the beginning of a whip-pan; it is also a very expensive class of problem to fix in post.
- **Proxy environments** - If the final database is not ready to go on shoot day or is too "heavy" for the game engines to render in real-time, the scene will

绿幕师、灯光师和电工被十几位技术奇才取代——更别提“不用把摄制组飞到巴西”这回事了。

2.9.3 问题

与任何强大的新技术一样，虚拟制作也有其特殊的一系列问题、限制和挑战。幸运的是，我的合成师同行们拥有大量工具、技巧和技术，几乎可以解决任何成像问题——即使是虚拟成像问题。以下是虚拟制作可能带来的一些激动人心的挑战：

- 摩尔纹 - 由于被拍摄的LED墙由离散的点光源组成，而非真实山景那样平滑连续的光源，在合适的相机距离下，LED的尺寸会与相机成像传感器感光元件的尺寸相匹配。当这种情况发生时，画面会爆发难以控制且后期无法修复的摩尔纹图案。

- Metamers - 由于LED墙的窄带尖峰光谱发射特性，各种道具和服装容易出现令人不适的色彩偏移。

- 实用地面——虚拟布景的地面一直延伸到LED墙，当颜色不匹配时，地面与墙面之间可能出现断层，必须在后期制作中修复。想象一个沙漠场景，布景地面上真实的沙子与CG墙中的沙子不匹配。

- 额外照明——如果已有顶置LED面板，可以解决顶部照明问题，但仍需设置灯光照亮主持人前方区域。这可能带来难题，因为要避免这些前侧灯光直射LED墙导致画面泛白。

- 延迟 - 如果场景内容过于复杂，实时渲染CG环境的游戏引擎可能会跟不上，导致摄像机移动时出现延迟（即滞后现象）。这不仅在快速摇摄开始时，演员连续数帧僵在原地会显得尴尬；而且这类问题在后期修复的成本也极其高昂。

- 代理环境 - 如果最终数据库在拍摄日尚未就绪，或对游戏引擎而言过于“沉重”无法实时渲染，场景将

be shot with proxies. This means the talent will have to be isolated (think roto) and then composited over the final CG when it becomes available.

- **Greenscreens** - When all else fails, they will actually put up a virtual greenscreen behind the talent and film the scene anyway. Of course, that shot will have to be greenscreen keyed and composited the old-fashioned way.

There are other issues and problems with virtual production, but I think we have made the point that even with this dazzling new tech, we are still going to need the folks in post-production to fix things. The moral of the story is, as usual: after reading the sales brochure for an exciting new tech destined to change the world, upon close inspection there will always be "gotchas."

2.10 What about 35 mm Film?

While the entire modern imaging pipeline has gone totally digital, some 27 movies were still shot on 35 mm film in 2020 because some directors simply love the look of film. Of course, after the film is developed, it is promptly digitized and projected digitally in the theater, it is an all-digital process. You could still be confronted with a project that was shot on film, so in this section, we will look at how the image files from a film capture differ from a digital camera capture.

2.10.1 Composition

Cinema and HDTV are normally shot with the composition that will be seen during display, although sometimes cinema is shot wider and then cropped down for display. 35 mm film is normally shot at full aperture and the film scanner scans the entire frame. In Figure 2.28, you can see the camera's rounded film gate corners in the raw film scan. It is then cropped for digital cinema or HDTV.

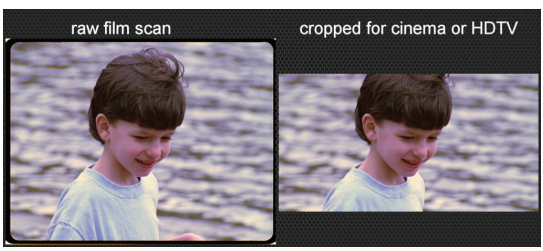


Figure 2.28 35mm full aperture film scan

使用代理拍摄。这意味着演员必须被隔离（想象成抠像），然后在最终CG可用时将其合成上去。

- 绿幕 - 当所有其他方法都失败时，他们实际上会在演员身后搭建一个虚拟绿幕，并无论如何拍摄场景。当然，那个镜头将不得不以传统方式进行绿幕抠像和合成。

虚拟制作还存在其他问题和挑战，但我认为我们已经阐明了一点：即便拥有这项令人眼花缭乱的新技术，我们仍然需要后期制作人员来修正问题。故事的寓意一如既往：在阅读了那些宣称将改变世界的炫酷新技术宣传册后，仔细审视之下总会发现其中存在“陷阱”。

2.10 关于35毫米胶片呢?

尽管整个现代成像流程已经完全数字化，但2020年仍有约27部电影采用35毫米胶片拍摄，因为有些导演就是偏爱胶片的质感。当然，胶片冲洗后会立即进行数字化处理，并在影院以数字方式放映——这本质上仍是全数字流程。但你仍可能遇到胶片拍摄的项目，因此本节将探讨胶片拍摄与数码相机拍摄的图像文件有何不同。

2.10.1 组成

电影和HDTV通常采用显示时可见的构图进行拍摄，尽管有时电影会拍摄得更宽，然后在显示时进行裁剪。35毫米胶片通常以全画幅拍摄，胶片扫描仪会扫描整个

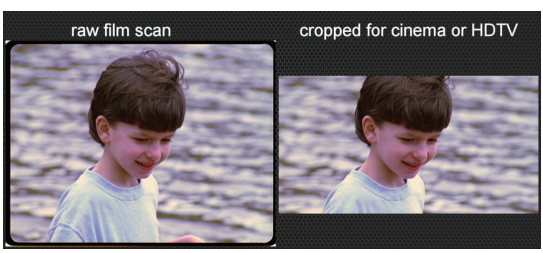


图 2.28 35毫米全画幅胶片扫描

在图2.28中，您可以在原始胶片扫描中看到相机圆形的片门角落。随后会为数字影院或高清电视进行裁剪。

2.10.2 Frame Rate

As you know, movies are filmed and projected at 24 fps. Chapter 5: Display Space discusses high frame rate movies of 48 and even 60 fps, but this chapter is about capture. Higher capture rates affect both motion blur and grain. A higher frame rate means a shorter shutter time, which in turn means less light and less motion blur. If the camera is "over-cranked" to, say, 100 fps for playback at 24 fps we get a slow-motion shot, so the reduced motion blur works fine. But a show filmed at 48 fps will have to be retimed to 24 fps for television by "skip printing," which means to drop out every other frame. The retained frames will have less motion blur than they otherwise should, so they run the risk of exhibiting motion strobing in some scenes.

The reduced shutter time also means a shorter exposure, of course, so a "faster" film stock will be needed. As we learned earlier, digital cameras are made "faster" by increasing the ISO number, which increases noise. Film is made faster by increasing the size of the silver halide crystals in the camera negative, which makes the film grainier. The bottom line is that films shot at high frame rates need more aggressive degraining, which can soften the image.

2.10.3 Color Gamut

Film is a subtractive color space, like print, so it has a misshapen color gamut compared to the tidy triangles of additive color spaces, which we will learn more about later. Figure 2.29 shows how truly wide the 35 mm color gamut is. While it does not cover 100% of the human visual capability, we will learn later that it is much wider than HDTV and even today's digital cinema.

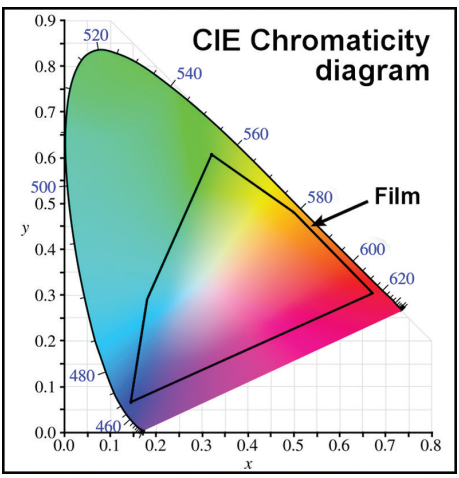


Figure 2.29 Color gamut of 35mm film

2.10.2 帧率

如你所知，电影以每秒24帧拍摄和放映。第五章《显示空间》讨论了48帧甚至60帧的高帧率电影，但本章关注的是拍摄环节。更高的拍摄帧率会影响动态模糊和画面颗粒感。帧率越高意味着快门时间越短，进而导致进光量减少、动态模糊减弱。若将摄影机“升格拍摄”至100帧并以24帧回放，就会得到慢镜头效果，此时减少的动态模糊并无问题。但一部以48帧拍摄的影片在电视播放时需通过“跳帧打印”技术调整为24帧，即每隔一帧丢弃一帧。保留下来的帧会比正常情况减少动态模糊，因此在某些场景中可能出现动态频闪的风险。

快门时间的缩短当然也意味着曝光时间更短，因此需要“感光度更高”的胶片。正如我们之前了解到的，数码相机通过提高ISO值来提升感光度，但这会增加噪点。而胶片则是通过增大底片上卤化银晶体的尺寸来提升感光度，这会使胶片颗粒更明显。关键在于，以高帧率拍摄的影片需要进行更彻底的降颗粒处理，但这可能会导致图像变软。

2.10.3 色域

胶片是一种减色色彩空间，类似于印刷，因此与加色色彩空间的规整三角形相比，其色域形状不规则，这一点我们稍后会进一步探讨。图2.29展示了35毫米胶片色域的实际宽广程度。虽然它未能覆盖100%的人类视觉能力，但后续我们会了解到，其色域范围远超高清电视，甚至比如今的数字影院更加广阔。

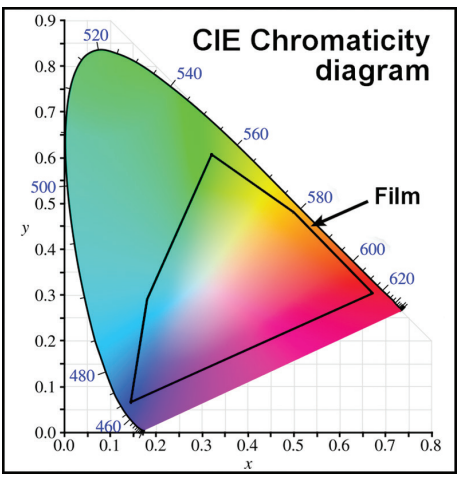


图 2.29 35毫米胶片的色域

2.10.4 Grain

Two interesting things about film grain: first, Kodak film research revealed that grain actually increases the perceived sharpness of the film without the "edgy" look of video; second, the audience loves it. Apparently, if the movie is too clean, it starts to feel like video. This is a learned behavior based on years of watching videos on TV and films in the theater, but it is nonetheless absolutely real to the audience.

A digital capture will have noise while a film capture will have grain—and they are not the same. To be fair, grain is also noise—the noise of film. 35 mm grain is usually larger and has a different structure than digital noise. I say "usually larger" because a beautifully exposed fine-grained film stock could have less noise than a poorly exposed digital capture where the director likes to crank up the ISO. The key point here is about de-noise tools. Quality de-noise tools will offer a choice between film grain and digital noise that allows the software to optimize its algorithm based on the actual type of noise to be removed. Properly optimized, the de-noise tools will remove most of the noise without too much loss of detail.

2.10.5 A Softer Look

A scene captured on film will look softer than if it were captured with a digital camera; not softer as in blurry, but softer as in not so hard-edged. Digital cameras—whether they are cinema cameras like an ARRI Alexa or video cameras like a Sony HXR—add sharpening to their output images, as mentioned above in the debayering discussion. Sharpening the camera output has been a standard practice for video from the beginning, so it has become associated with video. In fact, some audience members' candid reaction to high frame rates and high-resolution digital films, with their ultra-sharp appearance, is to say that it looks like video—a soul-crushing experience for the director.

The explanation for all of this is that an edge captured with film is made up of exposed film grains of irregular size and orientation, so an edge becomes a bit "vague," making it appear softer at speed. A video edge is made up of regularly spaced pixels in the imaging sensor which are also smaller than the film grain, producing a sharper edge natively, to which the camera then adds sharpening as part of color reconstruction.

2.10.4 谷物

关于胶片颗粒有两个有趣的现象：首先，柯达胶片研究表明，颗粒感实际上能提升胶片观感的锐度，同时避免视频那种“生硬”的视觉效果；其次，观众对此十分喜爱。显然，如果电影画面过于干净，反而会让人感觉像是在看视频。这种认知源于多年来在电视上看视频与在影院观影形成的对比经验，但对观众而言，这种感受无疑是真实存在的。

数字拍摄会产生噪点，而胶片拍摄则会产生颗粒——这两者并不相同。公平地说，颗粒也是一种噪点——即胶片的噪点。35毫米胶片的颗粒通常比数字噪点更大，且结构不同。我说“通常更大”是因为曝光良好、颗粒细腻的胶片可能比导演喜欢调高ISO、曝光不佳的数字拍摄产生的噪点更少。这里的关键在于降噪工具。优质的降噪工具会提供胶片颗粒与数字噪点的选项，让软件能根据实际需要去除的噪点类型优化算法。经过适当优化后，降噪工具能在去除大部分噪点的同时，不过多损失细节。

2.10.5 更柔和的视觉效果

胶片拍摄的场景看起来会比数码相机拍摄的更柔和；不是模糊的那种柔和，而是边缘不那么锐利的那种柔和。正如上文去马赛克讨论中提到的，数码相机——无论是像ARRI Alexa这样的电影摄影机，还是像索尼HXR这样的摄像机——都会对输出图像进行锐化处理。从一开始，锐化相机输出就是视频制作的标准做法，因此这种做法逐渐与视频联系在一起。事实上，一些观众对高帧率和高分辨率数字电影的直接反应是，它们看起来太锐利了，像视频——这对导演来说简直是毁灭性的打击。

这一切的解释是，胶片捕捉的边缘由不规则大小和方向的曝光胶片颗粒组成，因此边缘会变得有些“模糊”，使其在动态中显得更柔和。视频边缘则由成像传感器中规则排列的像素构成，这些像素也比胶片颗粒更小，天生就能产生更锐利的边缘，随后相机会在色彩重建过程中进一步添加锐化处理。

2.10.6 No Rolling Shutter

The digital camera's rolling shutter artifact, described earlier, is caused by the imaging sensor being scanned line by line from top to bottom while something moves during the scan. 35 mm film cameras do not exhibit rolling shutter because their shutters are mechanical rotating disks. The disk sits in front of the film gate with a big notch cut out of it—typically 180 degrees—and rotates, alternating between blocking and exposing the film frame in the gate. If objects move in front of this mechanical shutter, it just adds motion blur to the image because the film frame is being held steady in the gate. As a result, it does not do the goofy skewing that we see with a rolling shutter.

2.10.7 Color Temperature

We saw the importance of the color temperature of lights and how this affects the captured image. This is why digital cameras have a color balance adjustment which specifically deals with the color temperature of the light source on location. 35 mm film, however, is not adjustable. The way the film handles this is to make two versions of every film stock: one balanced for daylight at 5600K and the other balanced for indoor tungsten lights at 3200K. You have to choose which stock to use depending on whether you are shooting outdoors (daylight) or indoors (tungsten). Get it wrong by using the daylight stock indoors and the film will turn out too red. Use the tungsten stock outdoors and it will turn out too blue. When film folk need to correct for the color temperature of the light sources, they have to put colored filters in front of the lens. Barbaric.

Of the hundreds of movies produced in 2020, only 27 were shot on film. It is sad to say that even this last limited use of film will soon dwindle to nothing. There are several reasons for this. First, the infrastructure for film is closing down. There are fewer and fewer labs where it can be developed. There are fewer and fewer camera rentals where you can rent a 35 mm film camera. Film-based cinematography is being taught in fewer and fewer schools, so soon there will be no new talent to replace the old guard. Digital cameras offer both a cleaner production pipeline with direct-to-disk workflows and no need for labs—plus the instant gratification of seeing your shot immediately without waiting to watch dailies the following day. Finally, aside from the love for the look of film, digital cinema cameras are simply more flexible and

2.10.6 无卷帘快门

数码相机的卷帘快门伪影，如前所述，是由于成像传感器在扫描过程中自上而下逐行扫描时，物体发生移动所导致的。35毫米胶片相机不会出现卷帘快门现象，因为它们的快门是机械旋转盘。该圆盘位于胶片片门前，通常切出一个大缺口（约180度），通过旋转交替遮挡和曝光片门中的胶片帧。若物体在此机械快门前移动，只会给图像增添运动模糊，因为胶片帧在片门中保持稳定。因此，它不会产生我们在卷帘快门上看到的扭曲变形效果。

2.10.7 色温

我们认识到光源色温的重要性及其对成像的影响。因此数码相机配备了白平衡调节功能，专门用于应对现场光源的色温变化。然而35毫米胶片无法进行此类调整，胶片的解决方案是为每种胶片生产两种版本：一种针对5600K日光环境校色，另一种则针对3200K室内钨丝灯光校色。拍摄者必须根据户外（日光）或室内（钨丝灯）场景选择对应型号——若在室内误用日光型胶片，成像会严重偏红；在户外使用灯光型胶片则会严重偏蓝。当胶片使用者需要校正光源色温时，只能在镜头前加装彩色滤光片。这简直是原始手段。

在2020年生产的数百部电影中，仅有27部采用胶片拍摄。令人遗憾的是，即便是胶片这最后一点有限的应用，也即将消失殆尽。这背后有几个原因。首先，胶片相关的基础设施正在关闭。能够冲洗胶片的实验室越来越少；可供租赁35毫米胶片摄影机的器材店也日益减少。越来越少学校教授基于胶片的电影摄影技术，因此很快将没有新人才能接替老一辈的胶片摄影师。数码摄影机则提供了更简洁的制作流程——直接存储至硬盘的工作流无需依赖冲印厂，还能即时查看拍摄画面，无需等到次日观看样片。最后，除了对胶片质感的偏爱之外，数字电影摄影机确实更具灵活性，且

powerful than mechanical 35 mm cameras. And with a little love, the digital capture can be given a film look in post.

2.11 Conclusion

Camera capture is the first link in the digital imaging chain and is where the most data loss occurs, simply because World Space is vast and our little cameras are quite limited. Even with their limited dynamic range and restricted color gamuts, digital cameras can still make fantastically beautiful images for the human eye—for the simple reason that the human eye is similarly limited and digital cameras are exquisitely designed to match our color sensibilities. The key is to make certain that the camera does not miss image information that the eye will notice.

We saw how the Bayer array captures a scene, with each pixel only seeing one color, but after some software manipulation and color secret sauce each pixel ends up with three colors—red, green, and blue—and a darned nice picture from the manufacturer's admittedly made-up colors. I guess this demonstrates what a truly high-art camera design is.

As we saw, however, the effective resolution is really only about 70% of the advertised resolution. The camera artifacts of noise, rolling shutter, pixel vignetting, and anamorphic softening further degrade the captured image, underscoring the need to capture the highest possible quality in-camera so that you still have some picture left after dealing with them.

After the camera has captured a scene, the data must be exported to a hard drive for the downstream processing that we like to call "post-production." In the next chapter, we will navigate the treacherous terrain of Storage Space, where there is a vast array of file formats, codecs, compression schemes, and color spaces to choose from. This provides a terrific opportunity for the underinformed to do permanent harm to the quality of the project. This chapter aims to provide you with useful information so that you can make informed decisions that will protect the quality of the work for all time—which means setting things up to be ready for the next generation of display technologies, or "future-proofing."

比机械35毫米相机更强大。并且稍加用心，数字拍摄在后期处理中也能获得胶片般的视觉效果。

2.11 结论

相机捕捉是数字成像链中的第一环，也是数据损失最多的地方，这仅仅是因为世界空间浩瀚无垠，而我们的小相机却相当有限。尽管动态范围和色域受限，数码相机仍能为人眼创造出极其精美的图像——原因很简单：人眼同样存在局限，而数码相机经过精心设计，恰好契合了我们的色彩感知。关键在于确保相机不会遗漏人眼能够察觉的图像信息。

我们看到了拜耳阵列如何捕捉场景，每个像素最初只能感应一种颜色，但经过软件处理和色彩“秘制配方”的调整后，每个像素最终都拥有了红、绿、蓝三色信息——制造商凭借这些人为构建的色彩，竟能呈现出相当出色的画面。我想，这正体现了真正高水准相机设计的精髓所在。

然而，正如我们所见，实际有效分辨率仅为标称分辨率的70%左右。相机本身的噪点、卷帘快门、像素渐晕和变形镜头柔化等伪影会进一步劣化捕获的图像，这凸显了在机内捕捉尽可能高质量画面的必要性——以便在处理这些问题后，仍能保留部分可用的画面。

摄像机捕捉场景后，数据必须导出至硬盘，以便进行我们称之为“后期制作”的下游处理。在下一章中，我们将穿越存储空间这片险峻之地——这里充斥着海量的文件格式、编解码器、压缩方案与色彩空间可供选择。对信息掌握不足者而言，这无疑是永久损害项目质量的绝佳陷阱。本章旨在提供实用信息，助您做出明智决策，从而永久守护作品质量：这意味着建立适应下一代显示技术的准备体系，亦即实现“未来兼容性”。

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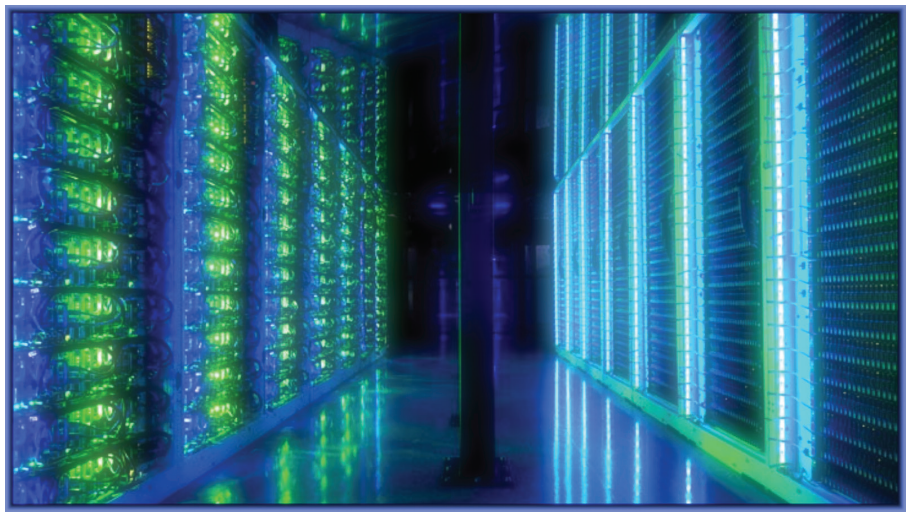

3 Storage Space

In the previous chapter, we saw how modern digital cameras produce a torrent of image data that needs to be stored for subsequent editing, visual effects, color correction, and display. How that data is exported from the camera and the file format chosen to store it in have a profound limiting effect on the retained quality of those captured images. The thrust of this chapter is to give the modern digital imaging artist an understanding of the consequences of the file format choices for both export and storage to avoid inadvertently degrading the project.

We will start by reaffirming the basics of image structure: pixels, bit depth, channels, deep data, and even metadata. This sets things up for a comprehensive discussion of compression strategies which involve reduced bit depth, color subsampling, and the awesome Discrete Cosine Transform (DCT)—the Holy Grail of all data compression. Since most modern cinema and photography cameras offer a log output option, we will take a close look at log images: what they are, when to use them, how they work, and what to watch out for.

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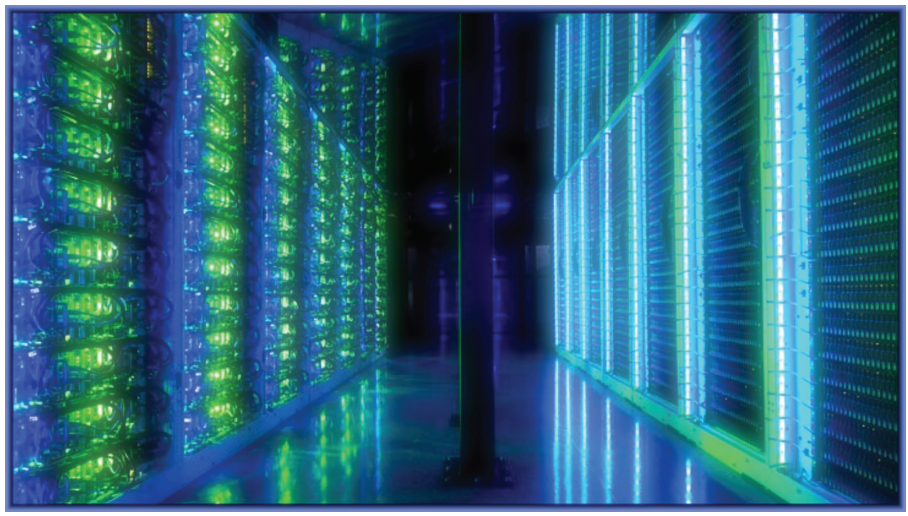

3 存储空间

在上一章中，我们了解到现代数码相机如何产生需要存储以供后续编辑、视觉效果、色彩校正和显示的大量图像数据。这些数据从相机导出的方式以及选择存储它们的文件格式，对捕获图像的保留质量有着深远的限制性影响。本章的重点是让现代数字影像艺术家理解导出和存储时文件格式选择的后果，以避免无意中降低项目质量。

我们将从重申图像结构的基础知识开始：像素、位深度、通道、深层数据，甚至元数据。这为全面讨论压缩策略奠定了基础，这些策略包括降低位深度、色彩子采样，以及强大的离散余弦变换（DCT）——数据压缩领域的圣杯。由于大多数现代电影和摄影相机都提供对数输出选项，我们将仔细研究对数图像：它们是什么、何时使用、如何工作，以及需要注意的事项。

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Both cinematography and stills photography have a wide range of file format options for storing their images. But as we will learn, it's not the file format that kills you; it's the codec. A deep dive into compression codecs will help you understand how much you can compress your work without excessive harm. This is followed by a grand tour of the most popular image storage file formats for both photography and cinematography and which of those codecs they use. The chapter ends with a frank discussion about the file formats that you should never use if you want to maintain quality work. Let's jump into Storage Space.

3.1 Image Structure

One of the essential functions of an image file format is to preserve the image structure, which is based on a precisely defined hierarchy of pixels, bit depth, and channels. Here we will explore image structure, starting with the mundane pixel—only to discover it can have awesome power when it goes deep. We will look at bit depth and the force multiplier of racking them up in longer and longer blocks. The pixels and bits are organized into channels, which is where the real magic begins for CG. We will then take a deep dive into deep data (the CG kind, not the AI kind) to understand its power and cost, before finishing up with a whirlwind tour of metadata.

3.1.1 Pixels

At this point in your career, you probably know what a pixel is, or you wouldn't be reading this book. You are undoubtedly keenly aware that a pixel is the smallest picture element (a.k.a "pixel") that a computer can address in an image or display and that the pixels are arranged in a two-dimensional array and accessed by their x-y coordinates. You have also probably encountered pixels that are three or four layers deep, with RGB or RGBA components. You may even have worked with pixels that are only one layer deep, such as a matte. Easy stuff.

But perhaps you have never encountered pixels that contained not color, but data about the pixels, such as a motion vector or surface normal. They are in the same two-dimensional array of identical size to match their colored

无论是电影摄影还是静态摄影，都有多种文件格式可供存储图像。但我们会发现，真正影响成败的并非文件格式本身，而是编解码器。深入探讨压缩编解码器将帮助您理解，在不造成过度损害的前提下，工作文件可以压缩到什么程度。接下来，我们将全面梳理摄影与电影摄影中最常用的图像存储格式，以及它们各自采用的编解码器。本章最后将坦诚探讨那些为保持作品质量应绝对禁用的文件格式。现在，让我们进入存储空间的探索。

3.1 图像结构

图像文件格式的基本功能之一是保留图像结构，这种结构建立在精确定义的像素层次、位深度和通道基础上。我们将从最基础的像素开始探索图像结构——最终发现当它深入发展时能迸发惊人力量。我们将考察位深度，以及通过将比特堆叠成越来越长的数据块所产生的倍增效应。像素与比特被组织成通道，这正是计算机图形学真正魔力的起点。随后我们将深入探究深度数据（计算机图形学意义上的深度，而非人工智能范畴），理解其威力与代价，最后以对元数据的快速巡览作为收尾。

3.1.1 像素

在职业生涯的这个阶段，你可能已经知道像素是什么，否则也不会读这本书。你无疑清楚地意识到，像素是计算机在图像或显示中可以寻址的最小图像元素（亦称“像素”），它们以二维阵列排列，并通过其x-y坐标进行访问。你可能也遇到过具有三到四层深度的像素，比如RGB或RGBA分量。甚至可能处理过仅有一层深度的像素，比如遮罩。这些都是基础知识。

但也许你从未遇到过这样的像素：它们包含的不是颜色，而是关于像素的数据，比如运动矢量或表面法线。它们位于同样大小的二维数组中，以匹配其彩色

brethren and are used to operate on and modify the color pixels. If you would like a visual, check out Figure 3.1, then return here. We will hear more about these "data pixels" in the very next section, where we will learn why our CG colleagues call them AOVs.

In a CG render, a given pixel may be on the geometry or off the geometry. If it's on the geometry, it gets the color of the geometry; and if it's off, it gets zero black. But what about a pixel that lands right on the edge of the geometry—half on, half off? In that case, it gets 50% of the color from the geometry. If 75% of it covers the geometry, it gets 0.75 of its color. If this were not done, the edge of the geometry would be made up of 100% on and 100% off pixels, resulting in a nasty, jagged edge that we would call "aliased." The process of assigning the pixel color value depending on what fraction of it is covered by the geometry is called "anti-aliasing," which eliminates the jaggies.

In summary, pixels are the basic units of digital image representation, playing a crucial role in how images and videos are captured, displayed, and seen in the digital domain.

3.1.2 Bit Depth

We first met bit depth in Chapter 2 in the context of camera capture, where cameras are limited to 10 to 14 bits or so. But when you are talking about image structure, the bits can go wild and we can have 32, 64, or even 128-bit images for some scientific applications. In the real world of VFX, we routinely store images at 16 bits per channel and then "promote" them to 32 bits when loading into memory for processing. Whether you are working with 8 bits or 64, the same binary math applies. It just gets bigger.

Digital electronics only have two states: on and off. Binary math represents that with a number—the bit (**b**inary dig**it**)—that can be either a one or zero. Incredibly, stacked together into larger and larger bundles, these humble bits can be made to represent any number from minus infinity to plus infinity; letters of the alphabet to make text; pixels to make images; data for audio and now artificial intelligence (AI). We'll see where that goes. But here we are focused on images and how bits are used to encode them.

The first bundled name for bits was the byte, which is a bundle of 8 bits and is the smallest addressable unit of memory. The original thought was

兄弟们，它们被用来操作和修改颜色像素。如果你想有个直观的了解，可以看看图3.1，然后再回到这里。我们将在下一节中更多地了解这些“数据像素”，届时我们将明白为什么我们的CG同事称它们为AOVs。

在CG渲染中，一个给定的像素可能位于几何体上，也可能位于几何体外。如果它在几何体上，就获取几何体的颜色；如果在几何体外，则显示为零值黑色。但如果一个像素恰好落在几何体边缘——一半在内、一半在外呢？这种情况下，它会获取几何体50%的颜色。如果像素有75%的面积覆盖几何体，则获取其75%的颜色。若不进行这样的处理，几何体边缘将完全由100%覆盖和0%覆盖的像素构成，从而产生我们称之为“锯齿”的粗糙锯齿状边缘。根据像素被几何体覆盖的比例来分配像素颜色值的过程，被称为“抗锯齿”，它能有效消除锯齿现象。

总而言之，像素是数字图像表示的基本单位，在数字领域中图像的捕捉、显示与观看方式中扮演着关键角色。

3.1.2 位深度

我们最初在第2章中讨论相机捕获时接触到了位深，当时相机的位深被限制在10到14位左右。但当谈及图像结构时，位深可以大幅提升，在某些科学应用中甚至可以使用32位、64位或128位的图像。在视觉特效的实际应用中，我们通常以每通道16位存储图像，然后在加载到内存进行处理时将其“提升”至32位。无论你处理的是8位还是64位图像，二进制运算的原理都是相同的，只是规模变大了。

数字电子学只有两种状态：开和关。二进制数学用一个数字——比特（二进制位）——来表示这种状态，它可以是1或0。令人惊叹的是，当这些微小的比特被层层堆叠成越来越大的组合时，它们便能表示从负无穷到正无穷的任何数字；构成文本的字母表中的字符；生成图像的像素；音频数据乃至如今的人工智能（AI）。我们将见证这一切的发展方向。但在此，我们专注于图像以及比特如何被用于编码图像。

比特的第一个捆绑名称是字节，它是一束8位比特，是内存中最小的可寻址单元。最初的想法是

to use the word "bite," but due to concern that typos would confuse "bits" and "bites," the unique spelling "byte" was established. With 8 bits, each capable of being in two different states (zero or one), a byte makes for a possible 256 unique numbers (2^8), ranging from 0 to 255. In the very early days of computers, the byte was the standard unit of data. After a few weeks of this, it was realized that larger numbers were needed, so they created a new digital bundle—called a "word"—that was two bytes long. Now, with 16 bits of binary numbers, the word made 65,536 possible numbers, ranging from 0 to 65,535. But soon, even this was not large enough.

While bytes are always 8 bits, a word can be 2 or more bytes long. The word is the fundamental unit of data for a computer and is used to represent not only data but memory addresses. The memory thing is very important. With 32-bit words, you can only address 4.3 billion memory locations. This was fine in 1970 when computers came with a few hundred megabytes of memory, but memory grew to many gigabytes (billions of bytes) which was more than 32 bits could handle, so in the 1990s, the world switched to 64-bit words. Indeed, today, computers are characterized by whether they are 32 or 64-bit systems. With a 64-bit (2^{64}) system, they can address 16.8 million terabytes of memory. That seems to be fine for now.

FUN FACT: When your esteemed author first started working with computers using the first commercial microprocessor, the Intel 4004, it had a word size of 4 bits. To differentiate it from the 8-bit byte, it was referred to as a "nibble." True story.

We frequently need to discuss bit depth with our colleagues, but there is a nomenclature problem here. The bit-depth discussion can become ambiguous when bit depth should never be ambiguous. Let's take a look at what someone might say vs. what they mean:

- **8-bit image** - This might refer to a one-channel grayscale 8-bit image or perhaps a three-channel RGB image with 8 bits per channel (more properly, a 24-bit image).
- **16-bit image** - This invariably refers to a three-channel RGB image with 16 bits per channel, or more properly, a 48-bit image. It could also refer

原本想用“bite”这个词，但由于担心拼写错误会混淆“bits”和“bites”，便确立了独特的拼写“byte”。一个字节包含8个比特，每个比特可处于两种不同状态（0或1），因此一个字节能表示256个唯一的数字（2^8），范围从0到255。在计算机的早期阶段，字节是标准的数据单位。几周后，人们意识到需要更大的数字，于是创建了一种新的数字组合——称为“字”——长度为两个字节。现在，有了16位二进制数，一个字可以表示65,536个可能的数字，范围从0到65,535。但很快，就连这个范围也不够大了。

虽然字节始终是8位，但一个字可以长达2个或更多字节。字是计算机数据的基本单位，不仅用于表示数据，还用于表示内存地址。内存这一点非常重要。使用32位字时，只能寻址43亿个内存位置。这在1970年计算机仅配备几百兆字节内存时没有问题，但随着内存增长到数十亿字节（数十亿字节），32位已无法处理，因此在1990年代，世界转向了64位字。事实上，如今计算机的特征在于它们是32位还是64位系统。使用64位（2{v*}）系统，可以寻址1680万太字节的内存。目前看来这已经足够了。

> 趣闻：当尊敬的作者首次使用第一代商用微处理器Intel 4004接触计算机时，其字长为4位。为区别于8位字节，它被称为“半字节”。确有其事。

我们经常需要与同事讨论位深，但这里存在一个术语问题。当位深本应毫无歧义时，关于位深的讨论却可能变得模糊不清。让我们看看人们可能说的话与他们实际想表达的意思之间的差异：

- 8位图像 - 这可能指单通道的8位灰度图像，也可能指每个通道8位的三通道RGB图像（更准确地说，是24位图像）。

- 16位图像 - 这通常指每个通道16位的三通道RGB图像，更准确地说，是48位图像。也可能指

to an extremely high dynamic range three or four-channel RGB/RGBA image with 16-bit half-float data per color channel. Floating point is 32 bits, so the 16-bit version is referred to as “half-float.”

- **24-bit image** - This invariably refers to a three-channel RGB 8-bit image.
- **32-bit image** - This invariably refers to a four-channel RGBA 8-bit image with an alpha channel. It could also refer to an extremely high dynamic range three or four-channel RGB/RGBA image with 32-bit float data per color channel.

These ambiguities could be resolved if we always referred to the bit depth of the entire image. This would mean calling a three-channel 8-bit image “24 bits” rather than sometimes referring to its per-channel bit depth (8 bits) and other times referring to its entire image bit depth (24 bits).

3.1.3 Channels

Images and their storage files are organized into channels. A channel is a two-dimensional array of pixels (color or data), and while most images are comprised of three channels—red, green, and blue—as we will see shortly, that rule is not ironclad. A roto can be a one-channel image while CG renders will normally have at least four channels—RGBA, with the “A” being the alpha channel. But the king of channels is the mighty EXR file.

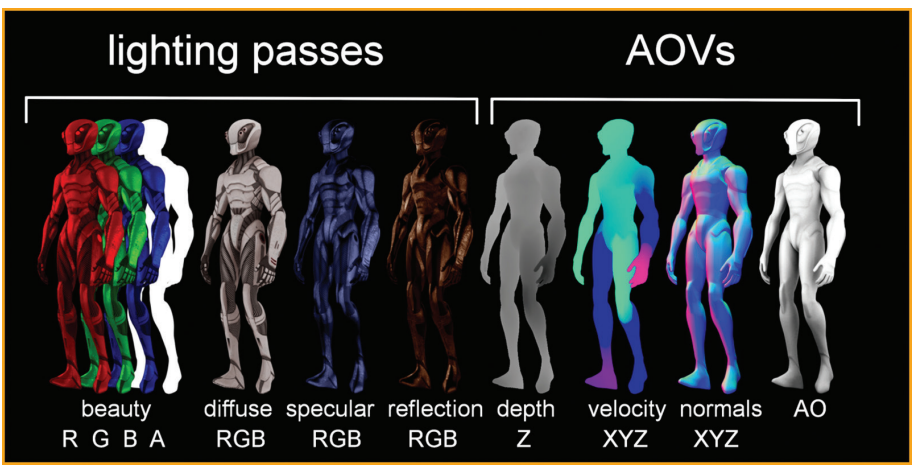


Figure 3.1 Layers and Channels

转换为具有极高动态范围的三或四通道RGB/RGBA图像，每个颜色通道采用16位半浮点数据。浮点数为32位，因此16位版本被称为“半浮点”。

- 24-b 它图像 - 这通常指的是三通道RGB 8 位图像。
- 32位图像 - 这通常指带有Alpha通道的四通道RGBA 8位图像。也可能指每个颜色通道采用32位浮点数据的极高动态范围三通道或四通道RGB/RGBA图像。

如果我们总是提及整个图像的位深度，这些歧义就可以得到解决。这意味着将一个三通道8位图像称为“24位”，而不是有时指其每个通道的位深度（8位），有时又指整个图像的位深度（24位）。

3.1.3 通道

图像及其存储文件被组织成通道。通道是一个二维像素阵列（颜色或数据），虽然大多数图像由三个通道——红、绿、蓝——组成，但正如我们稍后将看到的，这一规则并非铁律。遮罩绘制可以是一个单通道图像，而CG渲染通常至少包含四个通道——RGBA，其中“A”代表阿尔法通道。但通道之王是强大的EXR文件。

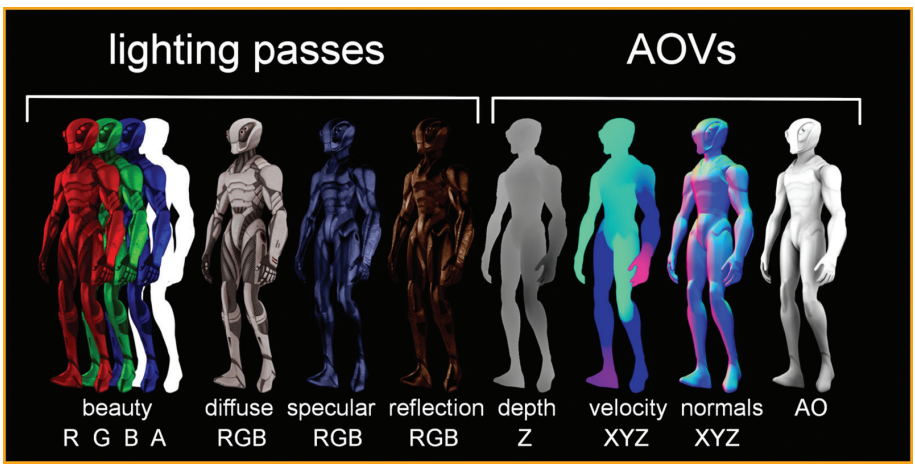


图 3.1 图层与通道

We delve into EXR files in Section 3.7.1 EXR, but here we are all about the channel structure. The EXR file format was developed specifically for visual effects. It has unique support for 35 mm film scans (yes, a few directors still shoot on film), but the heavy lifting is reserved for CG renders with their multiple layers and unlimited channels.

Figure 3.1 illustrates a typical CG render with layers and channels. Layers are a group of channels, such as the beauty layer comprised of four channels—RGBA. The diffuse, specular, and reflection layers are all three-channel RGB lighting passes. The alpha channel (A) is represented only once, typically as the fourth channel of the beauty pass. Referring again to Figure 3.1, the right group of images is AOVs, meaning "Arbitrary Output Variable." These are not lighting passes but instead are data about the lighting passes. A modern CG render may have 20-30 of these layers averaging three channels each, resulting in perhaps 90 channels—all in one frame! As a consequence, EXR files can be quite large.

The **depth pass** is a one-channel image named **Z** (depth.z) that represents the per-pixel distance from the camera to the surface of the geometry. The **velocity pass** contains three-channel vectors (XYZ) indicating per-pixel speed and direction for adding motion blur in post. The **normals pass** carries per-pixel three-channel vectors for the surface normals that moderate lighting. The **AO** (Ambient Occlusion) is a one-channel layer that contains shading information. As you can see, layers can be made up of one, three, four, or even more channels and those channels can contain many types of data, not just RGB information.

3.1.4 Deep Data

No, we are not talking about deep fakes or any other application of AI. Deep data is the name given to the pixels of deep images that are used for deep compositing. These are all familiar terms for VFX folks like CG artists and compositors, but Muggles (non-VFX folk) will find the concept quite unfamiliar. Deep compositing was designed to solve immensely complex CG scenes, like the example in Figure 3.2, which might consist of hundreds of billions of polygons and take many hours *per frame* to render. The problem is the holdout mattes for a shot with such great layering complexity.

我们在3.7.1节EXR中深入探讨了EXR文件，但这里我们主要关注其通道结构。EXR文件格式是专为视觉效果开发的。它对35毫米胶片扫描（是的，仍有少数导演使用胶片拍摄）具有独特的支持，但其核心优势在于处理CG渲染的多层结构和无限通道。

图3.1展示了一个典型的包含图层与通道的CG渲染结果。图层是一组通道的集合，例如由四个通道——RGBA——构成的美颜图层。漫反射、高光和反射图层均为三通道的RGB光照通道。阿尔法通道（A）通常仅出现一次，一般作为美颜通道的第四个通道。再次参考图3.1，右侧的图像组为AOVs，即“任意输出变量”。这些并非光照通道，而是关于光照通道的数据。现代CG渲染可能包含20-30个此类图层，每个图层平均有三个通道，这意味着单帧画面可能产生约90个通道！因此，EXR文件可能会非常庞大。

深度通道是一幅名为Z（depth.z）的单通道图像，表示从相机到几何体表面每个像素的距离。速度通道包含三通道矢量（XYZ），指示每个像素的速度和方向，用于后期添加运动模糊。法线通道携带每个像素的三通道矢量，表示表面法线，用于调节光照。AO（环境光遮蔽）是一个单通道层，包含着色信息。如您所见，图层可以由一个、三个、四个甚至更多通道组成，这些通道可以包含多种类型的数据，而不仅仅是RGB信息。

3.1.4 深度数据

不，我们讨论的并非深度伪造或任何其他AI应用。深度数据是指用于深度合成的深度图像像素。这些术语对于CG艺术家和合成师等视觉特效从业者来说耳熟能详，但“麻瓜”（非视觉特效从业者）会觉得这个概念相当陌生。深度合成技术旨在解决极其复杂的CG场景问题，例如图3.2中的示例场景可能包含数千亿个多边形，每帧渲染需耗时数小时。其核心难题在于处理具有超高分层复杂度的镜头遮罩。

Figure 3.2 A deep compositing image

To place the CG character (the T-Rex here) in the scene, it will invariably have to be in front of some objects and behind others. The objects it is behind (the foreground leaf in this case) require a holdout matte to be rendered for the composite. The holdout matte is a one-channel image that masks off the pixels that are occluded (hidden) by being behind another object in the scene. If the T-Rex is moved, the holdout mattes will have to be re-rendered. If the leaf is moved, the holdout mattes will have to be re-rendered. And, being one channel, the holdout matte is a binary mask: either the T-Rex is in front of or behind the leaf. But what if it is both, as in some motion blur on the leaf over the foot? The motion-blurred edge is partially transparent with smeared pixels. A binary mask is no longer adequate.

Figure 3.3 Deep pixel data

Enter deep data to solve both of these problems—namely, the changing holdout mattes and the blending of a motion-blurred element to the T-Rex. Deep data has no mattes. Each pixel is rendered with multiple samples of color, transparency, and depth. Whereas normal pixels have a triplet of RGB

图3.2 深度合成图像

要将CG角色（此处为霸王龙）置于场景中，它必然需要位于某些物体之前、另一些物体之后。对于被它遮挡的物体（本例中的前景树叶），需要渲染遮罩通道用于合成。遮罩通道是一种单通道图像，能屏蔽因位于场景中其他物体后方而被遮挡（隐藏）的像素。若移动霸王龙，遮罩通道需重新渲染；若移动树叶，遮罩通道也需重新渲染。由于是单通道图像，遮罩通道本质为二元遮罩：霸王龙要么在树叶前，要么在树叶后。但如果出现两者并存的情况呢？例如树叶在脚部产生运动模糊时，模糊边缘会形成半透明状态的拖影像素。此时二元遮罩便不再适用。

图3.3 深度像素数据

输入深度数据来解决这两个问题——即不断变化的遮罩和将运动模糊元素与霸王龙融合。深度数据没有遮罩。每个像素都通过颜色、透明度和深度的多重采样进行渲染。而普通像素只有RGB三元组

data plus the alpha channel, deep data has a series of multiple five-channel "quintets" of data, with each quintet containing RGBAZ—as in Figure 3.3, where one pixel has been expanded out to show its deep data. RGB is the color, A is the transparency, and Z is its depth in the scene. Each pixel actually holds a volume.

So, unlike conventional CG objects with polygonal surfaces of zero thickness, a deep pixel has a thickness indicated by the multiple Z channels in Figure 3.3. This particular example shows a pixel six layers deep, but there can be more or fewer layers as required. Now, here's the punchline—there are no longer any holdout mattes! Instead, every pixel "knows" its depth in the scene and sorts itself out with any pixels in front of or behind it. Move an object and the pixels re-sort themselves. No re-renders!

Deep data is ideal for complex surface interactions between volume effects such as fire, smoke, explosions, or dust and regular objects. But this awesome capability comes at a heavy price. Literally. Render times are extensive and file sizes are huge. Let's just say that deep data makes a regular EXR file look like a JPEG. To that point, EXR files are the only file format that supports deep data. Due to their heft and expense, deep images are normally only used on the "money shots" in big-budget films.

Watch deep compositing in action as a deep object is shifted in position and depth without re-rendering anything.

3.1.5 Metadata

Metadata is information about the image data and is embedded in the file structure along with the pixels. Metadata is data about the data and comes in three general types: technical, descriptive, and administrative. Imaging software reads the metadata and makes decisions on its display, orientation, and frame rate.

- **Technical metadata** - This contains information about the camera and its settings during image capture. There is also file information about the image itself—resolution, file format, color space, etc. Depending on the camera, there can also be GPS information pinpointing the time and location where the picture was taken.

数据加上阿尔法通道，深度数据拥有一系列多五通道的“五元组”数据，每个五元组包含RGBAZ——如图3.3所示，其中一个像素被展开以显示其深度数据。RGB是颜色，A是透明度，Z是其在场景中的深度。每个像素实际上承载着一个体积。

因此，与具有零厚度多边形表面的传统CG对象不同，深度像素具有厚度，如图3.3中的多个Z通道所示。这个特定示例展示了一个六层深的像素，但根据需求可以有更多或更少的层数。现在，关键点来了——不再需要任何遮罩！相反，每个像素都“知道”其在场景中的深度，并与前后任何像素自行排序。移动一个物体，像素就会重新排序。无需重新渲染！

深度数据非常适合处理体积效果（如火焰、烟雾、爆炸或尘埃）与常规物体之间复杂的表面交互。但这种令人惊叹的能力代价高昂——字面意义上的高昂。渲染时间漫长，文件体积巨大。可以说，深度数据让常规的EXR文件看起来像JPEG一样轻巧。正因如此，EXR是唯一支持深度数据的文件格式。由于其庞大的体量和处理成本，深度图像通常仅用于大制作影片中的“核心镜头”。

观看深度合成在实际操作中的应用，无需重新渲染任何内容，即可调整深层对象的位置和深度。

3.1.5 元数据

元数据是关于图像数据的信息，它嵌入在文件结构中，与像素并存。元数据即关于数据的数据，主要分为三类：技术性、描述性和管理性。成像软件读取元数据，并据此决定其显示方式、方向以及帧率。

- 技术元数据——包含拍摄时相机及其设置的相关信息，同时涵盖图像本身的文件信息，如分辨率、文件格式、色彩空间等。根据相机型号的不同，还可能包含记录拍摄时间与地点的GPS定位信息。

- **Descriptive metadata** - There can be a title, or a brief text description saved with the image describing its contents. Keywords and tags might be assigned by users to assist in searches on the image. Often there is copyright information about the holder and usage rights.
- **Administrative metadata** - This includes the dates and times the image was created or modified, and information about the photographer or creator of the image. There can also be licensing information along with restrictions for use.

Metadata has become increasingly important as software and image processing operations become more sophisticated, complex, and intimate with the image. It helps to organize large collections to make them more searchable and the tags have become essential for training AI systems. It is very important that the file format chosen preserves all the metadata and even updates it as the image file is accessed and operated on.

> FUN FACT: Virtually every image file format stores its metadata in an embedded XMP (Extensible Metadata Platform) file that was developed by Adobe Systems in 2001. As an industry standard, it can be read by most apps.

3.2 Compression Strategies

Image compression is definitely the dark undergrowth of the Storage Space terrain. Images are disk hogs, so we invariably want to compress them to take less disk space and file transfer times. However, compression also degrades the images, so there is a tradeoff. Too much compression and your images are irreparably degraded. Too little and file sizes get out of control, slowing production and raising costs. And remember—compression is forever. Once an image is compressed, there is no restoring the discarded data. You would have to go back and re-compress the original image with less compression if you want to restore discarded data.

There are three general approaches for compressing images:

- Reduce the number of bits per color channel (bit depth/color depth)
- Reduce the chrominance resolution (color subsampling)
- Remove unneeded detail in the image (DCT transforms).

- 描述性元数据——可以包含标题或简短文字说明，随图像保存以描述其内容。用户可能会分配关键词和标签，以协助图像搜索。通常还会包含关于持有者和使用权限的版权信息。

- 行政元数据 - 这包括图像创建或修改的日期和时间，以及摄影师或图像创作者的信息。还可能包含使用许可信息及使用限制。

随着软件和图像处理操作变得更加复杂、精密且与图像紧密融合，元数据的重要性日益凸显。它有助于组织庞大的图像库，使其更易于检索，并且这些标签已成为训练人工智能系统的关键要素。所选文件格式必须能够完整保留所有元数据，甚至在图像文件被访问和处理时更新这些信息，这一点至关重要。

> 有趣的事实：几乎每种图像文件格式都将其元数据存储在由Adobe Systems于2001年开发的嵌入式XMP（可扩展元数据平台）文件中。作为行业标准，大多数应用程序都能读取它。

3.2 压缩策略

图像压缩无疑是存储空间领域中的阴暗灌木丛。图像是磁盘空间的吞噬者，因此我们总是希望压缩它们以减少磁盘占用和文件传输时间。然而，压缩也会降低图像质量，这就存在一个权衡。压缩过度，图像将遭受不可逆的损伤；压缩不足，文件大小又会失控，拖慢生产进度并推高成本。请记住——压缩是永久性的。一旦图像被压缩，被丢弃的数据就无法恢复。若想恢复已丢弃的数据，只能回头用更低的压缩率重新压缩原始图像。

压缩图像有三种通用方法：

- 减少每个颜色通道的位数（位深度/颜色深度）
- 降低色度分辨率（颜色子采样）
- 移除图像中不必要的细节（DCT变换）。

Of course, all three methods can be combined, as we will soon see.

In visual effects, "compression" is a dirty word because we need every bit of image data to maintain quality to withstand the heavy processing in post that VFX often requires. Today's emphasis on quality and future-proofing the project demands the maximum data per frame, but again, that drives up storage requirements, transfer times, and production costs.

Camera raw refers to the unprocessed images captured by the imaging sensor which are saved in a proprietary file format unique to the camera manufacturer. As a result, these are the highest possible quality but often contain more information than is needed, which is inefficient for post-production. Not to mention that most apps in the post-production pipeline won't be able to read a proprietary camera raw file. If possible, export the camera raw as an OpenEXR or DNG file, which are file standards that can be read by other apps. The next best would be 10-bit log DPX files, which are 20%-30% smaller than EXR files and much smaller than the camera raw.

The first and most important storage decision in the data pipeline is to decide what file format to use when exporting the image out of the camera. Some cameras can export the chosen format directly, but for others, the camera raw is written to disk and then converted later. The chosen file format puts an upper limit on the quality of the production, so one must choose wisely. A mistake here is an unrecoverable error that ripples down the entire pipeline and throughout your career in perpetuity.

3.2.1 Reduced Bit Depth

File formats can come in 8, 10, 12, and 16 bits per color channel, so reducing the number of bits per channel is one method of data compression. But one must be careful here. Reducing the bit depth too much can introduce banding. Bit-depth reductions can be achieved in two different ways. The first is to eliminate unneeded bits—for example, by reducing 12-bit image sensor data to 8 bits for output to monitors that can't show more than 8 bits anyway. A second method is to reallocate the bits for more efficient data storage, such as converting the 12-bit linear image sensor data to a 10-bit log image. Not to worry—we have an entire section on log images where their wonders and mysteries are revealed. You too will learn to love log.

当然，这三种方法都可以结合使用，我们很快就会看到。

在视觉效果领域，“压缩”是个忌讳词，因为我们需要保留图像数据的每一个比特来维持画质，以承受后期制作中视觉特效常需的重度处理。如今对画质的重视以及项目的前瞻性需求，要求每帧画面都包含最大数据量，但这又推高了存储需求、传输时间和制作成本。

相机原始文件指的是由图像传感器捕获的未处理图像，这些图像以相机厂商独有的专有文件格式保存。因此，它们具有最高可能的画质，但通常包含比所需更多的信息，这对后期制作来说效率不高。更不用说后期制作流程中的大多数应用程序都无法读取专有的相机原始文件。如果可能，应将相机原始文件导出为OpenEXR或DNG文件，这两种文件标准可被其他应用程序读取。次优选择是10位对数DPX文件，它们比EXR文件小20%-30%，且比相机原始文件小得多。

数据管道中首要且最重要的存储决策，是确定将图像导出相机时应使用何种文件格式。部分相机可直接导出所选格式，但其他相机需先将原始数据写入磁盘再进行转换。所选文件格式决定了作品质量的上限，因此必须审慎选择。此处的失误将引发无法挽回的错误，其影响会如涟漪般贯穿整个工作流程，并可能永久伴随你的职业生涯。

3.2.1 降低位深度

文件格式可以支持每个颜色通道8、10、12和16位，因此减少每个通道的位数是一种数据压缩方法。但这里必须谨慎操作。过度降低位深度可能导致色带现象。位深度缩减可通过两种不同方式实现：第一种是舍弃不需要的位——例如将12位图像传感器数据降至8位，以适配最高仅支持8位显示的监视器。第二种方法是通过重新分配位来实现更高效的数据存储，例如将12位线性图像传感器数据转换为10位对数图像。不必担心——我们设有专门章节深入探讨对数图像，揭示其奇妙与奥秘。您也将学会欣赏对数的魅力。

Another term for describing the bit depth is “precision,” which refers to how small the brightness “steps” are from one code value to the next. If the precision is too low (too few bits), the brightness steps between code values are large and can result in visible color banding (a.k.a. contouring) as illustrated in Figure 3.4. An 8-bit image has 256 brightness steps per color channel which are close enough together that banding is normally not seen if viewed on a standard television or computer monitor.

Figure 3.4 Banding (contouring) resulting from low bit depth

Did you notice the evasive “normally not seen” phrasing in the last sentence? It means that banding could be seen if displayed on a higher dynamic range monitor; or it could be introduced by an ill-advised color grade, like in the right panel of Figure 3.4—the sky has erupted with banding while the clouds are just fine. So the lesson here is even with normally sufficient precision, there can be special areas of an image that still show banding.

Another case is where data bits are lost due to image processing operations. If the image is heavily processed, resulting in stretching, the distance between the code values banding can suddenly erupt. Consider the case of scaling an 8-bit integer gradient ranging from 0 to 250 down by 0.1 so it now ranges from 0 to 25. The scaled gradient will only have 6 bits left in it. If it is then restored to its original range of 0 to 250 by scaling the code values by 10, there will still only be 6 bits to span the range—banding! See Figure 3.5. Try this in Photoshop and see what happens.

This is why the 16-bit tiff was born. It has 65,535 shades of gray per channel and is also mapped to 0 to 1.0, so the same heavy processing will still leave the brightness steps between code values small enough that banding is not seen. Keep in mind that all digital images are intrinsically banded because of the discrete steps in brightness between the code values. The trick is to keep the banding below the threshold of visibility. This is true whether working in

描述比特深度的另一个术语是“精度”，它指的是从一个码值到下一个码值的亮度“步长”有多小。如果精度太低（比特数太少），码值之间的亮度步长会很大，可能导致可见的色彩条带（也称为轮廓效应），如图3.4所示。一幅8位图像每个颜色通道有256个亮度步长，这些步长足够接近，在标准电视或电脑显示器上观看时通常不会看到条带现象。

图3.4 低比特深度导致的条带（轮廓）效应

你是否注意到最后一句中那个含糊其辞的“通常不可见”的说法？这意味着如果在更高动态范围的显示器上展示，可能会看到条带；或者也可能由于不当的色彩分级而产生，如图3.4右侧面板所示——天空已经布满了条带，而云层却完好无损。因此这里的教训是，即使通常精度足够，图像中仍可能存在特殊区域会显示条带。

另一种情况是数据位因图像处理操作而丢失。如果图像经过重度处理导致拉伸，代码值之间的条带距离可能突然增大。考虑将一个从0到250的8位整数梯度按0.1缩放至0到25的情况。缩放后的梯度仅剩6位数据。若随后通过将代码值乘以10恢复至原始0到250的范围，仍然只有6位数据来覆盖整个范围——从而产生条带！参见图3.5。可在Photoshop中尝试此操作并观察结果。

这就是16位TIFF格式诞生的原因。它每个通道拥有65,535级灰度，同样被映射到0至1.0的范围，因此即使进行重度处理，亮度值之间的阶梯仍能保持足够细微，从而避免出现色带断层。请记住，由于编码值之间存在离散的亮度阶梯，所有数字图像本质上都存在断层现象。关键在于将断层控制在肉眼不可见的阈值之下。无论处理{v*}

integer or float. It's just that floating-point data has incredibly microscopic banding, so it is virtually impossible to be seen even after heavy abuse from processing.

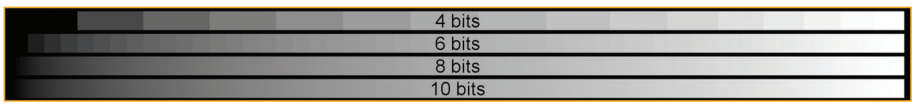


Figure 3.5 How bit depth affects banding

Figure 3.5 illustrates this relationship between bit depth and banding. As the bit depth increases from 4 to 10, the brightness steps between each code value get smaller and the gradient looks smoother. You can clearly see banding in the 4 and 6-bit gradients (16 and 64 brightness steps respectively) and depending on your monitor, you may or may not see banding in the 8-bit gradient (zoom in to the dark end of the 8-bit gradient). The 10-bit gradient has greater precision than your 8-bit monitor, so you should not see any banding there. Even though the 8-bit gradient may not show any banding now, if any significant processing is done it can suddenly develop banding, as we have seen.

3.2.2 Color Subsampling

Color subsampling means to sample (digitize) the color part of an image at less than 100% resolution. It is less a data compression scheme than a data reduction scheme. This is not just a semantic quibble. Data compression takes existing image data and squeezes information out of it to reduce file size. But if a problem occurs the original image could be compressed again with a lighter compression setting. Color subsampling digitizes the original image with less data to begin with, so it is unrecoverable.

Color subsampling is based on an attribute of human vision—namely, that our eyes are more sensitive to luminance (brightness) than color (chrominance). The image data retains 100% resolution for the luminance, but maybe 50% or even 25% resolution for the chrominance. But before we can intelligently discuss color subsampling, we first have to get clear on luminance and chrominance—the foundational principles behind color subsampling.

整数或浮点数。只是浮点数据具有极其微小的条带效应，因此即使在经过大量处理的重度滥用后，也几乎不可能被察觉。

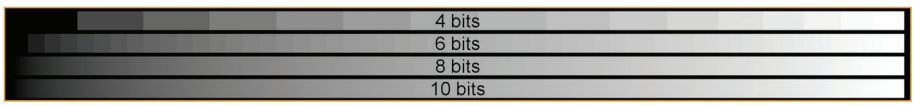


图 3.5 位深度如何影响色带效应

图3.5展示了位深度与色带效应之间的关系。随着位深度从4增加到10，每个代码值之间的亮度阶梯变得更小，渐变看起来更平滑。你可以清楚地看到4位和6位渐变（分别对应16和64级亮度阶梯）中的色带效应，而根据显示器的不同，你可能在8位渐变中看到色带效应，也可能看不到（请放大观察8位渐变的暗部区域）。10位渐变具有比8位显示器更高的精度，因此你在那里应该看不到任何色带效应。尽管8位渐变现在可能不显示任何色带效应，但如果进行任何重要的处理，它可能会突然出现色带效应，正如我们所见。

3.2.2 色彩二次采样

色彩子采样意味着以低于100%的分辨率对图像的色彩部分进行采样（数字化）。它与其说是一种数据压缩方案，不如说是一种数据缩减方案。这不仅仅是语义上的吹毛求疵。数据压缩是对现有图像数据进行处理，从中挤出信息以减少文件大小。但如果出现问题，原始图像可以用更轻的压缩设置重新压缩。而色彩子采样从一开始就用较少的数据对原始图像进行数字化，因此是不可恢复的。

颜色子采样基于人类视觉的一个特性——即我们的眼睛对亮度（明度）比对颜色（色度）更敏感。图像数据在亮度上保持100%的分辨率，但在色度上可能只有50%甚至25%的分辨率。但在我们深入讨论颜色子采样之前，首先需要明确亮度和色度——这是颜色子采样背后的基本原理。

3.2.2.1 Luminance and Chrominance

It all started in 1953 with the introduction of color television. When NTSC color television was developed, it became clear that the color signal would not fit within the bandwidth of the old black-and-white broadcast television signal. Taking advantage of the fact that our eyes are less sensitive to color, video engineers separated the chrominance (hue and saturation) from the luminance (brightness), allowing the chrominance to be broadcast at a lower resolution. The analog luminance/chrominance signal was named YUV, where Y was the luminance and U and V carried the chrominance. With Y at 100% resolution and U and V at 50% resolution, they were able to reduce the bandwidth requirements by one-third making color television practical.

Figure 3.6 YCbCr image

Fast forward to the digital age and the analog YUV was replaced by its digital counterpart, YCbCr. Using the exact same strategy, Y is the luminance and CbCr is the chrominance. A YCbCr image is shown in Figure 3.6. This color model is used to encode video data for efficient transmission, but its color representation is a bit difficult to visualize. The Y channel represents luminance while the chrominance is represented by the Cb and Cr channels, where Cb is blue minus Y and Cr is red minus Y. Teasing the colors apart like this into luminance and chrominance sets things up for the all-pervasive color subsampling image data reduction that was originally developed for broadcast television but is also used in many video and photographic compression schemes.

3.2.2.2 Subsampling Schemes

Camera manufacturers realized that they could cut the chrominance short and nobody would notice. This opened the door for all manner of digital chicanery, with color subsampling schemes for modern video cameras. The color subsampling is described with three numbers (e.g., 4:2:2 or 4:2:0) that represent how many data samples are taken in a row of 4 pixels. The first number is always 4 and represents four samples of luminance in the 4-pixel

3.2.2.1 Luminance and Chrominance

这一切始于1953年彩色电视的问世。当NTSC彩色电视技术被开发出来时，人们发现彩色信号无法适配原有黑白广播电视信号的带宽。利用人眼对色彩敏感度较低的特性，视频工程师将色度（色调与饱和度）从亮度信号中分离出来，使色度得以以较低分辨率进行传输。这种模拟亮度/色度信号被命名为YUV，其中Y代表亮度，U和V承载色度信息。通过将Y保持100%分辨率、U和V降至50%分辨率，他们成功将带宽需求降低了三分之一，从而使彩色电视得以实现。

图3.6 YCbCr图像

快进到数字时代，模拟YUV被其数字对应物YCbCr所取代。采用完全相同的策略，Y代表亮度，CbCr代表色度。图3.6展示了一张YCbCr图像。这种色彩模型用于编码视频数据以实现高效传输，但其色彩表现方式较难直观想象。Y通道表示亮度，而色度则由Cb和Cr通道表示——其中Cb是蓝色分量减去Y，Cr是红色分量减去Y。通过将色彩这样分解为亮度和色度，为无处不在的色彩子采样图像数据缩减技术奠定了基础。这项技术最初为广播电视开发，如今也广泛应用于多种视频和摄影压缩方案中。

3.2.2.2 子采样方案

相机制造商意识到，他们可以削减色度信息而无人察觉。这为各种数字技巧打开了大门，现代摄像机采用的色彩二次采样方案便是其中之一。色彩二次采样用三个数字（例如4:2:2或4:2:0）表示，这些数字代表在4个像素行中采集的数据样本数量。第一个数字始终为4，代表4个像素行中的四个亮度样本。

block or 100% resolution for luminance. The second number is how many samples are in the Cb channel and the third is the Cr channel. The insidious nature of this scheme is that it looks just fine on the monitor, but to the computer trying to pull a chromakey, it is seriously degraded. As a result, amateur videographers will shoot green screens using a camera that subsamples the color and then criticize you for pulling a lousy chromakey. When this happens to you (and it will), simply open the book to this section and smack the videographer upside the head with it.

Keyers use the color (chrominance) of the image to create their keys, so color subsampling degrades them simply because there is less color information to work with. Figure 3.7 demonstrates the chunky edges a key will inherit when the chrominance is subsampled to half resolution with 4:2:2 sampling. The left panel shows a close-up of the chin area that was keyed. The middle panel shows a key from the full-resolution chrominance sampling (4:4:4) which produced a nice smooth edge. The right panel shows a key from a 4:2:2 chrominance subsampled version which produced large chunky pixels along the keyed edge—plus, at no extra charge, a loss of fine hair detail.

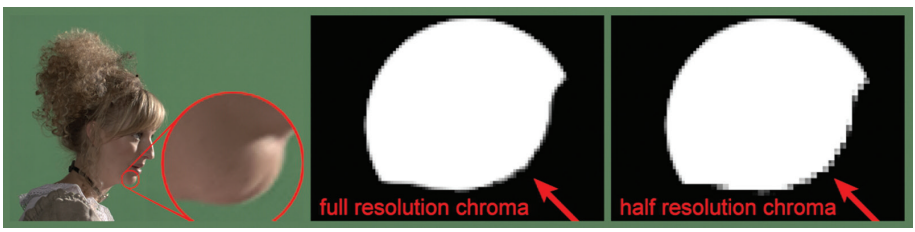


Figure 3.7 Reduced chrominance samples result in chunky edges in the key

Today there are many color subsampling schemes used by camera manufacturers. Figure 3.8 illustrates the range of subsampling schemes, and because this book is visual effects centric it also describes how it affects the ability to pull a good key for green or blue screens. Each scheme includes a checkerboard segment to illustrate the missing frame data.

块或100%的亮度分辨率。第二个数字表示Cb通道中有多少样本，第三个是Cr通道。这种方案的隐蔽性在于它在显示器上看起来很好，但对于试图进行色度键控的计算机来说，它严重降低了质量。因此，业余摄像师会使用对颜色进行子采样的摄像机拍摄绿幕，然后批评你拉出的色度键效果很差。当这种情况发生在你身上时（而且肯定会发生），只需打开本书的这一部分，用它敲打摄像师的脑袋。

键控器利用图像的颜色（色度）来创建键控，因此颜色子采样会降低键控质量，原因很简单：可供处理的颜色信息减少了。图3.7展示了当色度通过4:2:2采样降至一半分辨率时，键控边缘会出现的块状效应。左图显示了键控处理的下巴区域特写；中图展示了全分辨率色度采样（4:4:4）生成的键控，其边缘平滑自然；右图则呈现了4:2:2色度子采样版本产生的键控——不仅键控边缘出现大块像素，还额外损失了纤细发丝的细节。

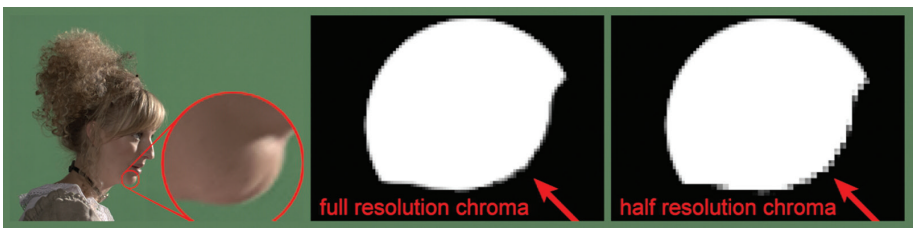


图3.7 色度样本减少导致键控边缘出现块状效应

如今，相机制造商采用了多种色彩子采样方案。图3.8展示了各类子采样方案的范围，由于本书以视觉效果为核心，同时也说明了这些方案如何影响从绿幕或蓝幕中提取优质键控的能力。每种方案均包含一个棋盘格片段，用以说明缺失的帧数据。

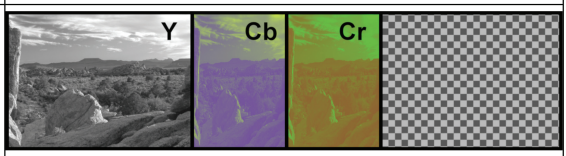
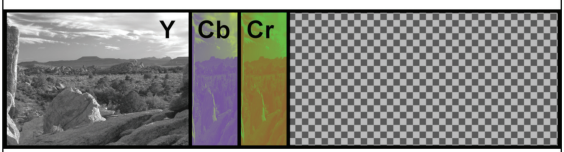
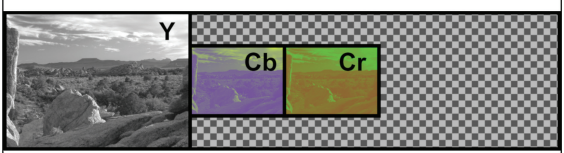
4:4:4 - This is the gold standard. There is no subsampling and all the color data is there. Excellent keys.	
4:2:2 - This is the video industry standard. 50% of the color information is gone so good keys are more difficult.	
4:1:1 - This is very poor-quality video from inexpensive cameras. 75% of the color information is gone and keys are very poor quality.	
4:2:0 - This is the sampling scheme for MPEG-2 which loses 75% of the color like 4:1:1 but uses a different sampling pattern. Still very poor.	

Figure 3.8 YCbCr color subsampling schemes

Sometimes you are stuck trying to key poor-quality video like 4:2:0, resulting in poor-quality edges. The technique shown here will help to salvage the composite.

3.2.3 Discrete Cosine Transform (DCT)

In 1972 Nasir Ahmed, working at Kansas State University, conceived of the Discrete Cosine Transform (DCT) as a method of lossy image compression, and the world was changed forever. It is now the basis of virtually all image compression methods and is the main engine of JPEG image compression, and it has even been adopted by camera manufacturers. It can achieve a 10:1 compression of a photograph with no visible degradation. Besides being fast to compute and very effective at compressing images, the DCT is also "adjustable," in that you can dial in the amount of compression you want.

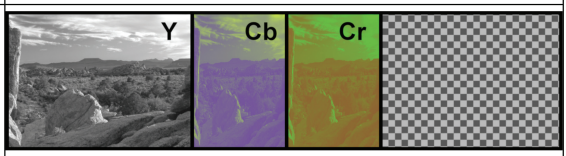
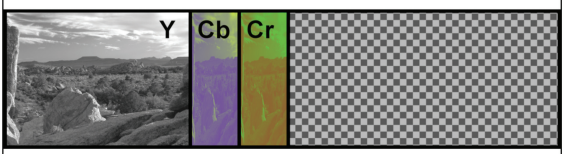
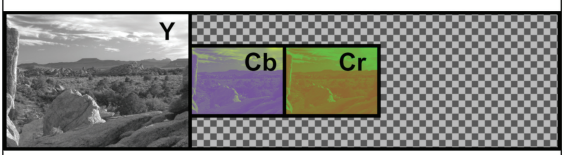
4:4:4 - This is the gold standard. There is no subsampling and all the color data is there. Excellent keys.	
4:2:2 - This is the video industry standard. 50% of the color information is gone so good keys are more difficult.	
4:1:1 - This is very poor-quality video from inexpensive cameras. 75% of the color information is gone and keys are very poor quality.	
4:2:0 - This is the sampling scheme for MPEG-2 which loses 75% of the color like 4:1:1 but uses a different sampling pattern. Still very poor.	

图3.8 YCbCr颜色子采样方案

有时你会因处理低质量视频（如4:2:0）而陷入困境，导致边缘效果不佳。这里展示的技术将有助于挽救合成效果。

3.2.3 离散余弦变换（DCT）

1972年，堪萨斯州立大学的纳西尔·艾哈迈德构想出离散余弦变换（DCT）作为一种有损图像压缩方法，世界从此彻底改变。如今，它几乎是所有图像压缩技术的基础，也是JPEG图像压缩的核心引擎，甚至已被相机制造商广泛采用。该技术能以10:1的压缩比处理照片，且视觉质量无明显下降。除了计算速度快、压缩效率高之外，DCT还具有“可调节性”——用户可根据需求精准控制压缩程度。

Astonishingly, if a DCT-compressed image is uncompressed for display and then compressed again to the same level, there will be no degradation. This is because the first compression removes picture detail down to a certain level; uncompressing it for display does not restore that detail, and then compressing it again to the same level finds the picture detail already gone—so, no change. How DCT compression works internally is revealed by the self-explanatory equation in Figure 3.9.

$$F(u,v) = \left(\frac{2}{N}\right)^{\frac{1}{2}} \left(\frac{2}{M}\right)^{\frac{1}{2}} \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} \Lambda(i).\Lambda(j).cos\left[\frac{\pi.u}{2.N}(2i+1)\right] cos\left[\frac{\pi.v}{2.M}(2j+1)\right].f(i,j)$$

Figure 3.9 The DCT equation

Or, if you prefer, I could just explain it in English. The short form of the story is that the DCT simply removes fine detail from an image and how much fine detail is removed is adjustable. This adjustability is a key part of its universal appeal. The original image is first transformed from its original 2D spatial domain to a frequency domain using Fourier transforms where the high frequencies can be discarded to reduce file size. After the high frequencies are eliminated, an inverse transform is used to convert the image back to its spatial domain, minus the fine detail. As a side note, the DCT transform can also be used to compress audio files.

To create a JPEG image, the original RGB image is first transformed to YCbCr so the luminance is separated from the chrominance. A DCT with settings for the desired level of detail removal is applied to the Y channel. A DCT with different settings is applied to the CbCr channels so the luminance (Y) is compressed differently than the chrominance (CbCr). For example, the luminance might be compressed by 90% while the chrominance is compressed by 97%, so we end up with more detail in the luminance than the chrominance to comport with the sensibilities of human vision, which is more sensitive to changes in luminance than chrominance.

This is very different than the color subsampling for the video described earlier, which is also done with YCbCr. Video color subsampling is fixed at the defined integer sampling values (4:2:2, 4:1:1, etc.), has no subsampling for the Y channel, does not use DCT, and is not adjustable. The color subsampled image is captured that way and there is nothing you can do about it. The DCT, however, takes an image and squeezes unwanted detail out of it.

令人惊讶的是，如果一张经过DCT压缩的图像被解压缩以用于显示，然后再次压缩到相同的水平，将不会有任何质量下降。这是因为第一次压缩会去除图像细节至某一特定程度；解压缩显示并不会恢复这些细节，而再次压缩到相同水平时，发现图像细节早已丢失——因此，图像不会发生变化。图3.9中一目了然的方程揭示了DCT压缩的内部工作原理。

$$F(u,v) = \left(\frac{2}{N}\right)^{\frac{1}{2}} \left(\frac{2}{M}\right)^{\frac{1}{2}} \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} \Lambda(i).\Lambda(j).cos\left[\frac{\pi.u}{2.N}(2i+1)\right] cos\left[\frac{\pi.v}{2.M}(2j+1)\right].f(i,j)$$

图 3.9 DCT 方程

或者，如果您愿意，我也可以用英语解释。简而言之，DCT 只是从图像中移除细节，而移除多少细节是可调的。这种可调性是其普遍吸引力的关键部分。原始图像首先通过傅里叶变换从其原始的二维空间域转换到频域，在此过程中可以丢弃高频部分以减少文件大小。消除高频后，使用逆变换将图像转换回空间域，但细节已减少。顺便提一下，DCT 变换也可用于压缩音频文件。

要创建JPEG图像，首先将原始的RGB图像转换为YCbCr格式，以便将亮度与色度分离。对Y通道应用离散余弦变换（DCT），并根据所需细节去除程度进行设置。对CbCr通道则采用不同设置的DCT，使得亮度（Y）与色度（CbCr）以不同方式压缩。例如，亮度可能压缩90%，而色度压缩97%，最终亮度保留的细节多于色度，这符合人类视觉的特性——人眼对亮度变化的敏感度高于色度变化。

这与之前描述的YCbCr视频颜色子采样大不相同。视频颜色子采样采用固定的整数采样值（如4:2:2、4:1:1等），Y通道不进行子采样，不使用DCT变换，且不可调整。图像在采集时即已进行颜色子采样，无法后期更改。而DCT变换则是对现有图像进行处理，通过压缩去除不必要的细节。

FUN FACT: The DCT applies to virtually all signal processing tasks far beyond photos and video, and is also used to compress audio, text, speech, seismic data, and biomedical signal processing.

3.2.4 MPEG

Once video graduated from being broadcast over radio waves to roof antennae to becoming a digital media for DVDs and internet transmission, it became immediately clear that the file sizes and bandwidth requirements were far more than what the tech of the day could deliver. The hunt was on for some kind of digital compression scheme that would dramatically lower the data requirements for moving video images. Industry was mobilized and teams were formed to address the problem. Finally, in 1988, the Motion Picture Experts Group (MPEG) was formed—and what they came up with was truly astonishing. They could squeeze an entire feature film down to around 5% of its original file size with no visible degradation.

One of the key requirements of this new compression scheme was that it had to be computationally cheap to decompress because it would have to be done on DVD players, set-top boxes, and other non-supercomputer systems thousands of times a day. However, the video signal only needed to be compressed once, so it was okay if the compression had to be done on large computers with highly paid "compressionists" who selected scene-by-scene compression strategies based on the scene content. Today, the compression is done with automation to analyze the scene content and choose compression algorithms—under human guidance, of course.

Here we will take a high-level view of the MPEG compression process, which uses a combination of both spatial and temporal compressions. It is so successful that it is used on virtually every DVD and Blu-Ray disk, streaming video, and internet video.

3.2.4.1 Spatial Compression

The starting point of MPEG compression is to remove noise because this high-frequency detail requires more data to describe. Second, DCT compression is done to reduce the file size of each frame. Performing this spatial

有趣的事实：DCT几乎适用于所有信号处理任务，远不止照片和视频，还用于压缩音频、文本、语音、地震数据以及生物医学信号处理。

3.2.4 MPEG

当视频从通过无线电波传输到屋顶天线，发展到成为DVD和互联网传输的数字媒体时，人们立刻意识到，文件大小和带宽需求远远超过了当时技术所能提供的水平。于是，寻找一种能够大幅降低视频图像数据传输需求的数字压缩方案成为当务之急。行业迅速动员起来，组建团队以解决这一问题。最终在1988年，运动图像专家组（MPEG）成立——他们所提出的方案确实令人惊叹。他们能将整部故事片的文件大小压缩至原始大小的约5%，且画质无明显损失。

这种新型压缩方案的关键要求之一，是解压过程必须计算成本低廉，因为它需要在DVD播放器、机顶盒和其他非超级计算机系统上每天执行数千次。然而，视频信号只需压缩一次，因此即使压缩工作必须在大型计算机上完成也无妨——这些计算机由高薪的“压缩师”操作，他们根据场景内容逐帧选择压缩策略。如今，压缩过程已实现自动化，系统会在人工指导下分析场景内容并选择压缩算法。

这里我们将从高层次视角审视MPEG压缩流程，该技术融合了空间压缩与时间压缩的双重策略。其成效如此卓著，以至于几乎每张DVD与蓝光光盘、所有流媒体视频及网络视频都采用了{v*}这一标准。

3.2.4.1 空间压缩

MPEG压缩的起点是去除噪声，因为这些高频细节需要更多数据来描述。其次，进行DCT压缩以减少每帧的文件大小。执行这种空间

compression reduces the file size by as much as 50% to start. We haven't done any temporal compression yet.

3.2.4.2 Temporal Compression

Studies showed that only about 4% of the pixels change frame-to-frame in a typical scene, so the overarching concept of MPEG compression is to encode only those changes. This is done by selecting key frames called I-frames to be kept intact, then generating in-between frames by encoding their differences from the keyframe. Further, those differences are encoded with motion vectors that instruct the system how to move a block of pixels from the I-frame to the synthesized frames.

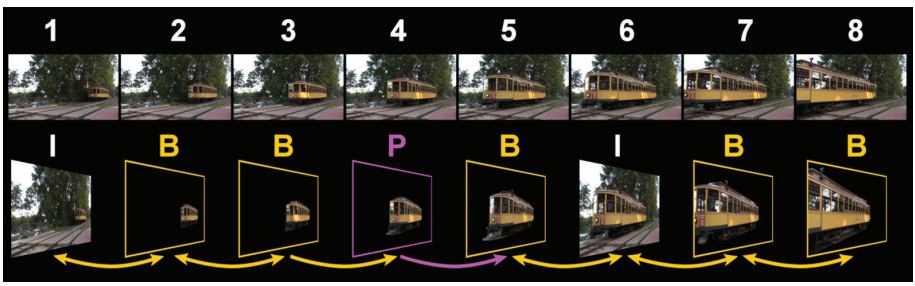


Figure 3.10 A conceptual model of MPEG compression

Figure 3.10 illustrates the basic idea of the motion estimation used by MPEG compression without the gory technical details. Frames 1 and 6 are **I-frames (Intra-coded frames)**, which are intact JPEG-type frames and serve as the keyframes for the synthesized in-between frames. Frame 4 is a **P-frame** (Predictive frame), which contains only the changes from the previous frame. Frames 2 and 3 are **B-frames** (Bi-directionally predictive frames), meaning they can provide motion vector information to the frames both before and after them. The I-frames occur typically every two to five seconds, depending on scene complexity, and are almost always found at the head of a cut to provide image data for the following frames to facilitate editing, streaming, and playback.

3.2.4.3 Summary

MPEG compression is intended for a streaming video application as if watching a VHS tape. While it can be edited, it takes special software to cope with the P and B-frames, as they are not whole frames like the I-frames.

压缩可将文件大小从一开始就减少多达50%。我们尚未进行任何时间压缩。

3.2.4.2 时间压缩

研究表明，在典型场景中，只有约4%的像素在帧与帧之间发生变化，因此MPEG压缩的核心思想是仅对这些变化进行编码。具体实现方式是选取称为I帧的关键帧保持完整，然后通过编码中间帧与关键帧的差异来生成这些中间帧。此外，这些差异会通过运动向量进行编码，运动向量指示系统如何将像素块从I帧移动到合成帧中。

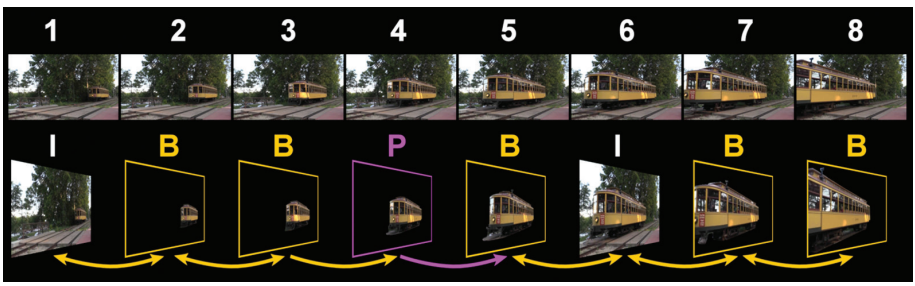


图 3.10 MPEG 压缩的概念模型

图3.10展示了MPEG压缩中运动估计的基本思想，省略了复杂的技术细节。帧1和帧6是I帧（帧内编码帧），它们是完整的JPEG类型帧，并作为合成中间帧的关键帧。帧4是P帧（预测帧），仅包含与前一帧相比的变化部分。帧2和帧3是B帧（双向预测帧），这意味着它们可以向前后帧提供运动矢量信息。I帧通常每两到五秒出现一次，具体取决于场景复杂度，并且几乎总是出现在剪辑的开头，为后续帧提供图像数据，以便于编辑、流媒体传输和播放。

3.2.4.3 总结

MPEG压缩技术专为流媒体视频应用设计，如同观看VHS录像带一般。虽然它支持编辑，但处理P帧和B帧需要特殊软件，因为它们不像I帧那样是完整的画面。

Also, after editing, the finished work must be recompressed back to MPEG, which can take significant computing power and result in a loss of quality. There are a great many MPEG compression standards, from very old to ultra-modern and efficient, but they all work on these general principles.

3.2.5 Graphics Lossless Compression

To round out the discussion of compression, we will take a quick look at lossless compression for graphics. Graphics can be compressed in a lossless way because, unlike natural photographic images, they contain long horizontal strips of identical code values. This repeating arrangement provides the opportunity to use statistical redundancy methods that are perfectly reversible—hence "lossless."

The officious term "statistical redundancy" is neatly illustrated in Figure 3.11 with the close-up inserts for both photographic and graphic examples. The photographic close-up on the left shows the eye and how the pixel values are essentially random, so no statistical redundancy. The graphic close-up on the right contains long rows of identical (redundant) pixels, which have (you guessed it!) statistical redundancy. Both RLE (Run-Length Encoding) and LZW (Lempel-Zic-Welch) compression take advantage of the repeating patterns of pixel values in graphic images but in different ways. Following is a brief description of how they work.

Figure 3.11 Photographic vs. graphic images

3.2.5.1 RLE Compression

Let's say the top scan line of the graphic close-up in Figure 3.11 is all the same value of orange for 100 pixels. Rather than repeating the same orange code value 100 times, we could encode the run length of orange as "100

此外，编辑完成后，成品必须重新压缩回MPEG格式，这一过程可能需要大量的计算能力，并可能导致质量损失。MPEG压缩标准种类繁多，从非常古老的到超现代且高效的都有，但它们都遵循这些基本原则。

3.2.5 图形无损压缩

为了完善对压缩的讨论，我们将简要探讨图形的无损压缩。图形之所以能够进行无损压缩，是因为与自然摄影图像不同，图形包含大量具有相同代码值的水平条带。这种重复排列为使用统计冗余方法提供了机会，这些方法完全可逆——因此称为“无损”。

“统计冗余”这一术语在图3.11中通过摄影与图形示例的特写插图得到了清晰阐释。左侧摄影特写展示了眼睛区域，其像素值本质上呈随机分布，因此不存在统计冗余。右侧图形特写则包含大量完全相同的（冗余）像素行——这正是统计冗余的体现。无论是RLE（游程编码）还是LZW（Lempel-Zic-Welch）压缩技术，都利用了图形图像中像素值的重复模式，但实现方式各有不同。以下将简要说明它们的工作原理。

图3.11 摄影图像与图形图像的对比

3.2.5.1 RLE压缩

假设图3.11中图形特写的顶部扫描线有100个像素都是相同的橙色值。与其重复橙色编码值100次，我们可以将橙色的游程长度编码为“100”。

orange." This is not only much smaller than the original scan line, but can also be perfectly reversed by taking the code "100 orange" and expanding it back to 100 actual pixels of orange, thus perfectly replicating the original graphic. This type of compression is called RLE. RLE is one of the compression options in TIFF, BMP, and PCX file formats and is the best choice for simple graphic images with large areas of the same color and limited variation.

3.2.5.2 LZW Compression

One-click up from simple RLE compression is LZW (named for its developers, Lempel, Ziv, and Welch), which is also totally lossless. The basic idea is to place each pixel color in a table the first time it is encountered. All subsequent instances of that pixel are replaced by a pointer to the original pixel in the table. The pointers are much smaller than the RGB values, so the file size is reduced. LZW is very fast to encode and decode, while the amount of compression is based on the image complexity. Indeed, compressing a photographic image with LZW will result in a larger file size. LZW compression is an option in GIF, TIFF, PNG, and PDF file formats. As an interesting side note, LZW compression is also commonly used to compress text files. LZW is your best choice for more complex graphics with more colors and finer detail.

File size impacts labor and render times, so most file formats will have compression choices. So, which is best, RLE or LZW? They both return a lossless result, so the discriminators are the resulting file size and speed. Unless you are compressing huge graphic images or thousands of them, speed will not be an issue so we will focus on the compressed file size. Which compresses more depends on image size and content, so if you really must choose one over the other run a test with a representative image and compare the resulting file sizes.

3.3 Codecs

The word "codec" is a portmanteau of **co**der/**dec**oder and is the heart and soul of image compression, which is very important to the efficient production of films, videos, and still images. Choose unwisely and you can either

橙色。”这不仅比原始扫描线小得多，而且可以通过取代码“100橙色”并将其扩展回100个实际橙色像素来完美逆转，从而完美复制原始图形。这种压缩类型称为RLE。RLE是TIFF、BMP和PCX文件格式中的压缩选项之一，对于具有大面积相同颜色且变化有限的简单图形图像来说，是最佳选择。

3.2.5.2 LZW压缩算法

从简单的RLE压缩一键升级而来的是LZW（以其开发者Lempel、Ziv和Welch命名），它同样完全无损。其基本思路是：首次遇到每种像素颜色时，将其存入表格。随后出现的相同像素均替换为指向表格中原始像素的指针。这些指针远小于RGB值，因此文件尺寸得以缩减。LZW编码与解码速度极快，而压缩量取决于图像复杂度。实际上，用LZW压缩摄影图像反而会导致文件变大。LZW压缩是GIF、TIFF、PNG和PDF文件格式的可选功能。有趣的是，LZW压缩也常用于压缩文本文件。对于色彩更丰富、细节更精细的复杂图形，LZW是您的最佳选择。

文件大小会影响工作量和渲染时间，因此大多数文件格式都会提供压缩选项。那么，RLE和LZW哪种更好呢？两者都能实现无损压缩，所以区别在于最终的文件大小和速度。除非你要压缩巨大的图形图像或成千上万的图像，否则速度不会成为问题，因此我们将重点关注压缩后的文件大小。哪种压缩率更高取决于图像尺寸和内容，所以如果你真的必须在两者中做出选择，最好用代表性图像进行测试，并比较生成的文件大小。

3.3 编解码器

“编解码器”一词是编码器/解码器的合成词，它是图像压缩的核心与灵魂，对于电影、视频和静态图像的高效制作至关重要。选择不当则可能导致

damage the quality of your images or waste disk space on bloated databases. To help you navigate the minefield of codecs, we will look at the three most popular lossy codecs: H.264, H.265, and JPEG 2000. ProRes and DNxHD have their own internal library of codecs which are discussed in their own section later. Here we are concerned with what might be called "public" codecs, in that you may find them as an option in many file formats.

FUN FACT: A "portmanteau" is a "blend word" formed by combining two closely associated words, such as "smog"—a blending of smoke and fog.

3.3.1 H.264

H.264 is an 8-bit lossy video codec based on the MPEG-4 Part 10 compression scheme briefly summarized in Section 3.2.4 MPEG. It is wildly popular, being used by some 90% of video industry developers simply because it can squeeze the file size of a video clip by up to 99% with little visible degradation. Another popular feature is that the amount of compression is variable depending on the settings during encoding. It was designed so that the compression operation, which is done only once, requires large computing power but the decompression for viewing can be done on very light hardware.

Table 3.1 lists both the bit rate and file sizes for one hour of uncompressed HD video, plus low compression and high compression. The actual numbers can be a bit squishy because the image content affects the compression results, but these are good averages. The bottom line is that you can get 90% percent of the picture with less than 1% of the data using the H.264 codec. So how in the world does it do that?

TABLE 3.1 Compression ratios for HD video using H.264 codec

One hour of video	Bit rate	File size	Compression ratio
No compression	1493 Mbps	660 GB	n/a
Low compression	10 Mbps	4.4 GB	149:1
High compression	2 Mbps	1 GB	746:1

损害图像质量或在臃肿的数据库上浪费磁盘空间。为了帮助您避开编解码器的雷区，我们将探讨三种最流行的有损编解码器：H.264、H.265和JPEG 2000。ProRes和DNxHD拥有各自的内部编解码器库，我们将在后续章节专门讨论。这里我们关注的是所谓的“公共”编解码器，因为您可能会在许多文件格式中找到它们作为可选选项。

> 有趣的事实：“portmanteau”是一种“混合词”，由两个紧密相关的单词组合而成，例如“smog”——即烟（smoke）和雾（fog）的混合。

3.3.1 H.264

H.264是一种基于MPEG-4第10部分压缩方案的8位有损视频编解码器，其技术概要已在3.2.4节MPEG中简述。该格式极为流行，约90%的视频行业开发者使用它，原因在于它能将视频文件大小压缩高达99%且几乎不产生视觉质量损失。另一大受欢迎的特性是其压缩率可通过编码设置灵活调整。该技术设计时注重压缩过程（仅需执行一次）需消耗大量计算资源，而解压播放则可在性能较低的硬件上轻松完成。

表3.1列出了未压缩高清视频一小时所需的比特率和文件大小，以及低压缩和高压缩下的数据。具体数值可能因图像内容对压缩效果的影响而略有浮动，但这些是较好的平均值。关键在于，使用H.264编解码器，你只需不到1%的数据量就能获得90%的画面效果。那么它究竟是如何做到这一点的呢?

表3.1 使用H.264编解码器的HD视频压缩比

One hour of video	Bit rate	File size	Compression ratio
No compression	1493 Mbps	660 GB	n/a
Low compression	10 Mbps	4.4 GB	149:1
High compression	2 Mbps	1 GB	746:1

These unbelievable compression ratios are more aggressive than what was described in Section 3.2.4 because that section covered the generic workings of MPEG. Here we are looking specifically at the MPEG-4 Part 10 implementation used in H.264.

3.3.2 H.265

H.265 is a second-gen implementation of H.264 offering from 25% to 50% better compression for the same data rate—or, conversely, much better-looking pictures at the same bit rate. For this reason, it is very popular for 8K media. It also offers 10-bit data making it just the thing for high dynamic range projects. In addition to using DCT, it uses DST (Discrete Sine Transform) with variable block sizes. It also has improved motion vector prediction and improved motion compensation filtering. All this fancy signal processing requires much more computing power to decode than H.264, but these days major computing power is very affordable.

However, the adoption of H.265 has been slow due to its computational requirements. This will eventually be addressed by Moore's Law, making H.265 more agreeable over time. Another adoption obstacle has been the greater cost and complexity of the patent licensing compared to H.264. However, H.265's clear advantage at quality compression of 4K and 8K video streams with significantly lower bandwidth requirements than H.264 means that H.265 will continue to be a strong contender in the compression game.

> FUN FACT: In 1965, Gordon Moore—founder of Intel—proclaimed that the number of transistors on a microchip would double every 18 months, a prediction which is now known as "Moore's Law." So far, so true.

3.3.3 JPEG 2000

While JPEG uses DCT to compress image data, JPEG 2000 uses wavelet compression. Its file extension is .jp2 instead of .jpg. One key feature is the ability to compress different regions of the frame at different compression levels. It can keep the area of interest in sharp focus while encoding the defocused background with less quality. It also offers fewer artifacts than regular JPEG.

这些令人难以置信的压缩比如第3.2.4节所述更为激进，因为该节涵盖了MPEG的通用工作原理。这里我们专门研究H.264中使用的MPEG-4第10部分实现。

3.3.2 H.265

H.265是H.264的第二代实现，在相同数据速率下可提供25%至50%的更高压缩率——或者说，在相同比特率下能呈现更优质的画面。因此，它在8K媒体领域非常受欢迎。该技术还支持10位数据，使其成为高动态范围项目的理想选择。除了采用DCT（离散余弦变换）外，它还结合了可变块大小的DST（离散正弦变换），并改进了运动矢量预测与运动补偿滤波。虽然这些复杂的信号处理技术相比H.264需要更强的解码计算能力，但如今高性能计算资源已非常普及。

然而，由于H.265对计算能力的要求较高，其普及速度一直较为缓慢。这一问题最终将随着摩尔定律的推进得到解决，使H.265逐渐变得更易推广。另一项普及障碍在于，与H.264相比，H.265的专利授权成本更高、流程更复杂。但H.265在4K和8K视频流的高质量压缩方面具有明显优势，其带宽需求显著低于H.264，这意味着H.265仍将在压缩技术领域保持强劲的竞争力。

> 有趣的事实：1965年，英特尔创始人戈登·摩尔预言，微芯片上的晶体管数量每18个月就会翻一番，这一预测后来被称为“摩尔定律”。迄今为止，这一预言依然成立。

3.3.3 JPEG 2000

JPEG 使用 DCT 压缩图像数据，而 JPEG 2000 采用小波压缩技术，其文件扩展名为 .jp2 而非 .jpg。其关键特性之一是能够对画面的不同区域进行差异化压缩：可将关注区域保持清晰锐利，同时以较低质量编码虚化背景。相比常规 JPEG，它还能减少伪影的产生。

While JPEG 2000 offers modest performance improvements over JPEG, its main claim to fame is flexible options when decoding the image. For example, a JPEG 2000 image may be read in at a lower resolution. However, these fancy decoding options utilize complex codecs that are computationally demanding.

3.3.4 ProRes

Apple ProRes is a very popular high-quality video file format specifically designed for editing with an internal family of codecs that offer a range of quality choices. The ProRes codecs range from a low of 422LT (342 megabits (MBits) per second) to a high of 4444XQ (1659 MBits per second) for a 4K UHD at 25 fps. Note that the ProRes codecs refer to bit rates rather than quality settings—perhaps because bit rates are much easier to quantify than quality. Famous for running smoothly on low-end hardware like laptops, even at 4K, Apple ProRes is excellent for editing where you need to jump to random places in the video and have it start playing immediately.

Apple's ProRes line of file formats also use DCTs, like the JPEG format. They are not strictly JPEGs because JPEG compression uses a pre-defined set of parameters. What Apple did was create its own set of DCT parameters so it could "tune" the process to its exact requirements of quality and streaming speed. Although each frame is not a true JPEG image, you may think of them as such if you don't tell Apple. Video frames are sampled at either 4:4:4:4 (if an alpha channel) or 4:2:2 (if ordinary video), transformed to YCbCr, and then given the customized DCT compression appropriate to the chosen ProRes file format. You should note with pride that you understood every word of the last sentence, which puts you in the top 1% of the class.

Now that we have had the MPEG discussion, I can tell you that ProRes uses I-frames only. This makes it highly editable (but of course!) and very fast to decode, skip frames, and play backward.

3.3.5 DNxHD

As everybody knows, DNxHD stands for "Digital Nonlinear Extensible High Definition," which is a proprietary compression standard brought to you by Avid. It is a video-based protocol with a wide range of choices for resolution,

尽管JPEG 2000相较于JPEG仅提供了有限的性能提升，但其主要优势在于解码图像时具备灵活的选项。例如，JPEG 2000图像可以以较低分辨率读取。然而，这些复杂的解码选项依赖于计算需求较高的编解码器。

3.3.4 ProRes

Apple ProRes 是一种非常流行的高质量视频文件格式，专为编辑而设计，其内部编解码器家族提供了一系列质量选择。ProRes 编解码器的范围从最低的 422 LT（每秒 342 兆比特）到最高的 4444XQ（每秒 1659 兆比特），适用于 25 帧/秒的 4K UHD 视频。需要注意的是，ProRes 编解码器指的是比特率而非质量设置——这可能是因为比特率比质量更容易量化。Apple ProRes 以在笔记本电脑等低端硬件上流畅运行而闻名，即使在 4K 分辨率下也是如此，非常适合需要随机跳转到视频中任意位置并立即开始播放的编辑工作。

苹果的ProRes系列文件格式也采用了DCT（离散余弦变换）技术，与JPEG格式类似。它们并非严格意义上的JPEG文件，因为JPEG压缩使用预定义的参数集。苹果的做法是创建了一套专属的DCT参数，从而能根据其对画质和流媒体速度的精确需求“微调”压缩过程。虽然每帧画面并非真正的JPEG图像，但只要不告诉苹果，你大可将它们视作此类格式。视频帧会按4:4:4:4（含Alpha通道时）或4:2:2（普通视频）进行采样，转换为YCbCr色彩空间，再根据所选ProRes格式进行定制化的DCT压缩。若你完全理解上一句话的每个词，请自豪地意识到——这已让你跻身顶尖1%的行列。

既然我们已经讨论了MPEG，我可以告诉你，ProRes仅使用I帧。这使得它具有极高的可编辑性（当然如此！），并且在解码、跳帧和反向播放时速度非常快。

3.3.5 DNxHD

众所周知，DNxHD代表“数字非线性可扩展高清”，这是由Avid推出的一种专有压缩标准。它是一种基于视频的协议，提供多种分辨率选择，

scan mode, and bit rate that range from high-end broadcast to online streaming. It even has a "visually lossless" compression option, meaning while it loses image detail for sure, you will hardly notice it most of the time. It is ideally suited for editing (meaning that others are not!) in real-time on modest hardware. It has been officially standardized by SMPTE as VC-3 and as a result, it is supported by a great many hardware and software products.

3.4 Containers

Containers, or wrappers, are a "shell" that contains a codec. The H.264 video codec may be "wrapped" in MOV, MP4, or AVI containers, for example. These containers, or wrappers, have internal support utilities to facilitate the operation of the codec and they do differ from one another in features—for example, the support for subtitles or chapters. Also, one app you are using might accept .mov files but not work with .avi files, even though both containers are holding an H.264 codec. Here's the breakdown.

3.4.1 MOV

Developed by Apple, the .mov file format is also known as a QuickTime movie. It is a multimedia container in that it holds video, audio, text for subtitles, timecode, and more. The .mov file format is supported by many apps on both Mac and Windows platforms. As a multimedia container, it could, for example, hold H.264 video with AAC or MP3 audio along with a closed-caption data stream. The important point here is that a .mov file can contain any number of codecs and is ideal for both editing and final delivery for display.

3.4.2 MP4

Officially known as MPEG-4, this is a container format for video, audio, subtitles, and even stills, which was developed by the Moving Pictures Expert Group (MPEG). Its popularity comes from its wide use on almost every platform and its balance between quality and file size. It has two unique features: progressive downloading and adaptive streaming. Progressive downloading allows the user to start watching the video even if the file is not finished downloading, while adaptive streaming allows the quality to be adapted to

扫描模式，以及从高端广播到在线流媒体的比特率范围。它甚至有一个“视觉无损”压缩选项，这意味着虽然它肯定会丢失图像细节，但在大多数情况下你几乎不会注意到。它非常适合在普通硬件上进行实时编辑（这意味着其他格式不适合！）。它已被SMPTE正式标准化为VC-3，因此得到了众多硬件和软件产品的支持。

3.4 容器

容器，或称封装器，是一种包含编解码器的“外壳”。例如，H.264视频编解码器可以被“封装”在MOV、MP4或AVI容器中。这些容器或封装器具有内部支持工具，以促进编解码器的运行，并且它们在功能上确实有所不同——例如，对字幕或章节的支持。此外，您正在使用的某个应用程序可能接受.mov文件，但无法处理.avi文件，即使这两个容器都包含H.264编解码器。以下是详细解析。

3.4.1 MOV

由苹果公司开发的.mov文件格式也被称为QuickTime电影。它是一种多媒体容器，可容纳视频、音频、字幕文本、时间码等。.mov文件格式在Mac和Windows平台上均受到众多应用程序的支持。作为多媒体容器，例如，它可以包含H.264视频、AAC或MP3音频以及隐藏字幕数据流。这里的关键在于，.mov文件可以包含任意数量的编解码器，非常适合编辑和最终展示交付。

3.4.2 MP4

正式名称为MPEG-4，这是一种由动态图像专家组（MPEG）开发的视频、音频、字幕甚至静态图像的容器格式。其普及得益于在几乎所有平台上的广泛使用以及在画质与文件大小之间的平衡。它具备两项独特功能：渐进式下载与自适应流媒体。渐进式下载允许用户在文件未完全下载时即可开始观看视频，而自适应流媒体则能根据网络状况动态调整画质。

the internet connection speed as it is playing. This makes .mp4 files very popular with streaming services.

3.4.3 AVI

This Microsoft file wrapper also know as Audio Video Interleave (AVI) is a multimedia container format for audio and video which synchronizes the audio to the video upon playback. While the .avi file format is well supported by Microsoft products (naturally), it is a bit less supported elsewhere than .mp4 or .mov file formats. It lacks some of the modern features of the previous wrappers, such as subtitles, menus, and metadata. Its support for lossless video compression codecs means that .avi file sizes can become ungainly. Did I mention that it works with Microsoft products?

3.4.4 Summary

While all these containers are prevalent throughout the industry, it is still true that the QuickTime MOV is an Apple product and works best with Apple products, while AVI is a Microsoft product so works best in a Microsoft environment. The MP4 container, however, is an industry-agnostic product created by the Motion Picture Expert Group, an industry committee, and is the most accepted and commonly used container of all.

3.5 Log Images

Log images are first and foremost a data compression scheme. Storing data in a log format allows for a much more efficient allocation of bits than the normal linear file formats like TIFF and PNG. It also allows for more black detail and avoids clipping the highlights. However, many don't really understand what log images actually are, so in this section, we will learn what they are, how they are created, and their strengths and weaknesses.

The log format is specifically designed for photographic images, as opposed to graphic images, which can be efficiently encoded with run-length

在播放时，互联网连接速度会随之调整。这使得 .mp4 文件在流媒体服务中非常受欢迎。

3.4.3 AVI

这份微软文件封装格式，亦称音频视频交错（AVI），是一种多媒体容器格式，用于同步音频与视频的播放。尽管.avi文件格式在微软产品中得到了良好支持（这很自然），但在其他平台上的支持度略逊于.mp4或.mov格式。它缺少先前封装格式的一些现代功能，例如字幕、菜单和元数据。其对无损视频压缩编解码器的支持意味着.avi文件体积可能变得臃肿。我是否提过它适用于微软产品?

3.4.4 总结

尽管这些容器格式在行业内普遍存在，但QuickTime MOV作为苹果公司的产品，在苹果生态系统中表现最佳；而AVI是微软的产品，因此在微软环境下运行最为顺畅。然而，MP4容器是由行业委员会——运动图像专家组（Motion Picture Expert Group）制定的行业通用格式，它是目前接受度最高、使用最广泛的容器格式。

3.5 日志图像

Log图像首先是一种数据压缩方案。以Log格式存储数据，相比TIFF和PNG这类常规线性文件格式，能实现更高效的比特分配。它还能保留更多暗部细节，并避免高光部分被裁切。然而，许多人并不真正理解Log图像的实际{v*}

因此，在本节中，我们将了解它们是什么、如何创建，以及它们的优缺点。

日志格式专为摄影图像设计，而非图形图像，后者可通过行程编码高效压缩。

encoding. Log images are not suitable as a workspace or for display, except for colorists that prefer to color grade in log. They are attractive for storage because they offer high dynamic range, no banding, and smaller file sizes. A 10-bit log file has more image information than a 12-bit linear file. However, you should know that all log images have a maximum code value. For a 10-bit log DPX file, for example, the maximum code value of 1023 linearizes to float 13.5. Unless you are shooting into the sun, very few shots would ever actually go that high.

Floating point file formats such as OpenEXR and TIFF 16-bit floating point also offer a high dynamic range with no banding in the darks. However, as they are 16-bit files, they have a larger file size than a 10 or 12-bit log file. So the log file formats offer low-loss data compression without banding while still supporting HDR images. The "low loss" here is simply the fact that the original image is put through a floating-point log operation that introduces microscopic round-off errors. That said, as time marches on, the larger file sizes of EXR images will become less and less of an issue.

3.5.1 Why Log?

Before tearing into understanding log images, we should understand what the problem is with linear images like TIFF and PNG when shooting high dynamic range content. The bottom line is that linear file formats do not store image data efficiently. You may recall from Chapter 1 that our visual system has a log response. We are very sensitive to small changes in brightness down in the darks, but our vision loses sensitivity as things get brighter. If we want to encode images efficiently, we need a bit allocation method that mimics our visual system's response curve with code values close together in the darks and further apart in the brights.

In Chapter 1, we learned the 1% rule for human vision, which means that the eye needs about a 1% difference in brightness between two adjacent color chips for us to see the difference. In the darks, a 1% difference is a small value. In the brights, a 1% difference is a large value because our eyes are losing sensitivity in the brights.

If the brightness steps in the darks are too far apart (1% or more), we will see banding. If the brightness steps in the brights are too close together, there will be wasted bits as we will not be able to see the difference between, for

编码。对数图像不适合作为工作空间或用于显示，除非是偏好在对数模式下进行色彩分级的调色师。它们对于存储具有吸引力，因为它们提供高动态范围、无条带效应以及更小的文件大小。一个10比特的对数文件比12比特的线性文件包含更多的图像信息。然而，您应该知道，所有对数图像都有一个最大编码值。例如，对于一个10比特的对数DPX文件，最大编码值1023线性化后对应浮点数13.5。除非您正对着太阳拍摄，否则很少有镜头会实际达到如此高的值。

诸如OpenEXR和TIFF 16位浮点等浮点文件格式同样提供了高动态范围，且暗部无条带效应。然而，由于它们是16位文件，其文件体积比10或12位对数文件更大。因此，对数文件格式在支持HDR图像的同时，能实现低损耗数据压缩且避免条带问题。此处的“低损耗”仅指原始图像经过浮点对数运算时会产生微小的舍入误差。值得一提的是，随着时间推移，EXR图像较大的文件体积将越来越不成问题。

3.5.1 为什么使用对数?

在深入理解对数图像之前，我们应当先弄清楚拍摄高动态范围内容时，像TIFF和PNG这类线性格式图像存在什么问题。关键在于线性文件格式无法高效存储图像数据。你可能还记得第一章提到，我们的视觉系统具有对数响应特性。我们对暗部区域的细微亮度变化极为敏感，但随着亮度提升，视觉敏感度会逐渐下降。若想高效编码图像，就需要一种能模拟视觉系统响应曲线的比特分配方法——使暗部区域的编码值更密集，而亮部区域的编码值间隔更宽。

在第一章中，我们学习了人类视觉的1%法则，这意味着眼睛需要相邻两个色块之间约有1%的亮度差异，我们才能看出区别。在暗部区域，1%的差异是一个很小的数值；而在亮部区域，1%的差异则是一个很大的数值，因为我们的眼睛在亮部会逐渐丧失敏感度。

如果暗部的亮度阶跃过大（1%或更多），我们将看到条带。如果亮部的亮度阶跃过于接近，将会浪费比特位，因为我们无法区分

example, 8-bit code values 250 and 253, as their on-screen pixel brightness difference is less than 1%. The problem with equally spaced brightness in linear files is that if the bits are close together to avoid banding in the darks, it wastes bits in the brights. Space the bits wider to avoid wasting them in the brights and we will get banding in the darks.

This conundrum is neatly illustrated in Figure 3.12. The *linear for darks* gradient at the top illustrates the linear code values close together to avoid banding in the darks. Unfortunately, they are also close together in the brights, so it will take several code values before there is a perceptible change in brightness to the eye, which results in wasted bits. Not an efficient encoding scheme.

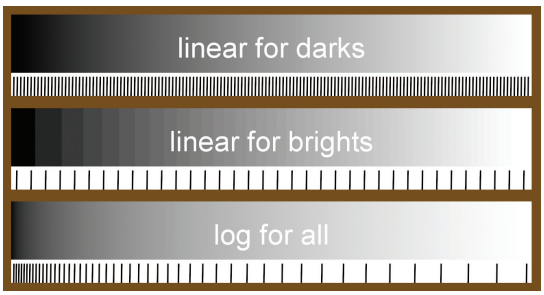


Figure 3.12 Bit allocation for linear vs. log images

The *linear for brights* gradient in the middle illustrates placing the linear code values far apart to avoid wasting bits in the brights. Unfortunately, the wider spaced bits result in banding in the darks which is quite visible in the gradient. Not a quality encoding scheme. Interestingly, the banding seems to disappear in the brights. So a linear encoding scheme representing equal brightness steps must invariably fail at one end or the other.

The *log for all* gradient at the bottom illustrates placing log code values with their variable spacing close together in the darks to avoid banding but further apart in the brights to avoid wasting bits. This is an efficient *and* quality encoding scheme. We just need a simple mathematical rule to describe this variable spacing, and that rule is log.

Besides data efficiency and no banding, there is a third benefit of log images: they can handle HDR images—at least up to a point. 10-bit log images (i.e., Cineon and DPX) cover the range of floating point values of 0 to 13.5, with other log formats having different upper limits. So log images deliver greater data packing efficiency along with support for HDR images. Better pictures in a smaller file size. And that is the point.

例如，8位码值250和253，它们在屏幕上的像素亮度差异小于1%。线性文件中亮度等间距的问题在于，如果比特间距过小以避免暗部出现条带，就会在亮部浪费比特。将比特间距调大以避免在亮部浪费，则暗部又会出现条带。

这个难题在图3.12中得到了清晰的展示。顶部的暗部线性渐变展示了为避免暗部出现条带效应而紧密排列的线性码值。遗憾的是，这些码值在亮部也同样排列紧密，因此需要经过多个码值才能实现

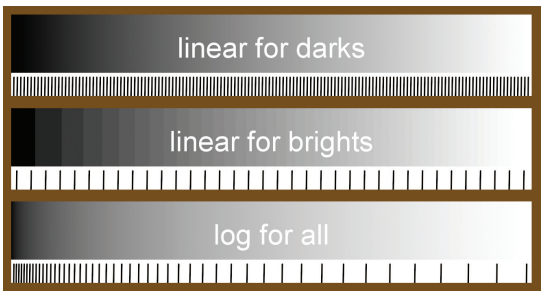


图 3.12 线性图像与对数图像的比特分配

亮度上人眼可察觉的变化，这导致了比特的浪费。并非一种高效的编码方案。

中间亮部梯度的线性编码展示了将线性码值间隔设置得较远，以避免在亮部浪费比特。然而，这种更宽的比特间隔导致了暗部的条带效应，这在梯度中相当明显。这并非一种优质的编码方案。有趣的是，条带效应似乎在亮部消失了。因此，一种代表等亮度步长的线性编码方案必然会在某一端失败。

底部所有梯度的对数图示表明，将对数代码值以变量间距紧密排列在暗部以避免条带，而在亮部则间距较大以避免浪费比特。这是一种高效且高质量的编码方案。我们只需要一个简单的数学规则来描述这种变量间距，而这个规则就是对数。

除了数据效率和没有条带效应外，对数图像还有第三个好处：它们可以处理HDR图像——至少在一定程度上。10位对数图像（即Cineon和DPX）覆盖了0到13.5的浮点数值范围，其他对数格式则有不同的上限。因此，对数图像在支持HDR图像的同时提供了更高的数据打包效率。更小的文件尺寸，更好的图像质量。这正是关键所在。

3.5.2 What is Log Data?

Author's note: This section is optional, so if you are math averse it can be skipped without permanent harm to your career. The rest of this chapter, however, is mandatory, starting back in Section 3.5.3. See you there …

Log images store the log *exponents* of the RGB values instead of the RGB values themselves. "Log" is short for "logarithmic," which refers to the math function of raising a number to some power, such as 2^3, where "2" is the base and "3" is the log exponent. This, of course, means to multiply the base (2) by itself three times (2 x 2 x 2), which will return the number 8. So we can design an encoding scheme that only stores the exponents as long as the base is defined.

Log file formats like Cineon and DPX are 10-bit integer, so their raw file data ranges from 0 to 1023. This integer data is put through a log-to-linear conversion to produce the floating-point equivalent to be used for floating-point applications. For example, 10-bit integer code value 596 becomes float 0.5. If the app is 8-bit integer, this floating-point number will subsequently be converted to integer 128. The maximum 10-bit code value of 1023 will be evaluated to float 13.5, the maximum brightness that the DPX 10-bit log supports.

To describe an RGB code value with 10-bit log, you will need three 10-bit fields for a total of 30 bits. This fits nicely within a single 32-bit computer "word," thus requiring only 4 bytes per pixel. For a 16-bit float like EXR, you need three 16-bit fields for a total of 48 bits. This requires 6 bytes per pixel—50% more bytes per pixel than the 10-bit log. However, as we saw earlier, the maximum code value DPX supports is 13.5, while EXR ranges up to 65,504. A searing number like this might be useful if you are rendering the interior of a blue supergiant star, but for more earthly needs, the DPX maximum will do fine.

Figure 3.13 DPX maximum code value of 13.5 satisfies earthly needs

3.5.2 What is Log Data?

作者注：本节为可选内容，因此如果您对数学感到不适，可以跳过，不会对您的职业生涯造成永久性损害。不过，本章其余部分为必读内容，从第3.5.3节开始。我们那里见……

对数图像存储的是RGB值的对数指数，而非RGB值本身。“Log”是“对数”的简称，指将某个数提升到特定幂次的数学函数，例如2^3，其中“2”是底数，“3”是对数指数。这意味着将底数（2）自乘三次（2×2×2），最终得到数字8。因此只要明确定义底数，我们就可以设计一种仅存储指数的编码方案。

像Cineon和DPX这样的日志文件格式采用10位整数存储，因此其原始文件数据的取值范围为0到1023。这些整数数据会经过对数到线性的转换，生成对应的浮点数以供浮点运算应用。例如，10位整数值596会转换为浮点数0.5。若应用程序使用8位整数，该浮点数随后会被转换为整数值128。最大的10位编码值1023将对应浮点数13.5，这也是DPX 10位对数格式支持的最高亮度值。

要用10位对数描述一个RGB编码值，你需要三个10位字段，总共30位。这恰好能容纳在单个32位计算机“字”中，因此每个像素只需4字节。对于像EXR这样的16位浮点数，你需要三个16位字段，总共48位。这需要每个像素6字节——比10位对数每个像素多50%的字节。然而，正如我们之前所见，DPX支持的最大编码值为13.5，而EXR的范围高达65,504。如果你正在渲染一颗蓝超巨星内部，这样惊人的数值或许有用，但对于更接地气的需求，DPX的最大值已经足够。

图 3.13 DPX最大码值13.5满足现实需求

So how much picture will the DPX maximum of 13.5 get you? Figure 3.13 shows a night shot of a burning building, which is an extreme example of dark detail with superbright fire pixels. The camera exposure was opened up to capture the night detail, so the fire is severely overexposed. However, even in this extreme example, the highest code value is 11.9 in the red channel—well below the maximum of 13.5. And just to prove the point, the right picture has had a gain of 0.1 applied to darken it way down to reveal that the fire detail is preserved with no clipping.

At this point, inquiring minds might want to know the difference between a log function and a gamma function. As we just saw, the log function raises a fixed base of 2 to the power of the exponent to describe the output log code values like this:

$$\textbf{log: output cv} = 2^{\textbf{exponent}}$$

A gamma function, however, is a power function that takes the input pixel code values (cv) and raises them to the power of gamma, like this:

$$\textbf{gamma: output cv} = \textbf{input cv}^{\textbf{gamma}}$$

where the gamma value depends on the color space. Figure 3.14 illustrates the very different shapes of a log curve vs. a gamma curve of 2.5. Now, aren't you glad you didn't skip this section?

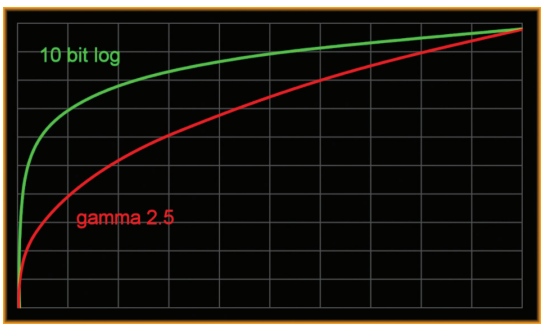


Figure 3.14 Log vs. gamma curves

3.5.3 Data Point Distribution

As we have seen, the whole magic behind log images is how the data points are distributed. Here we will visualize how the data samples are distributed after log images are linearized for the workspace. Figure 3.15 tells it all. On the left is the original linear image, such as the raw output of a camera. The data sample points are equally spaced from black to white—the very definition of a linear image. The middle log image was created by converting the linear image to log, then resampling (re-digitizing) when writing it to disk for storage.

那么DPX最大值13.5能为你带来多少画面细节？图3.13展示了一栋燃烧建筑的夜景照片，这是一个暗部细节与超亮火焰像素并存的极端案例。相机为捕捉夜景细节而延长曝光时间，导致火焰严重过曝。然而即便在此极端情况下，红色通道的最高编码值仅为11.9——远低于13.5的峰值上限。为验证此结论，右侧图片已施加0.1增益进行大幅压暗处理，可见火焰细节完整保留且未发生剪切。

此时，好奇的读者可能想知道对数函数与伽马函数之间的区别。正如我们刚才所见，对数函数将固定底数2提升到指数幂，以描述输出对数代码值，如下所示：

$$\text{日志：输出 cv} = 2^{\text{exponent}}$$

伽马函数是一种幂函数，它接收输入像素代码值（cv）并将其提升至伽马次幂，如下所示：

$$\text{gamma：输出 cv} = \text{输入 cv}^{\text{gamma}}$$

伽马值取决于色彩空间。图3.14展示了以2.5为参数的伽马曲线与对数曲线在形态上的显著差异。现在，您是否庆幸自己没有跳过这一节？

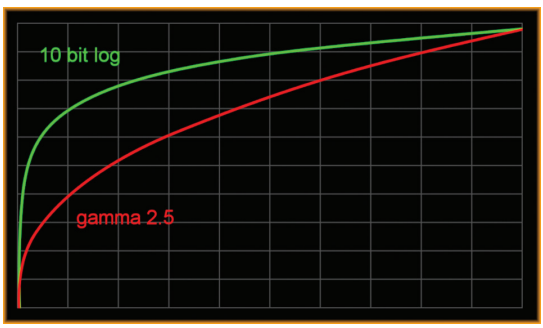


图 3.14 对数与伽马曲线

3.5.3 数据点分布

正如我们所见，对数图像背后的全部奥秘在于数据点的分布方式。这里我们将直观展示对数图像在工作区线性化后数据样本的分布情况。图3.15说明了一切：左侧是原始线性图像（例如相机的原始输出），数据采样点从黑到白均匀分布——这正是线性图像的定义。中间的对数图像是通过将线性图像转换为对数格式，并在写入磁盘存储时重新采样（重新数字化）而生成的。

When the log image is read in from disk and linearized for processing, you get the linearized log image on the right. Note that the original linear image and the linearized log image will be visually identical, but they will not be data identical.

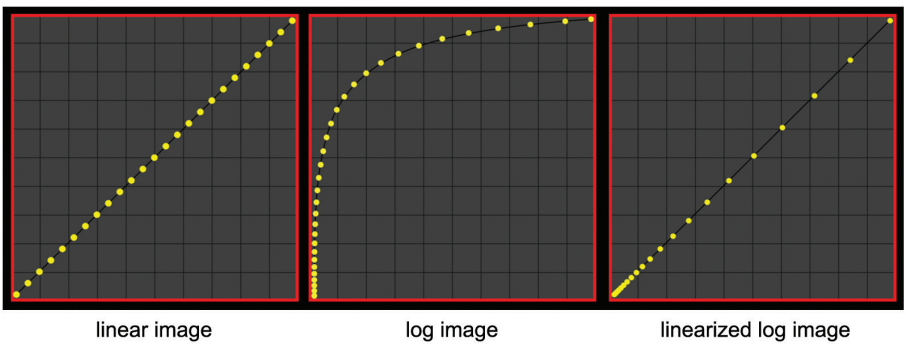


Figure 3.15 How log images distribute code values

So when a log image such as a DPX file is read in from disk, it is first linearized before any work can begin, resulting in the data sample points being distributed like the linearized log image in Figure 3.15. This reveals that the linear data points are now very close together in the darks and become spread out as you move up into the highlights. As a result, the tightly packed dark code values represent very small changes in brightness between them, while the spread-out code values in the highlights represent very large changes in brightness. And this is exactly what we want.

3.5.4 How Log Images Are Created

So far, we have focused on understanding what a log image is, its virtues, and how it works. Here we will look at how log images are created. They are created by passing a linear image through a simple 1D LUT (Look Up Table), which is quick, or it can be done computationally with a math function. Note that converting linear data to log is **not** a color space conversion. It is a data conversion function.

当从磁盘读取对数图像并进行线性化处理时，你会得到右侧的线性化对数图像。请注意，原始线性图像与线性化对数图像在视觉上完全相同，但数据层面并不一致。

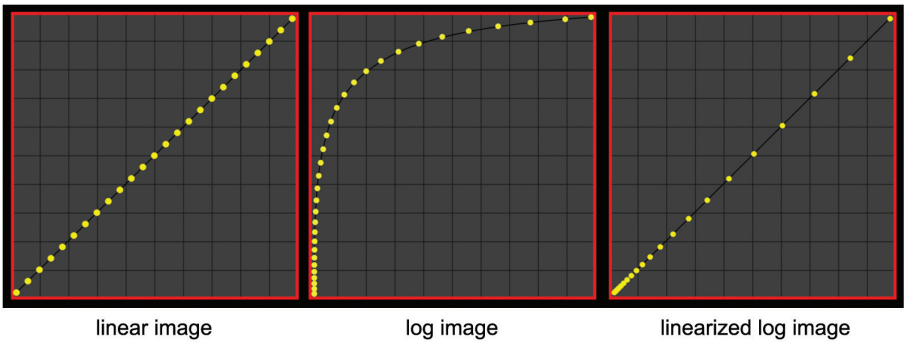


图3.15 对数图像如何分布代码值

因此，当从磁盘读取诸如DPX文件这样的对数图像时，首先需要对其进行线性化处理，然后才能开始任何工作，这导致数据采样点的分布类似于图3.15中的线性化对数图像。这表明线性数据点在暗部区域非常密集，随着向高光区域移动，它们逐渐变得分散。因此，紧密排列的暗部代码值代表它们之间非常微小的亮度变化，而高光区域分散的代码值则代表非常大的亮度变化。而这正是我们想要的效果。

3.5.4 日志图像如何创建

到目前为止，我们主要探讨了什么是对数图像、其优点以及工作原理。这里我们将了解对数图像的创建方式。它们是通过将线性图像经过一个简单的一维查找表（LUT）快速生成，或者也可以通过数学函数进行计算处理。请注意，将线性数据转换为对数并非色彩空间转换，而是一种数据转换函数。

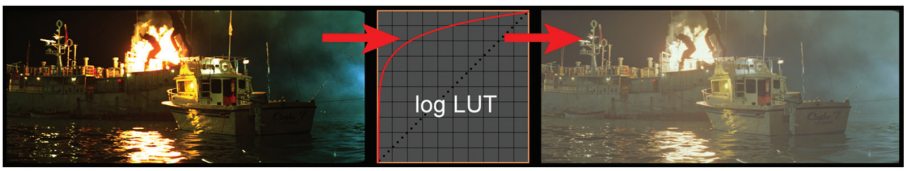


Figure 3.16 Digital camera workflow

Figure 3.16 illustrates the workflow from a digital camera. The camera raw image on the left is the debayered output of the imaging sensor, which is linear because imaging sensors are inherently linear devices. It is dark because it has not been gamma-corrected yet. It is passed through a log LUT (middle image), producing the log version on the right. You can now see why log images are all flat and low contrast by examining the log LUT curve. The darks are lifted way up and the brights are heavily compressed. After the log LUT, the image is resampled to typically 10 or 12 bits when writing to disk to produce the log image for storage.

Hopefully, by now you are quite comfortable with log images—what they are, their virtues, and their limitations. The last topic is how they fit into the overall digital imaging workflow, which is covered in the next chapter, Work Space.

3.5.5 Log for Photography

The wonders of log images have penetrated still photography with a slightly different spin than cinematography. For cinematography, the main interest lies in low-loss compression while maintaining dark detail and high dynamic range. Photographers are less interested in the data compression than in retaining the dark detail. Incorrectly referred to as "log recording," the images are still recorded linearly by the imaging sensor but are simply passed through a log LUT before exporting. As a result, they inherit the advantages of log images—namely dark detail and high dynamic range.

For years, photographers have been capturing digital images in an 8-bit sRGB format that immediately looks good on a monitor if properly exposed. The problem with this approach is that the captured image is in "display

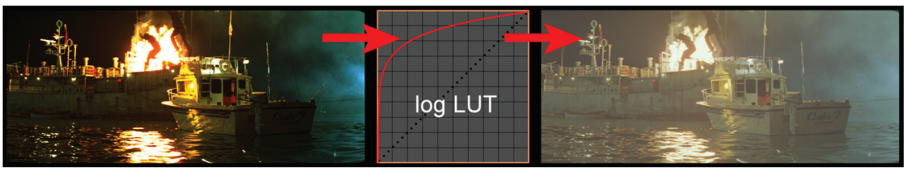


图 3.16 数码相机工作流程

图3.16展示了数码相机的工作流程。左侧的相机原始图像是成像传感器的去拜耳输出，由于成像传感器本质上是线性设备，因此该图像呈线性状态。图像显得暗淡是因为尚未进行伽马校正。它经过对数查找表处理（中间图像），生成右侧的对数版本。通过观察对数查找表曲线，你现在可以理解为何对数图像总是显得平坦且对比度低——暗部被大幅提升，而亮部则被严重压缩。经过对数查找表处理后，图像在写入磁盘时通常会重采样为10位或12位，以生成用于存储的对数图像。

希望到目前为止，你已经对对数图像——它们是什么、它们的优点以及它们的局限性——感到相当熟悉了。最后一个主题是它们如何融入整个数字成像工作流程，这将在下一章“工作空间”中介绍。

3.5.5 摄影日志

对数图像的奇迹已经以略微不同于电影摄影的方式渗透到静态摄影中。对于电影摄影，主要关注点在于低损耗压缩，同时保持暗部细节和高动态范围。摄影师对数据压缩的兴趣不如对保留暗部细节的兴趣浓厚。错误地

被称为“对数记录”，图像仍然由成像传感器线性记录，但在导出前仅通过一个对数查找表（LUT）处理。因此，它们继承了对数图像的优势——即暗部细节和高动态范围。

多年来，摄影师一直以8位sRGB格式捕捉数字图像，只要曝光得当，这些图像在显示器上立刻就能呈现出良好效果。这种方法的问题在于，捕捉到的图像处于“显示

space" (more on this in Chapter 5: Display Space) meaning it is designed to fit nicely on a display device but lacks extra latitude for major manipulation such as heavy color correction. The typical DSLR camera captures a scene in 12 to 15 bits linear, then crushes it to 8 bits upon export as a JPEG. With a log export, the original linear bits are converted to log before being written to the camera's memory card and then exported as a log image. The downside is that the log image does not immediately look good on a monitor because it must be converted to linear and color-corrected before viewing by the public, whereas the JPEG output offers instant gratification as it is immediately viewable.

Figure 3.17 illustrates the visual difference between a typical JPEG camera export and a log export after it has been color-corrected. The .jpeg export (left) has crushed black shadows with very little detail. You can see the preserved shadow detail in the log export (center), but the tone range is flat and low contrast, so it is not ready for primetime viewing. After color-correcting (right), the log image has a normal tone range, but the shadow detail has been preserved to the point you can see the tires distinct from the shadow. Attempting to color-correct the original JPEG by raising the blacks would have resulted in banding artifacts because the dark detail had been discarded.

Figure 3.17 Comparison of a JPEG export to a color-corrected log export

Let's be clear—there is no "shooting in log" or "log photography." There is only shooting in linear, as that is the nature of the imaging sensor. There is only exporting linear images with a conversion to log, whether that is done inside or outside of the camera. That said, I am sure that the practitioners of the digital arts will continue to use "shooting log" or something similar as it is engrained in the lexicon of the industry. Hell, I'll probably use it too, just to avoid continually having to explain myself. See what happens when people don't know color science?

“显示空间”（更多内容见第5章：显示空间）意味着它设计用于在显示设备上良好适配，但缺乏进行重大调整（如重度色彩校正）的额外宽容度。典型的数码单反相机以12至15位线性方式捕捉场景，然后在导出为JPEG时将其压缩至8位。通过日志导出，原始线性数据在写入相机存储卡之前会转换为对数格式，随后导出为日志图像。其缺点是日志图像在显示器上不会立即呈现良好效果，因为它必须先转换为线性格式并进行色彩校正，才能供公众观看；而JPEG输出则能提供即时满足感，因为它可直接查看。

图3.17展示了典型JPEG相机导出文件与经过色彩校正后的对数导出文件在视觉上的差异。左侧的.jpeg导出文件存在黑色阴影区域细节严重丢失的问题。中间的对数导出文件保留了阴影细节，但影调范围平坦、对比度低，因此不适合直接用于正式观看。经过色彩校正后（右侧），对数图像的影调范围恢复正常，同时阴影细节得以保留，甚至能清晰分辨轮胎与阴影的界限。若尝试通过提亮黑色区域来校正原始JPEG文件，则可能因暗部细节已被丢弃而产生色彩断层现象。

图 3.17 JPEG导出与色彩校正对数导出的对比

必须明确一点——不存在所谓的“log拍摄”或“log摄影”。成像传感器的本质决定了拍摄只能是线性的。所谓的log拍摄，实际上只是将线性图像通过转换输出为log格式，无论这个转换过程发生在相机内部还是外部。话虽如此，我相信数字艺术的从业者们仍会继续使用“拍摄log”这类说法，毕竟这已成为行业词汇的一部分。说实话，我可能也会继续这么用，毕竟谁想反复解释这些概念呢？看看人们不了解色彩科学时会发生什么就知道了？

See several examples of shadow detail preserved by log images but lost with JPEGS.

3.6 Cinema Export Options

Filmmakers and videographers want the maximum image quality in post-production with an eye toward archiving the job to future-proof it. The image quality is constrained by the file formats chosen in two places: the camera export and the file storage format. The camera export represents the amount of data you have to work with for the project, while the storage format represents how much of that data will be preserved during post. Here we will look at camera export options with their powers and limitations. We will look at the file format storage options in the following section.

The selection of what file format to export from the camera sets the upper limit of the image quality at the head end of the production. Camera RAW preserves the most image data but is not suitable for the pipeline. Camera log preserves the next most image data, but it too is not suitable for the pipeline. Selecting a compressed file format for export will save a lot of disk space but with a permanent loss of image quality. Here we will look closely at all three options so you may choose wisely. A mistake here will travel down the entire production pipeline and ricochet through your career like firing a pistol in a diving bell.

3.6.1 Camera RAW

Camera RAW is essentially a raw dump of the imaging sensor's 12 to 15-bit linear data. Some cameras export the Camera RAW debayered (maybe with some options), while others export the raw sensor data and provide tools for the colorist to dial in the debayering process itself for the best look. Either way, the advantage is a lossless workflow that retains the maximum image

查看几个由对数图像保留但JPEG格式丢失的阴影细节示例。

3.6 影院导出选项

电影制作人和摄像师希望在后期制作中获得最高的图像质量，并着眼于将工作存档以使其经得起未来考验。图像质量受到两个环节所选文件格式的限制：相机导出和文件存储格式。相机导出代表了你拥有的数据量

用于项目的处理，而存储格式则代表了在后期制作中这些数据将被保留多少。这里我们将探讨相机导出选项及其优势和限制。我们将在下一节中查看文件格式存储选项。

选择从相机导出何种文件格式，决定了制作流程起点的图像质量上限。相机RAW格式保留了最多的图像数据，但不适用于制作流水线。相机log格式保留了次多的图像数据，同样不适用于流水线。选择压缩格式进行导出能节省大量磁盘空间，但会永久损失图像质量。本文将详细剖析这三种选项，助您做出明智选择。此处的决策失误将贯穿整个制作流程，如同在潜水钟里开枪般，在您的职业生涯中反复回荡。

3.6.1 相机RAW

Camera RAW本质上是成像传感器12至15位线性数据的原始转储。部分相机导出经过去马赛克处理的Camera RAW（可能附带一些选项），而另一些则导出原始传感器数据，并为调色师提供工具，以便通过调整去马赛克处理流程本身来获得最佳视觉效果。无论哪种方式，其优势都在于实现无损工作流，从而保留最大限度的图像信息。

information. The disadvantage is that a lot of work and expertise are required before getting a useful image. One downside to exporting Camera RAW is that some apps may not be able to read the Camera RAW files plus cameras come and go, so even fewer will be able to read it in the future.

Camera RAW is, of course, the file format that preserves 100% of the camera data, but at a heavy cost in terms of storage space and transfer times. Considered the digital camera negative, some cinematographers insist on this maximum image data to provide maximum flexibility in post. The Camera RAW will be exported and then processed to select the "sweet spot" of the footage for conversion to the chosen file format for the job, like ProRes, DPX, or EXR. Some cameras offer the option to export files as an EXR file directly—but then you lose the option to select the "sweet spot" of the footage.

3.6.2 Camera Log

Just below the Camera RAW file in picture quality is the camera log—a good compromise between quality and file size. When camera log is selected for output, the camera has captured the scene in linear Camera RAW, then debayered it and passed it through a log LUT in the camera during export. The advantages are excellent data retention in the dark detail, high dynamic range, and reduced file size. The disadvantages are that it needs a color space conversion and color correction before getting a useful image, and there is also the maximum code value limitation described earlier. It is rare for the image content to exceed the maximum log code value, but it is a consideration before committing the entire project to a log export. Although camera log contains less information than Camera RAW, it represents the most picture information possible in a reasonable file size.

3.6.3 Compressed Exports

Today's digital cinema cameras, like RED and ARRI, also have options to export compressed formats such as ProRes and DNxHR, which puts them in display space right out of the camera. However, this means giving up control over what is the best part of the picture for conversion. These might be viable options for television where only light processing is to be done in post, but they are not really suitable for high-end feature film work.

信息。缺点是在获得有用图像之前需要大量的工作和专业知识。导出相机RAW的一个缺点是，某些应用程序可能无法读取相机RAW文件，而且相机型号不断更新换代，因此未来能够读取它的软件可能会更少。

相机RAW格式无疑是能100%保留相机数据的文件格式，但代价是巨大的存储空间和传输时间。它被视为数码底片，部分电影摄影师坚持使用这种最大化的图像数据，以便在后期制作中获得最大灵活性。Camera RAW文件将被导出并处理，从中选取素材的“最佳表现区间”转换为工作所需的指定格式，如ProRes、DPX或EXR。某些相机支持直接导出EXR文件——但这将导致无法选取素材的“最佳表现区间”。

3.6.2 相机日志

在画质方面仅次于Camera RAW文件的是相机log格式——这是画质与文件大小之间的良好折中方案。选择输出相机log时，相机会以线性Camera RAW格式捕捉场景，随后在导出过程中进行去马赛克处理，并通过相机内置的log查找表转换。其优势在于能出色保留暗部细节数据、具备高动态范围，同时减小文件体积。缺点在于需要经过色彩空间转换和色彩校正才能获得可用图像，且存在前文提及的最大编码值限制。虽然图像内容极少会超出log最大编码值，但在决定将整个项目以log格式导出前仍需考虑此因素。尽管相机log包含的信息量少于Camera RAW，但它以合理的文件大小实现了最大限度的图像信息保留。

3.6.3 压缩导出

如今，RED和ARRI等数字电影摄影机也支持导出如ProRes和DNxHR这类压缩格式，使其能够直接从摄影机进入显示空间。然而，这意味着放弃了在转换过程中对画面最佳部分的控制权。对于后期只需轻度处理的电视制作而言，这些或许是可行的选择，但它们并不真正适用于高端故事片制作。

There are several "flavors" of ProRes, ranging from low to high compression rates to support low to high bit rates, but even the highest-quality ProRes 4444XQ, at 12 bits per color channel (plus a 16-bit alpha channel), is great for video but only marginally acceptable for the low end of feature film work. The compression is less severe and has a file size for a single 4K DCP frame of 8.8 MB, compared to 35.2 MB for DPX or up to 50 MB for EXR. However, it still has a limited dynamic range even though it is "several times that of Rec. 709" (HDTV). Good luck trying to find out if "several times" means three times or 10 times Rec. 709—keeping in mind that Rec. 709 is not a very high bar to begin with having only about 5.2 stops (or 37:1) of dynamic range. ProRes is also specifically designed to be an efficient streaming file format that rolls off the disk quickly, so it is very popular with video editors.

FUN FACT: Rec. 709 refers to "Recommendation 709" for HDTV production and broadcast specs from the International Telecommunications Union (ITU), a specialized agency of the United Nations.

3.7 Cinema File Formats

File formats for film and broadcast come in two broad categories—individual files for a sequence of frames, such as EXR or DPX, and containers that hold entire sequences in one file, such as MOV or ProRes. Each file format has its strengths, weaknesses, and intended uses, and it is up to you to choose wisely. A bad choice can result in a bloated database that slows down production or deprecated image quality that robs you of that Oscar. This is important stuff!

In a given production job, you may have to navigate through a wide range of file formats depending on what stage you are at and what the mission is. Here is a quick list of the kinds of files and when they might be needed:

- Scan of 35mm film: DPX, EXR
- Camera output: RAW, DPX, EXR, ProRes
- Post-production: DPX, EXR, ProRes

ProRes有多种“风味”，从低到高的压缩率支持低到高的比特率，但即便是最高质量的ProRes 4444XQ——每色彩通道12位（外加16位Alpha通道）——虽对视频处理表现出色，在故事片制作的低端应用中却仅勉强合格。其压缩程度较轻，单帧4K DCP文件大小为8.8 MB，而DPX格式为35.2 MB，EXR格式甚至高达50 MB。然而，尽管其动态范围“数倍于Rec. 709”（HDTV标准），实际表现仍有限制。要知道Rec. 709本身动态范围仅约5.2档（即37:1），起点并不高，而“数倍”究竟指三倍还是十倍则难以查证。ProRes专为高效流式传输设计，能快速从磁盘读取数据，因此深受视频编辑者青睐。

> 有趣的事实：Rec. 709指的是国际电信联盟（ITU，联合国的一个专门机构）制定的高清电视制作和广播规范“建议书709”。

3.7 影院文件格式

电影和广播的文件格式大致分为两类——用于帧序列的独立文件，如EXR或DPX，以及将整个序列封装在单个文件中的容器格式，如MOV或ProRes。每种文件格式都有其优势、局限性和适用场景，需要您明智选择。不当的选择可能导致数据库臃肿拖慢制作进度，或图像质量受损让您与奥斯卡失之交臂。这至关重要！

在特定的制作任务中，根据所处阶段和任务目标，您可能需要处理多种文件格式。以下是一份简要的文件类型及其适用场景列表：

- 35毫米胶片扫描：DPX、EXR格式
- 相机输出：RAW、DPX、EXR、ProRes
- 后期制作：DPX、EXR、ProRes

- Send a clip to the director's tablet: MOV, MP4, AVI
- Send a still image to the director's tablet: JPEG, TIFF, PNG
- Digital Intermediate: DPX, EXR, ProRes
- Theatrical release: DCP: MXF container with JPEG 2000 frames
- HD and UHD broadcast: MOV, AVI, MP4, ProRes
- A frame for online or print articles: JPEG, PNG, TIFF.

Of course, your production may vary, and as time moves on, new file formats will be introduced and others will fade away through neglect. The point of this list is to show how much we depend on different file types for different situations—a file format for every occasion.

Choosing a file format requires choosing a compression scheme, so one must understand which compression scheme is used in which file format. In this section, we will pull together all that we learned in Section 3.2 about bit depth, linear, and floating point to look at the prevailing file formats used in post-production and their relative merits.

3.7.1 EXR

The EXR file format is the highest image quality file format available today and has become the preferred choice for visual effects and future-proofing feature films. It was developed by ILM specifically for film scans and visual effects work, so it has support for stereo, extensibility, deep data, unlimited channels, and multi-part—the ability to read in selected channels without loading the entire frame. It is a linear floating-point file format with 16 bits per color channel in a half float format with an enormous dynamic range of 30 stops (or over 1 billion to 1) with a maximum code value of 65,504 compared to 13.5 for DPX.

Being floating point, it has incredible precision in the darks and midtones and is utterly immune to banding. The encoding scheme even increases precision as you get closer to zero. But EXR is a disk hog. The file size for a three-channel RGB 4K DCP (4096 x 2160) will be up to 50 MB per frame, depending on image content and compression choice. It supports lossy, lossless, and no compression options. The RLE and ZIP options are

•发送一个片段到导演的平板：MOV、MP4、AVI

•向导演的平板发送静态图像：JPEG、TIFF、PNG

•数字中间片：DPX、EXR、ProRes

•影院发行：DCP：采用JPEG 2000帧的MXF容器格式

•高清和超高清广播：MOV、AVI、MP4、ProRes

•在线或印刷文章的框架格式：JPEG、PNG、TIFF。

当然，你的生产情况可能有所不同，随着时间的推移，新的文件格式会被引入，而其他格式则会因无人问津而逐渐消失。这份清单的意义在于展示我们在不同情境下对各种文件类型的依赖程度——可谓每种场合都有对应的文件格式。

选择文件格式需要选择压缩方案，因此必须了解哪种文件格式使用哪种压缩方案。在本节中，我们将结合第3.2节中学到的关于位深度、线性和浮点数的知识，探讨后期制作中常用的文件格式及其相对优劣。

3.7.1 EXR

EXR文件格式是当今可用的最高图像质量文件格式，已成为视觉特效和未来电影制作的优选格式。它由ILM专门为胶片扫描和视觉特效工作开发，因此支持立体影像、可扩展性、深度数据、无限通道以及多部分功能——能够在不加载整个帧的情况下读取选定通道。这是一种线性浮点文件格式，每个颜色通道采用16位半浮点格式，拥有高达30档（或超过10亿比1）的巨大动态范围，最大编码值为65,504，而DPX格式仅为13.5。

由于采用浮点格式，它在暗部与中间调区域拥有惊人的精度，且完全不会出现色带断层。其编码方案甚至会在数值趋近于零时进一步提升精度。但EXR格式极其占用存储空间——一个三通道RGB 4K DCP（4096 x 2160）的单帧文件大小可能高达50MB，具体取决于图像内容与压缩方式的选择。该格式支持有损压缩、无损压缩及不压缩三种模式，其中RLE与ZIP选项为

lossless, while the others are high-quality lossy compressions with minimal image degradation. Its file size could be larger than the original camera log, but it is an infinitely more useful file format for post-production since most apps can read EXR files.

3.7.2 DPX

DPX is an integer log file format that is an updated version of the old Cineon log file format and is much used in visual effects. For a three-channel RGB image, its file size (4 bytes per pixel) is one-third smaller than an EXR (6 bytes per pixel) and is designed to optimize disk space with minimal losses from the original image. It has a little over 11 stops of dynamic range, with a maximum code value of 13.5 and a file size for a single 4K DCP frame of 35.2 MB. One thing to watch out for when working with DPX is that it also supports 10-bit Rec. 709 video. So, when someone hands you a DPX file, it could have either a log image *or* video. Surprise!

3.7.3 ProRes

Apple ProRes is a high-quality "visually lossless" video compression format for editing and is not intended for end-user viewing. All ProRes codecs use the now-familiar DCT for compression. Designed specifically to be a light load for processors to run on laptops or other small systems with excellent support for random access, it supports up to 8K resolution. It offers a range of codec options, each designed to maintain a target bit rate while maintaining the intended compression quality. Here are some of the more important ones:

- **ProRes 422** - Standard Quality; 10 bits per channel; 4:2:2 chrominance subsampling; no alpha channel. A high-quality video standard that is better than broadcast.
- **ProRes 422 HQ** - High Quality; 10 bits per channel; 4:2:2 chrominance subsampling; no alpha channel. A higher quality version than ProRes 422 with less DCT compression but larger file sizes.
- **ProRes 4444** - Finishing Quality Video; 10 or 12 bits per channel; 4:4:4:4 sampling; 16-bit alpha channel. A very high-quality version with no subsampling of chrominance. Great for keying greenscreens.

无损，而其他格式则是高质量的有损压缩，图像质量下降极小。其文件大小可能比原始相机日志更大，但由于大多数后期制作软件都能读取EXR文件，因此它是无限更实用的文件格式。

3.7.2 DPX

DPX是一种整数对数文件格式，它是旧版Cineon对数文件格式的更新版本，在视觉效果领域应用广泛。对于三通道RGB图像，其文件大小（每像素4字节）比EXR格式（每像素6字节）小三分之一，旨在以最小化原始图像损失的方式优化磁盘空间。它拥有略高于11档的动态范围，最大编码值为13.5，单帧4K DCP的文件大小为35.2 MB。处理DPX时需注意：它也支持10位Rec. 709视频。因此，当有人给你DPX文件时，它可能是对数图像或视频——意外吧！

3.7.3 ProRes

Apple ProRes 是一种用于编辑的高质量“视觉无损”视频压缩格式，并非面向最终用户观看。所有 ProRes 编解码器均采用如今为人熟知的 DCT 进行压缩。该格式专为处理器在笔记本电脑或其他小型系统上轻负载运行而设计，具备出色的随机访问支持能力，最高可支持 8K 分辨率。它提供一系列编解码选项，每种选项均旨在保持目标比特率的同时维持预设的压缩质量。以下是一些较为重要的选项：

- ProRes 422 - 标准质量；每通道10位；4:2:2色度二次采样；无Alpha通道。这是一种优于广播标准的高质量视频格式。

- ProRes 422 HQ - 高质量；每通道10位；4:2:2色度二次采样；无Alpha通道。相比ProRes 422，此版本质量更高，DCT压缩更少，但文件体积更大。

- ProRes 4444 - 后期制作质量视频；每通道10或12位；4:4:4:4采样；16位Alpha通道。此版本画质极高，色度信息未经二次采样，非常适合绿幕抠像处理。

- **ProRes 4444 XQ** - Finishing Quality Cinema; 10 or 12 bits per channel; 4:4:4:4 sampling. Supports HDR and 16-bit alpha channel; even less DCT compression than ProRes 4444.

3.7.4 DNxHR

Avid DNxHR is a high-quality lossy video compression format designed for multi-generation compositing with reduced storage and bandwidth requirements using DCT compression. While used for HD projects, it was specifically designed to support high-resolution projects (hence the "HR") at 2K, 4K, and 8K resolutions.

- **DNxHR SQ** - Standard Quality; 8-bit 4:2:2 chrominance subsampling.
- **DNxHR HQ** - High Quality; 8-bit 4:2:2 chrominance subsampling, with less compression.
- **DNxHR HQX** - Extra High Quality; 10-bit 4:2:2 chrominance subsampling; broadcast quality.
- **DNxHR 444** - Finishing Quality; 12-bit 4:4:4 no chrominance subsampling; cinema quality delivery; supports HDR and an alpha channel.

3.7.5 Archival

Once the job is done, the question becomes what file format to use to archive the work. The post-production pipeline file format may have used EXR, DPX, ProRes, or some other, but now it is time to think about preserving it for the future. For television, DPX or ProRes will be adequate. There are data losses—albeit small—when converting back and forth to DPX or ProRes, so keep this in mind when considering "dusting off" an old project for some last-minute tweaks.

For feature film, EXR is the most robust and least lossy file format, so many use it for both the post pipeline and the archival format. However, some are converting the final show back to the Camera RAW with which it was originally exported. The big advantage here is that if a shot needs to be tweaked later, it will intercut and can easily be color-corrected to match the rest of the footage.

- ProRes 4444 XQ - 影视级精修质量；每通道 10 或 12 位；4:4:4:4 采样。支持 HDR 与 16 位 Alpha 通道；相比 ProRes 4444 采用更低的 DCT 压缩率。

3.7.4 DNxHR

Avid DNxHR 是一种高质量有损视频压缩格式，专为多代合成设计，采用DCT压缩技术以降低存储和带宽需求。该格式虽适用于高清项目，但其专门设计旨在支持2K、4K及8K分辨率的高清项目（因此得名“HR”）。

- DNxHR SQ - 标准质量；8位4:2:2色度二次采样。
- DNxHR HQ - 高质量；8位4:2:2色度二次采样，压缩率更低。
- DNxHR HQX - 超高画质；10位 4:2:2 色度二次采样；广播级质量。
- DNxHR 444 - 后期制作质量；12位4:4:4无色彩二次采样；影院级交付质量；支持HDR及Alpha通道。

3.7.5 归档

工作完成后，问题就变成了使用何种文件格式来归档作品。后期制作流程可能曾采用EXR、DPX、ProRes或其他格式，但现在需要考虑如何为未来保存这些文件。对于电视制作，DPX或ProRes已足够。需注意的是，与DPX或ProRes格式相互转换时会产生数据损失——尽管微小——因此在考虑对旧项目进行"翻新"做最后调整时，请牢记这一点。

对于故事片而言，EXR是最稳健且损耗最小的文件格式，因此许多制作流程在后期管线及归档时都采用此格式。然而，也有部分制作者会将最终成片转回最初导出时使用的相机RAW格式。这样做的一大优势在于，若后续需要对某个镜头进行调整，它能无缝剪辑，并可通过调色轻松匹配其余素材。

3.8 Photography Export Options

The first and most important file format consideration is what to export from the camera. The options are camera raw, log recording, and a compressed file like JPEG. Some cameras offer the raw and log options in a DNG file which can be read by other apps. Following is a description of each option, with its advantages and disadvantages.

3.8.1 Camera Raw

"Camera raw" means the same thing for photographers as cinematographers— it is simply a data dump of the imaging sensor, with little or no processing other than debayering in most cases. It is also done for the same reason: so that the maximum image information is available for processing in post. The downsides are that you must have a file reader that can read the camera raw for your brand of camera, and you won't have a nice picture until you have done a color space conversion and grading. Many cameras use the DNG file wrapper for their camera raw because DNG is supported by most software apps, so if their camera raw becomes obsolete one day, you will still be able to read the DNG file.

DNG or not, camera raw is very helpful for a print workflow, for the simple reason that if the camera export file is a compressed file type like JPEG, it will have been compressed and clipped. So, if you have grand designs for high-quality print work, exporting the camera raw and converting it to a large workspace like Adobe RGB will get you the prettiest printing pixels possible. Adobe RGB and other large color spaces are covered in detail in Chapter 4: Work Space.

3.8.2 Log Recording (Camera Log)

Log recording ("camera log" in cinematography speak) offers a nice compromise between camera raw and a compressed export file format. The virtues and vices of log recording for photography were covered in detail in Section 3.5.5 as part of the discussion of log images in general, so we don't need to repeat them here.

3.8 摄影导出选项

首要且最重要的文件格式考量是相机导出何种格式。可选方案包括相机原始格式、对数记录以及JPEG等压缩文件。部分相机提供以DNG文件形式存储的原始格式与对数记录选项，该格式可被其他应用程序读取。以下将逐一阐述各选项的优劣特性。

3.8.1 Camera Raw

“相机原始数据”对摄影师和电影摄影师而言意义相同——它本质上是图像传感器的数据转储，在大多数情况下除了去马赛克外几乎未经任何处理。这样做的原因也相同：为了在后期处理时能获得最大的图像信息。其缺点在于，你必须拥有能读取对应品牌相机原始数据的文件阅读器，并且在完成色彩空间转换与调色前，你无法得到理想的画面。许多相机采用DNG文件封装其原始数据，因为大多数软件应用程序都支持DNG格式，这样即使未来某天相机原始格式过时，你仍能读取DNG文件。

无论是否为DNG格式，相机原始文件对于打印工作流程都非常有帮助，原因很简单：如果相机导出的文件是像JPEG这样的压缩文件类型，它就已经被压缩和裁剪过了。因此，如果你对高质量打印作品有宏伟的规划，导出相机原始文件并将其转换为像Adobe RGB这样的大色彩空间，将能让你获得尽可能精美的打印像素。Adobe RGB及其他大色彩空间的详细内容将在第4章：工作空间中介绍。

3.8.2 日志记录（相机日志）

日志记录（在电影摄影术语中称为“相机日志”）在相机原始格式和压缩导出文件格式之间提供了一个很好的折中方案。关于摄影中日志记录的优缺点，已在第3.5.5节中作为对数图像讨论的一部分进行了详细说明，因此这里无需重复。

3.8.3 Compressed Exports

Virtually all DSLR cameras offer JPEG as an export option. The advantage is that you have an immediately useful picture (assuming good exposure) that looks nice on a monitor because it has been saved out in the monitor's display space. It does this by discarding image data upon output, so the discarded data is unrecoverable. The disadvantages are that the camera has made all the color decisions for you (rightly or wrongly), and you have precious little latitude to do any serious color-correcting.

3.9 Photography File Formats

The issues associated with file formats for stills photography are very different than those for cinematography. One obvious example is file size. Finding storage for a couple of hundred photographs pales in comparison to the 144,000 frames of a typical feature film, times three or four states of production. Another is disk access speed. Many of the cinema file formats are highly optimized for streaming UHDTV frames off a disk at 30 fps, while this feature does not even show up in a stills camera's sales brochure.

Stills photography is also much more demanding for resolution, quality, and sharpness because the intended customer will end up staring at a large print of a single gorgeous still image rather than watching a frenetic bar fight on the big screen in a theater. Tragically, the gorgeous high-resolution still images captured and color-corrected by loving photographers must endure the indignation of high compression to be viewed over the internet. The key is to work with as much color information as possible and then compress the final output for display.

In this section, we will look at file storage options for photos. Certain cameras can export some of these file formats directly, but here we are looking at them for storage on disk to load in for editing, display, or print. If a photo is exported as camera raw or a log recording, it will not be suitable for display until after it has been color space converted, color corrected, and written out in a display-friendly file format. Here are the virtues and vices of several of the most important file formats in use today.

3.8.3 Compressed Exports

几乎所有单反相机都提供JPEG作为导出选项。其优势在于，您能立即获得一张可直接使用的照片（假设曝光良好），由于它以显示器的色彩空间保存，因此在屏幕上看起来效果出色。这是通过在输出时丢弃部分图像数据实现的，因此被丢弃的数据无法恢复。缺点在于相机已为您做出了所有色彩决策（无论正确与否），而您几乎没有余地再进行任何严肃的色彩校正。

3.9 摄影文件格式

静态摄影所涉及的文件格式问题与电影摄影截然不同。一个明显的例子是文件大小：存储几百张照片的需求，与一部标准故事片14.4万帧画面（还需乘以三到四个制作阶段）相比简直微不足道。另一个区别是磁盘访问速度——许多电影文件格式都针对以30帧/秒的速度从磁盘流式传输超高清电视画面进行了深度优化，而这功能甚至不会出现在静态相机的销售手册中。

静态摄影对分辨率、画质和清晰度的要求也高得多，因为目标客户最终会凝视一张大幅印刷的唯美静态照片，而非在影院大银幕上观看一场激烈的酒吧打斗。可悲的是，摄影师精心拍摄并调色的高清静态影像，必须经过高压缩的折损才能在互联网上传播。关键在于尽可能保留丰富的色彩信息，再将最终输出压缩至适合展示的规格。

在本节中，我们将探讨照片的文件存储选项。某些相机可以直接导出其中部分文件格式，但此处我们关注的是将其存储在磁盘上以供编辑、显示或打印时加载。若照片以相机原始格式或对数记录格式导出，则需经过色彩空间转换、色彩校正，并写入便于显示的文件格式后，才适合用于展示。以下是当今几种最重要文件格式的优缺点。

3.9.1 DNG

This Adobe file format is considered a “digital negative” intended to retain maximum RGB image data, so there is no alpha channel support. Camera raw can be saved as a DNG file, making it portable to almost every photo editing app. It has both lossless and minimal lossy compression options and is either native or optional on many cameras to export directly.

3.9.2 JPEG

This file format is designed to reduce disk space and file transfer times by compressing the image file. The compression is both lossy and adjustable, so you can dial in the file size you want. It is not a high dynamic range format because it is specifically designed for viewing images on desktop monitors, laptops, and cell phones, so it stays strictly within its display limitations of three channels only and there is no alpha channel.

3.9.3 PSD

This is uncompressed and designed to preserve all of the Photoshop layers and filters you apply when editing a picture, then display the final composite of all those layers. It is a workspace file format rather than a distribution or display format because it is a large file and requires Adobe products to read it.

3.9.4 PNG

This is a display and transferring file format that can preserve transparency with an alpha channel. It is optimized for graphic rather than photographic images, as it supports lossless compression using RLE or LZW.

3.9.5 TIFF

This supports both lossy and lossless compression, plus an alpha channel. It is adaptable to both photographic and graphic image content. It supports different color standards (e.g., RGB, CMYK) and comes in variable bit depths, including a 32-bit floating point HDR option.

3.9.1 DNG

这种Adobe文件格式被视为一种“数字底片”，旨在保留最大限度的RGB图像数据，因此不支持Alpha通道。相机原始数据可保存为DNG文件，使其能在几乎所有照片编辑应用中通用。它提供无损和最小程度有损的压缩选项，并且在许多相机上可直接导出，或作为原生支持或可选功能。

3.9.2 JPEG

该文件格式旨在通过压缩图像文件来减少磁盘空间和文件传输时间。其压缩方式兼具有损与可调节特性，因此您可根据需求调整文件大小。它并非高动态范围格式，因其专为在桌面显示器、笔记本电脑及手机屏幕上查看图像而设计，故严格限定于仅三个通道的显示限制内，且不包含阿尔法通道。

3.9.3 PSD

这是未经压缩的格式，旨在保留编辑图片时应用的所有Photoshop图层和滤镜，然后显示所有这些图层的最终合成效果。它是一种工作空间文件格式，而非分发或展示格式，因为文件体积较大，且需要Adobe产品才能读取。

3.9.4 PNG

这是一种能够通过Alpha通道保留透明度的显示和传输文件格式。它特别适用于图形而非摄影图像，因为它支持使用RLE或LZW进行无损压缩。

3.9.5 TIFF

它同时支持有损和无损压缩，并带有alpha通道。无论是摄影图像还是图形图像内容，它都能适应。它支持不同的色彩标准（如RGB、CMYK），并提供可变的位深度选项，包括32位浮点高动态范围（HDR）选项。

3.9.6 WebP

WebP is a new web-savvy image file format designed to have higher quality with smaller file sizes than JPEG for faster download times. It supports both lossy and lossless compression schemes, as well as transparency. A lossless compressed image will be 26% smaller than a PNG and a lossy compressed image will be 24%-34% smaller than a JPEG of equal quality, but will look nicer.

3.9.7 HEIF

High Efficiency Image Format (HEIF) is another new file format that was introduced in 2015 with the aim of replacing JPEG which, after 25 years, is showing limitations. It touts half the file size for equivalent quality, bit depth up to 16 bits, multiple images, metadata, depth information, and thumbnails. HEIF has a very bright future with the trend toward 4K and HDR images. Importantly, it has industry support from Apple, Canon, Windows 10, Adobe Lightroom, Affinity Photo, GIMP, and several other photo editing apps.

3.9.8 Do Not Use

Many file formats are “display space” formats only—they are specifically designed to be swiftly read from disk and quickly unpacked to view on a computer monitor or TV set in real-time. The emphasis is on low bit rates and reduced disk space requirements. Quality is a second, even a third priority. But all display formats suffer from one major limitation—they are specifically designed to “fit” a given display device, so they contain no extra image information beyond what the intended display device can show. This is by design, to avoid making the file sizes larger than necessary, but it also means there is little extra image information that can tolerate processing without noticeable damage.

While useful as proxies, these display codecs must never be used for post-production pipeline storage. They are designed to be used at the end of the production pipeline to make a display space version of the project for viewing purposes. Key examples include H.264, MPEG-1, DNxHR, and the lower bit rate ProRes codecs like ProRes 422LT. Some—like Avid DNxHR—claim

3.9.6 WebP

WebP是一种新型的网页智能图像文件格式，旨在以比JPEG更小的文件大小提供更高质量的图像，从而实现更快的下载速度。它支持有损和无损压缩方案，以及透明度。无损压缩图像将比PNG小26%，而有损压缩图像在同等质量下将比JPEG小24%-34%，但看起来更美观。

3.9.7 HEIF

高效图像格式（HEIF）是另一种于2015年推出的新文件格式，旨在取代已使用25年、逐渐显现局限性的JPEG格式。它宣称在同等画质下文件体积减半，支持高达16位的色深，并能容纳多张图像、元数据、深度信息和缩略图。随着4K和HDR图像的发展趋势，HEIF前景十分广阔。重要的是，它获得了苹果、佳能、Windows 10、Adobe Lightroom、Affinity Photo、GIMP及其他多款图像编辑应用的行业支持。

3.9.8 请勿使用

许多文件格式仅为“显示空间”格式——它们专门设计用于从磁盘快速读取并迅速解压，以便在计算机显示器或电视上实时观看。其重点在于低比特率和减少磁盘空间需求，而质量则是次要甚至第三位的考虑。然而，所有显示格式都存在一个主要局限——它们专门设计为“适配”特定显示设备，因此除了目标显示设备能呈现的内容外，不包含任何额外的图像信息。这种设计旨在避免文件体积不必要地增大，但也意味着几乎没有多余的图像信息能承受处理而不造成明显损伤。

虽然这些显示编解码器作为代理很有用，但绝不能用于后期制作流程的存储。它们设计用于生产流程的末端，为项目创建显示空间版本以供查看。主要例子包括H.264、MPEG-1、DNxHR以及较低比特率的ProRes编解码器，如ProRes 422LT。有些——比如Avid DNxHR——声称

to be suitable for visual effects, but only for television, and I would not want to trust my feature film to the sales brochure claims of Avid.

Unfortunately, these codecs may be attractive because of the truly impressive compression ratios that save disk space. But if that is a concern, my advice is to buy more disk space. If you choose an overly compressed codec for your project, by the time it gets through grading, editing, visual effects, and color correction, you could find that you have an unmitigated disaster on your hands.

3.10 Conclusion

It's been a long trek through Storage Space, but you are now equipped to make informed choices about file formats, codecs, and wrappers. I'll bet you never knew the deep secrets of compression strategies before this. I'll also wager that you will never again get blindsided by compression artifacts. The key insight here is that there are file formats designed for a Work Space that retain all or most of the image data, and other file formats intended for a Display Space that hold just enough data to look good on the display and no more.

I hope you have developed an appreciation for log images and their role in the digital imaging pipeline as efficient keepers of high dynamic range image data and are keenly aware of both their limited high dynamic range and their virtues for shadow detail, efficient data storage, and rapid disk access. Of course, the king of rapid disk access is ProRes—the Apple wrapper that can stream 8K to a feeble laptop without breaking a sweat. The export options for both photography and cinematography are the gatekeepers of image quality because they are the first step on the journey to Storage Space and, if badly managed, the project is doomed for life.

In the next chapter, Work Space, we load the carefully saved image data into a workstation and begin operating on the data to make the final version for distribution and display. The chapter starts with a deep dive into color models and color spaces before laying out the decisions you have to make for your Work Space regarding integer or floating-point math (oh boy!).

适合用于视觉效果，但仅限于电视，我不会愿意把我的故事片托付给Avid销售手册上的承诺。

遗憾的是，这些编解码器可能因其真正令人印象深刻的压缩比而颇具吸引力，毕竟能节省磁盘空间。但如果这是你担忧的问题，我的建议是购买更多磁盘空间。如果你为项目选择了过度压缩的编解码器，等到它经历调色、剪辑、视觉特效和色彩校正等流程后，你可能会发现自己手上已是一场彻头彻尾的灾难。

3.10 结论

在存储空间的漫长跋涉后，你现在已能对文件格式、编解码器和封装方式做出明智选择。我敢打赌，在此之前你从未了解压缩策略的深层奥秘，也敢断言你今后再不会被压缩伪影杀个措手不及。其中的关键洞见在于：有些文件格式专为工作空间设计，能保留全部或大部分图像数据；另一些则针对显示空间，仅存储足以在屏幕上呈现良好效果的数据，绝不多留分毫。

我希望你已经对对数图像及其在数字成像流程中的作用有了深刻的认识——它们是高动态范围图像数据的高效保存者，并且敏锐地意识到它们既具备有限的高动态范围，又拥有在阴影细节、高效数据存储和快速磁盘访问方面的优势。当然，快速磁盘访问的王者当属ProRes——这款苹果开发的封装格式，能够轻松流畅地将8K视频传输到性能普通的笔记本电脑上。无论是摄影还是电影摄影，导出选项都是图像质量的守护者，因为它们是迈向存储空间之旅的第一步，如果处理不当，项目将注定终身受损。

在下一章《工作空间》中，我们将精心保存的图像数据加载到工作站，并开始对数据进行处理，以制作用于分发和展示的最终版本。本章首先深入探讨色彩模型与色彩空间，随后详细说明您在工作空间中必须做出的决策——包括选择整数或浮点数运算（这可真不简单！）。

Not to worry—like anything else, once you understand it, it's easy. There is even an exploration of the super-sized color spaces for photography, Adobe RGB and ProPhoto RGB. Even if you are not a photographer you will find the story informative and it will set you up for the granddaddy of all Work Spaces—ACES!

别担心——就像其他事情一样，一旦你理解了，就会觉得很简单。书中甚至探讨了摄影中使用的超大色彩空间，如Adobe RGB和ProPhoto RGB。即使你不是摄影师，也会发现这些内容很有启发性，并为你理解所有工作空间的鼻祖——ACES！做好铺垫。

4 Work Space

In the previous chapter, we explored the virtues of different image file formats for storing our digital images for cinematography and photography. Here we move to the next stage of the production pipeline by asking what happens when our thoughtfully stored images are loaded from disk into the Work Space, which is where all of the editing, VFX, and color grading operations will be done. This will serve as a guide toward choosing a workspace that will protect the quality of your work and offer future-proofing that prepares the job for any future display technology.

This chapter removes the confusion between color models and color spaces and then dives deep into what defines a color space: the primary chromaticities, the color gamut, white points, and transfer functions. Following that is a lively discussion of working in linear space vs. gamma-corrected space; then considerations of integer vs. floating point math are proffered. Once armed with all this foundational information, you are well-equipped to choose your Work Space.

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4 工作空间

在上一章中，我们探讨了不同图像文件格式在电影摄影和摄影领域存储数字图像的优点。现在我们将进入制作流程的下一阶段，探讨当精心存储的图像从磁盘加载到工作空间时会发生什么——所有编辑、视觉特效和调色操作都将在工作空间中完成。这将为选择工作空间提供指导，以保护作品质量并实现未来兼容性，使工作能够适应任何未来的显示技术。

本章消除了色彩模型与色彩空间之间的混淆，随后深入探讨了定义色彩空间的要素：原色色度、色域、白点及传递函数。接着生动讨论了在线性空间与伽马校正空间中的工作差异；继而提出了整数与浮点数运算的考量。掌握这些基础信息后，您便能游刃有余地选择适合您的工作空间。

DOI: 10.4324/9781003486084-5

An important trend in the modern Work Space is the non-destructive workflow. The idea is that the original image data is never actually altered; instead, there are modification layers on top of it. For example, the camera raw gets a LUT that makes it viewable on a monitor; then color grading adds a CDL (Color Decision List), and then editing adds the EDL (Edit Decision List). The camera raw is "rendered" through these layers to produce the final deliverable image, and if changes are needed, the modification layers are updated but the original camera raw is permanently preserved.

There are also sections devoted to photography, as photographers have special issues that those of us working in video and film don't have, like not being able to view their work properly on a monitor. The chapter ends with information on strategies for archiving your project, the final act in the Work Space. Since this chapter is thick and meaty, we will warm up with something easy—color theory.

4.1 Color Theory

Most readers will be familiar with additive and subtractive color systems, but since we will be referring to them frequently, a quick review is in order. Even if you know them "cold," I suggest you not skip this section just to make sure we are sharing the same terminology.

4.1.1 Additive Color

Additive color systems are used with light-emitting devices like TVs, monitors, digital projectors, tablets, and cell phones. They use RGB primaries and add them together (hence "additive") in various proportions to make secondary colors and all the colors in between (Figure 4.1). Mixing all three primaries at 100% produces white. The secondary colors cyan, magenta, and yellow form the primaries for subtractive colors.

现代工作空间的一个重要趋势是非破坏性工作流程。其核心理念是原始图像数据永不直接修改，而是在其基础上叠加调整层。例如，相机原始文件先加载LUT（查找表）使其能在显示器上正常显示；随后调色环节添加CDL（色彩决策表），剪辑环节再叠加EDL（剪辑决策表）。相机原始文件经过这些图层的"渲染"处理后生成最终交付影像，如需修改只需更新调整层，原始相机数据则被永久保留。

书中也有专门讨论摄影的章节，因为摄影师面临着我们从事视频和电影工作的人所没有的特殊问题，比如无法在显示器上正确查看他们的作品。本章最后介绍了项目归档策略的相关信息，这是工作空间中的最后一步。由于本章内容详实丰富，我们将从一些简单的部分开始热身——色彩理论。

4.1 色彩理论

大多数读者可能已经熟悉加色系统和减色系统，但鉴于我们将频繁提及它们，有必要快速回顾一下。即使你对此了如指掌，我仍建议你不要跳过本节，以确保我们使用相同的术语体系。

4.1.1 加色法

加色系统用于发光设备，如电视、显示器、数字投影仪、平板电脑和手机。它们使用RGB三原色，并按不同比例叠加（因此称为“加色”）以产生二次色及其中间的所有颜色（图4.1）。将三种原色以100%混合会产生白色。二次色青色、品红色和黄色则构成了减色系统的原色。

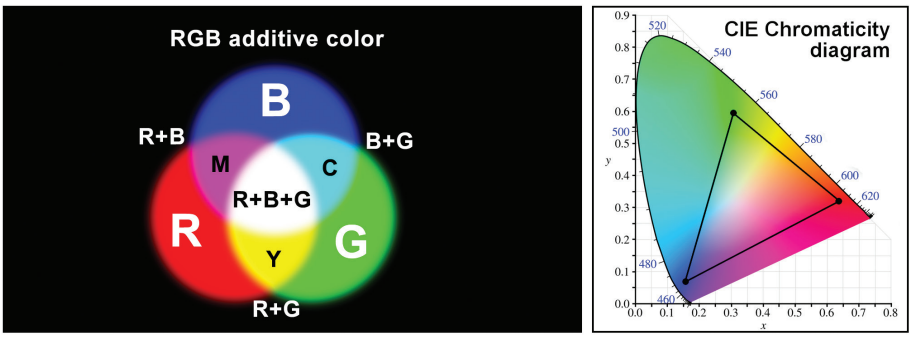


Figure 4.1 The additive color model with RGB gamut

Because the additive color system uses three primaries defined by their location on the CIE chromaticity diagram, the gamut of an RGB color system will be the familiar tidy triangle.

4.1.2 Subtractive Color

Subtractive color is used when making images for print, photographs, and 35 mm film. The dyes, inks, and paints create color the way nature does—by subtracting selected wavelengths of light (hence "subtractive") from a white light source. A subtractive color system uses CMY primaries and it is helpful to think of them as "color blockers." Referring to Figure 4.2, cyan reflects green and blue but blocks red, so it's a "red blocker." Magenta is a "green blocker" and yellow is a "blue blocker."

In a subtractive color system, the RGB becomes the secondary colors. Take green, for example, in Figure 4.2, which is the overlap of cyan and yellow. Starting with white light, cyan blocks the red, and yellow blocks the blue, so the surviving color is green.

With an additive color system, you get white by having all three primaries on at 100%. However, in a subtractive system, you don't get black by having all three primaries on at 100%. Instead, you get a muddy brown, for the simple reason that the inks and dyes are not 100% effective at blocking their respective wavelengths. So we need a little help here—which we get by adding one more pigment, black, represented by the letter K. Our subtractive primaries are now CMYK, as illustrated in Figure 4.2.

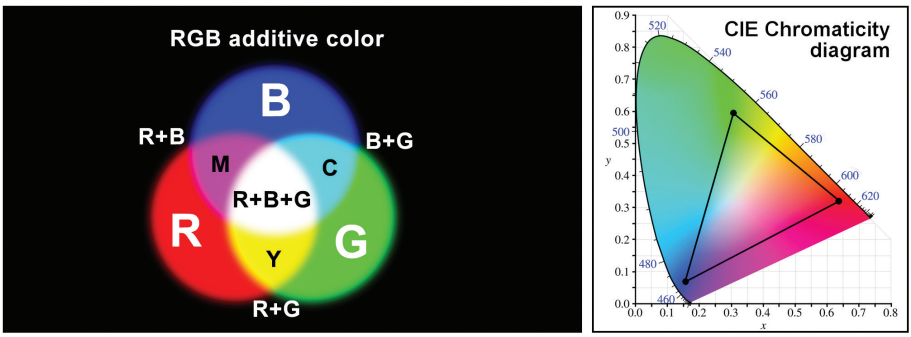


图4.1 采用RGB色域加色模型

由于加色系统使用三种原色，这些原色由它们在CIE色度图上的位置定义，因此RGB色彩系统的色域将是人们熟悉的整齐三角形。

4.1.2 减色法

减色法用于制作印刷品、照片和35毫米胶片的图像。染料、油墨和颜料通过从白色光源中减去特定波长的光（因此称为“减色”）来创造颜色，这与自然界的方式相同。减色系统使用CMY三原色，将其理解为“颜色阻挡剂”有助于理解。参考图4.2，青色反射绿色和蓝色但阻挡红色，因此它是“红色阻挡剂”。洋红色是“绿色阻挡剂”，黄色是“蓝色阻挡剂”。

在减色系统中，RGB成为次要颜色。以图4.2中的绿色为例，它是青色和黄色的重叠部分。从白光开始，青色阻挡红色，黄色阻挡蓝色，因此最终呈现的颜色是绿色。

在加色系统中，将三原色均设为100%即可得到白色。然而，在减色系统中，将三原色均设为100%并不会得到黑色，反而会呈现浑浊的棕色调。这主要是因为油墨和染料无法100%有效地阻挡其对应波长的光线。因此我们需要借助一点额外帮助——通过添加第四种颜料“黑色”（用字母K表示）来实现。于是我们的减色三原色体系就变成了如图4.2所示的CMYK系统。

However, the shape of the CMYK gamut is a bit irregular. The “bulges” are also due to the colorants not being 100% effective. So, the gamut’s green tip, for example, is produced by the cyan and yellow colorants, making it a bit off. As a result, the subtractive green tip is not at the perfect triangular location like an additive green primary.

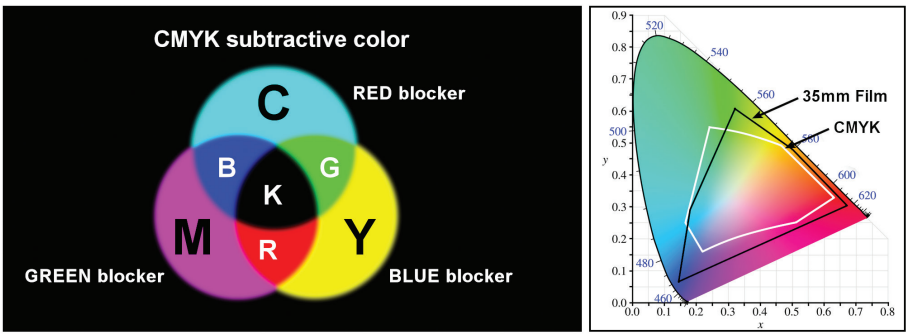


Figure 4.2 The subtractive color model with CMYK gamut

The dyes used in 35 mm film are more effective at blocking 100% of their light, so a separate black component is not needed. Due to the light-leaking inks and pigments of print, plus their use of a separate black ink for CMYK and the heavy dyes used for 35 mm film, their gamuts in Figure 4.2 are not a tidy triangle like they are for additive RGB systems.

4.2 Color Models

Color models are often confused with color spaces. A color model is the little brother of a color space. The color model is incomplete because it only describes the axis of the color space coordinate system, like Figure 4.3 for the RGB, HSL and YCbCr color models. The RGB color cube plots the location of each pixel of a three-channel image into a 3D cube with the grayscale line from the white corner down to the black corner. The HSL color model uses a cone, for the simple reason that the saturation axis gets shorter as the lightness gets darker at the bottom. The YCbCr cube holds the color information in an XY plane as color differences from Y, the luminance, which is represented by the grayscale line down the center, front to rear. Each color model has its advantages and the original three-channel image can be transformed to RGB, HSL, or YCbCr and back in a completely reversible process.

然而，CMYK色域的形状有些不太规则。这些“凸起”也是由于着色剂并非100%有效所致。例如，色域的绿色尖端是由青色和黄色着色剂产生的，这使得它有些偏离。因此，减色法的绿色尖端并不像加色法的绿色原色那样位于完美的三角形位置。

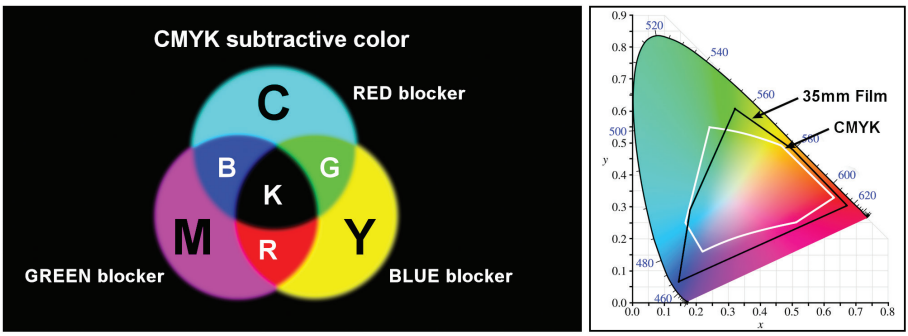


图4.2 CMYK色域下的减色模型

35毫米胶片中使用的染料能更有效地100%阻挡其光线，因此无需单独的黑色成分。由于印刷品存在漏光的油墨和颜料，加上CMYK使用单独的黑色油墨，以及35毫米胶片采用的重型染料，图4.2中它们的色域并非像加色法RGB系统那样呈现规整的三角形。

4.2 色彩模型

色彩模型常与色彩空间混淆。色彩模型是色彩空间的"小弟"，它并不完整，因为它只描述了色彩空间坐标系的轴向，如图4.3所示的RGB、HSL和YCbCr色彩模型。RGB色彩立方体将三通道图像的每个像素位置映射到三维立方体中，灰度线从白色角延伸至黑色角。HSL色彩模型采用圆锥体，原因很简单：随着底部明度变暗，饱和度轴会缩短。YCbCr立方体将色彩信息存储在XY平面中，作为与亮度Y的色差，而亮度Y由从前到后贯穿立方体中心的灰度线表示。每种色彩模型都有其优势，原始的三通道图像可以完全可逆地转换为RGB、HSL或YCbCr格式，并能无损还原。

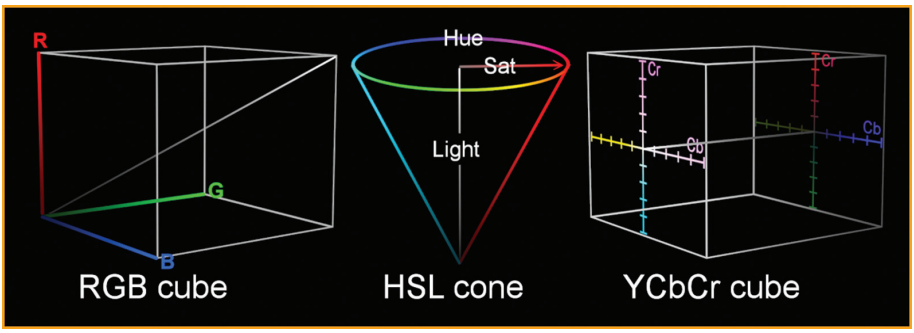


Figure 4.3 Prominent color models

But three-channel RGB images are not cubes, so what is the point of representing them as such? The point is that once an image is transformed into this cubic representation, suddenly we have a lot of mathematical tools like trigonometry that can be applied to it that could not be applied to the original three-channel image. While the number of things you can do in this arrangement is manifold and varied, one important use is to key greenscreens for visual effects.

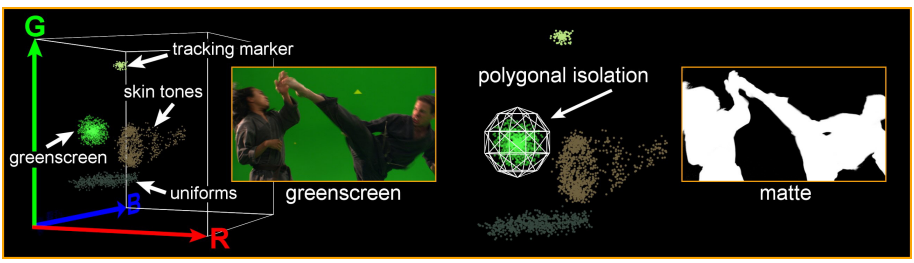


Figure 4.4 Primatte arranges pixels in a 3D color cube for keying

Primatte is a world-class keyer used in high-end feature film visual effects that uses a transformation of the three-channel image into a 3D color cube for creating a high-precision matte. Figure 4.4 starts on the left with the greenscreen pixels arranged in an RGB color cube. Note how they sort themselves out into clusters in 3D space, which separates the greenscreen pixels from the tracking marker, skin tones, and uniform pixels. The image's RGB values are transformed to XYZ locations in the cube, so similarly colored pixels become clustered together within the cube. The "polygonal isolation" in the middle of Figure 4.4 shows a close-up of how the greenscreen cluster is

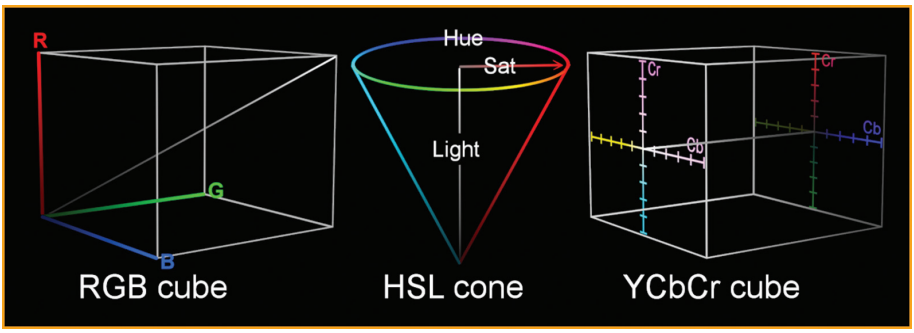


图4.3 主要颜色模型

但是三通道的RGB图像并非立方体，那么将其表示为立方体有何意义呢？关键在于，一旦图像被转换成这种立方体表示形式，我们就能突然运用大量数学工具（如三角学）来处理它，而这些工具原本无法应用于原始的三通道图像。虽然在这种排列方式下可进行的操作多种多样，但一个重要用途是为视觉特效进行绿幕抠像。

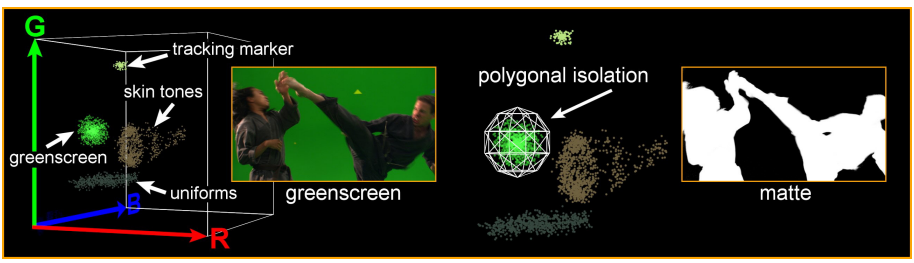


图4.4 Primatte在三维色彩立方体中排列像素以进行抠像

Primatte是一款世界级的抠像工具，广泛应用于高端电影视觉特效制作，它通过将三通道图像转换为三维色彩立方体来生成高精度蒙版。图4.4左侧展示了绿幕像素在RGB色彩立方体中的分布情况。值得注意的是，这些像素在三维空间中自然聚集成簇，从而将绿幕像素与跟踪标记、肤色及纯色像素区分开来。图像中的RGB值被转换为立方体内的XYZ坐标位置，使得颜色相近的像素在立方体内聚集。图4.4中部的“多边形隔离”示意图特写了绿幕像素簇如何被

wrapped in a 3D polygonal mesh, which in turn is used to produce a matte to key out the characters.

Different color models are optimized for different situations, so there are many of them. The general concept is that each color model has virtues that are useful for a certain situation. Here we will look at three of the most common color models, what their virtues are, and how they are useful.

4.2.1 RGB

The RGB color model is, of course, the primary color model because it literally represents the data that will be loaded into the display. No prep, no processing—just load and go. Its virtue is that it is literal and straightforward, not representational at all, and can be loaded directly into a display for viewing without processing. For that reason, it is the ideal color model for creating CG images and for primary color grading. The problem is that it speaks to the color of an image in a display-centric way, not in the image-centric way that we humans like to think of color.

Figure 4.5 The RGB color model

Allow me to illustrate with a simple example. With the RGB color model you can, for example, increase the amount of red by simply gaining up the red channel. Easy. But what if you want to increase the saturation? Or make the image warmer without altering the brightness? Or darker? Suddenly the RGB color model is clumsy, or completely inadequate, as it does not speak to the image the same way we humans think about it. For that, you need something like the HSL color model.

4.2.2 HSL

The HSL (Hue, Saturation, Lightness) color model speaks to images the same way we humans think about them, which is why it was developed.

包裹在三维多边形网格中，进而用于生成遮罩以抠出角色。

不同的色彩模型针对不同的场景进行了优化，因此种类繁多。普遍的观点是，每种色彩模型都有其适用于特定场景的优点。在此，我们将探讨三种最常见的色彩模型，它们的优势所在，以及如何发挥作用。

4.2.1 RGB

RGB色彩模型当然是主要的色彩模型，因为它直接代表了将要加载到显示器中的数据。无需准备，无需处理——只需加载即可。它的优点在于它是直接且直观的，完全不具代表性，可以直接加载到显示器中观看而无需处理。因此，它是创建CG图像和进行初级色彩分级的理想色彩模型。问题在于，它以一种以显示器为中心的方式描述图像的颜色，而不是我们人类喜欢的那种以图像为中心的思考颜色的方式。

图4.5 RGB颜色模型

让我用一个简单的例子来说明。使用RGB色彩模型时，你可以通过简单地提升红色通道来增加红色分量，这很容易。但如果你想提高饱和度呢？或者在不改变亮度的情况下让图像更暖？或者更暗？突然间，RGB色彩模型就显得笨拙甚至完全不够用了，因为它不符合我们人类对图像的直观理解。为此，你需要像HSL色彩模型这样的工具。

4.2.2 HSL

HSL（色相、饱和度、明度）色彩模型以我们人类思考图像的方式来描述它们，这正是它被开发出来的原因。

If we want to increase saturation, we dial up the saturation value. If we want the image to be warmer, we adjust the hue value. And if we want it lighter or darker, we adjust the lightness parameter. However, the system needs software that converts these HSL values back to RGB for viewing on the display. In sum, HSL is for humans and RGB is for computers.

Figure 4.6 The HSL color model

How the HSL color model represents color is illustrated in Figure 4.6. The original three-channel RGB image is "reorganized" by a color transformation into a three-channel HSL image where the hue channel is the color of each pixel represented by where it lies on the color wheel from zero to 360 degrees. The saturation channel is simply a grayscale image where each pixel's brightness represents its saturation, and the lightness channel is the actual luminance of each pixel.

Using HSL, we can now adjust the hue of an image without altering the saturation, or the lightness without disturbing the hue. To increase saturation, just gain up the saturation channel. These adjustments in turn change the exported RGB values to edit the image. While we have been looking at the HSL color model, note that there is a zoo of related color models, such as HSI (Hue, Saturation, Intensity), HSV (Hue, Saturation, Value), HSB (Hue, Saturation, Brightness), and so on, but they all work on this same principle, with the differences being how they calculate saturation and luminance. They also have different aims, such as being computationally simpler or responding more intuitively to editing—for example, doubling the brightness value also doubles its brightness appearance to the eye.

4.2.3 YCbCr

Like the HSL color model, the YCbCr color model separates the luminance from the chrominance, but in a different way and for a different reason. We met the YCbCr color model in Chapter 3: Storage Space in the context of

若想提升饱和度，我们便调高饱和度数值；若希望图像更显温暖，则调整色调值；若要增亮或减暗，便调节明度参数。然而，系统需要软件将这些HSL值转换回RGB格式，以便在显示屏上呈现。总而言之，HSL服务于人类感知，而RGB服务于计算机处理。

图4.6 HSL色彩模型

HSL色彩模型如何表示颜色如图4.6所示。原始的三通道RGB图像通过色彩变换被“重组”为三通道HSL图像，其中色相通道表示每个像素的颜色，由其在0到360度色轮上的位置决定。饱和度通道本质上是一幅灰度图像，每个像素的亮度代表其饱和度，而明度通道则表示每个像素的实际亮度。

使用HSL，我们现在可以调整图像的色调而不改变饱和度，或者调整亮度而不干扰色调。要增加饱和度，只需提升饱和度通道的增益。这些调整进而会改变导出的RGB值以编辑图像。虽然我们一直在讨论HSL色彩模型，但请注意，还存在一系列相关的色彩模型，例如HSI（色调、饱和度、强度）、HSV（色调、饱和度、明度）、HSB（色调、饱和度、亮度）等等，但它们都基于相同的原理运作，区别在于计算饱和度和亮度的方法不同。它们也有不同的目标，比如计算更简便，或者对编辑的响应更直观——例如，将亮度值加倍也会使其视觉亮度加倍。

4.2.3 YCbCr

与HSL色彩模型类似，YCbCr色彩模型也将亮度与色度分离开来，但方式不同且目的各异。我们在第三章：存储空间的相关内容中已接触过YCbCr色彩模型。

image compression. For that reason, it is hugely important as the native delivery format for broadcast, cable, DVD, and Blu-Ray players; plus it plays a role in post-production color correction.

Figure 4.7 The YCbCr color model

Colorists can grade in YCbCr because it allows them to adjust exposure and contrast with the Y channel without disturbing the overall color, especially skin tones. Some refer to the Y channel as the "detail" channel because it carries the high-frequency detail information of an image so it can, for example, be sharpened or denoised without disturbing any color.

By adjusting the chroma components separately, the colorist can achieve proper color balance, removing any color casts. For example, if a clip has an overabundance of blue, the Cb component can be reduced, restoring color balance by adding warmth to the image. By changing the amplitude of the CbCr components, saturation can be increased or decreased.

Understanding these three color models will be much easier after watching this video of all three in 3D action.

4.2.4 CMYK

CMYK (Cyan, Magenta, Yellow, Black) are the native primaries for print media. This is a subtractive color model (see Figure 4.2), so the inks, paints, or dyes are illuminated with white light and the CMYK layers subtract colors to reflect the RGB that our eyes are tuned to. The "K" in CMYK stands for "key"—a traditional printer's term for black ink because the color plates are all "keyed" or aligned to the black.

图像压缩。因此，它作为广播、有线电视、DVD和蓝光播放器的原生传输格式至关重要；同时，它在后期制作色彩校正中也发挥着作用。

图 4.7 YCbCr 色彩模型

调色师可以在YCbCr色彩空间中进行调色，因为这样他们可以通过Y通道调整曝光和对比度，而不会干扰整体色彩，尤其是肤色。有些人将Y通道称为“细节”通道，因为它承载图像的高频细节信息，因此可以对其进行锐化或降噪处理，而不会影响任何颜色。

通过分别调整色度分量，调色师可以实现正确的色彩平衡，消除任何色偏。例如，如果片段中蓝色过多，可以降低Cb分量，通过为图像增添暖色调来恢复色彩平衡。通过改变CbCr分量的幅度，可以增加或降低饱和度。

观看这段三种颜色模型在3D动态演示中的视频后，理解这三种颜色模型会变得容易得多。

4.2.4 CMYK

CMYK（青色、品红色、黄色、黑色）是印刷媒介的原生基色。这是一种减色模型（见图4.2），因此油墨、颜料或染料在白光照射下，CMYK各层通过减去颜色来反射我们人眼所适应的RGB色彩。CMYK中的“K”代表“关键色”——这是印刷业对黑色油墨的传统术语，因为所有色版都以黑色为“基准”或对齐标准。

Figure 4.8 The CMYK color model

4.3 Color Spaces

A color space is a mathematically precise description of a specific range of possible colors and luminance values and is typically arranged within a three-dimensional space. There are many different color spaces, each with specific virtues that are useful in specific situations. And, like the color models above, we can move between color spaces with color space transforms. One method is to use 3D LUTs, but a dedicated color space transform math expression gives better results and is perfectly reversible. The 3D LUT is not so reversible because there are round-off errors in both directions when converting to another color space and back.

It is helpful to think of camera-captured images (as opposed to CG images) as traversing these three distinct "spaces" on their way to final display:

- **Capture Space** - For image capture devices like cameras and flatbed scanners.
- **Work Space** - For image processing and color editing. This is where visual effects and color timing are done. Editing is not affected because it does not alter the image or its color.
- **Display Space** - Where images are formatted for specific displays like HDTV, digital cinema, and print.

Every project must navigate all three spaces without damage or distortion while preserving the artistic intent. There will be losses simply because cameras capture more information than is needed for the workspace, which in turn has more information than is needed for the display space. This chapter, of course, is about Work Space.

A color space must have these three attributes to be considered a true color space:

图4.8 CMYK颜色模型

4.3 色彩空间

色彩空间是对特定范围内可能颜色与亮度值的数学精确描述，通常构建于三维空间内。存在多种不同的色彩空间，各自具备适用于特定场景的独特优势。与上文提到的色彩模型类似，我们可通过色彩空间转换在不同色彩空间之间进行切换。一种方法是使用3D LUT，但专用的色彩空间转换数学表达式能提供更优效果且具备完全可逆性。而3D LUT的可逆性较弱，因为在转换至其他色彩空间并返回时，双向计算均存在舍入误差。

将相机捕捉的图像（与CG图像相对）视为在最终显示前穿越这三个不同的“空间”是有帮助的：

- 捕捉空间 - 适用于相机和平板扫描仪等图像捕捉设备。

- 工作空间 - 用于图像处理和色彩编辑。这里是进行视觉效果和色彩调校的地方。由于它不会改变图像或其色彩，因此不会影响剪辑工作。

- 显示空间 - 图像在此处针对特定显示设备（如高清电视、数字影院和印刷品）进行格式化的区域。

每个项目都必须在无损或不变形的情况下穿越这三个空间，同时保持艺术意图。损失是不可避免的，因为摄像机捕捉的信息量超过了工作空间所需，而工作空间的信息量又超过了显示空间所需。当然，本章讨论的是工作空间。

一个色彩空间必须具备这三个属性才能被视为真正的色彩空间：

- **Primary chromaticities** - Red, green, and blue primary colors are described by their locations in the CIE xy-chromaticity diagram. We will refer to these as "primaries" for short.
- **White point** - What is considered white, also located on the CIE xy-chromaticity diagram.
- **Transfer function** - Loosely referred to as the "gamma," this defines the relationship between the digital code values and their brightness on the screen.

In this section, we will delve into all three subjects. Upon emerging at the other end, you will have a very clear understanding of color spaces, which ones to use and when, and how to navigate between them.

4.3.1 CIE Chromaticity Diagram

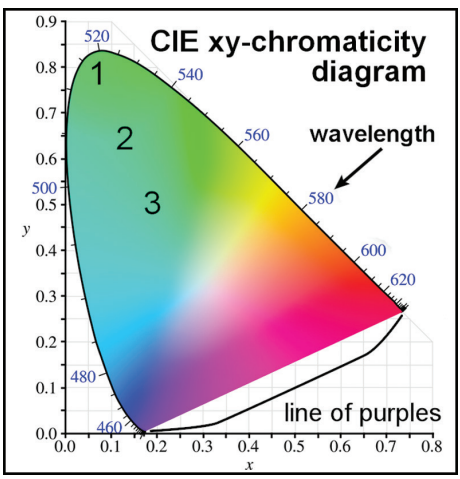


Figure 4.9 Color sample points

We met the CIE x-y chromaticity diagram in Chapter 1 as a map of the entire range of human vision, where colors can be identified by their x,y location in the diagram. The closer a color is located to the very edge of the horseshoe-shaped diagram, the purer (i.e., more saturated) it is. Another way to say this is that "color purity" means a very narrow range of wavelengths.

Figure 4.9 has three sample points numbered 1, 2, and 3 marked on a CIE diagram, with 1 near the edge and 3 further in. These three numbers map to the illustration in Figure 4.10, which shows how narrow the range of wavelengths are at each sample point.

- 主色度 - 红、绿、蓝三原色在CIE xy色度图中的位置描述了它们的特性。为简便起见，我们将这些称为“原色”。

- 白点 - 被视为白色的点，同样位于CIE xy色度图上。

- 传递函数 - 通常被称为“伽马”，它定义了数字代码值与其在屏幕上亮度之间的关系。

在本节中，我们将深入探讨所有三个主题。当您最终掌握这些内容时，将会对色彩空间、何时使用何种色彩空间以及如何在它们之间进行转换形成非常清晰的理解。

4.3.1 CIE 色度图

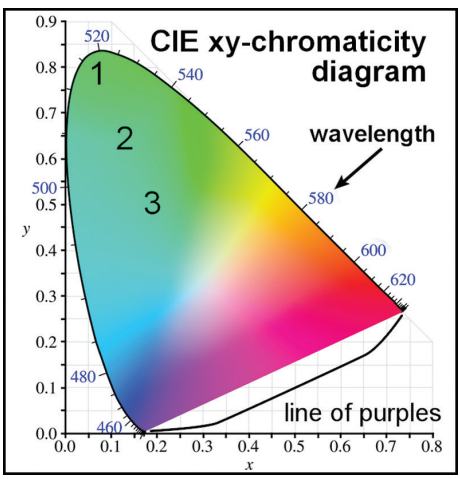


图4.9 颜色采样点

我们在第1章中遇到了CIE x-y色度图，它作为人类视觉全范围的地图，其中颜色可以通过它们在图表中的x,y位置来识别。颜色越靠近马蹄形图表的边缘，它就越纯净（即饱和度越高）。另一种说法是，“颜色纯度”意味着波长范围非常窄。

图4.9在CIE色度图上有三个标记为1、2和3的采样点，其中1靠近边缘，3更靠内。这三个数bers映射到图4.10中的示意图，该图展示了每个采样点处的波长范围有多窄。

In Figure 4.10, the left panel marked "1" illustrates green right at the edge of the graph at 520 nanometers. It has a very narrow wavelength, so the color spot is a very pure green. Panel 2 moves away from the edge toward the center and shows a wider range of wavelengths, which means the color is a less pure green. Panel 3 goes even further, showing an even wider range of wavelengths, which is even less pure. If we kept moving to the center of the graph, we would enter the "white zone" and lose all color.

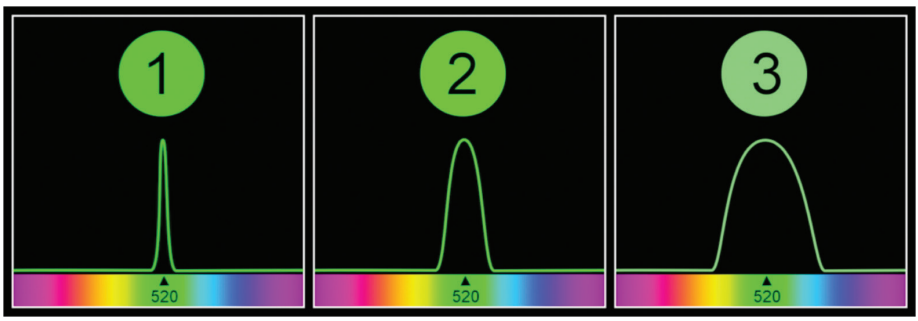


Figure 4.10 The further from the edge, the less pure the color

We need to point out a couple of things here. As explained above, the wavelengths in nanometers are printed around the perimeter and are, of course, natural colors, meaning that each frequency of light produces a specific color sensation in the brain. However, as we saw in Chapter 1, the bottom edge of the graph is known as the "line of purples," and those purples are not natural colors. No frequency of light would produce those purples. They are produced in the human brain by a *combination* of red and blue light.

This brings up another very important point. There are two ways to stimulate a color in the human brain: a single frequency of light or a combination of two other colors. A case in point is yellow at 570 nm (not labeled) or a combination of red and green. Having different physical colors produces the same color sensation in the brain as the metamers we covered in Chapter 1.

在图4.10中，左侧标记为“1”的面板展示了恰好位于图表边缘520纳米处的绿色。它具有非常窄的波长，因此色点是一种非常纯粹的绿色。面板2从边缘向中心移动，显示出更宽的波长范围，这意味着颜色是一种较不纯粹的绿色。面板3更进一步，显示出更宽的波长范围，纯度更低。如果我们继续向图表中心移动，就会进入“白色区域”，失去所有颜色。

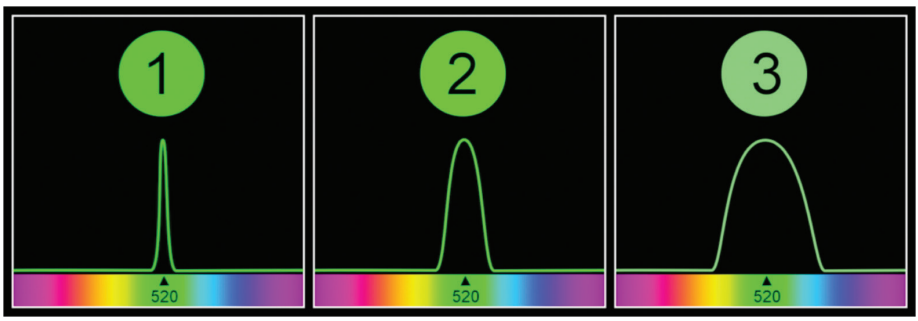


图4.10 离边缘越远，颜色越不纯

这里需要指出几点。如上所述，纳米单位的波长标注在圆周外围，这些当然是自然色，意味着每种频率的光都会在大脑中产生特定的颜色感知。然而，正如我们在第一章所见，图表底部的边缘被称为“紫色线”，而这些紫色并非自然色。没有任何频率的光能单独产生这些紫色。它们是由红光和蓝光混合后，在人类大脑中合成的结果。

这就引出了另一个非常重要的观点。刺激人脑感知颜色有两种方式：单一频率的光或两种其他颜色的组合。一个典型的例子是570纳米的黄色（图中未标注）或红绿两色的组合。不同的物理颜色在大脑中产生相同的颜色感觉，正如我们在第一章中讨论过的同色异谱现象。

4.3.2 Primary Chromaticities

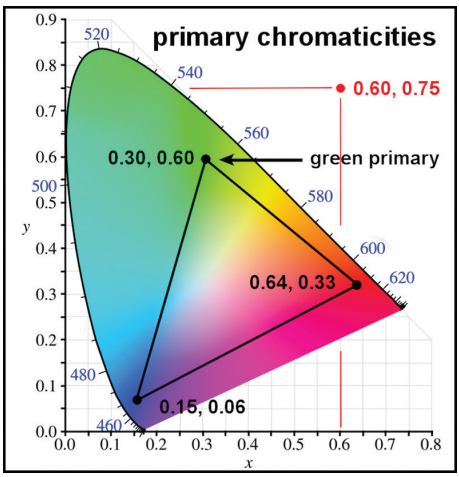


Figure 4.11 Primary chromaticities

RGB devices like monitors and projectors have three primary colors—red, green, and blue—that must be defined with great precision. This is done by specifying their color coordinates in the CIE xy-chromaticity diagram. How the primary colors are specified is illustrated in Figure 4.11. For example, the green primary is marked by the black dot at x = 0.30, y = 0.60. This is how the chromaticity (color) of the three primary colors of an RGB system is defined—hence "primary chromaticities."

So what about the red dot located at 0.60, 0.75 just outside of the CIE diagram? While it is a mathematically valid point, it falls outside the range of the human visual system, so it is not a visible color—it's an imaginary color. So what good is it? When performing mathematical operations on pixels, the math will sometimes produce overflow values outside of the normal range. We do not want to clip them and discard this information, since it may well be needed for subsequent calculations that would then be all wrong. We need to retain these values to maintain the integrity of our math operations. Also, I wanted to introduce the concept of these imaginary numbers for later discussions of supersized gamuts, where things can get a little weird.

4.3.3 White Points

Every color space has a white point, which is defined as the color of white you get when all three code values are at maximum—that is, 255 255 255 for integer folks or 1.0, 1.0, 1.0 for you floaters. Two very important color spaces—Rec. 709 for HDTV and sRGB for desktop monitors—share the

4.3.2 Primary Chromaticities

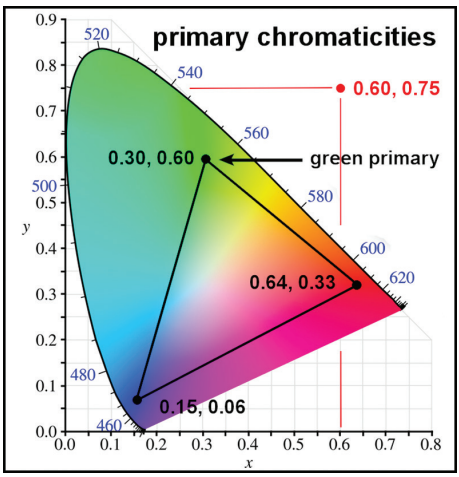


图4.11 主要色度

RGB设备（如显示器和投影仪）具有红、绿、蓝三原色，这些颜色必须被极其精确地定义。这是通过在CIE xy色度图中指定它们的色度坐标来实现的。图4.11展示了如何具体规定这些原色。例如，绿色原色由黑色圆点标记在x = 0.30, y = 0.60处。RGB系统的三原色色度（颜色）正是通过这种方式定义的——因此称为“原色色度”。

那么，位于CIE图外0.60、0.75处的红点又代表什么呢？虽然它在数学上是一个有效点，但超出了人类视觉系统的范围，因此并非可见颜色——它是一种假想颜色。这有什么意义呢？在对像素进行数学运算时，计算有时会产生超出正常范围的溢出值。我们不想裁剪这些值并丢弃信息，因为后续计算很可能需要这些数据，否则结果会完全错误。我们需要保留这些值以维持数学运算的完整性。此外，我想引入这种假想数字的概念，以便后续讨论超大色域时使用——届时情况可能会变得有些奇特。

4.3.3 白点

每个色彩空间都有一个白点，其定义为当所有三个编码值达到最大值时所呈现的白色——对整数用户而言是255、255、255，对浮点数用户则是1.0、1.0、1.0。两个非常重要的色彩空间——用于高清电视的Rec. 709和用于桌面显示器的sRGB——共享着

same primaries, so have identical color gamuts, as shown in Figure 4.12. They also share the same white point, namely D65, which is one of several CIE standard illuminants (covered in Chapter 1). Change the white point and you have a different color space.

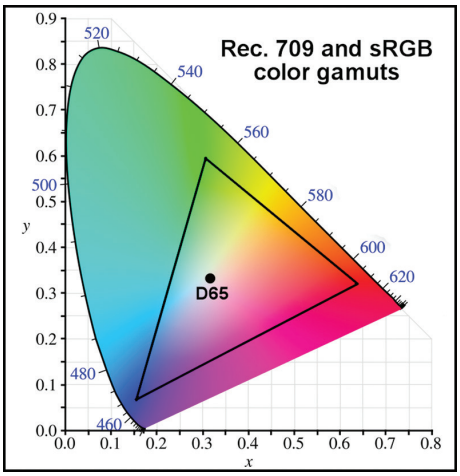


Figure 4.12 Rec. 709 and sRGB white point

Standard illuminants are important white points for many color spaces, so they have been formalized, specified, codified, and named. Harking back to the color temperature discussion in Chapter 1, D65 represents light with a color temperature of 6500K which is used for HDTV and digital cinema. It represents "outdoor daylight lighting," so it is cooler with a slight bluish cast to account for the blue skylight.

The D50 standard illuminant is used by the print industry and has a color temperature of 5000K. It represents "daylight horizon light" in the early morning or afternoon, so is warmer with a yellowish cast. There are others, but these are all that we will need for now. The important thing is that should you encounter an unfamiliar standard illuminant someday—say, D60—you will probably realize that its color temperature is 6000K and that it is a bit cooler than D65.

4.3.4 The Transfer Function

So far, we have seen that the primaries, gamut, and white points for the sRGB and Rec. 709 color spaces are identical. So, what's the difference between them? Why, the transfer function, of course. And you thought I had forgotten about the transfer function—no such luck. The transfer function for sRGB (gamma 2.2) is included in the general sRGB standards defined by IEC 61966-2-1 and the Rec. 709 transfer function (gamma 2.4) has been defined by a separate BT.1886 spec since 2011.

相同的原色，因此具有相同的色域，如图4.12所示。它们也共享相同的白点，即D65，这是CIE标准照明体之一（在第1章中已介绍）。改变白点，就会得到一个不同的色彩空间。

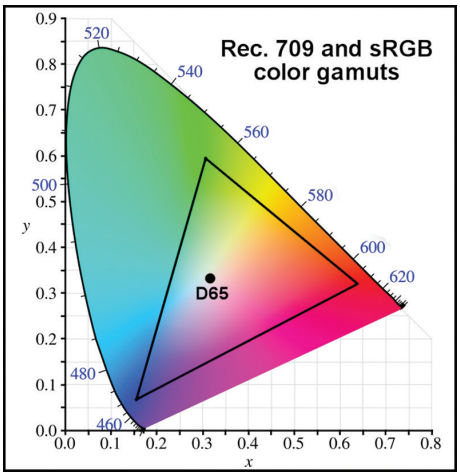


图 4.12 Rec. 709 与 sRGB 白点

标准光源是许多色彩空间的重要白点，因此它们已被规范化、具体化、编码化并命名。回顾第1章中关于色温的讨论，D65代表具有色温的光——温度为6500K，用于高清电视和数字影院。它代表“户外日光照明”，因此色调偏冷，略带蓝色调以模拟蓝天光的效果。

D50标准光源被印刷行业采用，其色温为5000K。它代表清晨或午后的“日光地平线光”，因此色调偏暖，带有淡黄色调。还有其他标准光源，但目前我们只需了解这些。重要的是，如果将来某天你遇到不熟悉的标准光源——比如D60——你或许能意识到它的色温是6000K，且比D65稍冷一些。

4.3.4 传递函数

到目前为止，我们已经看到sRGB和Rec. 709色彩空间的三原色、色域和白点都是相同的。那么，它们之间的区别是什么呢？当然是传递函数了。你以为我忘了传递函数——可没那么走运。sRGB的传递函数（伽马2.2）包含在IEC 61966-2-1定义的通用sRGB标准中，而Rec. 709的传递函数（伽马2.4）自2011年起已由单独的BT.1886规范定义。

FUN FACT: BT. 1886 is a spec released by the International Telecommunications Union (ITU) to standardize the transfer function for HDTV. Amazingly, there was no industry standard before its release in 2011.

So, what is the transfer function? It is a mathematical operation applied to the incoming signal of a monitor (the RGB code values of an image) to correct for the non-linearity of the monitor. The old CRT monitors had a non-linear response curve that darkened the image and the shape of that response curve is a gamma function. The transfer function is designed to compensate for the monitor darkening gamma by applying an opposite gamma to the image we like to call "gamma correction." Modern displays, of course, don't use CRTs and have a linear response. However, to maintain backward compatibility with the last 100 years of image-making, flat-panel displays have LUTs to replicate the old CRT response curve. It's a legacy issue.

As a first approximation of the transfer function, we can refer to it as simply the gamma of the color space. As mentioned earlier, for sRGB the gamma is 2.2 and for Rec. 709 it is 2.4. So, if it is just the gamma, then why do we have to lard it up with polysyllabic gobbledygook like "transfer function"? Because, of course, upon close inspection, there is more to it than just a simple gamma function.

Here's the story: if we were to use just a gamma of, say, 2.4 as the transfer function, then the curve would look like Figure 4.13. The problem with using a gamma curve alone is that at the head of the curve at 0,0 the tangent is a straight-up vertical line, which means it has a slope of infinity. An infinite slope is a fatal problem for real devices like video signal amplifiers because it means the signal is being amplified by infinity. Starting in the lower left-hand corner, the slope of the gamma curve is infinity—straight up. As you follow the curve up, the slope lowers to 1 billion to one, then 1 million to one, etc. These are

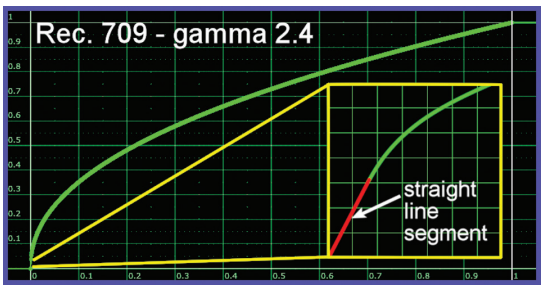


Figure 4.13 A transfer function

有趣的事实：BT. 1886是国际电信联盟（ITU）发布的一项规范，旨在标准化高清电视的传递函数。令人惊讶的是，在2011年该规范发布之前，行业内竟无统一标准。

那么，什么是传递函数？它是一种应用于显示器输入信号（图像的RGB编码值）的数学运算，旨在校正显示器的非线性特性。老式CRT显示器具有非线性响应曲线，会使图像变暗，这种响应曲线的形状就是伽马函数。传递函数的设计目的是通过对我们称之为“伽马校正”的图像施加相反的伽马值，来补偿显示器变暗的伽马效应。当然，现代显示器已不再使用CRT技术，并具备线性响应。然而，为了保持与过去百年图像制作历史的向后兼容性，平板显示器内置了查找表（LUT）来模拟旧式CRT的响应曲线。这属于历史遗留问题。

作为传递函数的初步近似，我们可以直接将其称为色彩空间的伽马值。如前所述，sRGB的伽马值为2.2，Rec. 709则为2.4。那么，如果它仅仅是伽马值，为何我们非得用“传递函数”这类多音节术语来复杂化呢？原因在于，仔细审视后便会发现，其内涵远不止一个简单的伽马函数。

故事是这样的：如果我们仅使用一个伽马值，比如2.4，作为传递函数，那么曲线就会如图4.13所示。单独使用伽马曲线的问题在于，在曲线起点(0,0)处的切线是一条完全垂直的直线，这意味着其斜率为无穷大。无穷大的斜率会

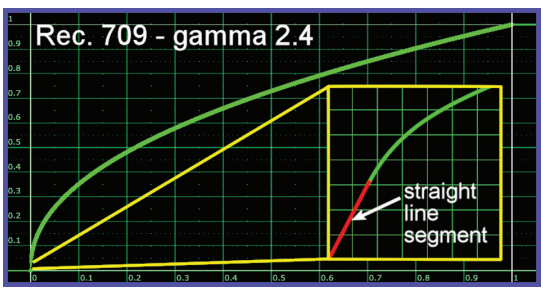


图 4.13 传递函数

对于视频信号放大器这类真实设备来说，这是一个致命问题，因为它意味着信号正被无限放大。从左下角开始，伽马曲线的斜率是无穷大——垂直上升。沿着曲线向上移动，斜率逐渐降低至十亿比一，接着是百万比一，以此类推。这些

utterly impractical numbers for amplification. Not to mention that the noise would be amplified by a billion as well. Not cool.

The solution to this pesky engineering problem is shown in the magnified insert window on the right of Figure 4.13. The engineers simply patched in a short straight-line segment (red) with a reasonable slope of 4.5 that starts at 0,0 and goes to the point where it merges with the gamma curve (green), and the actual gamma curve takes over from there on out. This straight-line segment deviates from the true gamma curve, but only by a tiny amount in the very darks, so is not perceptible. The sRGB transfer functions use the same approach but its straight-line segment has a slope of 12.92. The good news is that unless you are designing video hardware amplifiers, you can forget about the straight-line segment and simply use the gamma correction value for your imaging work.

To close the conversation on color models and color spaces, here is a useful summary list:

Important color models:
RGB, CMYK, CIE, YUV, YCbCr, HSL/HSV/HSB/HSI

Important color spaces:
sRGB, Rec. 709, Rec. 2020, Adobe RGB, ProPhoto RGB, DCI-P3, SWOP v2, CIEXYZ, CIELAB

4.3.5 Color Gamuts

The black dots mark the location of the primaries in the CIE diagram and when those dots are connected, they form a triangle that outlines all the possible colors that the color space can make. The outline of all the possible colors of a color space is called the “gamut” of that color space. The gamut of a color space is a very important attribute and affects much of our art. One example is creating colors while grading that are out of gamut for HDTV, which is why some software has the zebra pattern to mark out-of-gamut pixels. Gamuts are also key players when converting from one color space to another because each color space will have its own unique gamut, so a way

完全不切实际的放大数字。更不用说噪声也会被放大十亿倍。这一点都不酷。

这个恼人的工程问题的解决方案如图4.13右侧放大插入窗口所示。工程师们直接插入了一段短直线段（红色），该线段以合理的4.5斜率从原点(0,0)出发，延伸至与伽马曲线（绿色）交汇的点，此后便由实际伽马曲线接管。这段直线虽与真实伽马曲线存在偏差，但仅在极暗区域有微小偏离，因而难以察觉。sRGB传递函数采用相同方法，但其直线段斜率为12.92。好消息是，除非您正在设计视频硬件放大器，否则可以完全忽略这段直线部分，直接在影像处理工作中使用伽马校正值即可。

关于色彩模型和色彩空间的讨论即将结束，这里有一份实用的总结列表：

重要的颜色模型：RGB、CMYK、CIE 、YUV、YCbCr、HSL/HSV/HSB/HSI

重要的色彩空间：sRGB、Rec. 709、Rec. 2020 、Adobe RGB、ProPhoto RGB、DCI-P3、SWOP {v*}2、CIEXYZ、CIELAB

4.3.5 色域

黑色圆点标记了CIE色度图中三原色的位置，当这些点相连时，它们形成一个三角形，勾勒出该色彩空间所能生成的所有可能颜色。一个色彩空间所有可能颜色的轮廓被称为该色彩空间的“色域”。色域是色彩空间非常重要的属性，深刻影响着我们的艺术创作。例如，在调色时创建超出高清电视色域范围的颜色，这就是为什么有些软件会使用斑马纹图案来标记超色域像素。色域在色彩空间转换过程中也起着关键作用，因为每个色彩空间都有其独特的色域，因此需要一种方法

must be found to map the source gamut into the destination gamut. It turns out this is a non-trivial problem, as we will see shortly.

The color gamut created by its primaries is one of the key features of a color space. Figure 4.14 illustrates the color gamuts for sRGB, the color space for workstation monitors, and DCI-P3, the color space for digital cinema projectors. For additive color RGB devices like monitors and projectors, their gamuts are tidy little triangles because they are defined by three primaries connected by straight lines.

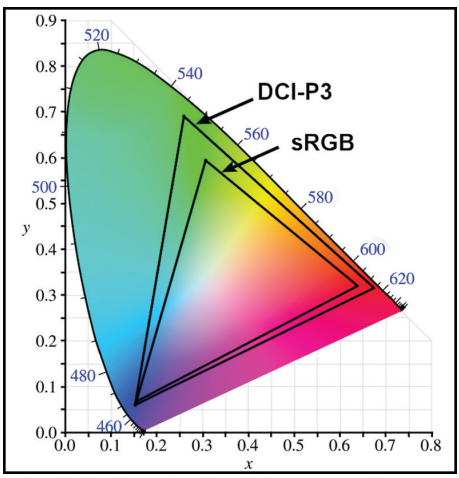


Figure 4.14 Additive color gamuts

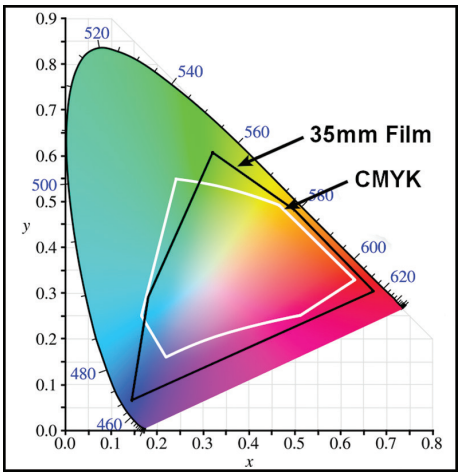


Figure 4.15 Subtractive color gamuts

Subtractive color devices like CMYK printing inks and 35 mm film do not have such tidy color gamuts. Figure 4.15 shows these two subtractive color spaces with their irregular shapes. Seeing their gamuts plotted together like this reveals that CMYK print can reproduce some greens that 35 mm film cannot, but 35 mm film is more robust in the blues, magentas, and reds. Such is the power of plotting the color gamuts of different color spaces.

FUN FACT: There are color spaces larger than the human capacity that use "imaginary" primaries to define huge gamuts. You are sure to encounter some later in this book.

必须找到一种方法将源色域映射到目标色域。事实证明，正如我们稍后将看到的，这是一个非平凡的问题。

由其原色创建的色域是色彩空间的关键特征之一。图4.14展示了sRGB（工作站显示器的色彩空间）和DCI-P3（数字电影投影机的色彩空间）的色域。对于像显示器和投影机这样的加色法RGB设备，它们的色域是整齐的小三角形，因为它们由三条直线连接的原色所定义。

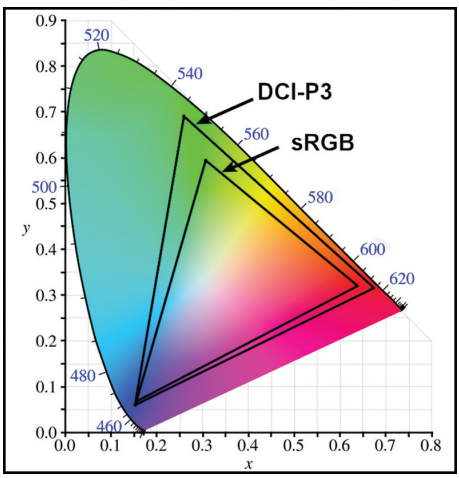


图 4.14 加色色域

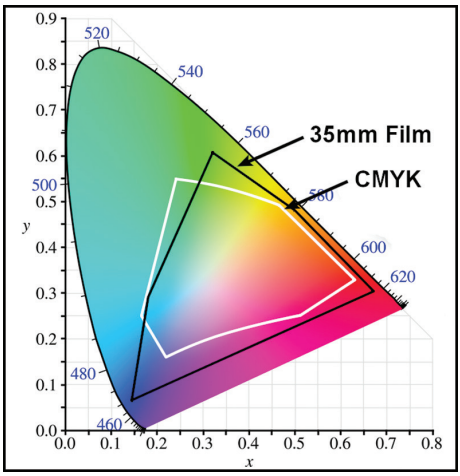


图 4.15 减色法色域

像CMYK印刷油墨和35毫米胶片这样的减色设备没有如此整齐的色域。图4.15展示了这两种具有不规则形状的减色空间。将它们色域如此绘制在一起可见，CMYK印刷能再现一些35毫米胶片无法呈现的绿色，而35毫米胶片在蓝色、洋红色和红色方面表现更出色。这便是绘制不同色彩空间色域图的力量所在。

有趣的事实：存在一些比人类感知能力更大的色彩空间，它们使用“虚构”的原色来定义巨大的色域。在本书后续内容中，你一定会遇到一些这样的例子。

4.3.6 Pointer's Gamut

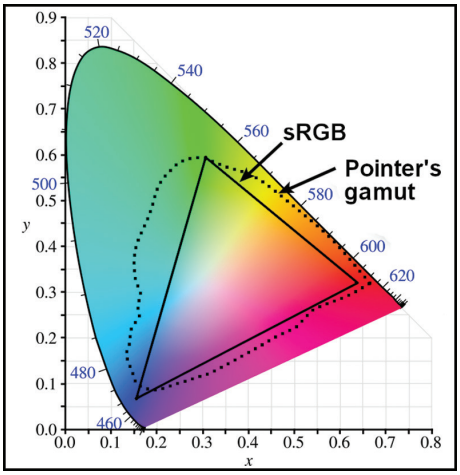


Figure 4.16 Pointer's and sRGB gamuts

Pointer's gamut is not part of any color space but rather an outline of naturally seen colors—the actual colors of diffuse reflective surfaces visible in the real world such as grass, rocks, cloth, and hair. In Figure 4.16, Pointer's gamut is plotted compared to the sRGB color gamut for monitors. Two things become immediately apparent. First, Pointer's gamut is much larger than the gamut of our monitors, which means our monitors cannot display all of the colors found in the real world. Second, it is irregularly shaped because it is derived not from three simple primaries, like sRGB, but from measurements of the reflected light of thousands of regular objects in the real world. It does not include specular reflections because those take on the color and intensity of the light source, not the materials that reflect the light.

Pointer's gamut is important because it is a summary of the range of colors produced by the real world, and it reveals that we would need a color space with a much larger gamut if we wanted to see all of the colors of the real world.

What a difference a gamut makes. Watch this video as the gamut is changed in real time and see the impact on the displayed image.

4.3.7 CIEXYZ and CIELAB (CIE L*a*b*)

You need not concern yourself with the CIEXYZ or CIELAB color spaces, as they tend to be used as closed black boxes. I mention them here simply

4.3.6 指针色域

Pointer色域并非任何色彩空间的一部分，而是自然可见色彩的轮廓——即现实世界中漫反射表面（如草地、岩石、布料和毛发）呈现的实际色彩。在图4.16中，Pointer色域与显示器sRGB色域被并列绘制。两点事实立刻显现：首先，Pointer色域远大于显示器色域，这意味着显示器无法呈现现实世界中的所有色彩；其次，因其基于{v*}

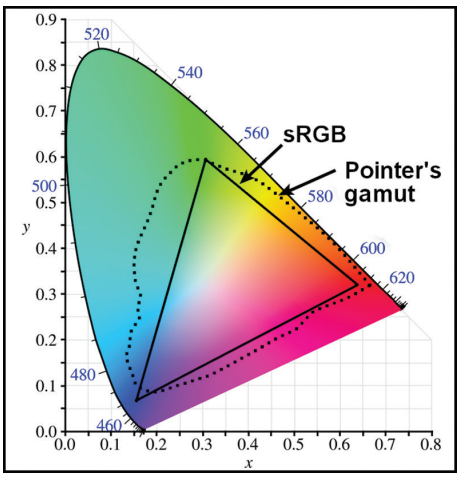


图 4.16 Pointer色域与sRGB色域

并非像sRGB那样源自三种简单的原色，而是基于对现实世界中数千种常规物体反射光的测量。它不包含镜面反射，因为镜面反射呈现的是光源的颜色和强度，而非反射光线的材料本身的特性。

指针色域之所以重要，是因为它概括了现实世界产生的色彩范围，并揭示出如果我们想要看到现实世界的所有颜色，就需要一个色域大得多的色彩空间。

色域带来的变化何其显著。观看这段视频，实时观察色域变化如何影响显示画面的效果。

4.3.7 CIEXYZ 与 CIELAB（CIE L*a*b*）

您无需关注CIEXYZ或CIELAB色彩空间，因为它们通常被当作封闭的黑箱使用。我在此提及它们仅仅

because you will trip over them frequently in your color management travels and if you are like me, you will want to have at least some idea of what the heck they are, even if you might never actually have to work with them. They are mathematically abstract color spaces not grounded in any physical devices, like Rec. 709 for HDTV or DCI-P3 for theatrical projectors. Rather, they are perceptual color spaces based on human colorimetry.

They were created as a workspace for convenient color computations and as such need to be very large to make sure they encompass all color gamuts. You will most likely encounter them when they are used as a connection space when converting between ICC profiles—something we will learn more about later. But you don't have to worry about working with them, as they are internal representations of color spaces used for calculations. Here is a brief explanation of them.

CIEXYZ is the original three-dimensional color space determined by the CIE in 1931 by measuring human observers to map out the full range of human vision. The CIE xy-chromaticity diagram is actually a two-dimensional slice of this 3D color space. CIEXYZ is named after the three imaginary primaries, XYZ, that define a color space large enough to contain all of the colors perceived by humans.

It is a luminance/chrominance color representation where the Y axis is luminance and the X and Z axes carry the chromaticity information. Because it is based on human visual sensibilities, it is referred to as a "colorimetric color space," as opposed to a device-dependent color space like Rec. 709 for HDTV sets. As such, it is most often used as a central "connection" space to convert between any two color spaces.

The reasons it was created are that it contains all visible colors, separates luminance from chrominance, and has no negative coordinates of the spectral colors (don't ask!). One of its main uses beyond a connection space is to report color samples from scientific instruments which can then be transformed into other color spaces for specific uses.

CIELAB (a.k.a. CIE *L*a*b**) is derived from the CIEXYZ color space. It is another device-independent luminance/chrominance color representation specifically designed to be perceptually uniform. Its main virtue is that a change to the color values will result in a matching perceptual change.

因为在色彩管理之旅中，你会经常遇到它们，而且如果你像我一样，即使可能永远不必实际使用它们，也至少会想对它们究竟是什么有个大概了解。它们是数学上抽象的色彩空间，不像用于高清电视的Rec. 709或用于影院投影机的DCI-P3那样基于物理设备，而是基于人类色度学的感知色彩空间。

它们被创建为一个便于进行颜色计算的工作空间，因此需要非常大以确保涵盖所有色域。您最有可能在ICC配置文件之间转换时遇到它们，此时它们被用作连接空间——这一点我们稍后会进一步了解。但您无需担心如何使用它们，因为它们只是用于计算的色彩空间内部表示。以下是关于它们的简要说明。

CIEXYZ是国际照明委员会（CIE）于1931年通过测量人类观察者而确定的原始三维色彩空间，旨在描绘人类视觉的完整范围。CIE xy色度图实际上是这个三维色彩空间的二维切片。CIEXYZ以三个假想原色XYZ命名，它们定义了一个足以涵盖人类所有感知色彩的色域。

它是一种亮度/色度颜色表示法，其中Y轴代表亮度，X轴和Z轴承载色度信息。由于基于人类视觉感知特性，它被称为“色度学色彩空间”，与设备依赖型色彩空间（如用于高清电视的Rec. 709标准）相对。因此，它最常被用作核心的“连接”空间，用于任意两个色彩空间之间的转换。

它被创建的原因在于它包含了所有可见颜色，将亮度与色度分离，并且没有光谱颜色的负坐标（别问！）。除了作为连接空间之外，它的一个主要用途是报告科学仪器中的颜色样本，这些样本随后可以转换为其他颜色空间以供特定用途。

CIELAB（亦称CIE L*a*b*）源自CIEXYZ色彩空间，是另一种设备无关的亮度/色度色彩表示体系，专门设计为感知均匀的色彩空间。其主要优点在于：色彩数值的变化会带来与之匹配的感知变化。

For example, a 2% increase in the code values will result in a 2% increase in perceived brightness in the darks as well as in the brights. Try this same operation in sRGB and you will find the 2% change shows up big in the darks and hardly at all in the brights because sRGB is not perceptually uniform, while CIELAB is.

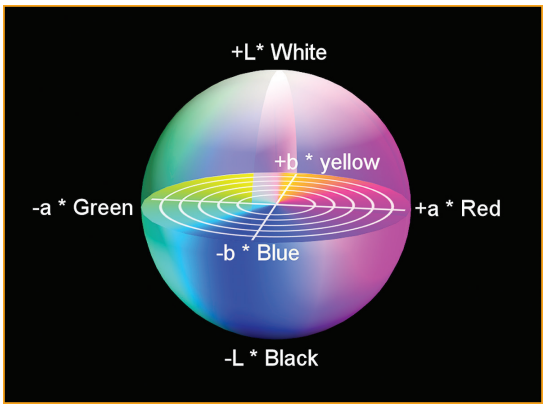


Figure 4.17 CIE L*a*b color space

Figure 4.17 shows the L*a*b coordinate system, with the +L luminance axis going to white and -L going to black. The "a" axis is green to red and the "b" axis is blue to yellow. Because it is perceptually uniform, it is very valuable in a range of industries where there is a need to specify colors with accuracy and repeatability. Examples include color management for print, photo editing, manufacturing of textiles, plastics, and automotives as well as matching paints at the paint store, keeping cosmetics the right color, and many more. It is also commonly used as the connection space for color space conversions—that is, to convert sRGB to CIELAB to CMYK for printing.

4.4 Linear vs. Gamma Corrected

A key decision when setting up your Work Space is whether to work in linear or a gamma-corrected space like sRGB or Rec. 709. A "gamma-corrected workspace" means your images are all gamma-corrected, with the display gamma curve baked into the RGB values. This makes them immediately displayable on monitors (sRGB) or HDTV (Rec. 709), so another term for this might be "working in display space." If you were working with video on an HDTV display, indeed, you would be working in display space—namely the color space of the HDTV. However, if you were working with Rec. 709 footage on a workstation monitor, which is sRGB, we should not say you were working in display space lest the Color Science Police suspend your software license.

例如，代码值增加2%会导致暗部和亮部的感知亮度均增加2%。在sRGB中尝试相同的操作，你会发现2%的变化在暗部表现明显，而在亮部几乎无法察觉，这是因为sRGB不具备感知均匀性，而CIELAB则具备这一特性。

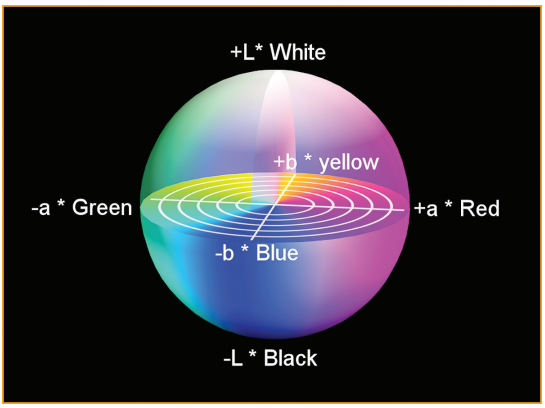


图 4.17 CIE L*a*b 色彩空间

图 4.17 显示了 L*a*b 坐标系，其中+L亮度轴指向白色，-L指向黑色。“a”轴从绿色到红色，“b”轴从蓝色到黄色。由于其感知均匀性，它在众多需要精确且可重复指定颜色的行业中极具价值。例如印刷色彩管理、照片编辑、纺织品、塑料和汽车制造，以及油漆店的颜色调配、保持化妆品颜色准确等众多领域。它也常被用作色彩空间转换的连接空间——例如，将sRGB转换为CIELAB，再转换为CMYK用于印刷。

4.4 线性与伽马校正

设置工作空间时的一个关键决策是选择在线性空间还是伽马校正空间（如sRGB或Rec. 709）中工作。“伽马校正工作空间”意味着您的所有图像都经过伽马校正，显示器的伽马曲线已嵌入RGB值中。这使得它们能直接在显示器（sRGB）或高清电视（Rec. 709）上显示，因此也可称为“在显示空间中工作”。如果您在高清电视上处理视频，确实是在显示空间——即高清电视的色彩空间中工作。然而，若您在工作站显示器（sRGB）上处理Rec. 709素材，则不应称其为在显示空间中工作，以免色彩科学警察吊销您的软件许可证。

A huge amount of work is done in gamma-corrected space, especially for video, with most artists not even realizing it or the impact that it has on their work. Whenever you work with video like ProRes, DNxHR, or MP4, you are using gamma-corrected images. If your project uses files like TIFF, PNG, JPEG, or BMP, you are also working with gamma-corrected images. While all of these can be linearized to work in linear space, most don't bother. To be fair, excellent results can still be achieved while working in gamma-corrected space, but as we shall shortly see, the results are not exactly correct. In this section, we will understand what it means to work in gamma-corrected space and its consequences.

Many file formats considered linear are in fact gamma-corrected linear, and this affects the results of our post-processing and color correction math operations. The difference between working with gamma-corrected images vs. true linear images is illustrated in Figure 4.18. These are simply two 50% gray patches summed together with an overlapping region.

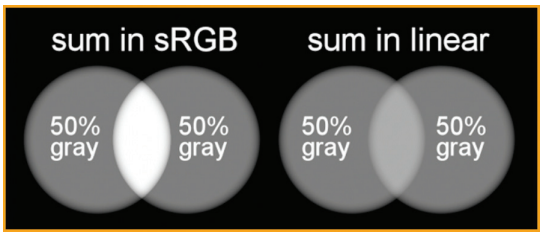


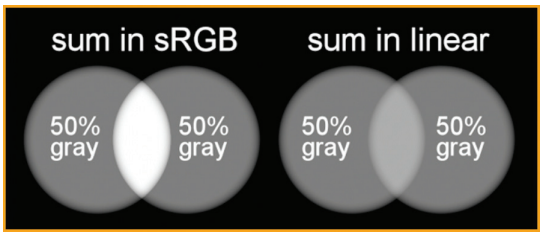
Figure 4.18 Math results differ in linear vs. sRGB

The "sum in sRGB" example on the left is the result of two 50% gray patches already in sRGB color space being summed together. The "sum in linear" example on the right sums two gray patches that have been linearized and then converts the results to sRGB for display. While both pairs of 50% gray patches are identical, the overlapped regions are not. The difference is the result of the sum occurring in sRGB color space vs. linear space.

Now here's the punchline: the "sum in linear" illustration is in fact what you would see in the real world if two spotlights were overlapped like this. The overlapped region would appear about 18% brighter (from Section 1.5.3 Our Visual Response) than each spotlight, not the 100% brighter result for the "sum in sRGB." This little summing demo is just the tip of the non-linear iceberg and neatly illustrates why you have to work in linear if you want your images to be truly correct. Other visual virtues of working in linear are more photorealistic glows, defocus, and blending operations.

大量工作是在伽马校正空间完成的，尤其是在视频领域，大多数艺术家甚至没有意识到这一点，也不了解这对他们作品的影响。每当你处理ProRes、DNxHR或MP4等视频时，你使用的就是经过伽马校正的图像。如果你的项目使用TIFF、PNG、JPEG或BMP等文件，你同样在处理伽马校正图像。虽然所有这些都可以线性化以在线性空间工作，但大多数人并不在意。公平地说，在伽马校正空间工作仍能取得出色的效果，但正如我们即将看到的，结果并不完全正确。在本节中，我们将理解在伽马校正空间工作的含义及其后果。

许多被认为是线性的文件格式实际上是经过伽马校正的线性格式，这会影响我们后处理和色彩校正数学运算的结果。图4.18展示了处理经过伽马校正的图像与真实线性图像之间的差异。图中仅是两个50%灰度色块叠加后形成的重叠区域。



左侧的“sRGB 求和”示例展示了两个已在 sRGB 色彩空间中的 50% 灰度色块相加的结果。右侧的“线性空间求和”示例则先将两个灰度色块线性化后相加，再转换回 {v*}。

图 4.18 线性与 sRGB 色彩空间中的数学结果差异

将结果转换为sRGB以进行显示。虽然两对50%灰度色块完全相同，但重叠区域却并非如此。这种差异是由于在sRGB色彩空间与线性空间中进行叠加运算所导致的。

现在关键点来了：如果现实世界中两个聚光灯像这样重叠，你实际看到的正是“线性求和”示意图所展示的效果。重叠区域会比每个聚光灯单独照射时亮约18%（根据1.5.3节《我们的视觉响应》），而非“sRGB求和”得出的亮度翻倍结果。这个简单的求和演示只是非线性冰山的一角，它清晰地解释了为何想要图像完全正确就必须在线性空间工作。在线性空间处理的其他视觉优势还包括更逼真的辉光、散焦和混合效果。

Color correcting is another task that is best done in linear, but of course, one can color correct gamma-corrected images like video (Rec. 709). It is done all the time. Some colorists prefer to color correct in log, but they are using tools specifically designed to work with log data.

Figure 4.19 A gain of 2 applied in three different color spaces

Just to prove the point, Figure 4.19 shows what happens when the same color grade is applied in three different color spaces. When a gain of 2 was applied to the linear version of the original image, the picture became quite presentable. When the same gain was applied to the Rec. 709 version, things started to get a bit blown out. I can't even look at the log image. The gain operation is designed for linear images, not log. Incredibly, if you want to double the brightness of a Cineon log image, you have to add 90 code values to the log data.

4.4.1 What is Gamma, Exactly?

Gamma was touched on briefly in Chapter 2, but here we will zoom in for a closer look because gamma is so pervasive in our world of digital images that it pops up everywhere. Mathematically speaking, gamma is a power function that takes some input value and raises it to some power, and we have simply named that power "gamma." To cite an example, the input value 3 raised to the power of 2 (3^2) produces an output value of 9. We could say that 3 had a gamma of 2 applied to it. The input value need not be an integer, of course. We could take an input value of 0.5 raised to the power of 2 (0.5^2) which produces an output value of 0.25. Further, the gamma value itself need not be an integer. So we could take the input value 0.5 raised to the power of 2.2 ($0.5^{2.2}$), which produces an output value of 0.21763764082 or so.

色彩校正是另一项最好在线性空间完成的任务，但当然，人们也可以对经过伽马校正的图像（如Rec.709标准的视频）进行色彩校正。这种做法一直很常见。有些调色师更喜欢在log空间进行色彩校正，但他们使用的是专门为处理log数据设计的工具。

图 4.19 在三种不同色彩空间中应用增益为 2 的效果

为了证明这一点，图4.19展示了同一色彩等级在三种不同色彩空间中的应用效果。当对原始图像的线性版本施加2倍增益时，画面变得相当可观；而对Rec. 709版本施加相同增益时，画面开始出现过曝现象。至于对数图像，我甚至无法直视——增益操作本是为线性图像设计，而非对数图像。令人惊讶的是，若要将Cineon对数图像的亮度提升一倍，必须在对数数据上增加90个编码值。

4.4.1 Gamma 究竟是什么?

伽马在第二章中略有提及，但这里我们将深入探讨，因为伽马在数字图像世界中无处不在，随处可见。从数学角度讲，伽马是一种幂函数，它接收某个输入值并将其提升到某个幂次，我们简单地将这个幂次称为“伽马”。举例来说，输入值3提升到2次幂（3^2）得到输出值9。我们可以说对3应用了伽马值为2的变换。当然，输入值不必是整数。我们可以取输入值0.5提升到2次幂（0.5^2），得到输出值0.25。此外，伽马值本身也不必是整数。因此，我们可以将输入值0.5提升到2.2次幂（$0.5^{2.2}$），得到输出值约为0.21763764082。

TABLE 4.1 Gamma calculations

Gamma	0.50	2.00
Code values	**Results**	
0.00	**0.00**	**0.00**
0.25	**0.50**	**0.06**
0.50	**0.71**	**0.25**
0.75	**0.87**	**0.56**
1.00	**1.00**	**1.00**

When working with images, gamma values must be applied to floating point numbers rather than integers. So code value 128 in an 8-bit integer image will first be converted to 0.5 to apply a gamma to it, and then the results will be converted back to an 8-bit integer. The same thing is done when multiplying or dividing integer numbers. Addition and subtraction, however, can be done on the original integer pixel values.

One more thing to know about gamma is that you can gamma up or gamma down. Referring to the gamma calculations in Table 4.1, the pixel input code values are given a gamma of 0.50 (green) which is less than 1, and 2.0 (red) which is greater than 1 for comparison. Note that with a gamma of 0.5, all of the results are larger than the original code value while for a gamma of 2.0, the results are smaller. Note also that for code values 0 and 1, the output values are unchanged ($0^n = 0$ and $1^n = 1$) and, of course, a gamma of 1 is no change ($n^1 = n$). This is the way of gamma.

In Figure 4.20, the gamma curves are plotted to illustrate the values from Table 4.1 graphically while Figure 4.21 illustrates them visually to show the effect of gamma on a gradient and an image. Gamma 1.0 is the “no change” gamma for both the curves and the gradients. The takeaway here is that applying a gamma of 0.5 brightened up the images while applying a gamma of 2.0 darkened them down.

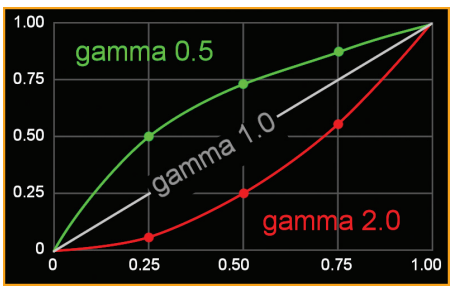


Figure 4.20 Gamma curves

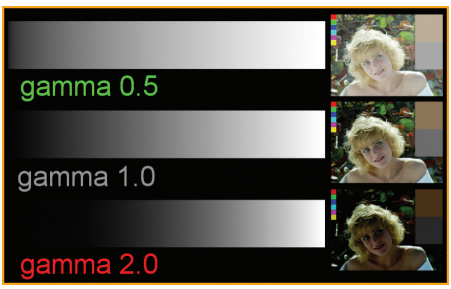


Figure 4.21 Gamma gradients

But wait—something’s wrong here. I am showing that a gamma greater than one darkens the image, but every time you apply a gamma greater than 1,

表 4.1 伽马计算

Gamma	0.50	2.00
Code values	**Results**	
0.00	**0.00**	**0.00**
0.25	**0.50**	**0.06**
0.50	**0.71**	**0.25**
0.75	**0.87**	**0.56**
1.00	**1.00**	**1.00**

在处理图像时，伽马值必须应用于浮点数而非整数。因此，8位整数图像中的代码值128会先转换为0.5以应用伽马校正，随后结果将转换回8位整数。在进行整数乘法或除法运算时，同样需要此类转换。而加法与减法操作则——然而，可以在原始整数像素值上完成。

关于伽马还有一点需要了解，那就是可以进行伽马提升或降低。参考表4.1中的伽马计算示例，像素输入码值分别采用了小于1的0.50伽马值（绿色）和大于1的2.0伽马值（红色）进行对比。值得注意的是，当伽马值为0.5时，所有计算结果都大于原始码值；而伽马值为2.0时，结果则更小。同时需注意，对于码值0和1，输出值保持不变（$0^n=0$ 与 $1^n=1$），且伽马值为1时自然也不产生变化（$n^1=n$）。这就是伽马的工作原理。

在图4.20中，绘制了伽马曲线以图形化展示表4.1中的数值，而图4.21则通过视觉化呈现，展示了伽马值对渐变和图像的影响。伽马值1.0对曲线和渐变而言都是“无变化”的基准值。此处得出的结论是：应用0.5的伽马值会使图像变亮，而应用2.0的伽马值则会使图像变暗。

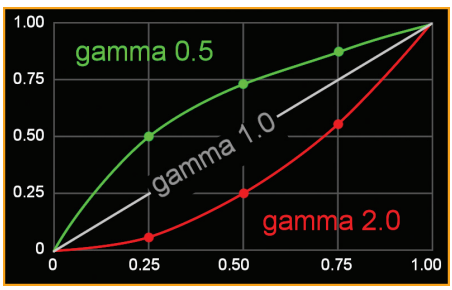


图 4.20 伽马曲线

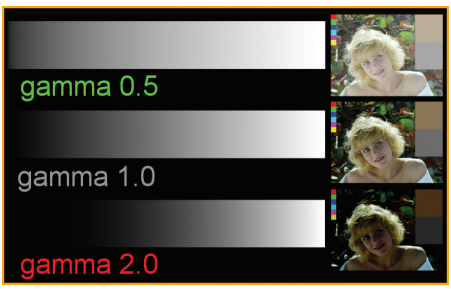


图 4.21 伽马梯度

但是等等——这里有点不对劲。我展示的是大于1的gamma值会使图像变暗，但每次你应用大于1的gamma时，

it brightens the image. So who is right: me or your software? Why, me, of course! What I am showing is mathematically correct, but it is also counter-intuitive—a larger gamma making the picture darker just seems wrong. For example, a Gain of 2 will definitely brighten an image, while a Gain of 0.5 will darken it. So virtually all software developers actually invert the number you enter (1/your gamma) so that large gamma values brighten and small values darken, just like nature intended. Now that we have established precisely what gamma is, both mathematically and visually, we are ready for the larger discussion about working with gamma in our Work Space.

4.4.2 Monitor Gamma

Due to the infernal internal physics of the electron guns used in CRTs (Cathode Ray Tubes) used for TV screens and monitors since the 1950s, the image displayed on the screen is seriously darkened by the monitor's gamma 2.4 response curve, as illustrated in Figure 4.22, with the *gamma darkened* image on the left. The inability of the CRT to correctly display the image brightness posed a serious engineering problem in the day. Something had to be done.

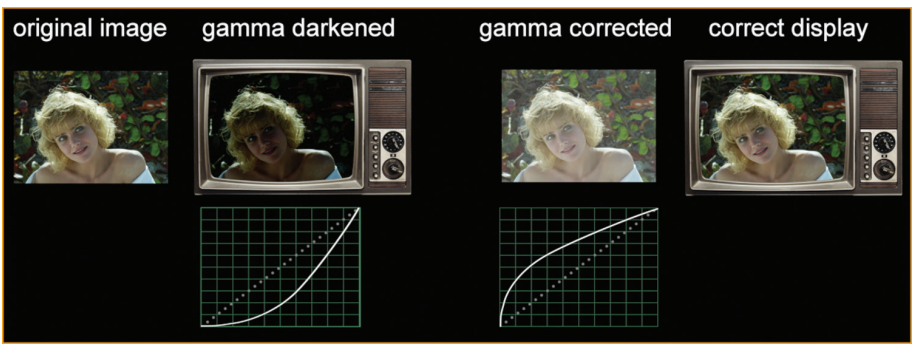


Figure 4.22 How the CRT gamma affected the displayed image

FUN FACT: The very first efforts to invent television in the 1920s were mechanical. They used a spinning disk with spiral holes to scan the scene to a photoelectric cell that transmitted a signal to a similar synchronized disk spinning at the receiving end.

它能使图像变亮。那么谁是对的：我还是你的软件？当然是我！我所展示的在数学上是正确的，但它也违反直觉——更大的伽马值让图片变暗似乎就是不对的。例如，增益为2肯定会让图像变亮，而增益为0.5则会使其变暗。因此，几乎所有软件开发人员实际上都会将你输入的数字取倒数（1/你的伽马值），这样一来，大的伽马值会增亮，小的伽马值会变暗，正如自然规律所预期的那样。既然我们已经从数学和视觉上精确地确立了伽马是什么，我们现在就可以准备进行关于在工作空间中处理伽马值的更广泛讨论了。

4.4.2 显示器伽马值

由于自20世纪50年代以来用于电视屏幕和显示器的阴极射线管（CRT）中电子枪的内部物理特性，屏幕显示的图像受到显示器伽马2.4响应曲线的严重影响而变暗，如图4.22所示，左侧为伽马变暗的图像。当时，CRT无法正确显示图像亮度构成了严重的工程问题。必须采取一些措施。

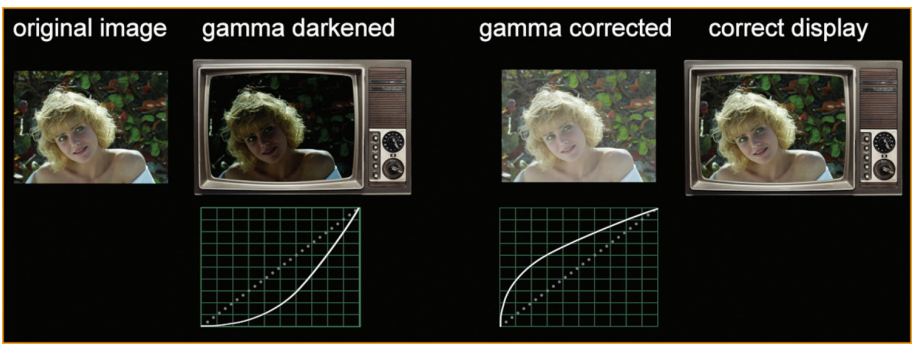


图4.22 CRT伽马值如何影响显示图像

有趣的事实：20世纪20年代发明电视的最初尝试是机械式的。他们使用带有螺旋孔的旋转圆盘将场景扫描到光电管上，光电管将信号传输到接收端一个类似同步旋转的圆盘上。

4.4.3 Gamma Correction

The engineering solution was to pre-brighten all images in-camera to gamma correct them with an inverse gamma. When this gamma-corrected image was displayed on a monitor, they would cancel each other out and the picture would look normal, as shown in the *correct display* in Figure 4.22.

For the mathletes among you, the inverse gamma is simply 1/monitor gamma—which in this case is 1/2.4 (one divided by 2.4) which is 0.42. This means we want to raise the code value 0.5 to the power of 0.42 to perform a gamma correction, which results in a code value of 0.75 ($0.5^{.42} = 0.75$). So the original linear code value of 0.5 has been pre-brightened with a gamma correction to 0.75, which then goes onto the CRT, which in turn darkens it by the monitor gamma of 2.4 resulting in an on-screen brightness of 0.5. The rest of us will just remember the gamma for Rec. 709 is 2.4 and the gamma for sRGB is 2.2 and get on with life.

But we don't use CRTs anymore, so why are we still doing this? For backward compatibility with all previous video. When the modern flat-panel displays were developed, it was decided to add a factory gamma LUT curve to match the old CRTs so that the billions of existing videos and images would look right on the new displays. This is why image file formats intended for display on monitors or TV are gamma-corrected and not truly linear.

4.4.4 Gamma-Corrected Images

Whenever you load a TIFF, PNG, JPEG, PSD, ProRes, or any other image file designed to be displayed on a monitor, it means the image has had a gamma correction baked into it, so it is not truly linear. These images are actually in the display space of the monitor (we will learn all about this in the next exciting chapter!), and as we saw in Figure 4.18, that affects the results of the math operations. It affects our work because, as mentioned before, all image processing and color correction operations want to be done on linear images for the results to be truly correct. To linearize a gamma-corrected image, the baked-in gamma correction must be backed out with an inverse LUT.

As mentioned, you can certainly work successfully with gamma-corrected images even if it's not quite right. Most of our video and web work is done this way because image processing and color correction operations still look

4.4.3 伽马校正

工程解决方案是在相机内将所有图像预亮化，通过逆伽马进行伽马校正。当这种经过伽马校正的图像在显示器上显示时，两者会相互抵消，画面看起来就会正常，如图4.22中的正确显示所示。

对于你们中的数学爱好者来说，逆伽马值就是1除以显示器伽马值——本例中为1/2.4（即1除以2.4），结果是0.42。这意味着我们需要将代码值0.5进行0.42次幂运算以实现伽马校正，最终得到代码值0.75（$0.5^{.42}$=0.75）。因此原始线性代码值0.5经过伽马校正预提亮至0.75，随后输入CRT显示器，再由显示器2.4的伽马值将其暗化，最终在屏幕上呈现0.5的亮度值。其余人士只需记住Rec.709的伽马值是2.4，sRGB的伽马值是2.2，然后继续享受生活即可。

但我们不再使用CRT显示器了，为什么还要这样做呢？这是为了向后兼容所有旧视频。当现代平板显示器被开发出来时，制造商决定添加出厂伽马查找表曲线来匹配旧CRT显示器的效果，以确保数十亿现有的视频和图像在新显示器上看起来正常。这就是为什么用于显示器或电视显示的图像文件格式都经过伽马校正，而非真正线性的原因。

4.4.4 伽马校正图像

每当你加载一个TIFF、PNG、JPEG、PSD、ProRes或其他任何设计用于在显示器上显示的图像文件时，就意味着该图像已经嵌入了伽马校正，因此它并非真正的线性图像。这些图像实际上处于显示器的显示空间（我们将在下一章中详细了解这一点！），并且正如我们在图4.18中所见，这会影响数学运算的结果。它影响我们的工作，因为如前所述，所有图像处理和色彩校正操作都应在线性图像上进行，结果才能完全正确。要将经过伽马校正的图像线性化，必须使用反向查找表（LUT）来移除嵌入的伽马校正。

如前所述，即使不完全准确，您也完全可以成功地处理伽马校正后的图像。我们的大部分视频和网络工作都是这样完成的，因为图像处理和色彩校正操作看起来仍然

reasonable. We paint Photoshop images in sRGB color space and color-correct video in Rec. 709 all the time. But for high-end VFX and CG work, "looking reasonable" is not good enough. To work at this level, it has to be right in the real world. And the only way for it to truly be right is to work in linear.

Since you have just finished reading this excellent description of gamma, you probably don't need to watch this video showing it in action.

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4.4.5 Linear Images

What makes an image linear is that it has a one-to-one relationship between its code values and the original image sensor data, which is itself a linear device. Linear images create a problem when they are displayed on a monitor. Recall that all monitors have a "legacy LUT" to give them a gamma 2.2 (sRGB) or 2.4 (Rec. 709) response like the old CRTs, which seriously darkens the displayed image (see Figure 4.22). For the linear image to display correctly on a monitor, we need to add a display LUT with a gamma of 2.2 or 2.4 to map it to the display screen correctly. Keep in mind that a display LUT changes the code values on its way to the monitor but leaves the original linear data intact.

Figure 4.23 Comparison of luminance code values sRGB vs. linear images

合理的。我们总是在sRGB色彩空间中绘制Photoshop图像，并在Rec. 709标准下进行视频色彩校正。但对于高端的视觉特效和计算机图形工作来说，“看起来合理”是远远不够的。要在这个级别上工作，它必须在现实世界中是正确的。而唯一能确保其真正正确的方法，就是在线性空间中进行工作。

既然你刚刚读完这篇关于伽马函数的精彩描述，可能就不需要观看这个展示其实际应用的视频了。

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4.4.5 线性图像

图像之所以呈线性，是因为其编码值与原始图像传感器数据之间存在一一对应关系，而传感器本身即是线性设备。当线性图像在显示器上呈现时会产生问题。请记住，所有显示器都配有“传统LUT”以模拟旧式CRT显示器的伽马2.2（sRGB）或2.4（Rec. 709）响应特性，这会导致显示图像严重变暗（见图4.22）。为使线性图像在显示器上正确显示，我们需要添加伽马值为2.2或2.4的显示LUT，将其准确映射至显示屏幕。需注意：显示LUT会在信号传输至显示器的过程中改变编码值，但原始线性数据本身保持不变。

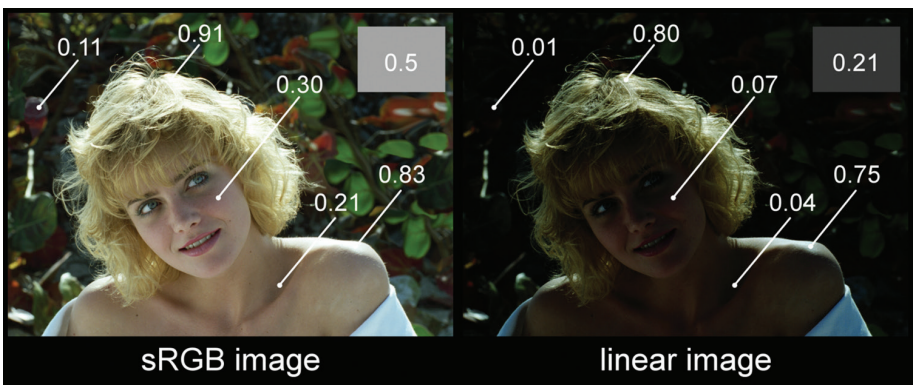

图 4.23 sRGB与线性图像亮度编码值对比

One irksome idiosyncrasy of linear images illustrated in Figure 4.23 is that without a gamma correction, they have disturbingly small code values. The sRGB image on the left has familiar floating-point code values for luminance typical of an image of this content and the 50% gray chip in the upper right-hand corner has a code value of 0.5, as expected.

However, checking the linear image on the right, the 50% gray chip now has a code value of 0.21. Compare the sRGB cheek at 0.30 to the 0.07 in linear—a huge drop in code value. However, the hair highlight in sRGB of 0.91 only drops to 0.80 in linear—not such a big drop. The reduction in code values is large in the darks and less in the brights because of the shape of the gamma curve, as illustrated in the *gamma-corrected* curve in Figure 4.22.

Each file type—such as TIFF, JPEG, ProRes, or DPX—has a built-in gamma or log LUT baked into it. There is only one file type, OpenEXR, that specifically calls out for linear images with no gamma or log LUTs baked in. Keeping the image data linear was one of the main design criteria when the OpenEXR file was developed. When an EXR image is displayed on a monitor, a viewing LUT must be added to make the image look right.

When you first start working with linear images, they can be hard on your head. This video will ease the pain for you.

4.5 Integer vs. Float

The next key decision point in choosing your workspace is whether to work in integer or floating point (float). Integer was faster, but float was of higher quality. However, with today's high-performance computers, one can work in float with no real speed penalty. Note that choosing a color space to work in does not dictate integer or float. You can work in integer or float in sRGB, Rec. 709, or linear color spaces.

The early days of image processing—as exemplified by Adobe Photoshop and After Effects—was 8-bit integer, simply because early computers had

线性图像的一个令人烦恼的特性如图4.23所示：未经伽马校正时，其代码值会小得令人不安。左侧的sRGB图像具有符合此类内容图像典型特征的熟悉浮点亮度代码值，右上角的50%灰度色块代码值如预期为0.5。

然而，检查右侧的线性图像，50%灰度的色块现在具有0.21的编码值。将sRGB模式下0.30的面颊值与线性模式下的0.07对比——编码值大幅下降。但sRGB模式下0.91的发丝高光在线性模式下仅降至0.80——降幅并不大。由于伽马曲线的形状，编码值的减少在暗部区域较大，在亮部区域较小，如图4.22中经伽马校正的曲线所示。

每种文件类型——如TIFF、JPEG、ProRes或DPX——都内置了伽马或对数查找表。只有一种文件类型，即OpenEXR，明确要求存储未经伽马或对数查找表处理的线性图像。保持图像数据的线性是开发OpenEXR文件时的主要设计准则之一。当EXR图像在显示器上显示时，必须添加一个查看查找表才能使图像看起来正常。

当你刚开始接触线性图像时，可能会感到头疼。这个视频将为你缓解这种不适。

4.5 整数与浮点数

选择工作空间的下一个关键决策点是使用整数还是浮点（float）格式。整数格式速度更快，但浮点格式质量更高。不过，凭借当今的高性能计算机，使用浮点格式工作并不会造成实质性的速度损失。需要注意的是，选择色彩空间并不决定必须使用整数或浮点格式。您可以在sRGB、Rec. 709或线性色彩空间中使用整数或浮点格式进行工作。

图像处理的早期——以Adobe Photoshop和After Effects为例——采用的是8位整数，这主要是因为早期计算机的

slow central processing units (CPUs) and limited storage space. It takes more disk space to store 16-bit images and much faster CPUs to do floating-point math. Early computers had to have an optional separate chip for floating point math, called a Floating Point Accelerator (FPA). Today's CPUs, with their billions of transistors, have an FPA built into the chip. Due to the vastly higher performance in today's computers, Photoshop and After Effects offer options to work in 16 bits and even floating point—for selected operations.

4.5.1 Integer

For an 8-bit integer workspace, the numbers will range from zero to 255 in whole number steps—0, 1, 2, 3 ... 253, 254, 255—with no decimal places. While addition and subtraction can be done with integer math directly, multiplication and division are converted to floating point behind the screen and the results are rounded off to the nearest integer, which introduces small round-off errors. These operations must be temporarily converted to floating point because, for example, a 50% gray pixel (integer 128) multiplied by another 50% gray pixel must be evaluated as 0.5 x 0.5 (returning 0.25) rather than 128 x 128 (returning 16,385). A normal workflow will include dozens of these integer-to-float/float-to-integer conversions, each adding a small round-off error that accumulates over the processing pipeline. There will also be clipping because, for example, code value 200 + 200 (returning 400) will be instantly clipped to 255.

Interestingly, 16-bit integer math, which ranges from zero to 65,535, is also normalized from zero to 1.0 in float; so while there is a great increase in precision to reduce round-off errors and avoid color banding, there is no increase in dynamic range to solve clipping problems. Rec. 709 video also supports an integer 10-bit option ranging from zero to 1023, which reduces color banding, but that too normalizes from zero to 1.0 in float. Integer math may be adequate for broadcast and web applications, but it falls short when it comes to feature film work or future-proofing your project.

4.5.2 Float

When working in float, an 8-bit integer number is first "normalized," meaning it is converted to a floating point number between zero and 1.0. For example, 8-bit integer 128 becomes float 0.5, which does not introduce a round-off

缓慢的中央处理器（CPU）和有限的存储空间。存储16位图像需要更多的磁盘空间，而进行浮点运算则需要快得多的CPU。早期的计算机不得不配备一个可选的独立芯片来处理浮点运算，这种芯片被称为浮点加速器（FPA）。如今，拥有数十亿晶体管的现代CPU已将FPA集成到芯片内部。由于当今计算机性能的大幅提升，Photoshop和After Effects提供了在16位甚至浮点模式下工作的选项——适用于特定操作。

4.5.1 整数

对于一个8位整数工作空间，数值范围将以整数步长从零到255——0、1、2、3……253、254、255——没有小数位。虽然加法和减法可以直接使用整数运算完成，但乘法和除法会在后台转换为浮点数进行计算，结果会四舍五入到最接近的整数，这会引入微小的舍入误差。这些运算必须临时转换为浮点数，因为例如，一个50%灰度的像素（整数值128）乘以另一个50%灰度的像素时，必须按0.5 × 0.5（返回0.25）计算，而不是128 × 128（返回16,385）。正常的工作流程会包含数十次这样的整数到浮点数/浮点数到整数的转换，每次转换都会增加微小的舍入误差，这些误差在处理流程中不断累积。此外还会出现数值裁剪，例如，代码值200 {v*} 200（返回400）会立即被裁剪到255。

有趣的是，16位整数运算（范围从0到65,535）在浮点数中也被归一化为0到1.0；因此，虽然精度大幅提升以减少舍入误差并避免色彩条带，但动态范围并未增加以解决裁剪问题。Rec. 709视频还支持从0到1023的10位整数选项，这减少了色彩条带，但同样在浮点数中归一化为0到1.0。整数运算对于广播和网络应用可能足够，但在处理故事片工作或为项目未来兼容性做准备时则显不足。

4.5.2 浮点数

在使用浮点数处理时，一个8位整数首先会被“归一化”，即转换为0到1.0之间的浮点数。例如，8位整数128会变成浮点数0.5，这一过程不会引入舍入误差。

error. Since all subsequent math operations stay in float, they do not accumulate visible round-off errors, but they do accumulate infinitesimally small, billionth-of-a-code value errors. A floating-point pixel value of 0.503180813 will be written to the frame buffer for display on screen as integer code value 128 because the frame buffer only takes integers, but the original floating-point number is retained internally for all processing. At the end of the processing pipeline, if the image is written to disk in an integer file format such as TIFF or DPX, there will be a small round-off error converting the final floating-point pixel values to integer. For example, float 0.5 will be converted to integer 128, but then again, so will float 0.503.

FUN FACT: It's called "floating point" because the decimal point is free to float to new positions based on the size of the number—for example, 3,563,179 could be written as 3.563179×10^6 or as 356.3179×10^4.

For the high-end work of feature film or the next-generation video standards, floating point math is an absolute requirement. Floating point math avoids image degradation from multiple image processing operations and also solves the dynamic range problem, as it comfortably handles code values well above 1.0. In an all-float pipeline, you might save your image data as OpenEXR files, for example, which are saved as 16-bit half-float values that are "promoted" to 32-bit float when imported into the computer for processing. The logic here is that storing the image as a 16-bit half-float retains far more data about the image than you will ever need, and then promoting it to 32-bit for processing adds so much more precision that you will never visibly degrade your images. Yes, there are "micro-degradations," but round-off errors of this infinitesimal small size are far below the visible threshold.

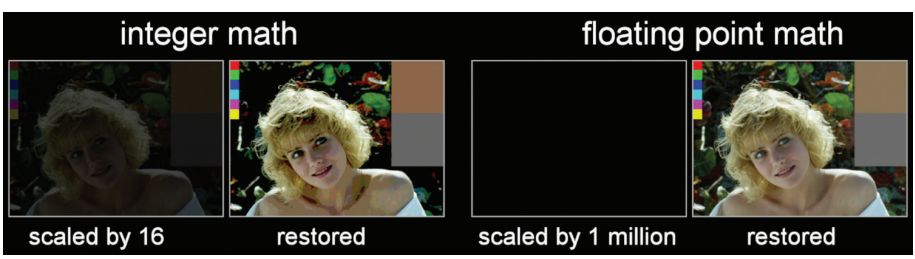

Figure 4.24 Difference between scaling RGB values in integer vs. float

错误。由于所有后续数学运算都保持在浮点数中进行，它们不会累积明显的舍入误差，但会累积极其微小的、十亿分之一码值的误差。一个浮点像素值0.503180813会被写入帧缓冲区，在屏幕上显示为整数码值128，因为帧缓冲区只接受整数，但原始的浮点数在内部所有处理过程中都会保留。在处理管道的末端，如果图像以整数文件格式（如TIFF或DPX）写入磁盘，将浮点像素值转换为整数时会产生微小的舍入误差。例如，浮点数0.5会被转换为整数128，但同样，浮点数0.503也会被转换为128。

> 有趣的事实：之所以称为“浮点数”，是因为小数点可以根据数字的大小自由浮动到新位置——例如，3,563,179 可以写作 $3.563179x10^6$ 或 $356.3179x10^4$。

对于电影特效或下一代视频标准的高端制作而言，浮点运算是不可或缺的。浮点运算能避免多重图像处理操作导致的画质衰减，同时解决动态范围问题——因为它能轻松处理远高于1.0的编码值。在全浮点处理流程中，您可以将图像数据存储为OpenEXR文件（例如），这类文件以16位半浮点值保存，导入计算机处理时会"升级"为32位浮点值。其原理在于：以16位半浮点格式存储的图像所保留的数据量远超实际需求，而升级至32位处理时增加的精度之高，足以确保图像永远不会出现肉眼可见的劣化。诚然存在"微观劣化"，但如此微小的舍入误差远低于视觉可辨的阈值。

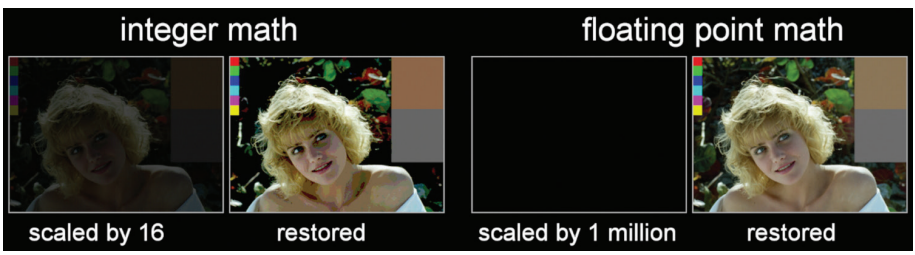

图4.24 RGB值在整数与浮点数缩放中的差异

I often demonstrate the image degradation problem with integer math using the demo in Figure 4.24. I take an 8-bit TIFF image in Photoshop and scale the RGB values down by a factor of 16 to nearly black, then restore it to its original brightness by the same factor of 16. Photoshop works in integer math, so the image erupts with hideous color banding artifacts in the skin tones and the blacks are crushed to a single value.

I then repeat the demo with floating-point math using the same 8-bit TIFF image, but this time I scale the RGB values down by 1 million and then restore it by scaling it back up by 1 million. The original 8-bit image is restored perfectly, with no visible degradation.

This video highlights the visual differences when working in float vs. integer.

4.6 LDR vs. HDR Images

We live in a bifurcated world where all images can be sorted into two distinct piles—LDR (Low Dynamic Range) and HDR (High Dynamic Range)—which seriously affects our choice of Work Space. LDR images are viewed on standard dynamic range displays of 500 nits or less—like what you find on the internet, cell phones and tablets, DVDs, HDTV, and print. Their data may be either an 8-bit or 16-bit integer or floating point between zero and 1.0. HDR images are intended for much brighter displays of 1000 nits or more and are found in Blu-Ray, UHDTV, gaming, and a few streaming services. Their data must be floating point and range well above 1.0.

And don't think it's all about bit depth. There are 8-bit and 10-bit LDR images, as well as 10 and 12-bit HDR images. There *shouldn't* be any 8-bit HDR images, since they are sure to exhibit banding on a 1000-nit display. In the next chapter, we will look at LDR and HDR displays. Here we are focused on the image data itself.

我经常用图4.24中的演示来说明整数运算导致的图像质量退化问题。在Photoshop中取一张8位TIFF图像，将RGB值缩小16倍至接近全黑，再用同样的16倍系数恢复其原始亮度。由于Photoshop采用整数运算，图像会爆发出可怕的肤色色带伪影，黑色部分则被压缩为单一数值。

随后，我使用同一张8位TIFF图像，以浮点运算重复演示，但这次我将RGB值缩小100万倍，然后再通过放大100万倍来恢复。原始的8位图像被完美还原，未见任何明显画质劣化。

本视频重点展示了使用浮点数与整数时视觉上的差异。

4.6 LDR与HDR图像对比

我们生活在一个二分化的世界里，所有图像都可以被归入两个截然不同的类别——LDR（低动态范围）和HDR（高动态范围）——这严重影响了我们对工作空间的选择。LDR图像适用于500尼特或更低的标准动态范围显示器，例如互联网、手机和平板电脑、DVD、高清电视以及印刷品中常见的显示设备。其数据可以是8位或16位整数，也可以是0到1.0之间的浮点数。HDR图像则面向亮度更高的显示器（1000尼特以上），常见于蓝光光盘、超高清电视、游戏及少数流媒体服务。其数据必须采用浮点数格式，且数值范围远高于1.0。

别以为这全是关于位深的问题。既有8位和10位的LDR图像，也有10位和12位的HDR图像。理论上不应存在8位的HDR图像，因为它们在1000尼特的显示屏上必然会出现色带。下一章我们将探讨LDR与HDR显示技术，而本章我们聚焦于图像数据本身。

As you know, with 8 bits per channel, you can get 16.7 million possible colors. With 10 bits, this explodes to over 1 billion. And we won't even talk about 12-bit color. It's not so much about how many billions of colors you can make with your bits, it's really about how fine each color gradient step is so that there is no banding or contouring on the very bright HDR displays. Streaming services like Netflix, Amazon Prime Video, and Disney+ offer HDR content at the time of writing. However, the world is drifting toward HDR, so over time there will no doubt be additions to this list.

4.7 Choosing Your Work Space

The choice of a workspace is fundamental to maintaining the quality of the work. The workspace should be larger than any of the color spaces used in the project, for the simple reason that some math operations can generate interim results with large numbers and if they fall outside of the workspace, they will get clipped and damage the image quality. As we will soon see, this is one of the main problems with using the display space as the workspace.

Armed with the knowledge of what a color space is, plus an understanding of the impact of working in linear or gamma-corrected images and integer or float, we are now fully equipped to tackle the question of the moment: what should your Work Space be? To answer this, you need to decide what your overall priorities are. Of course, the nice client will have the final say here, because there may be costs associated with producing the job for future-proofing and risks associated with not doing so. Here are some general guidelines:

- **LDR project with no need for future-proofing** - This might be episodic HDTV, commercials, or web content. The workspace can be the same as the display space (i.e., Rec. 709 or sRGB) so images are gamma-corrected and it's okay to use integer math—16 bits if you can.
- **LDR project but needs future-proofing** - This might be high-end episodic HDTV (e.g., *Game of Thrones*), so the producers want to be able to redistribute it in the future. The workspace needs to be linear and float—maybe even ACES.

如您所知，每个通道8位可呈现1670万种颜色。若采用10位色深，颜色数量将激增至超过10亿种。至于12位色深我们更无需多言。关键并非仅在于比特数能创造多少亿种色彩，更在于每个色彩渐变阶跃的精细程度——这能确保在极高亮度的HDR显示屏上不会出现色带或轮廓断层。截至本文撰写时，Netflix、Amazon Prime Video和Disney+等流媒体平台已提供HDR内容。随着全球影像技术向HDR趋势演进，未来无疑将有更多平台加入这一行列。

4.7 选择你的工作空间

工作空间的选择对于保持工作质量至关重要。工作空间应大于项目中使用的任何色彩空间，原因很简单：某些数学运算可能产生数值较大的中间结果，如果这些结果超出工作空间范围，就会被裁剪并损害图像质量。我们很快会看到，这正是将显示空间用作工作空间的主要问题之一。

掌握了色彩空间的知识，加上对在线性或伽马校正图像以及整数或浮点数格式下工作影响的理解，我们现在已全副武装，可以着手解决当前的核心问题：你的工作空间应该是什么？要回答这个问题，你需要确定自己的整体优先级。当然，好心的客户在这方面拥有最终决定权，因为为未来兼容性制作项目可能涉及成本，而不这样做也可能带来风险。以下是一些通用指导原则：

- 无需面向未来的LDR项目——这可能是剧集式HDTV、广告或网络内容。工作空间可以与显示空间相同（即Rec.709或sRGB），因此图像会进行伽马校正，使用整数运算也无妨——如果条件允许，可采用16位。

- LDR项目但需具备未来适应性——这可能属于高端系列剧HDTV（例如《权力的游戏》），因此制作方希望未来能够重新发行该内容。工作空间需为线性且浮点处理，甚至可能需要采用ACES标准。

- **HDR project but no need for future-proofing** - This might be for UHDTV and feature film, but the producers don't want to bear the cost of future-proofing. The workspace needs to be linear and float.
- **HDR and needs future-proofing** - This would be big-budget, high-end episodic UHDTV and most feature films. The workspace needs to be linear and float—preferably ACES.

4.7.1 Working in Linear

The first thing to clarify about working in linear is that linear is NOT a color space. It does not have assigned primaries, a white point, or a transfer function. If it's not a color space, then what is it? It's a Work Space.

In Section 4.4.5 Linear Images, we took a look at the attributes of linear images. In this section, we examine how those attributes affect the way we work with them. As we saw in the overlapping spotlight illustration in Figure 4.18, if it has to be right, it has to be linear. Linear is in fact how the real world works, and because of that physical fact, all CG renders are done in linear because CG is a simulation of the real world. However, when those linear renders are saved to disk in a file format other than OpenEXR, the software will bake in a gamma correction whether you want it to or not.

We also saw how VFX image processing operations and color corrections must also be done in linear for the results to be truly correct. A corollary to working in linear is that the workspace math must be floating point, and because it is floating point, it will have high dynamic range and support negative code values. These three attributes—linear, float, and HDR—are absolutely required to maintain maximum quality and to future-proof a project.

The general workflow for working in linear is to open an image file in an app, back out any log or gamma curves to make it linear, and then have a viewing LUT such as sRGB so the image will look nice on the monitor. We can follow the action in Figure 4.25 using the log image on the left as an example. The log image is read from the disk and then immediately converted to linear by applying the inverse of the log LUT that was baked into it. The image is now linear, and all image processing and color corrections are done on this version. The sRGB version is made on the fly by passing the linear version through the sRGB viewing LUT for the monitor.

- HDR项目但无需面向未来验证——这可能适用于UHDTV和故事片，但制片方不愿承担面向未来验证的成本。工作空间需为线性且浮点格式。

- HDR与未来验证需求——这将是大预算、高端剧集UHDTV及多数故事片的标准。工作空间需为线性且浮点处理，首选ACES。

4.7.1 线性工作

关于在线性空间工作的首要澄清点是，线性空间并非一个色彩空间。它没有指定的原色、白点或传递函数。既然不是色彩空间，那它究竟是什么？它是一个工作空间。

在4.4.5线性图像章节中，我们探讨了线性图像的属性。本节我们将研究这些属性如何影响我们处理它们的方式。正如我们在图4.18的重叠聚光灯示意图中所见，若要求结果准确，就必须采用线性处理。线性本质上是现实世界的运作方式，基于这一物理事实，所有计算机图形渲染都在线性空间中进行，因为CG是对现实世界的模拟。然而，当这些线性渲染结果以非OpenEXR格式保存到磁盘时，无论你是否需要，软件都会自动嵌入伽马校正。

我们还了解到，视觉特效图像处理操作和色彩校正也必须在线性空间中进行，才能确保结果真正准确。在线性空间工作的一个必然推论是，工作区的数学运算必须采用浮点数格式，而正因其浮点特性，它将具备高动态范围并支持负编码值。线性、浮点与高动态范围这三个属性，是保持最高画质并为项目预留未来适应性的绝对必要条件。

在线性模式下工作的一般流程是：在应用程序中打开图像文件，通过反向应用嵌入的log或gamma曲线使其转换为线性，然后使用如sRGB的查看LUT，以便图像在显示器上呈现良好效果。我们可以以图4.25左侧的log图像为例，跟随操作步骤进行说明。log图像从磁盘读取后，立即通过应用其内置log LUT的反向转换，将其转换为线性图像。此时图像已处于线性状态，所有的图像处理和色彩校正都在此版本上进行。通过将线性版本传递至显示器的sRGB查看LUT，即可实时生成sRGB版本。

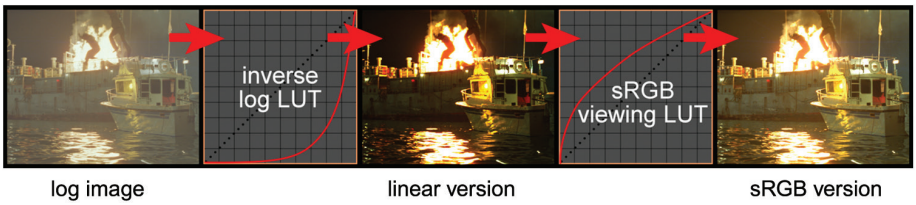

Figure 4.25 The linear workflow

After all processing and color correction, the final results are written to disk in the required file format for the job. At that time, the linear images get a log or gamma LUT applied, depending on the file format to be written. For example, if saved as a ProRes file, a Rec. 709 LUT will be baked in; if saved as a TIFF file, an sRGB LUT will be baked in, and if saved as a DPX, a Cineon log LUT will be baked in. To see your fine work on a monitor, the linear image must get a gamma correction like Figure 4.22. For viewing on a workstation monitor, the gamma correction LUT will be sRGB, and for video, it will be Rec. 709.

The single exception to this workflow is OpenEXR files which, by definition, are already linear, so they just load and go without any color space conversion. When writing to disk, they get no color space conversion, as they are already linear and are simply written out in the EXR file format. Using OpenEXR throughout the pipeline is the highest quality that preserves and protects your project for future-proofing.

With our new-found knowledge of what makes a color space, we can now place the linear workflow in its proper perspective. While its color model is defined (RGB), as well as its transfer function (linear), it is not a true color space of its own because it has no defined primaries—it simply adopts the color primaries of whatever image you linearize. Linear is still the preferred workspace regardless of whether the color space of the input images is sRGB, Rec. 709, DCI-P3, or the awesome new Rec. 2020—all of which will be covered in detail in the next chapter, Display Space.

4.7.2 Working in Display Space

Working in display space means working in the same color space as the intended display device—8 bits and gamma corrected—with no headroom, footroom, or extra precision. This is done all the time. If you are working on graphics intended for display on the web or a workstation monitor (an

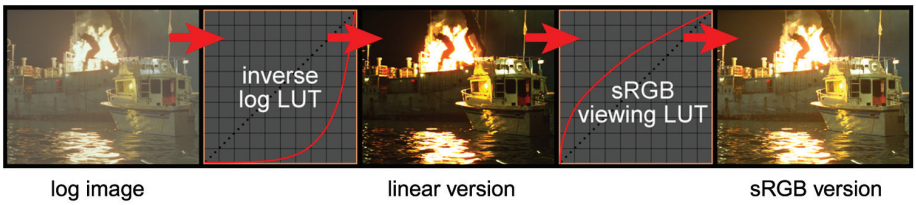

图 4.25 线性工作流

经过所有处理和色彩校正后，最终结果会以任务所需的文件格式写入磁盘。此时，线性图像会根据要写入的文件格式应用对数或伽马查找表。例如，若保存为ProRes文件，会嵌入Rec. 709查找表；若保存为TIFF文件，则嵌入sRGB查找表；若保存为DPX文件，则嵌入Cineon对数查找表。若要在显示器上呈现精细的视觉效果，线性图像必须进行如图4.22所示的伽马校正。在工作站显示器上查看时，伽马校正查找表采用sRGB标准，而用于视频时则采用Rec. 709标准。

此工作流程的唯一例外是OpenEXR文件，根据定义，它们已经是线性的，因此它们只需加载即可，无需任何色彩空间转换。写入磁盘时，它们也不会进行色彩空间转换，因为它们已经是线性的，只需以EXR文件格式直接写出。在整个流程中使用OpenEXR是最高质量的选择，能够为您的项目提供保护并确保未来兼容性。

凭借我们新获得的关于色彩空间构成的知识，现在可以将线性工作流程置于正确的视角中审视。虽然其色彩模型（RGB）和传递函数（线性）均已定义，但它本身并非一个真正的色彩空间，因为它没有明确指定的原色——它仅采用你所线性化的任何图像的原色。无论输入图像的色彩空间是sRGB、Rec.709、DCI-P3，还是令人惊叹的新标准Rec.2020（所有这些都将在下一章“显示空间”中详细探讨），线性空间仍然是首选的工作空间。

4.7.2 在显示空间工作

在显示空间工作意味着在与目标显示设备相同的色彩空间中操作——8位且经过伽马校正——没有头部空间、底部空间或额外精度。这是常见的做法。如果你正在为网页或工作站显示器（一种

sRGB display device), you can load images that are in sRGB color space into memory without converting them in any way. If you are working on a video project intended for broadcast, then your images will be in Rec. 709. They too are simply loaded into memory without conversion and are properly viewed with a broadcast monitor, which is a Rec. 709 display device. If your sRGB workstation does not have a broadcast monitor, you can fudge things a bit with a Rec. 709 LUT on your viewer that will make a fairly close approximation. This is how the vast majority of non-feature film work is done and it works reasonably well.

However, working in display space introduces several problems that you will need to properly manage:

- **Clipping** - Working in display space means that any operations that go over 1.0 (which is very easy to do) will get clipped, revealing some unpleasant hard edges. To alert you to clipped pixels, many software packages have a "zebra" display that marks all clipped pixels with a crosshatch pattern. There are two ways you can work here. First, avoid clipping in the first place by turning on the zebra display to carefully avoid driving any code values above 1.0. Alternatively, just blast away on the shot without worrying about clipping, then check at the end and introduce a soft clip as the last operation to soften the hard edges.

- **Banding** - Color banding can be introduced if using integer math. However, most systems support the ability to promote an 8-bit or 16-bit image to float internally or may even do it "automagically" for you. This is highly recommended, as we saw above the many evils that can befall an artist working in the bare minimum, 8-bit integer.

- **Image quality** - The most insidious of all, display spaces are gamma-corrected, so are not linear, and as we saw above, image processing ops done on gamma-corrected images are not mathematically correct and can introduce artifacts or degradation.

That said, you can safely work in display space if you keep the above caveats in mind and you do not need to future-proof your work.

4.7.3 Working in ACES

One cannot discuss workspaces without broaching ACES: the mother of all workspaces. ACES provides a super-wide gamut high dynamic range color

在sRGB显示设备上工作时，您可以将处于sRGB色彩空间的图像直接加载到内存中，无需进行任何转换。若您正在制作面向广播的视频项目，则图像会采用Rec.709标准。这些图像同样无需转换即可直接载入内存，并通过广播监视器（即Rec.709显示设备）正确显示。如果您的sRGB工作站未配备广播监视器，可以在查看器上使用Rec.709查找表进行近似模拟，获得相对接近的效果。绝大多数非电影长片的项目均采用此流程，其实际表现也相当可靠。

然而，在显示空间中进行工作会带来几个需要妥善管理的问题：

- 裁剪 - 在显示空间工作意味着任何超过1.0的操作（这很容易发生）都会被裁剪，暴露出一些不悦目的硬边。为了提醒你注意被裁剪的像素，许多软件包都设有“斑马纹”显示功能，用交叉线图案标记所有被裁剪的像素。这里有两种处理方法：首先，通过开启斑马纹显示来仔细避免任何代码值超过1.0，从而从一开始就避免裁剪。或者，先不考虑裁剪问题直接处理镜头，最后检查时引入软裁剪作为最终操作，以柔化硬边。

- 条带化 - 如果使用整数运算，可能会引入颜色条带。然而，大多数系统支持将8位或16位图像内部提升为浮点格式，甚至可能为您“自动魔法般”地完成这一转换。这非常推荐，正如我们之前所见，艺术家在最低限度的8位整数环境下工作可能遭遇诸多弊端。

- 图像质量——这是所有问题中最隐蔽的，显示空间经过伽马校正，因此是非线性的，正如我们上文所见，在伽马校正图像上进行的图像处理操作在数学上并不正确，可能会引入伪影或导致画质下降。

话虽如此，只要牢记上述注意事项，并且您不需要为作品预留未来兼容性，那么在显示空间中进行操作是安全的。

4.7.3 在ACES中工作

不讨论ACES就无法谈论工作空间：所有工作空间的母体。ACES提供了超宽色域的高动态范围色彩

space designed specifically for feature film work combining CG with live action. However, it is primarily a color management system, so it is covered in detail in Chapter 6: Color Management.

4.7.4 Working in Log

The only people actually allowed to work in log are colorists. As we have relentlessly argued, visual effects must be done in linear space. Many colorists like to work in log because it responds like film and they can apply color corrections to the log image to adjust exposure, white point, and latitude *before* it is converted to the display space like sRGB, Rec. 709, or DCI-P3, where the data will get clipped and deprecated. The color-correcting software uses math specifically designed for log data. For example, to double the exposure of a DPX 10-bit log (0 - 1023) image, it will add 90 code values to the log data instead of the gain operation used for linear or sRGB data.

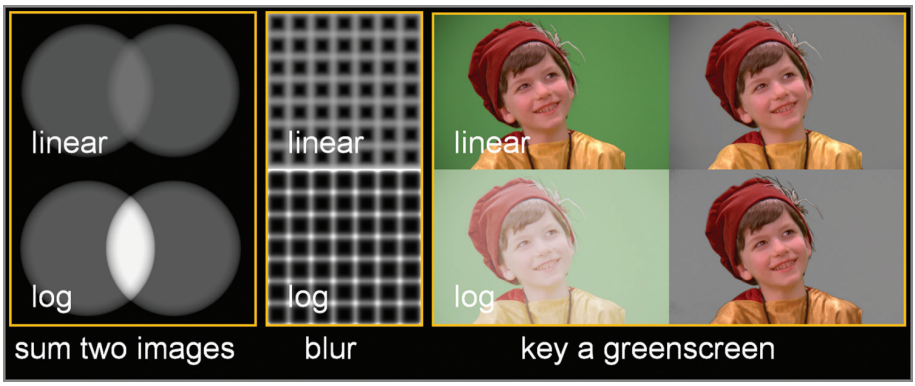

Figure 4.26 Examples of working in log

The reason we cannot work in log space is that the math for all image processing operations will be very, very wrong. Figure 4.26 offers some examples. The *sum of two images* on the left shows the log case to be very wrong with excessive brightness in the overlapped region. The *blur* of a grid example shows the log version is choked in everywhere, making the intersections look "knobby." In the *key a greenscreen* example, the log version has hideous dark edges and discolorations of the foreground character. The only exception would be to convert the linear image to log for sharpening, then back to linear. This avoids negative code value artifacts at high-contrast edges. Try it. You'll like it.

专为结合CG与实景拍摄的故事片工作而设计的空间。然而，它主要是一个色彩管理系统，因此将在第6章：色彩管理中详细阐述。

4.7.4 对数运算

唯一真正被允许在log空间工作的人是调色师。正如我们一直坚持的观点，视觉特效必须在线性空间中进行。许多调色师喜欢在log空间工作，因为它的响应特性类似于胶片，他们可以在log图像上应用色彩校正来调整曝光、白点和宽容度，然后再将其转换为sRGB、Rec. 709或DCI-P3等显示空间——转换后数据将被裁剪和降级。调色软件使用专门为log数据设计的数学运算。例如，要使DPX 10-bit log（0-1023）图像的曝光加倍，软件会向log数据添加90个码值，而不是对线性或sRGB数据使用的增益操作。

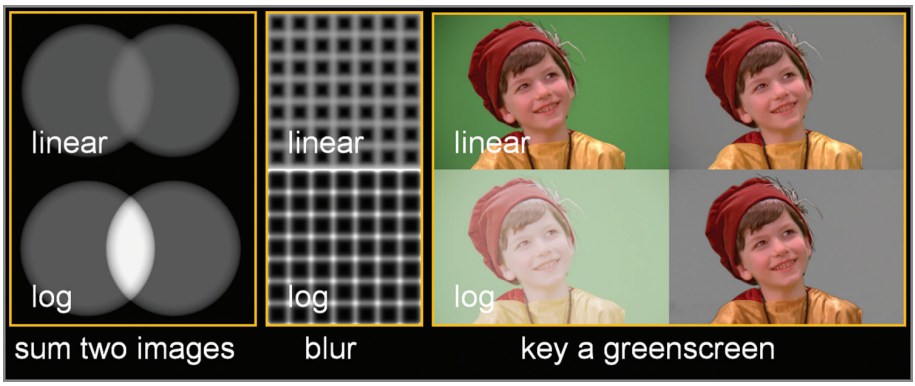

图 4.26 对数运算示例

我们之所以不能在对数空间下工作，是因为所有图像处理运算的数学结果都会变得非常、非常错误。图4.26提供了一些示例。左侧两张图像的叠加显示了对数情况下的结果严重失真，重叠区域亮度过高。网格图像的模糊示例显示对数版本处处显得阻塞，使交叉点看起来“多节状”。在关键的绿幕示例中，对数版本出现了难看的黑边和前景角色的颜色失真。唯一的例外是将线性图像转换为对数以进行锐化，然后再转回线性空间。这样可以避免高对比度边缘出现负编码值伪影。试试看，你会喜欢的。

4.7.4.1 Picking the Right Curve

In order to work in linear with camera-exported log images, they must first be "de-logged" to linearize them using a log LUT. Your first concern is to use the correct camera log LUT since there is a veritable zoo of log formats out there nowadays. If, for example, the original image is exported as REDlog, it would not do to linearize it with an AlexaLogC inverse LUT. The tragedy here is that if you use the wrong LUT, the results will still look pretty good and it becomes very confusing as to what is wrong. Sometimes you will be given clips without the information about which camera log LUT was used. If your software can read the image metadata, the answer might be there. If not, you might need to check with the comp super, the VFX super, the DP, the DIT, the producer, the colorist, the production assistant, or the guy in the cubicle next to you.

When I started compositing, there was only one log curve—Cineon (which is also used by DPX). Today every digital cinema camera manufacturer has its own camera log curve optimized for its cameras, plus there are a variety of other "industry" log LUTs available. You can see the issue in Figure 4.27 with a graph of several log LUTs on the left. You just have to figure out which LUT is needed for your shot because often the clips are just dropped off without camera information. Hopefully, the clip's metadata will reveal the camera and log LUT. There are also "industry" log formats like SLog, in addition to the camera logs like ViperLog and REDLog. So your first challenge is to use the right log curve to linearize your images. The red box in the lower left-hand corner of Figure 4.27 represents the next challenge, which is what happens near zero.

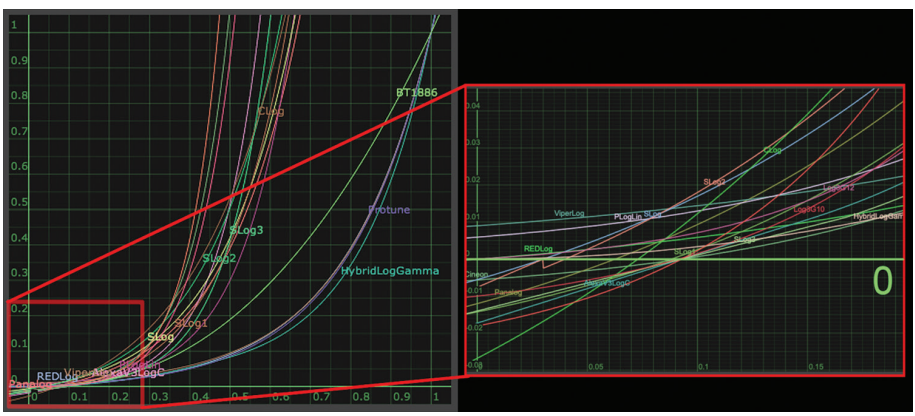

Figure 4.27 Log curves with a blowup window showing negative curves

4.7.4.1 选择正确的曲线

为了在后期处理中与相机导出的对数图像保持线性工作流程，首先必须使用对数查找表（LUT）对其进行“去对数化”以转为线性。首要关键是选用正确的相机对数LUT，因为如今市面上存在各式各样的对数格式。例如，若原始图像以REDlog格式导出，使用AlexaLogC的反向LUT进行线性化将导致错误。此处的棘手之处在于：即使选错LUT，结果看起来仍可能相当不错，从而令人难以察觉问题所在。有时您收到的素材可能未附带所用相机对数LUT的信息。若您的软件能读取图像元数据，答案或许藏于其中；若不能，您可能需要向合成总监、视效总监、摄影指导、数字影像工程师、制片人、调色师、制片助理，甚至隔壁工位的同事确认。

当我开始进行合成工作时，只有一种对数曲线——Cineon（DPX格式也使用它）。如今，每家数字电影摄影机制造商都有专为其摄影机优化的自有对数曲线，此外还有各种其他“行业”对数查找表可供选择。你可以在图4.27中看到这个问题，左侧图表展示了几种对数查找表。你只需要确定你的镜头需要哪种查找表，因为通常素材片段直接送来时并不附带摄影机信息。希望片段的元数据能揭示摄影机型号和对数查找表类型。除了像ViperLog和REDLog这样的摄影机对数格式外，还有SLog这类“行业”对数格式。因此你的首要挑战是使用正确的对数曲线将图像线性化。图4.27左下角的红框代表了下一个挑战，即接近零值时的处理问题。

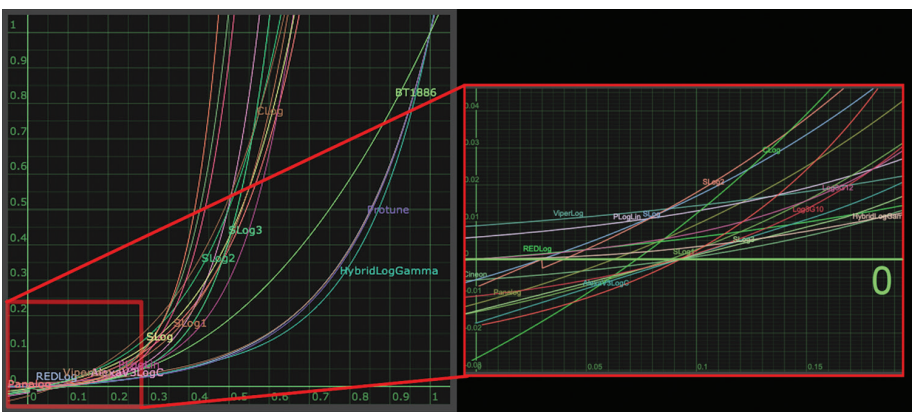

图4.27 带有放大窗口显示负曲线的测井曲线

4.7.4.2 Going Negative

Most (but not all) log curves contain negative code values. On the right of Figure 4.27 is a blowup of the red box showing the curves as they get near to zero. These negative code values must be preserved throughout the post-production pipeline or the blacks will get clipped. Integer workspaces do not support negative code values, so you will have to work in float if you want to preserve them.

So what picture information is contained in the negative code values of log images? For the most part, it is the black grain structure or, in the case of digital capture, the black noise. The top of Figure 4.28 shows the black grain structure from a 35 mm film scan (gained up to show the grain pattern,) and below that is a graph of the red channel grain. The dotted line is code value zero, with the grain pattern going both above and below zero. If the image is clipped at zero, the bottom half of the black grain pattern will be gone, as shown on the right. The clipped grain will look much different than the normal grain and will reveal a processed image.

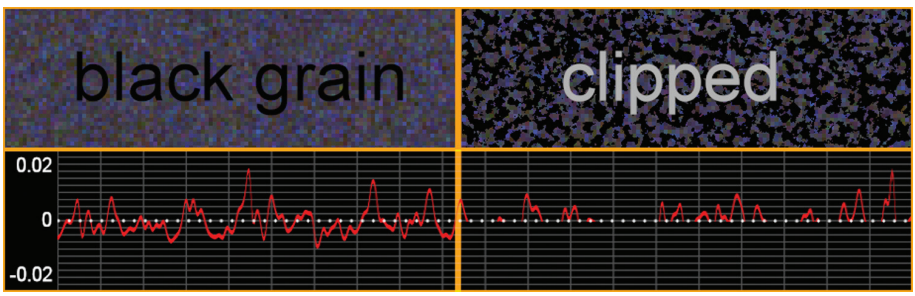

Figure 4.28 Black grain goes negative

4.7.4.3 Limited Dynamic Range

When working on a film, the job will be in float so the full dynamic range of the log image is preserved. However, when working in Rec. 709 television, it has a limited dynamic range between zero and 1.0, so something has got to go—and that is the high dynamic range content. This could be simple highlights, fiery content like explosions and village torches, or a camera pointed at light sources like the sun. As we saw above, there can also be a loss of black detail if the negative code values are clipped.

4.7.4.2 负向调整

大多数（但并非全部）对数曲线都包含负编码值。图4.27右侧是红框区域的放大图，展示了曲线趋近于零时的状态。这些负编码值必须在后期制作流程中全程保留，否则黑色区域将被裁切。整数工作空间不支持负编码值，因此若需保留这些数值，必须在浮点环境下操作。

那么，对数图像负码值中包含哪些图像信息呢？大部分情况下，它对应的是黑色颗粒结构，或者在数字拍摄中表现为黑色噪点。图4.28顶部展示了35毫米胶片扫描的黑色颗粒结构（已增益处理以显示颗粒图案），下方则是红色通道颗粒的曲线图。虚线代表码值零点，颗粒图案在零点的上下均有分布。如果图像在零点处被裁剪，黑色颗粒图案的下半部分将会丢失，如右图所示。被裁剪的颗粒看起来与正常颗粒差异显著，会暴露出图像经过处理的痕迹。

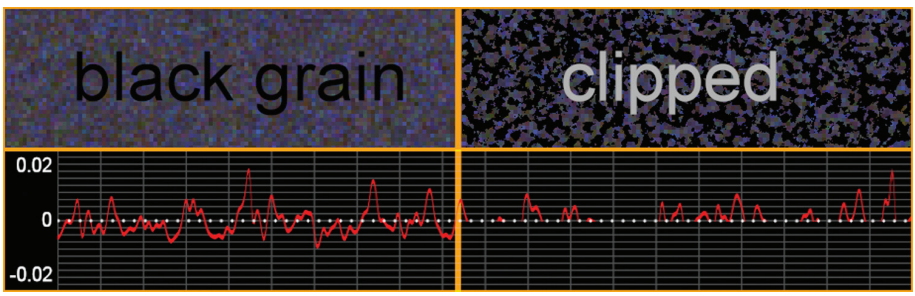

图 4.28 黑棋走负

4.7.4.3 有限动态范围

在电影制作中，工作流程会采用浮点处理，以保留对数图像的全部动态范围。然而，在Rec. 709电视标准下工作时，其动态范围被限制在0到1.0之间，因此必须舍弃部分内容——即高动态范围的部分。这可能包括简单的高光、爆炸和村庄火炬等炽热内容，或是相机直接对准太阳等光源的情况。正如前文所述，如果负片编码值被截断，还可能损失黑色细节。

Figure 4.29 shows an HDR log image on the left and the Rec. 709 version on the right, and below them are their histograms. The horizontal axis is the RGB code value and the vertical axis is the percentage of pixels that share that RGB value. A tall curve means there are many pixels of that color, while a low one means there are few. It has nothing to do with the pixels' location on the screen. We will see more about histograms in Chapter 6.

The log histogram in Figure 4.29 covers a range of floating point code values from zero to 9.0 for this particular image. Interestingly, the explosion shows up in two places: the red peak around 6.0 and the green around 9.0. This shows that the explosion has a green tint, but it's not noticeable because it is so bright. Even though the entire range of the code values is preserved, they will appear clipped on the monitor because the monitor only ranges from zero to 1.0. On the right is the limited dynamic range sRGB histogram showing that the code values have been redistributed between zero and 1.0, plus the massively clipped explosion with a spike of pixels piled up on the right side of the graph, indicating that a lot of pixels were clipped to the same code value.

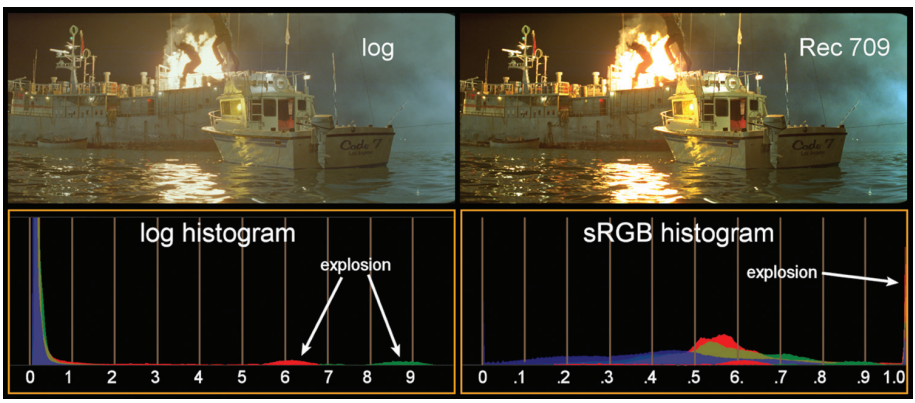

Figure 4.29 Histograms of log vs. Rec. 709 image

FUN FACT: Histograms are not just for images. They are also an important tool for statistics and data analysis.

图4.29左侧展示了一张HDR对数图像，右侧是其Rec. 709版本，下方则分别为两者的直方图。横轴代表RGB编码值，纵轴表示具有相同RGB值的像素百分比。曲线峰值越高表示该颜色像素数量越多，峰值越低则像素数量越少。这与像素在屏幕上的位置无关。我们将在第六章进一步探讨直方图的相关内容。

图4.29中的对数直方图显示了这张特定图像从零到9.0的浮点编码值范围。有趣的是，爆炸效果在两个位置显现：红色峰值约在6.0处，绿色峰值约在9.0处。这表明爆炸带有绿色调，但由于亮度极高而不易察觉。尽管编码值的整个范围被保留下来，它们在显示器上会显得被裁剪，因为显示器的范围仅为零到1.0。右侧是有限动态范围的sRGB直方图，显示编码值已在零到1.0之间重新分布，同时爆炸部分被大幅裁剪——图像右侧堆积的像素尖峰表明大量像素被裁剪至相同的编码值。

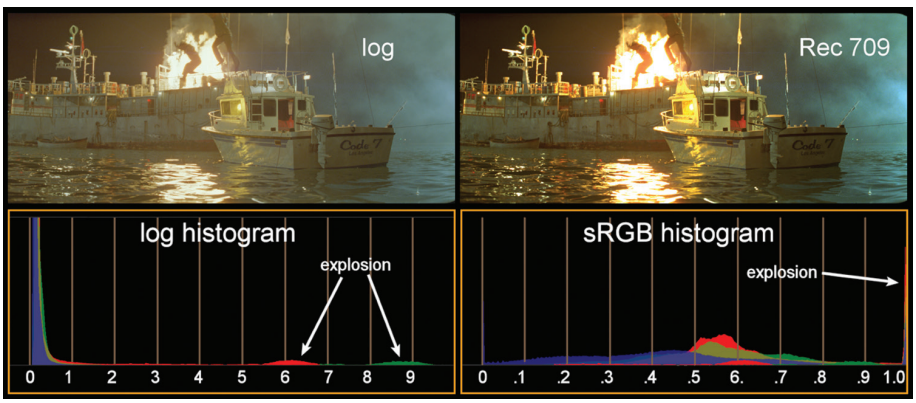

图4.29 对数图像与Rec. 709图像的直方图对比

有趣的事实：直方图不仅用于图像处理，它也是统计学和数据分析中的重要工具。

4.8 Photography Work Spaces

Digital cameras are another important source of digital images, but they do have their differences in available workspaces compared to video and feature film. So if you are a photographer, this section is for you. If you are not a photographer, you might want to eavesdrop on the conversation to learn about their unique high dynamic range and wide color gamut color spaces.

Photographers and graphic artists going to print suffer an indignity that artists working in visual effects or motion graphics, editors and colorists usually don't. The latter groups have displays that can show the true colors of their work, while photographers working with print do not. As we shall see in the next section, the typical sRGB monitor not only cannot show some colors that print can but also can display colors that print can't. This mismatch between what is seen on the screen and what ends up in print is an endless source of misery for those working with print.

The root cause is that print and monitors have very different gamuts that don't match well, because monitors use RGB primaries with additive color while print uses CMYK primaries with subtractive color. These produce very different gamuts with a lot of non-overlapping regions. The monitor cannot display the range of CMYK greens and cyans that fall outside of the sRGB gamut. Further, the CMYK print cannot properly display the dark blues and magentas on the monitor that fall outside of its gamut. Here we will explore the workarounds for this dilemma.

4.8.1 sRGB

Stills photographers can use sRGB as a workspace if all they need to do is deliver a nice JPG for display on a monitor. The photo can be exported from the camera as a JPG and then given a little crop and color correction. More ambitiously, the photo can be exported as camera raw or camera log and then "developed" for maximum picture quality that fits within sRGB. There is no mismatch here because the workspace is sRGB and the display space is sRGB. True, the camera raw can contain colors that will not fit within sRGB, but at least what you see on the screen is what you get.

4.8 摄影工作空间

数码相机是数字图像的另一个重要来源，但与视频和故事片相比，它们在可用工作空间方面确实存在差异。因此，如果您是摄影师，本节内容正适合您。如果您不是摄影师，或许也可以旁听一下这场对话，了解其独特的高动态范围和广色域色彩空间。

摄影师和平面艺术家在印刷时遭受的屈辱，通常是从事视觉效果或动态图形工作的艺术家、编辑和调色师所不会经历的。后者的显示器能够展现他们作品的真实色彩，而从事印刷工作的摄影师则没有这样的条件。正如我们将在下一节中看到的，典型的sRGB显示器不仅无法显示某些印刷能够呈现的色彩，反而能展示一些印刷无法实现的颜色。屏幕上所见与最终印刷成品之间的这种不匹配，给从事印刷工作的人们带来了无尽的痛苦。

根本原因在于印刷品和显示器的色域差异很大且难以匹配，因为显示器采用加色法的RGB三原色，而印刷采用减色法的CMYK四原色。这两种模式产生的色域存在大量非重叠区域。显示器无法呈现超出sRGB色域的CMYK绿色和青色范围；反之，CMYK印刷也无法准确再现超出其色域的显示器深蓝色与洋红色。下文将探讨这一困境的解决方案。

4.8.1 sRGB

静态摄影师如果只需要交付一张在显示器上展示的优质JPG图片，可以将sRGB作为工作空间。照片可以从相机以JPG格式导出，然后稍作裁剪和色彩校正。更理想的情况下，照片可以以相机原始格式或相机对数格式导出，随后进行“冲洗”处理，以获得适配sRGB范围内的最佳画质。这里不存在不匹配问题，因为工作空间是sRGB，显示空间也是sRGB。诚然，相机原始格式可能包含无法适配sRGB的色彩范围，但至少屏幕上所见即所得。

For photography destined for high-end printing, the deliverable to the printer must be CMYK. The problem is that the CMYK gamut of Figure 4.15 does not fit well within the sRGB gamut of Figure 4.14. To deal with this problem, several "super RGB" workspaces have been developed with much larger gamuts that, as a group, are referred to as "synthetic color spaces." They are synthetic because they don't map to any physical display devices. They are purely mathematical constructs that allow unconstrained image manipulation in the workspace that avoids any clipping of colors. The final results get a color space conversion to SWOP CMYK before going to the printer. Next, we will take a look at the two most important synthetic color spaces used for photographic workspaces—Adobe RGB and ProPhoto RGB—and then the SWOP color space.

4.8.2 Adobe RGB

The Adobe RGB gamut (Figure 4.30) is designed for print applications to provide a straightforward mapping from RGB to CMYK. It covers 50% of the visible colors and is large enough to fully encompass the CMYK gamut. The gamut increase was achieved simply by moving the sRGB green primary up into the more saturated greens. It uses a gamma of 2.2 and the D65 white point.

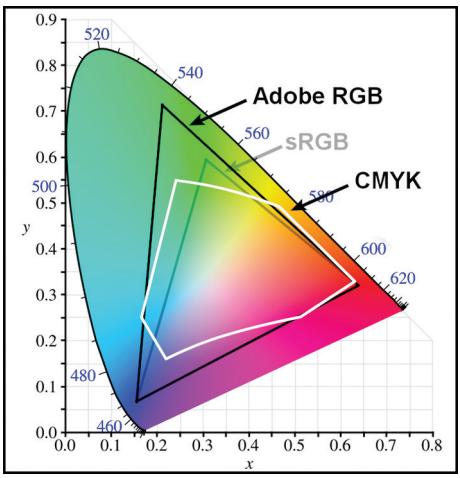

Figure 4.30 Adobe RGB gamut

When using this workspace, there will be no lost green/cyans, but the monitor can still display the blue/magentas that you won't get in the final CMYK output. Large gamuts like this "stretch" the 8-bit code values rather far, so if you can't work in float, then 16 bits is recommended to avoid color banding.

For this workspace to be of any use, a monitor that can display it will be needed, of course. There are many brands available, but they are more expensive than a standard sRGB monitor. Also, watch out for claims like

对于面向高端印刷的摄影作品，交付给印刷商的必须是CMYK格式。问题在于，图4.15中的CMYK色域无法很好地适配图4.14中的sRGB色域。为解决这一问题，业界开发了多种“超级RGB”工作空间，它们拥有更大的色域范围，这些空间被统称为“合成色彩空间”。之所以称为“合成”，是因为它们并不映射到任何物理显示设备。它们是纯粹的数学构造，允许在工作空间内进行无约束的图像处理，避免任何色彩裁剪。最终成果在交付印刷前，会经过色彩空间转换至SWOP CMYK。接下来，我们将探讨用于摄影工作空间的两个最重要的合成色彩空间——Adobe RGB和ProPhoto RGB，以及SWOP色彩空间。

4.8.2 Adobe RGB

Adobe RGB色域（图4.30）专为印刷应用设计，旨在提供从RGB到CMYK的直接映射。它覆盖了50%的可见色彩，其范围足以完全涵盖CMYK色域。该色域的扩展主要通过将sRGB的绿色原色向更高饱和度的绿色区域移动实现。它采用2.2的伽马值和D65白点。

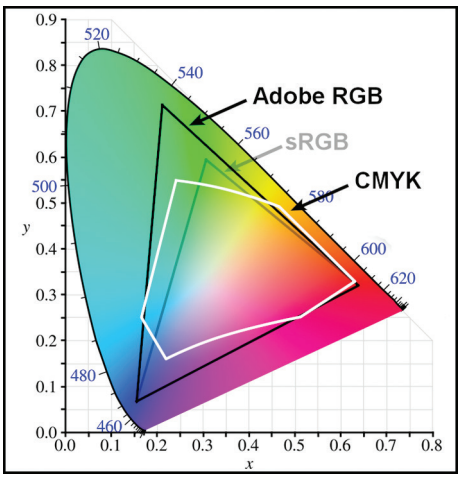

图 4.30 Adobe RGB 色域

使用此工作区时，不会丢失绿色/青色，但显示器仍可显示蓝色/

您将无法在最终的CMYK输出中获得这些洋红色。如此大的色域会将8位编码值“拉伸”得相当远，因此如果无法使用浮点运算，建议采用16位以避免色彩条带。

要使这个工作空间发挥任何作用，当然需要一个能够显示它的显示器。市面上有许多品牌可供选择，但它们比标准的sRGB显示器更昂贵。此外，还需警惕诸如{v*}之类的宣传。

"covers 90% of Adobe RGB." The missing 10% is invariably the 10% you really need for your job.

4.8.3 ProPhoto RGB

Originally developed by Kodak for film and photography prints, this huge synthetic workspace (Figure 4.31) uses a gamma of 1.8. It is so large that the green and blue primaries fall outside of the visible spectrum of the CIE xy-chromaticity chart, requiring the use of those imaginary numbers we encountered way back in Figure 4.11. This is done so the large gamut can still be defined by three primaries like other color spaces. These **imaginary primaries** are not realizable in the real world, but this is not a problem because the scene gamut of the photographic image will cluster well within the visible spectrum and not extend into the imaginary.

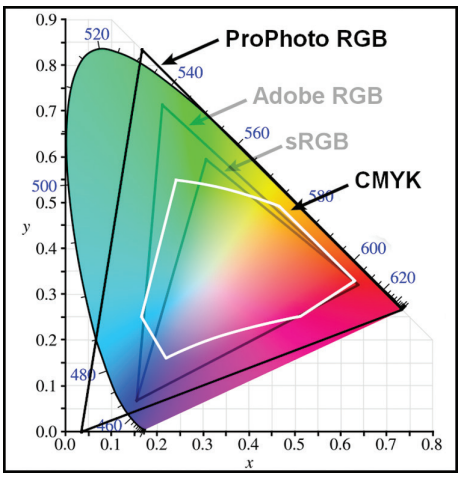

Figure 4.31 ProPhoto RGB gamut

The ProPhoto RGB workspace covers over 90% of the visible color spectrum and 100% of likely occurring natural colors in the real world, as it covers virtually the entire Pointer's gamut. This gamut is even larger than Adobe RGB, so working in float or 16 bits is a must to avoid color banding.

4.8.4 SWOP

SWOP is the "color space" that utilizes a CMYK color model for print. Technically, it's not a real color space like Rec. 709 with primaries, a white point, and a transfer function. While it does not have primaries, it does have specifications to precisely define the CMY ink "primaries." There is no white point per se, but there is a clear specification for the white light source (D55) used to illuminate the print. And it certainly does not have a gamma type

“覆盖了90%的Adobe RGB色域。”缺失的10%往往正是你工作中真正需要的那部分。

4.8.3 ProPhoto RGB

最初由柯达为胶片和摄影印刷开发的这一庞大合成工作空间（图4.31）采用了1.8的伽马值。其色域极为广阔，以至于绿色与蓝色原色超出了CIE xy色度图的可见光谱范围，必须借助我们在图4.11中曾遇到的虚数概念进行描述。这样设计是为了让宽广的色域仍能像其他色彩空间一样由三原色定义。这些虚拟原色在现实世界中无法实现，但这并不构成问题，因为场景色域

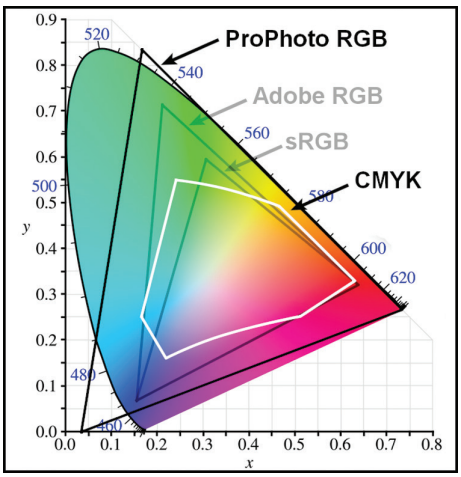

图 4.31 ProPhoto RGB 色域

摄影图像的色彩将很好地聚集在可见光谱范围内，而不会延伸至虚部。

ProPhoto RGB 工作空间覆盖了超过90%的可见色彩光谱以及现实世界中可能出现的100%自然色彩，因为它几乎涵盖了完整的Pointer色域。该色域甚至比Adobe RGB更广，因此必须使用浮点或16位深度进行处理，以避免色彩条带现象。

4.8.4 SWOP

SWOP 是印刷中使用的 CMYK 色彩模型所对应的“色彩空间”。从技术上讲，它并非像 Rec. 709 那样拥有原色、白点和传递函数的真实色彩空间。虽然它没有原色，但确实有精确定义 CMY 油墨“原色”的规范。它本身没有白点，但对用于照射印刷品的白色光源（D55）有明确规范。而且它肯定不具备伽马类型的 {v*}。

of transfer function, but it does have a dot-gain that roughly equates to a gamma of 1.8. Maybe we should call it a “quasi-color space.”

The ink colors are specified by **Specifications for Web Offset Publications**, abbreviated as **SWOP**. It is both an organization and a set of ink color specifications to standardize professionally printed materials. SWOP also publishes ICC profiles for converting RGB to CMYK colors for printing, which is covered in Chapter 6: Color Management. And no, “for Web Offset” does not refer to the internet but instead to high-speed printing on a continuous roll of paper rather than individual sheets.

SWOP provides complete documentation for the following:

- **Color Separation Guidelines** - Recommendations for preparing RGB images into the four-color separations of CMYK.
- **Ink Specifications** - Recommendations defining the types of inks to be used to ensure consistency in color reproduction. Sounds a bit like primaries to me.
- **Proofing Practices** - Guidelines for creating and evaluating preliminary prints (proofs) to ensure they accurately represent the final printed product.
- **Paper Specifications** - Recommendations for paper types and qualities that affect the final appearance of the printed colors.

When editing photos for print with your favorite Photoshop app, you might be tempted to work in RGB and then just convert to CMYK for printing when done. But there is a dastardly color space trap here, illustrated in Figure 4.32. Notice how the black crosshatch pattern marks two areas where the RGB gamut is larger than the CMYK? These represent colors that you will see on your monitor that will not print because those colors are outside of the CMYK gamut. Darn!

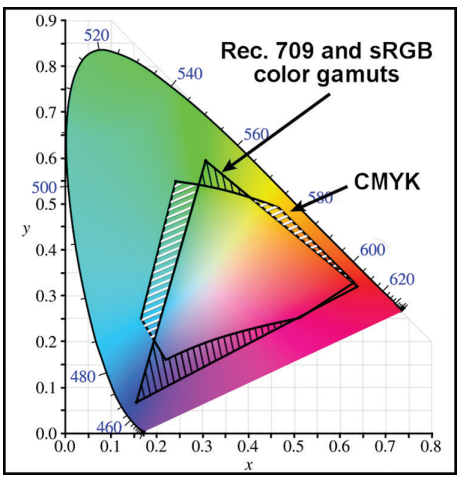

Figure 4.32 CMYK vs. RGB gamuts

传递函数，但它确实有一个网点增大，大致相当于1.8的伽马值。也许我们应该称之为“准色彩空间”。

油墨颜色由《网络胶印出版物规范》指定，简称SWOP。它既是一个组织，也是一套用于标准化专业印刷材料的油墨颜色规范。SWOP还发布了用于将RGB颜色转换为印刷用CMYK颜色的ICC配置文件，相关内容将在第6章：色彩管理中介绍。另外，“网络胶印”中的“网络”并非指互联网，而是指使用连续卷筒纸而非单张纸进行高速印刷的工艺。

SWOP为以下内容提供完整的文档

- 色彩分离指南 - 将RGB图像准备为CMYK四色分色的建议。

- 墨水规格 - 为确保色彩复现一致性而定义使用墨水类型的建议。这听起来有点像三原色的概念。

- 打样实践 - 创建和评估初印样张（打样）的指南，以确保其准确呈现最终印刷产品。

- 纸张规格 - 关于影响印刷色彩最终呈现效果的纸张类型与品质建议。

当您使用最喜爱的Photoshop应用程序编辑用于打印的照片时，可能会倾向于在RGB模式下工作，完成后直接转换为CMYK进行打印。但这里存在一个棘手的色彩空间陷阱，如图4.32所示。注意到黑色交叉线图案标记出的两个区域了吗？那里RGB色域大于CMYK色域。这些区域代表您能在显示器上看到却无法打印的颜色，因为它们超出了CMYK色域范围。真糟糕！

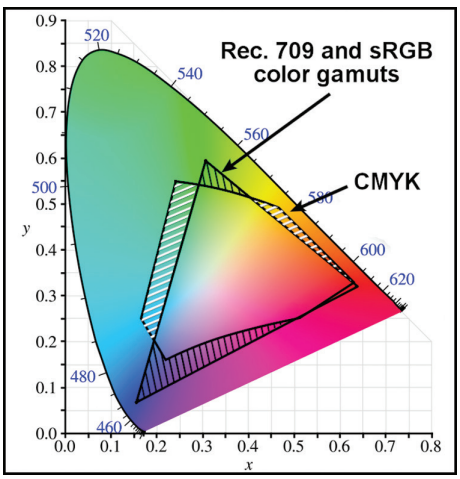

图 4.32 CMYK 与 RGB 色域对比

But wait, there's more! Do you see the white crosshatch patterns? They represent colors available in CMYK that you cannot see on your monitor because they are outside of the RGB gamut of your display. Drat! So, you have colors you can see that won't print and print colors that you cannot see. As we will learn in Chapter 5: Display Space, some special monitors can display the full gamut of CMYK, but they are super-expensive and you no doubt do not have one.

If you work in RGB and then convert to CMYK with an ICC profile for export, you will find your greens and deep blue-magentas (black crosshatch regions) have shifted hue and saturation as the ICC profile struggles to tuck in the RGB to the CMYK profile. You can check this by loading the output CMYK version next to the original RGB version to look for differences. If your system has a difference operation, that will help to identify any modified areas.

If, by chance, your picture is everywhere inside the union of the RGB and CMYK gamuts, you will not notice any difference. As for the regions where the CMYK gamut exceeds the RGB gamut (white crosshatch), all is lost. You have no way to create, much less see, those greens—unless you pony up for the expensive monitor, that is. For these reasons, many prefer to work in CMYK directly, even with an RGB monitor. Do keep in mind that CMYK is a color model, not a color space. SWOP is the color space that uses CMYK colors like Rec. 709 is the color space that uses the RGB colors.

4.9 Archiving

When the job is finished in the chosen workspace, the first question is: "Where shall we hold the wrap party?" The second question is: "In what file format should the job be archived?" Hopefully, the future fate of the project has been considered in the selection of the workspace, so it can now guide the choice of archive space. Yep: yet another space. There are several schools of thought on this, all heavily dependent on the future intentions for the project, budgets, and the desires of the producers. Some of those schools of thought are as follows.

但是等等，还有更多！你看到那些白色交叉线图案了吗？它们代表CMYK中可用的颜色，这些颜色在你的显示器上是看不到的，因为它们超出了你显示屏的RGB色域范围。真糟糕！所以，有些颜色你能看到却无法打印，而有些打印颜色你却无法看到。正如我们将在第五章“显示空间”中学到的，一些特殊显示器能够显示完整的CMYK色域，但它们极其昂贵，你肯定没有一台。

如果您在RGB模式下工作，然后使用ICC配置文件转换为CMYK以进行导出，您会发现绿色和深蓝-洋红色（黑色交叉阴影区域）的色调和饱和度发生了变化，因为ICC配置文件在将RGB压缩到CMYK配置文件中时遇到了困难。您可以通过将输出的CMYK版本与原始的RGB版本并排加载来检查差异。如果您的系统有差异操作功能，这将有助于识别任何被修改的区域。

如果您的图片恰好完全位于RGB和CMYK色域的并集之内，您将不会察觉到任何差异。至于CMYK色域超出RGB色域的区域（白色交叉线部分），所有信息都将丢失。您无法创建——更不用说看到——那些绿色调，除非您愿意为昂贵的显示器买单。因此，许多人更倾向于直接在CMYK模式下工作，即便使用的是RGB显示器。请务必记住，CMYK是一种色彩模型，而非色彩空间。SWOP是使用CMYK色彩的色彩空间，正如Rec. 709是使用RGB色彩的色彩空间一样。

4.9 归档

当工作在选定的工作空间完成时，第一个问题是：“我们在哪里举行杀青派对？”第二个问题是：“工作应该以什么文件格式存档？”希望在选择工作空间时已经考虑了项目的未来命运，因此现在它可以指导存档空间的选择。是的：又一个空间。关于这一点有几种思想流派，都严重依赖于项目的未来意图、预算和制片人的愿望。其中一些思想流派如下。

4.9.1 OpenEXR

This is the highest-precision form of the job with essentially unlimited dynamic range, so it contains the maximum information available. It is an excellent candidate for archiving, especially for an ACES job, with the only caveat being the file size. This becomes less of an issue over time, as the trajectory of technological progress is to make storage cheaper and transfer times faster.

4.9.2 DPX

If the file format for the job was DPX, it makes sense to archive off the final DPX files. If the job was done as an OpenEXR project and there was no need to future-proof the work, then DPX might be a candidate to save file size. However, depending on scene content, the DPX files could actually be larger than the EXRs because DPX has no compression, while EXR does a "light" compression by default. Other considerations are that there are small data losses converting from EXR float to DPX integer, and the DPX has an upper limit of a little over 11 stops, while EXR is essentially unlimited.

4.9.3 Camera Raw

Some productions will archive the job from EXR back to the original camera output format like ARRIRAW, which produces a smaller file size. Other advantages include no data loss compared to the camera original plus the ability to re-grade later and have the results cut in perfectly with the original project. One downside is obsolescence—cameras come and go, so some apps used today may be unable to read your particular camera's raw file format in the future. This will not happen when using industry-standard files like DPX or OpenEXR files.

4.10 Conclusion

As we have seen, the choice of workspace depends on the quality of the work you are trying to deliver plus archival considerations for future-proofing. Beyond choosing the color space to work in, there are also considerations

4.9.1 OpenEXR

这是工作中精度最高的形式，具有几乎无限的动态范围，因此包含了可用的最大信息量。它是存档的绝佳选择，特别是对于ACES工作而言，唯一的注意事项是文件大小。随着时间的推移，这个问题会逐渐减轻，因为技术进步的轨迹是使存储更便宜、传输时间更快。

4.9.2 DPX

如果作业的文件格式是DPX，那么归档最终的DPX文件是合理的。如果项目是以OpenEXR格式完成，且无需为未来兼容性做特别准备，那么DPX可能是节省文件大小的备选方案。然而，根据场景内容的不同，DPX文件实际上可能比EXR文件更大，因为DPX没有压缩，而EXR默认会进行“轻度”压缩。其他需要考虑的因素包括：从EXR浮点数转换为DPX整数时会有轻微数据损失，且DPX的动态范围上限约为11档左右，而EXR则基本没有限制。

4.9.3 Camera Raw

部分制作流程会将工作从EXR归档回原始相机输出格式，如ARRIRAW，从而生成更小的文件体积。其他优势包括相比相机原始文件无数据损失，以及日后可重新调色，并能将结果完美剪辑回原始项目中。一个缺点是过时风险——相机型号更迭频繁，因此当前使用的某些应用程序未来可能无法读取特定相机的原始文件格式。而使用行业标准文件（如DPX或OpenEXR文件）则不会出现此问题。

4.10 结论

正如我们所见，工作空间的选择取决于您试图交付的作品质量，以及为未来兼容性所做的归档考量。除了选择工作的色彩空间外，还需考虑

for working in integer vs. float. We also saw that it is common to use the display space as the workspace for standard dynamic range projects with no future-proofing requirement. However, for high dynamic range projects or standard dynamic range projects that require future-proofing, the requirement is to choose a workspace that is "larger" than the final deliverable display space. If this is not done, problems like clipping and image processing artifacts will degrade the final product.

For high-end work like feature films, the workspace must be linear, float, and HDR. While this will protect the work from clipping and processing artifacts, and offers some future-proofing, it is not the ultimate solution. For the ultimate in work quality and future-proofing, the ACES workflow is recommended, which we will cover in detail in Chapter 6.

The next stage of the digital imaging pipeline is the Display Space, where the images are viewed and where we will see the powerful effect that they have on the perception of our images. The first section of the next chapter is all about understanding the performance characteristics of displays, including gamuts, brightness, contrast ratios, frame rates, and more. This establishes the terminology for a detailed analysis of the major types of displays—workstation monitors, HDTV, and digital cinema—including details on their internal workings and how they generate light. We then look into the future to see what's coming soon for the office, home, and theater.

对于在整数与浮点数中工作的情况。我们还看到，对于没有未来验证需求的标准动态范围项目，通常将显示空间用作工作空间。然而，对于高动态范围项目或需要未来验证的标准动态范围项目，要求是选择一个比最终可交付的显示空间“更大”的工作空间。如果不这样做，像裁剪和图像处理伪影这样的问题会降低最终产品的质量。

对于高端工作，如故事片，工作空间必须是线性、浮点和HDR的。虽然这将保护作品免受裁剪和处理伪影的影响，并提供一定的未来适应性，但它并非终极解决方案。为了追求最高的工作质量和未来适应性，推荐使用ACES工作流程，我们将在第6章中详细讨论。

数字成像流程的下一阶段是显示空间，在这里图像被观看，我们将看到它们对我们图像感知的强大影响。下一章的第一部分全部关于理解显示器的性能特征，包括色域、亮度、对比度、帧率等。这为详细分析主要类型的显示器——工作站监视器、高清电视和数字影院——建立了术语体系，包括它们内部工作原理和如何产生光的细节。然后我们展望未来，看看办公室、家庭和影院即将迎来什么。

5 Display Space

Regardless of what Work Space was chosen for the job, it will be viewed in some Display Space during post-production, then eventually reformatted to what will likely be several different Display Spaces for distribution—HDTV, UHDTV, theatrical, web, etc. Unless the display space was also used as the Work Space, the project will likely have to be converted from a larger Work Space to a smaller display space. Work Space wants the maximum data, but Display Space only wants exactly what the specific display can handle. As we saw in Chapter 4, these conversions are fraught with danger, so a clear understanding of these display spaces—with their limitations and conversion issues—is essential to maintaining quality. The overarching mission is to maintain the original look of the work, even on lesser displays.

The current state of affairs is that for television work, our sRGB workstation displays are reasonably close to replicating the HDTV color space, look and feel. This is not true for feature film work, as our workstations fall far below DCI-P3 theatrical digital cinema projectors. But all of this is about to change. An awesome new color space is being deployed that not only approaches the limits of human vision, but will also be used for workstation monitors,

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5 显示空间

无论为工作选择了何种工作空间，在后期制作中它都会在某个显示空间中被查看，最终很可能会被重新格式化为几种不同的显示空间以供分发——高清电视、超高清电视、影院、网络等。除非显示空间也被用作工作空间，否则项目很可能需要从一个较大的工作空间转换到较小的显示空间。工作空间追求最大数据量，而显示空间只需要特定显示器能处理的精确数据。正如我们在第4章所见，这些转换充满风险，因此清晰理解这些显示空间——包括其局限性和转换问题——对于保持质量至关重要。首要任务是维持作品的原始观感，即使在较差的显示器上也是如此。

当前的情况是，对于电视制作，我们的sRGB工作站显示器在复制HDTV色彩空间及其观感方面已相当接近。然而，在电影长片制作中则不然，因为我们的工作站远未达到DCI-P3影院数字投影机的标准。但这一切即将改变。一种令人惊叹的新色彩空间正在部署中，它不仅逼近人类视觉的极限，还将应用于工作站显示器，

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broadcast television, and digital cinema. So, to understand the present and get ready for the future, read on.

We will start by understanding the key performance criteria that are used to measure the characteristics of display devices, such as spectral emissions and contrast ratios. We will then follow three main types of displays involved in digital imaging—workstation monitors, HDTV, and digital cinema—to see how their display characteristics stand up relative to each other. Then we cast our gaze to the future to see what is coming at us with the next generation of UHDTV and Digital Cinema 2.0.

5.1 Display Attributes

In Chapter 4, we learned about color spaces and how they are defined by their primaries, their gamuts, their transfer functions (i.e. gammas), and their white points. The color space is a key starting point for understanding a display device, but much more is needed. A color space is just a mathematical construct, but a display space is a physical device with physical characteristics. In this section, we will look at all of the different attributes that are required to characterize the performance of display devices. Then, in the following section, we will leverage that knowledge to understand the internal workings and attributes of a variety of key display devices that we use in our work.

5.1.1 Spectral Emissions

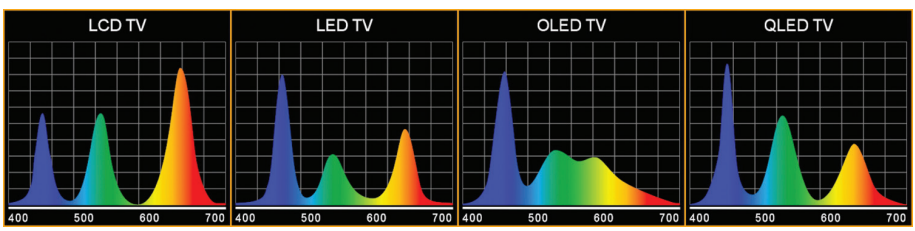


Figure 5.1 Spectral emissions of the major television display technologies

广播电视和数字影院。因此，要理解现在并为未来做好准备，请继续阅读。

我们将从理解用于衡量显示设备特性的关键性能标准开始，例如光谱发射和对比度。接着，我们将关注数字成像中涉及的三种主要显示类型——工作站显示器、高清电视和数字影院——以比较它们的显示特性如何相互对应。然后，我们将目光投向未来，展望下一代超高清电视和数字影院2.0将带来怎样的发展。

5.1 显示属性

在第四章中，我们学习了色彩空间及其如何通过三原色、色域、传递函数（即伽马值）和白点来定义。色彩空间是理解显示设备的关键起点，但所需知识远不止于此。色彩空间仅是数学构造，而显示空间则是具有物理特性的实体设备。本节将探讨表征显示设备性能所需的各种不同属性。随后，在下一节中，我们将运用这些知识来理解工作中使用的各类关键显示设备的内部工作原理与属性。

5.1.1 光谱发射

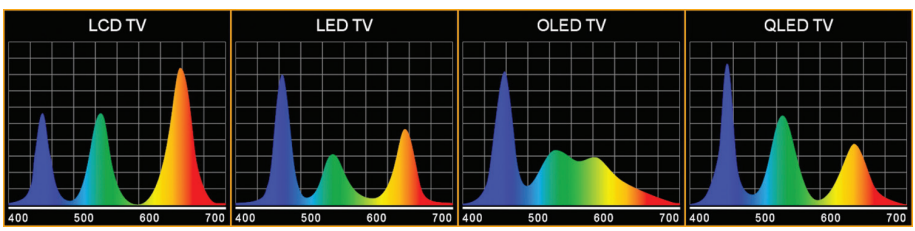


图5.1 主要电视显示技术的光谱发射

We now have a variety of technologies for making television displays and each technology has its own characteristic spectral emission—the wavelengths of light that it will emit. The interesting thing about this is that even though the four display types in Figure 5.1 have very different spectral emissions, and the images they display look remarkably similar. Each display emits somewhat different spectrums of light and in different proportions, yet the sky is still blue and the strawberries are still red. Remember the debayering process we looked at in Chapter 2: Camera Space, where it was revealed that the final output image of digital cameras is a combination of the camera's sensor data plus the manufacturer's "secret sauce" to make nice images? A similar process goes on here where the image processing logic now built into all TV sets "massages" the input signal on its way to the screen to rebalance the output to look nice based on its particular spectral emission pattern, plus a little help from metamerism.

5.1.2 Color Gamut

The color gamut of a display device is one of its most important features, as it determines what color spaces it can support. Current theatrical digital projectors support the DCI-P3 gamut, while current monitors support sRGB/Rec. 709 and some newer ones even support P3. Watch out for the fine print that says, "Covers 95% of P3," as your most critical colors will no doubt land in the missing 5%.

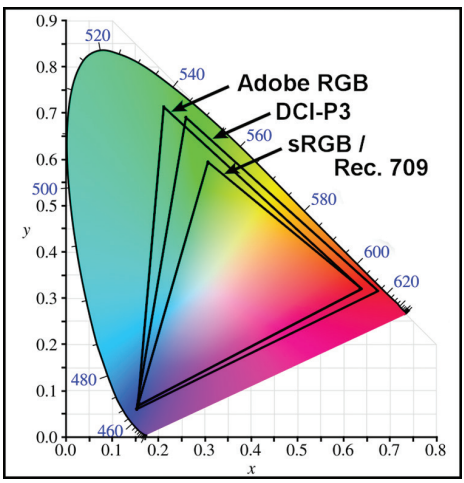


Figure 5.2 Major monitor color gamuts

For print applications, there are a few monitors that cover the huge Adobe RGB gamut we saw in Chapter 4, which fully encapsulates the SWOP CMYK gamut and comes with an equally impressive price tag. Photographers have expensive toys.

如今，我们拥有多种制造电视显示器的技术，每种技术都有其独特的光谱发射特性——即它将发射的光的波长。有趣的是，尽管图5.1中的四种显示器类型具有截然不同的光谱发射特性，但它们显示的图像看起来却惊人地相似。每种显示器发射的光谱略有不同，比例也各异，但天空依然是蓝色的，草莓依然是红色的。还记得我们在第二章“相机空间”中探讨的去马赛克过程吗？那揭示了数码相机的最终输出图像是相机传感器数据加上制造商的“秘制配方”以生成精美图像的结合体。类似的过程也在此发生：如今内置于所有电视机的图像处理逻辑会对输入信号进行“调校”，在其传输至屏幕的过程中，根据其特定的光谱发射模式重新平衡输出，使其看起来更美观，同时借助了同色异谱现象的一点帮助。

5.1.2 色域

显示设备的色域是其最重要的特性之一，因为它决定了设备能够支持何种色彩空间。目前影院数字投影仪支持DCI-P3色域，而现有显示器大多支持sRGB/Rec.709标准，部分新款显示器甚至能支持P3色域。需特别注意那些标注“覆盖95% P3色域”的说明，因为您最关键的色彩表现很可能恰好落在缺失的5%范围内。

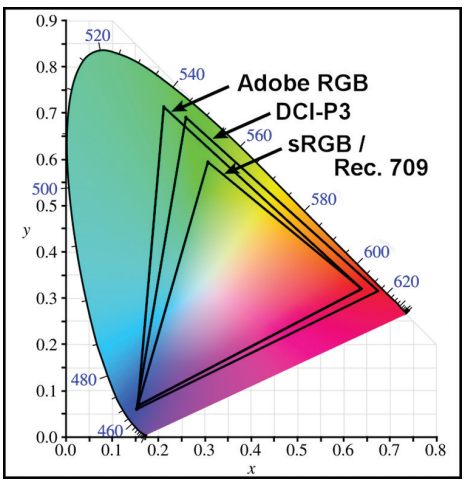


图 5.2 主要显示器色域

对于印刷应用来说，有几款显示器覆盖了巨大的
我们在第4章中看到的Adobe RGB色域，它完全囊括了SWOP CMYK色域，并伴随着同样令人印象深刻的价格标签。摄影师们拥有昂贵的玩具。

5.1.3 White Point

The white point of a color space refers to the xy-chromaticity coordinates or color temperature that represents pure white for that color space. While most displays can change their white points, the exception is the current generation of digital cinema projectors that use xenon lamps, which have a D65 color temperature. LED and LCD monitors don't use a single bulb as a light source like digital cinema projectors, so their white points are programmable. And since the white points of modern displays are programmable, you will want to be sure you are not working on a movie that will be projected at D65 with your monitor set to D50, so check your workstation's color temperature. But if you are a photographer and want to accurately view your photos before going to print, you will absolutely want to set your monitor to D50.

5.1.4 Transfer Function (Gamma)

The transfer function (colloquially known as the "gamma") was covered in detail in Chapter 4. While it was initially baked into the electronics of the old CRT television sets, all of today's monitors and digital cinema projectors are linear so they have no gamma—unless you want to call linear a gamma of 1.0—which you can, of course. However, they are all outfitted with viewing LUTs that impart a gamma to the displayed or projected image electronically. And since the gamma is defined in the software, it can be changed when a display changes its color space. Today the gamma of a modern display device is software configurable, which increases its flexibility.

Watch the transfer function in action as the straight-line segment fixes the noise problem caused by using just gamma alone.

5.1.5 Brightness

The brightness—or, more correctly, the luminance—of a flat-panel display is, of course, how much light it can project toward viewers. It is usually measured in "nits" (see Chapter 1), one of several metrics for measuring the brightness

5.1.3 白点

色彩空间的白点指的是代表该色彩空间中纯白色的xy色度坐标或色温。虽然大多数显示器可以调整其白点，但当前使用氙灯的数字电影投影机是个例外，它们具有D65色温。LED和LCD显示器不像数字电影投影机那样使用单一灯泡作为光源，因此它们的白点是可编程的。由于现代显示器的白点可调，您需要确保在制作一部将以D65色温投影的电影时，您的显示器没有设置为D50，因此请检查工作站的色温。但如果您是摄影师，并希望在打印前准确查看照片，那么您绝对应该将显示器设置为D50。

5.1.4 传递函数（Gamma）

传递函数（俗称“伽马”）在第4章中已详细讨论。虽然它最初被固化在老式CRT电视机的电子电路中，但如今所有显示器和数字影院投影仪都是线性的，因此它们没有伽马——除非你想将线性称为1.0的伽马，这当然也可以。不过，这些设备都配备了查看查找表，通过电子方式为显示或投影的图像赋予伽马值。由于伽马值在软件中定义，当显示器更改其色彩空间时，它可以被调整。如今现代显示设备的伽马值可通过软件配置，这增强了其灵活性。

观察传递函数如何发挥作用，直线段解决了仅使用伽马值{v*}单独导致的噪声问题。

5.1.5 亮度

平板显示器的亮度——或者更准确地说，是发光度——当然是指它能向观看者投射多少光。通常以“尼特”为单位进行测量（见第1章），这是衡量亮度的几种度量标准之一。

of a surface. Projectors, on the other hand, are measured in lumens (see Chapter 1) which is how much light they can pour onto a screen, rather than the brightness of the screen itself. This makes sense since the brightness of the screen changes with the distance to the projector. A 10,000-lumen projector might be used in a small theater, while 50,000 lumens would be needed in a larger theater where the projector is further away and the screen is much larger, spreading those lumens over more screen acreage.

5.1.6 Contrast Ratio

The contrast ratio of a display is the ratio of the brightest white to the darkest black that it can produce and is a key factor in having a bright and vibrant display (see Figure 5.3). The problem with contrast ratios is that they are not well defined, so monitor manufacturers devise clever ways to increase their scores. One such method is to report the *dynamic* contrast ratio. These can produce truly astonishing numbers of up to 1,000,000:1!

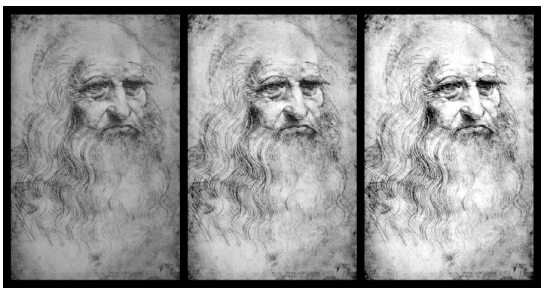


Figure 5.3 Increasing contrast ratios

Dynamic contrast ratios apply to backlit displays where the back-light LEDs can be turned on and off. To get the black number, they turn off the LEDs behind a black patch of picture, and to get the white number, they turn the LEDs on behind a white patch. This is how they get the truly astonishing dynamic contrast ratio numbers. However, in the real world, these numbers are highly dependent on picture content. But big numbers sell TVs.

The more meaningful contrast ratio is the *native* or *static* contrast ratio, which refers to the brightest white and darkest black on the screen *at the same time*, like with a large checkerboard pattern. So when you are being impressed with contrast ratios, be sure to check which type you are dealing with.

Now, here is the bizarre thing about contrast ratios—let's say a mythical monitor's black level measures 1 nit, and the white level measures 1000 nits. The contrast ratio is calculated as the white level divided by the black level

关于一个表面。另一方面，投影仪以流明（见第1章）来衡量，这是指它们能投射到屏幕上的光量，而非屏幕本身的亮度。这是合理的，因为屏幕的亮度会随着与投影仪的距离而变化。一个小型剧院可能使用10,000流明的投影仪，而在更大的剧院中，由于投影仪距离更远、屏幕更大，需要50,000流明才能将光线覆盖更广的屏幕面积。

5.1.6 对比度

显示器的对比度是指其能够产生的最亮白色与最暗黑色之间的比值，这是实现明亮鲜艳显示效果的关键因素（见图5.3）。对比度的问题在于其定义不够明确，因此显示器制造商会设计巧妙的方法

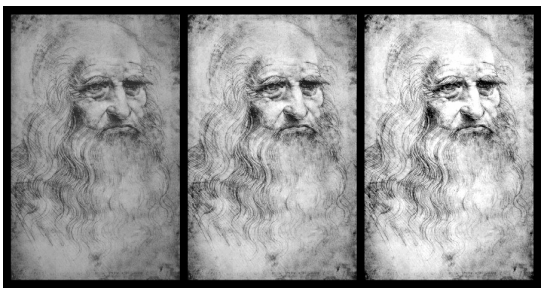


图 5.3 提升对比度比率

提高他们的分数。一种方法是报告动态对比度。这些可以产生高达1,000,000:1的真正惊人数字!

动态对比度适用于背光可开关的显示器。为了得到黑色数值，他们会关闭黑色图像区域背后的LED灯；而为了得到白色数值，则会点亮白色区域背后的LED灯。这正是他们获得惊人动态对比度数值的方法。然而在实际使用中，这些数值高度依赖于画面内容。但庞大的数字确实能促进电视销售。

更有意义的对比度是原生或静态对比度，它指的是屏幕上同时出现的最亮白色和最暗黑色，就像一个大棋盘图案那样。所以当你对对比度印象深刻时，务必确认你正在处理的是哪种类型。

现在，关于对比度有个奇怪的现象——假设一台神话般的显示器黑色水平测得1尼特，白色水平测得1000尼特。对比度的计算方式是白色水平除以黑色水平

(1000/1), which results in a 1000:1 contrast ratio. If you want to double the contrast ratio to 2000:1, you need to raise the whites to 2000 nits, right? Doubling the brightness of a display device can be done, but it will entail some engineering, more expensive components, and higher power consumption. However (here's the bizarre part), you can also double the contrast ratio by lowering the blacks down to 0.5 nits. Suddenly we now have a 2000:1 contrast ratio simply by making the blacks blacker. The takeaway here is that the contrast ratio has less to do with how bright the screen is, but rather how black the blacks are.

This is the industry trend for improving the contrast ratio. Today's monitor or HDTV does not have truly black blacks. Just watch some movie credits roll by and check out the black background. Not so black, is it? The reason is the LCD's display is backlit, and when the pixels are commanded to shut off completely to make it black, some of the backlight still leaks through. However, as we shall see shortly, there is some exciting new tech coming that makes much blacker blacks and therefore much greater contrast ratios with more vibrant displays.

5.1.7 Bit Depth

Figure 5.4 Increased bit depth

Some will call it "color depth," but bit depth means how many bits are allocated to each color channel. Today, 8 or 10 bits per color channel is the norm for displays, with a few supporting 12 bits. Figure 5.4 shows how increasing the bit depth increases the number of brightness steps from black to white and reduces the difference between each step to get a visually smooth gradient without banding. The banding in Figure 5.4 has been exaggerated to illustrate how the banding steps get smaller with increased bit depth. Always keep in mind that if an 8-bit image is pushed too hard in post, you might introduce color banding yourself.

Increased bit depth becomes more important the higher the contrast ratio of the display, simply because it "stretches" the bits over a larger brightness range so the steps between the bits get larger and can become visible. Note that a 10-bit display (1024 steps) will have 4 brightness steps for each step

(1000/1)，这带来了1000:1的对比度。若想将对比度翻倍至2000:1，就需要将白色提升至2000尼特，对吗？虽然显示设备的亮度翻倍可以实现，但这需要一些工程调整、更昂贵的组件以及更高的功耗。然而（奇怪之处来了），你也可以通过将黑色降低至0.5尼特来使对比度翻倍。突然间，我们仅通过让黑色更黑就实现了2000:1的对比度。这里的启示是：对比度与屏幕的亮度关系不大，而更取决于黑色有多深邃。

这是提升对比度的行业趋势。如今的显示器或高清电视无法呈现真正的纯黑。只需观看一些电影片尾滚动的字幕，观察其黑色背景——并非纯黑，对吗？原因在于液晶显示屏采用背光技术，当像素被指令完全关闭以显示黑色时，仍有部分背光会透出。不过我们很快会看到，令人振奋的新技术即将问世，它能实现更深沉的黑色，从而带来更高的对比度与更鲜艳生动的显示效果。

5.1.7 位深度

有些人会称之为“色深”，但位深指的是每个颜色通道分配了多少位。如今，每个颜色通道8位或10位已成为

图 5.4 增加的位深度

显示器通常支持12位色深。图5.4展示了增加位深度如何提升从黑到白的亮度阶数，并减小每阶之间的差异，从而获得视觉上平滑无断层（色带）的渐变效果。图中断层现象已被夸大，以说明随着位深度增加，色带阶梯会如何变得更细微。需始终注意：若对8位图像进行过度后期处理，可能会人为造成色彩断层。

增加位深在显示器的对比度越高时变得越重要，这仅仅是因为它在一个更大的亮度范围内“拉伸”了比特，使得比特之间的步长变大并可能变得可见。请注意，一个10位显示器（1024个步长）将为每个步长提供4个亮度步长。

in 8 bits (256 steps) and 12 bits (4096 steps) will have 16 steps for each one in 8 bits.

FUN FACT: Another term for banding is “contouring,” which occurs when an image is captured with too few bits or an image is re-mapped into fewer bits reducing the number of available colors.

5.1.8 Frame Rates

Both HDTV and digital cinema displays have specs for specific frame rates which define how many original images they can display in a second. I think of frame rates as coming in two flavors—reasonable frame rates, like 24, 25, 30, and 60 fps (frames per second), and unreasonable frame rates, like 23.976, 29.97, and 59.98 fps. An explanation of just the reasonable frame rates goes thusly:

- 24 fps - Standard 35 mm film frame rate
- 25 fps - PAL (European) television frame rate
- 30 fps - A “rounded up” video frame rate
- 60 fps - Double the 30 fps standard.

The unreasonable frame rates are a bit messy to explain. Early black-and-white TV used to be an honest 30 fps (yay!). But when the color signal was mixed with the original black and white signal, it generated an interference with the audio signal (darn!). To solve this problem, it was decided to slightly “detune” the video signal by 1 part in 1000 (drat!). This was well below the level that would disturb the old black-and-white TV electronics. If a 30-fps signal is slowed by 1/1000th, you get 29.97 for your basic TV frame rate (sigh!).

So that explains the tortured history of the 29.97 frame rate. The 59.94 frame rate is easy since it is exactly twice the 29.97 frame rate because in the interlaced video, there are two fields for every frame. The excuse for the 23.976 frame rate is that this is what you get if you map a 24-fps feature film to a 29.97 video signal to transfer a film to video.

在8位（256级）和12位（4096级）中，每8位将对应16级。

> 有趣的事实：条带效应的另一个术语是“轮廓化”，当图像以过少的比特数捕获或图像被重新映射到更少的比特数，从而减少可用颜色数量时，就会发生这种现象。

5.1.8 帧率

高清电视和数字影院显示器都有特定的帧率规格，这定义了它们每秒能显示多少原始图像。我认为帧率可分为两种类型——合理的帧率，如24、25、30和60 fps（每秒帧数），以及不合理的帧率，如23.976、29.97和59.98 fps。仅对合理帧率的解释如下：

- 24 帧每秒 - 标准 35 毫米胶片帧率
- 25 帧每秒 - PAL（欧洲）电视制式帧率
- 30 fps - 一种“向上取整”的视频帧率
- 60 帧每秒 - 是 30 帧每秒标准的两倍。

不合理的帧率解释起来有点混乱。早期的黑白电视原本是标准的30帧每秒（好耶！）。但当彩色信号与原始黑白信号混合时，却对音频信号产生了干扰（可恶！）。为了解决这个问题，人们决定将视频信号稍微“失调”千分之一（唉！）。这种调整幅度远低于干扰老旧黑白电视电子设备的水平。如果一个30帧每秒的信号被减慢1/1000th，就得到了电视的基本帧率29.97（唉！）。

这就解释了29.97帧率为何有一段曲折的历史。59.94帧率则很好理解，因为它正好是29.97帧率的两倍——在隔行扫描视频中，每帧画面包含两个场。而23.976帧率的出现，则是将24帧/秒的电影转换为29.97帧视频信号时产生的折衷方案，用于将胶片电影转录到视频介质。

One piece of advice when dealing with frame rates: don't assume that when you see 29.97 fps on your camera or clip, it is actually 30 or vice versa. The 30-fps frame rate is to be taken literally and not confused with 29.97. Modern digital displays have no trouble coping with all of these different frame rates, so they invariably support multiple frame rates. Which devices support which frame rates will be revealed shortly when we talk about each type of display device.

5.1.9 Refresh Rates

Workstation monitors and laptops are not spec'd to frame rates, but rather refresh rates—the maximum times per second a new image can be displayed. The actual frame rate you get depends on the video card driving the monitor. The refresh rate is measured in Hz (Hertz), which is cycles per second, so a 60 Hz refresh rate means the monitor is capable of displaying up to 60 frames per second. Gaming monitors support higher speeds of up to 240 Hz, but you will pay extra for the speed.

5.1.10 Calibration

First, the distinction between characterization vs. calibration of a monitor. Characterization is simply to measure the performance of a monitor. Calibration is to adjust the monitor to match a particular specification. If you work in a color-critical job (e.g., colorist, editor, or visual effects artist), color scientists recommend that you calibrate your monitor once a month. For the rest of us, modern display devices are stable enough not to warrant regular calibration. The calibration process produces an ICC profile for the monitor, but it is recommended that you make as many of the adjustments to the monitor itself as possible because some apps don't play well with ICC profiles.

To calibrate your own monitor, you will need to purchase a colorimeter for around $200 and then download the software that comes with it. Because every monitor is different, I am reduced here simply to telling you what to do, not how to do it. Modern monitors come with a long list of image enhancement features "designed to improve your viewing experience" that will spoof the calibration process because the colorimeter will be looking at the screen. These fancy features need to be disabled before starting. Disable features

关于帧率处理的一条建议是：不要以为在相机或片段上看到29.97 fps就等同于30 fps，反之亦然。30 fps的帧率应按字面意义理解，切勿与29.97混淆。现代数字显示设备能轻松应对各种不同帧率，因此它们普遍支持多种帧率标准。具体设备支持的帧率范围，我们将在后续讨论各类显示设备时详细说明。

5.1.9 刷新率

工作站显示器和笔记本电脑的规格并非以帧率衡量，而是刷新率——即每秒能够显示新图像的最大次数。实际获得的帧率取决于驱动显示器的显卡。刷新率以赫兹（Hz）为单位计量，即每秒周期数，因此60 Hz刷新率意味着显示器最高可支持每秒显示60帧。游戏显示器支持高达240 Hz的更高刷新速度，但您需要为这种速度支付额外费用。

5.1.10 校准

首先，区分显示器的特性描述与校准。特性描述仅仅是测量显示器的性能。校准则是指调整显示器以符合特定规格。如果你从事对色彩要求严格的工作（例如调色师、编辑或视觉特效艺术家），色彩科学家建议你每月校准一次显示器。对于我们其他人来说，现代显示设备足够稳定，无需定期校准。校准过程会为显示器生成一个ICC配置文件，但建议你尽可能在显示器本身进行尽可能多的调整，因为有些应用程序与ICC配置文件的兼容性不佳。

要校准您自己的显示器，您需要购买一个约200美元的色彩校准仪，然后下载其配套软件。由于每台显示器都不同，我在此只能告诉您该做什么，而非具体操作步骤。现代显示器配备了一系列“旨在提升观看体验”的图像增强功能，这些功能会干扰校准过程，因为色彩校准仪将直接读取屏幕显示。在开始前，请务必关闭这些花哨的功能。

like low blue light filter, shadow boosting, dynamic contrast, power saving, backlight strobing, and any other post-processing operations. If you have a wide gamut display, make sure the gamut clamps are off.

Before you begin, set the monitor brightness to a comfortable level because changing brightness after calibration can affect the results. Also, set the room as dark as possible to minimize any ambient light getting into the colorimeter. Attach the colorimeter to the center of the screen, then start the software and follow the directions. Once it gets rolling, it will start cycling through colors under the colorimeter to make a LUT that will go into its ICC profile.

5.1.11 Viewing Surround

A critical issue that affects the appearance of any display device is the "surround," which is the lighting environment in which a given display device is supposed to be viewed, including the color of the wall behind and around the display. It is part of the specs for the display device and comes in three versions:

- Dark surround - For movie theaters
- Dim surround - For viewing HDTV in the home
- Bright surround - Office lighting for workstation monitors.

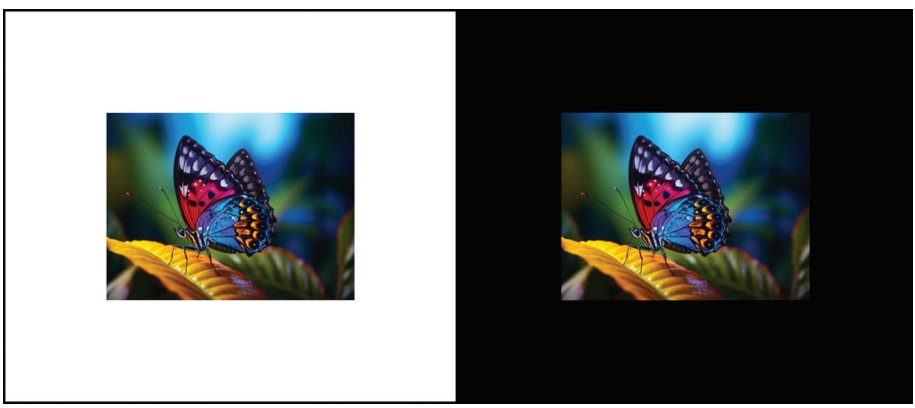


Figure 5.5 Lateral brightness adaptation alters apparent contrast, brightness, and colorfulness. Courtesy of Vedant Saxena

如低蓝光滤镜、阴影增强、动态对比度、节能、背光频闪以及其他任何后处理操作。如果您使用的是广色域显示器，请确保色域限制功能已关闭。

开始之前，请将显示器亮度调整至舒适水平，因为校准后更改亮度会影响结果。同时，尽量将房间调暗，以尽量减少进入色度计的环境光。将色度计附着在屏幕中央，然后启动软件并按照指示操作。一旦开始运行，它将开始在色度计下方循环显示颜色，以生成将写入其ICC配置文件的查找表（LUT）。

5.1.11 查看周边

影响任何显示设备外观的一个关键问题是“环境光”，即观看特定显示设备时所在的光照环境，包括显示器后方及周围墙壁的颜色。这是显示设备规格的一部分，分为三种版本：

- 暗黑环绕 - 专为电影院设计
- 环绕声 - 适用于家庭观看高清电视
- 明亮环绕 - 适用于工作站显示器的办公室照明。

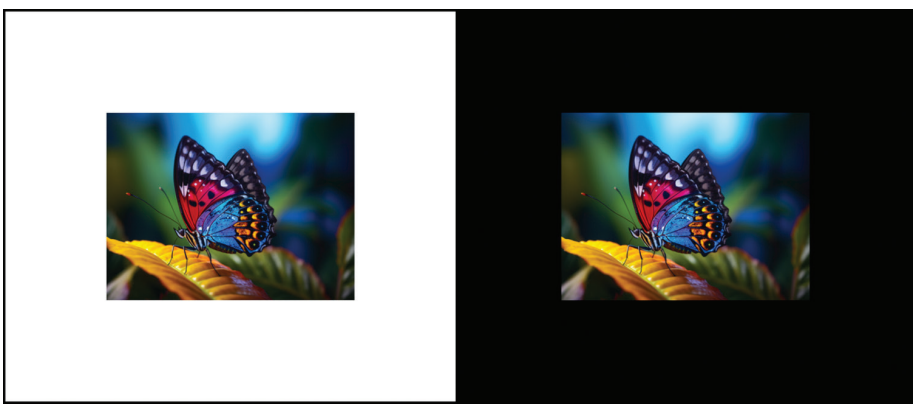


图5.5 横向亮度适应会改变表面对比度、亮度和色彩鲜艳度。由Vedant Saxena提供

There are two reasons why the surround is so important. The first is that the eye is affected by the brightness of the wall behind the screen by a process called lateral brightness adaptation, which we met in Chapter 1, but now we will view with color. The color pictures in Figure 5.5 are identical, but the right picture in the black surround appears to have greater contrast, brightness, and colorfulness compared to that on the white surround. Displays are designed for specific viewing surroundings to maintain consistency in their images. The second reason the surround is important is "flare," where room light falls onto the display screen and fogs up the blacks. Raising the black level even slightly drops the contrast ratio hugely.

Figure 5.6 The effect of flare on display appearance. Courtesy of Columbia Pictures

The images in Figure 5.6 illustrate how flare degrades the display. Starting on the left (A), viewing a theatrical image in its proper dark surround, it appears bright and the colors vibrant. Increase the flare (B) by raising the theater lighting to the dim surround specified for HDTV viewing and it looks dimmer and desaturated. Raise the theater lighting again (C) to the bright surround intended for workstation monitors (somewhat exaggerated) and you see how washed out and degraded the image would be.

FUN FACT: Kodak once observed that if everyone in a theater audience wore a white shirt, the resulting bounce light back to the screen would noticeably degrade the movie. Now you know why.

The dark surround of a theater is there for one very good reason—because of the response of the human visual system, it seriously reduces the required brightness for a movie, which saves a great deal of money. If the projector bulbs had to be ten times brighter, not only would electricity costs skyrocket but the heat generated would require the addition of cooling systems

环绕环境之所以如此重要，有两个原因。首先，眼睛会受到屏幕后方墙壁亮度的影响，这一过程称为侧向亮度适应，我们在第一章中已提及，但现在我们将从色彩角度来审视。图5.5中的彩色图片是相同的，但黑色环绕中的右侧图像与白色环绕中的相比，显得对比度更高、亮度更强、色彩更鲜艳。显示器是为特定的观看环境设计的，以保持图像的一致性。环绕环境重要的第二个原因是“眩光”，即室内光线照射到显示屏上，使黑色部分变得模糊。即使黑色水平略有提高，也会大幅降低对比度。

图5.6 眩光对显示效果的影响。由哥伦比亚电影公司提供

图5.6中的图像展示了眩光如何降低显示效果。从左图（A）开始，在适宜的黑暗环境中观看影院级图像，画面显得明亮且色彩鲜艳。将剧院照明提升至高清晰度电视观看所需的微光环境（B），增加眩光后，画面会显得更暗且饱和度降低。再次将剧院照明提升至工作站显示器设计所用的明亮环境（C）（略有夸张），你就能看到图像会变得多么暗淡和劣化。

有趣的事实：柯达公司曾发现，如果剧院里的每位观众都穿白衬衫，反射回屏幕的光线会明显影响电影画质。现在你明白原因了。

剧院黑暗的环境有一个非常重要的原因——由于人类视觉系统的反应，它能显著降低电影所需的亮度，从而节省大量资金。如果投影灯泡必须亮十倍，不仅电费会飙升，产生的热量还需要额外增加冷却系统。

to prevent the equipment from melting. Keep in mind that all the light you see on the entire theater screen has to come from one small bulb. As we will see in Section 5.4.4 Digital Cinema, a theatrical screen in a dark surround is 1/10th the brightness of a HDTV screen in a dim surround—all due to the viewing surround. You will be quite surprised at how dim theater screens really are.

Here are some mind-blowing animated optical illusions to demonstrate just how tricky our visual system truly is.

5.2 SDR vs. HDR

In the home television market, where 90% of the viewing audience will watch feature films and 100% will watch broadcast television, we are transitioning from Standard Dynamic Range (SDR) to High Dynamic Range (HDR) picture content. The actual specs for HDR content are discussed in detail in Section 5.5 Coming Soon. Here we will go over the Display Space requirements for these two viewing intensities.

The key difference between SDR and HDR displays is contrast ratios. SDR displays have a contrast ratio of less than 1000:1 and only use data that ranges between zero and 1.0. This means that you can work in display space and use either float or integer data. You can also use 8-bit file formats like JPEG to export images. It also means that a typical workstation monitor will do fine for development.

HDR displays are supposed to have a contrast ratio of 8000:1 or more and use data above 1.0. This means that you must work in float, should work in linear, and your typical workstation monitor will not do anymore since the UHDTV will be many times brighter than the typical monitor. You will also need to save HDR images in a HDR file format such as DPX, EXR, or TIFF float or use 12-bit sequence wrappers like ProRes 4444 or MOV using the HEVC (h.265) codec.

为了防止设备熔化。请记住，整个影院银幕上你看到的所有光线都必须来自一个小灯泡。正如我们将在5.4.4节《数字影院》中看到的那样，黑暗环境中的影院银幕亮度仅为昏暗环境中高清电视屏幕的1/10th——这完全是由于观看环境的影响。你会对影院银幕实际有多暗感到相当惊讶。

这里有一些令人惊叹的动画视觉错觉，展示了我们的视觉系统究竟有多复杂。

5.2 SDR 与 HDR

在家庭电视市场中，90%的观众会观看故事片，100%的观众会观看广播电视，我们正在从标准动态范围（SDR）向高动态范围（HDR）图像内容过渡。HDR内容的具体规格将在第5.5节“即将推出”中详细讨论。这里我们将探讨这两种观看强度对显示空间的要求。

SDR与HDR显示器的核心差异在于对比度。SDR显示器的对比度低于1000:1，且仅使用0到1.0范围内的数据。这意味着您可以在显示色彩空间中进行工作，并选择使用浮点数或整数数据。您还可以使用JPEG等8位文件格式导出图像。这也意味着，典型的工位显示器足以满足开发需求。

HDR显示器的对比度应达到8000:1或更高，并使用超过1.0的数据。这意味着您必须在浮点模式下工作，且应在线性色彩空间中进行操作，而您常用的工作站显示器将不再适用，因为超高清电视的亮度将远超普通显示器。您还需要将HDR图像保存为DPX、EXR或TIFF浮点等HDR文件格式，或使用HEVC（h.265）编解码器通过ProRes 4444或MOV等12位序列封装格式进行存储。

5.2.1 Cinema

Image data for digital cinema projection is HDR, but the brightness of a theatrical screen is only around 50 nits, which seems to fly in the face of what was stated earlier about HDR images on "big nit" displays of 1000 nits or more. The answer lies in the contrast ratio. A typical digital cinema will have a contrast ratio of up to 4000:1, which needs HDR data. How do they get such high contrast ratios with a paltry 50-nit screen? It's the blacks, of course. The dark surround of the theater is the darkest viewing surround, which drives the blacks down and the contrast ratio up.

5.3 HDR Display Formats

The transition from SDR television to HDR has introduced a serious new display problem—the lack of adherence to standards. While the specs for HDR are detailed and specific, TV manufacturers cannot yet meet the full standard for brightness and gamut. As a result, each television manufacturer supports HDR to a different degree. Some have 500-nit displays that they call HDR, while others offer up to 2000 nits. Who knows how many nits will be available tomorrow? As for the gamut, some support the Rec. 2020 wide color gamut by 60% or 70%, while others set their wide gamut target for DCI-P3 and call that a wide color gamut.

Here is the problem for content developers—how do you master a show that will display consistently on UHDTV sets with different HDR brightness ranges? The SDR approach—which uses one standard transfer function (gamma), gamut, and brightness for all HDTVs—doesn't work with HDR displays of differing brightness and gamut ranges. If a show is mastered for a 500-nit HDR display, it will not be able to take advantage of another TV set with a brightness of 2000 nits. If the show is mastered for a 2,000-nit TV, the picture will get clipped when displayed on a 500-nit TV. It's a moving target. Even if several versions of a show were mastered, there would be no way to get the right version to the right TV set in the home.

So far, two solutions have been deployed—HLG and PQ. HLG attempts to provide a solution that will work on both SDR *and* HDR, while PQ takes the high road and only supports HDR programming. A given HDR display will support one or more of these standards.

5.2.1 电影院

数字影院投影的图像数据是高动态范围（HDR）的，但影院银幕的亮度仅为50尼特左右，这似乎与先前关于HDR图像需要1000尼特以上“高亮度”显示器的说法相悖。答案在于对比度。典型的数字影院对比度可达4000:1，这正需要HDR数据支撑。那么如何用区区50尼特的银幕实现如此高的对比度？关键在于黑色表现。影院的黑暗环境提供了最暗的观看背景，这压低了黑位、提升了对比度。

5.3 HDR显示格式

从SDR电视向HDR的过渡引发了一个严峻的新显示问题——标准执行的不统一。尽管HDR的规格详尽而具体，但电视制造商目前仍无法完全达到亮度和色域的标准。因此，各家电视制造商对HDR的支持程度各不相同：有些将500尼特亮度的显示屏称为HDR，而另一些则提供高达2000尼特的亮度。谁知道明天又会涌现多少尼特的规格？至于色域，有些厂商以60%或70%的比例支持Rec.2020广色域标准，另一些则将DCI-P3设为目标并称之为广色域。

这就是内容开发者面临的问题——如何制作一部能在不同HDR亮度范围的超高清电视上保持显示一致性的节目？传统标准动态范围（SDR）方案——即对所有高清电视采用统一的标准传递函数（伽马）、色域和亮度——无法适配亮度与色域范围各异的HDR显示设备。若节目以500尼特HDR显示设备为基准制作，将无法充分发挥2000尼特亮度电视的显示潜力；若以2000尼特电视为基准制作，在500尼特电视上播放时画面又会出现裁切失真。这如同移动标靶般难以把握。即便制作多个节目版本，也无法确保家庭中的每台电视都能匹配到对应版本。

迄今为止，已部署两种解决方案——HLG与PQ。HLG试图提供一种能同时适用于SDR和HDR的解决方案，而PQ则选择高端路线，仅支持HDR节目制作。特定的HDR显示器将支持其中一种或多种标准。

5.3.1 Hybrid Log Gamma (HLG)

Hybrid Log Gamma (HLG) attempts to make the same show look good on both SDR and HDR displays. The problem is that HDR content cannot be displayed on SDR displays without looking awful. HLG attempts to bridge this gap. The idea is that the HLG transfer function will look good on both SDR and HDR displays without modification. It is a transfer function that is a blend of a gamma and a log curve.

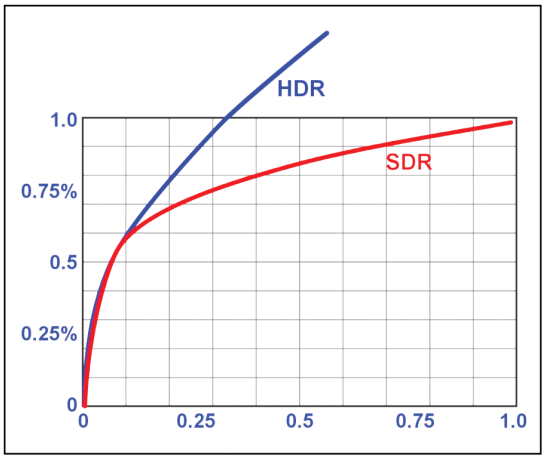


Figure 5.7 HLG curves for SDR and HDR

The first part of the curve is a standard gamma curve, which then transitions into a log curve. On an SDR display, the curve looks like the red line in Figure 5.7. The HDR pixels are gently squeezed down in the log portion of the curve and rolled off to avoid clipping. However, if the image is sent to an HDR display, it recognizes the HLG signal and reinterprets it to look like the blue HDR line—an extended range gamma curve that fits the brightness range of that TV. This way, the transfer function adapts to whatever HDTV it is displayed on. The downside to the HLG is that in the HDR mode, it does not display the image with full HDR vibrancy compared to other methods. However, being downward compatible with SDR TVs is a very valuable feature. An HDR show can be transmitted to homes without concern for whether the viewer has an SDR HDTV or an HDR UHDTV set.

5.3.2 The Perceptual Quantizer (PQ)

Yikes—it sounds like a deadly sci-fi weapon! But no, the Perceptual Quantizer (PQ) is a fancy new type of "adaptive" transfer function specifically for HDR displays because the usual solutions just don't cut it with the different brightness ranges available with HDR. The usual solutions—log or gamma functions—exhibit banding (contouring) at different luminance ranges with

5.3.1 Hybrid Log Gamma (HLG)

混合对数伽马（HLG）试图让同一节目在标准动态范围（SDR）和高动态范围（HDR）显示器上都能呈现良好效果。问题在于，HDR内容若未经处理直接在SDR显示器上播放，效果会非常糟糕。HLG正是为了弥合这一差距而生。其核心理念是：HLG传递函数无需修改即可在SDR和HDR显示器上均呈现优质画面。这是一种融合了伽马曲线与对数曲线的传递函数。

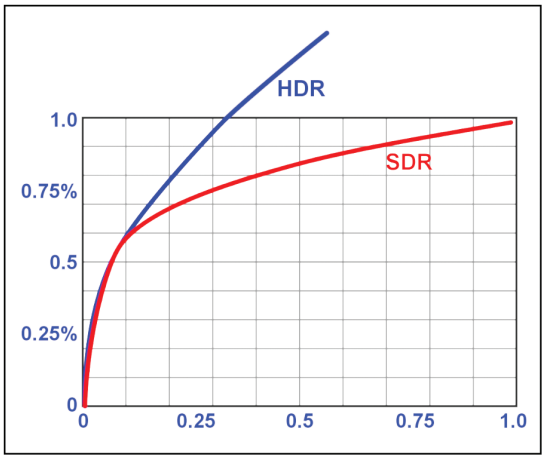


图5.7 SDR与HDR的HLG曲线

曲线的第一部分是标准的伽马曲线，随后过渡为对数曲线。在SDR显示器上，该曲线如图5.7中的红线所示。HDR像素在对数曲线部分被平缓压缩，并通过滚降处理避免剪切。然而，若图像传输至HDR显示器，设备会识别HLG信号并将其重新解读为蓝色HDR曲线——这是一条扩展范围的伽马曲线，可适配该电视的亮度范围。如此一来，传递函数能自适应任何播放的HDTV设备。HLG的不足之处在于，在HDR模式下，相比其他方法其图像显示的HDR生动性未能完全释放。但向下兼容SDR电视的特性极具价值。HDR节目可无需顾虑用户使用的是SDR HDTV还是HDR UHDTV设备，直接传输至家庭终端。

5.3.2 感知量化器（PQ）

哎呀——听起来像是一种致命的科幻武器！但并非如此，感知量化器（PQ）是一种专门为HDR显示器设计的“自适应”传递函数，因为常规解决方案无法应对HDR提供的不同亮度范围。常规解决方案——对数或伽马函数——在不同亮度范围内会出现条带（轮廓）现象。

high-brightness images. Linear light formats like OpenEXR eliminate banding but are not efficient, meaning they waste bits. A new approach was needed.

The New Approach Department at Sony was tasked with solving this problem, so they did. What they came up with is the Perceptual Quantizer, which is based on the properties of human vision, rather like the CIE xy-chromaticity chart is based on human vision. The basic idea is that metadata about the medium is transmitted along with the PQ. Each monitor looks at its own maximum brightness capability and performs a tone-mapping of the incoming video to optimize its appearance on that display. There are several standards, as follows:

- **HDR10** - This is the earliest and most common HDR standard, and also the least flexible. Static metadata about the show is transmitted that indicates its brightest pixel value as well as its average pixel value. This information is used to perform a tone mapping of the show to whatever that display's brightness is capable of. It assumes a maximum brightness of 1000 nits and uses the Rec. 2020 gamut and 10 bits per channel encoding.

  This was a major step forward, but HDR10 had two problems. First, static metadata meant that only one tone mapping was done for the entire show, which may not be appropriate for all scenes. Second, HDR displays started becoming available that were brighter than 1000 nits so it could not take full advantage of them.

- **HDR10+** - This implemented *dynamic metadata* and upped the maximum brightness to 4,000 nits and support for 8K resolution. Dynamic metadata provides scene-by-by-scene information for the brightest pixel and average brightness, so the tone-mapping is optimized for each scene. It is championed by Samsung, Panasonic, and 20th Century Fox, among others.

- **Dolby Vision** - Dolby came up with the highest-quality, most demanding standard that supported a maximum display brightness of 10,000 nits and it is the most common standard in the field today. Using dynamic metadata, it performs scene-by-scene tone mapping at 12 bits per channel. Too good for regular Blu-Ray, it only comes with *Ultra HD* Blu-Ray. Anything mastered in Dolby Vision is considered future-proofed.

高亮度图像。像OpenEXR这样的线性光格式消除了条带效应，但效率不高，意味着它们浪费了比特。因此需要一种新的方法。

索尼的新方法部门被委以解决这一问题的重任，他们做到了。他们提出的解决方案是感知量化器，它基于人类视觉的特性，类似于CIE xy色度图也是基于人类视觉设计的。其核心理念是，与PQ一同传输的还有关于媒介的元数据。每台显示器会根据自身的最大亮度能力，对输入视频进行色调映射，以在该显示屏上实现最佳视觉效果。现有以下几种标准：

- HDR10——这是最早且最常见的HDR标准，也是最不灵活的。它会传输关于节目的静态元数据，指示其最亮像素值以及平均像素值。这些信息用于根据显示器的亮度能力对节目进行色调映射。它假设最大亮度为1000尼特，并采用Rec. 2020色域和每通道10位编码。

  这是一个重大的进步，但HDR10存在两个问题。首先，静态元数据意味着整部节目只采用一种色调映射处理，这可能并不适合所有场景。其次，市面上开始出现亮度超过1000尼特的HDR显示器，而HDR10无法充分发挥其性能优势。

- HDR10+ - 该标准实现了动态元数据，并将最高亮度提升至4000尼特，同时支持8K分辨率。动态元数据能逐场景提供最亮像素与平均亮度的信息，从而优化每个场景的色调映射。该标准由三星、松下和20th世纪福克斯等公司共同倡导。

- 杜比视界 - 杜比公司制定了最高质量、要求最严苛的标准，支持最高10,000尼特的显示亮度，这也是当今领域中最常见的标准。它采用动态元数据，以每通道12位进行逐场景色调映射。由于标准过高，普通蓝光无法承载，仅适用于超高清蓝光。任何以杜比视界制作的内容都被视为具备未来兼容性。

5.4 Display Devices

Armed with the above dictionary of display attributes—like color gamut, white point, and gamma—we can now visit each display device in turn and understand its performance characteristics. Once you understand the different color spaces these devices use, it becomes clear why an image must get a color space conversion to have a consistent appearance on different types of displays. But first, let us pause to understand how the most pervasive display device—the flat-panel display—is constructed, as it is used throughout the discussion that follows.

5.4.1 Flat-Panel Display Construction

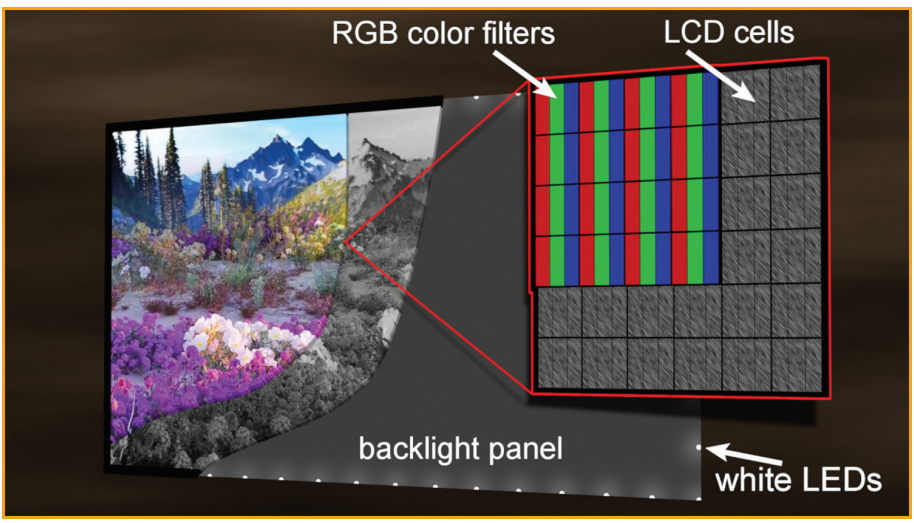


Figure 5.8 Flat-panel display construction

The design for modern LED/LCD monitors and TVs has converged around a colored LCD (Liquid Crystal Display) backlit by white LEDs. Older designs still use a fluorescent tube for backlighting. Referring to Figure 5.8, each pixel has a cluster of three LCD cells to attenuate the backlight from dark to light, and each LCD cell has a colored filter to make red, green, or blue light. The backlight panel provides the overall illumination, the LCD cells control the individual pixel brightness by blocking light, and the colored filter gives each pixel its color. Most use edge lighting with LEDs around the edge of

5.4 显示设备

掌握了上述显示属性词典——如色域、白点和伽马值——我们现在可以逐一审视每种显示设备，并理解其性能特征。一旦理解了这些设备所使用的不同色彩空间，就会明白为什么图像必须经过色彩空间转换，才能在不同类型的显示器上呈现一致的外观。但首先，让我们暂停一下，了解一下最普遍的显示设备——平板显示器——是如何构造的，因为接下来的讨论将贯穿其应用。

5.4.1 平板显示器结构

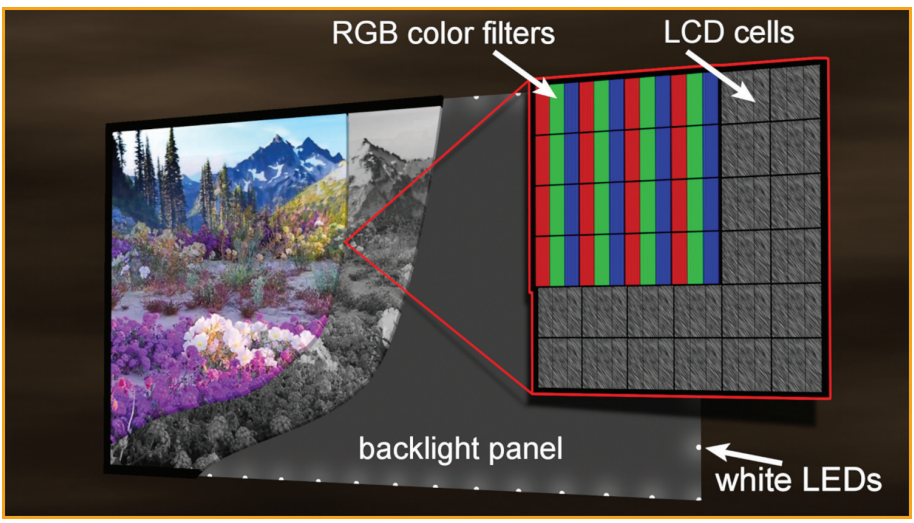


图5.8 平板显示器结构

现代LED/LCD显示器和电视的设计已统一采用白色LED背光的彩色LCD（液晶显示器）。旧式设计仍使用荧光管进行背光照明。如图5.8所示，每个像素包含三个LCD单元组，用于将背光从暗调节到亮；每个LCD单元配有彩色滤光片以生成红、绿或蓝色光。背光面板提供整体照明，LCD单元通过阻挡光线控制单个像素亮度，彩色滤光片则赋予每个像素颜色。多数产品采用边缘照明方式，将LED置于

the screen, with light diffusers to spread it uniformly over the entire screen, which makes for thin displays. A few use direct lighting with LEDs spaced across the entire backing screen, which increases thickness and weight, but also adds the capability for local dimming to help lower black levels.

The main issue with this design is that the LCD cells cannot completely shut off all the light from the backlight panel, so the blacks aren't totally black. Another problem is that the backlight panel might not light the screen uniformly resulting in uneven screen brightness. However, overall, this design is cost-effective and works well, so most display manufacturers use it.

5.4.2 Workstation Monitors

These are the display devices for our workstations in post-production for VFX, digital paint, motion graphics, gamers, video editing, colorists, and photographers. Video editors and colorists will often have a second broadcast monitor so they can view their work in the intended display space of Rec. 709 for television, but CG and VFX artists typically have a regular sRGB workstation monitor to ply their trade. Fortunately, an sRGB monitor is not too different from a Rec. 709 HDTV. The difference is that the HDTV will be brighter and have a heavier 2.4 gamma compared to the sRGB 2.2 gamma. Powerful laptops have become an important addition to workstation towers, and the good news is that their monitors work and look essentially the same as our desktop monitors.

Author's Note: The "Specs" sections below convey brightness and contrast ratios for different types of displays. The problem is that you get different numbers depending on who you ask. I researched these numbers from several sources, put those results in a food blender, and hit "frappe" to blend them into a useful average. Your results may vary.

5.4.2.1 Specs

There is no single spec for workstation monitor resolution, but they typically range from 1920 x 1080 to 2560 x 1440 or larger. They use the sRGB

屏幕采用光扩散器将光线均匀散布到整个屏幕，这使得显示器变得轻薄。少数采用直接照明方式，将LED灯分布在整个背光屏上，这增加了厚度和重量，但也增加了局部调光功能，有助于降低黑色水平。

这种设计的主要问题在于，液晶单元无法完全阻挡背光板的所有光线，因此黑色显示不够纯粹。另一个问题是背光板可能无法均匀照亮屏幕，导致屏幕亮度不均。但总体而言，这种设计性价比高且效果良好，因此大多数显示器制造商都采用这种方案。

5.4.2 工作站监视器

这些是我们工作站用于后期制作VFX、数字绘画、动态图形、游戏玩家、视频编辑、调色师和摄影师的显示设备。视频编辑和调色师通常会配备第二台广播监视器，以便在Rec. 709的电视标准显示空间中查看作品，但

CG和VFX艺术家通常使用标准的sRGB工作站显示器进行创作。幸运的是，sRGB显示器与Rec.709 HDTV的差异并不大。主要区别在于HDTV的亮度更高，且采用2.4伽马值（相较于sRGB的2.2伽马值更重）。高性能笔记本电脑已成为工作站台式机的重要补充，好消息是它们的显示器在功能和观感上与桌面显示器基本一致。

作者注：下面的“规格”部分列出了不同类型显示器的亮度和对比度比率。问题在于，不同来源给出的数据各不相同。我从多个渠道研究了这些数据，将结果放入“食物搅拌机”中，按下“混合”键，将它们融合成一个有用的平均值。您得到的结果可能有所不同。

5.4.2.1 规格

工作站显示器分辨率并无统一标准，但通常范围从1920 x 1080到2560 x 1440或更高。它们采用sRGB

color gamut illustrated in Figure 5.9, which is shared with Rec. 709 for HDTV. The difference is that monitors have a lower transfer function (gamma) and typically a lower brightness and contrast ratio than HDTV. The gamma is 2.2 and the brightness is generally around 250 nits but can go as high as 500. Contrast ratios for monitors are typically around 1000:1. Bit depth is normally 8 bits, but 10 and 12-bit monitors are available. Refresh rates for workstation monitors range from 60 to 120 Hz, but the actual frame rate depends on the video card driving it. Gaming monitors can go to 240 Hz but cost extra. The viewing environment is a bright office surround, and the viewing distance is 20 to 40 inches. Laptop displays typically share the same specs, except for maybe being a bit less bright and noticeably smaller.

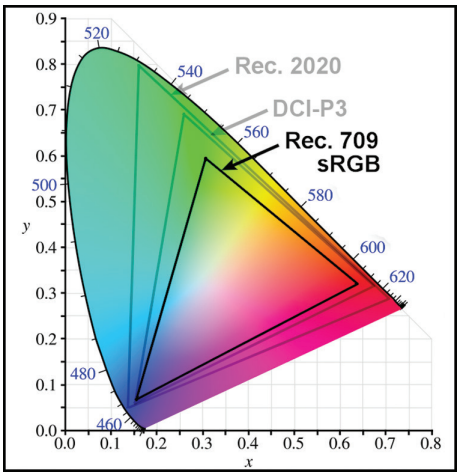


Figure 5.9 Rec. 709 / sRGB color gamut

5.4.3 HDTV

Let's face it—whether you work in broadcast or movies, the vast majority of your audience will end up seeing your work on an HDTV screen. Their construction is the same as the monitors above, with a backlight panel providing light, LCD cells providing brightness control, and the colored filters providing color. While HDTV shares the same color primaries as workstation monitors, what makes it different is that it is much brighter and has deeper blacks, a darker gamma, and therefore a greater contrast ratio.

HDTV sets have special image processing on board to do things like upres low-res signals, increase frame rates with motion compensation, deinterlacing, color space conversion, compression artifact mitigation, sharpening, and enhancing color and contrast. In other words, HDTV—unlike workstation monitors—messes with your picture. Some models also include ambient

图5.9所示的色域与高清电视Rec. 709标准共享。不同之处在于，显示器的传递函数（伽马值）较低，且通常比高清电视的亮度和对比度更低。伽马值为2.2，亮度一般在250尼特左右，最高可达500尼特。显示器的对比度通常约为1000:1。位深度通常为8位，但也有10位和12位的显示器可供选择。工作站显示器的刷新率范围为60至120赫兹，但实际帧率取决于{v*}。

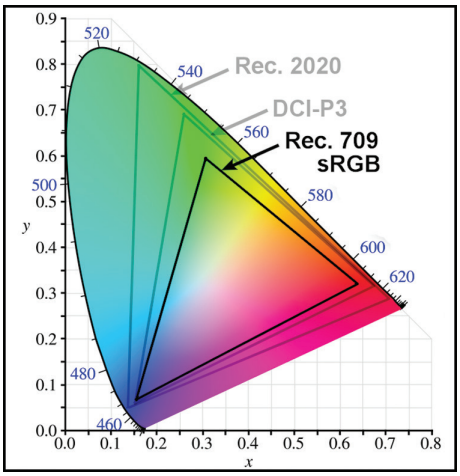


图 5.9 Rec. 709 / sRGB 色域

视频卡驱动它。游戏显示器可以达到240赫兹，但需要额外费用。观看环境是明亮的办公室环境，观看距离为20到40英寸。笔记本电脑显示屏通常具有相同的规格，除了可能亮度稍低且明显更小。

5.4.3 高清电视

让我们面对现实吧——无论你从事广播还是电影行业，绝大多数观众最终都会在HDTV屏幕上观看你的作品。它们的构造与上述显示器相同，背光面板提供光源，LCD单元控制亮度，彩色滤光片则负责呈现色彩。

色彩。尽管HDTV与工作站显示器共享相同的原色，但它的不同之处在于亮度更高、黑色更深、伽马值更暗，因此对比度也更高。

高清电视内置了特殊的图像处理功能，用于提升低分辨率信号、通过运动补偿增加帧率、去隔行扫描、色彩空间转换、减轻压缩伪影、锐化图像，以及增强色彩和对比度。换言之，与工作站显示器不同，高清电视会对画面进行干预处理。部分型号还具备环境光

light sensors to automatically adjust the brightness and contrast to the room lighting so the display's appearance will be more consistent with different surround levels.

5.4.3.1 Specs

HDTV resolution is 1920 x 1080 (aspect ratio 1.78), has square pixels, and its color primaries are shared with sRGB, as illustrated in Figure 5.9. Note that current HDTV specs use a gamma of 2.4 (BT.1886), but before 2011 the early HDTV gamma was 2.2 (Rec.709). HDTV brightness is typically 500 nits but can be up to 1000 if you are willing to pay. The contrast ratio is around 4000:1 and the bit depth is 8 or 10 bits. Frame rates are tightly spec'd out at 23.97, 24, 25, 29.97, 30, 50, 59.94, and 60 fps. The viewing environment is a dim surround, like a typical living room. The recommended viewing distance is around twice the screen size—so for a 50-inch screen, that's 100 inches (a little over 8 feet), and for a 65-inch TV that's 130 inches (nearly 11 feet).

5.4.4 Digital Cinema

The typical high-quality three-chip digital cinema projector today uses a xenon lamp as a white light source and then separates the white light into red, green, and blue with prisms. Each color beam is sent to its own DLP (Digital Light Processing) chip with about 8 million micro-mirrors, providing the per-pixel brightness modulation from dark to light. DLP chips are expensive, so there are lower-cost designs that only require one chip, but these are not as bright because they project the red, green, and blue colors sequentially rather than simultaneously. This can result in color strobing (flashing) artifacts during fast action sequences, which are dubbed the "rainbow effect" and shown in Figure 5.10.

Digital cinema projectors are truly amazing tech. This well-produced video shows the inner workings of a high-end Christie 4k projector.

光线传感器可根据环境光线自动调节屏幕的亮度和对比度，使显示屏在不同光照条件下呈现更一致的视觉效果。

5.4.3.1 规格

高清电视分辨率为1920×1080（宽高比1.78），采用方形像素，其色域基准与sRGB相同，如图5.9所示。需注意当前高清电视标准采用2.4伽马值（BT.1886），但2011年之前早期高清电视伽马值为2.2（Rec.709）。典型亮度为500尼特，若预算充足最高可达1000尼特。对比度约为4000:1，位深度为8位或10位。帧率严格规范为23.97、24、25、29.97、30、50、59.94及60帧/秒。观看环境为昏暗环绕光（如典型客厅场景），建议观看距离约为屏幕尺寸的两倍——以50英寸屏幕为例需保持100英寸（约8英尺）距离，65英寸电视则需130英寸（近11英尺）。

5.4.4 数字影院

如今典型的高质量三芯片数字电影投影机使用氙灯作为白光光源，然后通过棱镜将白光分离为红、绿、蓝三色。每束色光被发送到其专用的DLP（数字光处理）芯片。

拥有约800万个微镜，提供从暗到亮的像素级亮度调制。DLP芯片价格昂贵，因此存在仅需单芯片的低成本设计方案，但由于其采用红、绿、蓝三色顺序投射而非同步投射，亮度表现会有所不足。这可能导致快速动作场景中出现色彩频闪（闪烁）伪影，这种现象被称为“彩虹效应”，如图5.10所示。

数字电影放映机确实是令人惊叹的技术。这段制作精良的视频展示了一台高端科视4K投影机的内部工作原理。

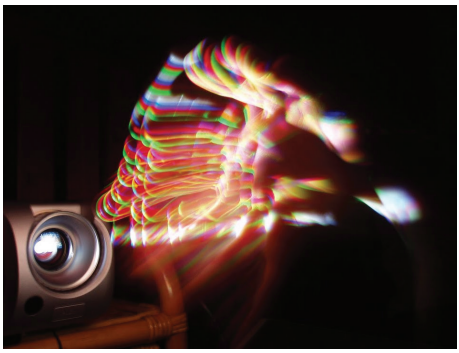


Figure 5.10 The rainbow effect

The weak link in current digital cinema projectors is the xenon lamp. They are power hogs, requiring up to 3000 watts, and like the rest of us, they get dimmer with age and are expensive to replace ($750+). The lifetime of a lamp is in the range of 500-1,500 hours, but an insensitive theater owner might be tempted to leave the lamp in too long to avoid the replacement cost at the expense of a dimmer movie for the audience. Worse, some will even lower the voltage of the lamp to extend its life, which also results in dim projection. While the lower voltage also shifts the color temperature of the lamp toward the reds, it is mitigated by the projector's sensor and feedback circuitry to keep the color temperature constant. As an added excitement, xenon lamps are a high-pressure device, at about 440 psi (almost 30 times atmospheric pressure!), so they represent an explosion risk, are packaged and shipped with an explosion shield, and are very scary to replace.

5.4.4.1 Specs

Digital cinema resolution is 2048 x 1080 (aspect ratio 1.89) and its color primaries are illustrated here as DCI-P3, which is noticeably larger than Rec. 709. The gamma is an honest 2.6 and the screen brightness is set to an anemic 48 nits—but in the dark surround of a movie theater, it looks quite bright to the dark-adapted eye. It is a 12-bit system with a bracing contrast ratio of 2000:1, simply because its blacks are really black. Frame rates are 24, 25, 30, 48, 50, and 60 fps, with none of the 29.97 silliness of HDTV. The viewing environment is a dark

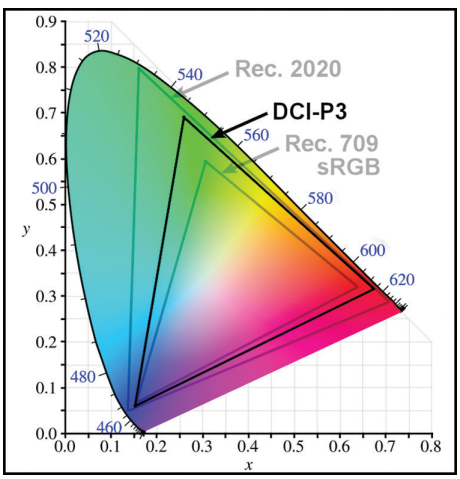
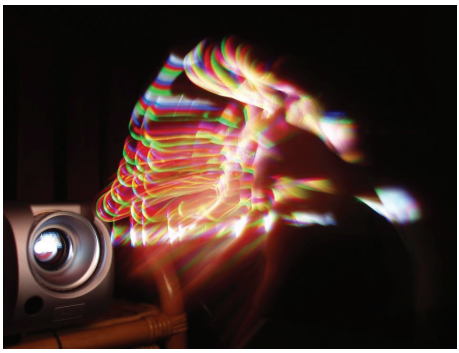


Figure 5.11 Digital cinema color gamut



当前数字影院投影机的薄弱环节是氙气灯。它们耗电量巨大，最高可达3000瓦，而且和我们人类一样，会随着使用年限增长而亮度衰减，更换成本高昂（750+美元）。一盏灯的使用寿命在500至1500小时之间，但有些对画质不敏感的影院经营者可能会为了节省更换成本而过度延长使用时间，导致影片放映亮度不足。

图 5.10 彩虹效应

对于观众来说，更糟糕的是，有些人甚至会降低灯泡的电压以延长其寿命，这同样会导致投影变暗。虽然较低的电压也会使灯泡的色温偏向红色，但投影仪的传感器和反馈电路会对此进行调节，以保持色温恒定。更令人紧张的是，氙气灯是一种高压设备，压力约为440 psi（几乎是大气压的30倍！），因此存在爆炸风险，包装和运输时都配有防爆罩，更换时也令人胆战心惊。

5.4.4.1 规格

数字影院的分辨率为2048 x 1080（宽高比1.89），其色域标准在此标示为DCI-P3，明显大于Rec.709。伽马值采用标准的2.6，屏幕亮度设定为偏暗的48尼特——但在电影院黑暗环境的衬托下，对于暗适应的眼睛而言仍显得相当明亮。该系统采用12位色深，并具备2000:1的强劲对比度，这完全得益于其真正的纯黑表现。帧率支持24、25、30、48、50及60 fps，完全避免了高清电视29.97帧的尴尬设定。观影环境为全暗

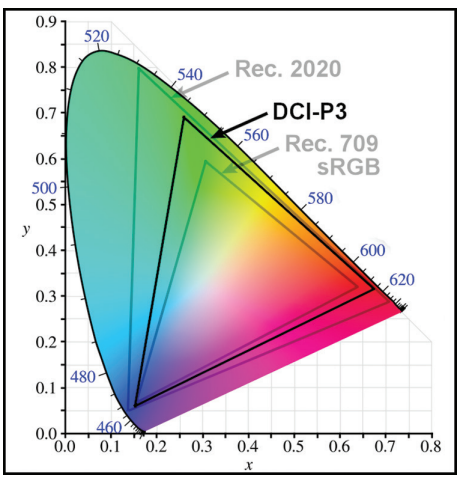


图 5.11 数字影院色域

surround with a D65 white point, and the viewing distance is based on what row you sit in.

We are warned not to watch HDTV in a dark room because of eye strain, but why is that when we routinely watch movies in the dark? It's because of the nits. The problem is that an HDTV screen is actually ten times brighter than a movie screen and would scorch your retinas in a dark surround with your eyes adapted with great big open pupils.

5.5 Coming Soon

As promised in the intro to this chapter, it's time to cast our steely gaze into the future. When we look to the future of UHDTV, we see the wide color gamut (WCG) and HDR of Rec. 2020, which uses the 2.4 gamma transfer function from BT.1886. There are several brands available now calling themselves UHDTV but read their specs carefully. It might be a few years before you can purchase a true UHDTV at a reasonable price. Movie theaters have Digital Cinema 2.0 coming soon, but theater owners have to get over the cost of the new projectors. The tech is ready, but their budgets are not.

5.5.1 UHDTV

We already have HDTV (High Definition TV), but the TV manufacturers needed more juice for the next generation of products, so they came up with UHDTV (Ultra-High Definition TV). It is the next generation of HDTV and features Rec. 2020—a vast new color space released in 2012 with extreme new primaries that push the boundaries of display devices to the very edge of human vision, as illustrated in Figure 5.12. The caveat with UHDTV is that to date, none of the

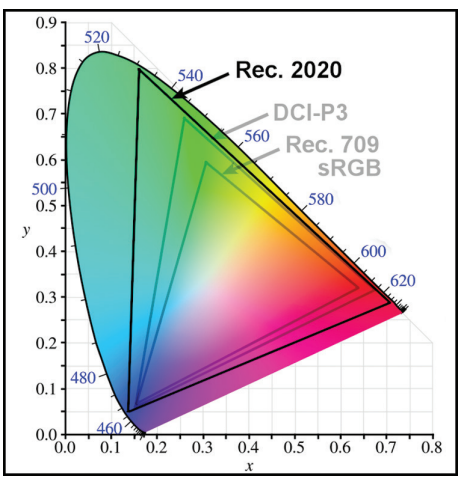


Figure 5.12 UHDTV Rec. 2020 color gamut

以D65白点环绕，观看距离则取决于您所坐的排数。

我们被警告不要在黑暗的房间里观看高清电视，以免眼睛疲劳，但为什么我们经常在黑暗中看电影却没事呢？这是因为尼特（亮度单位）的问题。问题在于，高清电视屏幕实际上比电影屏幕亮十倍，在黑暗环境中，当你的眼睛适应了环境、瞳孔放大时，它会灼伤你的视网膜。

5.5 即将推出

正如本章引言所承诺的，现在是时候将我们锐利的目光投向未来了。展望超高清电视的未来，我们看到采用BT.1886标准中2.4伽马传递函数的Rec.2020广色域（WCG）与高动态范围（HDR）技术。目前市面上已有多个品牌自称超高清电视，但请仔细阅读其规格参数。或许还需数年时间，真正的超高清电视才能以合理价格普及。影院即将迎来数字电影2.0时代，但影院经营者仍需克服新投影设备的成本压力——技术已然就绪，预算却尚未到位。

5.5.1 超高清电视

我们已经有了高清电视（HDTV），但电视制造商需要为下一代产品注入更多活力，于是他们推出了超高清电视（UHDTV）。这是高清电视的下一代产品，并采用了Rec. 2020标准——这是一个于2012年发布的广阔新色彩空间，其极端的原色将显示设备的边界推向了人类视觉的极限，如图5.12所示。关于超高清电视需要注意的是，迄今为止，尚无任何

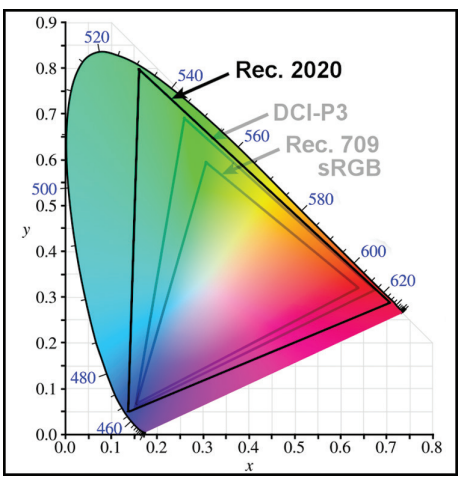


图 5.12 UHDTV Rec. 2020 色域

manufacturers can make a display that meets the specs outlined here. So what's coming soon for UHDTV is progressively greater brightness and ever wider gamuts. In addition to wider color gamuts, brightness and contrast ratios go way up, so unfortunately, we have some new acronyms to learn:

- **HDR (High Dynamic Range)** is supposed to have a contrast ratio of at least 6,000:1. Most of its large contrast ratio numbers come from deeper blacks, but the brightness is also increased from the typical 500 nits of HDTV to a much brighter 600 to 1000 for HDR displays.

- **WCG (Wide Color Gamut)** refers to the Rec. 2020 gamut being wider than Rec. 709 or even DCI-P3 (ghosted), although some manufacturers will refer to their P3-compliant monitors as WCG as well. To date, no UHDTV offers the full Rec. 2020 WCG, so you will see clever dodges like "85% Rec. 2020" instead.

- **OLED (Organic LED)** has incredible near-zero blacks raising their contrast ratios to nearly infinite, with brightness levels of around 540 nits. The micro-LED screens are further off due to current cost constraints, but they hold the promise of the brightest displays, purest color, and lowest power, and are just around the corner.

- **QLED (Quantum dot LED)** technology by Samsung is an upgrade of the current LED/LCD display designs. QLED displays replace the passive RGB filters used to generate color with active colored crystal semiconductors that pass more light from the backlight, giving them a wider color gamut and much greater brightness. QLED displays routinely produce 1000 nits of brightness and high-end models can even manage 4000 nits. Because of the zero blacks of these displays, they tout contrast ratios of infinity—a very unhelpful data point.

- **MicroLEDs** are millions of tiny RGB LEDs mounted on a flat panel. In active development by Samsung, LG, Sony, Apple, and others, they offer incredibly bright and pure colors, zero blacks, energy efficiency, and exceptional lifespans. The current obstacles are excessive manufacturing complexity and production costs. Once these obstacles have been overcome, microLED displays will become a major contender for market dominance due to their intrinsic advantages.

- **UHDTV** is often misleadingly touted as "4K" but is an exact doubling of HDTV in both dimensions, so the resolution is 3840 pixels wide and not the 4096 of real 4K. Is there no limit to the perversity of marketing

制造商可以制造出符合此处所述规格的显示屏。因此，超高清电视即将迎来的是逐步提升的亮度和更宽广的色域。除了更广的色域，亮度和对比度也大幅提升，遗憾的是，我们因此需要学习一些新的缩略词:

- HDR（高动态范围）的对比度至少应为6,000:1。其大部分高对比度数值源于更深的黑色表现，同时亮度也从传统HDTV的典型500尼特提升至HDR显示器更亮的600至1000尼特。

- WCG（广色域）指的是Rec. 2020色域比Rec. 709甚至DCI-P3（虚线框）更宽广，尽管部分厂商也会将其符合P3标准的显示器称为WCG。迄今为止，尚无任何UHDTV能提供完整的Rec. 2020广色域，因此你会看到诸如“85% Rec. 2020”这类巧妙的规避表述。

- OLED（有机发光二极管）拥有近乎零黑的惊人表现，将对比度提升至接近无限，亮度水平约为540尼特。由于当前成本限制，Micro-LED屏幕的普及尚需时日，但它们预示着最明亮的显示效果、最纯净的色彩和最低的功耗，并且已近在眼前。

- 三星的QLED（量子点发光二极管）技术是对现有LED/LCD显示设计的升级。QLED显示屏用主动发色的晶体半导体取代了用于生成色彩的被动RGB滤光片，这种半导体能让背光透过更多光线，从而提供更广的色域和更高的亮度。QLED显示屏通常能产生1000尼特的亮度，高端型号甚至可达4000尼特。由于这些显示屏能实现纯黑效果，其宣称的对比度为无限——这是一个实际参考意义不大的数据点。

- MicroLED 是由数百万个微型 RGB LED 组成的平面显示技术。三星、LG、索尼、苹果等公司正积极研发，其具备极高的亮度与纯净色彩、纯黑表现、节能特性以及超长寿命。当前的主要障碍在于制造工艺过于复杂且生产成本高昂。一旦突破这些瓶颈，凭借其内在优势，MicroLED 显示屏将成为市场竞争中的主导力量。

- UHDTV常被误导性地吹捧为“4K”，但其实际分辨率是HDTV在两个维度上的精确翻倍，因此宽度为3840像素，而非真正4K的4096像素。营销手段的扭曲程度难道没有上限吗?

departments? Early versions of UHDTV have been out for a couple of years using the current LED/LCD tech and some even have OLED displays. There are even several manufacturers trying to produce new microLED emissive screens that emit light directly.

Figure 5.13 Comparison of SDR image on HDTV to HDR image on UHDTV

Although it is tricky to display an HDR image with an LDR picture in a book, Figure 5.13 attempts to convey the visual impact that HDR would bring to the display if you were to set the two monitors side by side. The HDTV on the left with an SDR (Standard Dynamic Range) image would look fine until you glanced to the right at the HDR image on UHDTV. Now you can never go back. Midtones like clothes and cars would look similar, but the blacks would be incredibly deep and the brights would really pop with amazing color purity. Keep in mind that these displays are designed for a dim surround so, as we learned in Section 5.1.11 Viewing Surround, if the lighting is too bright, the blacks will be fogged up and these incredible contrast ratios will drop.

FUN FACT: Rec. 2020 is the production and broadcast standards published by the International Telecommunications Union (ITU), covering resolution, primaries, gamma, bit depth, frame rates, and viewing surround for UHDTV.

5.5.1.1 Specs

UHDTV resolution is 3840 x 2160 (4K) and 7680 x 4320 (8K) and uses the Rec. 2020 color primaries mentioned above. The transfer function is BT.1886, which is the same as for Rec. 709, so the effective gamma is also 2.4. As mentioned above, HDR contrast ratios are 6,000:1 or more, but this can be

部门？超高清电视的早期版本已经推出几年了，它们采用了当前的LED/LCD技术，有些甚至配备了OLED显示屏。甚至有几家制造商正试图生产新型的微米LED自发光屏幕，这种屏幕能直接发光。

图5.13 HDTV上的SDR图像与UHDTV上的HDR图像对比

尽管在书中用LDR图片展示HDR图像颇具挑战，但图5.13试图呈现当两台显示器并排摆放时，HDR将为显示效果带来的视觉冲击。左侧播放SDR（标准动态范围）图像的高清电视原本效果尚可——直到你瞥见右侧UHD电视上的HDR图像。从此便再难回头。衣物、车辆等中间色调看起来或许相近，但黑色会深邃得令人惊叹，亮部则将以惊人的色彩纯度跃然眼前。需注意这些显示器专为昏暗环境设计，正如5.1.11节“观看环境”所述：若环境光线过亮，黑色会泛灰，这些惊人的对比度也将大打折扣。

有趣的事实：Rec. 2020是国际电信联盟（ITU）发布的制作和广播标准，涵盖超高清电视的分辨率、原色、伽马、位深度、帧率以及观看环境。

5.5.1.1 规格

UHDTV的分辨率为3840 x 2160（4K）和7680 x 4320（8K），并采用上文提到的Rec. 2020原色标准。其传递函数为BT.1886，与Rec. 709相同，因此有效伽马值也为2.4。如前所述，HDR对比度为6,000:1或更高，但这可以

achieved with either a brightness of 1000 or more with normal black levels or a lower brightness with zero black levels. Bit depths are 10 or 12 bits (compared to Rec. 709's 8 or 10 bits), but frame rates, viewing surround, and viewing distances are the same as HDTV. The problem is, nobody can make one yet that fulfills 100% of the specs.

5.5.2 Digital Cinema 2.0

Digital cinema is constantly being driven by the relentless advances in home television. Today, the current Rec. 709 HDTV specs are noticeably less than the current digital cinema DCI-P3 specs. That is all about to change. As today's TVs have wider gamuts, brighter colors, and higher contrast ratios than ever, it is getting harder and harder for theaters to give you a better viewing experience than your home TV. With UHDTV resolution paired with Rec. 2020 color space, theaters will fall behind the home TV experience, so they will have to up their game as well.

Exotic new display technologies are being developed for theaters that will dramatically increase screen brightness, color purity, and contrast ratios. Recalling that zero blacks are the key to high contrast ratios, cinema has the advantage that when they turn off the lights, the blacks are really black. To take advantage of this, they are developing two projection technologies and one emissive technology. The three competing technologies for next-gen blazing light displays are—RGB lasers, laser phosphor, and LED walls.

5.5.2.1 RGB Lasers

RGB lasers are the logical choice because the light can be made very bright, with very pure colors. The three lasers each go to a DLP chip, like the current tech, so they are essentially replacing the xenon lamp with lasers. Lasers are not only brighter but have lifetimes of 10,000 to 20,000 hours compared to a xenon lamp's 500-1,500 hours. The big hurdle, of course, is price. There are already several laser digital cinema projectors on the market and it's only a matter of time before they penetrate the installed base.

5.5.2.2 Laser Phosphor

The laser phosphor system addresses the RGB laser cost problem by being less expensive, at the expense of performance. Its brightness is only

可通过两种方式实现：要么在正常黑位下亮度达到1000或更高，要么在零黑位下亮度较低。位深度为10或12位（而Rec. 709标准为8或10位），但帧率、观看环境及观看距离与高清电视相同。问题在于，目前尚无人能制造出完全符合100%规格的产品。

5.5.2 数字影院 2.0

数字影院正不断受到家用电视技术迅猛发展的推动。如今，现行的Rec. 709高清电视标准明显低于当前数字影院的DCI-P3标准。但这一切即将改变。随着当今电视拥有比以往更广的色域、更鲜艳的色彩和更高的对比度，影院越来越难以提供比家用电视更佳的观影体验。当超高清电视分辨率与Rec. 2020色彩空间相结合时，影院体验将落后于家庭电视，因此它们也必须提升自身水准。

新型的显示技术正在为影院开发，这些技术将显著提高屏幕亮度、色彩纯度和对比度。回想一下，纯黑是实现高对比度的关键，电影院在关灯时能够实现真正的黑色，这为其带来了优势。为了利用这一点，他们正在开发两种投影技术和一种自发光技术。下一代高亮度显示的三种竞争技术分别是——RGB激光、激光荧光和LED墙。

5.5.2.1 RGB激光器

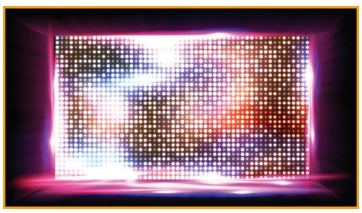
RGB激光是合乎逻辑的选择，因为其光线可以做到非常明亮，色彩极为纯净。三束激光分别投射到DLP芯片上，与现有技术类似，实质上是用激光取代了氙气灯。激光不仅更亮，而且寿命长达10,000至20,000小时，而氙气灯仅为500-1,500小时。当然，最大的障碍在于价格。市场上已有多款激光数字电影放映机，它们渗透现有设备基础只是时间问题。

5.5.2.2 激光荧光粉

激光荧光系统通过降低性能来解决RGB激光成本问题，其价格更为低廉，但亮度仅为

comparable to a lamp projector, but the big win is 30,000 hours of service life and half the electricity bill. It uses a single blue laser (big savings there!) to illuminate a phosphor surface, which in turn emits yellow light. The yellow light provides the red and green, so just add the blue from the laser to get RGB.

5.5.2.3 LED Screens



LED movie screens made up of an array of LED panels are in development and a few have been installed in the field, with stunning results. They are ten times brighter than lamp projectors, have fabulous color purity, and have great uniformity across the face of the screen. The problem is cost. One projector covers an entire screen, and for a bigger screen, you just purchase the next larger projector. However, with LED panels, if you double the screen size, you quadruple the number of panels to purchase. The industry will need a price breakthrough for this one to succeed, but it is ultimately the answer because it is essentially direct electricity to light—no lasers, no DLPs, and no intervening mechanisms.

5.5.2.4 High Frame Rate Movies

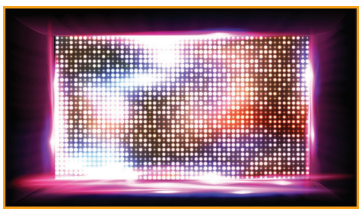
High frame rate (HFR) movies are very feasible today because of the speed and brightness of these projectors, so if HFR catches on, the new digital cinemas will be ready for them. In 2012, Peter Jackson began filming *The Hobbit: An Unexpected Journey* at 48 fps. I screened it once and it was indeed stunning. James Cameron is a great fan of stereo features, but the stereo effect breaks down during fast action at 24 fps, so Cameron has been pushing for higher frame rates for years. He would like to see 48 and even 60-fps presentations. Modern digital cinema cameras and digital cinema projectors can handle these frame rates easily, so we may see more HFR projects in the future. However, a 100-minute movie will have 144,000 frames at 24 fps, but at 60 fps it will require 360,000 frames, so there is a price to be paid.

However, some do not agree with James Cameron, Peter Jackson, and Ang Lee about the virtues of HFR movies. Their objections are based on psychology—our engrained perceptions of movies and television over the last

可比作一台灯式投影仪，但最大的优势在于3万小时的使用寿命和减半的电费。它采用单一蓝色激光（此处节省巨大！）照射荧光表面，进而发出黄光。黄光提供红色和绿色，只需加上激光的蓝色即可获得RGB。

5.5.2.3 LED显示屏

由LED面板阵列组成的LED电影屏幕正在开发中，少数已在实地安装，效果令人惊叹。它们比灯泡投影仪亮十倍，色彩纯度极佳，屏幕表面均匀性极好。问题在于成本。



投影仪覆盖整个屏幕，要获得更大的屏幕，只需购买下一款更大的投影仪。然而，对于LED面板，如果将屏幕尺寸翻倍，需要购买的面板数量将增至四倍。这一技术要想成功，行业需要在价格上取得突破，但它终究是最终的解决方案，因为它本质上是将电能直接转化为光——无需激光、无需DLP，也没有任何中间机械装置。

5.5.2.4 高帧率电影

高帧率（HFR）电影如今已非常可行，这得益于投影仪的速度和亮度。因此，如果HFR流行起来，新建的数字影院将能轻松应对。2012年，彼得·杰克逊开始以48帧/秒拍摄《霍比特人：意外之旅》。我曾观看过一次，效果确实令人震撼。詹姆斯·卡梅隆是立体特效的忠实拥护者，但在24帧/秒下，快速动作场景的立体效果会失效，因此卡梅隆多年来一直致力于推动更高帧率。他希望看到48甚至60帧/秒的呈现方式。现代数字电影摄影机和数字电影投影仪可以轻松处理这些帧率，未来我们可能会看到更多HFR项目。然而，一部100分钟的电影在24帧/秒下包含144,000帧，在60帧/秒下则需要360,000帧，这意味着更高的制作成本。

然而，并非所有人都认同詹姆斯·卡梅隆、彼得·杰克逊和李安关于高帧率电影优点的看法。他们的反对意见基于心理学——即我们过去数十年间对电影和电视根深蒂固的视觉认知。

nearly 100 years. The 24-fps experience of movies, with its motion blur and softer appearance, is associated with entertainment and high production values. The 30-fps experience of videos is associated with the crisp smoothness of video cameras, camcorders, sitcoms, and news shows. These are learned responses but are nevertheless powerful deterrents for many viewers because they feel that HFR films look like videos. It must be galling to a movie producer to make an expensive 4K HFR movie only to have the audience think it looks like video. Ouch!

5.5.2.5 Specs

Next-gen digital cinema resolution is an honest 4K at 4096 x 2160 (aspect ratio 1.89) and will use the Rec. 2020 color gamut discussed above. The effective gamma is 2.4 and the brightness will vary depending on the projector type, ranging from a low of 48 nits for lamp and laser phosphor projectors to the RGB laser projectors that can do 100 nits or more, with honest contrast ratios of up to 1 million to one. It is a 12-bit system and supports frame rates of 24, 25, 30, 48, 50, and 60 fps. The viewing environment is the dark surround, of course, and the viewing distance depends on your seat.

5.6 Conclusion

Understanding the various display spaces and technologies is obviously essential to understanding how our work will ultimately appear to the viewer. But the real question is: how will this affect you? Life is a bit easier for digital artists working in television because all that is needed is a broadcast-quality monitor to see their work exactly like it will appear in the field. With the current Rec. 709 specs, even a plain sRGB monitor is a fairly close match. But when Rec. 2020 is deployed, the old sRGB monitors will no longer cut it because the difference will simply be too great.

Feature film artists will also have to upgrade to HDR displays to have any hope of seeing things close to how they will appear in the theater. One problem will be the surround, as the work environment will be at least a dim surround, while the theater has a dark surround that dramatically boosts contrast ratios and apparent brightness and color vibrancy. We might compensate for working in the wrong surround by using an aggressive viewing

近100年来，电影以24帧每秒的体验——带着动态模糊和更柔和的画面质感——始终与娱乐性和高制作水准紧密相连。而30帧每秒的视频体验，则关联着摄像机、摄像机的清晰流畅感，以及情景喜剧和新闻节目的观感。这些虽是后天习得的反应，却对许多观众构成了强大的心理阻碍，因为他们觉得高帧率电影看起来像视频。电影制片人斥巨资制作4K高帧率电影，结果观众却认为它像视频，这想必令人懊恼不已。唉！

5.5.2.5 规格

下一代数字影院分辨率是真正的4K，即4096 x 2160（宽高比1.89），并将使用上文讨论的Rec. 2020色域。有效伽马值为2.4，亮度将根据投影机类型而变化，范围从灯源和激光荧光粉投影机的最低48尼特，到可实现100尼特或更高的RGB激光投影机，真实对比度高达100万:1。这是一个12位系统，支持24、25、30、48、50和60 fps的帧率。观看环境当然是暗室，观看距离则取决于您的座位位置。

5.6 结论

理解各种显示空间和技术对于了解我们的作品最终如何呈现给观众至关重要。但真正的问题是：这将如何影响你？对于从事电视行业的数字艺术家来说，生活稍微轻松一些，因为他们只需要一台广播级质量的显示器，就能看到作品在实地播放时的真实效果。按照现行的Rec. 709标准，即便是普通的sRGB显示器也能提供相当接近的匹配效果。然而，一旦Rec. 2020标准得到广泛应用，旧的sRGB显示器将无法满足需求，因为两者之间的差异实在太大。

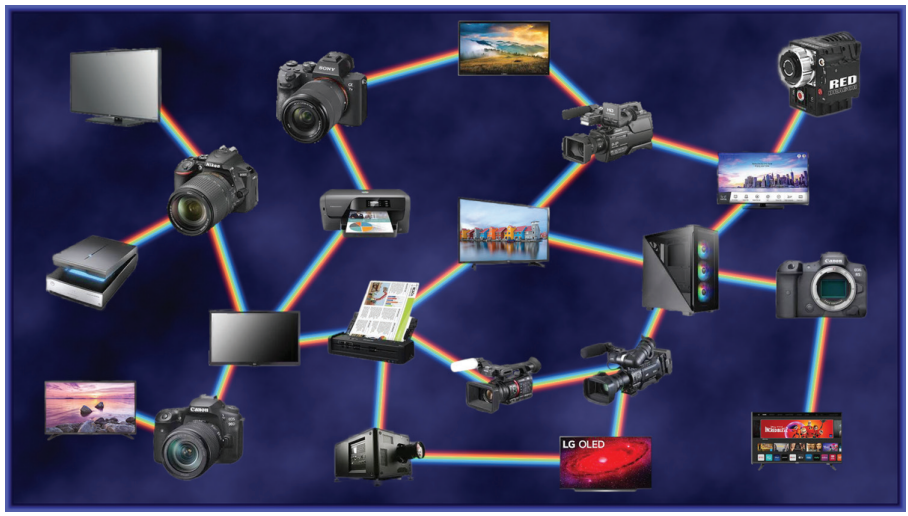
电影艺术家们也必须升级到HDR显示器，才能有望看到接近影院放映效果的画面。其中一个问题在于环境光——工作环境至少是微光环境，而影院则采用暗场环境，能显著提升对比度、视觉亮度与色彩鲜活度。我们或许可以通过采用激进的观看模式来弥补工作环境光不匹配的问题。

LUT or by purchasing ultra-bright displays to better approximate the appearance in the digital cinema. The colorist working in DI (Digital Intermediate) already has a dark surround with a high-quality projector to color time the movie.

What the future portends is a world of high-performance displays in the home, office, and theater. Whether you are a cinematographer, graphic artist, colorist, visual effects artist, or photographer, this puts your work under much greater "display stress" where even the smallest defect becomes visible, even exaggerated. There is also the transition problem where (say) 50% of your target audience is viewing with early version displays, while the other 50% is using the latest versions. Now what do you do? Play to the lowest common denominator and look bad on the high-end displays or go for the max and lose the low end of the audience? What you will need is a large dose of color management to cope with all of this—which, fortunately, is what the last chapter is all about.

LUT或通过购买超高亮度显示器来更好地模拟数字影院中的视觉效果。在DI（数字中间片）环节工作的调色师已经拥有一个配备高质量投影仪的暗室环境，用于对电影进行色彩定时。

未来预示着一个家庭、办公室和影院中高性能显示器的世界。无论您是电影摄影师、平面艺术家、调色师、视觉特效艺术家还是摄影师，这都将使您的作品承受更大的“显示压力”，即使是最微小的缺陷也会变得可见，甚至被放大。此外还存在过渡问题，例如，50%的目标观众使用早期版本的显示器观看，而另外50%则使用最新版本。那么您该怎么办？是迎合最低标准，让作品在高端显示器上表现不佳，还是追求极致，却失去低端观众？您将需要大量的色彩管理来应对这一切——幸运的是，这正是最后一章要讨论的内容。



6 Color Management

Now that we have trekked through the digital imaging universe together from scene to screen, it's time to pull it all together where the rubber meets the road—color management. The prime directives of color management are as follows:

- Preserve the original artistic intent
- Provide consistent appearance across all display platforms.

Preserving the artistic intent entails maintaining the look of the original captured image through the entire digital imaging pipeline to the final display as it moves through multiple operations at multiple locations using multiple tools. For this, we have OCIO.

Providing a consistent appearance across all display platforms entails creating several versions that have the same appearance even though they are on different platforms with different capabilities. For this, we have ACES.

The rapid proliferation of display technologies is making color management solutions progressively more daunting. In Chapter 5, we saw how the tech

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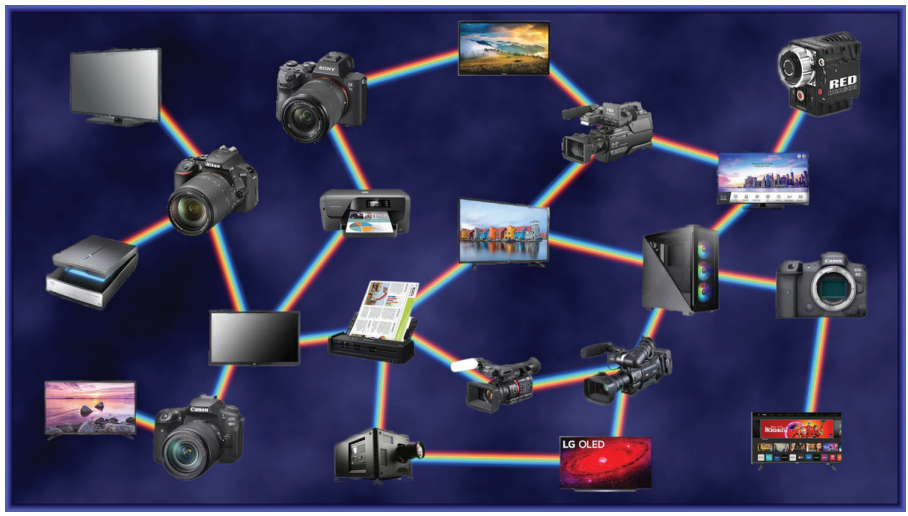

6 色彩管理

既然我们已经一同从场景到屏幕穿越了数字成像的宇宙，现在是将所有内容整合起来、让理论付诸实践的时候了——色彩管理。色彩管理的主要指导原则如下：

- 保留原始艺术意图
- 在所有显示平台上提供一致的外观。

保持艺术意图意味着在数字成像流程中，从原始捕捉图像到最终显示，当它通过多个工具在多个位置进行多次操作时，始终保持原始图像的外观。为此，我们有了OCIO。

在所有显示平台上提供一致的外观意味着需要创建多个版本，这些版本即使处于能力各异的不同平台上，也能保持相同的外观。为此，我们推出了ACES。

显示技术的快速普及使得色彩管理解决方案变得越来越令人望而生畏。在第5章中，我们看到了这项技术

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has just begun the early stages of "self-aware" images with the Perceptual Quantizer, which can adjust their appearance based on the capabilities of whatever display they happen to be on in order to maintain a consistent appearance without human intervention. It will be very interesting to see where this goes in the future.

We begin the color management story by looking at color grading operations in detail, then follow them right into the Digital Intermediate (DI) suite for color timing, where the film will get its final look. To manage color space conversions, we have ICC profiles, which have some embarrassing secrets to reveal, which will lead us to supersized gamuts and how to cope with them. With our tools and techniques established, we will look at color pipelines, how to set them up, and what can be expected from them.

With the post-production essentials behind us, we then explore OCIO—Sony's gift to visual effects and animation that maintains tight color management discipline between software applications, different departments, and even far-flung studios. And then there is ACES—the mother of all color management systems—which is almost too awesome to believe. But it is my job to make it believable, so we shall see how well I do.

6.1 Color Grading

Color correction and color grading are often used interchangeably, but they shouldn't be. "Color correction" is really correcting for misfires in the camera capture—overexposed, black levels, white point, etc. It basically aims to produce a properly exposed color-balanced plate to work with. The term "color grading" should be reserved for applying an aesthetic "look" to the work. "Color timing" refers to color-correcting 35 mm film in a lab by adjusting the timing (duration) of RGB exposure lights. Virtually obsolete today, the term is still hanging around being misapplied, so I just wanted to clear that up for the record.

Regardless of your actual role in working with digital images, editing their color is certain to be a key task. Here we will look at color grading using 1D and 3D LUTs, and then take a rare inside look at the operations lift, gamma, gain, and offset from the standpoint of their operational effects on the pixel

感知量化器刚刚开启了“自我感知”图像的早期阶段，它能够根据图像所在显示器的性能自动调整外观，以保持一致性，无需人工干预。未来这一技术的发展方向将非常值得关注。

我们从详细探讨调色操作开始色彩管理的故事，随后直接进入数字中间片（DI）流程进行配光，电影将在此获得最终视觉效果。为了管理色彩空间转换，我们使用ICC配置文件，其中揭示了一些令人尴尬的秘密，这将引导我们探索超大色域及其应对方法。在确立工具与技术后，我们将研究色彩流水线：如何搭建它们，以及能从中获得怎样的效果。

后期制作的基本要素介绍完毕后，我们接着探讨OCIO——这是索尼赠予视觉特效和动画行业的礼物，它能在不同软件应用、不同部门乃至相距遥远的工作室之间，保持严格的色彩管理规范。然后是ACES——所有色彩管理系统之母——其强大程度几乎令人难以置信。但我的职责就是让它变得可信，所以让我们看看我能做得多好。

6.1 色彩分级

色彩校正与色彩分级常被混用，但二者本不应等同。“色彩校正”实质是修正摄影机拍摄中的失误——曝光过度、黑电平、白点等，其核心目标是生成曝光准确、色彩平衡的底片以供后续创作。而“色彩分级”应专指为作品赋予美学“风格”的过程。“配光”一词原指实验室中通过调节RGB曝光光源的时长来校正35毫米胶片色彩的技术。尽管该术语如今已基本过时，却仍被误用流传，特此说明以正视听。

无论你在处理数字图像时的实际角色是什么，编辑色彩都必定是关键任务。这里我们将探讨使用一维和三维查找表进行调色，然后罕见地从操作效果的角度，深入剖析提升、伽马、增益和偏移对像素的具体作用。

data, along with the change in the appearance of the image. We do not address the subject of "looks" or other artistic whimsy because that is not what this book is about. This book addresses the tough stuff—the tech. Also, there is already an overabundance of books on how to make your color grading gorgeous, so you don't need yet another one from me.

6.1.1 1D LUTs

"LUT" is an abbreviation for Look Up Table. It remaps input RGB values by looking them up in a table to determine what the output value should be. LUTs are fast, flexible, and precise. Figure 6.1 illustrates the input/output relationship defined by the LUT curve. The input numbers are on the bottom row, so in this case an input of 0.5 generates an output of 0.6. LUTs are flexible because the curve can be drawn in many different ways to create an infinite variety of color corrections. Figure 6.1 is what is called a 1D LUT (one-dimensional)—which is strange, since it clearly has a 2D curve on it. Oh well. A typical 1D LUT will have separate curves for R, G, B, and A, and is useful for color grading. It is also used to create log images. Note that converting an image from linear to log or log to linear is NOT a color space conversion. It's really just a data transformation to put it into a more useful form. Primaries and white points are unaffected.

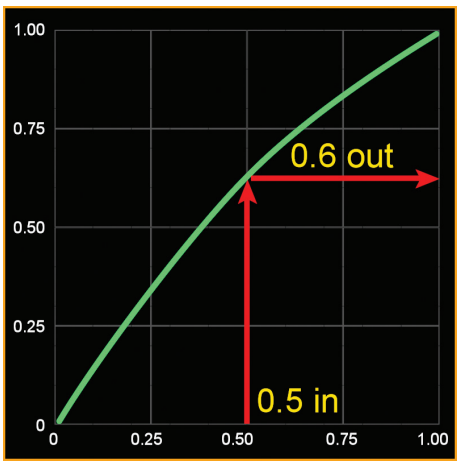


Figure 6.1 The humble 1D LUT

6.1.2 3D LUTs

The key virtue of the 3D LUT is that it can produce output colors that are a mix of two or more channels, whereas the 1D LUT only operates within each color channel. 3D LUTs are used for color space conversions and for cross-channel color grading operations like desaturation since that operation must act on all three channels of a pixel. However, the main application for 3D LUTs is color space conversions.

数据，以及图像外观的变化。我们并不探讨“观感”或其他艺术性的奇思妙想，因为这不是本书的主题。本书关注的是硬核内容——技术。此外，市面上已有大量关于如何让调色效果惊艳的书籍，因此你并不需要我再多添一本。

6.1.1 一维查找表

“LUT”是查找表的缩写。它通过查表将输入的RGB值重新映射，以确定输出值应为多少。LUT具有快速、灵活且精确的特点。图6.1展示了由LUT曲线定义的输入/输出关系。输入数值位于底部一行，因此在此示例中，输入0.5会生成输出0.6。LUT的灵活性在于其曲线可通过多种不同方式绘制，从而创造出无限多样的色彩校正效果。图6.1展示的是

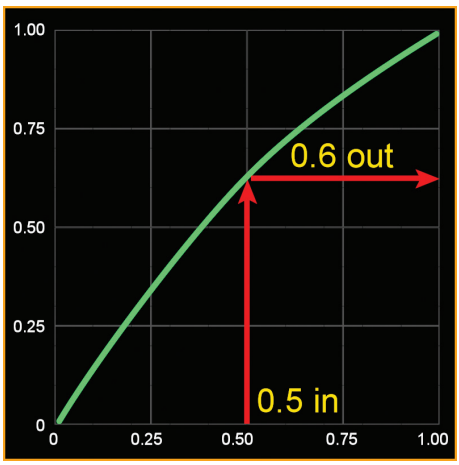


图6.1 朴素的1D LUT

称为一维LUT（一维查找表）——这有点奇怪，因为它上面明明有一条二维曲线。不过算了。典型的一维LUT会为R、G、B和A分别设置独立的曲线，常用于调色。它也被用来创建对数图像。请注意，将图像从线性转换为对数或从对数转换为线性，并非色彩空间转换。这实际上只是一种数据变换，目的是让数据处于更有用的形式。原色和白点不受影响。

6.1.2 三维查找表

3D LUT的关键优势在于它能生成混合两个或多个通道的输出色彩，而1D LUT仅能在每个色彩通道内独立运作。3D LUT被用于色彩空间转换以及跨通道的色彩分级操作（例如去饱和度），因为这类操作必须同时作用于像素的所有三个通道。然而，3D LUT最主要的应用场景仍是色彩空间转换。

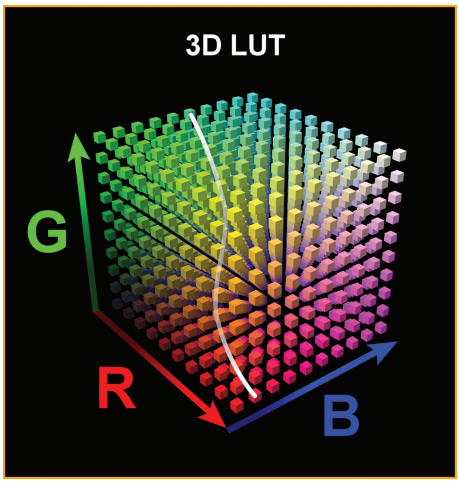


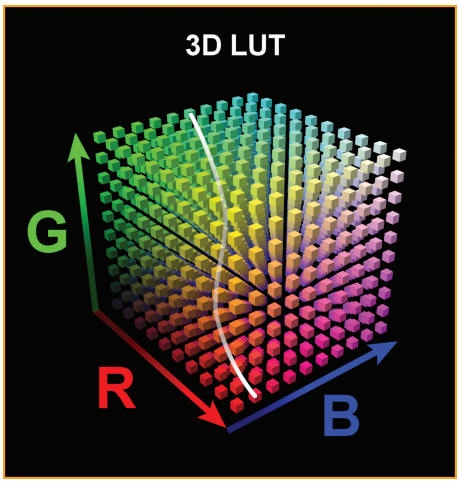
Figure 6.2 The awesome 3D LUT

A 3D LUT is represented as a cubic space with pre-computed data points distributed throughout, as in Figure 6.2. Recalling the RGB cube from Chapter 4: 4.2 Color Models, where RGB values are located inside a 3D cube, for the 3D LUT an input RGB triplet locates a pre-computed data point containing the output RGB values. When an input RGB triplet lands between the pre-computed data points, it is interpolated. Because it is interpolated, this process is not "invertible"—meaning it is not reversible without round-off errors. Converting the output image back to the original source will NOT produce an exact mathematical restoration. However, if the color space conversion is done with a math expression, the process is invertible.

> FUN FACT: 3D LUTs are the "Swiss Army knife" of color. They can be used for color grading, camera matching, and display calibration.

6.1.3 Grading Operations

While most artists have a good idea of what the color grading operations do to the appearance of an image, many are unclear on the math behind them and what they actually do to the image data. Using Figure 6.3, we will take an intimate look at lift, gamma, gain, and offset to understand what happens to the code values. Each grading operation will be illustrated with a LUT as a learning aid so you can visualize the operation, but in the real world, these ops are done with simple math expressions. LUTs are easier on your head than math expressions, plus it is much easier to visualize their operation.



三维查找表（3D LUT）表现为一个立方空间，其中分布着预先计算的数据点，如图6.2所示。回顾第4章“4.2 色彩模型”中的RGB立方体：RGB值位于三维立方体内，而对于3D LUT，输入的RGB三元组会定位到一个包含输出RGB值的预先计算数据点。当输入的RGB三元组落在预先计算的数据点之间时，会进行插值处理。由于经过插值，这一过程是“不可逆”的——意味着若无舍入误差则无法还原。转换

图6.2 令人惊叹的3D LUT

将输出图像还原回原始源图像不会产生精确的数学还原。然而，如果色彩空间转换是通过数学表达式完成的，该过程是可逆的。

> 有趣的事实：3D LUT是色彩的“瑞士军刀”。它们可用于色彩分级、相机匹配和显示器校准。

6.1.3 分级操作

尽管大多数艺术家对调色操作如何影响图像外观有很好的理解，但许多人对其背后的数学原理以及它们实际对图像数据产生的作用并不清楚。借助图6.3，我们将深入探讨提升、伽马、增益和偏移，以理解它们对编码值的影响。每种调色操作都将以查找表（LUT）作为学习工具进行演示，以便您直观理解操作过程，但在实际应用中，这些操作是通过简单的数学表达式完成的。查找表比数学表达式更易于理解，且能更直观地展现其运作方式。

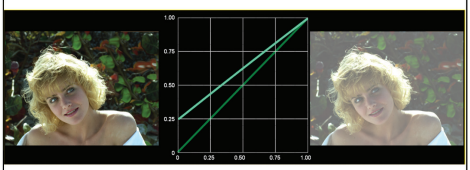
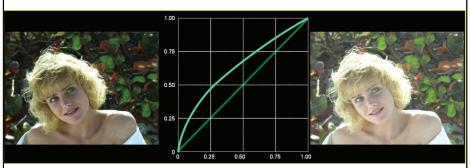
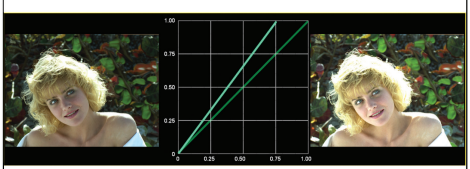
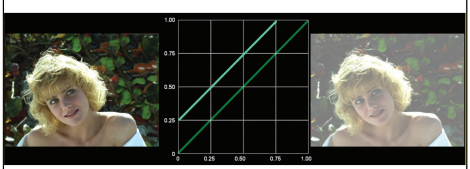
	**Lift** - scales the RGB values pivoting around 1.0. The LUT shows the blacks being raised (or lowered) in order to mostly affect the black level. The effect is diminished toward the highlights and code value 1 is unchanged.
	**Gamma** - the math is covered in Chapter 4, section 4.4.1. The LUT shows the shadows and midtones being raised (or lowered) using a gamma curve. Code values 0 and 1 are unchanged.
	**Gain** - scales the RGB values pivoting around 0. The LUT shows the highlights being raised (or lowered) in order to affect overall picture brightness or exposure. The effect is diminished towards the shadows and code value 0 is unchanged.
	**Offset** - adds a constant value to all code values. The LUT shows all code values being raised (or lowered) equally in order to affect the overall picture level from shadows to highlights.

Figure 6.3 Color grading operations internally with effect on image

Saturation: Sits apart from the others because the results of one color channel depend on the values in the other two channels. With a gain on the red channel, for example, the output red values depend only on the red gain value. With a saturation operation, the red channel's output depends on the color values of the green and blue channels too. Saturation operations may be performed with a math expression that affects the entire image or a 3D LUT which can be adjusted to only affect specific regions of the image.

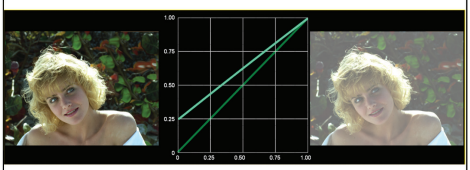
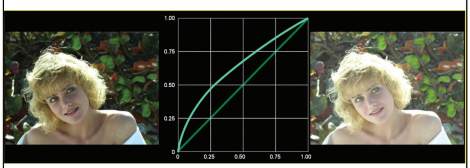
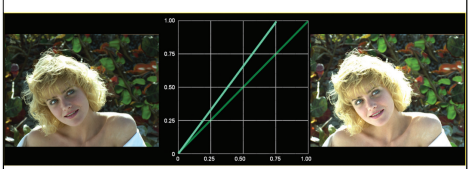
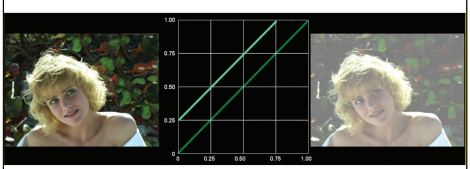
	**Lift** - scales the RGB values pivoting around 1.0. The LUT shows the blacks being raised (or lowered) in order to mostly affect the black level. The effect is diminished toward the highlights and code value 1 is unchanged.
	**Gamma** - the math is covered in Chapter 4, section 4.4.1. The LUT shows the shadows and midtones being raised (or lowered) using a gamma curve. Code values 0 and 1 are unchanged.
	**Gain** - scales the RGB values pivoting around 0. The LUT shows the highlights being raised (or lowered) in order to affect overall picture brightness or exposure. The effect is diminished towards the shadows and code value 0 is unchanged.
	**Offset** - adds a constant value to all code values. The LUT shows all code values being raised (or lowered) equally in order to affect the overall picture level from shadows to highlights.

图6.3 内部调色操作对图像的影响

饱和度：之所以与其他参数不同，是因为一个颜色通道的结果取决于另外两个通道的数值。例如，红色通道的增益调整仅影响红色输出值；而饱和度操作中，红色通道的输出还会受到绿色和蓝色通道数值的影响。饱和度调整可通过影响整幅图像的数学表达式实现，也可通过三维查找表进行精细控制，使其仅作用于图像的特定区域。

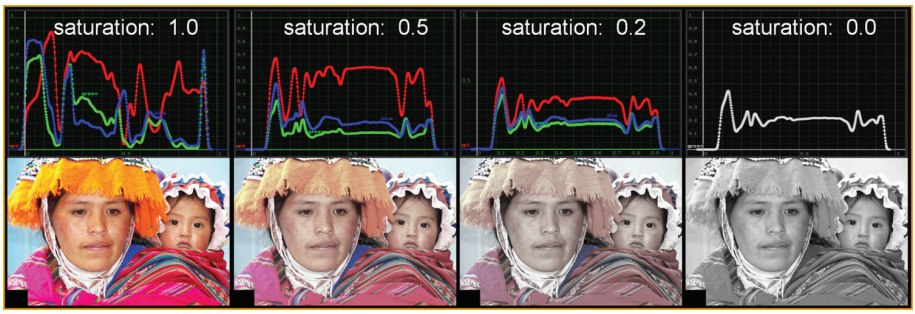


Figure 6.4 Graph of the effect of desaturation on RGB code values

Figure 6.4 illustrates a horizontal row of RGB values sampled from roughly halfway up the picture and then graphed above. The graphs illustrate how the RGB values move closer together as the saturation goes down. Finally, in saturation 0.0, they become identical and land right on top of each other, producing a pure luminance image. Or do they? Beware—some software will produce a proper luminance image when desaturated 100%, but other software will not. What does your software do?

Watch lift, gamma, gain, offset, and saturation in action as the image is graded before your very eyes, including clipping alerts.

6.1.4 Range Controls

While most grading operations affect one range of brightness more than others, it is clear from the examples above that they all affect the entire range of code values to one degree or another. So when you do a lift to correct the black levels it will also slightly affect the midtones and so on. This means that the color grading process becomes iterative because one adjustment affects the others. This results in a lot of time-consuming back and forth in the color grading process.

To cope with this, sophisticated color correctors will have three sets of lift, gamma, and gain controls, each isolated to affect just the shadows,

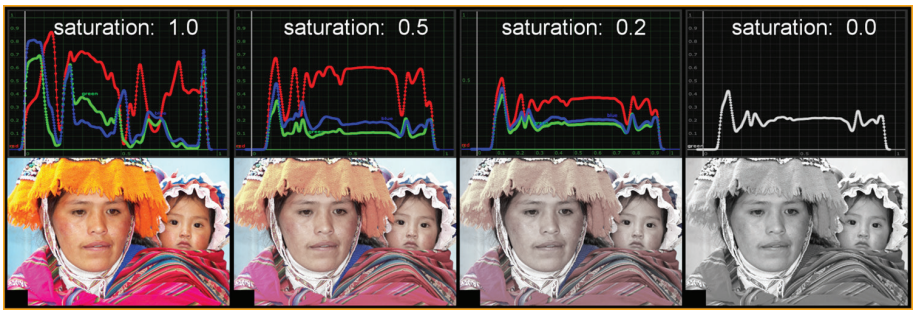


图6.4 去饱和度对RGB码值影响的曲线图

图6.4展示了从图片大约中间高度处采样的一行水平RGB值，并将其绘制在上方。图表说明了随着饱和度降低，RGB值如何逐渐靠近。最终，在饱和度0.0时，它们变得完全相同并完全重叠，生成一个纯亮度图像。但果真如此吗？请注意——有些软件在完全去饱和度（100%）时会生成正确的亮度图像，而其他软件则不会。您的软件表现如何？

观看图像在您眼前进行分级时，{v*} 提升、伽马、增益、偏移和饱和度{v*} 的实际效果，包括剪切警报。

6.1.4 范围控制

虽然大多数调色操作对某一亮度范围的影响大于其他范围，但从以上示例可以看出，所有操作都会在不同程度上影响整个代码值范围。因此，当你通过提升操作修正黑位时，它也会轻微影响中间调等区域。这意味着调色过程会变得迭代化，因为一项调整会牵动其他部分。这导致调色过程中需要耗费大量时间进行反复调整。

为此，高级色彩校正器将配备三组提升、伽马和增益控制，每组独立作用于阴影、

midtones, or highlights. Figure 6.5 shows a three-ball color grading panel where each trackball is assigned a range. So you could, for example, apply a lift to the shadows, a gamma to the midtones, and a gain to the highlights. These grading operations can also be applied to individual color channels to adjust hue—for example, gaining down the blue channel to make the picture warmer.

Figure 6.5 Three-ball color grading panel

6.1.5 Secondary Color Grading

Primary color grading, or what some filmic folk might refer to as a "one light," is intended to "even up" the exposure and color of a film project or TV show to eliminate jarring cut-to-cut differences. It is a sort of baseline starting point for the image's journey to final grading and ultimate display in the home or theater. The key points here are that the primary grade is global, so affects the entire image overall, and it operates on the three primaries—red, green, and blue.

Secondary color grading is more surgical and isolates selected pixels for focused treatment. These pixels can be isolated based on either their value (hue, saturation, or lightness) or their physical location on the screen with masks and keys—fondly referred to as "power windows." Power windows need to follow their targets around the screen, so sophisticated color correctors come with trackers that will keep the power window on target for the colorist. Following is a grand tour of the secondary color grading methods used by the pros:

- **Zones** - With "zone control," the image is segregated by luminance to adjust each zone individually. This is where the darks, midtones, and highlights get treated separately, which is why a three-ball control panel has three balls—one for each zone. This kind of natty control is perfect for color grading a composite where the foreground image needs to match the background plate very closely in order to make a convincing composite. Getting this wrong screams "composite" and is the short path to a new career in either the food delivery or home care industry.

中间调，或高光。图6.5展示了一个三球色彩分级面板，其中每个轨迹球被分配了一个范围。因此，例如，您可以对阴影应用提升，对中间调应用伽马，对高光应用增益。这些分级操作也可以应用于单独的颜色通道以调整色调——例如，降低蓝色通道的增益以使画面更温暖。

图6.5 三球色彩分级面板

6.1.5 二级调色

初级调色，或一些电影界人士可能称之为“单光调色”，旨在“平衡”电影项目或电视节目的曝光和色彩，以消除镜头切换间令人不适的差异。这是图像迈向最终调色、并呈现在家庭或影院屏幕前的基准起点。此处的关键在于：初级调色是全局性的，因此会整体影响整个画面，且作用于三原色——红、绿、蓝。

二级调色更为精准，能隔离选定像素进行针对性处理。这些像素可根据其数值（色相、饱和度或明度）或通过遮罩与键控——常被亲切称为“强力窗口”——按屏幕上的物理位置进行隔离。强力窗口需跟随目标在屏幕上移动，因此高级调色系统配备追踪器，能确保调色师使用的窗口始终锁定目标。以下是专业调色师采用的二级调色方法全景导览：

- 区域 - 通过“区域控制”，图像会根据亮度进行分区，以便单独调整每个区域。这正是暗部、中间调和高光被分别处理的原因，所以三球控制面板会有三个球——每个区域一个。这种精细的控制非常适合对合成画面进行色彩分级，其中前景图像需要与背景板非常匹配，才能制作出令人信服的合成效果。如果处理不当，就会明显暴露出“合成”痕迹，这将是转行送餐或家政服务的捷径。

- **Secondaries** - The secondary colors of cyan, magenta, and yellow can be isolated and adjusted without affecting the primaries (red, green, and blue). This is done internally by generating a hue mask that isolates the color in question without affecting nearby colors. This is very important for skin tones. Sophisticated color correctors like DaVinci Resolve can generate saturation masks, but they are less common.

- **Keys** - This technique creates a key to isolate an item of interest for separate grading based on some attribute of its color. An obvious example is a lumakey, but there are also hue keys and even saturation keys to work with. I remember one scene salvage job where a woman with a graying streak in her hair reminiscent of the Bride of Dracula (the streak, not her) was appalled by her on-camera appearance as the camera bleached the streak white. She asked me if I could fix it, which I did. Realizing that the white streak had no chroma, and therefore no saturation, I was able to use a saturation key to isolate and tastefully recolor her bleach streak to a matching lovely blond color.

- **Masks** - Masks or "power windows" are generated shapes used to isolate a color operation to some area of the image. Some systems will allow you to throw circles or squares on the screen and then dial in their size, shape, softness, and falloff. More sophisticated systems will allow you to draw arbitrary shapes as a mask and even keyframe it to stay on a moving target. Even more sophisticated systems offer trackers that will track the shape over the moving target for you. If it's a greenscreen composite or a CG shot, there will be mattes created for the VFX, which are much in demand by colorists since they can be used to refine the color grading of specific elements in the shot.

- **Gradients and vignettes** - These are machine-generated as very soft masks for subtle color grading operations designed to call attention towards or away from items of interest or to convey a certain mood. A vertical linear gradient might be used to enhance a colorful sunset. A circular gradient can be used to softly highlight a lovely face. Conversely, undesirable elements in the frame may be darkened or subdued to make them less noticeable.

- **HSL adjustments** - Rather than using a physical mask to isolate a color grade, HSL adjustments use a color model isolation. The hue is adjusted by rotating the image's color wheel, leaving brightness and saturation unaffected—very useful for correcting skin tones. Saturation can easily

- 次要色 - 青色、品红色和黄色的次要色可以被单独分离和调整，而不会影响主要色（红色、绿色和蓝色）。这是通过内部生成一个色调遮罩来实现的，该遮罩可以隔离目标颜色，同时不影响邻近颜色。这对于肤色处理尤为重要。像DaVinci Resolve这样复杂的色彩校正工具可以生成饱和度遮罩，但这类功能较为少见。

- 键控技术 - 这种技术通过创建键来根据颜色属性分离出需要单独调色的目标元素。最典型的例子是亮度键，但也有色相键甚至饱和度键可供使用。我记得有一次场景修复工作中，一位女士头发中有一缕灰白发丝（让人联想到德古拉的新娘——仅指那缕发丝而非她本人），由于摄像机将这缕发丝拍得惨白，她对镜头中的形象感到震惊。她问我能否修复，我做到了。意识到这缕白发没有色度因而缺乏饱和度后，我运用饱和度键将其分离出来，并精心重新着色为相配的金色，效果自然雅致。

- 遮罩 - 遮罩或“局部调色窗口”是通过生成形状来将色彩操作限定在图像的特定区域。有些系统允许您在屏幕上放置圆形或方形遮罩，并通过参数调节其尺寸、形状、柔化度及衰减范围。更先进的系统支持绘制任意形状的遮罩，甚至可通过关键帧使其跟随运动目标。尖端系统还配备跟踪器功能，可自动追踪运动目标上的遮罩形状。对于绿幕合成或CG镜头，视觉特效会生成专用蒙版，这些蒙版深受调色师青睐，因为它们能用于精细化调整镜头中特定元素的色彩分级。

- 渐变与虚光遮罩 - 这些由机器生成的柔和遮罩，用于精细的色彩分级操作，旨在引导视线聚焦或远离兴趣点，或传递特定氛围。垂直线性渐变可用于增强绚烂的日落效果，圆形渐变则能柔和地凸显美丽面容。反之，画面中不需要的元素可通过加深或柔化处理降低其存在感。

- HSL调整 - 不同于使用物理遮罩来隔离色彩分级，HSL调整采用色彩模型进行隔离。通过旋转图像的色轮来调整色调，同时保持亮度和饱和度不变——这对于校正肤色非常有用。饱和度可以轻松

be increased or decreased with the HSL color model, but when using the RGB color model, adjusting saturation is virtually impossible. RGB adjustments affect just the one channel selected, while saturation is a cross-channel operation. The “L” in HSL stands for “lightness” and is one way to calculate the luminance of an image. Raising or lowering the lightness will make the image lighter or darker without disturbing the hue or saturation.

Secondary color grading is demonstrated with zones, keys, masks, and gradients, and HSL adjustments are made while you watch.

6.1.6 Scopes

Scopes are familiar tools for video folks like editors and colorists, but many of the rest of us have had scant contact with them. Your software may even have scopes, but perhaps you have not deployed them simply because you are not familiar with what they are good for. We will fix that right now, so you can up your game by adding scopes to your toolset. If you already know and love your scopes, feel free to skip this section. I am sure there are no buried gems of knowledge in here. Move along. These are not the droids you are looking for.

As we learned in Chapter 1, human vision is both adaptable and variable because our optical wetware cannot be calibrated. Your eyes may lie, but a scope never will. Scopes offer a repeatable way to “view” an image that is independent of our vision or the display. Put an image on two monitors that are adjusted differently and the scopes will still give the same readings because they are looking at the data, not its appearance on the screen. They provide statistics about the images and these stats have both technical and artistic value. On the technical side, they are useful for troubleshooting color problems and confirming that everything is in spec and in gamut. On the artistic side, they are useful for color-matching clips. High-end cameras will even have them built in to monitor the shoot.

在HSL色彩模型中，饱和度可以增加或减少，但使用RGB色彩模型时，调整饱和度几乎是不可能的。RGB调整仅影响所选的一个通道，而饱和度是一种跨通道操作。HSL中的“L”代表“明度”，是计算图像亮度的一种方式。提高或降低明度会使图像变亮或变暗，而不会干扰色调或饱和度。

二级调色通过区域、关键帧、遮罩和渐变来演示，并在您观看的同时进行HSL调整。

6.1.6 作用域

示波器是剪辑师和调色师等视频工作者熟悉的工具，但许多其他人却很少接触它们。你的软件可能也有示波器，但或许你从未使用过，仅仅因为不熟悉它们的用途。我们现在就来解决这个问题，让你通过将示波器加入工具库来提升技能。如果你已经了解并喜爱示波器，可以跳过本节。我敢保证这里没有隐藏的知识宝藏。继续前进吧，这不是你要找的机器人。

正如我们在第一章所学，人类视觉既具适应性又多变，因为我们的光学“湿件”无法被校准。你的眼睛可能会欺骗你，但示波器永远不会。示波器提供了一种可重复的“查看”图像的方式，这种方式独立于我们的视觉或显示设备。将同一图像放在两台调整不同的显示器上，示波器仍会给出相同的读数，因为它们观察的是数据本身，而非其在屏幕上的外观。它们提供关于图像的统计数据，这些数据兼具技术与艺术价值。在技术层面，它们有助于排查色彩问题，并确认一切符合规格且在色域范围内。在艺术层面，它们可用于色彩匹配剪辑。高端相机甚至内置示波器以监控拍摄过程。

The interesting thing about scopes is that they were designed to monitor early analog TV signals, so they are "whole frame" data samples. They offer a grand overview, making it easy to spot the extremes or out-of-spec data. They are also a child of video, so only work on data between zero and 1.0. Each type of scope specializes in one "point of view" on the image data so we will look at all three (Figure 6.6)—the waveform monitor, the vectorscope, and the histogram—to understand what they reveal and how to use them for color correction.

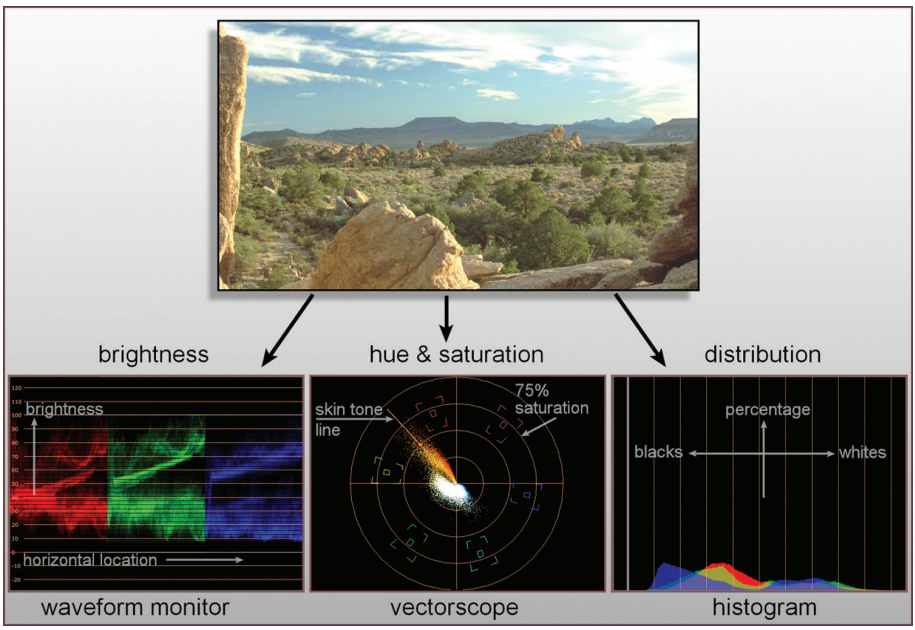


Figure 6.6 Scopes for image analysis

6.1.6.1 Waveform Monitor

This shows the brightness level (exposure) of the RGB channels. The example in Figure 6.6 is in "parade" mode, with each channel shown side by side, but there is also an "overlay" mode where all three channels are superimposed. There will also be a "luma" option for plotting just luminance. The horizontal scale correlates to the horizontal location on screen and the vertical scale indicates brightness from 0 to 100 IRE, while some will also have 0 to 255 for us binary types.

示波器有趣的一点在于，它们最初是为监测早期模拟电视信号而设计的，因此采集的是“整帧”数据样本。它们提供宏观视野，便于捕捉极端值或超规数据。同时它们脱胎于视频领域，故仅能处理0到1.0之间的数据。每种示波器专精于图像数据的特定“观察视角”，因此我们将通过波形监视器、矢量示波器和直方图（图6.6）这三种工具，解析它们揭示的信息及其在色彩校正中的应用方法。

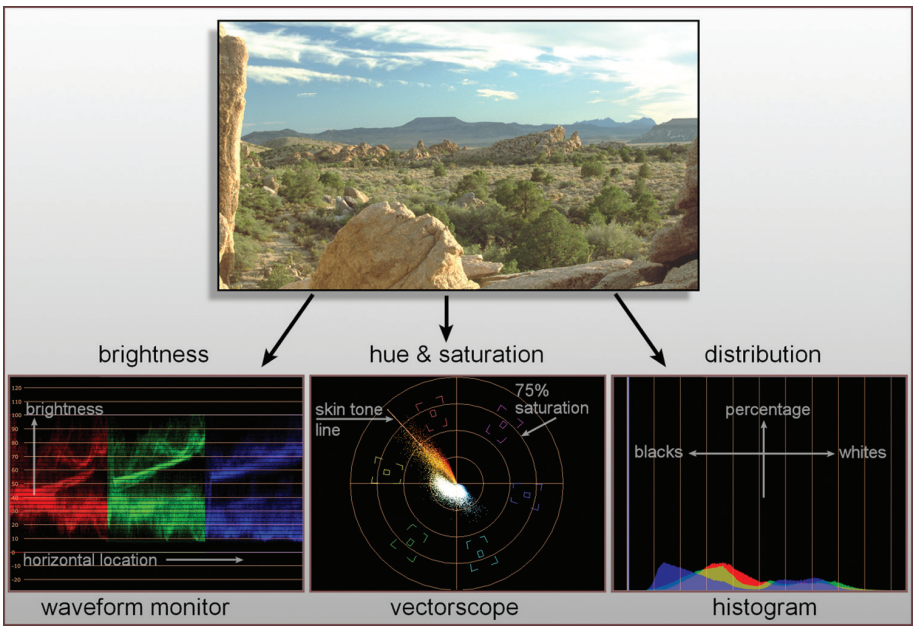


图6.6 图像分析的范围

6.1.6.1 波形监视器

这显示了RGB通道的亮度水平（曝光）。图6.6中的示例为“并列”模式，各通道并排显示，但还有一种“叠加”模式，三个通道会重叠显示。此外还会有仅绘制亮度的“亮度”选项。水平刻度对应屏幕上的水平位置，垂直刻度表示0至100 IRE的亮度范围，部分示波器还会为二进制用户提供0至255的刻度显示。

Some useful waveform monitor references include the following:

- Skin tones should have a luma of around 50 IRE
- 18% gray will be around 40 IRE
- For a greenscreen plate to get a good key, the green channel should be around 70 IRE, with red and blue at or below 40
- 7.5 to 100 IRE is broadcast safe.

Tip—for a normally color-balanced shot the three parade graphs will be about equal height.

6.1.6.2 Vectorscope

This shows hue and saturation. Like a color wheel, the hue is measured around the circle in degrees, while saturation is the distance from the center. Many vectorscopes have a "skin tone" line that shows where to place skin tone hues. Interestingly, this reference works for all skin types, regardless of race. For broadcast work, saturation should not exceed 75%. Without a vectorscope, a digital artist might find it very difficult to confirm the maximum saturation in an image and where it is located.

Besides getting skin tones right, when humans create elements like matte paintings and CG elements, we tend to make them too saturated. Remember the Kodak color memory tests mentioned in Chapter 2? Use the vectorscope to compare the saturation of the compositing element to the original live-action plate. You will no doubt often find that you need to add a desaturation operation to the element to balance it with the main plate.

6.1.6.3 Histogram

This shows the distribution of brightness—that is, what percentage of the pixels are in the shadows, what percentage are in the highlights, etc. The horizontal scale is luminance, black to white from left to right, while the vertical scale is self-scaling and represents the percentage relative to the whole picture. The histograms in Figure 6.7 show three different exposures of the picture. On the far left, the underexposed histogram is shifted to the left with black pixels piled up on the left edge, indicating that they are clipped. In the center, the normal exposure shows the histogram evenly distributed,

一些有用的波形监视器参考包括以下内容:

- 肤色应具有约50 IRE的亮度值
- 18% 灰阶大约对应 40 IRE
- 为了使绿幕板获得良好的抠像效果，绿色通道应保持在约70 IRE左右，而红色和蓝色通道需维持在40 IRE或以下。
- 7.5至100 IRE为广播安全范围。

提示——对于一张通常色彩平衡的镜头，三个波形图的柱状高度会大致相等。

6.1.6.2 矢量示波器

这展示了色相和饱和度。如同色轮，色相以角度为单位沿圆周测量，而饱和度则是与圆心的距离。许多矢量示波器都有一条“肤色线”，用于指示肤色色相应处的位置。有趣的是，这一参考标准适用于所有肤色类型，无论人种如何。在广播制作中，饱和度不应超过75%。若无矢量示波器，数字艺术家可能很难确认图像中的最大饱和度及其所在位置。

除了准确还原肤色，人类在制作遮罩绘画和CG元素时，往往会让它们过于饱和。还记得第二章提到的柯达色彩记忆测试吗？使用矢量示波器将合成元素的饱和度与原始实拍画面进行对比。你无疑会经常发现，需要对元素进行去饱和度处理，才能使其与主画面相协调。

6.1.6.3 直方图

这显示了亮度的分布——即阴影中的像素占多少百分比，高光中的像素占多少百分比等。横坐标表示亮度，从左到右由黑到白；纵坐标则是自适应的，表示相对于整张图片的百分比。图6.7中的直方图展示了图片的三种不同曝光情况。最左侧的曝光不足直方图整体左移，黑色像素堆积在左边缘，表明这些像素已被剪切。中间的正常曝光直方图则显示分布均匀，

indicating that the code values fill the full range of the color space. On the right is the overexposed example, with the histogram shifted to the right with white pixels piled up on the right edge, indicating that they too are clipped.

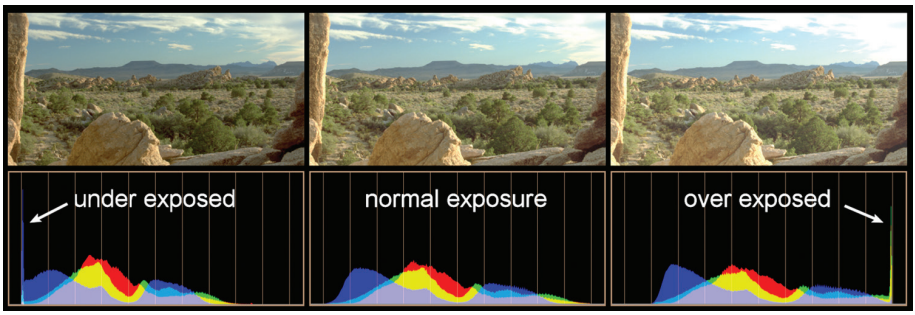


Figure 6.7 Histogram showing clipped sRGB pixels

Automatic color matching operations use histogram matching to match picture A to picture B, but you must be careful that the content of the two pictures is very similar. If the pictures are too dissimilar, their histograms will also be dissimilar, so when picture B is applied to picture A it will actually discolor it. You can also check white balance with a histogram by finding a bright white element in the frame and making sure that all three colors peak at the same point horizontally.

Histograms provide great insight into the color distribution in an image. Watch as various images are adjusted to see the impact on their histograms, including a log to Rec. 709 transition.

6.1.7 Color Grading in Log

Log images offer increased black detail and preserved highlights, retaining much more picture information for the colorist. Of course, the image data must be exported from the camera as log for this to work. If it is exported as a JPEG or Rec. 709, the blacks are already crushed and the highlights are already clipped, so there is no point in converting the damaged footage to log.

表明代码值填满了色彩空间的整个范围。右侧是曝光过度的示例，直方图向右偏移，白色像素堆积在右边缘，表明它们也被裁剪了。

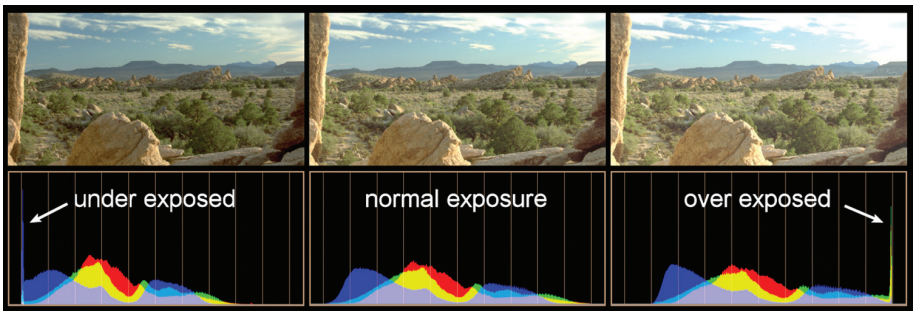


图6.7 显示裁剪后sRGB像素的直方图

自动色彩匹配操作使用直方图匹配将图片A与图片B进行匹配，但必须注意两张图片的内容应非常相似。若图片差异过大，其直方图也会存在显著差异，此时将图片B的直方图应用于图片A反而会导致其色彩失真。此外，您还可以通过直方图检查白平衡：在画面中寻找明亮的白色元素，并确保红、绿、蓝三色通道的峰值在水平方向上处于同一位置。

直方图能深入揭示图像中的色彩分布。观察不同图像调整时对其直方图的影响，包括对数到Rec.709转换的过程。

6.1.7 Log色彩分级

Log图像提供了增强的黑色细节并保留了高光，为调色师保留了更多的画面信息。当然，图像数据必须从相机中以log格式导出才能实现这一效果。如果以JPEG或Rec. 709格式导出，黑色部分已经压缩，高光部分已经裁剪，因此将受损的素材转换为log格式毫无意义。

Figure 6.8 Color grading log images

Figure 6.8 shows why colorists love log. In the top row is a Rec. 709 image. To its right are the crushed blacks in the shrubs when a gamma-up attempt was made to show more dark detail. There is none. To the right of that are the clipped fire highlights made obvious by applying a gain down on the picture. The fire has no internal detail. The log image in the bottom row, however, reveals plenty of dark detail in the bushes when a gamma-up is applied. Then there are the preserved highlights in the fire where you can see every detail of the blaze and window frames.

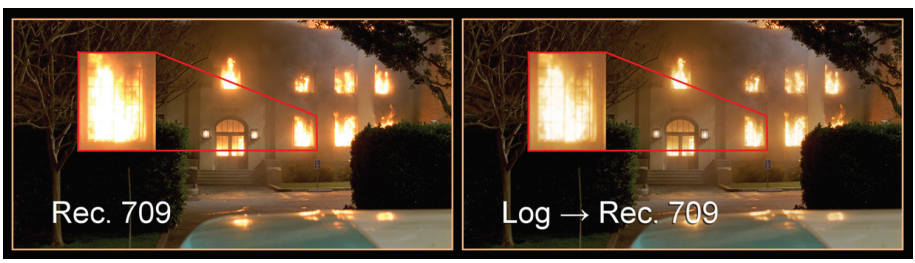


Figure 6.9 Using a soft clip in log preserves flame detail

Figure 6.9 illustrates what can be done with the preserved highlights in Figure 6.8. The Rec. 709 plate on the left was exported from the camera as Rec. 709 and the close-up shows the clipped flame highlights. The plate on the right started with a log camera export followed by a soft clip to the log data to gently lower the flame brightness, then converted to Rec. 709. The close-up shows the retained fine detail in the flame and window frame that was blown out and clipped in the Rec. 709 capture.

图6.8 对数图像的色彩分级

图6.8展示了调色师为何钟爱对数格式。顶行是Rec. 709图像。其右侧灌木丛中死黑的暗部，是在尝试提升伽马以显示更多暗部细节时产生的——实际上并无细节可显。再往右，通过降低增益使过曝的火光高光区显露无遗，火焰内部毫无细节层次。然而底行的对数图像在提升伽马后，灌木丛中呈现出丰富的暗部细节。火焰高光也得到完整保留，烈焰与窗框的每一处细节都清晰可辨。

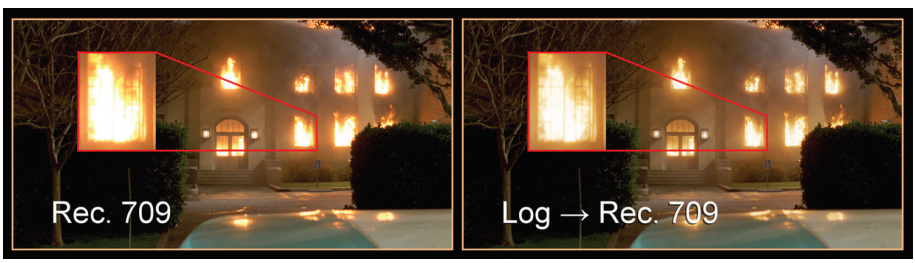


图6.9 在日志中使用软剪裁可保留火焰细节

图6.9展示了图6.8中保留的高光部分所能实现的效果。左侧的Rec. 709图像直接从相机以Rec. 709格式导出，特写画面显示火焰高光部分已裁切。右侧图像则先以对数格式从相机导出，随后对对数数据进行软裁切以平缓降低火焰亮度，再转换为Rec. 709格式。特写画面可见火焰和窗框处保留了丰富的细节层次，而这些细节在Rec. 709直出时因过曝裁切已丢失。

It is important to understand that color grading log data is done with dedicated software designed to work with log data. The same color grade operation applied to a log image and a linear image produces startlingly different results. For example, if you want to increase exposure on a log image, you add an offset. If you add an offset to a linear image, it will just fog up. To increase the exposure of a linear image, you would apply a gain operation, but a gain operation on log data will wildly increase the brights compared to the midtones and darks. Don't do that.

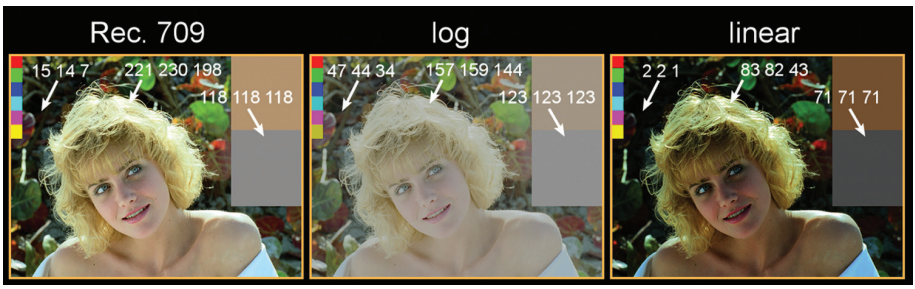


Figure 6.10 Code value ranges are different in different color spaces

Ranges for darks, midtones, and highlights are another big difference between log and other color metrics. Photoshop is an sRGB/Rec. 709 tool. So if you are familiar with color values in Photoshop, you will be quite surprised when you work with log or even linear images at how different the code values are. Figure 6.10 compares the code values for darks, midtones, and highlight ranges for three different color spaces using familiar Photoshop 8-bit code values. For Rec. 709, darks are around 12; for log, that's about 40 and linear, a mere 2. Midtones for Rec. 709 are around 118, log 123, and linear 71. The highlights for Rec. 709 are about 210; for log, this is 155 and for linear it is 71. This may take some getting used to.

6.1.8 Using Histograms

We were introduced to histograms briefly in Section 6.1.6, so you should have a good understanding of how they work. Here we will take a look at how to use them to solve certain color-correcting problems.

重要的是要理解，对数数据的色彩分级需要使用专门设计用于处理对数数据的软件来完成。同样的色彩分级操作应用于对数图像和线性图像会产生截然不同的结果。例如，如果你想增加对数图像的曝光度，你需要添加一个偏移量。如果你对线性图像添加偏移量，只会让图像变得模糊不清。要增加线性图像的曝光度，你应该应用增益操作，但对对数数据使用增益操作会极大地增加亮部，而中间调和暗部则变化不大。切勿这样做。

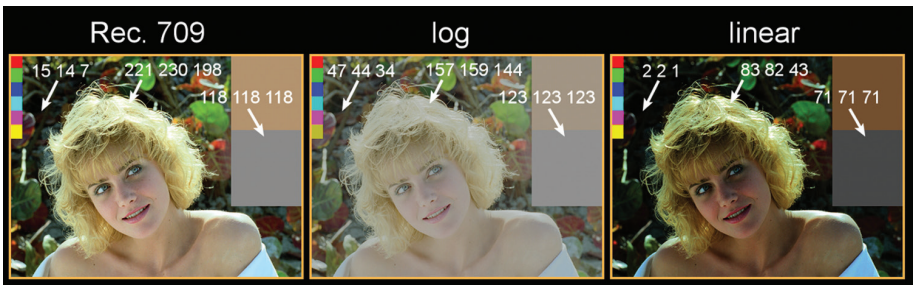


图6.10 不同色彩空间中的代码值范围不同

暗部、中间调和高光的范围是log与其他色彩度量之间的另一大差异。Photoshop是一款基于sRGB/Rec. 709标准的工具。因此，如果你熟悉Photoshop中的色彩数值，在处理log甚至线性图像时，你会惊讶地发现其编码值差异巨大。图6.10通过熟悉的Photoshop 8位编码值，对比了三种不同色彩空间中暗部、中间调和高光范围的编码值。在Rec. 709标准下，暗部数值约为12；log模式下约为40，而线性模式下仅为2。中间调在Rec. 709中约为118，log为123，线性为71。高光部分在Rec. 709中约为210；log为155，线性为71。这可能需要一些时间来适应。

6.1.8 使用直方图

我们在6.1.6节中简要介绍了直方图，因此你应该对其工作原理有很好的理解。这里我们将探讨如何利用它们来解决某些色彩校正问题。

6.1.8.1 Color Matching

Let's say you have two clips from an outdoor shoot but at different times of the day. One is warmer and brighter than the other, so they need to be made to match. There are color-matching operations that will take image A and magically match it to image B. How is this done? Behind the black curtain, the software makes a histogram of both images and then mathematically adjusts image A until its histogram matches the histogram from image B. This is very helpful unless image B does not have very similar content to image A. If their content differs significantly, then their histograms will differ significantly, and forcing image B to match the wrong target will result in a hideously discolored image B. Frankly, the best use of these color-matching operations is when you have a graded image and you want to impart that same grade to an ungraded version.

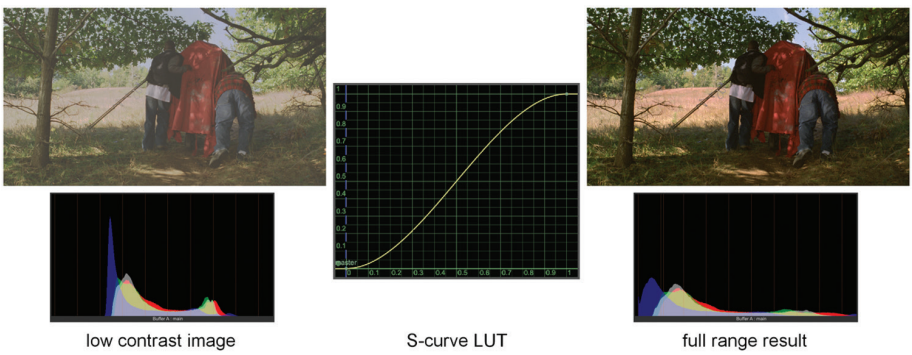


Figure 6.11 Increasing contrast with histograms

6.1.8.2 Increasing Contrast

Histograms are great for increasing contrast without introducing clipping, simply because you can see any clipping in the new histogram. Take Figure 6.11, for example. On the left is a low-contrast image with lifted shadows and a dull sky, with its histogram below. As you can see, the histogram is bunched up in the center and does not cover the full range of colors. This is typical of a low-contrast image. By applying the famous "S" curve LUT (center), contrast is restored as the image is now stretched to fill the full range of color (right). Importantly, the histogram reveals that the full range has indeed been filled, but it also shows no clipping. As we saw in earlier histograms (Figure 6.7), any clipping in the blacks or whites introduces a telltale spike at the left or right edge.

6.1.8.1 Color Matching

假设你有两段户外拍摄的片段，但拍摄时间不同。一段比另一段更温暖明亮，因此需要将它们调整到一致。有些色彩匹配操作能神奇地将图像A匹配到图像B的效果。这是如何实现的呢？在幕后，软件会为两张图像分别生成直方图，然后通过数学计算调整图像A，直到其直方图与图像B的直方图吻合。这种方法非常有效——除非图像B的内容与图像A差异过大。如果两者内容明显不同，它们的直方图也会显著不同，此时强行将图像B匹配到错误的目标，会导致图像B产生难看的色偏。坦白说，这类色彩匹配功能的最佳使用场景是：当你已有一张调色完成的图像，并希望将同样的色调风格应用到未调色的版本上。

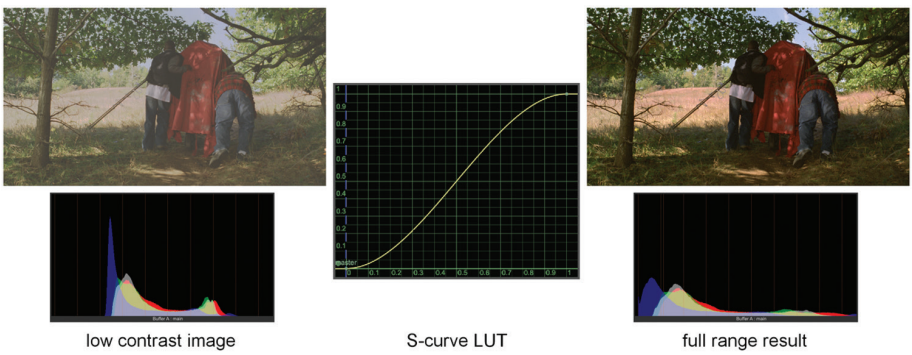


图 6.11 使用直方图增强对比度

6.1.8.2 增加对比度

直方图非常适合在不引入裁剪的情况下增加对比度，仅仅因为您可以在新的直方图中看到任何裁剪。以图6.11为例，左侧是一张低对比度图像，阴影部分被提亮，天空显得沉闷，其直方图位于下方。如您所见，直方图集中在中心区域，未能覆盖完整的色彩范围。这是低对比度图像的典型特征。通过应用著名的“S”曲线查找表（中间），对比度得以恢复，图像现在被拉伸以填满整个色彩范围（右侧）。重要的是，直方图显示完整范围确实已被填充，但也表明没有发生裁剪。正如我们在之前的直方图（图6.7）中所见，任何黑色或白色区域的裁剪都会在左侧或右侧边缘产生明显的尖峰。

6.2 Digital Intermediate (DI)

The high temple for color grading is, of course, the DI suite. Cinematographers, colorists, and editors are very familiar with the DI suite and the mysterious goings-on there. However, many artists in the imaging pipeline—such as off-line editors, gamers, motion graphics, CG artists, and VFX—may have never seen the inside of a DI suite, with its dark gray walls, acoustic baffling, huge digital projector, and cushy recliner chairs in front of a vast control panel of dials, buttons, and knobs. The vast control panel is for the colorist; the huge projector is to replicate the viewing conditions of a theater, and the cushy chairs are for the director, DP, and studio executives. Color grading a movie is a spectator sport.

Digital Intermediate, popularly called DI, is the process of applying the final creative color-graded "look" to a feature film, considered one of the most important steps in the making of a movie. The term "Digital Intermediate" is a bit arcane because it inherits its name from the glory days of 35 mm film. In those days, the edited film negative was transferred to an intermediate film stock, which is where "intermediate" comes from. The transfer was done using RGB lights that were timed to adjust exposure and color per shot, which is where the term "color timing" comes from. This color-timed intermediate film was used to strike the prints that were sent to theaters for projection. Then one fine day, Kodak developed film scanners and film recorders, so the camera negative could be digitized to produce the digital intermediate for digital color timing.

"Color timing" was the 35 mm film term. For the digital pipeline, it is now called "color grading," but you will hear people in casual conversation refer to it as "color timing" as well. Here we will take an inside look at the entire DI process, so you can see how this critical movie-making process works and how your work fits in.

6.2.1 Conforming

The modern digital cinema production workflow has the DIT delivering all of the day's takes to the DI facility to be loaded onto disk, where "proxies" (low-resolution stand-ins for the full-resolution frames) will automatically be generated. Later, the director will review them (likely on their laptop) to make the "selects"—the chosen takes that will be used in the movie. Once the

6.2 数字中间片（DI）

调色的圣殿，当然就是数字中间片调色室。摄影师、调色师和剪辑师对调色室及其中的神秘操作都非常熟悉。然而，影像制作流程中的许多艺术家——比如离线剪辑师、游戏设计师、动态图形师、CG艺术家和视觉特效师——可能从未踏足过调色室内部：那里有深灰色的墙壁、吸音隔板、巨大的数字投影仪，以及面对布满旋钮、按钮和拨杆的庞大控制面板的舒适躺椅。庞大的控制面板是为调色师准备的；巨大的投影仪是为了模拟影院观看环境；舒适的座椅则是为导演、摄影指导和制片厂高管准备的。电影调色，是一场供人观赏的技艺表演。

数字中间片，俗称DI，是将最终的创意调色“外观”应用于故事片的过程，被认为是电影制作中最重要的步骤之一。“数字中间片”这个术语有点晦涩，因为它继承自35毫米胶片的光辉岁月。在那个时代，剪辑后的胶片负片会被转移到中间胶片上，这便是“中间片”一词的由来。转移过程使用RGB光源，通过逐镜头调整曝光和色彩的时间控制来完成，这也正是“色彩配光”一词的起源。这种经过配光的中间胶片被用来制作送往影院放映的拷贝。后来有一天，柯达公司开发出了胶片扫描仪和胶片记录仪，于是摄影负片得以数字化，从而生成用于数字色彩配光的数字中间片。

“色彩配光”是35毫米胶片时代的术语。在数字流程中，现在被称为“色彩分级”，但在日常对话中你仍会听到人们称其为“色彩配光”。接下来我们将深入解析整个数字中间片（DI）流程，让你了解这一关键电影制作环节如何运作，以及你的工作如何融入其中。

6.2.1 符合性

现代数字电影制作流程中，数字影像技师会将当日所有拍摄素材送至数字中间片制作机构载入磁盘，系统将自动生成“代理文件”（即全分辨率画格的低分辨率替代文件）。随后导演会审阅这些文件（通常通过笔记本电脑）进行“镜头遴选”——即选定用于影片的拍摄素材。一旦

selects have been made, the editor can get to work editing the show offline on a laptop using the proxies. Shots with pending VFX are cut in with temps prepared on the editor's system as a placeholder, to be replaced later in the timeline with the final VFX shots as they become available.

The editor's system produces the Advanced Authoring Format (AAF) file, which contains data on which clips were selected, edits, transitions, speed changes, color LUTs, shot metadata, and more. The AAF file is then loaded into the DI colorist's system and is used to automatically assemble the full-resolution versions of the clips onto the timeline which "conforms" it to match the proxies edited on a laptop.

We now have a full-resolution version of the movie made up of un-timed (not color-graded) clips. Recalling that the DIT from Chapter 2: Camera Space worked with the director on-set to produce a preliminary color grade of each shot, this preliminary color grade will also be loaded into the colorist's system as a starting point for the final color timing of the entire movie during Mastering.

This is now an entirely non-destructive pipeline, meaning that the original camera clips remain intact and unmodified with their color and editing operations as a modification layer above the raw clips. With the camera raw clips permanently preserved like this, there is no loss of quality as the color, editing, transitions, and speed changes are endlessly iterated, revised, and re-rendered until the director runs out of money or schedule.

6.2.2 The Mastering Process

The first step is to create the color-graded 24p Master, which is the main theatrical release. The first color pass is to even up all of the shots to correct for over-exposed, under-exposed, hue shifts from available light color temperatures, and so on. Then a "hero" shot will be selected for each scene (e.g., there might be 12 shots in the bedroom scene) which is given a final color grade. These hero shots are compared from scene to scene to ensure color consistency across the project.

Once the bedroom scene hero shot is locked down, the other 11 shots are graded to match. Keep in mind that the 12 shots in the bedroom scene have 12 different color starting points, so must have 12 different grading

选择完成后，剪辑师就可以开始使用代理文件在笔记本电脑上离线编辑节目。待处理视觉特效的镜头会先用剪辑师系统中准备的临时素材进行剪辑，作为占位符，待最终视觉特效镜头完成后，再在时间线上替换这些{v*}镜头。

编辑系统生成高级制作格式（AAF）文件，其中包含所选片段、剪辑、转场、速度变化、色彩查找表（LUT）、镜头元数据等信息。随后，AAF文件被加载到数字中间片（DI）调色师的系统中，用于自动将全分辨率版本的片段组装到时间线上，从而“套底”匹配在笔记本电脑上编辑的代理文件。

我们现在拥有一个由未经调色（未进行色彩分级）的片段组成的全分辨率版本电影。回顾第二章：摄影空间中的数字影像技师（DIT）曾在拍摄现场与导演合作，为每个镜头制作了初步色彩分级，这个初步色彩分级也将被载入调色师的系统中，作为整个电影在母版制作期间最终色彩配光的起点。

现在，这已是一个完全非破坏性的处理流程，意味着原始摄像机片段保持完整且未经修改，其色彩和编辑操作作为原始片段之上的修改层得以保留。通过这种方式永久保存摄像机原始片段，在色彩、编辑、转场和速度变化被无限次迭代、修改和重新渲染的过程中，画质不会有任何损失——直到导演耗尽预算或时间。

6.2.2 母带处理流程

第一步是创建色彩分级的24p母版，这是主要的影院发行版本。首次色彩调整旨在平衡所有镜头，校正曝光过度、曝光不足、可用光色温导致的色调偏移等问题。随后，每个场景会选出一个“英雄镜头”（例如卧室场景可能包含12个镜头）进行最终色彩分级。这些英雄镜头会在场景间相互比对，以确保整个项目的色彩一致性。

一旦卧室场景的英雄镜头确定下来，其他11个镜头就要进行调色以匹配。请记住，卧室场景的12个镜头有12个不同的色彩起点，因此必须进行12种不同的调色处理。

operations applied to give them one consistent look. This is a non-trivial skill that colorists develop.

Beyond consistency, there is also the creative "look"—the real reason we are here. The colorist will make adjustments to exposure, contrast, and saturation to achieve some desired visual impact. Again, it is the talent of the colorist to know how to do that. In general, the effort is to make the scenes "punchy," with vibrant colors and deep shadows. Remember the Kodak color memory experiments from Chapter 2? That's what's in play here.

Once the 24p Master is developed, there follow several additional theatrical versions of it to accommodate different projection systems such as IMAX and Dolby, as well as 3D vs. 2D projection. From the 24p Master, various "trim passes" will be made to make separate low dynamic range DVD versions for HDTV and high dynamic range Blu-ray versions for UHDTV. Yet another grade for broadband like NetFlix and Amazon. The reason for all these versions is the color management prime directive—to maintain a consistent appearance across all displays. This is getting harder to do as display technology advances and diverges over time.

FUN FACT: In their relentless pursuit of the most extreme movie experience possible, IMAX theaters have introduced dual 4K laser projectors.

6.2.3 Visual Effects

For VFX artists delivering shots for DI, there are two main considerations. The first is to realize that in the DI suite, the projection system replicates the theater so the shot will undergo a dramatic increase in brightness and contrast which can suddenly reveal flaws that the compositor did not see because the shot was developed on a standard low dynamic range sRGB monitor. You NEVER want your shots to be kicked back by the colorist and this is why I teach "gamma-slamming"—the shot inspection technique of swinging the display gamma up and down to "stress" the shot and reveal problems before it is turned in for DI.

对它们应用操作以赋予一致的外观。这是调色师培养的一项非平凡技能。

除了保持一致性，还有创造性的“观感”——这才是我们在此的真正原因。调色师会调整曝光、对比度和饱和度，以达到某种期望的视觉冲击力。这同样考验调色师的才华，懂得如何实现这些调整。总的来说，目的是让场景“富有冲击力”，色彩鲜明、阴影深邃。还记得第二章中柯达色彩记忆实验吗？这正是此处所运用的理念。

一旦24p母版制作完成，后续会衍生出多个适应不同放映系统的影院版本，例如IMAX和杜比格式，以及3D与2D放映版本。基于24p母版，将通过多次“修剪处理”分别制作适用于HDTV的低动态范围DVD版本，以及适用于UHDTV的高动态范围蓝光版本。此外还会为Netflix和亚马逊等流媒体平台单独调色。这些版本存在的根本原因在于色彩管理的核心准则——确保所有显示设备上的视觉效果保持一致。随着显示技术的不断进步与分化，实现这一目标正变得愈发困难。

趣闻：IMAX影院为了追求极致的观影体验，引入了双4K激光投影仪。

6.2.3 视觉效果

对于为数字中间片（DI）交付镜头的视觉特效艺术家来说，有两个主要考量点。首先是要认识到，在DI调色棚中，投影系统会模拟影院环境，因此镜头会经历亮度和对比度的急剧提升，这可能突然暴露出合成师未曾发现的问题——因为镜头是在标准的低动态范围sRGB监视器上制作的。你绝不想自己的镜头被调色师退回，这正是我教授“伽马冲击”检查法的原因：通过大幅上下调整显示伽马值来“加压”测试镜头，在提交DI前发现问题。

Figure 6.12 Gamma-slamming to reveal defects

Figure 6.12 illustrates a composited shot with the original comp in panel A. Panel B has the gamma slammed down to reveal the highlights are too hot with too much red (arrow). Panel C has the gamma slammed up, which shows the black shadow got clipped (arrow). Slamming the gamma down shows problems in the highlights and slamming it up reveals problems in the darks.

The other big issue is the mattes. When CG elements are composited into the background plate, the compositor does a yeoman's job of color-correcting it to match the plate. However, the colorist will always want to "fine-tune" the work, so will need the mattes for the CG or greenscreen elements. If it is an EXR job, then it is most convenient to put all of the mattes into EXR layers with the comp. That way, they can never get lost or out of sync with the clip.

6.2.4 Deliverable File Formats

One of the key data management questions in the modern digital pipeline is what file format should be used to deliver the footage to the DI facility. The shots for a project may be delivered to DI either as sequential frames like DPX or as clips in a container such as DNxHR. Here is a list of the current popular file formats for DI:

- **DPX** - A popular format for film scans (digital intermediates) and visual effects work; the log format is uncompressed and very high quality. With support for 10, 12, and 16-bit depth, it is suitable for very high-end grading and finishing.
- **EXR** - Used industry-wide for its exceptional quality for high dynamic range content and arbitrary channels such as for mattes. Storing linear data at great precision, it is exceptionally suitable for complex VFX pipelines and HDR content.

图 6.12 伽马冲击法揭示缺陷

图6.12展示了合成镜头的示例，其中A面板为原始合成画面。B面板将伽马值大幅调低，显示出高光区域过曝且红色过多（箭头所指）。C面板则将伽马值大幅调高，可见黑色阴影部分出现裁切（箭头所指）。大幅降低伽马值可暴露高光区域的问题，而大幅提高伽马值则能揭示暗部细节的问题。

另一个大问题是遮罩。当CG元素合成到背景板中时，合成师会进行繁重的色彩校正工作以匹配背景板。然而，调色师总是希望“微调”这项工作，因此需要CG或绿屏元素的遮罩。如果是EXR格式的工作，最方便的做法是将所有遮罩与合成一起放入EXR图层中。这样，它们永远不会丢失或与剪辑不同步。

6.2.4 可交付文件格式

现代数字流程中一个关键的数据管理问题是：应采用何种文件格式将素材交付给数字中间片（DI）机构。项目镜头可以以序列帧（如DPX）形式交付，也可以以封装格式（如DNxHR）的片段形式交付。以下是当前DI环节常用的文件格式列表：

- DPX - 一种用于胶片扫描（数字中间片）和视觉特效工作的流行格式；其对数格式未经压缩且质量极高。支持10位、12位和16位色深，适用于非常高端的调色和最终制作。

- EXR - 因其在高动态范围内容和任意通道（如遮罩）方面的卓越品质而在行业内广泛应用。它以极高的精度存储线性数据，特别适合复杂的视觉特效流程和高动态范围内容。

- **ProRes** - Offers a good balance between image quality and file size. The high-end codecs offer very low losses in compression and support for one alpha channel. A great choice for DI in an Apple ecosystem or when disk space is at a premium.
- **DNxHR** - Like ProRes, this codec offers several quality and file size tradeoffs and is widely used for its efficiency and compatibility with the Avid editing system.
- **ARRIRAW** - For projects shot on ARRI cameras (and a LOT of projects are shot on ARRI cameras), this proprietary raw format offers absolute maximum quality right out of the camera. It can be read directly into DI color correctors and allows for maximum range and creative control in DI for grading and finishing.
- **MOV** - The QuickTime movie is a container or "wrapper" that can hold many types of codecs, like ProRes and DNxHR.

FUN FACT: DI suites have some of the most advanced tech in the movie-making industry, with high-powered workstations, high-precision color grading panels, state-of-the-art digital projectors, and real cushy recliners.

6.3 ICC Profiles

One key aspect of color management is to convert (transform) images from one color space to another. The main tool for that is ICC profiles, which contain information about color spaces and devices like monitors and printers. ICC profiles are sort of the Gen 1.0 of color management, with ACES being Gen 2.0, but we must start somewhere. Besides, while initially released in 1993, ICC profiles are still in wide use today. One of the most common uses of ICC profiles is converting RGB images to CMYK for print.

The primary goal of color management is to maintain a consistent appearance of an image as it moves from Camera Space to Work Space to Display Space, and even between multiple different workspaces and displays in the post-production pipeline. In other words, it should look the same no matter where and how you are viewing it. This is a non-trivial problem, for two

- ProRes - 在图像质量和文件大小之间提供了良好的平衡。高端编解码器在压缩过程中损失极低，并支持一个alpha通道。对于苹果生态系统中的数字中间片制作或磁盘空间紧张的情况来说，这是一个绝佳选择。

- DNxHR - 与 ProRes 类似，该编解码器在画质与文件大小之间提供多种权衡方案，因其高效性及与 Avid 编辑系统的广泛兼容性而被普遍采用。

- ARRIRAW - 对于使用ARRI摄像机拍摄的项目（许多项目都采用ARRI摄像机拍摄），这种专有的原始格式能提供摄像机直出的绝对最高画质。它可直接导入数字中间片调色系统，并在数字中间片的调色与最终成片环节中，实现最大范围的动态调整和创意控制。

- MOV - QuickTime 电影是一种容器或“包装器”，可以容纳多种类型的编解码器，例如 ProRes 和 DNxHR。

> 趣闻：DI套间拥有电影制作行业中最先进的技术，包括高性能工作站、高精度调色面板、尖端数字投影仪以及非常舒适的真皮躺椅。

6.3 ICC配置文件

色彩管理的一个关键方面是将图像从一个色彩空间转换（变换）到另一个色彩空间。实现这一功能的主要工具是ICC配置文件，其中包含有关色彩空间以及显示器和打印机等设备的信息。ICC配置文件可视为色彩管理的第一代技术，而ACES则代表第二代，但我们必须从某个起点开始。此外，尽管ICC配置文件最初于1993年发布，至今仍被广泛使用。ICC配置文件最常见的用途之一是将RGB图像转换为用于印刷的CMYK格式。

色彩管理的主要目标是在图像从相机空间转移到工作空间再到显示空间，甚至在后期制作流程中多个不同工作空间和显示器之间移动时，保持图像外观的一致性。换句话说，无论你在何处以及如何查看图像，它看起来都应该相同。这是一个非平凡的问题，原因有二

reasons. First, a given display may not be able to show all the colors of an image, so a perfect match simply becomes physically impossible. Second, as we saw in the last chapter, different devices will have different gamuts, and converting an image between two different gamuts is intrinsically problematic to the point of having to choose the rendering intent that will do the least damage to the new image.

To aid in this process, the International Color Consortium (ICC) published color management tools used by many equipment manufacturers and software developers. First, the tools provide a wide gamut **connection space** used as a workspace for color space transformations, and second, they provide a library of color space and equipment **profiles** that precisely define their color characteristics.

6.3.1 How ICC Profiles Work

Millions of digital imaging artists use ICC profiles to convert their images between color spaces without the foggiest idea of how this works. But you really should know how this works. More than just mastering your craft, if you understand how this works, you stand a chance of fixing things when they go wrong. With the "power of knowledge" motivational lecture out of the way, let us trace the inner processes of an ICC profile conversion.

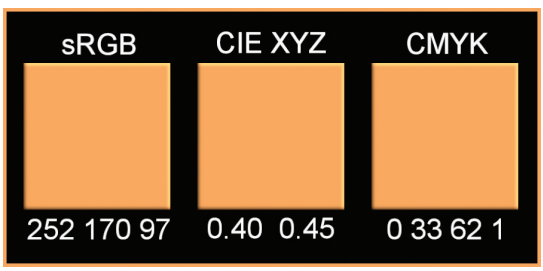


Figure 6.13 Same color chip in three color spaces

Let's say we want to convert an sRGB image from a properly calibrated workstation monitor to CMYK for print. You can follow the action by referring to Figure 6.13, starting with the left chip labeled "sRGB." Note its color and below it the RGB code values that it is made from. Now look at the right chip, our destination, labeled "CMYK." It is exactly the same color but has entirely different code values. This is the mission of color management—to move the chip from the sRGB color space into the CMYK color space but retain the exact same color—even though the code values will change dramatically.

原因有二。首先，给定的显示器可能无法呈现图像的所有颜色，因此完美匹配在物理上就变得不可能。其次，正如我们在上一章所见，不同设备具有不同的色域，将图像在不同色域间转换本质上就存在问题，以至于必须选择对图像损害最小的渲染意图。

为了协助这一过程，国际色彩联盟（ICC）发布了被许多设备制造商和软件开发者使用的色彩管理工具。首先，这些工具提供了一个宽广色域连接空间，用作色彩空间转换的工作空间；其次，它们提供了一个色彩空间和设备配置文件的库，这些配置文件精确地定义了其色彩特性。

6.3.1 ICC配置文件的工作原理

数以百万计的数字影像艺术家使用ICC配置文件在不同色彩空间之间转换他们的图像，却对其工作原理一无所知。但你真的应该了解这背后的机制。不仅仅是精通技艺，如果你明白其运作方式，当出现问题时，你就有机会修复它。抛开“知识的力量”这类激励性说教，让我们来追溯ICC配置文件转换的内部过程。

假设我们想要将来自经过适当校准的工作站显示器的sRGB图像转换为用于印刷的CMYK。你可以参照图6.13来跟随操作，从左侧标有“sRGB”的色块开始。注意它的颜色以及下方对应的RGB代码值。

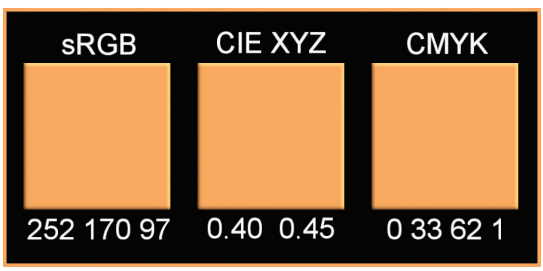


图 6.13 同一色块在三种色彩空间中的呈现

现在来看右边的色块，我们的目标，标记为“CMYK”。它完全是相同的颜色，但代码值却完全不同。这就是色彩管理的使命——将色块从sRGB色彩空间转移到CMYK色彩空间，同时保持完全相同的颜色——尽管代码值会发生巨大变化。

We need two things to get started. First, the pixel array that is the image, of course, but we also need an sRGB ICC profile to inform the **Color Management Module** (CMM) what color space the image is currently in. It is the color space that gives the code values meaning and identifies a particular color. If we told the CMM that the chip was DCI-P3, for example, those same numbers would produce a different color. The necessary ICC profile is usually embedded within the image automatically by the software that generated the image. If not, the CMM may apply a default profile or ask for one.

The pixel array with its profile gets a color space transformation that converts the sRGB chip to CIE XYZ color space which is the ICC profile connection space. We learned in Chapter 4 that the XYZ color space is very large and encompasses all colors visible to the eye by measuring observers. Note that while the XYZ code values are very different than the sRGB numbers, we are still looking at the exact same color inside the ICC connection space.

From the XYZ connection space, it takes the pixel array that is the image plus the ICC profile of the CMYK destination color space and performs a color space transformation from XYZ to CMYK to produce the chip on the right. Again, it is the exact same color as the sRGB and XYZ chips, but now with CMYK code values. Transferred to a properly calibrated printer and printed on paper, the chip could be held up to a properly calibrated sRGB monitor and the two chips would appear identical. This is color management.

Let's circle back to the "properly calibrated" monitor and printer for a minute. The reason they must be calibrated is because it is calibration that aligns the correct color to the code values. If not calibrated, the sRGB code values will produce a different color on the screen and the printer will print an off-color CMYK pixel. So the ICC profile conversion process must be "bookended" by calibrated equipment for it to be valid.

我们需要两样东西来开始。首先，当然是作为图像的像素数组，但我们也需要一个sRGB ICC配置文件，以告知色彩管理模块（CMM）图像当前所处的色彩空间。正是色彩空间赋予了代码值意义，并识别出特定的颜色。例如，如果我们告诉CMM芯片是DCI-P3色彩空间，同样的数字会产生不同的颜色。必要的ICC配置文件通常由生成图像的软件自动嵌入图像中。如果没有，CMM可能会应用默认配置文件或要求提供一个。

像素阵列及其配置文件经过色彩空间转换，将sRGB芯片转换为CIE XYZ色彩空间，即ICC配置文件的连接空间。我们在第4章了解到，XYZ色彩空间非常广阔，通过测量观察者涵盖了人眼可见的所有颜色。请注意，尽管XYZ编码值与sRGB数值差异很大，但在ICC连接空间内，我们看到的仍然是完全相同的颜色。

从XYZ连接空间出发，它获取作为图像的像素阵列以及CMYK目标色彩空间的ICC配置文件，并执行从XYZ到CMYK的色彩空间转换，从而生成右侧的色块。同样，它与sRGB和XYZ色块的颜色完全相同，但现在是以CMYK编码值呈现。若将其传输至经过正确校准的打印机并在纸张上印刷，该色块可被置于经过正确校准的sRGB显示器旁，两者将呈现完全一致的视觉效果。这便是色彩管理。

让我们再回到“经过适当校准”的显示器和打印机这个话题上。它们必须校准的原因在于，正是校准过程将正确的颜色与代码值对应起来。如果不进行校准，sRGB代码值会在屏幕上呈现不同的颜色，而打印机则会输出偏色的CMYK像素。因此，ICC配置文件转换过程必须由经过校准的设备“前后把关”，才能确保其有效性。

6.3.2 Color Space Conversions

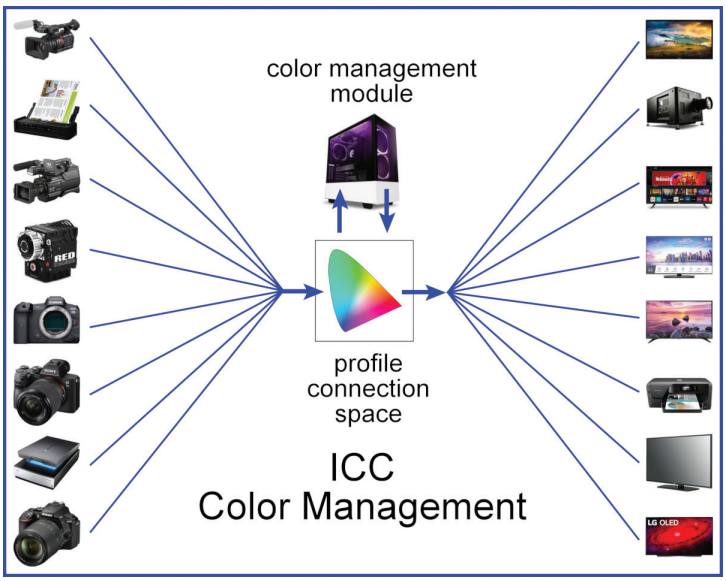


Figure 6.14 ICC color management overview

One fundamental problem of color space conversions is the "many in/many out" problem. Let's say we have just eight input devices and eight output devices that we want to convert between, so we will need 8 x 8 or 64 conversions. If we add one more device, we must add eight more conversions, so as we add more devices, the number of required conversions skyrockets. A better approach is illustrated in Figure 6.14, where each device has an ICC profile that describes its color characteristics. Those profiles are converted to the profile connection space—the special wide gamut color space described previously where the CMM performs the color transformation (color space conversion) between the source and destination profiles. So we only need 16 profiles for the 16 devices in Figure 6.14 instead of 64 conversions. And if we add one more device, we only need to add one more profile. Smart.

6.3.3 Gamut Mapping

Gamut mapping is an essential process in color space conversions because the gamuts of any two color spaces are invariably different; sometimes quite different. Gamut mapping attempts the transformation of the colors of an image to fully exploit the color range of the destination color space while

6.3.2 Color Space Conversions

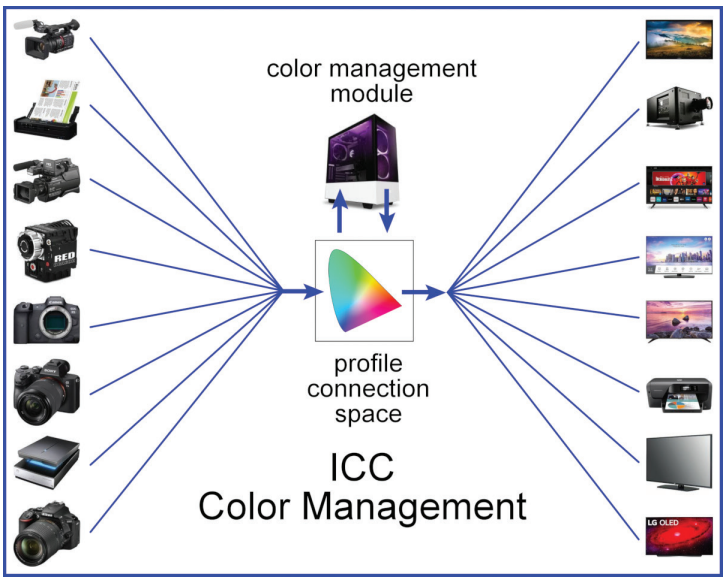


图 6.14 ICC 色彩管理概述

色彩空间转换的一个基本问题是“多进多出”问题。假设我们只有八台输入设备和八台输出设备需要相互转换，那么就需要8×8即64种转换方案。若增加一台设备，就必须额外增加八种转换方案——随着设备数量的增加，所需转换方案的数量会急剧上升。图6.14展示了一种更优方案：每台设备都配有描述其色彩特性的ICC配置文件，这些配置文件可转换为配置文件连接空间（即前文所述的专用广色域色彩空间），色彩管理模块在该空间内执行源配置文件与目标配置文件之间的色彩转换（色彩空间转换）。因此，图6.14中的16台设备仅需16个配置文件，而非64种转换方案。若新增一台设备，只需增加一个配置文件即可。非常巧妙。

6.3.3 色域映射

色域映射是色彩空间转换中必不可少的过程，因为任意两个色彩空间的色域总是不同，有时甚至差异巨大。色域映射旨在将图像色彩进行转换，以充分利用目标色彩空间的色彩范围，同时

maintaining the original artistic intent of its creator. One example is trying to view the ARRI or RED wide-gamut camera footage on a typical workstation monitor. It turns out that this is a non-trivial problem. It is rather like moving all of the furniture and fixtures from one living room to another that is not the same size or shape. The placement in the new room simply cannot be identical to the original living room. Compromises must be made.

Gamut mapping between color spaces with ICC Profiles has issues. If the destination gamut is smaller than the original gamut, it will get clipped. If the destination gamut is larger than the source gamut, it will not fill its full-color potential. Doing the obvious by simply stretching a smaller one into a larger one or squeezing a larger one down to a smaller one invariably introduces hue shifts and brightness changes all over the image. This is a very messy problem with no "right" answer. In an attempt to manage these artifacts, ICC Profiles provide four "rendering intents" with different approaches to the conversion with the intent of preserving the most important aspects of the original look to the degree possible in the new color space. Here is a summary of those rendering intents:

- **Perceptual** - Stretches or squeezes the source into the destination gamut so all colors are affected. Intended for photographs and images, it is usually the default setting as it preserves the overall relationship between the colors. However, going from a large gamut to a small one results in an overall loss of saturation even though the relative saturation between the colors is preserved.
- **Saturation** - Similar to Perceptual but retains the original saturation at the expense of hue and lightness. Like I said, a messy problem.
- **Relative colorimetric** - Maps the source into the destination gamut without squeezing or stretching for pre-press proofing. Colors outside the destination are adjusted to the closest color without disturbing the rest of the image. As a result, two different colors in the original could end up being nearly the same. Does not preserve the white point.
- **Absolute colorimetric** - Used for pre-press proofing, like relative colorimetric, but it preserves the white point, unlike Relative Colorimetric.

When mapping a small gamut to a larger gamut, some colorists will simply drop it into the larger color gamut without remapping. While this does not take full advantage of the larger color space, it does avoid problems.

保持创作者原有的艺术意图。一个例子是尝试在典型的工作站显示器上查看ARRI或RED广色域摄像机拍摄的素材。事实证明，这是一个不简单的问题。这就像把所有的家具和固定装置从一个客厅搬到另一个尺寸或形状不同的客厅。新房间的布局根本无法与原客厅完全相同。必须做出妥协。

使用ICC配置文件在不同色彩空间之间进行色域映射存在一些问题。如果目标色域小于原始色域，色彩会被裁剪；如果目标色域大于源色域，则无法充分发挥其全彩潜力。若采用简单直接的方法——将较小色域拉伸至较大色域，或将较大色域压缩至较小色域——都会不可避免地导致图像整体出现色相偏移和亮度变化。这是个非常棘手的难题，不存在“正确”的解决方案。为了控制这些伪影，ICC配置文件提供了四种“渲染意图”，通过不同的转换方法，尽可能在新色彩空间中保留原始视觉效果最重要的特征。以下是这些渲染意图的概述：

- 感知 - 将源色域拉伸或压缩至目标色域，使所有颜色均受影响。此模式专为照片和图像设计，通常作为默认设置，因为它能保持颜色间的整体关系。然而，当从大色域转换至小色域时，即使颜色间的相对饱和度得以保留，整体饱和度仍会降低。

- 饱和度 - 与感知模式类似，但以牺牲色调和明度为代价保留原始饱和度。正如我所说，这是个棘手的问题。

- 相对色度映射 - 将源色域映射到目标色域，不进行压缩或拉伸，适用于印前打样。目标色域外的颜色会调整为最接近的颜色，且不影响图像其余部分。因此，原始图像中两种不同的颜色最终可能变得几乎相同。不保留白点。

- 绝对色度 - 用于印前打样，与相对色度类似，但不同于相对色度的是，它会保留白点。

将小色域映射到大色域时，部分调色师会直接将其置入大色域而不进行重映射。这种做法虽未能充分利用更大的色彩空间，却能避免一些问题。

FUN FACT: The ICC Perceptual rendering intent is the most sophisticated, as it literally adjusts all the colors of an image in an attempt to fit a larger gamut into a smaller gamut by mapping the out-of-gamut colors to the nearest reproducible color in such a way as to maintain the overall visual relationship between the colors.

6.4 Working with Supersized Gamuts

Working with the photography supersized work spaces Adobe RGB and ProPhoto RGB from Chapter 4 introduces two non-trivial problems. The first is simply the ability to see your work. Since standard sRGB monitors don't fit the CMYK gamut very well, some colors can be displayed but not printed, while other colors can be printed but not displayed. This is mighty vexing for a photographer trying to deliver high-quality images for print. The second problem is that these gamuts are not themselves printable, so the work must be converted from the supersized workspace and shrunk to the much more limited display space of the final print process. And as we have seen, that conversion is fraught with danger.

6.4.1 Viewing Supersized Gamuts

There are two approaches to viewing the supersized gamuts. The first is to use software that tries to squeeze them into the gamut of a regular sRGB monitor and special monitors that can actually display them. There are special wide-gamut Adobe RGB print monitors available that can correctly display all of the SWOP CMYK colors, but they are very expensive. However, as of the time of writing, there are no monitors available that fully support ProPhoto RGB. It is simply beyond the current display technology. The second approach is to use special software that converts the work to your display's color space for sRGB or Adobe RGB monitors. These offer additional controls over the basic ICC Profile settings and attempt to simulate the actual appearance of the final print, even factoring in the paper stock to be used.

The techniques for downsizing the workspace into the print display space are tone mapping and gamut mapping. Many recommend tone mapping

有趣的事实：ICC感知渲染意图是最为复杂的，因为它实际上会调整图像的所有颜色，试图通过将超出色域的颜色映射到最接近的可再现颜色，从而将更大的色域适配到更小的色域中，同时保持颜色之间的整体视觉关系。

6.4 处理超大色域

使用第4章中介绍的Adobe RGB和ProPhoto RGB这两种超大型摄影工作空间会带来两个不小的问题。首先是能否看到你的作品。由于标准的sRGB显示器与CMYK色域匹配得并不理想，有些颜色可以显示但无法打印，而另一些颜色可以打印却无法显示。

显示出来。这对于试图提供高质量打印图像的摄影师来说非常令人烦恼。第二个问题是这些色域本身无法打印，因此作品必须从超大的工作空间转换，并缩小到最终打印过程中更为有限的显示空间。正如我们所看到的，这种转换充满了风险。

6.4.1 查看超大色域

有两种方法可以查看超大色域。第一种是使用软件尝试将它们压缩到普通sRGB显示器的色域范围内，或者使用能够实际显示这些色域的特殊显示器。市面上有专门的广色域Adobe RGB印刷显示器可以正确显示所有SWOP CMYK颜色，但价格非常昂贵。然而，截至本文撰写时，尚无任何显示器能完全支持ProPhoto RGB色域，这已超出当前显示技术的极限。第二种方法是使用特殊软件，将作品转换到您显示器（sRGB或Adobe RGB显示器）的色彩空间。这些软件提供了超出基本ICC配置文件设置的控制选项，并尝试模拟最终印刷品的实际外观，甚至将所用纸张类型也纳入考量因素。

将工作空间缩小至打印显示空间的技术包括色调映射与色域映射。多数建议采用色调映射

first, followed by gamut mapping. However, there are religious differences on this subject—as evidenced by the many white papers and books available on the subject, each claiming superior results.

6.4.2 Remapping Supersized Gamuts

You have gone through all of the trouble of editing your photos in the supersized Adobe RGB color space for the best quality, but now it's time to send it to the printer. As made clear from the gamut maps in Figure 6.15, the supersized Adobe RGB gamut will have to be remapped to the small and awkward SWOP CMYK gamut using the ICC Profiles outlined above, yet retain the appearance of the original work. Easier said than done. The fundamental reason this is such a difficult problem is that the objective is to maintain the same visual appearance from the monitor to the print media, knowing all the while from the different gamuts that they don't fit each other. The Adobe RGB has saturated greens and deep blue magentas that print simply cannot replicate.

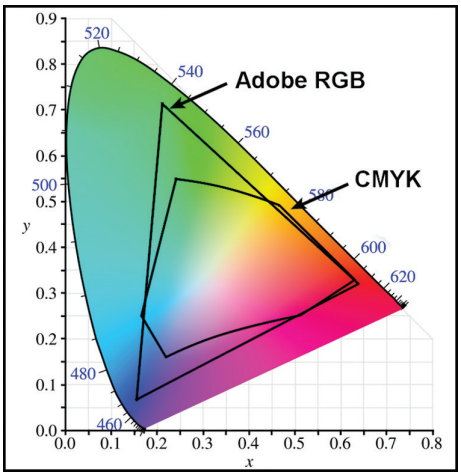


Figure 6.15 Adobe RGB gamut compared to SWOP CMYK

6.4.2.1 Large to Small Gamuts

There are two approaches to remapping large to small gamuts. The first is clipping—just chop out the extra color, and the second is compression—squeeze the extra color back into the destination gamut. The problem with clipping is that the clipped pixels can't go away—they are remapped to the nearest boundary color. This can lead to a loss of detail in saturated areas. The problem with compression is that nearby colors that are in gamut can be altered. Detail is not lost, but hue and saturation may take a hit. As I said, this is a non-trivial problem. All of these issues can be avoided if the image in the source color space does not extend beyond the destination color space

首先，进行色域映射。然而，在这一问题上存在诸多不同的观点——正如该主题下众多白皮书和书籍所展示的那样，每份资料都声称自己的方法能取得更优的结果。

6.4.2 重映射超大色域

您已经历了在超大Adobe RGB色彩空间中编辑照片以获得最佳质量的所有麻烦，但现在是将它发送到打印机的时候了。如图6.15中的色域图所示，超大的Adobe RGB色域必须使用上述概述的ICC配置文件，重新映射到小而笨拙的SWOP CMYK色域，同时保留原始作品的外观。说起来容易做起来难。这之所以成为一个如此棘手的问题，根本原因在于目标是在保持相同视觉{v*}

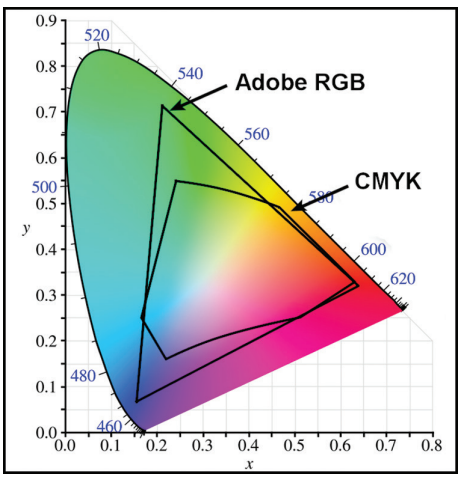


图6.15 Adobe RGB色域与SWOP CMYK色域对比

从显示器到印刷媒体的外观呈现，我们始终清楚，由于不同的色域范围，它们并不相互匹配。Adobe RGB拥有饱和的绿色和深邃的蓝紫色，这些色彩是印刷根本无法复制的。

6.4.2.1 从大色域到小色域

将大色域映射到小色域有两种方法。第一种是**裁剪**——直接切除超出范围的色彩；第二种是**压缩**——将超出范围的色彩挤压回目标色域内。裁剪的问题在于被裁切的像素并不会消失——它们会被重新映射到最接近的边界色，这可能导致高饱和度区域的细节丢失。压缩的问题则在于原本处于色域内的邻近色彩也可能被改变，虽然细节得以保留，但色调与饱和度可能受损。正如前文所言，这是个需要慎重处理的难题。如果源色彩空间中的图像未超出目标色彩空间的范围，所有这些问题都可以避免。

to begin with. The win here is that the Adobe RGB monitor will display those colors that are outside of the sRGB gamut (see Chapter 4, Figure 4.32).

6.4.2.2 Small to Large Gamuts

Earlier in Section 6.3.3, we ran through the four ICC Profile rendering intents—Adobe's answer to mismatched gamuts. But there are other options available in Color Management Modules (CMMs) in operating systems, many software packages like Photoshop and DaVinci Resolve, and standalone apps that go beyond the four rendering intents.

There are a variety of approaches to remapping a small gamut image into a larger gamut such as sRGB to DCI-P3. The first and easiest approach is to do nothing—just import the image as is into the large gamut color space and move on. While it can produce acceptable results, this approach obviously does not take advantage of the display potential of the larger destination gamut. As we learned, dropping the same code values into a new color space results in changing the color of the image (per Section 6.3.1).

Changing the color of the source image is inherently unavoidable but you can choose how the color will be affected. Here are some of those choices:

- **Perceptual scaling** - Scales the colors in such a way as to maintain the same overall "look" of the source image. Applicable when maintaining the overall look is more important than preserving exact color values.
- **Selective color enhancement** - The operator chooses which colors to expand over others. Normally colors are selected that are known to be more vibrant in the real world, such as blue skies and green grasses.
- **Saturation mapping** - Increases saturation (color "purity") to better utilize the extended color palette of the destination gamut. This can be done globally or in selected regions, being careful to not introduce unnaturally vivid colors.
- **Dynamic range expansion** - A color space with a larger gamut will invariably also have a greater dynamic range, so increasing the dynamic range with tone mapping is very effective. Oh look—tone mapping is the very next section!

首先，这里的优势在于Adobe RGB显示器能够显示那些超出sRGB色域范围的颜色（参见第4章图4.32）。

6.4.2.2 小到大的色域

早在6.3.3节中，我们介绍了四种ICC配置文件的渲染意图——这是Adobe针对色域不匹配问题提出的解决方案。但除了这四种渲染意图外，操作系统中的色彩管理模块（CMM）、许多软件包（如Photoshop和DaVinci Resolve）以及独立应用程序还提供了其他选项。

将小色域图像重新映射到更大色域（例如从sRGB到DCI-P3）存在多种方法。第一种也是最简单的方法是保持原状——直接将图像导入大色域色彩空间并继续处理。虽然这种方法可能产生可接受的结果，但显然无法充分利用目标大色域的显示潜力。正如我们所知，将相同的编码值直接置入新色彩空间会导致图像颜色发生变化（参见第6.3.1节）。

改变源图像的颜色本质上是不可避免的，但你可以选择颜色将如何受到影响。以下是一些可选方案：

- 感知缩放 - 以保持源图像整体"观感"不变的方式调整颜色。适用于维持整体视觉效果比保留精确颜色值更重要的场景。

- 选择性色彩增强 - 操作者选择将哪些颜色扩展至其他颜色之上。通常选取的是现实中已知更为鲜艳的色彩，例如蓝天和绿草。

- 饱和度映射 - 提升饱和度（色彩“纯度”），以更好地利用目标色域的扩展色彩范围。此操作可全局应用或针对选定区域实施，需注意避免引入不自然的鲜艳色彩。

- 动态范围扩展——色域更大的色彩空间必然也拥有更大的动态范围，因此通过色调映射来增加动态范围非常有效。看啊——色调映射正是下一节的内容！

- **3D LUTs** - Particularly useful for remapping colors from a smaller to larger gamut because they allow color grading based on hue, saturation, and luminance levels. The image can be "dialed in" to look fabulous in the larger destination color space.
- **Manual adjustments** - Basically, going in and color grading the output to taste, usually after using one of the above procedures to do most of the work. Just be careful to maintain natural skin tones and avoid over-saturation.

The bottom line is that the remapping of gamuts often does not work very well and will usually need some post-conversion color grading to restore the original artistic intent.

6.4.3 Tone Mapping

Tone mapping adjusts the luminance of a high dynamic range image to fit within a lower dynamic range while retaining the original visual intent. An example would be to adjust a DSLR photograph with 12 stops of dynamic range to fit into an sRGB monitor with 8 stops of dynamic range without clipping, like the example in Figure 6.16. Note that this is a dynamic range change, not a color space change, so no effect on the gamut. Changing the dynamic range is simply another way of saying "to make it brighter or darker," which is why it has no effect on the gamut. Of course, these techniques could also be used to remap an image from a small to a larger dynamic range, but the majority of use cases will be from HDR to LDR.

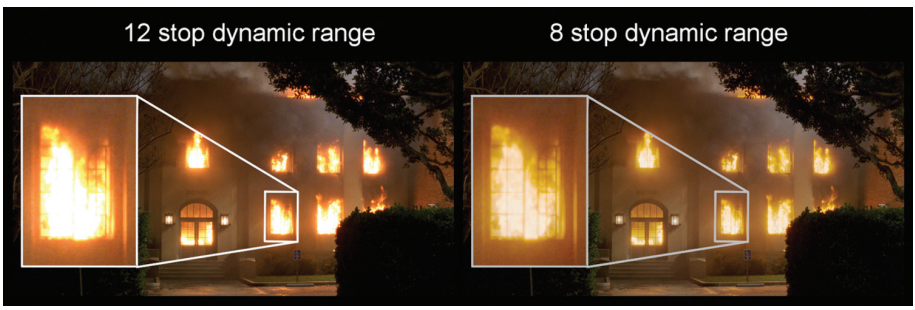


Figure 6.16 A HDR image tone mapped to a lower dynamic range

The fire in the 12-stop image on the left (see close-up window) is horribly clipped, resulting in a solid block of white completely lacking any detail plus it displays orange color banding. The tone-mapped version on the right has

- 3D LUT（查找表）——特别适用于将色彩从小色域重新映射到大色域，因为它们允许基于色调、饱和度和亮度级别进行色彩分级。通过调整，图像可以在更大的目标色彩空间中呈现出惊艳效果。

- 手动调整 - 基本上，就是在使用上述任一流程完成大部分工作后，根据个人喜好对输出画面进行调色。只需注意保持自然的肤色，并避免过度饱和。

关键在于，色域重映射通常效果不佳，往往需要后期转换的色彩分级来还原原始的艺术意图。

6.4.3 色调映射

色调映射调整高动态范围图像的亮度，使其适应较低动态范围，同时保留原始视觉意图。例如，将具有12档动态范围的单反相机照片调整至适合仅8档动态范围的sRGB显示器，且不产生裁切，如图6.16所示。需注意这是动态范围的变化，而非色彩空间的转换，因此不会影响色域。改变动态范围本质上等同于“调亮或调暗图像”，这正是其不影响色域的原因。当然，这些技术也可用于将图像从小动态范围重新映射到大动态范围，但大多数应用场景仍是从高动态范围转换到低动态范围。

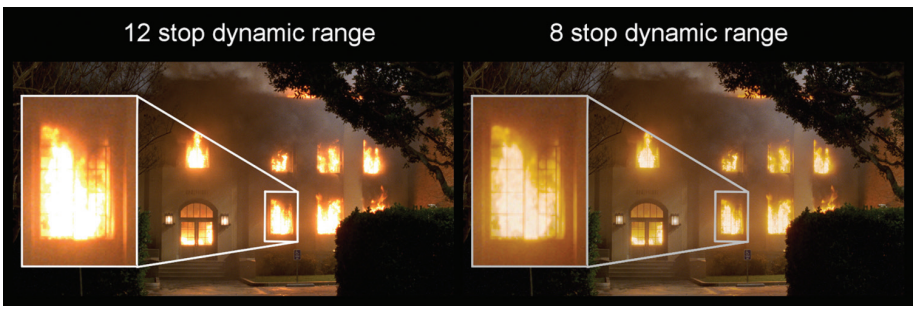


图6.16 一张高动态范围图像色调映射至较低动态范围

左侧12档图像中的火焰（见特写窗口）被严重裁剪，导致形成一块完全缺乏细节的纯白色块，并且还显示出橙色的色带。右侧经过色调映射的版本则

been brought down to 8 stops, which fits nicely on a standard sRGB monitor. There is plenty of detail in the fire, no color banding, and the window bars are completely visible—all while retaining almost all of the dark and midtone detail unchanged.

Both photo editing software for photography and image processing software for visual effects have tone mapping tools. Some tone mapping tools are referred to as "global" operators as they affect the overall image without consideration of local details. "Local" operators consider the nearby pixels to calculate the new luminance so are adaptive to the picture content. When allocating some of the luminance headroom to the high dynamic range elements, you may have to give up some of the midtone space for it.

6.4.3.1 Soft Clip LUT

Tone mapping from high dynamic range to lower dynamic range requires the services of a soft clip LUT to gently "bend" the tone scale down to fit within the lower dynamic range. If you don't have tone mapping software or prefer the personal touch of doing your own thing, you can create your own soft clip with a standard LUT curve available in virtually all software. Since we are dealing with code values greater than 1.0, you must be working in float, of course. As you may recall from Section 6.1.1, LUT curves remap the input code values along the bottom axis to the output code values on the vertical axis. The soft clip LUT remaps the largest code value in the image down to 1.0, the upper limit of the display, whether that display is a monitor or a print. This is a three-channel LUT, so R, G, and B are getting the same squeeze.

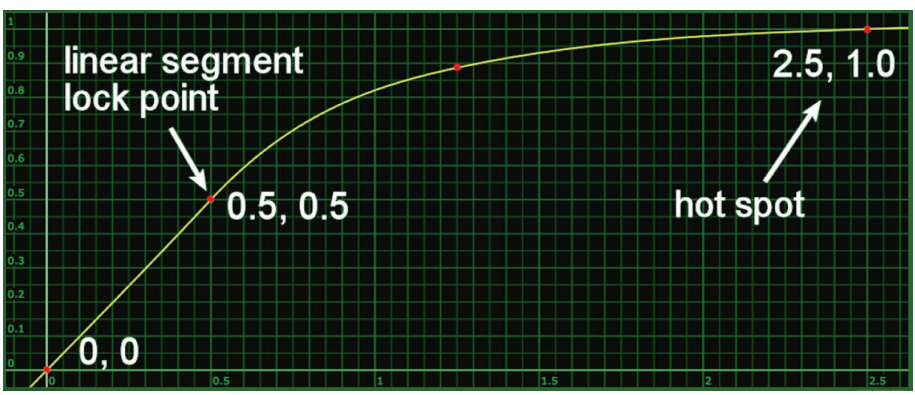


Figure 6.17 Soft clip LUT for tone mapping

已降至8档，这恰好适配标准的sRGB显示器。火焰中保留了丰富的细节，没有色彩条带，窗框也完全清晰可见——同时几乎完整保留了所有暗部与中间调的细节。

无论是用于摄影的照片编辑软件，还是用于视觉特效的图像处理软件，都配备了色调映射工具。部分色调映射工具被称为“全局”算子，因为它们会影响整个图像，而不考虑局部细节。“局部”算子则会考虑邻近像素来计算新的亮度，从而适应图片内容。当将部分亮度余量分配给高动态范围元素时，你可能需要为此放弃一些中间调空间。

6.4.3.1 软剪裁查找表

从高动态范围到低动态范围的色调映射需要软剪辑查找表的服务，以温和地“弯曲”色调尺度，使其适应较低的动态范围。如果你没有色调映射软件，或者更喜欢亲自动手打造个性化效果，你可以利用几乎所有软件中都提供的标准查找表曲线来创建自己的软剪辑。由于我们处理的是大于1.0的编码值，你当然必须在浮点环境下工作。正如你可能从第6.1.1节回忆起的，查找表曲线将底部轴上的输入编码值重新映射到垂直轴上的输出编码值。软剪辑查找表将图像中的最大编码值重新映射至1.0——这是显示器（无论是监视器还是打印品）的上限。这是一个三通道查找表，因此红、绿、蓝通道会受到相同的压缩处理。

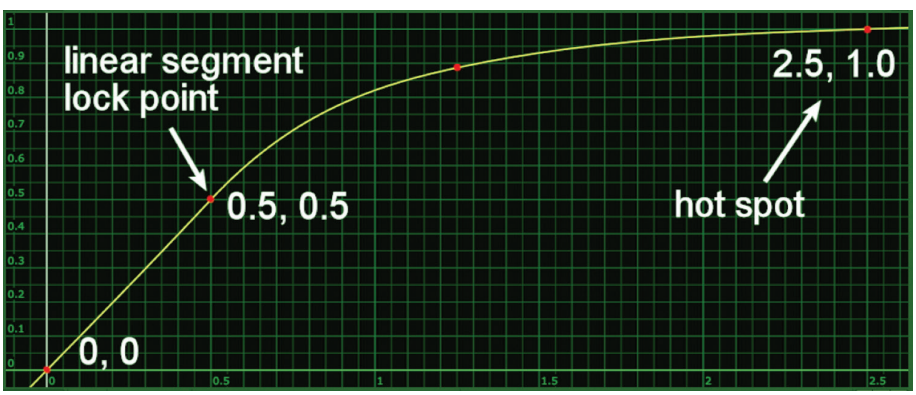


图 6.17 用于色调映射的软裁剪查找表

Using the burning building photo in Figure 6.16 as an example, first determine the "hot spot," meaning the brightest code value to be preserved—in this case, it's code value 2.5—which is easily found with a waveform monitor or pixel analyzing tools. You can narrow the search by slamming the viewer gamma down to zero so all pixels black out except for the hot ones. Referring to the soft clip LUT in Figure 6.17, plant a control point at the hot spot—namely 2.5, 1.0 which pulls the code value 2.5 down to 1.0. The image gets darker but is no longer clipped and the RGB values in the highlight region will converge toward the curve, reducing their saturation.

To retain normal brightness for the rest of the image, plant a control point at the upper end of the code values that are to be left unchanged. The objective here is to create a linear segment—the linear portion of the LUT that will not alter brightness. In this example, the upper limit was 0.5, so the linear segment lock point is set to 0.5, 0.5. All code values between 0 and 0.5 are now protected from change. Keep in mind that in linear space 50% gray is code value 0.18. It will likely be necessary to add another control point or two between the linear lock point and the hot spot in order to shape or sculpt the falloff of the curve for best results. A tip here is to make sure that there is still some slope to the line at the last control point—2.5, 1.0 in this case—to retain some detail. If the curve goes horizontal here, the hot spots will all get the same code values, resulting in a flat slab of solid color with no internal detail.

See how to create a 1D LUT to tone map a high dynamic range a low dynamic range.

6.5 Color Pipelines

Color pipelines address the question of how the color is managed throughout the job, and there are two general cases. The first case is Display Space, where the entire pipeline stays in a display space such as Rec. 709 for video or sRGB for the web. The other case is Color Managed, where the pipeline has multiple color spaces and grade files that are cobbled together specific to a show. Here we will look at both types.

以图6.16中的燃烧建筑照片为例，首先确定“热点”，即需要保留的最亮代码值——本例中为代码值2.5——可通过波形监视器或像素分析工具轻松定位。通过将查看器伽马值骤降至零，可使除热点外的所有像素变黑，从而缩小查找范围。参照图6.17中的软剪辑查找表，在热点处（即2.5, 1.0坐标）设置控制点，将代码值2.5下拉至1.0。图像会变暗但不再被裁剪，高光区域的RGB值将向曲线收敛，从而降低其饱和度。

为了保持图像其余部分的正常亮度，在需要保持不变的代码值上端设置一个控制点。此处的目标是创建一个线性段——即查找表（LUT）中不会改变亮度的线性部分。在本例中，上限设为0.5，因此线性段锁定点设置为0.5、0.5。现在，0到0.5之间的所有代码值都受到保护而不会被更改。请注意，在线性空间中，50%灰度的代码值为0.18。通常需要在线性锁定点与高光区域之间再添加一两个控制点，以调整曲线的衰减形态，从而获得最佳效果。这里有个技巧：确保最后一个控制点（本例中为2.5、1.0）处的曲线仍保留一定斜率，以维持部分细节。若曲线在此处变为水平，所有高光区域将获得相同的代码值，导致形成缺乏内部细节的纯色平板区域。

了解如何创建一维LUT对高动态范围图像进行色调映射低动态范围。

6.5 色彩处理管线

色彩管线解决了色彩在整个作业过程中如何管理的问题，通常存在两种通用情况。第一种情况是显示空间，整个管线保持在显示空间内，例如视频采用Rec.709标准，网络内容采用sRGB标准。另一种情况是色彩管理，管线涉及多个色彩空间和针对特定节目拼接而成的调色文件。接下来我们将探讨这两种类型。

6.5.1 Display Space

The biggest example of a Display Space color pipeline is video. A Rec. 709 video camera shoots the scene; then the video is revised (e.g., edited, color corrected, painted, VFX) on a Rec. 709 workspace and displayed on a Rec. 709 display device like HDTV. The artist can blithely ignore all color management issues (assuming a calibrated monitor), as these have been taken care of by the very smart color scientists who designed the video camera and the display monitor so you don't have to. You, the intrepid artist, are absolved. The only color management issue you need to worry about is generating out-of-gamut colors while color grading or compositing. To save you from even this minor exposure, most software will have scopes and or zebra stripes to alert you to out of gamut colors.

Another use case for working in Display Space is sRGB for the web. If you are a graphic designer or a motion graphics artist with the web as your Display Space, you can work in sRGB Display Space for the entire job. Like Rec. 709, you are working in a consistent color pipeline, so don't have anything to worry about. Except yourself, of course. You can easily introduce over-gamut colors with color grading and compositing effects which will become clipped and look awful.

Photographers have it fairly easy too. If the photograph is headed for display on the web, the picture can be exported from the camera in sRGB color space, worked on in sRGB color space, and then displayed on an sRGB monitor. Suppose the picture is to go to high-end print. In that case, the photographer will typically export the images out of the camera as Adobe RGB and edit them in an Adobe RGB workspace, while (hopefully) viewing them on an Adobe RGB display. Again, all in the same color space—namely Adobe RGB. The only color space conversion will be to convert the final RGB image to CMYK for the printer.

6.5.2 Color Managed

However, if your work is in visual effects for feature films or episodic TV, you cannot work in a Display Space color pipeline. Here the quality requirements are much higher and the desire for creative control and future-proofing means that the camera raw will go through many color stages that must be managed very carefully. Big-budget episodic TV often uses the same types

6.5.1 显示空间

显示空间色彩流程的最大示例是视频。Rec. 709摄像机拍摄场景；随后视频在Rec. 709工作空间中进行修订（如剪辑、调色、绘制、视觉特效），并在Rec. 709显示设备（如高清电视）上播放。艺术家可以全然忽略所有色彩管理问题（假设显示器已校准），因为这些已由设计摄像机和显示器的聪明色彩科学家妥善处理，无需您费心。您，这位无畏的艺术家，就此免责。唯一需要担心的色彩管理问题是在调色或合成时产生超色域色彩。为避免您连这点风险都需承担，多数软件会配备示波器和/或斑马纹警示超色域色彩。

在显示空间工作的另一个用例是用于网络的sRGB。如果您是一名图形设计师或动态图形艺术家，将网络作为您的显示空间，您可以在整个工作中使用sRGB显示空间。与Rec.709类似，您在一个一致的颜色管道中工作，因此无需担心任何问题。当然，除了您自己。您很容易通过色彩分级和合成效果引入超出色域的颜色，这些颜色将被裁剪并显得非常糟糕。

摄影师的工作也相对轻松。如果照片要用于网络展示，可以直接从相机中以sRGB色彩空间导出，在sRGB色彩空间中进行处理，最后在sRGB显示器上显示。假设照片需要用于高端印刷，摄影师通常会以Adobe RGB格式从相机导出图像，并在Adobe RGB工作空间中编辑，同时（最好）在Adobe RGB显示器上查看。整个过程始终处于同一色彩空间——即Adobe RGB。唯一的色彩空间转换仅发生在最终将RGB图像转为印刷所需的CMYK格式时。

6.5.2 色彩管理

然而，如果你的工作涉及电影长片或电视剧集的视觉特效，你就无法在显示空间色彩流程中工作。这里的质量要求要高得多，对创意控制和未来适用性的追求意味着相机原始数据需经过多个必须精心管理的色彩处理阶段。大预算的电视剧集通常采用相同类型的

of cameras and color pipeline that are used for feature films. Following is a common workflow I have encountered for Paramount and Amazon productions, but there could be other variations.

Assuming an ARRI Alexa camera for this example, the camera raw will be creatively debayered by the colorist and then the linear images exported to an OpenEXR file. You can follow the action in Figure 6.18. The colorist will include the AlexaLogC LUT, so the color grading may be done in log space. Of course, every camera manufacturer has developed its own log format optimized to its camera and guaranteed to give you a much better image than the other guy's camera, but we are taking AlexaLogC as a popular working example. Following the Log LUT is the Color Decision List (CDL), which contains the color grade specific to that shot. Then finally, the Show LUT is added, which contains global "look" LUTs for the entire show plus the color space conversion to Rec. 709 or P3 for digital cinema.

Figure 6.18 A color pipeline using OCIO

The log LUT, CDL, and Show LUT are not baked into the OpenEXR images but instead are sent along with the EXR files, hoping the downstream party knows how to apply them and in what order. Each downstream artist will combine the LUTs and the CDL into a **Viewer LUT** to show the final color-graded image. This way, the original data is never altered and the viewer LUT can be updated later and shared with others.

This color managed pipeline is very powerful and flexible, allowing the production to set up any desired color space and controls. And therein lies the problem. Every production is unique, so the chances of mistakes and confusion are rife—especially as the project moves between departments and even between studios. Troubleshooting color problems is very time-consuming, so some sort of structure and control is essential, and that is what OpenColorIO is all about.

用于故事片的相机和色彩流程。以下是我在派拉蒙和亚马逊制作中遇到的常见工作流程，但也可能存在其他变体。

以ARRI Alexa相机为例，调色师会对相机原始素材进行创意性去马赛克处理，随后将线性图像导出为OpenEXR文件。您可参照图6.18的流程。调色师会加载AlexaLogC LUT，以便在对数空间进行调色。当然，每家相机厂商都开发了适配自身设备的专属对数格式，并保证其成像效果远胜竞品，此处我们仅以广泛应用的AlexaLogC作为示例。对数LUT之后是色彩决策表（CDL），其中包含针对特定镜头的调色参数。最后加载的是项目LUT，它既包含适用于整部作品的全局“风格”LUT，也包含转换为Rec.709或P3数字影院色彩空间的映射。

图 6.18 使用 OCIO 的色彩处理管线

对数LUT、CDL和显示LUT并未直接烘焙到OpenEXR图像中，而是随EXR文件一同发送，期望下游方知晓如何应用及处理顺序。每位下游艺术家会将LUT与CDL合并为查看器LUT，以呈现最终调色画面。这种方式既保留了原始数据不被修改，又便于后续更新查看器LUT并与他人共享。

这种色彩管理流程非常强大且灵活，允许制作团队设置任何所需的色彩空间和控制参数。但问题也正源于此。每个制作项目都具有独特性，因此极易出现错误和混淆——尤其是当项目在不同部门甚至不同工作室之间流转时。排查色彩问题极其耗时，因此建立某种规范化的结构和控制机制至关重要，而这正是OpenColorIO的核心价值所在。

6.6 OpenColorIO (OCIO)

The color managed story with the ARRI Alexa shot neatly illustrates the need for OCIO. That workflow required the AlexaLogC LUT, a Color Decision List, and a Show LUT to be passed to downstream artists and then correctly applied to the work. This is a complicated and error-prone workflow. OCIO solves this problem by centralizing color management and making it accessible to the downstream artists. It also offers a standardized library of LUTs and math operations used by the downstream players so that color operations will be consistent between them.

In the production of visual effects and CG animation, the work migrates between several different software apps, departments, and even studios, each managing color its own way. For example, one app may convert images from log to linear with a math expression while another app uses a LUT, so the results are not identical. Also, color space conversions such as converting AlexaLogC to Rec. 709 are not consistent between studios. Everybody has their own math, LUTs, and tables. What was needed for the modern distributed production pipeline was for everybody to use the same math for the same operations. Enter OpenColorIO (OCIO), developed by Sony Picture Imageworks, then gifted to the Academy Software Foundation (ASF) as open-source code to propagate throughout the industry. And propagate it has.

6.6.1 Implementation

Today, all major software providers—such as Nuke, Maya, Houdini, Silhouette, DaVinci, Unreal Engine, After Effects, Photoshop, and many others—support OCIO. The "OpenColorIO-configs" file is downloaded free from GitHub and then connected to the software, which contains code to interface with the OCIO library. OCIO color operations and LUTs can be called from within the software app instead of using the app's native color operations. Now, when an image is converted from AlexaLogC to Rec. 709 in one studio, it will be identical to the conversion in another studio because we have all used the same OCIO library for the conversion.

The color managed workflow above correctly depicted the right color operations in the right order, but the problem was that the LUTs and CDL were loose files passed around to the various artists for each of them to build their own viewer LUT. Not a good arrangement. With OCIO, those files are

6.6 OpenColorIO (OCIO)

ARRI Alexa拍摄的色彩管理案例清晰地说明了OCIO的必要性。该流程需要将AlexaLogC LUT、色彩决策表和放映LUT传递给下游制作人员，并确保其正确应用于作品中。这是一个复杂且易出错的工作流程。OCIO通过集中管理色彩配置，并使其对下游制作人员可访问，从而解决了这一问题。它还提供了下游使用方通用的标准化LUT库与数学运算方法，确保各环节的色彩处理保持一致。

在视觉效果和CG动画制作中，工作流程需要在多个不同的软件应用、部门甚至工作室之间迁移，每个环节都以各自的方式管理色彩。例如，某个应用可能通过数学表达式将图像从对数转换为线性，而另一个应用则使用LUT（查找表），导致结果并不一致。此外，像将AlexaLogC转换为Rec.709这类色彩空间转换在不同工作室间也存在差异——每家都有自己独有的数学公式、LUT和转换表。现代分布式制作流程需要的是让所有环节对相同操作使用统一的数学转换标准。于是OpenColorIO（OCIO）应运而生，它由索尼影业图像工作室开发，后作为开源代码赠予学院软件基金会（ASF），旨在全行业推广统一标准。而这一标准确实已得到广泛传播。

6.6.1 实现

如今，所有主流软件供应商——如Nuke、Maya、Houdini、Silhouette、DaVinci、Unreal Engine、After Effects、Photoshop等众多产品——均已支持OCIO。用户可从GitHub免费下载“OpenColorIO-configs”配置文件，并将其关联至相应软件，该文件内含与OCIO库对接的代码。此后，在软件应用程序中可直接调用OCIO色彩运算与LUT，而无需依赖软件原生的色彩处理功能。这意味着当图像在某工作室从AlexaLogC转换为Rec.709时，其转换结果会与其他工作室完全一致——因为我们都在使用相同的OCIO库进行转换。

上述色彩管理流程正确地描绘了正确的色彩操作顺序，但问题在于LUT和CDL是松散的文件，分发给各位艺术家，让他们各自构建自己的查看器LUT。这不是一个好的安排。使用OCIO后，这些文件是

bundled together by the colorist and/or a pipeline guy and named so the artist simply selects “Shot 0038” from a list as their viewer LUT. Done.

When the shot is finalized, then and only then are all of the LUTs and color grades baked into the image to produce the display-referred version, which is Rec. 709 for HDTV in this example. To create the P3 version for digital cinema projection, the Rec. 709 LUT is simply replaced by a P3 LUT. OCIO is used to apply the LUTs from its standard library and the color grade from an OCIO CDL (Color Decision List) created by the colorist. Here we will see how all this is done.

6.6.2 Color Space Conversions

OCIO comes with a large library of color space conversions called “profiles” (not the same as ICC Profiles) that handle the color space transformations. The profiles are used when importing images, for display LUTs, during processing for coloring and compositing, and the final output is deliverable to a variety of display devices, each requiring its own profile.

Importantly, OCIO enforces a strict policy of **invertibility** for color space transforms such that an image may, for example, be converted from linear to DCI-P3 and back many times with no degradation of the original data. Non-invertible operations do have their place. One example of non-invertible operations is 3D LUTs. Non-invertible transforms are used only for final output and display since they do not have to be reversed.

6.6.3 Roles

Let’s say that when importing a matte painting, it needs a color space conversion from sRGB to linear. Rather than trying to train all the artists to use sRGB for all matte paintings, Rec. 709 for all video, and linear for all EXR files, OCIO allows you to define custom “role” names for each of these. Now the artist simply uses the role names “matte painting,” “video,” or “OpenEXR” when converting these files.

Not only does this avoid an artist from using the wrong conversion operation, but the role names can be redefined for each project. For example, if the next project uses Rec. 2020 video instead of Rec. 709, the artists do not have to be retrained to remember to use Rec. 2020 instead of Rec. 709. They simply use the “video” role, which has been redefined as Rec. 2020 for that project.

由调色师和/或流程人员打包在一起并命名，这样艺术家只需从列表中选择“Shot 0038”作为其监看LUT。完成。

当镜头最终确定时，所有LUT和色彩分级才会被烘焙到图像中，生成显示参考版本，本例中即为用于高清电视的Rec. 709标准。若要创建数字影院投影所需的P3版本，只需将Rec. 709 LUT替换为P3 LUT。OCIO用于从其标准库中应用LUT，并加载调色师创建的OCIO CDL（色彩决策列表）中的色彩分级数据。接下来我们将具体演示这一完整流程。

6.6.2 色彩空间转换

OCIO附带一个庞大的色彩空间转换库，称为“配置文件”（不同于ICC配置文件），用于处理色彩空间转换。这些配置文件在导入图像、应用显示查找表、进行调色与合成处理时被使用，最终输出可适配多种显示设备，每种设备都需要其专属配置文件。

重要的是，OCIO强制执行色彩空间转换的可逆性严格政策，确保图像能够多次在例如线性空间与DCI-P3之间转换而不会造成原始数据损失。不可逆操作也有其适用场景，例如3D LUTs就属于不可逆操作。不可逆转换仅用于最终输出和显示环节，因为这些环节无需进行反向转换。

6.6.3 角色

假设在导入遮罩绘画时，需要将色彩空间从sRGB转换为线性。与其试图培训所有艺术家对遮罩绘画统一使用sRGB、对视频统一使用Rec. 709、对EXR文件统一使用线性空间，OCIO允许您为每种类型定义自定义的“角色”名称。现在艺术家在转换这些文件时，只需使用“遮罩绘画”、“视频”或“OpenEXR”等角色名称即可。

这不仅避免了艺术家使用错误的转换操作，而且角色名称可以在每个项目中重新定义。例如，如果下一个项目使用Rec. 2020视频而非Rec. 709，艺术家无需重新培训以记住使用Rec. 2020而非Rec. 709。他们只需使用“视频”角色，该角色在该项目中已被重新定义为Rec. 2020。

Eliminating mistakes—especially color mistakes—can save a lot of time, money, and gnashing of teeth.

6.6.4 Display Transforms

What if Studio A is using sRGB monitors and Studio B is using Rec. 709 monitors? No worries—OCIO has got you covered. Studio A will load the OCIO sRGB display LUT for its viewer, while Studio B will use the OCIO Rec.709 LUT for its viewer. Within the physical limits of the two different displays, the images will appear identical—assuming calibrated monitors.

6.6.5 Look Dev

When the colorist wants to create the "look" for a project, it is given a text name like **Placid Fury** and placed in an XML file that supports color grading and LUTs. This file is then deposited in the central OCIO CONFIG FILE. Downstream artists can now view the project with the new look by loading the **Placid Fury** look transform. Also, the look can simply be turned off when the downstream artist needs to see the original footage. Having central control over look development also prevents some rogue artist from intervening in the color pipeline, either accidentally or deliberately. It also absolves the working artist from having to worry about color management.

6.6.6 Custom LUTs

If a colorist wants to create a custom LUT within an app, it is exported as an OCIO Common LUT Format (CLF) XML file. At the next stage of production, the clip is loaded in and OCIO applies the custom LUT using the same math as the colorist, so they are seeing exactly the same thing the colorist saw. Again, the model is that OCIO reads the XML file and performs the color operations instead of the hosting software.

6.6.7 Color Grading

But what about color grading with lift, gamma, gain, and offset? OCIO offers interoperable consistency by using this standard equation regardless of what app is doing the grade:

$$\text{pixel}_{out} = ((\text{pixel}_{in} * s) + o)\ \hat{}\ p$$

消除错误——尤其是颜色错误——可以节省大量时间、金钱，并避免令人咬牙切齿的烦恼。

6.6.4 显示变换

如果工作室A使用sRGB显示器，而工作室B使用Rec. 709显示器呢？别担心——OCIO已经为您考虑周全。工作室A将在其查看器中加载OCIO sRGB显示LUT，而工作室B则会使用OCIO Rec.709 LUT。在这两种不同显示器的物理限制范围内，只要显示器经过校准，图像看起来将完全一致。

6.6.5 外观开发

当调色师想要为项目创建“外观”时，会为其指定一个文本名称，例如Placid Fury，并将其置于支持色彩分级和LUT的XML文件中。随后，该文件会被存入中央OCIO配置文件中。下游艺术家现在可以通过加载Placid Fury外观变换来查看具有新外观的项目。此外，当下游艺术家需要查看原始素材时，只需关闭此外观即可。对外观开发进行集中控制还能防止某些不守规矩的艺术家意外或故意干预色彩流程。这也让工作中的艺术家无需再为色彩管理而担忧。

6.6.6 自定义查找表

如果调色师想在应用程序内创建自定义LUT，它会以OCIO通用LUT格式（CLF）XML文件的形式导出。在制作的下一阶段，片段被加载进来，OCIO会使用与调色师相同的数学运算来应用自定义LUT，因此他们看到的与调色师所见完全相同。同样，该模型是OCIO读取XML文件并执行色彩操作，而非由宿主软件完成。

6.6.7 色彩分级

但是，使用提升、伽马、增益和偏移进行色彩分级又如何呢？OCIO通过使用这个标准方程提供了互操作性的一致性，无论进行分级的应用程序是什么：

$$pixel_{out} = ((pixel_{in} * s) + o) ^ p$$

In plain English, this says the output pixel is equal to the input pixel times "s" (slope or gain) plus "o" (offset), all raised to the power of "p" (gamma). This equation directly calculates gain, offset, and gamma, and by combining gain and offset, it can also derive a lift operation. OCIO also has a standardized saturation routine to call whenever you want to adjust saturation. The values for s, o, p, and saturation are placed in a little XML file called a CDL (Color Decision List) that goes along with the footage, so that when Studio A applies the grade, it will appear identical to Studio B.

6.6.8 Discipline

OCIO also imposes color management discipline. For example, all texture maps are to be converted to linear prior to mipmap generation. And non-image data such as depth maps, normals, or motion vectors never get any color space conversions for the simple reason they do not represent color. Visual effects and CG animation projects can be safely distributed between apps and studios with confidence that the color intent of the original creators will be preserved because the definition of all the color ops is collected at a single point. Another key virtue of OCIO is that it plays nicely with ACES.

6.7 ACES Color Management

The mightiest of all color management systems is ACES. Released in 2014, it was developed by the Science and Technology Council of the Academy of Motion Picture Arts and Sciences to unify and standardize color management practices, to make it easier to maintain original creative intent and color accuracy across diverse production environments and distribution formats. It supports both live-action and CG imagery in a unified environment.

It is an end-to-end system from capture to display for managing color and protecting the quality of the image. It allows for the seamless integration of multiple images regardless of source—a critical need in today's production, where footage from ARRI and RED cameras may be combined along with some CG, a Photoshop digital matte painting, plus a JPEG from the internet. Even if a project had no CG and was shot with one camera, ACES would still be essential for comprehensive color management.

通俗地说，这意味着输出像素等于输入像素乘以“s”（斜率或增益）加上“o”（偏移量），然后整体再取“p”（伽马）次幂。该方程直接计算增益、偏移和伽马值，通过结合增益和偏移，它还能推导出提升操作。OCIO还提供了一个标准化的饱和度调整例程，可供随时调用以调整饱和度。s、o、p和饱和度的数值存储在一个名为CDL（色彩决策列表）的小型XML文件中，该文件随素材一起传递，因此当A工作室应用调色时，其效果会与B工作室呈现得完全一致。

6.6.8 纪律

OCIO 还强制实施色彩管理规范。例如，所有纹理贴图在生成mipmap前都必须转换为线性空间。而深度图、法线或运动向量等非图像数据则从不进行任何色彩空间转换，原因很简单：它们不代表色彩。视觉特效和CG动画项目可以在应用程序与工作室之间安全流转，因为所有色彩操作的定义都集中在单一节点，原始创作者的色彩意图得以可靠保留。OCIO的另一关键优势在于它能与ACES系统良好兼容。

6.7 ACES色彩管理

最强大的色彩管理系统当属ACES。它于2014年发布，由美国电影艺术与科学学院科学技术委员会开发，旨在统一和标准化色彩管理实践，使在不同制作环境和发行格式中保持原始创作意图和色彩准确性变得更加容易。它在统一环境中同时支持实拍影像和CG图像。

这是一个从采集到显示的端到端系统，用于管理色彩并保护图像质量。它能够无缝整合多种来源的图像——这在当今制作流程中至关重要，因为ARRI和RED摄影机的素材可能需要与CG特效、Photoshop数字绘景以及来自互联网的JPEG图像结合使用。即使项目没有CG特效且仅使用一台摄影机拍摄，ACES系统对于全面的色彩管理仍然不可或缺。

To date, hundreds of movies have been done in ACES and industry participation increases every year. It is used by all of the major studios, such as ILM and Weta Digital, and is steadily penetrating deeper into the smaller studios. ACES is not just for feature films but is being used for episodic TV and even some TV commercials. The reasons for its popularity are that it is end-to-end color management, uses rock-solid color science, future-proofs the work, and is supported by virtually all professional software providers. Plus, it is free.

ACES consists of two parts: a color encoding specification using a super-wide gamut color space, and a large library of transformations for converting to and from different color spaces, displays, and cameras. In this regard, it is conceptually similar to ICC Profiles. The super-wide color gamut ensures that all image sources, regardless of origin, will fit within it and be accommodated without clipping. It has a linear gamma with 30 stops of exposure range and a D60 white point. The D60 white point was chosen because experiments showed that viewers subjectively preferred it over D65 because it was closer to the white point of the motion picture system. The library of transformations ensures that any capture or display device can be transformed into and out of the ACES color space at the highest possible quality. Further, if any new devices are invented, all that will be needed is a new transformation.

ACES developed the concept of image data being “referred” to the color space that defines that data. We will learn more about this very important concept shortly, but here is a quick preview:

- **Capture referred** - This is the raw image data right out of the camera sensor (Camera Raw). It is “capture referred” because it still has the capturing camera’s fingerprints all over it.
- **Scene referred** - This is the “cleaned” image data that has had all capture influences removed and represents the luminance and color of the original real-world scene. CG renders are considered to be “scene referred” because they are simulations of real-world lighting.
- **Display referred** - This is image data that has been prepared for a specific display device like Rec. 709 for TV or P3 for digital cinema. Compared to scene referred data, it has reduced brightness, dynamic range, and bit depth; plus it has a transfer function (gamma curve) applied to optimize it for a particular display.

迄今为止，已有数百部电影采用ACES制作，行业参与度逐年提升。所有大型工作室，如工业光魔（ILM）和维塔数码（Weta Digital）均在使用该标准，且其正稳步渗透至小型工作室。ACES不仅适用于故事片，也被用于系列电视剧甚至部分电视广告。其受欢迎的原因在于：它提供端到端的色彩管理，采用坚实可靠的色彩科学，能确保作品适应未来技术发展，并得到几乎所有专业软件供应商的支持。此外，它还是免费使用的。

ACES由两部分组成：一是采用超广色域的色彩编码规范，二是用于在不同色彩空间、显示设备和摄像机之间进行转换的大型变换库。从概念上讲，它类似于ICC配置文件。超广色域确保所有图像源，无论来源如何，都能容纳其中且不会发生裁剪。它采用线性伽马，具有30档曝光范围，并采用D60白点。选择D60白点是因为实验表明，观众主观上更喜欢它而非D65，因为它更接近电影系统的白点。变换库确保任何采集或显示设备都能以最高质量转换至ACES色彩空间或从该空间转换出来。此外，如果发明任何新设备，只需新增相应的变换即可。

ACES 提出了图像数据“参考”到定义该数据的色彩空间的概念。我们很快会进一步了解这个非常重要的概念，但这里先快速预览一下：

- 捕获参考 - 这是直接从相机传感器输出的原始图像数据（相机原始数据）。它被称为“捕获参考”，是因为它仍处处带有拍摄相机的特征印记。

- 场景参考 - 这是经过“清理”的图像数据，已去除所有拍摄影响，代表原始现实世界场景的亮度和颜色。CG渲染被视为“场景参考”，因为它们是对现实世界光照的模拟。

- 显示参考 - 这是为特定显示设备（如电视的Rec. 709或数字影院的P3）准备的图像数据。与场景参考数据相比，它具有降低的亮度、动态范围和位深度；此外还应用了传递函数（伽马曲线）以针对特定显示设备进行优化。

ACES has abstracted a color encoding system that is not limited by any equipment or color standard and was developed specifically to integrate live-action film scans with computer-generated images. While the math and color science behind ACES is beyond the scope of this book, understanding what it is and how it works is not. Most of the professional apps that manipulate images support ACES, so after understanding more about it, you should be able to utilize it in the software you use. I encourage you to become friends with ACES and adopt it into your workflow as it is clearly becoming the standard color management system for the entertainment industry worldwide.

6.7.1 The ACES Workflow

We learned earlier that working in linear space is the best way to go. However, while linear is *almost* the ideal workspace, it does have one shortcoming, which is that when cameras photograph a scene, they bake their attributes into their images, like we saw at the top of Chapter 2: Capture Space. This is why the same scene captured by several different camera models will look slightly different: one camera may be slightly more sensitive in the reds, another may record less detail in the shadows, and so on.

When a camera captures a scene, it has "baked in" its personal feelings about color, giving the image data a bias. This biased image data is called "capture referred" and must be backed out so that the data refers to the actual light from the original scene. This is called "scene referred" data. It is linear, float, and HDR, just like ordinary linear space, but with the added distinction of representing the original scene without any camera bias. The linear scene referred data is then transformed into the ACES workspace, which has a huge gamut for its color calculations—far in excess of the human visual system or any physically realizable display device today or in the future. It is the ultimate in future-proofing. ACES is rapidly gaining acceptance in the world of VFX and Digital Intermediate, and it won't be long before you will have to confront it if you haven't already—so let's take a closer look at how it works.

ACES 抽象出了一套不受任何设备或色彩标准限制的色彩编码系统，它是专门为整合实景拍摄胶片扫描与计算机生成图像而开发的。虽然 ACES 背后的数学原理和色彩科学超出了本书的讨论范围，但理解它是什么以及如何运作并不困难。大多数处理图像的专业应用程序都支持 ACES，因此在深入了解它之后，你应该能在自己使用的软件中运用它。我鼓励你与 ACES 成为朋友，并将其纳入你的工作流程，因为它显然正在成为全球娱乐行业的标准色彩管理系统。

6.7.1 ACES工作流程

我们之前了解到，在线性空间中进行处理是最佳选择。然而，尽管线性空间近乎理想的工作环境，它确实存在一个缺点：当相机拍摄场景时，它们会将自身属性融入图像中，正如我们在第二章“捕捉空间”开头所看到的那样。这就是为什么同一场景由几款不同相机型号拍摄后，看起来会略有差异：某款相机可能对红色更敏感一些，另一款可能在阴影处记录较少细节，等等。

当相机捕捉场景时，它会将自身对色彩的“主观感受”融入其中，使图像数据产生偏差。这种带有偏差的图像数据被称为“捕获参考数据”，必须经过校正，使数据回归到原始场景的实际光线状态。校正后的数据称为“场景参考数据”。它具备线性、浮点和高动态范围特性，与普通线性空间类似，但额外强调其去除了相机偏差、真实还原原始场景的特点。线性场景参考数据随后会转换至ACES工作空间——该空间拥有远超人类视觉系统或任何现有及未来物理可实现显示设备的色域范围，专为色彩计算而设计，堪称未来兼容性的终极解决方案。ACES在视觉特效和数字中间片领域正迅速获得认可，无论你是否已接触过它，很快都将直面这一技术体系。现在，让我们深入解析其运作原理。

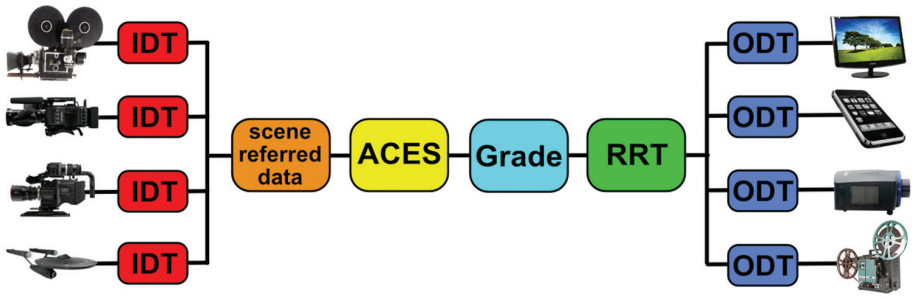


Figure 6.19 The ACES workflow

Figure 6.19 illustrates the ACES workflow. On the far left are various image generation and capture devices. Each manufacturer publishes a profile of the camera or scanner describing its color response characteristics. The **IDT** (Input Device Transform) uses that information to back out the camera's "fingerprint" to produce the **scene referred data**. The scene referred data then goes through a transformation into the **ACES** workspace (wide gamut, linear, float, and HDR) for all image processing and color grading.

The **Grade** and any compositing are done in the ACES linear scene space because this is most like affecting the lights in the original scene. Following the grade, the **RRT** (Reference Rendering Transform**)** imparts a "film look" to the graded data using an "S" curve LUT to give it a 35 mm film print look. The image data then goes to the **ODT** (Output Display Transform), where it becomes "display referred"—meaning that it is converted to the output device's color space and gets the highlight roll-off to bring down code values and avoid clipping. The job of the ODT is to ensure that the image will appear the same regardless of the display device—within the technical limits of those displays, of course. It accounts for the color primaries, white points, and transfer functions of the different display devices.

6.7.2 ACES Gamuts

ACES has two gamuts and a number of special color spaces. The main gamut, ACES 2065-1, uses the AP0 primaries—all imaginary colors—to create the huge ACES gamut described next. The second gamut, ACEScg, uses the AP1 primaries to define a cut-down gamut with far fewer imaginary

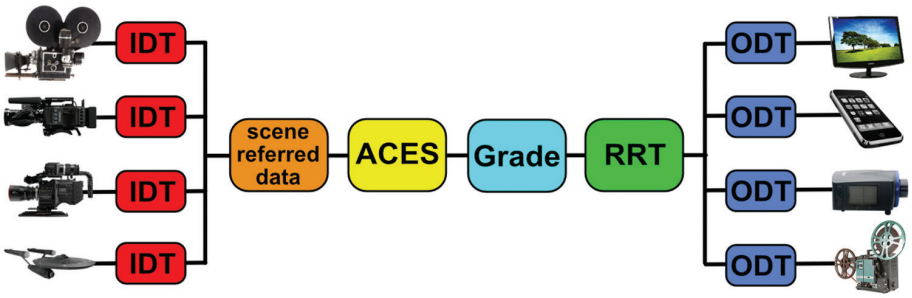


图 6.19 ACES 工作流程

图6.19展示了ACES工作流程。最左侧是各种图像生成与采集设备。每个制造商会发布相机或扫描仪的配置文件，描述其色彩响应特性。IDT（输入设备变换）利用该信息还原相机的“指纹”，以生成场景参考数据。随后，场景参考数据会经过转换进入ACES工作空间（广色域、线性、浮点、高动态范围），用于所有图像处理和色彩分级。

调色及所有合成操作均在ACES线性场景空间中进行，因为这最接近影响原始场景中的灯光。调色完成后，RRT（参考渲染变换）通过"S"曲线查找表为调色数据赋予"胶片质感"，使其呈现35毫米胶片印刷效果。随后图像数据进入ODT（输出显示变换），在此转换为"显示参考"状态——即转译至输出设备的色彩空间，并通过高光滚降处理降低码值以避免裁切。ODT的核心任务是确保图像在不同显示设备上呈现一致观感（当然需在设备技术限制范围内），它会适配不同显示设备的色域原色、白点及传递函数特性。

6.7.2 ACES色域

ACES 拥有两个色域和若干特殊色彩空间。主色域 ACES 2065-1 采用 AP0 原色——均为虚构色彩——以构建下文将描述的庞大 ACES 色域。第二色域 ACEScg 则采用 AP1 原色，定义了一个大幅缩减的色域，其中虚构色彩数量显著减少。

colors designed specifically for work with CG and VFX. Both of these gamuts use the D60 white point.

The ACEScg reduced gamut is also used in several alternate ACES workspaces, each designed for specific production environments. For colorists who want to color grade in a film-like log space, we have ACEScc. You can even add the film toe by working in ACEScct. For an on-set ACES workspace, we have ACESproxy, which provides real-time camera-to-monitor ACES support with preliminary grading. Read all about it in the following sections.

6.7.2.1 ACES 2065-1

In the beginning, there was but one ACES gamut: huge, weird, and wonderful, as shown in Figure 6.20. It was so large that it encompassed not only every other color space gamut, but also the entire human visual gamut, and then some. In order to get this full coverage, two of the ACES primaries (the tips of the gamut triangle) are **imaginary** by being placed outside of the CIE xy-chromaticity diagram. While a mathematically valid thing to do, they obviously do not represent visible colors to the human eye. They also introduce the empty spaces between the gamut lines and CIE diagram, which are filled with **imaginary colors**. Their presence is for the sole purpose of making the math internally consistent.

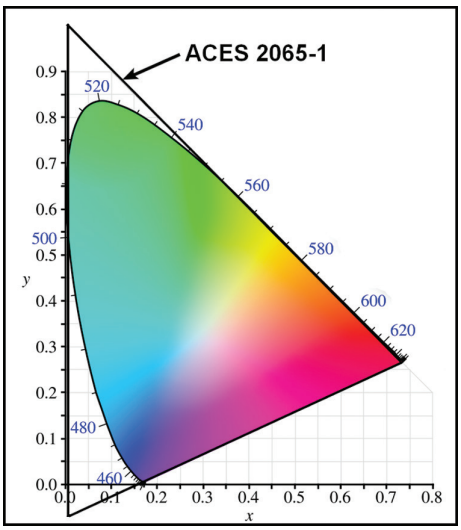


Figure 6.20 The ACES gamut

What are we to make of these imaginary primaries? What do they actually mean? Consider the DCI-P3 gamut in Figure 6.21. Its primaries are comfortably located within the CIE xy-chromaticity diagram, so each primary is located at some xy location which represents the RGB color at that location. Now let's imagine an image in P3 color space that filled some, but not all, of its gamut. The physical location of the three primaries does not bother the

专为CG和VFX工作设计的色彩。这两种色域均采用D60白点。

ACEScg 缩减色域也被用于多个替代 ACES 工作空间，每个空间都针对特定的制作环境而设计。对于希望在类似胶片对数空间中进行调色的调色师，我们提供了 ACEScc。您甚至可以通过在 ACEScct 中工作来添加胶片趾部曲线。对于现场 ACES 工作空间，我们提供了 ACESproxy，它支持实时从摄像机到监视器的 ACES 流程，并包含初步调色功能。您可以在以下章节中阅读所有相关内容。

6.7.2.1 ACES 2065-1

起初，只有一个ACES色域：巨大、奇特而美妙，如图6.20所示。它如此之大，不仅涵盖了所有其他色彩空间的色域，还包括了整个人类视觉色域，甚至更多。为了获得这种全覆盖，ACES的两个基色（色域三角形的顶点）被置于CIE xy色度图之外，因此是虚构的。虽然这在数学上是有效的，但它们显然不代表人眼可见的颜色。它们还在色域边界线与CIE色度图之间引入了空白区域。

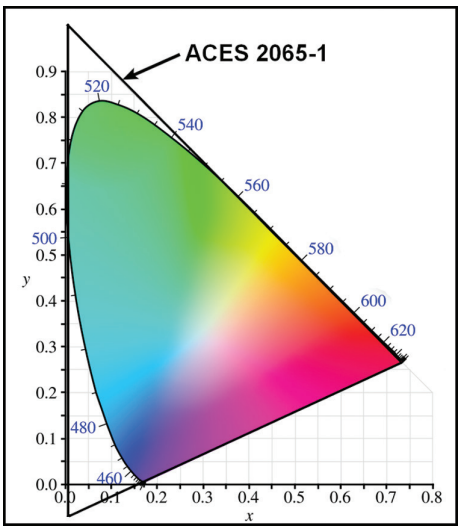


图 6.20 ACES 色域

图表中充满了假想的颜色。它们的存在仅仅是为了使数学在内部保持一致。

我们该如何理解这些虚构的原色？它们实际上意味着什么？考虑图6.21中的DCI-P3色域。它的原色舒适地位于CIE xy色度图内，因此每个原色都位于某个xy位置，该位置代表了该处的RGB颜色。现在，想象一个P3色彩空间中的图像，它填充了部分但非全部的色域。这三个原色的物理位置并不影响

image at all. The primaries simply provide the outline of their gamut and the image lives somewhere inside of that gamut. Now imagine that we grabbed the primary at the top of the triangle and dragged it straight up past the edge of the CIE gamut. It would end up in the imaginary color space and expand the gamut, but the image would be unaltered. Its pixels would still reside at their same xy locations, which would represent the same colors as before we moved the primary.

As it turned out, in practice, the ACES 2065-1 color space was a bit too large for some types of work. While color scientists loved it, artists didn't. It was not artist-friendly, it was hard to interpret, and it had too many imaginary colors. When pixels landed in the imaginary color zones, it became difficult to know how to deal with them. To address these concerns, ACES added several specialty color spaces designed for specific situations.

6.7.2.2 ACEScg

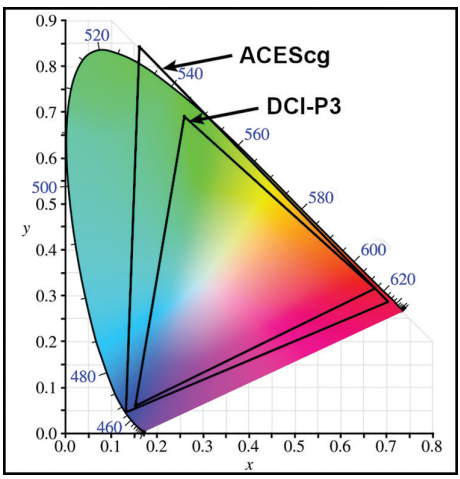


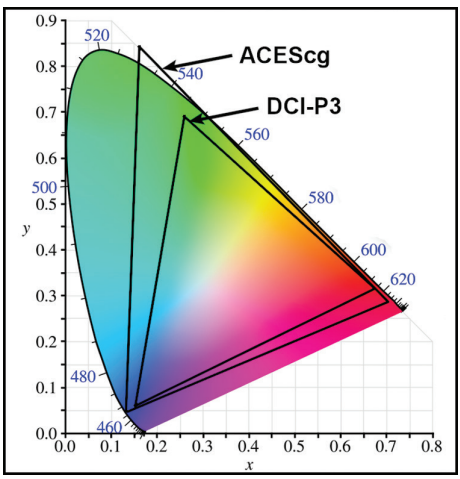
Figure 6.21 The ACEScg gamut

ACEScg (Figure 6.21) was developed specifically to solve the original ACES 2065-1 color space problems outlined above in order to work better for CG. ACEScg was released in December 2014 and is designed to be as large as possible while keeping the imaginary colors to a minimum by using different primaries than ACES 2065-1. Notice that only one primary in Figure 6.21 is slightly out of the CIE gamut. It is still much larger than the DCI-P3 gamut for digital cinema, but only slightly larger than Rec. 2020. ACEScg is ideal for texture mapping, painting, rendering, and compositing. Texture maps are converted to ACEScg before being imported to the renderer, but again, data such as normals, depth maps, and motion vectors are never converted because they are not color data.

完全不是图像本身。原色只是提供了它们色域的轮廓，而图像则位于该色域内部的某个位置。现在想象一下，我们抓住三角形顶部的原色，将其直接向上拖拽，越过CIE色域的边界。它最终会进入虚色空间并扩展色域，但图像本身不会改变。其像素仍将停留在它们原来的xy坐标位置，这些位置所代表的颜色，与我们移动原色之前完全相同。

事实证明，在实践中，ACES 2065-1色彩空间对于某些类型的工作来说有点太大了。虽然色彩科学家喜欢它，但艺术家们并不喜欢。它不够艺术家友好，难以解读，并且含有太多{v*}想象色彩。当像素落入想象色彩区域时，处理它们就变得困难。为了解决这些问题，ACES增加了几种专为特定场景设计的专业色彩空间。

6.7.2.2 ACEScg



ACEScg（图6.21）的研发初衷是为了解决上述原始ACES 2065-1色彩空间存在的问题，从而更好地服务于计算机图形领域。该色彩空间于2014年12月发布，其设计目标是在尽可能扩大色域范围的同时，通过采用与ACES 2065-1不同的基色来最大限度减少虚色。值得注意的是，图6.21中仅有一种基色略微超出CIE色域范围。虽然它仍远大于数字影院使用的DCI-P3色域，但仅比Rec. 2020色域略大一些。

图 6.21 ACEScg 色域

ACEScg是纹理贴图、绘画、渲染和合成的理想选择。纹理贴图在导入渲染器之前会转换为ACEScg，但法线、深度图和运动矢量等数据永远不会被转换，因为它们并非色彩数据。

6.7.2.3 ACEScc

Many colorists prefer to color grade in log space, so ACEScc is a log space just for them. It uses the same gamut as ACEScg, the only difference being that the data has been converted to log to closely approximate film behavior. **ACEScct** is the same color grading log space as ACEScc, but it has a "toe" added to the LUT to simulate actual film stock behavior.

The idea is that a colorist would convert the work to ACEScc for color grading only, then revert back to ACES 2065-1 for output to different display formats and final archiving. It is not a deliverable format. It is a color-grading workspace only.

6.7.2.4 ACESproxy

This is a special ACES workspace designed for on-set use during film production to perform real-time conversion of camera data for monitoring and preliminary grading. It's a DIT's delight. It is designed as a low-bandwidth, low-resolution format to facilitate real-time operations by using integer code values. It is a video range encoded signal to be directly compatible with monitors on set and supports fast preliminary color grading. One of its great elegances is that this preliminary color grade can be imported into the full ACES environment, where grading can be further refined.

6.7.3 VFX with ACEScg

The ACEScg color space was described earlier, so here we will focus on how to integrate it into the VFX workflow. The above ACES workflows are typical for filming, editing, color timing, and distributing a movie, an episodic TV show, or a commercial. But, as mentioned above, one of the prime motivations behind ACES was to seamlessly integrate CG into live action, and this is done by having both the CG and live action in the ACES color space for both viewing and compositing. Since the image data is linear, a Viewer LUT is needed to view it correctly. If using an sRGB monitor, an ACESsRGB LUT is needed, and for a Rec.709 monitor, use an ACESRec709 LUT.

The CG can get into the ACES color space in two different ways. First, it can be rendered in ACES by the 3D app. Second, it can be rendered in a scene referred color space like Rec. 709 and then put through an IDT to convert

6.7.2.3 ACEScc

许多调色师倾向于在log空间中进行色彩分级，因此ACEScc正是为他们设计的log空间。它使用与ACEScg相同的色域，唯一区别在于数据已转换为对数形式，以更贴近胶片特性。ACEScct是与ACEScc相同的色彩分级log空间，但其在LUT中增加了“趾部”曲线，以模拟实际胶片素材的特性。

其理念在于，调色师仅将作品转换为ACEScc格式进行色彩分级，随后再转回ACES 2065-1格式，以输出至不同的显示格式并完成最终归档。它并非交付格式，而仅作为色彩分级的工作空间。

6.7.2.4 ACESproxy

这是一个专为电影制作现场使用而设计的特殊ACES工作空间，用于实时转换摄像机数据以进行监看和初步调色。它是DIT（数字影像技师）的得力助手。该格式采用低带宽、低分辨率设计，通过使用整数值编码来促进实时操作。它是一种视频范围编码信号，可直接与现场监视器兼容，并支持快速初步色彩调整。其一大精妙之处在于，这种初步调色方案可以导入完整的ACES环境中，在此可进一步精细化调整色彩。

6.7.3 使用ACEScg的视觉特效

ACEScg色彩空间之前已经介绍过，这里我们将重点讨论如何将其整合到视觉特效工作流程中。上述ACES工作流程适用于电影、剧集电视节目或广告的拍摄、剪辑、调色和发行环节。但如前所述，ACES的核心动机之一是实现CG与实拍画面的无缝融合，这需要让CG和实拍内容都处于ACES色彩空间中进行观看与合成。由于图像数据是线性的，需要借助查看器LUT才能正确显示。若使用sRGB显示器，需加载ACESsRGB LUT；若使用Rec.709显示器，则需使用ACESRec709 LUT。

CG可以通过两种不同的方式进入ACES色彩空间。首先，它可以通过3D应用程序在ACES中渲染。其次，它可以在场景参考色彩空间（如Rec. 709）中渲染，然后通过IDT转换。

it to ACES for compositing. The best quality is, of course, to render the CG in ACES directly. If the CG is to be rendered in ACES, all texture maps are converted to ACES—a linear and HDR format—*prior* to mipmapping (pyramids). This only applies to maps that represent color, of course. A normal map (surface normals), for example, does not represent color so does not get converted to ACES.

For compositing, all of the different layers need to be converted to ACES color space, the compositing workspace must be linear, and an ACES Viewer LUT must be used for the monitor. If using an sRGB monitor, the ACESsRGB LUT is used, and for Rec. 709 monitors, it's the ACESRec709 that is needed. Once all of the image layers are in ACES, they can be composited together into a single scene. This ACES composited version of the VFX scene is then rendered out as an **ACES-compliant EXR file**, ready to go on to an ACES Digital Intermediate system for color timing and mastering. Writing it out as an ACES-compliant EXR file alerts the downstream app that the data is indeed ACES and not just a generic linear EXR file. If the downstream team is not working in ACES, it still improves quality for the VFX artists to work in ACES internally and then convert back to the original color space for delivery.

6.7.4 OCIO and ACES

OpencolorIO's large library of color space profiles includes the ACES color spaces. In this section, we will see how to use OCIO as a sort of "control panel" for working in ACEScg with virtually any visual effects app—whether that is CG, compositing, painting, editing, or rotoscoping. OCIO is used to convert the incoming images to ACEScg color space, to create a viewer LUT from ACEScg to sRGB, and to export images from ACEScg to Rec. 709 for final delivery, for example. While every software app will implement things differently, all follow this same general workflow steps:

1. **Installation** - If not already installed in your software, install the config.ocio file in the app's designated location from github.com and set an environmental variable to point to it. This tells the software where to look for the OCIO CONFIGURATION FILE that contains all the LUTs and tables.

2. **Rendering** - Set the rendering color space to ACEScg. You are not likely to need the full ACES 2065-1 color space.

将其导入ACES进行合成。当然，最高质量的做法是直接在ACES中渲染CG。若要在ACES中渲染CG，则所有纹理贴图在生成mipmap（金字塔）前都需转换为ACES——这是一种线性HDR格式。这当然仅适用于表示颜色的贴图。例如法线贴图（表面法线）并不表示颜色，因此无需转换为ACES。

在合成过程中，所有不同的图层都需要转换为ACES色彩空间，合成工作空间必须是线性的，并且显示器必须使用ACES Viewer LUT。如果使用sRGB显示器，则需使用ACESsRGB LUT；对于Rec. 709显示器，则需要ACESRec709 LUT。一旦所有图像图层都处于ACES色彩空间中，它们就可以合成为一个单一场景。这个VFX场景的ACES合成版本随后会被渲染为符合ACES标准的EXR文件，准备进入ACES数字中间片系统进行调色和母版制作。将其输出为符合ACES标准的EXR文件可以提醒下游应用程序，该数据确实是ACES格式，而不仅仅是通用的线性EXR文件。即使下游团队不使用ACES工作流程，视觉特效艺术家在内部使用ACES工作，最后再转换回原始色彩空间进行交付，仍能提升工作质量。

6.7.4 OCIO与ACES

OpenColorIO庞大的色彩空间配置文件库中包含了ACES色彩空间。在本节中，我们将了解如何将OCIO作为一种“控制面板”，在几乎所有视觉特效应用中使用ACEScg色彩空间——无论是CG制作、合成、绘画、剪辑还是逐帧描摹。例如，OCIO可用于将输入图像转换为ACEScg色彩空间，创建从ACEScg到sRGB的查看器LUT，并将图像从ACEScg导出为Rec.709格式以供最终交付。尽管每个软件的具体实现方式不同，但都遵循以下通用工作流程步骤：

1. 安装 - 若软件中尚未安装，请从 github .c om 将 config .oc io 文件安装至应用程序的指定位置，并设置环境变量指向该文件。这用于告知软件在何处查找包含所有 LUT 和表格的 OCIO 配置文件。

2. 渲染 - 将渲染色彩空间设置为ACEScg。您通常不需要使用完整的ACES 2065-1色彩空间。

3. **Viewer** - Set the viewer transform to sRGB if using a workstation monitor or Rec. 709 for HDTV monitors. If you work at a sexy big shop, it may have even upgraded to the latest P3 monitors, so tell OCIO that.

4. **Textures** - When importing textures for CG renders, use OCIO to convert them to ACEScg color space and save them to a separate directory. Use only the ACEScg version for rendering. If the files are just data such as normals, they don't get a color space conversion simply because they don't represent color.

5. **Write** - For the write operation to disk of the finished rendered frames, use the OCIO color space conversion from ACEScg to sRGB for workstation/web destined images, to Rec. 709 or Rec. 2020 for video, or DCI-P3 for digital cinema.

6. **Linear** - There are now two "flavors" of linear images—sRGB and ACEScg. If an sRGB image is simply converted to linear, it is an **sRGB linear** image. If that same sRGB image is converted to ACEScg, it is now an **ACEScg linear** image. These are not the same because the sRGB linear file will have used the sRGB primaries and white point, while the ACEScg linear will have used the ACEScg primaries and white point. When outputting EXR files to return to a client after compositing, for example, be sure it is the same type as the client gave you.

6.8 Historical Perspective

Color management has a long and storied history that is inspiring to look back on. Except for the old black-and-white TV standards, every one of these developments has starred somewhere in this book. Here is a timeline of the major milestones in the long march toward comprehensive color management for the modern digital imaging pipeline:

- **1931** - The CIE 1931 color space was one of the first mathematically defined color spaces (CIEXYZ). It was established by the International Commission on Illumination (CIE) by measuring human visual responses and is still fundamental to color science today.

- **1941** - The first black and white analog TV standards were published by the National Television System Committee (NTSC) of the United States, which was sponsored by the Federal Communications Commission (FCC).

3. 查看器 - 若使用工作站显示器，请将查看器转换设置为sRGB；若使用高清电视显示器，则设为Rec. 709。若您在一家实力雄厚的大型机构工作，可能已升级至最新的P3显示器，此时需告知OCIO此配置。

4. 纹理 - 导入用于CG渲染的纹理时，使用OCIO将其转换为ACEScg色彩空间并保存至独立目录。渲染时仅使用ACEScg版本。若文件仅为法线等数据类纹理，则无需进行色彩空间转换，因其本身不代表色彩信息。

5. 写入 - 对于已完成渲染帧的磁盘写入操作，根据目标用途使用OCIO色彩空间转换：工作站/网页用途的图像从ACEScg转换为sRGB，视频用途转换为Rec. 709或Rec. 2020，数字影院用途则转换为DCI-P3。

6. 线性 - 现在线性图像有两种“类型”——sRGB和ACEScg。如果sRGB图像直接转换为线性，它就是sRGB线性图像。如果同一张sRGB图像转换为ACEScg，则成为ACEScg线性图像。这两者并不相同，因为sRGB线性文件会使用sRGB原色和白点，而ACEScg线性文件则使用ACEScg原色和白点。例如，在合成后输出EXR文件返还给客户时，请务必确保其类型与客户提供的文件一致。

6.8 历史视角

色彩管理有着悠久而传奇的历史，回顾起来令人深受启发。除了古老的黑白电视标准外，本书涵盖了所有这些发展的关键节点。以下是现代数字成像流程走向全面色彩管理的漫长征程中，主要里程碑的时间线：

- 1931年 - CIE 1931色彩空间是最早以数学方式定义的色彩空间之一（CIEXYZ）。它由国际照明委员会（CIE）通过测量人类视觉响应而建立，至今仍是色彩科学的基础。

- 1941年——美国联邦通信委员会（FCC）支持的国家电视系统委员会（NTSC）发布了首批黑白模拟电视标准。

- **1976** - The perceptually uniform CIELAB color space improved on the CIE 1931 XYZ color space and was released by the CIE.
- **1990** - HDTV digital video standards were established by the release of Rec. 709 by the International Telecommunications Union (ITU).
- **1994** - ICC Profiles were released by the International Color Consortium (ICC) to provide a comprehensive system of color standards and color transformations.
- **1996** - sRGB was developed by Hewlett-Packard and Microsoft to provide the first color specifications for printers, monitors, and digital cameras. It was designed around the capabilities of CRT monitors.
- **2010** - OCIO—an open-source color management framework for visual effects and animation—was developed by Sony, then later transferred to the Academy Software Foundation (ASF) for maintenance and further development.
- **2012** - Rec. 2020 was released by the ITU to define the second generation of digital television standards with expanded gamut, dynamic range, and 4K and 8K resolutions.
- **2014** - The ACES Color Management system was released by the Academy of Motion Pictures Arts and Sciences to provide standardized color management workflows for film and animation.

The future trends in color management are for ever-more new technologies and developments. As the future of display technologies grows and subdivides, the subject of color management will become even more complicated. My hope is that with the foundational knowledge from this book, you will be well equipped to adapt to whatever the future throws at you.

6.9 Conclusion

This exploration of color science and the digital imaging pipeline has taken you on a transformative ride that started in World Space to uncover the mysteries behind light, cameras, storage, workspaces, and displays. Armed with the knowledge gained from this book, you now possess a deeper understanding of the intricate world of color science. You have unlocked the

- 1976年 - 感知均匀的CIELAB色彩空间改进了CIE 1931 XYZ色彩空间，并由国际照明委员会（CIE）正式发布。
- 1990年 - 国际电信联盟（ITU）通过发布Rec. 709标准，确立了HDTV数字视频规范。
- 1994年 - 国际色彩联盟（ICC）发布了ICC配置文件，旨在提供一个全面的色彩标准和色彩转换系统。
- 1996年 - sRGB由惠普和微软共同开发，旨在为打印机、显示器和数码相机提供首个色彩规范。其设计主要围绕CRT显示器的性能特点展开。
- 2010年 - OCIO——一个用于视觉效果和动画的开源色彩管理框架——由索尼开发，随后移交至学院软件基金会（ASF）进行维护和进一步开发。
- 2012年 - 国际电信联盟发布了Rec. 2020标准，旨在定义第二代数字电视规范，该标准扩展了色域、动态范围，并支持4K和8K分辨率。
- 2014年 - 美国电影艺术与科学学院发布了ACES色彩管理系统，旨在为电影和动画提供标准化的色彩管理工作流程。

色彩管理的未来趋势将不断涌现新技术与发展。随着显示技术的未来不断增长和细分，色彩管理这一主题将变得更加复杂。我希望通过本书的基础知识，您能充分准备好应对未来可能出现的任何挑战。

6.9 结论

对色彩科学与数字成像流程的探索，带您经历了一场从世界空间开始的变革之旅，揭示了光线、相机、存储、工作空间和显示器背后的奥秘。凭借从本书中获得的知识，您现在对色彩科学这一复杂领域有了更深刻的理解。您已解锁了

secrets of emission spectrums, color temperature, and standard illuminants, enabling you to appreciate the complexity of the color that surrounds us.

Delving into Camera Space, you have witnessed the magic of turning light into numbers, with the Bayer array turning photons into color. You now comprehend the limitations and nuances of cameras, equipping you with the power to capture and transform the world around you with more control. The voyage continued into Storage Space, where you mastered the art of preserving your professional images. With an understanding of file formats, codecs, and the hidden dangers they pose, you can now safeguard your artistic creations with confidence.

The Work Space has become your playground as you navigate the realms of color models, color spaces, and transfer functions. Armed with this knowledge, you can unleash your creativity in a controlled, precise manner, choosing the perfect space to preserve your vision. Finally, the grand finale in Display Space has unveiled the wonders of projection systems, monitors, and HDTVs. You stand at the forefront of cutting-edge technology, ready to embrace the future of visual experiences and share your creations with the world.

As you close this final chapter, remember that this journey is only the beginning. Armed with newfound knowledge of the tools of color science and color management, you are now poised to embark on your own extraordinary artistic adventures. Embrace the power of light, let your creativity shine, and paint the world with the vibrant hues of your imagination. The digital imaging pipeline is your canvas, and you are the master of its transformative magic.

Join me and Arthur C. Clarke for a spectacular deep dive into a full-color Mandelbrot set to visualize mathematical infinity. Produced in IMAX as a promo film for the planned documentary, Search for Infinity, it is a visual feast.

发射光谱、色温与标准光源的奥秘，让你领略环绕我们周遭色彩的复杂性。

深入相机空间，您已见证将光线转化为数字的魔力，拜耳阵列将光子变为色彩。如今您理解了相机的局限与精妙之处，从而能更自如地掌控捕捉与重塑周遭世界的能力。旅程延伸至存储空间，在此您掌握了保存专业影像的技艺。通过对文件格式、编解码器及其潜在风险的透彻认知，您现在能充满信心地守护自己的艺术创作。

工作空间已成为你的游乐场，你在这里探索色彩模型、色彩空间和传递函数的领域。掌握了这些知识，你就能以可控而精准的方式释放创造力，选择完美的空间来保存你的视觉构想。最终，显示空间的压轴大戏揭示了投影系统、显示器和高清电视的奇妙之处。你站在尖端技术的最前沿，准备好拥抱视觉体验的未来，并与世界分享你的创作。

当你合上这最后一章，请记住这段旅程仅仅是个开始。掌握了色彩科学与色彩管理工具的新知，你已准备好踏上属于自己的非凡艺术探索之旅。拥抱光的力量，让创造力绽放光芒，用想象中跃动的斑斓色彩描绘世界。数字影像流程是你的画布，而你正是施展其变革魔力的主宰。

与我及亚瑟·C·克拉克一同潜入绚丽多彩的曼德博集合，以可视化数学的无穷之境。这部为计划中的纪录片《寻找无穷》制作的IMAX宣传短片，是一场视觉盛宴。

Color science is an eclectic discipline utilizing math, science, and tech from many fields, so a robust glossary is necessary. Some of the definitions contain words in bold, indicating that the word is also defined in the glossary.

Numerical

8-bit image – A one-channel grayscale 8-bit image or a three-channel RGB image with 8 bits per channel, also referred to as a **24-bit image**.

16-bit image – A three-channel RGB image with 16 bits per channel, also referred to as a **48-bit image**.

24-bit image – A three-channel 8-bit RGB image.

24p Master – a color-graded master copy of the movie for theatrical release from which all other versions are derived.

32-bit image – A four-channel 8-bit RGBA image with an alpha channel.

4:1:1 – Video digitized with four samples of **luminance** and one sample of **chrominance** every 4 pixels.

4:2:0 – Video digitized with four samples of **luminance** and one sample of **chrominance** every 4 pixels.

4:2:2 – Video digitized with four samples of **luminance** and two samples of **chrominance** every 4 pixels.

4:4:4 – Video digitized with four samples of **luminance** and four samples of **chrominance** every 4 pixels.

色彩科学是一门融合数学、科学及多领域技术的折衷学科，因此需要一份详实的术语表。部分定义中**加粗**的词汇表示该术语同样收录于本词汇表内。

数值

8位图像——指单通道的8位灰度图像或每通道8位的三通道RGB图像，也称为24位图像。

16位图像——一种每个通道16位的三通道RGB图像，也称为48位图像。

24位图像——一种三通道8位RGB图像。

24p母版——一种经过色彩分级处理的电影母版拷贝，专为影院发行而制作，所有其他版本均由此衍生而来。

32位图像——一种带有Alpha通道的四通道8位RGBA图像。

4:1:1 – 视频数字化时，每4个像素采用4个亮度样本和1个色度样本。

4:2:0 – 视频数字化时，每4个像素采用4个亮度样本和1个色度样本。

4:2:2 – 视频数字化时，每4个像素采用4个亮度样本和2个色度样本。

4:4:4 – 视频数字化时，每4个像素采用4个亮度样本和4个色度样本。

A

ACES (Academy Color Encoding System) Color Management – A color management system that converts all image data to a large gamut linear light space where the capturing camera influences are removed so that it represents the original scene illumination.

Adaptation – The ability of the eye to adjust to different viewing conditions to maintain a constancy of color regardless of the lighting conditions.

Additive color – The method of creating colors by adding together red, green, and blue primaries.

Administrative metadata – The dates and times an image was created or modified.

Adobe RGB – Designed for print applications to provide a straightforward mapping from RGB to CMYK.

Aliasing – The pixilated, jagged lines that are introduced into a computer image by failing to compute an image with subpixel accuracy. Results in edge pixels that are either on or off instead of smoothly blended.

Ambient occlusion – A one-channel computer-generated layer that contains shading information.

Amplitude – The height of the electromagnetic wave that represents the wave's brightness and is independent of the wavelength.

Anamorphic – A non-square pixel image that is squeezed horizontally when captured and stretched horizontally for projection or display purposes.

Anti-aliasing – The mathematical operation that eliminates aliasing in CG images to give objects smooth edges.

AOV (Arbitrary Output Variable) – A CG render pass that contains data about the image, not the image itself.

ARRIRAW – A proprietary delivery file format of the raw camera captured from the ARRI camera having maximum quality.

AVI (Audio Video Interleave) – A Microsoft Windows multimedia container format for audio and video with a .avi file extension that synchronizes the audio and video on playback.

A

ACES（学院色彩编码系统）色彩管理——一种色彩管理系统，它将所有图像数据转换为大色域线性光空间，在此空间中移除拍摄相机的影响，从而准确再现原始场景的照明效果。

适应——眼睛在不同观看条件下调整以保持颜色恒常性的能力，无论光照条件如何。

加色法——通过将红色、绿色和蓝色原色相加来创建颜色的方法。

管理元数据——图像创建或修改的日期和时间。

Adobe RGB – 专为印刷应用设计，提供从RGB到CMYK的直接映射。

混叠——由于未能以亚像素精度计算图像而引入计算机图像中的像素化、锯齿状线条。导致边缘像素要么全开要么全关，而不是平滑过渡。

环境光遮蔽——一种单通道计算机生成层，包含着色信息。

振幅——电磁波的高度，代表波的亮度，且与波长无关。

变形宽银幕——一种非正方形像素图像，在拍摄时被水平压缩，而在投影或显示时则水平拉伸。

抗锯齿——消除计算机图形图像中锯齿的数学运算，使物体边缘平滑。

AOV（任意输出变量）——一种CG渲染通道，包含关于图像的数据，而非图像本身。

ARRIRAW – 一种专有的交付文件格式，源自ARRI相机拍摄的原始相机数据，拥有最高品质。

AVI（音频视频交错格式）——一种Microsoft Windows多媒体容器格式，用于音频和视频，文件扩展名为.avi，可在播放时同步音频和视频。

B

B-frame (Bi-directionally predictive frame) – A frame that provides motion vector information to the frames before and after it.

Banding – The appearance of steps or contouring in the image over regions that have smooth gradients. Also called **contouring**.

Bayer array – A checkerboard of colored filters sitting on top of the camera's imaging sensor which lets only selected wavelengths of light through to each photosite.

Bit – The smallest unit of digital data; its value is either zero or one.

Bit depth – The number of bits used per channel, also called **color depth**.

Bokeh – In photography, the appearance of point-of-light sources in an out-of-focus area of an image produced by a camera lens using a shallow depth of field. The shape of the bokeh is based on the camera's aperture.

Brain bar – A collection of artists and technicians behind the virtual production set who keep the disk drives, GPUs, and scene databases both synchronous and harmonious.

Brightness – A human being's subjective perception of luminance. Biological and unreliable.

C

Camera RAW – The unprocessed images captured by the imaging sensor are saved in a proprietary format unique to the camera's manufacturer. Also called **capture referred**.

Camera Space – The technology, specifications, and color spaces that define how a camera operates and is used.

Candela – The unit of measure for the brightness of a light source.

Capture Referred – The ACES term for a scene captured by a camera but without the camera's imprint removed.

CCD (Charged Coupled Device) – A global shutter chip with excellent low-light sensitivity and less noise.

B

B帧（双向预测帧）——一种为前后帧提供运动矢量信息{v*}的帧。

条带——在具有平滑渐变的图像区域中出现阶梯状或轮廓状的现象。也称为轮廓效应。

拜耳阵列——位于相机成像传感器顶部的一个彩色滤光片棋盘，它只允许特定波长的光通过每个感光点。

比特——数字数据的最小单位；其值为零或一。

位深度 – 每个通道使用的位数，也称为色彩深度。

散景（Bokeh）——在摄影中，指使用浅景深的相机镜头拍摄时，图像失焦区域中点状光源呈现的视觉效果。散景的形状取决于相机光圈的结构。

大脑吧——虚拟制作场景背后的一群艺术家和技术人员，他们确保磁盘驱动器、GPU和场景数据库既同步又和谐。

亮度——人类对光强的主观感知。具有生物性且不可靠。

C

相机RAW——成像传感器捕获的未经处理的图像以相机制造商独有的专有格式保存。亦称为原始捕获格式。

相机空间——定义相机操作和使用方式的技术、规格和色彩空间。

坎德拉——衡量光源亮度的单位。

捕获参考——ACES术语，指由摄像机捕获但未移除摄像机印记的场景。

CCD（电荷耦合器件）——一种全局快门芯片，具有出色的低光灵敏度和更低的噪声。

CDL (Color Decision List) – The standardized format for the list of color corrections made to footage that can be read by other color-correcting systems.

Channel – One of the image planes in a digital image that makes up the color image.

Chromatic aberration – The fanning out of the color channels near the edges of the frame due to refraction.

Chromatic adaptation – The ability of the eye to adapt to the color of the light source to turn it white to maintain color constancy.

Chrominance (chroma) – The color portion of a video signal that is separated into chrominance (color) and luminance (brightness).

CIE xy-chromaticity diagram – A foundational reference for the color of visible light that uses an xy coordinate system to identify the color and its saturation value for mathematical manipulations.

Cinemascope (Cscope) – A 2.39 aspect ratio projection format with squeezed pixels.

Clipping – When the RGB values exceed the maximum code value supported by the workspace.

CMOS (Complementary Metal Oxide Semiconductor) – A low-cost imaging sensor technology that can also produce the rolling shutter effect.

Codec (coder**/dec**oder**)** – A hardware or software-based lossy or lossless compression technology having two components—an encoder to compress the files for transmission and a decoder to decompress them back to the original for display.

Coherent light – An unnatural light constructed as in a laser where the phase and wavelength are precisely aligned.

Color correction – Altering an image to match another image by adjusting it separately in three brightness zones—the shadows, midtones, and highlights.

Color depth – Determines how fine the color steps are from zero to maximum. Another term for **bit depth**.

Color gamut – The range of possible colors a display can produce, defined by its primary chromaticities (primaries) and the outline drawn between them.

Color interference – When two wavelengths of light interact either constructively to gain amplitude or destructively to lessen amplitude.

CDL（色彩决策列表）——一种标准化的格式，用于记录对素材进行的色彩校正列表，可供其他色彩校正系统读取。

通道——数字图像中构成彩色图像的图像平面之一。

色差 – 由于折射导致画面边缘附近颜色通道呈扇形散开。

色适应——眼睛适应光源颜色，将其转变为白色以保持颜色恒常性的能力。

色度（chroma）——视频信号中分离出的颜色部分，分为色度（颜色）和亮度（明度）。

CIE xy色度图——一种用于可见光颜色的基础参考，它采用xy坐标系来标识颜色及其饱和度值，以便进行数学处理。

Cinemascope (Cscope) – 一种2.39宽高比的投影格式，采用压缩像素。

裁剪——当RGB值超过工作空间支持的最大编码值时。

CMOS（互补金属氧化物半导体）——一种低成本的成像传感器技术，也能产生滚动快门效应。

编解码器（coder/decoder）——一种基于硬件或软件的有损或无损压缩技术，包含两个组件：编码器将文件压缩以进行传输，解码器将其解压缩还原为原始状态以供显示。

相干光——一种非自然光，其构建方式如激光器那样，相位和波长被精确对齐。

色彩校正——通过分别调整图像的三个亮度区域——阴影、中间调和高光，使其与另一张图像相匹配。

色深——决定从零到最大值的颜色阶跃有多精细。位深的另一种说法。

色域——显示器能够产生的可能颜色范围，由其主色度（原色）以及它们之间绘制的轮廓线所定义。

色彩干扰——当两种波长的光相互作用时，要么发生建设性干涉以增强振幅，要么发生破坏性干涉以减弱振幅。

Color Management Module (CMM) – The processing engine of the **ICC Profile** that physically calculates the output color space.

Color perception – The ability of our eyes to respond to select frequency ranges of light that our brains turn into red, green, and blue color sensations.

Color reconstruction – The production of the output RGB image from the camera raw that has the Bayer array baked in.

Color space – A precise definition of a range of colors defined by their primaries, transfer function, and white point.

Color stimulus – A color science term for the observed light reflected from a colored surface lit by a light source.

Color subsampling – A compression scheme that reduces the image size by reducing its chrominance resolution.

Color temperature – The hue of the light source measured in degrees Kelvin (K).

Compression – The process applied to files to minimize their size in bytes.

Conforming – The process of reading an Edit Decision List (EDL) defining a project's edits and then automatically duplicating those edits with the full-resolution media.

Constructive interference – Two light waves that are in phase so their energy stacks up when they merge to create a greater amplitude photon.

Container – A wrapper or a “shell” that contains a **codec**.

Contrast ratio – The ratio of the brightest white to the darkest black that a display can produce.

D

D50—5000K – The white point used in photography and print.

D60—6000K – The white point used for ACES Digital Cinema P3 color.

D65—6500K – The white point for television which corresponds to daylight.

DCI-P3 – The color space of digital cinema.

色彩管理模块（CMM）——ICC配置文件的处理引擎，负责实际计算输出色彩空间。

色彩感知——我们的眼睛对特定频率范围的光线做出反应的能力，这些光线被我们的大脑转化为红、绿、蓝的色彩感觉。

色彩重建——从已嵌入拜耳阵列的相机原始数据生成输出RGB图像。

色彩空间——一种由原色、传递函数和白点定义的色彩范围的精确定义。

颜色刺激——一个颜色科学术语，指在光源照射下从有色表面反射出的观察到的光。

色彩子采样——一种通过降低色度分辨率来减小图像尺寸的压缩方案。

色温——以开尔文度（K）为单位测量的光源色调。

压缩 – 应用于文件以最小化其字节大小的过程。

一致性匹配——读取定义项目剪辑的编辑决策列表（EDL），随后自动使用全分辨率媒体复制这些剪辑的过程。

相长干涉——两束光波相位相同，当它们合并时能量叠加，产生振幅更大的光子。

容器——一种包含编解码器的包装器或“外壳”。

对比度——显示器能够产生的最亮白色与最暗黑色之间的比值。

D

D50—5000K – 摄影和印刷中使用的白点。

D60—6000K – 用于ACES数字影院P3色彩的白点。

D65—6500K – 电视的白点，对应日光。

DCI-P3 – 数字电影的色彩空间。

DCT (Discrete Cosine Transform) – A compression scheme that removes fine detail from an image with the amount of fine detail removal being adjustable.

Debayering – The production of the output RGB image from the camera raw that has the Bayer array baked in.

Deep compositing – Compositing with **deep images**.

Deep data – The name given to the pixels in a deep image that are used for deep compositing, with each pixel holding a volume of data and knowing its depth position in three-space.

Deep images – Images comprised of deep data containing multiple colors, transparency, and depth samples for each pixel.

Depth of field – The range of distances along the axis of an optical instrument, usually a camera lens, through which an object will produce an in-focus image.

Depth pass – A one-channel image named Z (depth.z) that represents the per-pixel distance from the camera to the surface of the geometry.

Descriptive metadata – A title or brief description saved with an image describing its contents.

Destructive interference – When two light waves are 180 degrees out of phase, thereby canceling each other out.

DI (Digital Intermediate) – The process of applying the final creative color graded "look" to a feature film.

Diffuse reflections – The light that is reflected from dull, non-shiny surfaces like cloth that scatter the light in many directions.

Display LUT – A Look Up Table that transforms image data in memory to look correct on a monitor or display.

Display Referred – Image data that has been formatted for a specific display device like Rec. 709 for TV.

Display Space – The devices, technology, and color spaces that comprise the display devices used for viewing images.

DIT (Digital Imaging Technician) – A member of the camera department with responsibilities such as organizing footage, on-set color correction, and signal integrity, as well as liaising between the production and post-production teams.

DCT（离散余弦变换）——一种压缩方案，通过可调节的精细细节去除量来移除图像中的细微细节。

去马赛克——从相机原始数据中生成输出RGB图像的过程，该原始数据已内置拜耳阵列。

深度合成——使用深度图像进行合成。

深度数据——这是指深度图像中用于深度合成的像素，每个像素承载着大量数据，并知晓其在三维空间中的深度位置。

深度图像——由深度数据组成的图像，每个像素包含多种颜色、透明度以及深度样本。

景深——沿光学仪器（通常是相机镜头）轴线的一段距离范围，在此范围内物体将产生对焦清晰的图像。

深度通道——一个名为Z（depth.z）的单通道图像，表示从相机到几何体表面每个像素的距离。

描述性元数据——与图像一同保存的标题或简短描述，用于说明其内容。

相消干涉——当两束光波的相位相差180度时，它们会相互抵消。

DI（数字中间片）——将最终创意调色的“外观”应用于故事片的过程。

漫反射——光线从无光泽、不闪亮的表面（如布料）反射，这些表面将光线散射到多个方向。

显示LUT – 一种查找表，可将内存中的图像数据转换为在监视器或显示器上正确显示的效果。

显示参考 – 为特定显示设备（如电视的Rec. 709）格式化的图像数据。

显示空间——指用于观看图像的显示设备所包含的设备、技术和色彩空间。

DIT（数字影像技术员）——摄影部门的一员，负责整理素材、现场色彩校正和信号完整性维护，并在制作与后期制作团队之间进行联络协调。

DNG – An Adobe file format considered to be a "digital negative" intended to retain the maximum RGB image data. It does not support an alpha channel.

DNxHD (Digital Nonlinear Extensible High Definition) – A video-based proprietary Avid compression standard with a wide range of resolutions, scan modes, and bit rates used for high-end broadcast to online streaming.

DPX (Digital Picture Exchange) – A file format that can hold 10-bit log film scans as well as 10-bit linear high-definition (HD) video frames.

Dynamic range – The luminance range for the darkest to the brightest part of the scene.

E

Emission spectrum – The wavelengths and intensities of light emitted by light sources.

EXR file – A file format developed by Industrial Light and Magic that features 16-bit floating point high dynamic range image data and multiple user-defined channels.

F

Film look – A creative target for images that make them look like they were shot on film.

Flare – When ambient light falls on a screen or inside a lens and fogs up the blacks.

Flicker Fusion Threshold (FFT) – The speed at which a flickering light will appear to be steady.

Focal point – The point at which light converges on the lens.

Footcandles – A measure for the number of lumens per square foot.

Frame rates – Defines how many original images can be displayed per second.

Fresnel Effect – A real-world phenomenon that makes dull, diffuse surfaces shiny based on the viewing angle of the light in the scene.

DNG – 一种Adobe文件格式，被视为“数字底片”，旨在保留最多的RGB图像数据。它不支持alpha通道。

DNxHD（数字非线性可扩展高清）——一种基于视频的Avid专有压缩标准，提供从高端广播到在线流媒体的广泛分辨率、扫描模式和比特率选择。

DPX（数字图像交换）——一种能够存储10位对数胶片扫描以及10位线性高清（HD）视频帧的文件格式。

动态范围 – 场景中最暗部分到最亮部分的亮度范围。

E

发射光谱——光源发出的光的波长和强度。

EXR 文件 – 一种由工业光魔公司开发的文件格式，其特点是采用16位浮点高动态范围图像数据，并支持多个用户自定义通道。

F

胶片感——一种创意目标，使图像看起来像是用胶片拍摄的。

眩光 – 当环境光线照射到屏幕或镜头内部时，会使黑色区域变得模糊。

闪烁融合阈值（FFT）——闪烁光源看起来变为稳定状态时的速度。

焦点——光线在镜头上汇聚的点。

英尺烛光——每平方英尺流明数的度量单位。

帧率——定义了每秒可以显示多少原始图像。

菲涅尔效应——一种真实世界的现象，它根据场景中光线的观察角度，使暗淡、漫反射的表面变得闪亮。

G

Gamma – A power function that takes some input value and raises it to some power and that power is called “gamma.”

Gamut – The full range of colors defined by the primaries of a color space.

Gamut mapping – The process of squeezing a large gamut into a smaller one or stretching a small gamut into a larger one.

Global shutter – When the camera captures the entire frame in the same instant.

H

H.264 – An 8-bit lossy video codec based on the MPEG-4 Part 10 variable encoding rate compression scheme.

H.265 – A second-gen implementation of H.264 offering from 25% to 50% better compression for the same data rate or producing better-looking pictures at the same bit rate.

HDR display – A high dynamic range display that has a contrast ratio of 8000:1 or more and can use data above 1.0.

HDR images – High dynamic range images that contain data over 1.0 and are used for capturing scenes with extremes in brightness.

HDTV 1.33 – A squeezed anamorphic HD video resolution of 1440x1080 which is stretched to fill the full HD frame.

HFR (High Frame Rate) – Film frame rates greater than the standard 24 frames per second.

Histogram – A scope that shows the distribution of brightness of the entire frame of an image.

HLG (Hybrid Log Gamma) – A transfer function that is a blend of gamma and a log curve that attempts to make the same show look good on both standard and high dynamic range display devices.

Holdout matte – A one-channel image that masks off the pixels that are occluded by being behind another object in the scene.

G

Gamma – 一种幂函数，它接收某个输入值并将其提升到某个幂次，该幂次被称为“gamma”。

色域——由色彩空间的原色定义的完整颜色范围。

色域映射——将大色域压缩到小色域，或将小色域拉伸到大色域的过程。

全局快门——当相机在同一瞬间捕捉整个画面时。

H

H.264 – 一种基于MPEG-4 Part 10可变编码率压缩方案的8位有损视频编解码器。

H.265——作为H.264的第二代实现，在相同数据速率下可提升25%至50%的压缩效率，或在相同比特率下生成画质更优的图像。

HDR 显示器 – 一种高动态范围显示器，其对比度达到 8000:1 或更高，并能使用超过 1.0 的数据。

HDR图像——高动态范围图像，其包含超过1.0的数据，用于捕捉亮度极端变化的场景。

HDTV 1.33 – 一种压缩变形的1440x1080高清视频分辨率，通过拉伸以填满整个高清画框。

高帧率（HFR）——指高于标准每秒24帧的电影帧率。

直方图——一种显示图像整个画面亮度分布的示波器。

HLG（混合对数伽马）——一种结合了伽马曲线和对数曲线的传递函数，旨在使同一节目在标准动态范围和高动态范围显示设备上都能呈现良好效果。

保留遮罩——一种单通道图像，用于遮罩场景中被其他物体遮挡的像素。

I

ICC Profiles – A color management tool that converts one color space into another or describes the color characteristics of a monitor or printer.

IDT (Input Device Transform) – The ACES process of using a device's **profile** to back out its "fingerprint" to produce **scene referred** data.

I-frame (Intra-coded frame) – An intact JPEG-type frame that serves as a keyframe.

Incident light – The light that falls on the surfaces of a scene.

Interlaced scan – A video scanning method where the frame is scanned twice from top to bottom for every other scan line to create two fields, which are then integrated to make one frame.

Iron Triangle (a.k.a exposure triangle) – The diagram that shows the correlation between ISO, shutter speed, and aperture size.

ISO – The sensitivity or "speed" of the imaging sensor.

J

JPEG – A compression scheme used on still images.

JPEG 2000 – A wavelet compression scheme with the ability to compress different regions of the frame at different compression levels with fewer artifacts.

Judder – A perceived uneven or jerky video playback arising from the movement of objects from one frame to the next. Also referred to as **motion strobing**.

K

L

LDR images – Low dynamic range images that are either 8-bit, 16-bit, or floating point images containing image data between 0 and 1.0 and used for displays of lesser brightness.

I

ICC配置文件——一种色彩管理工具，可将一种色彩空间转换为另一种色彩空间，或用于描述显示器或打印机的色彩特性。

IDT（输入设备变换）—— ACES流程中利用设备配置文件来反推其“指纹”，以生成场景参考数据的过程。

I帧（帧内编码帧）——一种完整的JPEG类型帧，用作关键帧。

入射光——落在场景表面的光线。

隔行扫描——一种视频扫描方法，其中每帧画面从上至下分两次扫描，每次扫描间隔一行，从而生成两个场，随后这两个场被整合为一帧。

铁三角（又称曝光三角）——展示ISO、快门速度和光圈大小之间关系的示意图。

ISO – 成像传感器的灵敏度或“速度”。

J

JPEG – 一种用于静态图像的压缩方案。

JPEG 2000——一种小波压缩方案，能够以不同的压缩级别对帧的不同区域进行压缩，且伪影更少。

抖动——指物体从一帧移动到下一帧时，视频播放出现感知上的不均匀或卡顿现象。亦称为运动频闪。

K

L

LDR 图像 – 低动态范围图像，指包含0至1.0之间图像数据的8位、16位或浮点图像，用于亮度较低的显示设备。

Lens distortion – An artifact of the camera lens that can alter the shape or color of a captured image.

Linear images – Images that do not have a gamma or log LUT baked into them and have a one-to-one relationship between their code values and luminance.

Log – Short for "logarithmic."

Log images – Images that are represented by log data as opposed to linear data. Each step in the data represents a uniform change of apparent brightness to the eye instead of a uniform change of actual brightness.

Logarithmic – Where the numbers represent exponents of numbers rather than the numbers themselves.

Lumen – The measurement of how much light falls onto a scene's surface.

Luminance – The measurement of the amount of light.

LUT (Look Up Table) – An array of values used to convert data from an input value to a new output value.

Lux – The measurement of how much light is being reflected off a surface.

LZW (Lempel, Ziv, and Welch) – A lossless compression scheme used primarily for graphic images that places the first instance of a pixel color encountered into a table, and all subsequent instances of that pixel are replaced by a pointer to the original pixel in the table.

M

Metadata – Data embedded within an image that describes its resolution, date of capture, ISO, etc. See also "Technical metadata," "Descriptive metadata," and "Administrative metadata."

Metamer – Two different light source emission spectrums that produce the same perceived color.

Mipmap – A CG process of pre-scaling texture maps so that they can be resized rapidly during rendering.

Motion blur – The blurring of an image as the subject moves through the frame during exposure.

镜头畸变——相机镜头的一种伪影，可能改变拍摄图像的形状或颜色。

线性图像——指那些未嵌入伽马或对数查找表，且其编码值与亮度之间存在一对一对应关系的图像。

Log – “对数”的缩写。

对数图像——指由对数数据而非线性数据表示的图像。数据中的每一步代表的是人眼感知亮度的均匀变化，而非实际亮度的均匀变化。

对数式——其中的数字代表的是数字的指数，而非数字本身。

流明 – T 测量有多少光线落在场景的s 表面。

亮度——对光量的测量。

LUT（查找表）——一个用于将数据从输入值转换为新输出值的数值数组。

勒克斯——测量从表面反射的光量的单位。

LZW（Lempel、Ziv和Welch）——一种主要用于图形图像的无损压缩方案，该方案将首次遇到的像素颜色存入表格，之后所有相同像素的实例均替换为指向表格中原始像素的指针。

M

元数据——嵌入在图像中描述其分辨率、拍摄日期、ISO等的数据。另请参阅“技术元数据”“描述性元数据”和“管理元数据”。

同色异谱——两种不同的光源发射光谱，产生相同的感知颜色。

Mipmap – 一种计算机图形学中的预处理技术，通过预先缩放纹理贴图，使其在渲染过程中能够快速调整尺寸。

运动模糊——拍摄对象在曝光期间移动穿过画面时，图像产生的模糊效果。

Motion strobing – The apparent jerkiness in the movement of a subject or the staccato movement across frames of images with moving elements that have insufficient motion blur.

MOV (QuickTime) – A multimedia container developed by Apple supported on Mac and Windows platforms that can hold video, audio, subtitles, timecode, and more. The .mov file is ideal for both editing and final delivery for display.

MP4 (MPEG-4) – A container for video, audio, subtitles, and stills developed by the Motion Picture Expert Group (MPEG). It has progressive downloading and adaptive streaming features, making the .mp4 file popular for streaming services, and is supported on Mac and Windows platforms.

MPEG – A compression scheme used to lower the data rates for moving images.

N

Nits – The measurement of the brightness of a monitor indicating how much light the screen emits.

Normals pass – A CG render pass containing surface normal vectors used to moderate lighting.

O

OCIO (OpenColorIO) – A centralized color management system with a standardized library of LUTs and math operations used by artists to keep color operations consistent for a show between departments and studios.

ODT (Output Display Transform) – The LUT that converts image data to the output device's color space to ensure consistency regardless of the display device.

One light – A primary color grade given to the film or TV project to even up the exposure and eliminate the jarring color difference between scene cuts.

OpenEXR file – See "EXR file."

Overlay mode – A setting on a waveform monitor that shows all three color channels superimposed.

运动频闪——指拍摄对象运动时出现的明显卡顿感，或是画面中移动元素因动态模糊不足而在连续帧间产生的断续跳动现象。

MOV（QuickTime）——由苹果开发的多媒体容器，支持Mac和Windows平台，可容纳视频、音频、字幕、时间码等。.mov文件非常适合编辑和最终展示交付。

MP4（MPEG-4）——由运动图像专家组（MPEG）开发的一种用于视频、音频、字幕和静态图像的容器格式。它具备渐进式下载和自适应流媒体功能，使得.mp4 文件在流媒体服务中广受欢迎，并兼容 Mac 和 Windows 平台。

MPEG – 一种用于降低动态图像数据率的压缩方案。

N

尼特——衡量显示器亮度的单位，表示屏幕发出的光量。

法线通道——一种包含表面法向量的CG渲染通道，用于调节光照效果。

O

OCIO（OpenColorIO）——一个集中式色彩管理系统，拥有标准化的LUT库和数学运算工具，艺术家通过它确保影视项目的色彩操作在不同部门与工作室间保持一致。

ODT（输出显示转换）——将图像数据转换为输出设备色彩空间的查找表，以确保无论显示设备如何，都能保持一致性。

一光——赋予电影或电视项目的一种主色调等级，用于平衡曝光并消除场景切换间刺眼的色彩差异。

OpenEXR 文件 – 参见“EXR 文件”。

叠加模式 – 波形监视器上的一种设置，显示所有三个颜色通道叠加在一起。

P

Parade mode – A setting on a waveform monitor that shows each color channel side-by-side.

Particles – The nature of light that is separate from the light wave and is responsible for the **photoelectric effect**.

P-frame (Predictive frame) – A frame that contains only the changes from the previous frame.

Phase – The timing of the wavelengths measured in degrees, with 360 degrees being one full wavelength.

Photoelectric effect – Occurs when photons of light strike a metal surface and eject electrons. Foundational physics of imaging devices like digital cameras.

Photon – A tiny particle (packet of light) made up of electromagnetic waves that have no mass and no charge.

Photosites – A light receptor on a digital camera imaging sensor.

Pixel – The smallest picture element that a computer can address in an image or display.

Pointer's gamut – The range of colors produced by the real world from diffuse reflective surfaces, not a color space.

Polarization – The orientation of light as it oscillates.

Polarized light – Light waves that all share the same orientation.

Power window – A mask that isolates a color grade to a specific area of the image.

PQ (Perceptual Quantizer) – An "adaptive" transfer function based on human vision that uses the metadata transmitted with the video to optimize its appearance on that display.

Precision – How small the brightness steps are from one code value to the next.

Primary chromaticities – Red, green, and blue primary colors described by their locations in the CIE xy-chromaticity diagram.

P

游行模式——波形监视器上的一种设置，将每个颜色通道并排显示。

粒子——与光波分离并负责光电效应的光的本质。

P帧（预测帧）——一种仅包含与前一帧差异的帧。

相位——以度为单位测量的波长时序，360度代表一个完整的波长。

光电效应——当光子撞击金属表面并击出电子时发生。这是数码相机等成像设备的基础物理原理。

光子——一种由电磁波组成的微小粒子（光包），没有质量和电荷。

感光点——数码相机成像传感器上的光接收器。

像素——计算机在图像或显示器中可以寻址的最小图像元素。

指针色域——由现实世界中漫反射表面产生的颜色范围，而非一个色彩空间。

偏振——光在振荡时的方向。

偏振光——所有光波都具有相同的振动方向。

电动窗口——一种将色彩分级隔离到图像特定区域的遮罩。

PQ（感知量化器）——一种基于人类视觉的“自适应”传递函数，它利用随视频传输的元数据来优化其在显示器上的呈现效果。

精度——从一个码值到下一个码值的亮度步长有多小。

主色度——红、绿、蓝三原色在CIE xy色度图中的位置坐标。

Profile – The color description of a device or color space used for color space transformations.

Progressive scan – In video, a raster scanning pattern that moves from top to bottom without skipping any lines.

ProPhoto RGB – Developed by Kodak for film and photography prints; covers over 90% of the visible color spectrum and 100% of likely occurring natural colors in the real world.

ProRes – A popular high-quality video file format specifically designed for editing by Apple with a family of codes and a range of quality choices.

Proxies – A low-resolution copy of the original footage that is faster to load and operate on.

Q

QuickTime – See "MOV."

R

Reflection pass – A CG render pass that contains only the reflections on a 3D object.

Refresh rate – The maximum times per second a new image can be displayed on a workstation monitor or laptop.

Resolution – The number of pixels in the image both horizontally and vertically.

RLE (Run Length Encoded) – A lossless compression scheme primarily used for graphics with large areas of the same color and limited variation.

Role – An OCIO prepared color operation given a unique name so the artist can invoke it by name and not worry about how it works.

Rolling shutter – An artifact produced by digital cameras caused by scanning the image sensor from top to bottom that introduces bizarre distortions in moving objects.

RRT (Reference Rendering Transform) – In ACES, the "film look" that is imparted to the graded data using an "S" curve LUT to give it a 35mm film print look.

配置文件 – 用于色彩空间转换的设备或色彩空间的颜色描述。

逐行扫描——在视频中，一种从上到下移动且不跳过任何行的光栅扫描模式。

ProPhoto RGB – 由柯达为胶片和摄影打印开发；覆盖超过90%的可见光谱以及现实世界中可能出现的100%自然色彩。

ProRes – 一种流行的高质量视频文件格式，由苹果公司专门为编辑设计，拥有系列编解码器及多种画质选择。

代理——原始素材的低分辨率副本，加载和操作速度更快。

Q

QuickTime – 参见“MOV”。

R

反射通道 – 一种仅包含3D物体上反射效果的CG渲染通道。

刷新率 – 指工作站显示器或笔记本电脑每秒能够显示新图像的最大次数。

分辨率 – 图像在水平和垂直方向上的像素数量。

RLE（行程长度编码）——一种无损压缩方案，主要用于具有大面积相同颜色且变化有限的图形。

角色 – 一个由OCIO准备的色彩操作，被赋予唯一名称，以便艺术家可以通过名称调用它，而无需担心其工作原理。

卷帘快门——一种由数码相机产生的伪像，其成因是从上至下扫描图像传感器，这会在移动物体中引入奇异的畸变。

RRT（参考渲染变换）——在ACES中，通过"S"曲线LUT对调色数据施加的"胶片观感"，使其呈现35毫米胶片印刷的视觉效果。

S

Scattering – Light waves are sent in random directions after impacting a particle when traveling through a medium like water or air.

Scene referred – Camera data that has had the camera's "fingerprint" removed so the data represents the original scene illumination.

SDR Display – A standard dynamic range display with a contrast ratio of less than 1,000:1.

Secondaries – In an RGB color model, they are the cyan, magenta, and yellow colors between the primary red, green, and blue colors.

Sensor crop – Where only a portion of the imaging sensor is used to capture an image.

Shutter – The electronic process that allows light onto the imaging sensor, thereby controlling motion blur and depth of field.

Spectral reflectance – The wavelengths of light that are reflected from an object's colored surface.

Spectral response – To respond to light based on its wavelength.

Spectral sensitivity – How sensitive a camera's imaging sensor is to specific wavelengths of light.

Specular pass – A CG render pass that contains only the specular highlights of a 3D object.

Specular reflections – The light that reflects off a highly shiny surface like a mirror at the same angle from which it came.

Spherical aberration – A defocused glow around the brightest elements of a scene concentrated around the outer edges of the frame.

sRGB – The official color space for computer monitors that share the same primary chromaticities as Rec. 709 for HDTV but have a different gamma and contrast ratio.

Standard illuminants – A standard of defined light sources based on their color temperature.

Stops – A doubling or halving of the light for a scene's exposure and a shorthand for quantifying relative exposure.

S

散射——当光波通过水或空气等介质传播时，撞击粒子后会朝随机方向发散。

场景参考——已移除相机“指纹”的相机数据，因此该数据代表原始场景照明。

SDR 显示器——一种标准动态范围显示器，其对比度低于 1,000:1。

二次色——在RGB颜色模型中，它们是位于原色红、绿、蓝之间的青色、品红色和黄色。

传感器裁剪——仅使用成像传感器的一部分来捕获图像。

快门——允许光线进入成像传感器的电子过程，从而控制运动模糊和景深。

光谱反射率——从物体彩色表面反射的光的波长。

光谱响应——根据光的波长来响应。

光谱灵敏度——指相机成像传感器对特定波长光的敏感程度。

高光通道 – 一种CG渲染通道，仅包含3D物体的镜面高光部分。

镜面反射——光线从高度光滑的表面（如镜子）以与入射角相同的角度反射出去。

球面像差——场景中最亮元素周围的一种失焦光晕，集中在画面外边缘。

sRGB——计算机显示器的官方色彩空间，它与HDTV的Rec. 709标准共享相同的主色度，但具有不同的伽马值和对比度。

标准光源——基于色温定义的标准化光源。

档位——场景曝光时光量的加倍或减半，以及量化相对曝光的简略表达方式。

Storage Space – The devices, technology and file formats used for the storage of image data.

Subtractive color – The method of creating color with inks, dyes, and paint that produces RGB colors by subtracting from white light. Uses CMYK colors.

SWOP – A quasi-color space that utilizes the CMYK color model for print.

T

Technical metadata – Information about the camera and its settings during image capture, which can also contain GPS information.

Tone mapping – Adjusts the luminance of a high dynamic range image to fit within a lower dynamic range while retaining the original visual intent.

Transfer function – Loosely referred to as the "gamma"; defines the relationship between the digital code values and their brightness on the screen.

U

UHDTV (Ultra-High Definition TV) – A new TV standard that has doubled and quadrupled the resolution of HDTV, with much greater brightness and dynamic range.

Unreal Engine – The world's most advanced real-time 3D creation tool for photoreal visuals and immersive experiences.

V

Vectorscope – A scope that shows the hue and saturation of an image being displayed using a radial color wheel type of display.

Veiling glare – The haze of light added over a photographed image due to non-parallel light rays, lens imperfections, internal reflections, and other optical anomalies.

存储空间——用于存储图像数据的设备、技术和文件格式。

减色法——通过从白光中减去某些颜色来产生RGB色彩，使用墨水、染料和颜料创建色彩的方法。采用CMYK色彩体系。

SWOP – 一种利用CMYK色彩模型进行印刷的准色彩空间。

T

技术元数据——关于相机及其在图像捕获过程中的设置信息，其中也可能包含GPS信息。

色调映射——调整高动态范围图像的亮度，使其适应较低的动态范围，同时保留原始视觉意图。

传递函数——通常被称为“伽马”；定义了数字码值与其在屏幕上亮度之间的关系。

U

UHDTV（超高清电视）——一种新的电视标准，其分辨率比HDTV提高了一倍和四倍，并拥有更高的亮度和动态范围。

虚幻引擎——世界上最先进的实时3D创作工具，用于制作照片级真实视觉效果和沉浸式体验。

V

矢量示波器——一种采用径向色轮式显示器，展示图像色调和饱和度的示波器。

杂散光——由于非平行光线、镜头缺陷、内部反射和其他光学异常，在拍摄图像上叠加的光线雾霾。

Velocity pass – Contains three-channel vectors (XYZ) indicating per-pixel speed and direction for adding motion blur in post.

Viewing surround – The lighting environment in which a given display is supposed to be viewed.

Vignetting – The darkening at the corners of an image.

Virtual production – A "light stage" made up of LED panels that surround the talent on two or three sides upon which backgrounds are displayed and then filmed.

Visible light – Those wavelengths of light for which we humans have evolved photoreceptors, roughly between 400 nanometers (nm) and 700 nm.

W

Waveform monitor – A scope that shows the brightness level or exposure of the RGB channels of the entire frame.

Wavelength – The distance from peak to peak in a light wave that is measured in nanometers.

WCG (Wide Color Gamut) – The vast new Rec. 2020 color space with extreme primaries for display devices like UHDTV.

White point – What is considered white in a colorspace, also located on the CIE xy-chromaticity diagram.

Work Space – The mathematical and colorspace environment in which all image processing and color grading is done.

World Space – Where the original scene is captured.

X

Y

YCbCr – A color space that converts RGB values to Y (**luminance**) and Cb (blue-difference) and Cr (red difference) **chroma** components.

速度通道 – 包含三通道矢量（XYZ），用于指示每个像素的速度和方向，以便在后期添加运动模糊效果。

观看环境——指预期观看特定显示器的照明环境。

暗角 – 图像角落变暗的现象。

虚拟制作——一种由LED面板组成的“光舞台”，将演员围绕在两到三面，背景显示在这些面板上并进行拍摄。

可见光——即人类进化出感光细胞所能感知的光波波长，大致在400纳米（nm）至700纳米之间。

W

波形监视器——一种显示整个帧的RGB通道亮度水平或曝光的示波器。

波长——光波中波峰到波峰的距离，以纳米为单位测量。

WCG（广色域）——全新的Rec. 2020色彩空间，为UHDTV等显示设备提供了极端的原色范围。

白点——在色彩空间中什么被认为是白色，同样位于CIE xy色度图上。

工作空间 – 所有图像处理和色彩分级在其中进行的数学与色彩空间环境。

世界空间——原始场景被捕捉的地方。

XY

YCbCr – 一种将RGB值转换为Y（亮度）以及Cb（蓝色差）和Cr（红色差）色度分量的色彩空间。

Z

Zebra stripes – White lines on a camera's viewfinder display screen that indicate where the image is overexposed or is out of gamut so it can be corrected.

Zone control – Adjusting an image restricted to the shadows, midtones, or highlights.

Z

斑马条纹——相机取景器显示屏上的白色线条，用于指示图像过曝或超出色域范围，以便进行校正。

区域控制——调整仅限于阴影、中间调或高光的图像。

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